

The Observation Theory Encyclopedia

Companion to *Data Mining as Observation*, 239 entries

The observation theory program, ahb-sjsu

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Preface

This is the encyclopedia of the observation theory program, the companion to the textbook *Data Mining as Observation*. It has 239 entries, one for every headword of the book's glossary and three that were filled by hand before the generator existed. Of the entries, 157 are concepts, 71 are instruments, 7 are results, 3 are corrections, and 1 is a reference.

Where the entries come from. An entry is built from the program's records rather than written beside them. The claims ledger in the repository geometric-observation is the authority for what a result is worth. A campaign track file is the authority for what was measured. The book's sources tables are the authority for every number the book states. The generator, `encyclopedia/generate.py`, gathers what those records say about one thing, keyed by their identifiers, so that a correction in a record propagates to the entry on the next rebuild and never has to be chased by hand. The curated parts of an entry, its definition, its conditions, and the list of related entries, live in `encyclopedia/entries.toml`.

How to read an entry. Every entry carries the same headings in the same order. A heading with nothing to say keeps the word none, so that a missing field and an empty field cannot be confused.

Field	What it holds	Source of truth
id	the file name without extension, lowercase, hyphenated, never reused	this directory
kind	one of concept, instrument, result, correction	this directory
definition	one or two sentences a reader can act on, with the symbols the equation uses	the volume or paper that first stated it

Field	What it holds	Source of truth
equation	the displayed form, with the book equation number where the book states it	the volume, the book
ledger	every claims-ledger row that bears on the entry, with its class in brackets, [proved] [demonstrated] [replicated] [predicted] [exploratory] [refuted]	geometric-observation/claims/LEDGER.md
first stated	where the idea first appears in the record, with a DOI or commit	papers, volumes
measurements	a table of the numbers, each with the file and line range and the commit it was read at	campaign tracks, experiment results
failures and corrections	every retraction, refuted bar, vacuous pass, and erratum that touches the entry, dated, at the same prominence as the measurements	ledger, ERRATA.md files
conditions	the assumptions under which the theorem or law holds, stated so a reader can tell when it does not apply	the volume, the book's revision notes
machine checked	the Lean file and theorem names, or none	observation-data-mining/lean/, ot-lean
used in	papers by DOI, book chapters by number, instruments by repository	the book's sources tables and index
related	other entry ids	this directory
status	the date the entry was last rebuilt or checked, and the commits of the records it was read from	this directory

Conventions. A ledger row carries its class in brackets, proved, demonstrated, replicated, predicted, exploratory, or refuted, and a refuted row sits in the entry at the same size as the results it bounds. A number names its repository, file, line range, and commit, exactly as the book's sources tables do, and a number without a row was removed. The machine-checked heading names the Lean file and theorems that check the entry's core against Mathlib, built with no sorry and the standard axioms only, and the book's appendix C says what each check does and does not cover. Book equation numbers, chapter numbers, and section numbers refer to *Data Mining as Observation*, draft 0.2.

How to cite. Cite the repository `ahb-sjsu/observation-theory-campaigns` at commit `43db613`, the entry by its id, and, for a number, the source row the entry gives. The provenance section at the end names the commit of every record the entries were read from.

Cross-references. Each entry ends with its related entries, and each reference carries the page it is found on. The back matter lists the entries by kind, by chapter of the book, by the Lean file that checks them, and by the ledger row that bears on them.

A

abstention

concept. Entry id abstention.

DEFINITION

The verdict a group receives when it has too few rows or too few queries to score. It is reported as a verdict, not dropped. Chapter 10.

EQUATION

Book equation 10.7.

$$\text{verdict} = \min_{i: n_i \geq n_{\min}, |Q_i| \geq q_{\min}} \text{score}_i, \quad \text{ABSTAIN otherwise.}$$

LEDGER

- NEG-14. (GO-P-2026-037, prospective) `a2_probe.median_unit_displacement` is a single-statistic predictor of the flip regime (`unit_disp` $\geq 1.0 \Rightarrow$ generic-polar-flip vs needs-blind-probe). [refuted]. `geometric-observation/claims/LEDGER.md:108` at 9f3829f.

FIRST STATED

The strata design in `turboquant-pro`, `turboquant-pro/docs/STRATA_RFC.md:24-98`, and chapter 10 section 10.4 of *Data Mining as Observation*.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 10 section 10.3	abstain below 2.5 k	<code>turboquant-pro/docs/HUBNESS_PRIMER.md:140-160</code>
chapter 10 section 10.4	area map, boundary rule, hash, refuse not warn, intra and transit counts, area classes, abstention rule	<code>turboquant-pro/docs/STRATA_RFC.md:24-98</code>
chapter 10 section 10.4	second prediction inverted, transit 0.389 vs 0.291, centrality difference signs, share 0.473 vs 0.391, seven of thirteen backbone	<code>turboquant-pro/docs/RESULTS_multilingual_strata.md:55-90</code> ; <code>turboquant-pro/docs/STRATA_RFC.md:98-130</code>

FAILURES AND CORRECTIONS

- NEG-14, [refuted]. (GO-P-2026-037, prospective) `a2_probe.median_unit_displacement` is a single-statistic predictor of the flip regime (`unit_disp` $\geq 1.0 \Rightarrow$ generic-polar-flip vs needs-blind-probe).

CONDITIONS

- A stratum with too few rows or too few queries to score receives the verdict abstain, which is reported under a registered cause and excluded from the minimum. Abstain is not a pass, and when no stratum is eligible the verdict itself is abstain.
- Raising the eligibility bar can only remove strata from the minimum and so can only raise the verdict, which is why every abstention is printed beside the verdict rather than dropped.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

lean/DataMiningAsObservation/Abstention.lean, theorems abstain_not_passes, verdict_eq_none_iff, passes_verdict_iff, inf_le_of_subset, at observation-data-mining af776fd.

USED IN

Data Mining as Observation chapters 0, 10, 11, 14.

RELATED

[min-over-strata](#) (p. 217); [anti-hub](#) (p. 17); [certificate](#) (p. 55); [coverage](#) (p. 86).

aggregation

concept. Entry id aggregation.

DEFINITION

Combining rows or groups into one number. An average lies between its members, and an aggregate over strata can pass a bar that one stratum fails and can reverse the sign every group shows. Chapter 2 section 2.7 and chapter 10 section 10.4.

EQUATION

Book equation 10.7.

$$\text{verdict} = \min_{i: n_i \geq n_{\min}, |Q_i| \geq q_{\min}} \text{score}_i, \quad \text{ABSTAIN otherwise.}$$

Book equation 14.5.

$$\text{FC}_g = \Pr[y = \text{violation} \mid \hat{y} = \text{clear}, g], \quad \text{verdict} = \max_{g: n_g \geq n_{\min}} \text{FC}_g, \quad \text{ABSTAIN otherwise.}$$

Book equation 8.2.

$$\widehat{\text{Var}}_{\text{NB}} = \left(\frac{1}{J} + \frac{n_{\text{test}}}{n_{\text{train}}} \right) \hat{\sigma}^2,$$

$$J = 1000, \frac{n_{\text{test}}}{n_{\text{train}}} = \frac{1}{4} \Rightarrow 1 + 250 = 251, \sqrt{251} = 15.84.$$

LEDGER

- NEG-14. (GO-P-2026-037, prospective) a2_probe.median_unit_displacement is a single-statistic predictor of the flip regime (unit_disp $\geq 1.0 \Rightarrow$ generic-polar-flip vs needs-blind-probe). [refuted]. geometric-observation/claims/LEDGER.md:108 at 9f3829f.
- GO-B-Llama. Trained frontier LLM (Llama-3.2-3B), softmax-attention consumer — blind probe on real post-RoPE keys [predicted]. geometric-observation/claims/LEDGER.md:115 at 9f3829f.
- GO-B-Llama-rematch. Trained frontier LLM (Llama-3.2-3B), softmax-attention consumer — reconnected dissociation on real post-RoPE keys [predicted]. geometric-observation/claims/LEDGER.md:118 at 9f3829f.

FIRST STATED

Chapter 2 section 2.7 and chapter 10 section 10.4 of *Data Mining as Observation*, with the aggregate staleness rate of chapter 13 section 13.3.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 2 section 2.7	the aggregate staleness rate describing neither reader	chapter 13 of this book, section 13.3
chapter 10 section 10.2	anti-hubs as where compressed indexes fail first, aggregate recall barely moves	turboquant- pro\docs\HUBNESS_PRIMER.md:86-131
chapter 10 section 10.4	area map, boundary rule, hash, refuse not warn, intra and transit counts, area classes, abstention rule	turboquant- pro\docs\STRATA_RFC.md:24-98
chapter 10 section 10.4	second prediction inverted, transit 0.389 vs 0.291, centrality difference signs, share 0.473 vs 0.391, seven of thirteen backbone	turboquant- pro\docs\RESULTS_multilingual_strata.md:55-90; turboquant- pro\docs\STRATA_RFC.md:98-130
chapter 11 section 11.6	anti-hub recall, p05, hub-rank correlation, hub-set overlap, the build gate	turboquant- pro\docs\HUBNESS_PRIMER.md:86-131

FAILURES AND CORRECTIONS

- NEG-14, [refuted]. (GO-P-2026-037, prospective) `a2_probe.median_unit_displacement` is a single-statistic predictor of the flip regime ($\text{unit_disp} \geq 1.0 \Rightarrow \text{generic-polar-flip vs needs-blind-probe}$).

CONDITIONS

- Combining rows or groups into one number, a mean, a sum, or a rate. An average lies between the smallest and largest member, a weighted aggregate over strata can pass a bar while one stratum fails it, and an aggregate can reverse the sign every group shows.
- Aggregate recall barely moves while anti-hubs fail, the aggregate staleness rate describes neither reader, and the report takes the worst group and counts the groups too thin to score.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/Ensemble.lean`, theorems `average_between`, `sq_average_le`, `average_const`, at `observation-data-mining af776fd`.

`lean/DataMiningAsObservation/Simpson.lean`, theorems `pooled_eq_weighted`, `reversal`, `no_reversal_of_equal_sizes`, at `observation-data-mining af776fd`.

`lean/DataMiningAsObservation/RecallAtK.lean`, theorems `recallAtK_mem_unit`, `recallAtK_eq_one_iff`, `aggregate_le_of_failing`, `aggregate_example`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 2, 5, 8, 10, 11, 12, 13.

RELATED

[min-over-strata](#) (p. 217); [Simpson's paradox](#) (p. 325); [stratification](#) (p. 339); [anti-hub recall](#) (p. 19); [sampling](#) (p. 313).

alignment

concept. Entry id alignment.

DEFINITION

The normalized overlap between a consumer's average read operator and the covariance of the data on the read subspace, written κ , between zero and one. Near one the read subspace and the high-variance subspace coincide and no flip is possible. Chapter 4, equation 4.6.

EQUATION

Book equation 4.6.

$$\kappa = \text{tr} \left(\bar{P}_C \bar{\Sigma}_x \right) \in [0, 1], \quad \kappa \rightarrow 1 \text{ is the coupling null (no flip).}$$

LEDGER

- GO-4. Fixed-budget verdicts invert under budget-matched observation, per the wavelength mechanism. [replicated]. [geometric-observation/claims/LEDGER.md:66 at 9f3829f](#).

FIRST STATED

Volume 14, chapter 12, [geometric-observation/chapters/ch12_failure_taxonomy_and_kappa.md:56-82](#), DOI 10.5281/zenodo.21776291, where the alignment law is a retrospective fit carried as exploratory with one prospective point.

MEASUREMENTS

Where the book states it	Numbers, as the book’s sources table records them	Source
chapter 4 section 4.4	alignment law, retrospective fit exploratory, one prospective point	geometric-observation\chapters\ch12_failure_taxonomy_and_kappa.md:82
chapter 4 section 4.4	GO-4 budget inversion, fixed m 10 rises, matched m 121, 126, 159 collapses, 3 seeds	geometric-observation\claims\LEDGER.md row GO-4

FAILURES AND CORRECTIONS

none

CONDITIONS

- Defined as the normalized overlap between the averaged read operator and the source covariance on the read subspace, between zero and one, and so a statement about one workload.
- The alignment law that relates it to the size of the flip is a retrospective fit with a single prospective point and is carried as exploratory, not as a theorem.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/Alignment.lean`, theorems `overlap_sq_le`, `alignment_le_one`, `alignment_nonneg`, `alignment_eq_one_of_proportional`, `coupling_null`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 2, 4, 7, 11, 14.

RELATED

[the flip](#) (p. 138); [read distortion](#) (p. 281); [coupling null](#) (p. 84); [water-filling](#) (p. 365).

allocation

instrument. Entry id allocation.

DEFINITION

The assignment of a bit budget across directions. Water-filling gives directions below the water no bits, and the allocation report warns below effective rank two. Chapter 4 section 4.2.

EQUATION

Book equation 4.2.

$$D(b) = \sum_i s_i \sigma_i^2 2^{-2b_i}, \quad s_i = v_i^\top P_C v_i, \quad b_i^* = \max\left(0, \frac{1}{2} \log_2 \frac{s_i \sigma_i^2}{\theta}\right), \quad \sum_i b_i^* = B,$$

Book equation 0.7.

$$r_{\text{eff}} = \frac{(\sum_i \lambda_i)^2}{\sum_i \lambda_i^2}.$$

Book equation 4.6.

$$\kappa = \text{tr}(\bar{P}_C \bar{\Sigma}_x) \in [0, 1], \quad \kappa \rightarrow 1 \text{ is the coupling null (no flip).}$$

LEDGER

- GO-4. Fixed-budget verdicts invert under budget-matched observation, per the wavelength mechanism. [replicated]. `geometric-observation/claims/LEDGER.md:66` at 9f3829f.
- GO-6. At matched rate, output coding $\leq$ surrogate $\leq$ reconstruction on the consumer metric; the output-reconstruction gap is governed by the $\ker P_C$ entropy share, and the surrogate-output gap vanishes as rate grows. [demonstrated]. `geometric-observation/claims/LEDGER.md:68` at 9f3829f.

FIRST STATED

Chapter 4 section 4.2 of *Data Mining as Observation*, with the allocation report in `turboquant-pro/turboquant_pro/read_allocation.307`.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 4 section 4.2	water-filling formula, directions below the water get no bits, the surrogate caveat	<code>readscope\readscope\allocate.py:1-100</code>
chapter 4 section 4.2	allocation report, gain over uniform, concentration caution below effective rank 2	<code>turboquant-pro\turboquant_pro\read_allocation.py:244-307</code>
chapter 10 section 10.5	fragile-first allocation 0.7118 vs 0.7251, targets up 0.004 to 0.034, redesigned allocator passed	<code>turboquant-pro\docs\RESULTS_strata_phase23_gates.md:110-125</code>

FAILURES AND CORRECTIONS

none

CONDITIONS

- The assignment of a bit budget across directions. Water-filling gives directions below the water no bits, the allocation report states its gain over uniform and warns below effective rank two, and a spectrum with half its mass in one direction has effective rank at most four.
- The fragile-first allocation read 0.7118 against 0.7251 and was redesigned, and the budget inversion of GO-4 shows an allocation at fixed m rising while the matched-m allocation collapses.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

lean/DataMiningAsObservation/WaterFilling.lean, theorems contribution_eq_water, terms_mul, two_sqrt_le, distortion2_ge, distortion2_eq_of_equal, at observation-data-mining af776fd.

lean/DataMiningAsObservation/EffectiveRank.lean, theorems sq_sum_le, effRank_le, sum_sq_le_sq_sum, one_le_effRank, effRank_const, effRank_single, at observation-data-mining af776fd.

USED IN

Data Mining as Observation chapters 0, 1, 2, 4, 10, 11.

RELATED

[water-filling](#) (p. 365); [budget](#) (p. 46); [effective rank](#) (p. 115); [concentrated](#) (p. 69); [bit](#) (p. 38).

anisotropic

concept. Entry id anisotropic.

DEFINITION

Of a covariance, having different variances in different directions. Chapter 0.

EQUATION

Book equation 0.5.

$$\Sigma v_i = \lambda_i v_i, \quad \Sigma = \sum_{i=1}^d \lambda_i v_i v_i^\top, \quad v_i \cdot v_j = 0 \ (i \neq j).$$

Book equation 3.1.

$$d_O(u) = u^\top \Sigma u, \quad u = (\cos 15^\circ, \sin 15^\circ), \\ \Sigma_1 = \text{diag}(0.3, 1.7), \Sigma_2 = \text{diag}(1.7, 0.3), \quad d_O = 0.394 \text{ vs } 1.606.$$

Book equation 4.6.

$$\kappa = \text{tr}(\bar{P}_C \bar{\Sigma}_x) \in [0, 1], \quad \kappa \rightarrow 1 \text{ is the coupling null (no flip).}$$

LEDGER

- GO-2 (neg. half: not reconstruction). At matched bits, downstream preservation is not controlled by reconstruction error. [demonstrated]. [geometric-observation/claims/LEDGER.md:63](#) at 9f3829f.
- GO-B-legal (035→036). Legal-citation retrieval (CourtListener), cosine-ranking consumer, LaBSE embeddings — real large corpus, non-physical consumer [predicted]. [geometric-observation/claims/LEDGER.md:119](#) at 9f3829f.

FIRST STATED

Chapter 0 section 0.4 of *Data Mining as Observation*, with the program's centring measurement in [geometric-observation/claims/LEDGER.md](#) row GO-B-legal and the anisotropy channel in [lebse/README.md:40-75](#).

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 11 section 11.8	0.765 to 0.971 with interval 0.190 to 0.223, 0.545 to 0.562 with interval 0.004 to 0.031, v1 0.340 with interval negative 0.214 to negative 0.081, anisotropy 0.570 to 0.259	lebse\README.md:40-75; lebse\PAPER.md:1-15,60-66; lebse\MODEL_CARD.md:48-52
chapter 12 section 12.3	uncompressed 0.79, centred 0.84	geometric-observation\claims\LEDGER.md row GO-B-legal
chapter 12 section 12.4	0.765 to 0.971, 0.545 to 0.562, v1 0.340, cosine 0.570 to 0.259	lebse\README.md:40-75; lebse\PAPER.md:1-15,60-66

FAILURES AND CORRECTIONS

none

CONDITIONS

- Of a covariance, having different variances in different directions, which is to say unequal eigenvalues. Two readers at different angles to an anisotropic covariance read different variances, and the flip between two codes exists exactly when the covariance is anisotropic along the directions that separate them.
- Centring an embedding raised a ranker from 0.79 to 0.84 on the legal corpus, which is the condition under which a learned quotient and a variance quotient disagree most, and an isotropic control collapses the flip.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/Isotropy.lean`, `theorems isotropic_reads_same, isotropic_no_flip, anisotropic_readers_differ, flip_iff_anisotropic`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 3, 9, 11, 12.

RELATED

[isotropic](#) (p. 179); [covariance matrix](#) (p. 85); [spectrum](#) (p. 334); [whitening](#) (p. 368).

anti arm

instrument. Entry id `anti-arm`.

DEFINITION

In a flip comparison, a third code built to destroy the consumer's read subspace at the same bits. It is expected to score worst, and a comparison in which it does not is not a flip. Chapter 4.

EQUATION

Book equation 4.5.

$$\text{flip} : \quad \text{task}(O) > \text{task}(R) \text{ and } \text{tr } M_\delta^O > \text{tr } M_\delta^R, \quad \text{task}(\text{anti}) < \text{task}(R), \quad \text{bits}(O) = \text{bits}(R).$$

LEDGER

- GO-2 (neg. half: not reconstruction). At matched bits, downstream preservation is not controlled by reconstruction error. [demonstrated]. `geometric-observation/claims/LEDGER.md:63` at `9f3829f`.
- NEG-16 (KV serving, end-task). *At matched bits and matched reconstruction error, steering KV quantization error into the attention read subspace degrades LongBench task score by the registered floors on a*

deployed-class model. Refuted at its registered effect sizes on this model and task. [refuted]. `geometric-observation/claims/LEDGER.md:92 at 9f3829f`.

- GO-B-whale (038). Sperm-whale coda dialect (DSWP/Sharma 2024), Clan classifier — cetacean communication; promotes the exploratory (A2) verdict to a sealed flip [predicted]. `geometric-observation/claims/LEDGER.md:120 at 9f3829f`.
- GO-B-blind (041). Blind, NON-ORACLE prospective flip on a fresh untouched domain (20 Newsgroups sci.space vs rec.autos), logistic classifier — the reviewer's decisive test: recover P_C non-oracle and commit the winning code + sign + magnitude *before* opening the test split [predicted]. `geometric-observation/claims/LEDGER.md:121 at 9f3829f`.

FIRST STATED

Volume 14, chapter 8, `geometric-observation/chapters/ch08_value.md:1-30`, DOI 10.5281/zenodo.21776291, as the third arm of every registered flip.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 1 section 1.4	the flip sealed in twelve domains and three physics, held in at least five domains and all three	<code>geometric-observation\chapters\ch08_value.md:40-70</code> ; <code>geometric-observation\claims\LEDGER.md</code> ROWS GO-2, GO-B-AV163, D3, 038
chapter 2 section 2.5	whitened code wins at every budget	<code>geometric-observation\chapters\ch08_value.md:84-97</code>
chapter 4 section 4.3	the flip definition, flip versus (A2) verdict	<code>geometric-observation\chapters\ch08_value.md:1-30,100-108</code>
chapter 4 section 4.3	GO-2 0.0934 vs 0.0938, 2.53 times, 12 of 12, anti 21 times; retrieval 0.0964, negative 4.70 and positive 4.65, recon 0.40	<code>geometric-observation\claims\LEDGER.md</code> ROWS GO-2 negative and positive halves
chapter 4 section 4.3	acoustic 148 of 201, 201 of 201, 152 of 201; seismic 13 of 17, 17 of 17, 13 of 17; whale 0.934 vs 0.883, 2 times, 300 of 300; at least 5 domains and 3 physics; battery prediction met	<code>geometric-observation\chapters\ch08_value.md:40-70</code> ; <code>geometric-observation\claims\LEDGER.md</code> ROWS GO-B-AV163, D3, 038
chapter 4 section 4.3	whitened code on whale 0.83, 0.85, 0.97 vs 0.41, 0.80, 0.89	<code>geometric-observation\chapters\ch08_value.md:84-97</code>
chapter 4 section 4.4	gradient compression anti 300 of 300, flip 27 percent, coupling boundary	<code>geometric-observation\chapters\ch08_value.md:108-116</code>
chapter 6 section 6.2	whale clan classifier, 8718 codas, 0.934 vs 0.883, reconstructs twice as well, 300 of 300	<code>geometric-observation\chapters\ch08_value.md:40-70</code> ; <code>geometric-observation\claims\LEDGER.md</code> ROW 038

FAILURES AND CORRECTIONS

- NEG-16 (KV serving, end-task), [refuted]. *At matched bits and matched reconstruction error, steering KV quantization error into the attention read subspace degrades LongBench task score by the registered floors on a deployed-class model.* Refuted at its registered effect sizes on this model and task.

CONDITIONS

- The anti arm is matched on bits with the other two codes, and its purpose is to show that the read subspace is what the comparison is about, since destroying it should be worst.
- A comparison in which the anti arm is not worst has not shown a flip, whatever the other two arms did.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/AntiArm.lean`, `theorems readDiag_le_max`, `readDiag_le_total`, `anti_arm_worst`, `at observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 4, 6, 7, 12.

RELATED

[the flip](#) (p. 138); [read distortion](#) (p. 281); [read operator](#) (p. 282).

anti-hub

concept. Entry id `anti-hub`.

DEFINITION

A row that is never or almost never retrieved as anyone's nearest neighbour. Chapter 10 names five kinds and shows one is manufactured by the query budget. Chapters 3 and 10.

EQUATION

Book equation 10.3.

$$N_k(x \mid Q) = |\{q \in Q : x \in \text{top}_k(q)\}|, \quad \text{anti-hub} : N_k(x) = 0.$$

Book equation 10.7.

$$\text{verdict} = \min_{i: n_i \geq n_{\min}, |Q_i| \geq q_{\min}} \text{score}_i, \quad \text{ABSTAIN otherwise.}$$

LEDGER

- NEG-11. (GO-5, prospective ×4) The $\alpha=1$ density/hubness quotient decisively and density-specifically restores invariant fidelity in a non-spectral domain. [refuted]. `geometric-observation/claims/LEDGER.md:104 at 9f3829f`.
- NEG-14. (GO-P-2026-037, prospective) `a2_probe.median_unit_displacement` is a single-statistic predictor of the flip regime ($\text{unit_disp} \geq 1.0 \Rightarrow \text{generic-polar-flip vs needs-blind-probe}$). [refuted]. `geometric-observation/claims/LEDGER.md:108 at 9f3829f`.

FIRST STATED

Radovanović, Nanopoulos, and Ivanović, hubs in space, 2010, as chapter 3 cites it, with the program's five kinds in `turboquant-pro`, `turboquant-pro/docs/HUBNESS_PRIMER.md:86-131`, and chapter 10 section 10.2 of *Data Mining as Observation*.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 3 section 3.5	78 vs 369, density correlation about 0.67, 8 percent vs 34 percent, centering vs mutual-proximity rescaling	turboquant-pro\docs\HUBNESS_PRIMER.md:38-59,140-160
chapter 8 section 8.10	the AI-assistant boundary, fresh-context passes	geometric-observation\chapters\ch15_registration_first.md; geometric-observation\chapters\ch17_open_problems_and_the_key_96
chapter 10 section 10.2	anti-hubs as where compressed indexes fail first, aggregate recall barely moves	turboquant-pro\docs\HUBNESS_PRIMER.md:86-131
chapter 10 section 10.3	abstain below 2.5 k	turboquant-pro\docs\HUBNESS_PRIMER.md:140-160
chapter 10 section 10.4	area map, boundary rule, hash, refuse not warn, intra and transit counts, area classes, abstention rule	turboquant-pro\docs\STRATA_RFC.md:24-98
chapter 10 section 10.4	first run, 350000 of 2391361 rows, 7 of 14 eligible, abstentions with 2 and 1 rows, Robin Hood 0.3447 to 0.4469 ratio 1.30 against 1.5, skew 2.68 to 4.40 ratio 1.64 against 3, 63 times sample range	turboquant-pro\docs\RESULTS_multilingual_strata.md:1-55
chapter 10 section 10.4	second prediction inverted, transit 0.389 vs 0.291, centrality difference signs, share 0.473 vs 0.391, seven of thirteen backbone	turboquant-pro\docs\RESULTS_multilingual_strata.md:55-90; turboquant-pro\docs\STRATA_RFC.md:98-130
chapter 10 section 10.5	Gate A design error, A prime skew 3.970 to 3.177, max 287 to 213, Robin Hood 0.372 to 0.261, fraction 0.117 to 0.079, compressed path 0.663 vs 0.90, seven strata, 0.62 to 0.69 vs 0.76 to 0.84	turboquant-pro\docs\RESULTS_strata_phase23_gates.md:1-45
chapter 10 section 10.5	rerank identical 0.6627, candidate coverage, depth curve 51 to 501, coverage 0.696 to 0.926, fidelity 0.663 to 0.702, originals rerank 0.923, 4096 bytes on 148	turboquant-pro\docs\RESULTS_strata_phase23_gates.md:45-110
chapter 10 section 10.5	fragile-first allocation 0.7118 vs 0.7251, targets up 0.004 to 0.034, redesigned allocator passed	turboquant-pro\docs\RESULTS_strata_phase23_gates.md:110-125
chapter 11 section 11.6	count of ten, hubs and anti-hubs, max 78 vs 369, density correlation about 0.67, 8 percent vs 34 percent, abstain below 2.5k, centering vs mutual-proximity rescaling	turboquant-pro\docs\HUBNESS_PRIMER.md:1-60,60-170
chapter 11 section 11.6	anti-hub recall, p05, hub-rank correlation, hub-set overlap, the build gate	turboquant-pro\docs\HUBNESS_PRIMER.md:86-131
chapter 11 section 11.6	the centroid-injection attack	turboquant-pro\docs\HUBNESS_PRIMER.md:168-197, citing arXiv 2604.05480
chapter 12 section 12.4	seven of thirteen backbone areas under the translation-trained encoder, all local-hub under the general one	turboquant-pro\docs\RESULTS_multilingual_strata.md:55-90

FAILURES AND CORRECTIONS

- NEG-11, [refuted]. (GO-5, prospective ×4) The $\alpha=1$ density/hubness quotient decisively and density-specifically restores invariant fidelity in a non-spectral domain.
- NEG-14, [refuted]. (GO-P-2026-037, prospective) `a2_probe.median_unit_displacement` is a single-statistic predictor of the flip regime ($\text{unit_disp} \geq 1.0 \Rightarrow \text{generic-polar-flip vs needs-blind-probe}$).

CONDITIONS

- An anti-hub is a row no query retrieves, and chapter 10 names five kinds, of which one is manufactured by the query budget rather than by the corpus.
- Anti-hubs are where compressed indexes fail first, since quantization rounds away the fine distinctions by which they are found, while aggregate recall barely moves, which is why recall is reported per stratum with the anti-hub stratum named.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/Hubness.lean`, theorems `sum_count`, `sum_count_eq`, `count_congr`, `antiHub_iff`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 3, 10, 11, 12.

RELATED

[hubness](#) (p. 168); [Poisson ceiling](#) (p. 253); [min-over-strata](#) (p. 217); [rank certificate](#) (p. 275).

anti-hub recall

instrument. Entry id `anti-hub-recall`.

DEFINITION

Recall at k restricted to the rows that are rarely retrieved, the observer's outliers, reported as the minimum over strata so that a mean cannot hide the tail. Chapter 10.

EQUATION

Book equation 10.7.

$$\text{verdict} = \min_{i: n_i \geq n_{\min}, |Q_i| \geq q_{\min}} \text{score}_i, \quad \text{ABSTAIN otherwise.}$$

Book equation 11.3.

$$\text{recall@}k = \frac{|\text{returned top-}k \cap \text{true top-}k|}{k},$$

true top- k computed from the uncompressed vectors.

Book equation 0.35.

$$\text{RH} = \sum_i \max\left(0, \frac{N_i}{\sum_j N_j} - \frac{1}{n}\right).$$

LEDGER

- NEG-14. (GO-P-2026-037, prospective) `a2_probe.median_unit_displacement` is a single-statistic predictor of the flip regime ($\text{unit_disp} \geq 1.0 \Rightarrow \text{generic-polar-flip vs needs-blind-probe}$). [refuted]. `geometric-observation/claims/LEDGER.md:108` at `9f3829f`.

FIRST STATED

Chapter 10 section 10.2 of *Data Mining as Observation*, with the program's gate in `turboquant-pro/docs/HUBNESS_PRIMER.md:86-131` and the strata result in `turboquant-pro/docs/RESULTS_strata_phase23_gates.md`.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 10 section 10.2	anti-hubs as where compressed indexes fail first, aggregate recall barely moves	turboquant-pro/docs/HUBNESS_PRIMER.md:86-131
chapter 10 section 10.4	first run, 350000 of 2391361 rows, 7 of 14 eligible, abstentions with 2 and 1 rows, Robin Hood 0.3447 to 0.4469 ratio 1.30 against 1.5, skew 2.68 to 4.40 ratio 1.64 against 3, 63 times sample range	turboquant-pro/docs/RESULTS_multilingual_strata.md:1-55
chapter 10 section 10.4	second prediction inverted, transit 0.389 vs 0.291, centrality difference signs, share 0.473 vs 0.391, seven of thirteen backbone	turboquant-pro/docs/RESULTS_multilingual_strata.md:55-90; turboquant-pro/docs/STRATA_RFC.md:98-130
chapter 10 section 10.5	Gate A design error, A prime skew 3.970 to 3.177, max 287 to 213, Robin Hood 0.372 to 0.261, fraction 0.117 to 0.079, compressed path 0.663 vs 0.90, seven strata, 0.62 to 0.69 vs 0.76 to 0.84	turboquant-pro/docs/RESULTS_strata_phase23_gates.md:1-45
chapter 10 section 10.5	rerank identical 0.6627, candidate coverage, depth curve 51 to 501, coverage 0.696 to 0.926, fidelity 0.663 to 0.702, originals rerank 0.923, 4096 bytes on 148	turboquant-pro/docs/RESULTS_strata_phase23_gates.md:45-110
chapter 10 section 10.5	fragile-first allocation 0.7118 vs 0.7251, targets up 0.004 to 0.034, redesigned allocator passed	turboquant-pro/docs/RESULTS_strata_phase23_gates.md:110-125
chapter 11 section 11.6	anti-hub recall, p05, hub-rank correlation, hub-set overlap, the build gate	turboquant-pro/docs/HUBNESS_PRIMER.md:86-131
chapter 12 section 12.4	seven of thirteen backbone areas under the translation-trained encoder; all local-hub under the general one	turboquant-pro/docs/RESULTS_multilingual_strata.md:55-90

FAILURES AND CORRECTIONS

- NEG-14, [refuted]. (GO-P-2026-037, prospective) `a2_probe.median_unit_displacement` is a single-statistic predictor of the flip regime ($\text{unit_disp} \geq 1.0 \Rightarrow \text{generic-polar-flip vs needs-blind-probe}$).

CONDITIONS

- Recall at k restricted to the rows that are rarely retrieved, the observer's outliers, reported as the minimum over strata so that a mean cannot hide the tail. It lies in the unit interval and is one exactly when every anti-hub's true neighbours are returned.
- Compressed indexes fail on anti-hubs first while aggregate recall barely moves, and the remedy that promotes isolated rows is the remedy that compression damages first, which is why the gate is a build gate and not a report line.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

lean/DataMiningAsObservation/RecallAtK.lean, theorems recallAtK_mem_unit, recallAtK_eq_one_iff, aggregate_le_of_failing, aggregate_example, at observation-data-mining af776fd.

lean/DataMiningAsObservation/Hubness.lean, theorems sum_count, sum_count_eq, count_congr, antiHub_iff, at observation-data-mining af776fd.

USED IN

Data Mining as Observation chapters 10, 11, 12.

RELATED

[anti-hub](#) (p. 17); [recall at k](#) (p. 286); [min-over-strata](#) (p. 217); [hub](#) (p. 166); [stratification](#) (p. 339).

anti-monotonicity

concept. Entry id anti-monotonicity.

DEFINITION

The property that support cannot increase when an itemset grows, which licenses Apriori to prune every superset of an infrequent itemset without loss. Chapter 5.

EQUATION

Book equation 5.1.

$$s(X) = \frac{\sigma(X)}{N}, \quad c(X \rightarrow Y) = \frac{\sigma(X \cup Y)}{\sigma(X)} = \frac{s(X \cup Y)}{s(X)}.$$

Book equation 5.3.

$$X \subseteq Y \implies s(X) \geq s(Y).$$

LEDGER

none

FIRST STATED

Agrawal and Srikant, fast algorithms for mining association rules, 1994, as chapter 5 section 5.3 of *Data Mining as Observation* reads it.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 5 section 5.1 to 5.3, 5.5	support, confidence, lift, the Apriori principle, FP-growth, closed and maximal itemsets, objective measures, Simpson's paradox, cross-support	TSK 2e chapter 5
chapter 5 section 5.3	the transferable idea as a course learning outcome	instructor working documents, not public [@bond2026course], ECE_514-01_FA26_session-outlines.md:117-136

FAILURES AND CORRECTIONS

none

CONDITIONS

- The property that support cannot increase when an itemset grows, since every transaction containing the larger set contains the smaller. It licenses Apriori to prune every superset of an infrequent itemset without loss, because no frequent itemset can lie above an infrequent one.
- The license is a statement about the measure and not about the data, and the book's sidebar carries it to thresholds outside itemsets, where the analogous license is the ceiling on a score that a Youden bound supplies.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/SafePruning.lean`, theorems `support_anti`, `apriori`, `subset_of_frequent`, at `observation-data-mining af776fd`.

`lean/DataMiningAsObservation/ItemSet.lean`, theorems `supportFrac_mem_unit`, `supportFrac_empty`, `supportFrac_anti`, `supportFrac_union_le`, `closed_of_maximal`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 5.

RELATED

[support](#) (p. 343); [safe pruning](#) (p. 311); [itemset](#) (p. 180).

Apriori principle

result. Entry id `apriori`.

DEFINITION

The statement that every subset of a frequent itemset is frequent, the contrapositive of anti-monotonicity. Chapter 5.

EQUATION

Book equation 5.3.

$$X \subseteq Y \implies s(X) \geq s(Y).$$

Book equation 5.1.

$$s(X) = \frac{\sigma(X)}{N}, \quad c(X \rightarrow Y) = \frac{\sigma(X \cup Y)}{\sigma(X)} = \frac{s(X \cup Y)}{s(X)}.$$

LEDGER

none

FIRST STATED

Agrawal and Srikant, fast algorithms for mining association rules, 1994, as TSK chapter 5 presents it, and chapter 5 section 5.3 of *Data Mining as Observation*.

MEASUREMENTS

Where the book states it	Numbers, as the book’s sources table records them	Source
chapter 5 section 5.1 to 5.3, 5.5	support, confidence, lift, the Apriori principle, FP-growth, closed and maximal itemsets, objective measures, Simpson’s paradox, cross-support	TSK 2e chapter 5

FAILURES AND CORRECTIONS

none

CONDITIONS

- Support cannot increase when an itemset grows, so an infrequent itemset has only infrequent supersets and every subset of a frequent itemset is frequent. This holds for any finite collection of transactions.
- The principle licenses pruning the lattice above an infrequent itemset without counting, and the count of candidates rather than the count of transactions decides whether the run finishes.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/SafePruning.lean`, `theorems support_anti`, `apriori`, `subset_of_frequent`, `at observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 5.

RELATED

[safe pruning](#) (p. 311); [Monotone Invariance Theorem](#) (p. 220).

attention

concept. Entry id attention.

DEFINITION

The operation inside a language model that lets each token look at earlier ones by comparing its query to their keys and averaging their values. Chapter 0 section 0.11.

EQUATION

Book equation 0.23.

$$\text{softmax}(z)_i = \frac{e^{z_i}}{\sum_j e^{z_j}}, \quad \text{output} = \sum_i \text{softmax}\left(\frac{q \cdot k_i}{\sqrt{d}}\right)_i v_i.$$

Book equation 11.1.

$$\cos(k, \hat{k}) = 0.995 \quad \text{while} \quad \text{PPL} : 12.24 \rightarrow 10643.$$

LEDGER

- G0-2/G0-12/G0-13 operational (KV serving, 077). Consumer-relative access width measured on a production serving stack (Qwen2.5-7B KV-cache eviction, matched budget): task quality tracks measured predictive uncertainty u about the consumer’s future reads, not nominal scorer width — the 32-query snapshot beats the 1024-query scorer $+0.4375 \pm 0.070$ at 5% keep (bar 0.30) and survives 97% eviction with zero drop, while wide-window scoring is statistically indistinguishable from random eviction at extreme budgets; the recency-hoarding starvation signature replicated across three disjoint prompt sets

(oracle-miss gap 0.370 vs bar 0.25). The novel equal-uncertainty analytic-equality control (degradation-titrated, constructible by design) REFUTED its own equality prediction on the pre-registered branch: with calibration health $4\times-130\times$ inside gates, equal scalar u did NOT give equal quality (V4 +0.078 vs 0.0625 tolerance; the pre-registered $p=0.03$ contrast firmed to $+0.359\pm0.068$, 5.3 SE) — equal scalar uncertainty is insufficient, error structure matters, consistent with GO-13 Theorem 1's own $r\geq 2$ scoping of equal-q universality (a scalar-context privilege). Successor arc: 056 honest miss $\rightarrow$ 075 ID burned on a disclosed design failure $\rightarrow$ 077 sealed and split-verdict. [demonstrated]. `geometric-observation/claims/LEDGER.md:81` at 9f3829f.

- NEG-2. Reconstruction cosine as a proxy for key quality. [refuted]. `geometric-observation/claims/LEDGER.md:95` at 9f3829f.

FIRST STATED

Chapter 0 section 0.11 and chapter 11 section 11.1 of *Data Mining as Observation*, with the program's head-level measurements in `turboquant-pro/docs/KV_KEYS_FINDING.md:1-49` and the serving-stack ledger row.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 1 section 1.4	cosine 0.995 and perplexity of order ten thousand, the recalibration negative	<code>geometric-observation\chapters\ch02_failure_of_observer_free60;</code> <code>geometric-observation\chapters\ch16_honest_negatives.md</code> NEG-2 and NEG-4; <code>turboquant-pro\docs\KV_KEYS_FINDING.md:1-49</code>
chapter 2 section 2.5	cosine 0.995 and the softmax reader	<code>turboquant-pro\docs\KV_KEYS_FINDING.md:1-49</code>
chapter 3 section 3.2	condition (A2), tangential displacement	<code>turboquant-pro\docs\KV_KEYS_FINDING.md:61-86;</code> <code>geometric-observation\chapters\ch09_legibility.md:42-54</code>
chapter 8 section 8.2	cosine 0.995, perplexity near $1e4$	<code>turboquant-pro\docs\KV_KEYS_FINDING.md:1-49;</code> <code>geometric-observation\chapters\ch16_honest_negatives.md</code> NEG-2
chapter 11 section 11.1	condition (A2), cosine satisfies it, post-rotary keys do not, the cone below cell size	<code>turboquant-pro\docs\KV_KEYS_FINDING.md:61-86;</code> <code>the-angular-observer\README.md:26-31</code>
chapter 11 section 11.2	fp16 12.24, values-only 13.12, PolarQuant K4 10643 and 0.095, per-channel uniform K4 14.91 and 0.062, per-channel NUQ K3 15.77 and 0.148, 2.4x and 670x, pre-rotary near 22000, 2 key heads serve 12 query heads	<code>turboquant-pro\docs\KV_KEYS_FINDING.md:1-49</code>
chapter 11 section 11.2	the calibration-time probe and the streaming monitor	<code>turboquant-pro\turboquant_pro\A2_probe.py:315-438;</code> <code>turboquant-pro\tests\test_A2_probe.py</code>

FAILURES AND CORRECTIONS

- NEG-2, [refuted]. Reconstruction cosine as a proxy for key quality.

CONDITIONS

- A head weights each earlier token's value by the softmax of its query-key score and sums. For scalar values the output lies between the smallest and largest value, and the head reads the keys only through their scores against the query, so a key change the query does not read leaves the output unchanged however large it is.
- That is the head's read subspace, a few query-weighted directions of each key, and the reason a key reconstruction at cosine 0.995 raised the perplexity by three orders of magnitude.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/Attention.lean`, `theorems output_le_max, min_le_output, output_congr, output_nuisance`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 1, 2, 4, 6, 8, 11, 13.

RELATED

[softmax](#) (p. 327); [read subspace](#) (p. 285); [nuisance](#) (p. 233); [the flip](#) (p. 138).

attribute type

concept. Entry id `attribute-type`.

DEFINITION

The declaration of which transformations of a column keep its meaning: any injective relabelling for nominal, a strictly increasing map for ordinal, an affine map for interval, a positive rescaling for ratio, nested in that order. Chapter 2 section 2.1.

EQUATION

Book equation 2.1.

$$P_{C_2 \circ C_1}(x) = J_1(x)^\top P_{C_2}(C_1(x)) J_1(x), \quad \text{rank } P_{C_2 \circ C_1}(x) \leq \text{rank } P_{C_2}(C_1(x)) \quad \text{at each row } x.$$

Book equation 0.12a.

$$x \sim_C x' \iff d_G(C(x), C(x')) = 0.$$

LEDGER

none

FIRST STATED

Stevens, on the theory of scales of measurement, 1946, as chapter 2 section 2.1 of *Data Mining as Observation* reads it after TSK section 2.1.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 2 section 2.1	the four attribute types and permitted transformations	TSK 2e section 2.1

FAILURES AND CORRECTIONS

none

CONDITIONS

- The declaration of which transformations of a column do not change its meaning. A ratio scale permits a positive rescaling, an interval scale an affine map with positive slope, an ordinal scale a strictly increasing map, and a nominal scale any injective relabelling, and the four classes are nested in that order.
- A rescaling preserves ratios and an affine map preserves ratios of differences, so a consumer that reads a ratio of two interval-scale columns reads the origin the scale said was arbitrary.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

lean/DataMiningAsObservation/AttributeType.lean, theorems scale_affine, affine_strictMono, strictMono_injective, scale_preserves_ratio, affine_preserves_difference_ratio, at observation-data-mining af776fd.

USED IN

Data Mining as Observation chapters 2, 13.

RELATED

[standardization](#) (p. 338); [discretization](#) (p. 105); [quotient](#) (p. 271).

attribution

instrument. Entry id attribution.

DEFINITION

An assignment to each feature of a share of one prediction. A gradient-based attribution estimates the consumer's sensitivity, and so a diagonal entry of the read operator when averaged. Shapley, permutation, and partial-dependence methods measure other things and are named for them. Chapter 14.

EQUATION

Book equation 0.8.

$$g_j \approx \frac{C(x + h e_j) - C(x - h e_j)}{2h}, \quad j = 1, \dots, d.$$

Book equation 0.9.

$$P_C = \mathbb{E}[g g^\top], \quad g = \nabla C(x).$$

Book equation 14.5.

$$\text{FC}_g = \Pr[y = \text{violation} \mid \hat{y} = \text{clear}, g], \quad \text{verdict} = \max_{g: n_g \geq n_{\min}} \text{FC}_g, \quad \text{ABSTAIN otherwise.}$$

LEDGER

- GO-1. The consumer's invariant/nuisance split is identifiable ex ante from the consumer functional. [predicted]. `geometric-observation/claims/LEDGER.md:62` at 9f3829f.
- GO-EC-3. A read operator recovered from a black-box consumer by query-only finite-difference probing, composed with the Kalman covariance as $\text{tr}(\hat{P}_C \Sigma)$, prospectively selects sensors that improve the held-out consumer at matched budgets with probe cost charged — capturing 94.6% of the known analytic optimum's gain on the positive-control arm (gate $\geq 75\%$) and improving 16.3% over the best consumer-agnostic policy on non-analytic consumers (gate $\geq 8\%$), with trace-matched ordering carried by the com-

position at 86.9% over 61 pairs (gate $\geq 65\%$). [predicted]. geometric-observation/claims/LEDGER.md:162 at 9f3829f.

FIRST STATED

Lundberg and Lee, a unified approach to interpreting model predictions, 2017, as chapter 14 section 14.5 of *Data Mining as Observation* reads it, with the program's contraction formula in gtc-prototype/docs/SPECTRUM_FINDINGS.md:90-99.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 11 section 11.7	GO-1 overlap 0.936 vs 0.059, flip 12 of 12, reconstruction 0.40	geometric-observation\claims\LEDGER.md row GO-1; geometric-observation\chapters\ch10_the_blind_probe.md:34-52
chapter 14 section 14.5	the contraction formula fairness minus the general component	gtc-prototype\docs\SPECTRUM_FINDINGS.md:90-99

FAILURES AND CORRECTIONS

none

CONDITIONS

- An assignment to each feature of a share of one prediction. A gradient-based attribution gives coordinate i the share g_i times the move along it, those shares sum exactly to the change in output for an affine consumer, a coordinate the consumer does not read gets share zero, and the average of the squared sensitivity is a diagonal entry of the read operator, so an averaged gradient attribution estimates the read operator.
- Shapley, permutation, and partial-dependence methods measure other things and are named for them, and for a tree the sensitivity is zero almost everywhere, so importances that count splits are the right reading and finite differences the wrong one.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

lean/DataMiningAsObservation/Attribution.lean, theorems `attr_sum_affine`, `attr_unread`, `sq_sensitivity_eq_readOp_diag`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 8, 14.

RELATED

[sensitivity](#) (p. 319); [read operator](#) (p. 282); [finite difference](#) (p. 137); [consumer](#) (p. 73).

audit

instrument. Entry id audit.

DEFINITION

A check of a number or system against its record by a reader who did not produce it, bound to a sealed file. Chapter 8 section 8.9 and chapter 14 section 14.1.

EQUATION

Book equation 8.1.

$$O_{\text{harness}} = (C_{\text{score}}, G_{\text{metric}}, B = \text{folds} \times \text{samples} \times \text{seeds}).$$

Book equation 14.2.

$$\text{coverage difference}_c = \text{AUROC}_c(\text{embedding}) - \text{AUROC}_c(\text{validated axes}), \quad \text{floor} \approx 0.08 \text{ to } 0.12.$$

Book equation 13.2.

$$T_{\text{coh}} = \frac{0.423}{f_D},$$

refuted fit: $\phi \approx 0.177 T_{\text{coh}} (R^2 = 0.915)$, sealed: $\phi \leq 6$ TTI at 10 Hz, 1 TTI at ≥ 50 Hz.

LEDGER

- NEG-15 (Bell boundary). *Query-conditioned hubness supplies a mechanism for Bell-inequality violation without action at a distance.* Refuted as a mechanism; the settings-as-queries reframing survives only as vocabulary. [demonstrated]. `geometric-observation/claims/LEDGER.md:93` at 9f3829f.
- GO-B-legal (035→036). Legal-citation retrieval (CourtListener), cosine-ranking consumer, LaBSE embeddings — real large corpus, non-physical consumer [predicted]. `geometric-observation/claims/LEDGER.md:119` at 9f3829f.

FIRST STATED

Chapter 8 section 8.9 and chapter 14 section 14.1 of *Data Mining as Observation*, with the audit binding in `xbse/README.md:195-219` and the Bell audit in `geometric-observation/articles/2026-08-03`.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 3 section 3.6	Bell audit, max S 2.00000 across 72 configurations, d 3 to 128, correlation negative 0.036, post-selected 2.7308, controls 2.748, 2.386, 3.174, seed 20260817, sealed at 6e825d8	<code>geometric-observation/articles/2026-08-03-hubness-does-not-weaken-bell.md:1-60</code> ; <code>geometric-observation/claims/LEDGER.md</code> row NEG-15; <code>geometric-observation/results/GO-bell-geometry-audit.json</code>
chapter 8 section 8.9	twenty-seven errors, four in checking tools	<code>constraint-gap-measurements/README.md:273-282</code>
chapter 14 section 14.1	the three questions, the trust beats, the audit binding	<code>gtc-prototype/README.md:10-58</code> ; <code>xbse/README.md:195-219</code>

FAILURES AND CORRECTIONS

none

CONDITIONS

- A check of a published number or system against its record, by a reader who did not produce it. An audit binds a verdict to a sealed file whose digest proves it unchanged, and a witness's disagreement with a certificate is the audit's finding.
- The audit of the measurement notes found twenty-seven errors, four in the checking tools, and the constraint-first Bell audit reached a maximum of 2.00000 across 72 configurations.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

lean/DataMiningAsObservation/Seal.lean, theorems changed_of_hash_ne, hash_eq_of_eq, exists_collision, at observation-data-mining af776fd.

lean/DataMiningAsObservation/Certificate.lean, theorems falseClear_mul_coverage, coverage_empty, falseClear_mem_unit, minOverStrata_passes_iff, minOverStrata_le_weighted_mean, at observation-data-mining af776fd.

USED IN

Data Mining as Observation chapters 1, 2, 3, 4, 8, 12, 14.

RELATED

[sealed](#) (p. 316); [witness](#) (p. 370); [certificate](#) (p. 55); [false-clear rate](#) (p. 133); [registered](#) (p. 296).

AUROC

concept. Entry id auroc.

DEFINITION

The area under the receiver operating characteristic curve, equal to the probability that a random positive scores above a random negative, ties counted as one half. Equation 0.16.

EQUATION

Book equation 0.16.

$$\text{AUROC} = \Pr[s^+ > s^-] + \frac{1}{2} \Pr[s^+ = s^-].$$

Book equation 0.38.

$$w = \max(0, 2 \cdot \text{AUROC} - 1).$$

Book equation 12.3.

$$\text{validated} \iff \text{AUROC}_{\text{cross}} - \max(\text{AUROC}_{\text{untrained}}, \text{AUROC}_{\text{BoW}}) \geq 0.10.$$

LEDGER

- GO-B-legal (035→036). Legal-citation retrieval (CourtListener), cosine-ranking consumer, LaBSE embeddings — real large corpus, non-physical consumer [predicted]. `geometric-observation/claims/LEDGER.md:119` at 9f3829f.
- GO-B-whale (038). Sperm-whale coda dialect (DSWP/Sharma 2024), Clan classifier — cetacean communication; promotes the exploratory (A2) verdict to a sealed flip [predicted]. `geometric-observation/claims/LEDGER.md:120` at 9f3829f.

FIRST STATED

Chapter 0 section 0.8 of *Data Mining as Observation*, the pairwise definition with ties counted one half, and the held-out AUROC of every encoder gate and flip in the program.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

CONDITIONS

- The probability that a random positive scores above a random negative, ties counted one half. It lies in the unit interval, is one when every positive scores above every negative and zero when every negative scores

above every positive, is one half for a constant score, and is unchanged by a strictly monotone transform of the score.

- It reads the ranking and not the values, so it is the discrimination half of every encoder's validation and says nothing about calibration.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/Auroc.lean`, `theorems pair_nonneg`, `auroc_nonneg`, `auroc_le_one`, `auroc_perfect`, `auroc_reversed`, `auroc_chance`, `auroc_monotone_invariant`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 5, 6, 8, 11, 12, 14.

RELATED

[Monotone Invariance Theorem](#) (p. 220); [reliability weight](#) (p. 298); [chance level](#) (p. 58); [Youden F1 bound](#) (p. 372); [cross-corpus gate](#) (p. 87).

B

bagging

instrument. Entry id bagging.

DEFINITION

Fitting the same model on bootstrap resamples and averaging. The average’s squared error is at most the members’ mean squared error, which is how it reduces variance. Chapter 7.

EQUATION

Book equation 0.18.

$$\widehat{\text{Var}}_{\text{NB}} = \left(\frac{1}{J} + \frac{n_{\text{test}}}{n_{\text{train}}}\right)\hat{\sigma}^2, \quad \frac{\widehat{\text{Var}}_{\text{NB}}}{\hat{\sigma}^2/J} = 1 + J \frac{n_{\text{test}}}{n_{\text{train}}}.$$

Book equation 8.2.

$$\widehat{\text{Var}}_{\text{NB}} = \left(\frac{1}{J} + \frac{n_{\text{test}}}{n_{\text{train}}}\right)\hat{\sigma}^2, \\ J = 1000, \frac{n_{\text{test}}}{n_{\text{train}}} = \frac{1}{4} \Rightarrow 1 + 250 = 251, \sqrt{251} = 15.84.$$

LEDGER

none

FIRST STATED

Breiman, bagging predictors, 1996, as chapter 7 section 7.1 of *Data Mining as Observation* reads it.

MEASUREMENTS

Where the book states it	Numbers, as the book’s sources table records them	Source
chapter 7 section 7.1	bagging variance, out-of-bag estimation, random forests, importances, AdaBoost	Hastie, Tibshirani, Friedman, ESL 2e chapters 15 and 10.1; TSK 2e 4.10

FAILURES AND CORRECTIONS

none

CONDITIONS

- Fitting the same model on bootstrap resamples of the rows and averaging. The average’s squared error is at most the mean of the members’ squared errors, which is the arithmetic by which bagging reduces variance, and the reduction is largest when the members covary least.
- Bagging reduces variance and boosting reduces bias, and neither changes what the base model reads.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/Ensemble.lean`, theorems `average_between`, `sq_average_le`, `average_const`, at `observation-data-mining af776fd`.

`lean/DataMiningAsObservation/Bootstrap.lean`, theorems `mean_sub`, `var_sub`, `paired_lt_iff`, `cov_comm`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 7.

RELATED

[ensemble](#) (p. 121); [bootstrap](#) (p. 45); [random forest](#) (p. 273); [standard error](#) (p. 337); [variance](#) (p. 361).

bag of words

concept. Entry id bag-of-words.

DEFINITION

A representation of a document as a vector with one coordinate per vocabulary term holding that term's count. Chapter 12.

EQUATION

Book equation 0.36.

$$w_{t,d} = \text{tf}_{t,d} \cdot \ln \frac{N}{\text{df}_t}, \quad \text{tf}_{t,d} = \frac{\text{count of } t \text{ in } d}{\text{length of } d}, \quad \text{df}_t = \text{documents containing } t.$$

Book equation 12.3.

$$\text{validated} \iff \text{AUROC}_{\text{cross}} - \max(\text{AUROC}_{\text{untrained}}, \text{AUROC}_{\text{BoW}}) \geq 0.10.$$

LEDGER

none

FIRST STATED

Chapter 12 section 12.1 of *Data Mining as Observation*, with the program's bag-of-words null in `xbse/README.md:104-130` and the chapter 14 scorecard.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

CONDITIONS

- A document as the vector of its term counts. Two documents that are rearrangements of each other have the same bag, so the bag is a quotient that declares word order irrelevant, the counts are nonnegative, and they sum to the document's length.
- As a null model it reads only counts, and an encoder is validated only when its held-out AUROC clears the bag's by the preregistered margin. The scorecard's bag scores lie between 0.46 and 0.54, near chance.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/BagOfWords.lean`, theorems `bag_perm`, `bag_example`, `bag_sum`, `bag_absent`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 12, 14.

RELATED

[TF-IDF](#) (p. 350); [cross-corpus gate](#) (p. 87); [quotient](#) (p. 271); [nuisance](#) (p. 233).

balanced accuracy

concept. Entry id balanced-accuracy.

DEFINITION

The mean of the per-class recalls, so that a large class cannot hide a small one. Equation 0.34.

EQUATION

Book equation 0.34.

$$BA = \frac{1}{K} \sum_{k=1}^K \frac{\text{correct in class } k}{\text{rows in class } k}.$$

Book equation 10.7.

$$\text{verdict} = \min_{i: n_i \geq n_{\min}, |Q_i| \geq q_{\min}} \text{score}_i, \quad \text{ABSTAIN otherwise.}$$

LEDGER

- NEG-11. (GO-5, prospective ×4) The $\alpha=1$ density/hubness quotient decisively and density-specifically restores invariant fidelity in a non-spectral domain. [refuted]. `geometric-observation/claims/LEDGER.md:104` at 9f3829f.

FIRST STATED

Chapter 0 section 0.14 of *Data Mining as Observation*, with the program's use in the anti-hub category table, `openvector-bench/results/R13_STAGE1_RESULTS.md`.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 3 section 3.5	query mass best single feature at every K for all four responses, seven features add at most 20 percent, threshold 1.25 on three of four, zero of four, 1000 real queries, 1024 dimensions	<code>openvector-bench/results/R13_STAGE0_RESULT.md:40-50</code> ; <code>openvector-bench/results/R13_STAGE1_RESULT.md:1-40</code>
chapter 10 section 10.2	five kinds, balanced accuracy 0.611, 0.718, 0.684, the category table, per-category 0.57 to 0.82, the sweep, pigeonhole floor	<code>openvector-bench/results/R13_STAGE1_RESULT.md:40-75</code>
chapter 10 section 10.2	half 2 failed, at most 1.28 times against 2, withdrawal and its scope	<code>openvector-bench/results/R13_STAGE1_RESULT.md:75-110</code>
chapter 11 section 11.6	query mass best single feature at every K for all four responses, seven features add at most 20 percent at 12 leaves, threshold 1.25 on three of four, zero of four, 1000 real queries, 1024 dimensions	<code>openvector-bench/results/R13_STAGE0_RESULT.md:40-50</code> ; <code>openvector-bench/results/R13_STAGE1_RESULT.md:1-40</code>

FAILURES AND CORRECTIONS

- NEG-11, [refuted]. (GO-5, prospective ×4) The $\alpha=1$ density/hubness quotient decisively and density-specifically restores invariant fidelity in a non-spectral domain.

CONDITIONS

- The mean of the per-class recalls, between the worst class and the best. Plain accuracy is the size-weighted mean of the same recalls, so a class of 990 rows recalled perfectly and a class of 10 never recalled give accuracy 0.99 and balanced accuracy 0.5.
- The book prefers the min over classes to the mean where a verdict is at stake, since a mean can still hide one failing class among many.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/BalancedAccuracy.lean`, `theorems min_le_balanced`, `balanced_le_max`, `hidden_class`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 10.

RELATED

[min-over-strata](#) (p. 217); [anti-hub](#) (p. 17); [abstention](#) (p. 9); [harness](#) (p. 159).

bar

instrument. Entry id `bar`.

DEFINITION

A threshold a statistic must reach, written down before the measurement. A bar discriminates only when the null fails it and the real system passes it, and a bar the null passes is vacuous. Chapter 8 section 8.3.

EQUATION

Book equation 8.3.

$$\Pr[\text{at least one of } m \text{ null tests passes}] = 1 - (1 - \alpha)^m, \quad \alpha_{\text{Bonferroni}} = \frac{\alpha}{m}.$$

Book equation 8.1.

$$O_{\text{harness}} = (C_{\text{score}}, G_{\text{metric}}, B = \text{folds} \times \text{samples} \times \text{seeds}).$$

Book equation 10.7.

$$\text{verdict} = \min_{i: n_i \geq n_{\min}, |Q_i| \geq q_{\min}} \text{score}_i, \quad \text{ABSTAIN otherwise.}$$

LEDGER

- OT-11. Feedback-free staleness: streaming-retrieval damage tracks measured drift; derived-cadence re-allocation removes it. `[void]. geometric-observation/claims/LEDGER.md:48 at 9f3829f`.
- NEG-14. (GO-P-2026-037, prospective) `a2_probe.median_unit_displacement` is a single-statistic predictor of the flip regime (`unit_disp ≥ 1.0 ⇒ generic-polar-flip vs needs-blind-probe`). `[refuted]. geometric-observation/claims/LEDGER.md:108 at 9f3829f`.

FIRST STATED

Chapter 8 section 8.3 of *Data Mining as Observation*, with the anti-vacuity bar quoted from `observation-theory-campaigns/experiments/PREREG-PF5-002.md:79-86`.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 8 section 8.3	PF5-001 vacuous pass (8 cells, 20000 census, 6.1e-7 vs 1e-5, 96 members, 160000 transmitted, zero reversing, record sha 073f6e692253)	observation-theory-campaigns\experiments\CAMPAIGN.md:255-277
chapter 8 section 8.3	anti-vacuity bar, quoted	observation-theory-campaigns\experiments\PREREG-PF5-002.md:79-86
chapter 9 section 9.2	the battery, three new templates, frozen ratios, code hash, bars at least 10 of 12 and the dimension ordering, eccentricity scope	the-angular-observer\PREREG_RECOGNIZER_BATTERY.md:1-80

FAILURES AND CORRECTIONS

- NEG-14, [refuted]. (GO-P-2026-037, prospective) `a2_probe.median_unit_displacement` is a single-statistic predictor of the flip regime (`unit_disp ≥ 1.0 ⇒ generic-polar-flip vs needs-blind-probe`).

CONDITIONS

- A threshold a statistic must reach, written down before the measurement. Passing a higher bar passes every lower one, a larger statistic passes every bar a smaller one passes, a bar discriminates exactly when it sits strictly above the null and at or below the real system, no bar discriminates when the null scores at least as high, and a bar the null passes is vacuous.
- A pass on eight cells at six parts in ten million against a bar of one in a hundred thousand was vacuous because the null cleared the bar too, which is where the anti-vacuity bar came from.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/Bar.lean`, `theorems passes_anti`, `passes_mono`, `discriminates_iff`, `no_bar_of_null_ge`, `exists_bar_of_lt`, `vacuous_of_null_passes`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 1, 2, 3, 4, 6, 7, 8, 9, 10, 11, 12, 13, 14.

RELATED

[gate](#) (p. 151); [vacuity threshold](#) (p. 359); [null model](#) (p. 234); [preregistration](#) (p. 258); [verdict](#) (p. 362).

baseline

instrument. Entry id baseline.

DEFINITION

The simplest scorer a claim must beat under the same protocol, chance, a constant, an untrained encoder, or a single model. A score below it is a finding. Chapter 6 section 6.3 and chapter 7 section 7.4.

EQUATION

Book equation 8.2.

$$\widehat{\text{Var}}_{\text{NB}} = \left(\frac{1}{J} + \frac{n_{\text{test}}}{n_{\text{train}}} \right) \hat{\sigma}^2,$$

$$J = 1000, \frac{n_{\text{test}}}{n_{\text{train}}} = \frac{1}{4} \Rightarrow 1 + 250 = 251, \sqrt{251} = 15.84.$$

Book equation 0.28.

$$P = \frac{TP}{TP + FP}, \quad R = \frac{TP}{TP + FN}, \quad F_1 = \frac{2PR}{P + R}.$$

Book equation 6.2.

$$\max_{\tau, \text{dir}} F_1(\mathbf{1}[\text{dir}(g(f(X)), \tau)], y) = \max_{\tau, \text{dir}} F_1(\mathbf{1}[\text{dir}(f(X), \tau)], y)$$

for every strictly monotone g .

LEDGER

- NEG-4. Lightweight online (Lloyd) key calibration beats the calibration-free default on softmax-KL. [refuted]. `geometric-observation/claims/LEDGER.md:97` at 9f3829f.
- GO-B-legal (035→036). Legal-citation retrieval (CourtListener), cosine-ranking consumer, LaBSE embeddings — real large corpus, non-physical consumer [predicted]. `geometric-observation/claims/LEDGER.md:119` at 9f3829f.

FIRST STATED

Chapter 6 section 6.3 and chapter 7 section 7.4 of *Data Mining as Observation*, with the single-model baseline in `constraint-gap/review/INDETERMINATES.md:1-40`.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 6 section 6.3	the five-dataset table, breast cancer 0.955 to 0.963, the pattern, DOI 10.5281/zenodo.20660206	<code>theory-radar\README.md:100-140</code>
chapter 7 section 7.3	the five-dataset outcomes	<code>theory-radar\README.md:100-140</code>
chapter 7 section 7.4	nine to three, seventeen indeterminate, five analyses zero resolved, three re-encode the correction	<code>constraint-gap\review\INDETERMINATES.md:1-40</code>
chapter 8 section 8.6	rights encoder 0.467 below untrained baseline, the AUROC lesson, cross-corpus same-sign fix	<code>xbse\README.md:160-172</code> ; <code>xbse\experiments\rights_r6_summary.json</code>

FAILURES AND CORRECTIONS

- NEG-4, [refuted]. Lightweight online (Lloyd) key calibration beats the calibration-free default on softmax-KL.

CONDITIONS

- The simplest scorer a claim must beat, chance, a constant, an untrained encoder, or a single model, run under the same protocol. A baseline that passes the bar makes the bar vacuous, and a score below the baseline is a finding.
- The rights encoder read 0.467, below the untrained baseline, and the formula's comparison with the ensemble is fair only against the single-model baseline inside one fold protocol.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/ChanceLevel.lean`, `theorems aurocNum_const`, `auroc_const`, `weight_const`, `tau_const`, at `observation-data-mining af776fd`.

lean/DataMiningAsObservation/Bar.lean, theorems passes_anti, passes_mono, discriminates_iff, no_bar_of_null_ge, exists_bar_of_lt, vacuous_of_null_passes, at observation-data-mining af776fd.

USED IN

Data Mining as Observation chapters 5, 6, 7, 8, 10, 11, 12, 13, 14.

RELATED

[chance level](#) (p. 58); [null model](#) (p. 234); [bar](#) (p. 34); [ensemble](#) (p. 121).

bi-Lipschitz

concept. Entry id bi-lipschitz.

DEFINITION

Of a map between two spaces, neither stretching nor shrinking any distance by more than a fixed factor. Equation 0.21.

EQUATION

Book equation 0.21.

$$\frac{1}{K} d(x, y) \leq d'(f(x), f(y)) \leq K d(x, y) \quad \text{for all } x, y.$$

LEDGER

- NEG-1. Fixed-scale, uniform-in-m bi-Lipschitz for the commute filter. [refuted]. [geometric-observation/claims/LEDGER.md:94 at 9f3829f](#).

FIRST STATED

Chapter 0 section 0.10 of *Data Mining as Observation*, with the refuted uniform bi-Lipschitz claim for the commute filter in Volume 14's honest negatives, ledger row NEG-1.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 3 section 3.2	condition (A2), tangential displacement	turboquant-pro\docs\KV_KEYS_FINDING.md:61-86; geometric-observation\chapters\ch09_legibility.md:42-54
chapter 3 section 3.3	row normalization is the projection onto the read subspace of the geodesic-rank consumer	geometric-observation\chapters\ch09_legibility.md:31-41; geometric-observation\chapters\ch03_historical_precursors.md:110
chapter 9 section 9.1	the geodesic-rank reader discards the radius, row normalization is the angular projection	geometric-observation\chapters\ch09_legibility.md:10-41; the-angular-observer\theorem.md:110-146

FAILURES AND CORRECTIONS

- NEG-1, [refuted]. Fixed-scale, uniform-in-m bi-Lipschitz for the commute filter.

CONDITIONS

- A map is bi-Lipschitz with factor K when it neither stretches nor shrinks any distance by more than K . The factor is at least one, compositions multiply the factors, and the map preserves a nearest neighbour whenever the runner-up is more than K squared times as far; the rank certificate's per-pair fact.
- It does not preserve every ordering, so it is not rank-faithful, and the program's fixed-scale uniform bi-Lipschitz claim for the commute filter is refuted and carried in the ledger.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/BiLipschitz.lean`, `theorems one_le_factor`, `within_comp`, `nn_preserved`, `not_rank_faithful`, `at observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 3.

RELATED

[rank-faithful](#) (p. 277); [rank certificate](#) (p. 275); [recognizer](#) (p. 288); [quotient](#) (p. 271).

bit

concept. Entry id `bit`.

DEFINITION

The unit of a budget. A direction quantized with b bits is stored at one of two to the power b levels. Chapter 0 section 0.7.

EQUATION

Book equation 0.13.

$$D(b) = \sum_i s_i v_i 4^{-b_i}, \quad b_i = \max\left(0, \frac{1}{2} \log_2 \frac{s_i v_i}{\theta}\right), \quad \sum_i b_i = B.$$

Book equation 4.2.

$$D(b) = \sum_i s_i \sigma_i^2 2^{-2b_i}, \quad s_i = v_i^\top P_C v_i, \quad b_i^* = \max\left(0, \frac{1}{2} \log_2 \frac{s_i \sigma_i^2}{\theta}\right), \quad \sum_i b_i^* = B,$$

LEDGER

- GO-4. Fixed-budget verdicts invert under budget-matched observation, per the wavelength mechanism. [replicated]. `geometric-observation/claims/LEDGER.md:66` at `9f3829f`.
- GO-7. A stored description's description rate and its conditional Landauer reset content are operationally separate resources: the same finite- n code index needing $\hat{R} \approx 0.67$ bits/symbol to describe is fully recoverable from retained side information at bin rate $0.26 = 0.39\hat{R}$, fails increasingly below its conditional content, and fails absolutely (err 1.00 at every bin rate) without S . [replicated]. `geometric-observation/claims/LEDGER.md:69` at `9f3829f`.

FIRST STATED

Chapter 0 section 0.7 of *Data Mining as Observation*, with the program's matched-bits comparisons in Volume 14 and `turboquant-pro`.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

CONDITIONS

- The unit of a budget. A direction quantized with b bits is stored at one of two to the b levels, so one more bit doubles the levels, halves a uniform quantizer's step, and quarters the squared error of the high-rate model, and bits add across directions since the levels multiply.
- Every flip comparison is at matched bits, since a code that spends more bits is a different budget and not a better reader.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/Bit.lean`, `theorems levels_succ`, `levels_add`, `step_succ`, `sqError_succ`, `sqError_antitone`, `levels_zero`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 1, 2, 4, 6, 10, 11, 12, 13, 14.

RELATED

[budget](#) (p. 46); [water-filling](#) (p. 365); [the flip](#) (p. 138); [Landauer's principle](#) (p. 194).

blind probe

instrument. Entry id `blind-probe`.

DEFINITION

An instrument that recovers a consumer's read operator from calls to the consumer alone, without access to its gradients or its code. Chapters 11 and 12.

EQUATION

Book equation 0.8.

$$g_j \approx \frac{C(x + h e_j) - C(x - h e_j)}{2h}, \quad j = 1, \dots, d.$$

Book equation 0.9.

$$P_C = \mathbb{E}[g g^\top], \quad g = \nabla C(x).$$

Book equation 11.4.

$$\text{directions resolved}(k) = \begin{cases} 1 \text{ or } 2, & k < d \\ \text{rank } P_C, & k \geq d \end{cases} \quad \text{cost} = 2d \text{ consumer calls per operating point.}$$

LEDGER

- OT-3. Under subspace-confined second-order transcripts, fewer than d directions cannot identify a hidden leading eigenspace (theorem, adaptive to $d-2$ / oblivious to $d-1$); a known k_0 -dim exclusion moves the cliff to exactly $d-k_0$ and never softens it. [demonstrated]. `geometric-observation/claims/LEDGER.md:31 at 9f3829f`.
- OT-6. The laws transfer outside compression with zero modification: blind-recovered P_C of a ranking consumer over embeddings; equal-Euclidean-energy perturbations; the trace picks the ranking-destroyer. [predicted]. `geometric-observation/claims/LEDGER.md:35 at 9f3829f`.

- NEG-12. (Gate B, prospective, real LLM) On a trained frontier attention layer the blind probe recovers the read operator above the sealed bar *and* projection beats reconstruction. [refuted]. `geometric-observation/claims/LEDGER.md:105` at 9f3829f.
- GO-B-Llama. Trained frontier LLM (Llama-3.2-3B), softmax-attention consumer — blind probe on real post-RoPE keys [predicted]. `geometric-observation/claims/LEDGER.md:115` at 9f3829f.
- GO-B-Llama-rematch. Trained frontier LLM (Llama-3.2-3B), softmax-attention consumer — reconnected dissociation on real post-RoPE keys [predicted]. `geometric-observation/claims/LEDGER.md:118` at 9f3829f.
- GO-B-Llama-rematch. Trained frontier LLM (Llama-3.2-3B), softmax-attention consumer — reconnected dissociation on real post-RoPE keys [predicted]. `geometric-observation/claims/LEDGER.md:118` at 9f3829f.
- GO-B-legal (035→036). Legal-citation retrieval (CourtListener), cosine-ranking consumer, LaBSE embeddings — real large corpus, non-physical consumer [predicted]. `geometric-observation/claims/LEDGER.md:119` at 9f3829f.
- GO-EC-3. A read operator recovered from a black-box consumer by query-only finite-difference probing, composed with the Kalman covariance as $\text{tr}(\hat{P}_C \Sigma)$, prospectively selects sensors that improve the held-out consumer at matched budgets with probe cost charged — capturing 94.6% of the known analytic optimum's gain on the positive-control arm (gate $\geq 75\%$) and improving 16.3% over the best consumer-agnostic policy on non-analytic consumers (gate $\geq 8\%$), with trace-matched ordering carried by the composition at 86.9% over 61 pairs (gate $\geq 65\%$). [predicted]. `geometric-observation/claims/LEDGER.md:162` at 9f3829f.

FIRST STATED

readscope, the blind probe as a specified instrument, PyPI `readscope`, `readscope/PRINCIPLES.md` and `readscope/SPEC.md`, and Volume 14 chapter 10.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 2 section 2.3	refusal regimes, selection consumers read order, recurrences compound	<code>readscope\readscope\regimes.py:1-60</code>
chapter 2 section 2.4	C-7, sixteen points in 128 dimensions, rank fifteen, near ten to the eleventh nats, refuses when samples do not exceed dimension, warns below five per dimension, loading is a property of two distributions	<code>readscope\SPEC.md:653-680</code>
chapter 2 section 2.4	C-7b, 92.1 percent, fit range 0.89 to 91.64, 15 percent inside, maximum 1.9e12, endpoint attenuation 0.437, 202 of 240 at 1.0, refuses to extrapolate, loading is a warning	<code>readscope\SPEC.md:680-720</code> ; <code>readscope\readscope\loading.py:136-219</code>
chapter 2 section 2.6	0.647 published, 1.000 recovered, weighted vs unweighted median 0.796 range 0.678 to 0.985, probe 1.000 on every cell, 36 cells, one cell at 0.985	<code>readscope\SPEC.md:410-450</code>
chapter 2 section 2.6	C-10, 24 to 576 queries, reference rank 24 to 128, 0.821 to 0.703 vs 0.647, 68 percent of distance closed, 0.174 to 0.056, probe 1.000000 on 16 cells, residual uncontrolled	<code>readscope\SPEC.md:450-500</code>

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 3 section 3.4	chance overlap rank over dim reported with every reading	readscope\README.md:200-222; readscope\readscope\metrics.py:31-99
chapter 4 section 4.2	water-filling formula, directions below the water get no bits, the surrogate caveat	readscope\readscope\allocate.py:1-100
chapter 4 section 4.2	effective rank as participation ratio, energy rank	readscope\readscope\spectrum.py:35-70
chapter 6 section 6.1	selection consumers have zero sensitivity almost everywhere	readscope\readscope\regimes.py:1-60
chapter 8 section 8.3	C-8 D5 mean absolute error 0.0000, every reading 1.000, C-7b saturation, the F-21 rule	readscope\CALIBRATION.md F-21 (about line 519)
chapter 8 section 8.4	C-11c paired null, rank 2 positional 0.385 vs null 0.572, 0.615 vs 0.224, 14 of 16 cells, scope 16 cells one 3B model 192 positions	readscope\SPEC.md:806-857; readscope\calibration\records\c11c-operator-drift.json
chapter 11 section 11.7	k over d table 1, 2, 2, 16, 16, 16	readscope\README.md:118-141; readscope\SPEC.md:243-285, record readscope\calibration\records\c2e-budget-law.json
chapter 11 section 11.7	rank-independent, 0.646 vs 0.366 at half dimension, never 0.90 below k equals d	readscope\CALIBRATION.md:419-434 F-15
chapter 11 section 11.7	C-15 equal budget, 3072 observations, medians 0.02, 0.11, 0.05, 0.04, 1.0, 1.0, margin 0.883 vs 0.3, 0.062 to 0.057 at 8x	readscope\SPEC.md:285-323, record readscope\calibration\records\c15-budget-surface.json, readscope\calibration\DECLARATION-C15.md
chapter 11 section 11.7	confinement theorem, side information moves the cliff to d minus k0, noisy cliff proved then measured	readscope\PRINCIPLES.md:112-142; geometric- observation\crucible\OT3- THEOREM.md; geometric- observation\crucible\OT3-NOISY- THEOREM.md
chapter 11 section 11.7	124 cells at resolution 1.000 at k over d 1.25, five families, linearity partial	readscope\README.md:200-222
chapter 11 section 11.9	drift at rank one 0.667 vs null 0.933, sixteen cells	readscope\CALIBRATION.md:600-660 F-24; readscope\calibration\records\c11c-operator-drift.json
chapter 11 section 11.9	C-12 four bars, 40 documents, 512 tokens, 13.4 point difference, teacher forcing removes it, negative 0.015 vs 0.005, Spearman negative 0.13 at p 0.45, sign test p 0.42, verdict FAIL, feedback compounding	readscope\calibration\records\c12-longgen-drift-sym.json; readscope\calibration\DECLARATION-C12.md at commit 90e2ce2; readscope\SPEC.md:806-825

FAILURES AND CORRECTIONS

- NEG-12, [refuted]. (Gate B, prospective, real LLM) On a trained frontier attention layer the blind probe recovers the read operator above the sealed bar *and* projection beats reconstruction.
- readscope/README.md:102-116 at c8d0289. What none of this changes: **the budget law, in its proven scope**. The cliff at $k = d$ is a property of consumer calls, not FLOPs — a faster backend buys speed, never admission. The theorem behind it (PRINCIPLES.md, P3; OT-3) covers **subspace-confined directional designs at an operating point**, which is what this probe’s per-point estimators are; it does not cover every allocation of calls across many operating points, and the sketch expectation $(1+1/k) \cdot S + \text{tr}(S)/k \cdot I$ shares S ’s eigenspaces at every k — so whether many cheap points can average their way back to the population operator was a **sample-complexity question, not a proven impossibility** — and C-15 has now measured it: at equal total consumer calls, sub-dimensional budgets do not catch up, at any graded rank, within $8\times$ the full-dimension spend (SPEC.md, C-15). The cliff is a property of total calls in the measured range; only the far asymptotic regime remains open.

CONDITIONS

- The probe recovers the read operator at one operating point from calls to the consumer alone, two calls per input dimension for a central difference.
- Its budget law is a theorem for subspace-confined designs at an operating point, which is what its per-point estimators are, and does not cover allocations of calls across many operating points.
- The probe refuses when samples do not exceed the dimension and warns below a stated margin, since the identifiability miss of chapter 2 came from reading it past that point.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/ProbeCliff.lean`, `theorems centralDiff_affine, centralDiff_basis, exists_blind_direction, indistinguishable, budget_cliff`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 1, 2, 4, 6, 7, 11, 12, 14.

RELATED

[read operator](#) (p. 282); [budget cliff](#) (p. 49); [read distortion](#) (p. 281); [the flip](#) (p. 138).

block error rate

concept. Entry id `block-error-rate`.

DEFINITION

The fraction of transmitted radio blocks the receiver cannot decode, abbreviated BLER. Chapter 13.

EQUATION

Book equation 13.2.

$$T_{\text{coh}} = \frac{0.423}{f_D},$$

refuted fit: $\phi \approx 0.177 T_{\text{coh}} (R^2 = 0.915)$, sealed: $\phi \leq 6 \text{ TTI at } 10 \text{ Hz}, 1 \text{ TTI at } \geq 50 \text{ Hz}$.

Book equation 0.25.

$$T_{\text{coh}} = \frac{0.423}{f_D}.$$

LEDGER

- OT-4. Operator drift predicts a real long-generation degradation and a derived refresh intervention moves it. [refuted]. `geometric-observation/claims/LEDGER.md:36 at 9f3829f`.
- OT-11. Feedback-free staleness: streaming-retrieval damage tracks measured drift; derived-cadence re-allocation removes it. [void]. `geometric-observation/claims/LEDGER.md:48 at 9f3829f`.

FIRST STATED

Chapter 13 section 13.4 of *Data Mining as Observation*, with the radio freshness track, `observation-theory-campaigns/experiments/RADIO-FRESHNESS-TRACK.md:20-35`, and the sealed sweep observation-theory-campaigns/analysis/csi/PREREG-XPROTO-CSI-SWEEP2.md.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 13 section 13.3	radio 0.27 to 0.42 naive to 0.055 to 0.13, neural reconstruction 0.28 to 0.13	<code>observation-theory-campaigns/experiments/RADIO-FRESHNESS-TRACK.md:20-35</code>
chapter 13 section 13.4	refuted fit 0.177 T_{coh} R^2 0.915, 0.15 threshold vs 0.10 claim, fresh baseline 0.113, do not cite	<code>observation-theory-campaigns/analysis/csi/CSI-refreshfloor.json:2,111-114</code> ; <code>observation-theory-campaigns/experiments/RADIO-FRESHNESS-TRACK.md:41-47</code>
chapter 13 section 13.4	sealed correction, 0.5 dB calibration, floors 4/6/4, 2/2/3, 1, slopes 0.1008, 0.1398, 0.1086, R^2 0.7376, 0.9218, 0.6174, bars B1 to B4 and MC1 to MC4 all true, seeds 20260827 to 20260829, sealed 1d3de4a	<code>observation-theory-campaigns/analysis/csi/PREREG-XPROTO-CSI-SWEEP2.md:1-60</code> ; <code>observation-theory-campaigns/analysis/csi/XPROTO-CSI-SWEEP2-graded.json</code> ; <code>observation-theory-campaigns/experiments/SEALS.md:90</code>

FAILURES AND CORRECTIONS

- OT-4, [refuted]. Operator drift predicts a real long-generation degradation and a derived refresh intervention moves it.
- `observation-theory-campaigns/experiments/RADIO-FRESHNESS-TRACK.md:41-50 at 9d86211`. **The age horizon** (`analysis/csi/`, XPROTO-CSI-SWEEP2 sealed 2026-08-27, graded PASS): at the calibrated 0.10 budget the largest compliant report period is a few TTI at 10 Hz and 1 TTI at 50 Hz and above, so there is no proportionality law to fit. The earlier `CSI-refreshfloor.*` claim of a floor $\approx 0.177 \cdot T_{coh}$ ($R^2=0.915$) was an **unsealed exploration** measured at a relaxed 0.15 threshold against the 0.10 target, with a fresh baseline that never met the budget. It is refuted; the record is kept, not cited. Separately, the optimal linear predictor cannot beat the one-coherence-time wall (Gaussian fading $\rightarrow$ Wiener optimal), and that wall sits well outside the budget horizon. The **RAN governor** (`analysis/ran`) is the reference O-RAN rApp core: observe $\rightarrow$ measure false-clear $\rightarrow$ refresh at the floor $\rightarrow$ escalate (diversity / re-route).

CONDITIONS

- The fraction of transmitted radio blocks the receiver cannot decode, in the unit interval. A block of n symbols each lost independently with probability q fails with probability one minus one minus q to the n , at most $n q$, never falling as the block grows, and equal to q for a single symbol.
- The measured rates of the radio track and the refuted refresh law fit against them are the ledger's numbers, with the sealed sweep's calibrated baseline as the correction.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

lean/DataMiningAsObservation/BlockErrorRate.lean, theorems bler_mem_unit, blockFail_le, blockFail_mono, blockFail_one, blockFail_zero, at observation-data-mining af776fd.

USED IN

Data Mining as Observation chapters 0.

RELATED

[coherence time](#) (p. 63); [refresh floor](#) (p. 292); [freshness](#) (p. 149); [false-clear rate](#) (p. 133).

boosting

instrument. Entry id boosting.

DEFINITION

Fitting a sequence of weak scorers, each on the rows the previous ones got wrong, and summing them. Its step size is positive exactly when the weak scorer beats chance, and the reweighting makes the last scorer's weighted error one half. Chapter 7.

EQUATION

Book equation 6.1.

$$s(x) = w \cdot x + b, \quad P_C = \mathbb{E}[\sigma'(s)^2] w w^\top, \quad \text{rank } P_C = 1.$$

Book equation 0.8.

$$g_j \approx \frac{C(x + h e_j) - C(x - h e_j)}{2h}, \quad j = 1, \dots, d.$$

Book equation 0.9.

$$P_C = \mathbb{E}[g g^\top], \quad g = \nabla C(x).$$

LEDGER

- GO-B-optim-D4 (034 · D4). Optimization — gradient compression, curvature (Hessian) read operator, on a REAL model (logistic regression); optional stretch [predicted]. [geometric-observation/claims/LEDGER.md:122](#) at 9f3829f.

FIRST STATED

Freund and Schapire, a decision-theoretic generalization of on-line learning, 1997, as chapter 7 section 7.1 of *Data Mining as Observation* reads it, with the curvature reader in [geometric-observation/claims/LEDGER.md](#) row GO-B-optim-D4.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 6 section 6.3	the five-dataset table, breast cancer 0.955 to 0.963, the pattern, DOI 10.5281/zenodo.20660206	theory-radar\README.md:100-140
chapter 7 section 7.1	bagging variance, out-of-bag estimation, random forests, importances, AdaBoost	Hastie, Tibshirani, Friedman, ESL 2e chapters 15 and 10.1; TSK 2e 4.10
chapter 7 section 7.2	second-order leaf values, logistic curvature	ESL 2e 10.9 to 10.13; XGBoost introduction to boosted trees

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 7 section 7.2	real logistic model, exact Hessian, anti 300 of 300, flip 82 of 300, coupling diagnosis, bound not refutation	geometric-observation\claims\LEDGER.md row GO-B-optim-D4
chapter 7 section 7.3	the five-dataset outcomes	theory-radar\README.md:100-140

FAILURES AND CORRECTIONS

none

CONDITIONS

- Fitting a sequence of weak scorers, each on the rows the previous ones got wrong, and summing them with a step size set by the weighted error. The step size is positive exactly when the weak scorer beats chance, and after the rows are reweighted the weighted error of the scorer just fitted is one half, so the next scorer must find something new.
- Gradient boosting reads the loss through its curvature, so the optimizer is a consumer whose read operator is the Hessian, and on the author's five-dataset benchmark the ensemble wins when the boundary needs many features and a formula wins when it needs few.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/Boosting.lean`, `theorems alpha_pos_iff`, `alpha_half`, `reweighted_error_half`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 1, 6, 7, 8.

RELATED

[ensemble](#) (p. 121); [curvature](#) (p. 91); [decision tree](#) (p. 95); [importance](#) (p. 172).

bootstrap

instrument. Entry id bootstrap.

DEFINITION

An estimate of a confidence interval made by resampling the data with replacement many times and recomputing the statistic each time. A paired bootstrap resamples the same rows for two methods at once so that shared sampling variation cancels. Chapter 0 section 0.9.

EQUATION

Book equation 8.2.

$$\widehat{\text{Var}}_{\text{NB}} = \left(\frac{1}{J} + \frac{n_{\text{test}}}{n_{\text{train}}} \right) \hat{\sigma}^2,$$

$$J = 1000, \frac{n_{\text{test}}}{n_{\text{train}}} = \frac{1}{4} \Rightarrow 1 + 250 = 251, \sqrt{251} = 15.84.$$

Book equation 0.18.

$$\widehat{\text{Var}}_{\text{NB}} = \left(\frac{1}{J} + \frac{n_{\text{test}}}{n_{\text{train}}} \right) \hat{\sigma}^2, \quad \frac{\widehat{\text{Var}}_{\text{NB}}}{\hat{\sigma}^2/J} = 1 + J \frac{n_{\text{test}}}{n_{\text{train}}}.$$

LEDGER

none

FIRST STATED

Efron, bootstrap methods, 1979, as chapter 0 section 0.9 states it, with the program's paired form in the flip comparisons of Volume 14 and the constraint-gap review.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 6 section 6.4	the uncorrected t stored as sigma, 251 and 15.84, nine wins to three, seven, eight, 3 wins 17 ties 11 losses on 31 datasets, the three-way inconsistency	constraint-gap\review\FINDINGS.md:1-35; constraint-gap\README.md:57-76
chapter 8 section 8.5	variance inflation 251, SE inflation 15.84, J_eff 3.98, about 62 needed, 3 of 9 at full, 7 at half, 8 at a third	constraint-gap\review\FINDINGS.md:1-35

FAILURES AND CORRECTIONS

none

CONDITIONS

- An interval estimated by resampling the data with replacement and recomputing the statistic each time. A paired bootstrap resamples the same rows for two methods at once, and the variance of the difference is the sum of the two variances minus twice their covariance, so pairing narrows the interval exactly when the two scores covary positively across rows.
- Resampling rows does not undo dependence between folds, which is the correction's job, and the bootstrap's interval is only as wide as the sample it resamples.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/Bootstrap.lean`, `theorems mean_sub`, `var_sub`, `paired_lt_iff`, `cov_comm`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 6, 7, 11, 12, 14.

RELATED

[confidence interval](#) (p. 71); [standard error](#) (p. 337); [Nadeau and Bengio correction](#) (p. 225); [harness](#) (p. 159).

budget

concept. Entry id budget.

DEFINITION

The third element of an observer. The bits, rows, calls, seconds, or dollars a consumer may spend. Chapter 1.

EQUATION

Book equation 1.1.

$$O = (C, G, B).$$

Book equation 0.13.

$$D(b) = \sum_i s_i v_i 4^{-b_i}, \quad b_i = \max\left(0, \frac{1}{2} \log_2 \frac{s_i v_i}{\theta}\right), \quad \sum_i b_i = B.$$

Book equation 11.4.

$$\text{directions resolved}(k) = \begin{cases} 1 \text{ or } 2, & k < d \\ \text{rank } P_C, & k \geq d \end{cases} \quad \text{cost} = 2d \text{ consumer calls per operating point.}$$

LEDGER

- OT-3. Under subspace-confined second-order transcripts, fewer than d directions cannot identify a hidden leading eigenspace (theorem, adaptive to $d-2$ / oblivious to $d-1$); a known k_0 -dim exclusion moves the cliff to exactly $d-k_0$ and never softens it. [demonstrated]. `geometric-observation/claims/LEDGER.md:31` at 9f3829f.
- GO-4. Fixed-budget verdicts invert under budget-matched observation, per the wavelength mechanism. [replicated]. `geometric-observation/claims/LEDGER.md:66` at 9f3829f.
- GO-12. The dynamic region's opening control — staleness is access width, not delay: with full context-path access, pure-delay aging is information-free (the recoding identity $(Y, V, S^{(\Delta)}) \stackrel{d}{=} (Y, V, P^\Delta S^{(0)})$ makes every $\sigma(S)$ -measurable conditional functional exactly Δ -invariant; circulant-exact, $O(1/n)$ edge leakage on finite windows); with time-local slice access the tax is GO-11's static quadratic with the substitution set by the *encoder's* access — single-letter (Y_t, V_t) records at $(\rho, s/a^{2\Delta})$, context-epoch-latent records at the strictly smaller $(\rho a^\Delta, \tau^2)$ (gap up to 0.053 bits) — both strictly increasing in Δ with common limit $\frac{1}{2} \log_2(1/D)$. GO-8's age-dependence is the slice regime's operational face. Theorem 1 (the conditional-variance reduction) settles the causal-prefix eraser in the single-record setting: every observation-subset σ -algebra enters through $q_{\mathcal{G}} = \text{Var}(V_t \mid \mathcal{G})$ alone; slice/prefix/path = single-sample/Kalman-fixed-lag/noncausal-Wiener variances; strict interpolation at every finite lag with exact gap $C^{2\Delta}(P_f - q_{\text{path}})$; $W(\Delta) = kT \ln 2 \cdot L_{\text{prefix}}$ is the dynamic conditional-Landauer curve. Theorem 2-spectral (070): the spectral conditional RDF — at the work endpoint, $L(D)$ is the equal-slope allocation of per-frequency static quadratics over the circulant spectrum, with classical reverse water-filling ($\tau^2 \rightarrow \infty$) and the static theorem (flat spectrum) exact; per-mode convexity proven; the third promotion of the water level completed for the single-consumer face; Toeplitz transfer scoped imported-with-lemmas, $O(1/n)$ numerics. The weighted spectral theorem (071) closes Conjecture 1 at circulant scope: J_w decomposes per-frequency at every w , each mode carrying Theorem 3's two-water-level system $(\gamma_0(\omega), \gamma_1(\omega))$ under ONE common distortion price (the conjectured 'pair of prices' refuted-as-phrased, rescoped to $\$(w, \mu)\$$); convexity via the full-region moment argument + perturbation. The third promotion is complete on BOTH faces. [predicted]. `geometric-observation/claims/LEDGER.md:78'` at 9f3829f.

FIRST STATED

Volume 14, chapter 4 for the triple and chapter 7 for cost, `geometric-observation/chapters/ch07_cost.md`, DOI 10.5281/zenodo.21776291.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 4 section 4.2	GO-6, d 8 and r 4, output at or below surrogate at or below reconstruction at every rate, about 500 times, gap 0.41 to 0.005, isotropic control collapses	<code>geometric-observation/claims/LEDGER.md</code> row GO-6; <code>geometric-observation/chapters/ch07_cost.md</code>

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 4 section 4.2	commission tax of order half $r \log M$ over m , omission floor, unbounded corrected to floor, naive projector 30 percent over, VI-6	geometric-observation\chapters\ch07_cost.md section on mismatch; geometric-observation\claims\LEDGER.md VI-6
chapter 4 section 4.2	floor measured on 16 of 16 heads, naive assumption off 20 to 60 percent, about 100000 times, 0.10 nats against about 6e-7, first gate missed 0 of 16, NEG-13	geometric-observation\chapters\ch07_cost.md; geometric-observation\claims\LEDGER.md GO-P-2026-027 and NEG-13
chapter 4 section 4.4	GO-4 budget inversion, fixed m 10 rises, matched m 121, 126, 159 collapses, 3 seeds	geometric-observation\claims\LEDGER.md row GO-4
chapter 11 section 11.7	confinement theorem, side information moves the cliff to d minus k_0 , noisy cliff proved then measured	readscope\PRINCIPLES.md:112-142; geometric-observation\crucible\OT3-THEOREM.md; geometric-observation\crucible\OT3-NOISY-THEOREM.md
chapter 13 section 13.6	attempt three, windows 1024, 256, 32, uncertainty 0.982 to 0.892, 5 of 6, 0.4375 with SE 0.070 at 5 percent keep vs 0.30, 97 percent eviction, oracle-miss 0.370 vs 0.25, V4 0.078 vs 0.0625, contrast 0.359 with SE 0.068, n 64, seed 20260812, 89 duty cycles	geometric-observation\claims\LEDGER.md row GO-2/GO-12/GO-13 operational; geometric-observation\prereg\GO-P-2026-077-kv-consumer-relative.md; geometric-observation\results\GO13-kvaw2-governed.json

FAILURES AND CORRECTIONS

none

CONDITIONS

- The budget is the bits, rows, calls, seconds, or dollars a consumer may spend. It bounds what of the read operator can be measured and used, and changes the operator only when it changes the consumer.
- Comparisons are made at matched budget. A verdict at a fixed budget can invert under budget matching, and the ledger carries the case.
- For a probe the budget is consumer calls, and recovery of the read operator is a cliff at the full dimension of the space.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/ProbeCliff.lean`, `theorems centralDiff_affine`, `centralDiff_basis`, `exists_blind_direction`, `indistinguishable`, `budget_cliff`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13.

RELATED

[observer](#) (p. 236); [water-filling](#) (p. 365); [budget cliff](#) (p. 49); [coverage](#) (p. 86).

budget cliff

result. Entry id budget-cliff.

DEFINITION

The finding that recovery of a read operator by a probe is a cliff at the full dimension of the space rather than a slope, and that the cliff does not move with the operator's rank. Chapter 11.

EQUATION

Book equation 11.4.

$$\text{directions resolved}(k) = \begin{cases} 1 \text{ or } 2, & k < d \\ \text{rank } P_C, & k \geq d \end{cases} \quad \text{cost} = 2d \text{ consumer calls per operating point.}$$

LEDGER

- OT-3. Under subspace-confined second-order transcripts, fewer than d directions cannot identify a hidden leading eigenspace (theorem, adaptive to $d-2$ / oblivious to $d-1$); a known k_0 -dim exclusion moves the cliff to exactly $d-k_0$ and never softens it. [demonstrated]. `geometric-observation/claims/LEDGER.md:31 at 9f3829f`.
- OT-10. The noisy cliff: noise floors accuracy at a derived level; the cliff's location never moves. [**[refuted-as-sealed]** warning]. `geometric-observation/claims/LEDGER.md:45 at 9f3829f`.

FIRST STATED

readscope, predicted then measured, and proved as the confinement theorem in Volume 14, `geometric-observation/crucible/OT3-THEOREM.md` and `geometric-observation/crucible/OT3-NOISY-THEOREM.md`, DOI 10.5281/zenodo.21776291.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 11 section 11.7	confinement theorem, side information moves the cliff to d minus k_0 , noisy cliff proved then measured	<code>readscope\PRINCIPLES.md:112-142</code> ; <code>geometric-observation\crucible\OT3-THEOREM.md</code> ; <code>geometric-observation\crucible\OT3-NOISY-THEOREM.md</code>

FAILURES AND CORRECTIONS

- `readscope/README.md:102-116 at c8d0289`. What none of this changes: **the budget law, in its proven scope**. The cliff at $k = d$ is a property of consumer calls, not FLOPs — a faster backend buys speed, never admission. The theorem behind it (`PRINCIPLES.md`, P3; OT-3) covers **subspace-confined directional designs at an operating point**, which is what this probe's per-point estimators are; it does not cover every allocation of calls across many operating points, and the sketch expectation $(1+1/k) \cdot S + \text{tr}(S)/k \cdot I$ shares S 's eigenspaces at every k — so whether many cheap points can average their way back to the population operator was a **sample-complexity question, not a proven impossibility** — and C-15 has now measured it: at equal total consumer calls, sub-dimensional budgets do not catch up, at any graded rank, within $8\times$ the full-dimension spend (`SPEC.md`, C-15). The cliff is a property of total calls in the measured range; only the far asymptotic regime remains open.

CONDITIONS

- The reader receives responses confined to the k directions it chose, the read operator is planted in a fixed subspace, and noise enters as the noisy theorem models it. Under those conditions the cliff at k equal to

the dimension is a theorem and is rank-independent.

- Known side information about a k_0 -dimensional exclusion moves the cliff to d minus k_0 and never softens it.
- Whether many cheap points spread across operating points can average their way back to the population operator was measured within a budget range and is not a theorem.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/ProbeCliff.lean`, theorems `centralDiff_affine`, `centralDiff_basis`, `exists_blind_direction`, `indistinguishable`, `budget_cliff`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 1, 4, 11.

RELATED

[blind probe](#) (p. 39); [read operator](#) (p. 282).

C

cache

concept. Entry id cache.

DEFINITION

A store of recently used data kept so that it need not be fetched again. A cold measurement is taken before the cache holds anything useful, a warm one after. Chapter 13.

EQUATION

Book equation 13.1.

$$\text{FC} = \Pr[W \text{ refutes} \mid \mathcal{C}_t \text{ clears}], \quad d_O(\Delta) = \text{tr}(P_C M_{\text{drift}}(\Delta)), \quad M_{\text{drift}}(\Delta) = \mathbb{E}[\delta_\Delta \delta_\Delta^\top].$$

Book equation 0.26.

$$\text{FC} = \Pr[W \text{ refutes} \mid \mathcal{C}_t \text{ clears}], \quad \text{coverage} = \Pr[\mathcal{C}_t \text{ clears}].$$

LEDGER

- OT-11. Feedback-free staleness: streaming-retrieval damage tracks measured drift; derived-cadence re-allocation removes it. [void]. `geometric-observation/claims/LEDGER.md:48` at 9f3829f.
- GO-12. The dynamic region's opening control — staleness is access width, not delay: with full context-path access, pure-delay aging is information-free (the recoding identity $(Y, V, S^{(\Delta)}) \stackrel{d}{=} (Y, V, P^\Delta S^{(0)})$ makes every $\sigma(S)$ -measurable conditional functional exactly Δ -invariant; circulant-exact, $O(1/n)$ edge leakage on finite windows); with time-local slice access the tax is GO-11's static quadratic with the substitution set by the *encoder's* access — single-letter (Y_t, V_t) records at $(\rho, s/a^{2\Delta})$, context-epoch-latent records at the strictly smaller $(\rho a^\Delta, \tau^2)$ (gap up to 0.053 bits) — both strictly increasing in Δ with common limit $\frac{1}{2} \log_2(1/D)$. GO-8's age-dependence is the slice regime's operational face. Theorem 1 (the conditional-variance reduction) settles the causal-prefix eraser in the single-record setting: every observation-subset σ -algebra enters through $q_{\mathcal{G}} = \text{Var}(V_t \mid \mathcal{G})$ alone; slice/prefix/path = single-sample/Kalman-fixed-lag/noncausal-Wiener variances; strict interpolation at every finite lag with exact gap $C^{2\Delta}(P_f - q_{\text{path}})$; $W(\Delta) = kT \ln 2 \cdot L_{\text{prefix}}$ is the dynamic conditional-Landauer curve. Theorem 2-spectral (070): the spectral conditional RDF — at the work endpoint, $L(D)$ is the equal-slope allocation of per-frequency static quadratics over the circulant spectrum, with classical reverse water-filling ($\tau^2 \rightarrow \infty$) and the static theorem (flat spectrum) exact; per-mode convexity proven; the third promotion of the water level completed for the single-consumer face; Toeplitz transfer scoped imported-with-lemmas, $O(1/n)$ numerics. The weighted spectral theorem (071) closes Conjecture 1 at circulant scope: J_w decomposes per-frequency at every w , each mode carrying Theorem 3's two-water-level system $(\gamma_0(\omega), \gamma_1(\omega))$ under ONE common distortion price (the conjectured 'pair of prices' refuted-as-phrased, rescoped to $\$(w, \mu)\$$); convexity via the full-region moment argument + perturbation. The third promotion is complete on BOTH faces. [predicted]. `geometric-observation/claims/LEDGER.md:78'` at 9f3829f.

FIRST STATED

Chapter 13 section 13.5 of *Data Mining as Observation*, with the cache reader row of chapter 1's consumer table and the hot-and-cold substrate measurements of the freshness track.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 8 section 8.9	13.7 ROUGE-L, 0.25/4.19/9.60/13.7 at 64/128/256/512, re-validation negative 0.31 n=40, 26.64 under symmetric nf4, _quant_nf4a_group unchanged since 289bdfc before 4f7baab	turboquant-pro\CLAIMS.md:63-81; turboquant-pro\benchmarks\kvquant_matrix\REVAL-2026-08-08.md

FAILURES AND CORRECTIONS

none

CONDITIONS

- A store of recently used data kept so that it need not be fetched again. A cache reader reads the entries in its footprint and nothing else, so its hit rate over a stream of requests lies in the unit interval, is zero cold and one warm, never falls when the cache holds more, and is unchanged by entries outside the requests.
- A cold and a warm measurement of the same system are two observers of it. The ZooKeeper substrate's false-clear rate was 0.99 hot and 0.01 cold, which is the case that a cache's number is the requests' number.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

lean/DataMiningAsObservation/Cache.lean, theorems hitRate_mem_unit, hitRate_cold, hitRate_warm, hitRate_mono, hitRate_footprint, at observation-data-mining af776fd.

USED IN

Data Mining as Observation chapters 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14.

RELATED

[KV cache](#) (p. 190); [eviction](#) (p. 126); [freshness](#) (p. 149); [replica](#) (p. 299); [consumer](#) (p. 73).

calibration

concept. Entry id calibration.

DEFINITION

The property of a score that a value of 0.8 means the row is positive eighty percent of the time. Measured by expected calibration error, equation 0.29.

EQUATION

Book equation 0.29.

$$\text{ECE} = \sum_b \frac{n_b}{n} |\bar{s}_b - \bar{y}_b|.$$

Book equation 0.38.

$$w = \max(0, 2 \cdot \text{AUROC} - 1).$$

LEDGER

- NEG-4. Lightweight online (Lloyd) key calibration beats the calibration-free default on softmax-KL. [re-futed]. `geometric-observation/claims/LEDGER.md:97` at 9f3829f.

FIRST STATED

Chapter 0 section 0.14 of *Data Mining as Observation*, with the program's calibrated authority in `gtc-prototype/docs/CALIBRATED_AUTHORITY.md:19-63` and the recalibration negative of ledger row NEG-4.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 12 section 12.5	ECE 0.018 to 0.101 vs raw up to 0.223, reliability weight, audit binding	<code>xbse\README.md:175-195</code> ; <code>gtc-prototype/docs\CALIBRATED_AUTHORITY.md:19-63</code>
chapter 14 section 14.2	reliability weights and calibration errors per axis, the collapsed family's mean weight 0.559 against the general valence channel's own 0.735, the three design rules, 0.048 to 0.049 and 0.089 to 0.101 at 696 pairs	<code>gtc-prototype/docs\CALIBRATED_AUTHORITY.md:1-65</code>

FAILURES AND CORRECTIONS

- NEG-4, [refuted]. Lightweight online (Lloyd) key calibration beats the calibration-free default on softmax-KL.

CONDITIONS

- A score is calibrated when a value of 0.8 means the row is positive eighty percent of the time. The expected calibration error, the bin-weighted mean absolute gap between mean score and fraction of positives, lies in the unit interval and is zero exactly when every weighted bin is calibrated.
- Calibration is a property of the score values and not of their ranking, so a recalibration can change it while leaving AUROC, the reliability weight, and every threshold-swept metric unchanged. The two are reported side by side and neither stands in for the other.
- Recalibrating a reconstruction toward the input improved the reconstruction and worsened the consumer, the standing negative that keeps calibration off the list of acceptance metrics for a code.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/Calibration.lean`, `theorems ece_nonneg, ece_le_one, ece_eq_zero_iff, ece_of_calibrated`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14.

RELATED

[reliability weight](#) (p. 298); [cross-corpus gate](#) (p. 87); [Monotone Invariance Theorem](#) (p. 220); [identity reader](#) (p. 171).

capacity

concept. Entry id capacity.

DEFINITION

The number of distinctions a model class can draw, the model-side half of the budget. Chapter 6.

EQUATION

Book equation 1.1.

$$O = (C, G, B).$$

Book equation 6.1.

$$s(x) = w \cdot x + b, \quad P_C = \mathbb{E}[\sigma'(s)^2] w w^\top, \quad \text{rank } P_C = 1.$$

Book equation 11.4.

$$\text{directions resolved}(k) = \begin{cases} 1 \text{ or } 2, & k < d \\ \text{rank } P_C, & k \geq d \end{cases} \quad \text{cost} = 2d \text{ consumer calls per operating point.}$$

LEDGER

- GO-4. Fixed-budget verdicts invert under budget-matched observation, per the wavelength mechanism. [replicated]. [geometric-observation/claims/LEDGER.md:66 at 9f3829f](#).

FIRST STATED

Chapter 6 section 6.1 of *Data Mining as Observation*, with the budget inversion in [geometric-observation/claims/LEDGER.md](#) row GO-4.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 4 section 4.4	GO-4 budget inversion, fixed m 10 rises, matched m 121, 126, 159 collapses, 3 seeds	geometric-observation/claims/LEDGER.md row GO-4
chapter 6 section 6.3	depth three, ten binary and eight unary operations, exact optimal F1 by sort and sweep, beam, projections	theory-radar/README.md:50-100,140-200

FAILURES AND CORRECTIONS

none

CONDITIONS

- The number of distinctions a model class can draw, the budget's model-side half. A read operator's rank is at most the dimension, and a probe with fewer directions than the dimension cannot resolve the operator, which is the cliff.
- A formula of depth three over eighteen operations has a capacity the ensemble exceeds, and the ensemble wins exactly when the boundary needs that excess.

Conditions are curated in [entries.toml](#) rather than read from a record.

MACHINE CHECKED

[lean/DataMiningAsObservation/ProbeCliff.lean](#), [theorems centralDiff_affine, centralDiff_basis, exists_blind_direction, indistinguishable, budget_cliff](#), at [observation-data-mining af776fd](#).

[lean/DataMiningAsObservation/Rank.lean](#), [theorems rank_le_width, rank_le_height, rank_outer_le_one, rank_mul_le, rank_zero, rank_transpose](#), at [observation-data-mining af776fd](#).

USED IN

Data Mining as Observation chapters 6, 7.

RELATED

[budget](#) (p. 46); [budget cliff](#) (p. 49); [rank](#) (p. 274); [ensemble](#) (p. 121).

certificate

concept. Entry id certificate.

DEFINITION

A claim, made at a time, that something is safe to act on. A clustering, a benchmark score, a freshness guarantee, and a rank bound are all certificates. Chapters 9 and 13.

EQUATION

Book equation 0.26.

$$\text{FC} = \Pr [W \text{ refutes } | \mathcal{C}_t \text{ clears}], \quad \text{coverage} = \Pr [\mathcal{C}_t \text{ clears}].$$

Book equation 9.3.

$$\hat{\mu} = \frac{\bar{s}_{\text{true}} - \bar{s}_{\text{distr}}}{\sigma_{\text{distr}}}, \quad \mu_{\text{crit}} = \mathbb{E} \left[\max_{N-1} \mathcal{N}(0, 1) \right], \quad \rho = \frac{\hat{\mu}}{\mu_{\text{crit}}}, \quad \rho = 1 \text{ vacuous.}$$

Book equation 11.2.

$$r = \frac{d_{\text{compressed}}}{d_{\text{exact}}}, \quad \kappa_{\text{strict}} = \frac{\max r}{\min r}, \quad \tau \geq 1 - 2\hat{\mu}(\kappa_{\text{strict}}), \quad \rho_S \geq 1 - 3\hat{\mu}(\kappa_{\text{strict}}),$$

$$\kappa_{97.5/2.5} = \frac{q_{97.5}(r)}{q_{2.5}(r)} \text{ gives the same two expressions as estimates, not floors.}$$

Book equation 13.1.

$$\text{FC} = \Pr [W \text{ refutes } | \mathcal{C}_t \text{ clears}], \quad d_O(\Delta) = \text{tr} (P_C M_{\text{drift}}(\Delta)), \quad M_{\text{drift}}(\Delta) = \mathbb{E} [\delta_{\Delta} \delta_{\Delta}^{\top}].$$

LEDGER

- GO-3. The certificate's vacuity threshold predicts where single-stage retrieval dies. [demonstrated]. [geometric-observation/claims/LEDGER.md:65 at 9f3829f](#).

FIRST STATED

Volume 14, chapter 19 for the certificate that ages and chapter 9 of *Data Mining as Observation* for the clustering certificate. The rank certificate is specified in [turboquant-pro/docs/CERTIFICATE_SPEC.md](#), DOI 10.5281/zenodo.20660087.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 9 section 9.3	the margin certificate, mu crit as the expected maximum of N minus 1 standard normals, rho, death at 0.948 within 6 percent, Spearman 0.991 vs 0.873, fourteen corpora, six gates, the v1 to v3 path, the standing correction	geometric-observation/experiments/G03-certificate-vacuity-v3-NOTES.md:1-60 ; geometric-observation/claims/LEDGER.md row GO-3

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 11 section 11.3	tau floor 1 minus 2 mu, Spearman floor 1 minus 3 mu, example kappa 1.0148, mu 0.0664, floor 0.8671, 19900 pairs, max certifiable 1.83, vacuous means exact rerank, sha256 binding, reference field and the 0.3 overlap difference	turboquant-pro\docs\CERTIFICATE_SPEC.md:1-80
chapter 12 section 12.6	XPROTO-LLM, benchmark 0.909, 0.920, 0.909, thirty slices, target 0.8, naive 0.333, aware 0.033, spread 0.380, deployment mean 0.736, six bars on three seeds, sealed 2026-08-25 at b61f7f1	observation-theory-campaigns\experiments\LLM-EVAL-TRACK.md:1-50; observation-theory-campaigns\analysis\llm\XPROTO-LLM-graded.json; observation-theory-campaigns\experiments\SEALS.md:85
chapter 13 section 13.2	the grammar, certificate, witness, refresh floor, false-clear rate, vacuity, the witness table	geometric-observation\chapters\ch19_the_certificate_that_ages 95; observation-theory-campaigns\experiments\FRESHNESS-PROGRAM.md:1-40

FAILURES AND CORRECTIONS

- observation-theory-campaigns/experiments/RADIO-FRESHNESS-TRACK.md:33-39 at 9d86211. warning **XPROTO-AICSI scope correction (2026-08-25)**. The v1 seal stands for what it tested, but the reconstruction-vs-consumer dissociation does **not** survive the community-standard substrate. On real 3GPP CDL-C with a CsiNet-class codec the NMSE-optimal codec reconstructs near-perfectly (NMSE $\approx$ 0.03) and false-clears 0.0 on all seeds; the pre-registered kill fired. See analysis/aicsi/PREREG-XPROTO-AICSI-V2.md (REFUTED AT SHAKEDOWN, NOT SEALED, kept negative). This is a scope correction, not a retraction. Do not headline the AICSI row; WCNC §III-C is dropped.

CONDITIONS

- A certificate is graded only where a witness exists, and where none exists it is unassessed rather than safe.
- A certificate is bound to what it certified, the bytes, the anchor sample, the footprint, and the time, and carries none of it to another corpus, workload, or moment.
- A vacuous certificate proves nothing about the object. It does not prove that no code preserves rank or that no structure exists, only that this test could not certify one.
- The rank certificate's floor is a guarantee in its strict setting and an estimate in its percentile setting.

Conditions are curated in entries.toml rather than read from a record.

MACHINE CHECKED

lean/DataMiningAsObservation/Certificate.lean, theorems falseClear_mul_coverage, coverage_empty, falseClear_mem_unit, minOverStrata_passes_iff, minOverStrata_le_weighted_mean, at observation-data-mining af776fd.

USED IN

Data Mining as Observation chapters 0, 1, 9, 11, 12, 13, 14.

RELATED

witness (p. 370); false-clear rate (p. 133); coverage (p. 86); rank certificate (p. 275); vacuity threshold (p. 359); coherence time (p. 63).

challenge set

instrument. Entry id challenge-set.

DEFINITION

A test collection built so that a shortcut which passes the benchmark fails on it. Chapter 12.

EQUATION

Book equation 14.2.

$$\text{coverage difference}_c = \text{AUROC}_c(\text{embedding}) - \text{AUROC}_c(\text{validated axes}), \quad \text{floor} \approx 0.08 \text{ to } 0.12.$$

Book equation 12.3.

$$\text{validated} \iff \text{AUROC}_{\text{cross}} - \max(\text{AUROC}_{\text{untrained}}, \text{AUROC}_{\text{BoW}}) \geq 0.10.$$

LEDGER

- NEG-15 (Bell boundary). *Query-conditioned hubness supplies a mechanism for Bell-inequality violation without action at a distance*. Refuted as a mechanism; the settings-as-queries reframing survives only as vocabulary. [demonstrated]. [geometric-observation/claims/LEDGER.md:93 at 9f3829f](#).

FIRST STATED

Chapter 12 section 12.6 and chapter 14 section 14.3 of *Data Mining as Observation*, with the program's identity-attack and euphemism sets in [gtc-prototype/docs/SPECTRUM_FINDINGS.md:66-99](#).

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 6 section 6.4	five analyses negative, 0.2 percent of floor, twelve times the band, learning-curve redesign	constraint-gap\README.md:57-76 ; constraint-gap\review\REDESIGN.md:1-45
chapter 7 section 7.3	correlation negative 0.59 at p 4.8e-4 with size, negative 0.562 at p 0.001 with the ensemble's score, predictive 0.345 and 0.589, interval negative 0.165 to 0.125, the one-line summary	constraint-gap\review\INDETERMINATES.md:60-80 ; constraint-gap\review\REDESIGN.md:45-70
chapter 7 section 7.4	the redesign, grid 50 to N, ten folds, twenty repetitions, crossing size, the 800-row boundary declared in advance, seventeen at median 0.006 and p 0.964, store the folds	constraint-gap\review\REDESIGN.md:1-110
chapter 8 section 8.5	9/3/19 to 3/17/11, five negative analyses, 0.2 percent of floor, twelve times the band, learning-curve redesign	constraint-gap\README.md:57-76 ; constraint-gap\review\REDESIGN.md

FAILURES AND CORRECTIONS

none

CONDITIONS

- A test collection built so that a shortcut which passes the benchmark fails on it. Every finite benchmark admits a shortcut, the lookup that returns the stored label for every benchmark row, and two scorers that agree on every benchmark row have the same benchmark error whatever they do elsewhere.

- A challenge set is any collection on which the shortcut and the intended rule disagree, which can only lie outside the benchmark, and on it the shortcut's error is positive while its benchmark error is zero.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/ChallengeSet.lean`, theorems `shortcut_passes`, `shortcut_fails`, `benchmark_blind`, `challenge_outside`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 9, 11, 12.

RELATED

[leakage](#) (p. 203); [harness](#) (p. 159); [deployment mismatch](#) (p. 101); [preregistration](#) (p. 258).

chance level

concept. Entry id `chance-level`.

DEFINITION

The value a rank statistic reaches on average under a random ranking. Zero for a correlation, one half for AUROC. Chapter 0 section 0.8.

EQUATION

Book equation 0.16.

$$\text{AUROC} = \Pr[s^+ > s^-] + \frac{1}{2} \Pr[s^+ = s^-].$$

Book equation 0.15.

$$\tau = \frac{\#\{\text{concordant pairs}\} - \#\{\text{discordant pairs}\}}{n(n-1)/2}.$$

Book equation 0.38.

$$w = \max(0, 2 \cdot \text{AUROC} - 1).$$

LEDGER

none

FIRST STATED

Chapter 0 section 0.8 of *Data Mining as Observation*, with the program's chance-overlap reporting in `readscope/readscope/metrics.py:31-99`.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

CONDITIONS

- The value a rank statistic reaches under a ranking that carries no information. A constant score ties every pair, so its AUROC is one half when both classes are present, its Kendall correlation with any score is zero, and its reliability weight is zero.
- Every reading the probe reports comes with its chance overlap beside it, and a gate subtracts the chance level before it counts a margin.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/ChanceLevel.lean`, `theorems aurocNum_const`, `auroc_const`, `weight_const`, `tau_const`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 3, 6, 11.

RELATED

[reliability weight](#) (p. 298); [cross-corpus gate](#) (p. 87); [Kendall correlation](#) (p. 185); [harness](#) (p. 159).

chunk

concept. Entry id chunk.

DEFINITION

A piece of a document, the unit a retrieval pipeline encodes and indexes. Chunking is a discrete stage with no Jacobian. Chapter 12.

EQUATION

Book equation 12.3.

$$\text{validated} \iff \text{AUROC}_{\text{cross}} - \max(\text{AUROC}_{\text{untrained}}, \text{AUROC}_{\text{BoW}}) \geq 0.10.$$

Book equation 0.36.

$$w_{t,d} = \text{tf}_{t,d} \cdot \ln \frac{N}{\text{df}_t}, \quad \text{tf}_{t,d} = \frac{\text{count of } t \text{ in } d}{\text{length of } d}, \quad \text{df}_t = \text{documents containing } t.$$

LEDGER

- GO-B-legal (035→036). Legal-citation retrieval (`CourtListener`), cosine-ranking consumer, LaBSE embeddings — real large corpus, non-physical consumer [predicted]. `geometric-observation/claims/LEDGER.md:119` at `9f3829f`.

FIRST STATED

Chapter 12 section 12.2 of *Data Mining as Observation*, with the pullback composition in `geometric-observation/chapters/ch06_mathematical_preliminaries.md:10-27`.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 2 section 2.2	read subspace small, operator local, pullback composition, rank cannot increase	<code>geometric-observation\chapters\ch05_the_read_metric_and_the_c</code> 48; <code>geometric-observation\chapters\ch06_mathematical_preliminaries</code> 27
chapter 11 section 11.4	R80 table, 25x probe depth, 92 percent own cell, one third cross-article, 0.53 matches same-article share, relative contrast discriminates nothing	<code>openvector-bench\results\R80_ANN.md:1-40</code> ; <code>turboquant-pro\docs\RESEARCH_ROADMAP.md:133-160</code>

Where the book states it	Numbers, as the book’s sources table records them	Source
chapter 12 section 12.2	pullback composition and the rank bound	geometric-observation\chapters\ch06_mathematical_preliminaries; chapter 2 of this book

FAILURES AND CORRECTIONS

none

CONDITIONS

- A piece of a document, the unit a retrieval pipeline encodes and indexes. Chunking is a discrete stage with no Jacobian, and a stage that identifies two inputs identifies them for every stage after it, so what a chunker merges no later stage can separate.
- A chunk’s vector is read by the index against a query vector, and the probe depth, cell, and cross-article share of the candidates are properties of the chunks and the query set together.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/Pipeline.lean`, `theorems quotient_inherited`, `quotient_inherited_chain`, `rank_comp_le_first`, `rank_comp_le_second`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 12.

RELATED

[retrieval-augmented pipeline](#) (p. 303); [Jacobian](#) (p. 183); [encoder](#) (p. 120); [inverted file](#) (p. 177).

classifier

concept. Entry `id classifier`.

DEFINITION

A consumer that produces a score per row and turns it into a decision with a threshold. Chapter 6.

EQUATION

Book equation 6.1.

$$s(x) = w \cdot x + b, \quad P_C = \mathbb{E}[\sigma'(s)^2] w w^\top, \quad \text{rank } P_C = 1.$$

Book equation 1.1.

$$O = (C, G, B).$$

LEDGER

- GO-1. The consumer’s invariant/nuisance split is identifiable ex ante from the consumer functional. [predicted]. `geometric-observation/claims/LEDGER.md:62` at `9f3829f`.
- GO-B-whale (038). Sperm-whale coda dialect (DSWP/Sharma 2024), Clan classifier — cetacean communication; promotes the exploratory (A2) verdict to a sealed flip [predicted]. `geometric-observation/claims/LEDGER.md:120` at `9f3829f`.

FIRST STATED

Chapter 6 section 6.1 of *Data Mining as Observation*, with the classifier row of Volume 14’s consumer table and the whale clan classifier of ledger row GO-B-whale.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 2 section 2.3	refusal regimes, selection consumers read order, recurrences compound	readscope\readscope\regimes.py:1-60
chapter 6 section 6.1	selection consumers have zero sensitivity almost everywhere	readscope\readscope\regimes.py:1-60

FAILURES AND CORRECTIONS

none

CONDITIONS

- A consumer that produces a score per row and turns it into a decision with a threshold. For a linear classifier the sensitivity is the link's slope times the weight direction, so the read operator is the workload mean of the squared slope times the outer product of the weights, sends every vector to a multiple of the weights, and reads nothing orthogonal to them.
- The score is affine, so chapter 0's exact quotient applies before the link. The whale clan classifier is the flip's classifier case, held-out AUROC 0.934 against 0.883 for a code that reconstructs twice as well.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/Classifier.lean`, `theorems readOp_classifier`, `readOp_classifier_mulVec`, `readOp_classifier_orth`, `score_orth`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 1, 2, 4, 5, 6, 7, 8, 10, 11, 12, 14.

RELATED

[consumer](#) (p. 73); [read operator](#) (p. 282); [threshold](#) (p. 350); [decision boundary](#) (p. 94); [margin](#) (p. 213).

codebook

concept. Entry id codebook.

DEFINITION

The small set of allowed values a quantizer replaces each number with. Chapter 0 section 0.11.

EQUATION

Book equation 4.2.

$$D(b) = \sum_i s_i \sigma_i^2 2^{-2b_i}, \quad s_i = v_i^\top P_C v_i, \quad b_i^* = \max\left(0, \frac{1}{2} \log_2 \frac{s_i \sigma_i^2}{\theta}\right), \quad \sum_i b_i^* = B,$$

Book equation 0.13.

$$D(b) = \sum_i s_i v_i 4^{-b_i}, \quad b_i = \max\left(0, \frac{1}{2} \log_2 \frac{s_i v_i}{\theta}\right), \quad \sum_i b_i = B.$$

LEDGER

- G0-7. A stored description's description rate and its conditional Landauer reset content are operationally separate resources: the same finite- n code index needing $\hat{R} \approx 0.67$ bits/symbol to describe is fully

recoverable from retained side information at bin rate $0.26 = 0.39\hat{R}$, fails increasingly below its conditional content, and fails absolutely (err 1.00 at every bin rate) without S . [replicated]. `geometric-observation/claims/LEDGER.md:69` at 9f3829f.

- NEG-4. Lightweight online (Lloyd) key calibration beats the calibration-free default on softmax-KL. [re-futed]. `geometric-observation/claims/LEDGER.md:97` at 9f3829f.

FIRST STATED

Chapter 0 section 0.11 and chapter 4 section 4.2 of *Data Mining as Observation*, with the six codebooks of ledger row GO-7 and the codebook confound of the honest negatives.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 4 section 4.2	GO-6, d 8 and r 4, output at or below surrogate at or below reconstruction at every rate, about 500 times, gap 0.41 to 0.005, isotropic control collapses	<code>geometric-observation/claims/LEDGER.md</code> row GO-6; <code>geometric-observation/chapters/ch07_cost.md</code>
chapter 4 section 4.4	GO-4 budget inversion, fixed m 10 rises, matched m 121, 126, 159 collapses, 3 seeds	<code>geometric-observation/claims/LEDGER.md</code> row GO-4
chapter 6 section 6.2	planted probe, overlap 0.936 vs 0.059, twelve of twelve, reconstruction 0.40, five of five	<code>geometric-observation/claims/LEDGER.md</code> row GO-1
chapter 7 section 7.2	curvature reader, identical reconstruction 0.298, flip twelve of twelve both, projected variance twelve of twelve, reconstruction Spearman negative 0.20, identity curvature tie	<code>geometric-observation/experiments/G02-gradient-curvature-NOTES.md:1-40</code> ; <code>geometric-observation/claims/LEDGER.md</code> row GO-B
chapter 7 section 7.2	real logistic model, exact Hessian, anti 300 of 300, flip 82 of 300, coupling diagnosis, bound not refutation	<code>geometric-observation/claims/LEDGER.md</code> row GO-B-optim-D4
chapter 9 section 9.3	the margin certificate, mu crit as the expected maximum of N minus 1 standard normals, rho, death at 0.948 within 6 percent, Spearman 0.991 vs 0.873, fourteen corpora, six gates, the v1 to v3 path, the standing correction	<code>geometric-observation/experiments/G03-certificate-vacuity-v3-NOTES.md:1-60</code> ; <code>geometric-observation/claims/LEDGER.md</code> row GO-3
chapter 11 section 11.7	GO-1 overlap 0.936 vs 0.059, flip 12 of 12, reconstruction 0.40	<code>geometric-observation/claims/LEDGER.md</code> row GO-1; <code>geometric-observation/chapters/ch10_the_blind_probe.md:34-52</code>
chapter 12 section 12.2	recall 0.999 vs 0.592, the derived death point within 6 percent across fourteen corpora	<code>openvector-bench/README.md:60-96</code> ; <code>geometric-observation/claims/LEDGER.md</code> row GO-3
chapter 12 section 12.3	uncompressed 0.79, centred 0.84	<code>geometric-observation/claims/LEDGER.md</code> row GO-B-legal

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 12 section 12.3	041 non-oracle, frozen LSA TF-IDF to SVD 100 train-only, AUROC 0.975 vs 0.910, flip tied, magnitude overshoot, partial	geometric-observation\chapters\ch10_the_blind_probe.md:100-118; geometric-observation\claims\LEDGER.md row GO-B-blind 041
chapter 13 section 13.5	GO-7, 0.67 bits per symbol, bin rate 0.26 equals 0.39, error 1.00 without side information, two families, six codebooks	geometric-observation\claims\LEDGER.md row GO-7
chapter 13 section 13.5	GO-8, 0.10 to 0.55 across ages 0 to 64, flip probability 0.05, 1 percent to 100 percent at age 32, Gaussian pass 5 of 5, the control-statistic caveat	geometric-observation\claims\LEDGER.md row GO-8; geometric-observation\experiments\GO-landauer-gaussian-secondsettings-NOTES.md
chapter 13 section 13.6	attempt three, windows 1024, 256, 32, uncertainty 0.982 to 0.892, 5 of 6, 0.4375 with SE 0.070 at 5 percent keep vs 0.30, 97 percent eviction, oracle-miss 0.370 vs 0.25, V4 0.078 vs 0.0625, contrast 0.359 with SE 0.068, n 64, seed 20260812, 89 duty cycles	geometric-observation\claims\LEDGER.md row GO-2/GO-12/GO-13 operational; geometric-observation\prereg\GO-P-2026-077-kv-consumer-relative.md; geometric-observation\results\GO13-kvaw2-governed.json

FAILURES AND CORRECTIONS

- NEG-4, [refuted]. Lightweight online (Lloyd) key calibration beats the calibration-free default on softmax-KL.

CONDITIONS

- The small set of allowed values a quantizer replaces each input with, under the nearest-codeword rule. The distortion of an input is its distance to the nearest codeword, nonnegative, zero when the input is a codeword, achieved by some codeword, and never larger for a larger codebook.
- Two arms compared at unequal codebooks are not a flip comparison, which is the twenty-five percent codebook confound the ledger carries, and the reset content of a stored description is separate from its rate, which is row GO-7's finding across six codebooks.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/Codebook.lean`, `theorems distortion_nonneg`, `distortion_codeword`, `distortion_anti`, `exists_nearest`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 4, 8, 11, 13.

RELATED

[quantization](#) (p. 266); [product quantization](#) (p. 261); [confound](#) (p. 72); [the flip](#) (p. 138).

coherence time

concept. Entry id coherence-time.

DEFINITION

The interval over which a changing quantity stays correlated with itself. A value measured longer ago than the coherence time is a guess. Equation 0.25.

EQUATION

Book equation 0.25.

$$T_{\text{coh}} = \frac{0.423}{f_D}.$$

Book equation 13.2.

$$T_{\text{coh}} = \frac{0.423}{f_D},$$

refuted fit: $\phi \approx 0.177 T_{\text{coh}} (R^2 = 0.915)$, sealed: $\phi \leq 6$ TTI at 10 Hz, 1 TTI at ≥ 50 Hz.

LEDGER

- OT-4. Operator drift predicts a real long-generation degradation and a derived refresh intervention moves it. [refuted]. geometric-observation/claims/LEDGER.md:36 at 9f3829f.

FIRST STATED

Clarke's model of mobile-radio reception for the formula, and Volume 14 chapter 19 for its role as the interval past which a certificate has decorrelated.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 13 section 13.3	radio 0.27 to 0.42 naive to 0.055 to 0.13, neural reconstruction 0.28 to 0.13	observation-theory-campaigns\experiments\RADIO-FRESHNESS-TRACK.md:20-35
chapter 13 section 13.4	refuted fit 0.177 T_{coh} R^2 0.915, 0.15 threshold vs 0.10 claim, fresh baseline 0.113, do not cite	observation-theory-campaigns\analysis\csi\CSI-refreshfloor.json:2,111-114; observation-theory-campaigns\experiments\RADIO-FRESHNESS-TRACK.md:41-47
chapter 13 section 13.4	sealed correction, 0.5 dB calibration, floors 4/6/4, 2/2/3, 1, slopes 0.1008, 0.1398, 0.1086, R^2 0.7376, 0.9218, 0.6174, bars B1 to B4 and MC1 to MC4 all true, seeds 20260827 to 20260829, sealed 1d3de4a	observation-theory-campaigns\analysis\csi\PREREG-XPROTO-CSI-SWEEP2.md:1-60; observation-theory-campaigns\analysis\csi\XPROTO-CSI-SWEEP2-graded.json; observation-theory-campaigns\experiments\SEALS.md:90

FAILURES AND CORRECTIONS

- OT-4, [refuted]. Operator drift predicts a real long-generation degradation and a derived refresh intervention moves it.
- observation-theory-campaigns/experiments/RADIO-FRESHNESS-TRACK.md:41-50 at 9d86211. **The age horizon** (analysis/csi/, XPROTO-CSI-SWEEP2 sealed 2026-08-27, graded PASS): at the calibrated 0.10 budget the largest compliant report period is a few TTI at 10 Hz and 1 TTI at 50 Hz and above, so there is no proportionality law to fit. The earlier CSI-refreshfloor.* claim of a floor $\approx 0.177 \cdot T_{\text{coh}}$ ($R^2=0.915$) was

an **unsealed exploration** measured at a relaxed 0.15 threshold against the 0.10 target, with a fresh baseline that never met the budget. It is refuted; the record is kept, not cited. Separately, the optimal linear predictor cannot beat the one-coherence-time wall (Gaussian fading → Wiener optimal), and that wall sits well outside the budget horizon. The **RAN governor** (`analysis/ran`) is the reference O-RAN rApp core: observe→measure false-clear→refresh at the floor→escalate (diversity / re-route).

CONDITIONS

- Clarke’s formula gives the coherence time from the Doppler frequency alone and assumes the fading model behind it.
- A value measured longer ago than the coherence time is a guess, and a certificate issued that long ago has decorrelated from the state it certified.
- The proportional refresh-floor law that was fit to it was refuted and is not cited. Its sealed replacement calibrated the baseline first and found floors at a few transmission intervals.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/CoherenceTime.lean, theorems clarke_to_three_decimals, examples, refuted_law_exceeds_sealed,`
at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 13.

RELATED

[certificate](#) (p. 55); [false-clear rate](#) (p. 133); [refresh floor](#) (p. 292); [witness](#) (p. 370).

cold, warm

concept. Entry id `cold-warm`.

DEFINITION

A cold measurement is taken before the cache holds anything useful, a warm one after. A cost figure without one of the two words beside it is not a cost figure. Chapter 0 section 0.13 and chapter 13.

EQUATION

Book equation 0.26.

$$FC = \Pr [W \text{ refutes} \mid \mathcal{C}_t \text{ clears}], \quad \text{coverage} = \Pr [\mathcal{C}_t \text{ clears}].$$

Book equation 13.2.

$$T_{\text{coh}} = \frac{0.423}{f_D},$$

refuted fit: $\phi \approx 0.177 T_{\text{coh}} (R^2 = 0.915)$, sealed: $\phi \leq 6 \text{ TTI at } 10 \text{ Hz, } 1 \text{ TTI at } \geq 50 \text{ Hz}$.

LEDGER

- OT-11. Feedback-free staleness: streaming-retrieval damage tracks measured drift; derived-cadence re-allocation removes it. `[void]. geometric-observation/claims/LEDGER.md:48 at 9f3829f`.
- GO-12. The dynamic region’s opening control — staleness is access width, not delay: with full context-path access, pure-delay aging is information-free (the recoding identity $(Y, V, S^{(\Delta)}) \stackrel{d}{=} (Y, V, P^\Delta S^{(0)})$ makes every $\sigma(S)$ -measurable conditional functional exactly Δ -invariant; circulant-exact, $O(1/n)$ edge leakage on finite windows); with time-local slice access the tax is GO-11’s static quadratic with the substitution set by the *encoder’s* access — single-letter (Y_t, V_t) records at $(\rho, s/a^{2\Delta})$, context-epoch-latent records at the strictly smaller $(\rho a^\Delta, \tau^2)$ (gap up to 0.053 bits) — both strictly increasing in Δ with common limit $\frac{1}{2} \log_2(1/D)$. GO-8’s age-dependence is the slice regime’s operational face.

Theorem 1 (the conditional-variance reduction) settles the causal-prefix eraser in the single-record setting: every observation-subset σ -algebra enters through $q_{\mathcal{G}} = \text{Var}(V_t \mid \mathcal{G})$ alone; slice/prefix/path = single-sample/Kalman-fixed-lag/noncausal-Wiener variances; strict interpolation at every finite lag with exact gap $C^{2\Delta}(P_f - q_{\text{path}})$; $W(\Delta) = kT \ln 2 \cdot L_{\text{prefix}}$ is the dynamic conditional-Landauer curve. Theorem 2-spectral (070): the spectral conditional RDF — at the work endpoint, $L(D)$ is the equal-slope allocation of per-frequency static quadratics over the circulant spectrum, with classical reverse water-filling ($\tau^2 \rightarrow \infty$) and the static theorem (flat spectrum) exact; per-mode convexity proven; the third promotion of the water level completed for the single-consumer face; Toeplitz transfer scoped imported-with-lemmas, $O(1/n)$ numerics. The weighted spectral theorem (071) closes Conjecture 1 at circulant scope: J_w decomposes per-frequency at every w , each mode carrying Theorem 3's two-water-level system $(\gamma_0(\omega), \gamma_1(\omega))$ under ONE common distortion price (the conjectured 'pair of prices' refuted-as-phrased, rescoped to $\$(w, \mu)\$$); convexity via the full-region moment argument + perturbation. The third promotion is complete on BOTH faces.[predicted].geometric-observation/claims/LEDGER.md:78' at 9f3829f.

FIRST STATED

Chapter 0 section 0.13 of *Data Mining as Observation*, with the placement study's cold measurement in turboquant-pro/docs/RESEARCH_ROADMAP.md:133-160.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 8 section 8.4	gate meta-rule, relative contrast discriminates nothing	openvector-bench\openvector_bench\score_rc1.py:1-14; turboquant-pro\docs\RESEARCH_ROADMAP.md:133-160
chapter 11 section 11.4	R80 table, 25x probe depth, 92 percent own cell, one third cross-article, 0.53 matches same-article share, relative contrast discriminates nothing	openvector-bench\results\R80_ANN.md:1-40; turboquant-pro\docs\RESEARCH_ROADMAP.md:133-160
chapter 13 section 13.1	22.30 s cold vs 0.835 s warm at eight probes, every prior number warm, fragmentation 63.7 bracketing 42 to 47, 64 shards, routing sparsity computed and discarded	turboquant-pro\docs\RESEARCH_ROADMAP.md:133-160

FAILURES AND CORRECTIONS

none

CONDITIONS

- A cold measurement is taken before the cache holds anything useful, a warm one after. The hit rate is zero on an empty cache, one when the cache holds the whole footprint, grows as the cache grows, and does not move when entries outside the footprint are added.
- A single cold measurement at eight probes read 22.30 seconds against 0.835 seconds warm, and every previously reported placement number was warm, so a cost figure without the word warm or cold beside it is not a cost figure.

Conditions are curated in entries.toml rather than read from a record.

MACHINE CHECKED

lean/DataMiningAsObservation/Cache.lean, theorems hitRate_mem_unit, hitRate_cold, hitRate_warm, hitRate_mono, hitRate_footprint, at observation-data-mining af776fd.

USED IN

Data Mining as Observation chapters 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14.

RELATED

[cache](#) (p. 51); [eviction](#) (p. 126); [replica](#) (p. 299); [budget](#) (p. 46); [certificate](#) (p. 55).

commit hash

instrument. Entry id commit-hash.

DEFINITION

The fingerprint git assigns to a snapshot of a repository, which fixes the time at which a file existed in that state. Chapter 0 section 0.12.

EQUATION

Book equation 8.3.

$$\Pr[\text{at least one of } m \text{ null tests passes}] = 1 - (1 - \alpha)^m, \quad \alpha_{\text{Bonferroni}} = \frac{\alpha}{m}.$$

LEDGER

- OT-7. The damage form, trace pairing, rank, and loading covariance are GL(d)-invariant under $P' = A^{-T}PA^{-1}$, while spectrum, effective rank, principal angles, and water-filling are O(d)-only — consumer-weighted damage is a geometric scalar, and P3's cliff cannot be bought down by reparameterization. [demonstrated].
geometric-observation/claims/LEDGER.md:30 at 9f3829f.

FIRST STATED

Chapter 0 section 0.12 of *Data Mining as Observation*, with the program's seal ledger, observation-theory-campaigns/experiments/SEALS.md:1-10, where every seal is a registration id, a sealing commit, and a hash.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

CONDITIONS

- The fingerprint git assigns to a snapshot of a repository, which fixes the time at which a file existed in that state. A changed hash proves a changed file and an equal hash is evidence, since a fixed-length digest of longer inputs cannot be injective.
- Every row of a sources table cites its commit, every seal records its sealing commit, and the book's own colophon names the commit it was built from, so that a number can be reproduced at the state it was measured in.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

lean/DataMiningAsObservation/Seal.lean, theorems changed_of_hash_ne, hash_eq_of_eq, exists_collision, at observation-data-mining af776fd.

USED IN

Data Mining as Observation chapters 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14.

RELATED

[hash](#) (p. 161); [sealed](#) (p. 316); [sources table](#) (p. 329); [preregistration](#) (p. 258).

commute time

concept. Entry id commute-time.

DEFINITION

The expected number of steps a random walk on a graph needs to go from one node to another and back. A particular scaling of the spectral embedding makes Euclidean distance equal it. Equation 0.20.

EQUATION

Book equation 0.20.

$$\Psi_i = \left(\frac{u_k(i)}{\sqrt{\lambda_k d_i}} \right)_{k \geq 2}, \quad \|\Psi_i - \Psi_j\|^2 = R(i, j) = \frac{C(i, j)}{\text{vol}(G)}.$$

Book equation 0.19.

$$L = I - D^{-1/2} A D^{-1/2}, \quad D = \text{diag}(d_1, \dots, d_n), \quad L u_k = \lambda_k u_k, \quad 0 = \lambda_1 \leq \lambda_2 \leq \dots.$$

Book equation 3.3.

$$X_i = r_i \theta_i, \quad r_i \rightarrow \frac{1}{\sqrt{d_i}}, \quad \theta_i = \frac{X_i}{\|X_i\|} \in S^{m-1}.$$

LEDGER

- NEG-1. Fixed-scale, uniform-in-m bi-Lipschitz for the commute filter. [refuted]. [geometric-observation/claims/LEDGER.md:94 at 9f3829f](#).

FIRST STATED

Chapter 0 section 0.10 of *Data Mining as Observation*, equation 0.20, with von Luxburg, Radl, and Hein, 2014, read in chapter 3 section 3.3 and the program's collapse proof in [the-angular-observer/theorem.md:110-146](#).

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 3 section 3.3	resistance converges to 1 over d_i plus 1 over d_j, simplex argument, radius converges to 1 over root d_i	the-angular-observer/theorem.md:110-146
chapter 3 section 3.3	the self-refuted v0.8 claim, NEG-1	the-angular-observer\README.md:170-175 ; geometric-observation\chapters\ch16_honest_negatives.md NEG-1
chapter 9 section 9.1	the geodesic-rank reader discards the radius, row normalization is the angular projection	geometric-observation\chapters\ch09_legibility.md:10-41 ; the-angular-observer/theorem.md:110-146

FAILURES AND CORRECTIONS

- NEG-1, [refuted]. Fixed-scale, uniform-in-m bi-Lipschitz for the commute filter.

CONDITIONS

- The expected number of steps a random walk on a graph needs to go from one node to another and back, the hitting time there plus the hitting time back, so it is symmetric and zero from a node to itself. Its scaling by the graph's volume is the resistance, and a particular scaling of the spectral embedding makes Euclidean distance equal it.
- On a large neighbourhood graph of a shape of dimension three or more the resistance converges to one over each degree, so it depends on two nodes only through their degrees and carries no geometry, which is why the magnitude of the spectral embedding is degree noise. The program's own first claim of a commute-metric form was self-refuted, NEG-1.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/CommuteTime.lean`, `theorems commute_symm`, `commute_self`, `resistance_symm`, `commute_eq_resistance_mul`, `collapsed_congr`, `collapsed_const`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 3.

RELATED

[spectral embedding](#) (p. 332); [Laplacian](#) (p. 199); [geodesic distance](#) (p. 154); [degree](#) (p. 97); [graph](#) (p. 156).

concentrated

concept. Entry `id concentrated`.

DEFINITION

Of a spectrum, having a few eigenvalues that carry most of the total. Chapter 0 section 0.4.

EQUATION

Book equation 0.7.

$$r_{\text{eff}} = \frac{(\sum_i \lambda_i)^2}{\sum_i \lambda_i^2}.$$

Book equation 0.5.

$$\Sigma v_i = \lambda_i v_i, \quad \Sigma = \sum_{i=1}^d \lambda_i v_i v_i^\top, \quad v_i \cdot v_j = 0 \ (i \neq j).$$

LEDGER

- OT-7. The damage form, trace pairing, rank, and loading covariance are $\text{GL}(d)$ -invariant under $P' = A^{-\top} P A^{-1}$, while spectrum, effective rank, principal angles, and water-filling are $O(d)$ -only — consumer-weighted damage is a geometric scalar, and P3's cliff cannot be bought down by reparameterization. [demonstrated]. `geometric-observation/claims/LEDGER.md:30` at `9f3829f`.

FIRST STATED

Chapter 0 section 0.4 of *Data Mining as Observation*, with the allocation report's concentration caution in `turboquant-pro/turboquant_pro/read_allocation.py:244-307`.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 4 section 4.2	effective rank as participation ratio, energy rank	readscope\readscope\spectrum.py:35-70
chapter 4 section 4.2	allocation report, gain over uniform, concentration caution below effective rank 2	turboquant-pro\turboquant_pro\read_allocation.py:244-307

FAILURES AND CORRECTIONS

none

CONDITIONS

- Of a spectrum, having a few eigenvalues that carry most of the total. If one eigenvalue carries a fraction f of the total, the effective rank is at most one over f squared, so half the mass in one direction gives effective rank at most four.
- The allocation report warns below effective rank two, which is the same statement read from the rank side, and an allocation over a concentrated spectrum spends almost everything on one direction.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/Concentrated.lean`, `theorems sq_le_sum_sq`, `effRank_le_of_fraction`, `effRank_le_four`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 3, 4, 6, 10, 11, 13.

RELATED

[spectrum](#) (p. 334); [effective rank](#) (p. 115); [water-filling](#) (p. 365); [isotropic](#) (p. 179).

confidence

concept. Entry id confidence.

DEFINITION

Of a rule, the support of the rule's itemset divided by the support of its antecedent. It is not symmetric and it misleads when the consequent is common. Chapter 5.

EQUATION

Book equation 5.1.

$$s(X) = \frac{\sigma(X)}{N}, \quad c(X \rightarrow Y) = \frac{\sigma(X \cup Y)}{\sigma(X)} = \frac{s(X \cup Y)}{s(X)}.$$

Book equation 5.2.

$$\text{lift}(X \rightarrow Y) = \frac{c(X \rightarrow Y)}{s(Y)} = \frac{s(X \cup Y)}{s(X) s(Y)}.$$

LEDGER

none

FIRST STATED

TSK chapter 5, as chapter 5 section 5.1 of *Data Mining as Observation* states it.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 5 section 5.1 to 5.3, 5.5	support, confidence, lift, the Apriori principle, FP-growth, closed and maximal itemsets, objective measures, Simpson's paradox, cross-support	TSK 2e chapter 5

FAILURES AND CORRECTIONS

none

CONDITIONS

- The support of a rule's itemset over the support of its antecedent, in the unit interval and not symmetric. Under independence the confidence equals the consequent's support whatever the antecedent, so a consequent in nine of ten transactions gives every rule into it confidence 0.9 with no association at all.
- Lift is confidence over the consequent's support and is one in that case, which is why chapter 5 sends rule evaluation to chapter 8's nulls.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/Confidence.lean`, `theorems confidence_mem_unit`, `confidence_asymm`, `confidence_indep`, `common_consequent`, `lift_of_indep`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 5, 6, 8, 9, 11, 13, 14.

RELATED

[lift](#) (p. 205); [support](#) (p. 343); [itemset](#) (p. 180); [multiple comparisons](#) (p. 221).

confidence interval

concept. Entry id `confidence-interval`.

DEFINITION

A range that would contain the true value in a stated fraction of repeated samples, usually 95 percent. Chapter 0 section 0.9.

EQUATION

Book equation 8.3.

$$\Pr[\text{at least one of } m \text{ null tests passes}] = 1 - (1 - \alpha)^m, \quad \alpha_{\text{Bonferroni}} = \frac{\alpha}{m}.$$

Book equation 8.2.

$$\widehat{\text{Var}}_{\text{NB}} = \left(\frac{1}{J} + \frac{n_{\text{test}}}{n_{\text{train}}} \right) \hat{\sigma}^2,$$

$$J = 1000, \frac{n_{\text{test}}}{n_{\text{train}}} = \frac{1}{4} \Rightarrow 1 + 250 = 251, \sqrt{251} = 15.84.$$

LEDGER

none

FIRST STATED

Chapter 0 section 0.9 of *Data Mining as Observation*, with the program's intervals in the re-gate table of chapter 14 and the hubness intervals of chapter 11.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

CONDITIONS

- A range that would contain the true value in a stated fraction of repeated samples. An interval with coverage one minus alpha misses in a fraction alpha, over m independent intervals the expected misses are m alpha and the chance that all cover is one minus alpha to the m, and the chance that at least one misses is at most m alpha.
- Nine intervals at ninety-five percent have a chance near seven percent of two or more misses, which is how the re-gate table's seven of nine inside the interval is read.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

lean/DataMiningAsObservation/ConfidenceInterval.lean, theorems `missRate_eq`, `some_miss_le`, `nine_intervals`, `nine_expected`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 6, 8, 14.

RELATED

[bootstrap](#) (p. 45); [standard error](#) (p. 337); [p-value](#) (p. 264); [preregistration](#) (p. 258).

confound

concept. Entry id `confound`.

DEFINITION

A variable that moves with both the treatment and the outcome so that a measured difference cannot be attributed. Chapter 8 requires controls before a claim.

EQUATION

Book equation 8.3.

$$\Pr[\text{at least one of } m \text{ null tests passes}] = 1 - (1 - \alpha)^m, \quad \alpha_{\text{Bonferroni}} = \frac{\alpha}{m}.$$

Book equation 8.2.

$$\widehat{\text{Var}}_{\text{NB}} = \left(\frac{1}{J} + \frac{n_{\text{test}}}{n_{\text{train}}} \right) \hat{\sigma}^2,$$

$$J = 1000, \frac{n_{\text{test}}}{n_{\text{train}}} = \frac{1}{4} \Rightarrow 1 + 250 = 251, \sqrt{251} = 15.84.$$

LEDGER

- NEG-4. Lightweight online (Lloyd) key calibration beats the calibration-free default on softmax-KL. [re-futed]. `geometric-observation/claims/LEDGER.md:97` at `9f3829f`.
- NEG-5. GO-P-2026-001 as registered: on KV keys, invariant-preserving (asym-NF4) beats reconstruction-optimal (per-block Lloyd) at matched bits, and tangential distortion dominates reconstruction in predict-

ing softmax-KL. [refuted]. [geometric-observation/claims/LEDGER.md:98](#) at 9f3829f.

- NEG-6. Relative per-channel-demeaned error norm is the quotient-tangential quantity that controls softmax-KL. [refuted]. [geometric-observation/claims/LEDGER.md:99](#) at 9f3829f.

FIRST STATED

Chapter 8 section 8.6 of *Data Mining as Observation*, with the program's codebook confound in Volume 14's honest negatives and the negative controls of the constraint-gap review.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 8 section 8.4	real 1.038 [0.958, 1.118], rotated 1.019 [0.903, 1.353], smooth 2.804 withdrawn, second attempt 0.184	constraint-gap-measurements\notes\negative_control.md:1-50
chapter 8 section 8.5	three-way inconsistency, lines 544, 548 to 556, 755	constraint-gap\review\FINDINGS.md:7-31 against theory- radar\paper\theory_radar_v6_submission.tex

FAILURES AND CORRECTIONS

- NEG-4, [refuted]. Lightweight online (Lloyd) key calibration beats the calibration-free default on softmax-KL.
- NEG-5, [refuted]. GO-P-2026-001 as registered: on KV keys, invariant-preserving (asym-NF4) beats reconstruction-optimal (per-block Lloyd) at matched bits, and tangential distortion dominates reconstruction in predicting softmax-KL.
- NEG-6, [refuted]. Relative per-channel-demeaned error norm is the quotient-tangential quantity that controls softmax-KL.

CONDITIONS

- A variable that moves with both the treatment and the outcome. In the linear case the naive difference of group means is the treatment effect plus the confound's effect times the difference of the confound's means between the groups, so it equals the effect exactly when the confound is balanced or has no effect, and the bias can have either sign and any size.
- Controls come before claims. The twenty-five percent codebook confound in the flip's early arms and the smooth-perturbation control that was withdrawn are the program's cases, each carried in the ledger.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/Confound.lean`, `theorems mean_outcome`, `naive_diff`, `naive_diff_of_balanced`, `naive_diff_of_no_effect`, `bias_unbounded`, at [observation-data-mining af776fd](#).

USED IN

Data Mining as Observation chapters 2, 3, 4, 6, 7, 8, 9, 12, 13, 14.

RELATED

[harness](#) (p. 159); [null model](#) (p. 234); [Simpson's paradox](#) (p. 325); [preregistration](#) (p. 258).

consumer

concept. Entry id consumer.

DEFINITION

The function that acts on a vector, the first element of an observer. A softmax, a classifier, an index, a person reading a report. Chapter 1.

EQUATION

Book equation 1.1.

$$O = (C, G, B).$$

Book equation 0.8.

$$g_j \approx \frac{C(x + h e_j) - C(x - h e_j)}{2h}, \quad j = 1, \dots, d.$$

LEDGER

- GO-1. The consumer's invariant/nuisance split is identifiable ex ante from the consumer functional. [predicted]. `geometric-observation/claims/LEDGER.md:62` at 9f3829f.
- GO-EC-3. A read operator recovered from a black-box consumer by query-only finite-difference probing, composed with the Kalman covariance as $\text{tr}(\hat{P}_C \Sigma)$, prospectively selects sensors that improve the held-out consumer at matched budgets with probe cost charged — capturing 94.6% of the known analytic optimum's gain on the positive-control arm (gate $\geq 75\%$) and improving 16.3% over the best consumer-agnostic policy on non-analytic consumers (gate $\geq 8\%$), with trace-matched ordering carried by the composition at 86.9% over 61 pairs (gate $\geq 65\%$). [predicted]. `geometric-observation/claims/LEDGER.md:162` at 9f3829f.

FIRST STATED

Volume 14, chapter 4, `geometric-observation/chapters/ch04_the_observer_triple.md:9-60` and its consumer table at lines 60-135, DOI 10.5281/zenodo.21776291.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 1 section 1.2	the observer triple, read subspace, nuisance, same read operator means same read geometry	<code>geometric-observation/chapters/ch04_the_observer_triple.md:9-60</code> ; <code>geometric-observation/OBSERVATION.md:1-10</code>
chapter 2 section 2.3	refusal regimes, selection consumers read order, recurrences compound	<code>readscope/readscope/regimes.py:1-60</code>
chapter 6 section 6.1	the classifier row of the consumer table, the output metric makes a different observer	<code>geometric-observation/chapters/ch04_the_observer_triple.md:60-135</code>
chapter 6 section 6.1	selection consumers have zero sensitivity almost everywhere	<code>readscope/readscope/regimes.py:1-60</code>
chapter 6 section 6.2	planted probe, overlap 0.936 vs 0.059, twelve of twelve, reconstruction 0.40, five of five	<code>geometric-observation/claims/LEDGER.md</code> row GO-1
chapter 11 section 11.7	GO-1 overlap 0.936 vs 0.059, flip 12 of 12, reconstruction 0.40	<code>geometric-observation/claims/LEDGER.md</code> row GO-1; <code>geometric-observation/chapters/ch10_the_blind_probe.md:34-52</code>

FAILURES AND CORRECTIONS

none

CONDITIONS

- A consumer is any computation that takes a vector in and gives an output out, a softmax, a classifier, an index, a person reading a report. The book's quantities are functions of the consumer, and two consumers with the same read operator are not thereby the same consumer.
- A selection consumer, a tree or a threshold, has zero sensitivity almost everywhere and is read by counting where it selects rather than by finite differences.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/ReadOperator.lean`, `theorems rank_one_reads_one_direction`, `readOp_mulVec`, `quad_readOp`, `quad_readOp_nonneg`, `readOp_mulVec_eq_zero_iff`, `readOp_diag`, `readOp_offdiag`, `readOp_symm`, `readOp_neg`, `affine_const_along_nuisance`, `readOp_affine`, `readOp_sqLength_basis`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14.

RELATED

[observer](#) (p. 236); [read operator](#) (p. 282); [sensitivity](#) (p. 319); [budget](#) (p. 46).

contraction

instrument. Entry id contraction.

DEFINITION

A formula that reduces several axes to one verdict. The instrument's contraction from ten axes is fairness minus the general valence component. Chapter 14 section 14.5.

EQUATION

Book equation 14.5.

$$FC_g = \Pr[y = \text{violation} \mid \hat{y} = \text{clear}, g], \quad \text{verdict} = \max_{g: n_g \geq n_{\min}} FC_g, \quad \text{ABSTAIN otherwise.}$$

Book equation 0.9.

$$P_C = \mathbb{E}[g g^\top], \quad g = \nabla C(x).$$

LEDGER

- GO-EC-3. A read operator recovered from a black-box consumer by query-only finite-difference probing, composed with the Kalman covariance as $\text{tr}(\hat{P}_C \Sigma)$, prospectively selects sensors that improve the held-out consumer at matched budgets with probe cost charged — capturing 94.6% of the known analytic optimum's gain on the positive-control arm (gate $\geq 75\%$) and improving 16.3% over the best consumer-agnostic policy on non-analytic consumers (gate $\geq 8\%$), with trace-matched ordering carried by the composition at 86.9% over 61 pairs (gate $\geq 65\%$). [predicted]. `geometric-observation/claims/LEDGER.md:162 at 9f3829f`.

FIRST STATED

Chapter 14 section 14.5 of *Data Mining as Observation*, with the formula in `gtc-prototype/docs/SPECTRUM_FINDINGS.md:90-99`.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 14 section 14.3	contraction 0.779 to 0.863, 77 of 1600, 4.8 percent, 0.872 to 0.863, weight 2.69	gtc-prototype\docs\SPECTRUM_FINDINGS.md:66-99
chapter 14 section 14.5	the contraction formula fairness minus the general component	gtc-prototype\docs\SPECTRUM_FINDINGS.md:90-99

FAILURES AND CORRECTIONS

none

CONDITIONS

- A formula that reduces several axes to one verdict, here ten axes to a toxicity verdict. An affine contraction has a rank-one read operator, the outer product of its weights, and its attributions sum exactly to the change in verdict.
- The contraction found by chapter 6's tool is fairness minus the general valence component, moved accuracy from 0.779 to 0.863 on 1600 items, and is a consumer a person can read.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/ReadOperator.lean`, `theorems rank_one_reads_one_direction`, `readOp_mulVec`, `quad_readOp`, `quad_readOp_nonneg`, `readOp_mulVec_eq_zero_iff`, `readOp_diag`, `readOp_offdiag`, `readOp_symm`, `readOp_neg`, `affine_const_along_nuisance`, `readOp_affine`, `readOp_sqLength_basis`, at `observation-data-mining af776fd`.

`lean/DataMiningAsObservation/Attribution.lean`, `theorems attr_sum_affine`, `attr_unread`, `sq_sensitivity_eq_readOp_diag`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 14.

RELATED

[attribution](#) (p. 26); [read operator](#) (p. 282); [residualization](#) (p. 302); [importance](#) (p. 172).

contrastive objective

concept. Entry id `contrastive-objective`.

DEFINITION

A training rule that pulls pairs declared similar together and pushes other pairs apart. The declared pairs are where an encoder learns what similar means. Chapter 12.

EQUATION

Book equation 12.3.

$$\text{validated} \iff \text{AUROC}_{\text{cross}} - \max(\text{AUROC}_{\text{untrained}}, \text{AUROC}_{\text{BoW}}) \geq 0.10.$$

Book equation 11.1.

$$\cos(k, \hat{k}) = 0.995 \quad \text{while} \quad \text{PPL} : 12.24 \rightarrow 10643.$$

LEDGER

- NEG-2. Reconstruction cosine as a proxy for key quality. [refuted]. `geometric-observation/claims/LEDGER.md:95` at 9f3829f.
- GO-B-legal (035→036). Legal-citation retrieval (CourtListener), cosine-ranking consumer, LaBSE embeddings — real large corpus, non-physical consumer [predicted]. `geometric-observation/claims/LEDGER.md:119` at 9f3829f.

FIRST STATED

Chapter 12 section 12.4 of *Data Mining as Observation*, with the program's encoders in `xbse`, `xbse/README.md:104-130`.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 12 section 12.5	within 0.75 to 0.955 collapsing to 0.47 to 0.55, cross-corpus positives as the fix, adversary not load-bearing	<code>xbse\README.md:104-130,160-172</code>

FAILURES AND CORRECTIONS

- NEG-2, [refuted]. Reconstruction cosine as a proxy for key quality.

CONDITIONS

- A training rule that pulls pairs declared similar together and pushes other pairs apart. With a margin loss the loss is nonnegative, zero exactly when every declared pair scores above every other pair by the margin, and never lower for a larger margin.
- Swapping which pairs are declared similar reverses that condition, and with a positive margin no scorer satisfies both, so the declared pairs are where an encoder learns what similar means. Fine-tuning moves the reader, and the cross-corpus positives were the fix when within-corpus scores collapsed.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/Contrastive.lean`, `theorems loss_nonneg, loss_eq_zero_iff, not_both, loss_mono_margin`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 11, 12.

RELATED

[encoder](#) (p. 120); [embedding](#) (p. 118); [cross-corpus gate](#) (p. 87); [quotient](#) (p. 271).

control

instrument. Entry `id control`.

DEFINITION

A comparison arm that should not show the effect, run beside the arm that should. A control that passes the bar makes the bar vacuous. Chapter 8 section 8.4.

EQUATION

Book equation 8.3.

$$\Pr[\text{at least one of } m \text{ null tests passes}] = 1 - (1 - \alpha)^m, \quad \alpha_{\text{Bonferroni}} = \frac{\alpha}{m}.$$

Book equation 0.17.

$$\Pr[X \geq c] = 1 - \sum_{i < c} e^{-\mu} \frac{\mu^i}{i!}, \quad c^* = \max\{c : n \Pr[X \geq c] \geq 1\}, \quad \mu = \frac{n_q k}{n}.$$

Book equation 4.6.

$$\kappa = \text{tr}(\bar{P}_C \bar{\Sigma}_x) \in [0, 1], \quad \kappa \rightarrow 1 \text{ is the coupling null (no flip).}$$

LEDGER

- GO-6. At matched rate, output coding $\leq$ surrogate $\leq$ reconstruction on the consumer metric; the output–reconstruction gap is governed by the ker P_C entropy share, and the surrogate–output gap vanishes as rate grows. [demonstrated]. `geometric-observation/claims/LEDGER.md:68 at 9f3829f`.
- NEG-2. Reconstruction cosine as a proxy for key quality. [refuted]. `geometric-observation/claims/LEDGER.md:95 at 9f3829f`.
- NEG-4. Lightweight online (Lloyd) key calibration beats the calibration-free default on softmax-KL. [refuted]. `geometric-observation/claims/LEDGER.md:97 at 9f3829f`.

FIRST STATED

Chapter 8 section 8.4 of *Data Mining as Observation*, with the negative control in `constraint-gap-measurements/notes/negative_control.md:50`.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 4 section 4.2	GO-6, d 8 and r 4, output at or below surrogate at or below reconstruction at every rate, about 500 times, gap 0.41 to 0.005, isotropic control collapses	<code>geometric-observation/claims/LEDGER.md row GO-6; geometric-observation/chapters/ch07_cost.md</code>
chapter 8 section 8.4	real 1.038 [0.958, 1.118], rotated 1.019 [0.903, 1.353], smooth 2.804 withdrawn, second attempt 0.184	<code>constraint-gap-measurements/notes/negative_control.md:1-50</code>
chapter 8 section 8.4	gate meta-rule, relative contrast discriminates nothing	<code>openvector-bench/openvector_bench/score_rc1.py:1-14; turboquant-pro/docs/RESEARCH_ROADMAP.md:133-160</code>

FAILURES AND CORRECTIONS

- NEG-2, [refuted]. Reconstruction cosine as a proxy for key quality.
- NEG-4, [refuted]. Lightweight online (Lloyd) key calibration beats the calibration-free default on softmax-KL.

CONDITIONS

- A comparison arm that should not show the effect, run beside the arm that should. A control that passes the bar makes the bar vacuous, so a claim names its control and reports both, and an isotropic control collapses the flip because there is no direction to read.
- The first smooth-field control read 2.804 and was controlling for nothing, because the gradient it was compared against was piecewise constant, and the second attempt read 0.184.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

lean/DataMiningAsObservation/NullModel.lean, theorems card_pos_perm, scores_perm, stat_of_counts_const, excess_eq_zero_iff, at observation-data-mining af776fd.

lean/DataMiningAsObservation/Bar.lean, theorems passes_anti, passes_mono, discriminates_iff, no_bar_of_null_ge, exists_bar_of_lt, vacuous_of_null_passes, at observation-data-mining af776fd.

USED IN

Data Mining as Observation chapters 2, 3, 4, 6, 7, 8, 9, 12, 13.

RELATED

[null model](#) (p. 234); [bar](#) (p. 34); [chance level](#) (p. 58); [confound](#) (p. 72); [isotropic](#) (p. 179).

correction

concept. Entry id correction.

DEFINITION

A change to a published number or claim that the record keeps beside the original, naming what was wrong. A statistical correction rescales a variance for a dependence the harness ignored. Chapter 2 section 2.4 and chapter 8 section 8.5.

EQUATION

Book equation 8.2.

$$\widehat{\text{Var}}_{\text{NB}} = \left(\frac{1}{J} + \frac{n_{\text{test}}}{n_{\text{train}}} \right) \hat{\sigma}^2,$$

$$J = 1000, \frac{n_{\text{test}}}{n_{\text{train}}} = \frac{1}{4} \Rightarrow 1 + 250 = 251, \sqrt{251} = 15.84.$$

Book equation 8.1.

$$O_{\text{harness}} = (C_{\text{score}}, G_{\text{metric}}, B = \text{folds} \times \text{samples} \times \text{seeds}).$$

Book equation 2.1.

$$P_{C_2 \circ C_1}(x) = J_1(x)^\top P_{C_2}(C_1(x)) J_1(x), \quad \text{rank } P_{C_2 \circ C_1}(x) \leq \text{rank } P_{C_2}(C_1(x)) \quad \text{at each row } x.$$

LEDGER

- NEG-9. The projected-variance trace $\text{tr}(P_C \Sigma_\delta)$ is a complete rank statistic for softmax-KL across all arms. [refuted]. [geometric-observation/claims/LEDGER.md:102](#) at 9f3829f.
- NEG-13 → resolved. (GO-P-2026-026, prospective, real LLM) Appendix-E's omission floor is a rate-irreducible downstream wall on trained Llama read operators. [missed]. [geometric-observation/claims/LEDGER.md:106](#) at 9f3829f.

FIRST STATED

Chapter 2 section 2.4 and chapter 8 section 8.5 of *Data Mining as Observation*, with the standing corrections in [geometric-observation/claims/LEDGER.md](#).

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 6 section 6.4	the uncorrected t stored as sigma, 251 and 15.84, nine wins to three, seven, eight, 3 wins 17 ties 11 losses on 31 datasets, the three-way inconsistency	constraint-gap\review\FINDINGS.md:1-35; constraint-gap\README.md:57-76
chapter 6 section 6.4	the reviewer's revision plan, drop sigma, bounded claim, 19 at 200 by 5 vs 11 at 20 by 5, rerun in progress	theory-radar\paper\REVISION_PLAN.md:1-60
chapter 6 section 6.5	loading weights and stability across folds requested	theory-radar\paper\REVISION_PLAN.md issue 6
chapter 8 section 8.5	variance inflation 251, SE inflation 15.84, J_{eff} 3.98, about 62 needed, 3 of 9 at full, 7 at half, 8 at a third	constraint-gap\review\FINDINGS.md:1-35

FAILURES AND CORRECTIONS

- NEG-9, [refuted]. The projected-variance trace $\text{tr}(P_C \Sigma \delta)$ is a complete rank statistic for softmax-KL across all arms.

CONDITIONS

- A change to a published number or claim that the record keeps beside the original, naming what was wrong and what replaced it. A statistical correction rescales a variance for a dependence the harness ignored, and a changed digest proves a changed file, so a correction is itself a sealed object.
- A correction can be the wrong shape. The uncorrected t was stored as a sigma, the fitted correction that transferred with mean absolute error zero was the identity, and the rows stay visible with the correction pointing at them.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/NadeauBengio.lean`, theorems `correctedVar_eq`, `naiveVar_le_correctedVar`, `inflation_unbounded`, `book_numbers`, at `observation-data-mining af776fd`.

`lean/DataMiningAsObservation/Seal.lean`, theorems `changed_of_hash_ne`, `hash_eq_of_eq`, `exists_collision`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 13, 14.

RELATED

[Nadeau and Bengio correction](#) (p. 225); [ledger class](#) (p. 204); [sealed](#) (p. 316); [harness](#) (p. 159).

correlation

concept. Entry id correlation.

DEFINITION

The cosine of two centred columns, between minus one and one, unchanged by shifting or rescaling either. Its distance lives on the quotient that discards shift and scale. Chapter 0 section 0.6.

EQUATION

Book equation 0.1.

$$x \cdot y = \sum_{i=1}^d x_i y_i, \quad \|x\| = \sqrt{x \cdot x}, \quad \cos \theta = \frac{x \cdot y}{\|x\| \|y\|}.$$

Book equation 3.1.

$$d_O(u) = u^\top \Sigma u, \quad u = (\cos 15^\circ, \sin 15^\circ), \\ \Sigma_1 = \text{diag}(0.3, 1.7), \Sigma_2 = \text{diag}(1.7, 0.3), \quad d_O = 0.394 \text{ vs } 1.606.$$

Book equation 0.18.

$$\widehat{\text{Var}}_{\text{NB}} = \left(\frac{1}{J} + \frac{n_{\text{test}}}{n_{\text{train}}} \right) \hat{\sigma}^2, \quad \widehat{\text{Var}}_{\text{NB}} = 1 + J \frac{n_{\text{test}}}{n_{\text{train}}}.$$

LEDGER

- NEG-9. The projected-variance trace $\text{tr}(\text{P_C} \cdot \Sigma_\delta)$ is a complete rank statistic for softmax-KL across all arms. [refuted]. [geometric-observation/claims/LEDGER.md:102 at 9f3829f](#).

FIRST STATED

Pearson, 1895, as chapter 0 section 0.6 and chapter 3 section 3.1 of *Data Mining as Observation* read it, with the radius-against-degree correlation in the-[angular-observer/README.md:135-139](#).

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 3 section 3.3	radius against degree correlation 0.92 to 0.99	the-angular-observer\README.md:135-139
chapter 6 section 6.4	the uncorrected t stored as sigma, 251 and 15.84, nine wins to three, seven, eight, 3 wins 17 ties 11 losses on 31 datasets, the three-way inconsistency	constraint-gap\review\FINDINGS.md:1-35; constraint-gap\README.md:57-76
chapter 6 section 6.4	five analyses negative, 0.2 percent of floor, twelve times the band, learning-curve redesign	constraint-gap\README.md:57-76; constraint-gap\review\REDESIGN.md:1-45
chapter 8 section 8.5	9/3/19 to 3/17/11, five negative analyses, 0.2 percent of floor, twelve times the band, learning-curve redesign	constraint-gap\README.md:57-76; constraint-gap\review\REDESIGN.md

FAILURES AND CORRECTIONS

- NEG-9, [refuted]. The projected-variance trace $\text{tr}(\text{P_C} \cdot \Sigma_\delta)$ is a complete rank statistic for softmax-KL across all arms.

CONDITIONS

- The cosine of two centred columns. It lies between minus one and one, is one for a column against itself, is symmetric, and is unchanged by shifting or positively rescaling either column, so correlation distance lives on the quotient that discards shift and scale.
- Rank correlations read the order and discard the values, and a read distortion that correlates with a downstream ranking at 0.80 is a control and not a complete rank statistic, NEG-9.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

lean/DataMiningAsObservation/Correlation.lean, theorems abs_corr_le_one, corr_self, cosine_comm, corr_comm, center_shift, center_smul, corr_shift, corr_smul, at observation-data-mining af776fd.

USED IN

Data Mining as Observation chapters 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14.

RELATED

[cosine](#) (p. 82); [Spearman correlation](#) (p. 330); [Kendall correlation](#) (p. 185); [covariance matrix](#) (p. 85); [quotient](#) (p. 271).

cosine

concept. Entry id cosine.

DEFINITION

The dot product divided by the product of the two lengths, between minus one and one, unchanged by rescaling either vector. Its distance is the angle. Equation 0.1.

EQUATION

Book equation 0.1.

$$x \cdot y = \sum_{i=1}^d x_i y_i, \quad \|x\| = \sqrt{x \cdot x}, \quad \cos \theta = \frac{x \cdot y}{\|x\| \|y\|}.$$

Book equation 3.1.

$$d_O(u) = u^\top \Sigma u, \quad u = (\cos 15^\circ, \sin 15^\circ), \\ \Sigma_1 = \text{diag}(0.3, 1.7), \quad \Sigma_2 = \text{diag}(1.7, 0.3), \quad d_O = 0.394 \text{ vs } 1.606.$$

Book equation 3.3.

$$X_i = r_i \theta_i, \quad r_i \rightarrow \frac{1}{\sqrt{d_i}}, \quad \theta_i = \frac{X_i}{\|X_i\|} \in S^{m-1}.$$

LEDGER

- NEG-2. Reconstruction cosine as a proxy for key quality. [refuted]. [geometric-observation/claims/LEDGER.md:95](#) at 9f3829f.
- GO-B-legal (035→036). Legal-citation retrieval (CourtListener), cosine-ranking consumer, LaBSE embeddings — real large corpus, non-physical consumer [predicted]. [geometric-observation/claims/LEDGER.md:119](#) at 9f3829f.

FIRST STATED

Chapter 0 section 0.1 of *Data Mining as Observation*, equation 0.1, with the program's cosine-against-consumer negative in [turboquant-pro/docs/KV_KEYS_FINDING.md:1-49](#).

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 1 section 1.4	cosine 0.995 and perplexity of order ten thousand, the recalibration negative	geometric-observation\chapters\ch02_failure_of_observer_free_60; geometric-observation\chapters\ch16_honest_negatives.md NEG-2 and NEG-4; turboquant-pro\docs\KV_KEYS_FINDING.md:1-49
chapter 2 section 2.5	cosine 0.995 and the softmax reader	turboquant-pro\docs\KV_KEYS_FINDING.md:1-49
chapter 3 section 3.3	resistance converges to 1 over d_i plus 1 over d_j , simplex argument, radius converges to 1 over root d_i	the-angular-observer\theorem.md:110-146
chapter 8 section 8.2	cosine 0.995, perplexity near $1e4$	turboquant-pro\docs\KV_KEYS_FINDING.md:1-49; geometric-observation\chapters\ch16_honest_negatives.md NEG-2
chapter 9 section 9.1	the geodesic-rank reader discards the radius, row normalization is the angular projection	geometric-observation\chapters\ch09_legibility.md:10-41; the-angular-observer\theorem.md:110-146
chapter 11 section 11.2	fp16 12.24, values-only 13.12, PolarQuant K4 10643 and 0.095, per-channel uniform K4 14.91 and 0.062, per-channel NUQ K3 15.77 and 0.148, 2.4x and 670x, pre-rotary near 22000, 2 key heads serve 12 query heads	turboquant-pro\docs\KV_KEYS_FINDING.md:1-49
chapter 12 section 12.3	uncompressed 0.79, centred 0.84	geometric-observation\claims\LEDGER.md row GO-B-legal

FAILURES AND CORRECTIONS

- NEG-2, [refuted]. Reconstruction cosine as a proxy for key quality.

CONDITIONS

- The dot product divided by the product of the two lengths. By Cauchy and Schwarz it lies between minus one and one, a vector has cosine one with itself, rescaling either vector by a positive factor does not change it, and it is the dot product of the two row-normalized vectors, so its distance is the angle on the quotient that discards length.
- A reconstruction cosine of 0.995 sat beside a consumer's perplexity of ten thousand, so the cosine is never an acceptance metric for a code, and on a spectral embedding the cosine is the right similarity because the length has converged to degree noise.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

lean/DataMiningAsObservation/Cosine.lean, theorems `abs_dot_le`, `cosine_le_one`, `neg_one_le_cosine`, `cosine_self`, `cosine_smul`, `cosine_eq_dot_rowNormalize`, at observation-data-mining af776fd.

USED IN

Data Mining as Observation chapters 0, 1, 2, 3, 5, 6, 8, 10, 11, 12.

RELATED

[dot product](#) (p. 110); [Euclidean distance](#) (p. 124); [spectral embedding](#) (p. 332); [quotient](#) (p. 271); [identity reader](#) (p. 171).

coupling null

concept. Entry id coupling-null.

DEFINITION

The case, alignment near one, in which the read subspace and the high-variance subspace coincide and the reconstruction-optimal code is also the read-preserving one. The flip's stated boundary. Chapter 4.

EQUATION

Book equation 4.6.

$$\kappa = \text{tr}(\bar{P}_C \bar{\Sigma}_x) \in [0, 1], \quad \kappa \rightarrow 1 \text{ is the coupling null (no flip).}$$

LEDGER

- GO-4. Fixed-budget verdicts invert under budget-matched observation, per the wavelength mechanism. [replicated]. [geometric-observation/claims/LEDGER.md:66 at 9f3829f](#).
- GO-B-optim-D4 (034 · D4). Optimization — gradient compression, curvature (Hessian) read operator, on a REAL model (logistic regression); optional stretch [predicted]. [geometric-observation/claims/LEDGER.md:122 at 9f3829f](#).

FIRST STATED

Volume 14, chapter 8, [geometric-observation/chapters/ch08_value.md:108-116](#), the gradient-compression case, and chapter 12 for the alignment dial, DOI 10.5281/zenodo.21776291.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 4 section 4.4	gradient compression anti 300 of 300, flip 27 percent, coupling boundary	geometric-observation\chapters\ch08_value.md:108-116
chapter 4 section 4.4	alignment law, retrospective fit exploratory, one prospective point	geometric-observation\chapters\ch12_failure_taxonomy_and_kappa82
chapter 7 section 7.2	real logistic model, exact Hessian, anti 300 of 300, flip 82 of 300, coupling diagnosis, bound not refutation	geometric-observation\claims\LEDGER.md row GO-B-optim-D4

FAILURES AND CORRECTIONS

none

CONDITIONS

- The coupling null is the case in which the read subspace and the high-variance subspace coincide, alignment near one, so that the reconstruction-optimal code is also the read-preserving one and no flip is possible.

- It is the flip's stated boundary, and a comparison that finds no flip there has confirmed the boundary rather than refuted the flip.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/Alignment.lean`, theorems `overlap_sq_le`, `alignment_le_one`, `alignment_nonneg`, `alignment_eq_one_of_proportional`, `coupling_null`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 4.

RELATED

[alignment](#) (p. 11); [the flip](#) (p. 138); [read distortion](#) (p. 281).

covariance matrix

concept. Entry id `covariance-matrix`.

DEFINITION

The matrix of pairwise covariances of a set of rows, whose trace is the total variance. Equation 0.3.

EQUATION

Book equation 0.3.

$$\Sigma_{ij} = \mathbb{E}[(x_i - \mu_i)(x_j - \mu_j)], \quad \text{tr } \Sigma = \sum_i \Sigma_{ii}.$$

Book equation 0.4.

$$\text{Var}(u \cdot x) = u^\top \Sigma u.$$

Book equation 0.5.

$$\Sigma v_i = \lambda_i v_i, \quad \Sigma = \sum_{i=1}^d \lambda_i v_i v_i^\top, \quad v_i \cdot v_j = 0 \ (i \neq j).$$

LEDGER

- OT-7. The damage form, trace pairing, rank, and loading covariance are $\text{GL}(d)$ -invariant under $P' = A^{-\top} P A^{-1}$, while spectrum, effective rank, principal angles, and water-filling are $O(d)$ -only — consumer-weighted damage is a geometric scalar, and P3's cliff cannot be bought down by reparameterization. [demonstrated]. `geometric-observation/claims/LEDGER.md:30 at 9f3829f`.
- OT-2. Loading is a covariance, not a distance: reading error under a change of measure is priced by $\varepsilon \cdot \|E[h \cdot A]\|$ — predicted from the base measure alone — and a full-magnitude shift orthogonal to the operator's variation does nothing. [predicted]. `geometric-observation/claims/LEDGER.md:33 at 9f3829f`.

FIRST STATED

Chapter 0 section 0.2 of *Data Mining as Observation*, paired with the read operator in chapter 4, and Volume 14 chapter 5, `geometric-observation/chapters/ch05_the_read_metric_and_the_quotient.md:7-48`.

MEASUREMENTS

Where the book states it	Numbers, as the book’s sources table records them	Source
chapter 2 section 2.2	read subspace small, operator local, pullback composition, rank cannot increase	geometric-observation\chapters\ch05_the_read_metric_and_the_o48; geometric-observation\chapters\ch06_mathematical_preliminaries27
chapter 2 section 2.2	read distortion controls but is not a complete rank statistic, twelve of twelve, one middle pair misordered, NEG-9	geometric-observation\chapters\ch05_the_read_metric_and_the_o75
chapter 4 section 4.2	effective rank as participation ratio, energy rank	readscope\readscope\spectrum.py:35-70
chapter 4 section 4.2	allocation report, gain over uniform, concentration caution below effective rank 2	turboquant-pro\turboquant_pro\read_allocation.py:244-307

FAILURES AND CORRECTIONS

none

CONDITIONS

- The weighted average of the centred rows’ outer products, the same construction as the read operator with the centred row in place of the sensitivity. Its quadratic form along a unit direction is the variance of the projection onto it, its trace is the total variance, and it is positive semidefinite.
- Chapter 4 pairs it with the read operator in one basis. Where the two are proportional there is no flip, and where they are not the water-filling allocation reads both.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/Covariance.lean`, `theorems var_eq_quad`, `trace_eq_total`, `quad_nonneg`, `cov_symm`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 1, 2, 3, 4, 9, 10, 11, 12.

RELATED

[read operator](#) (p. 282); [whitening](#) (p. 368); [water-filling](#) (p. 365); [alignment](#) (p. 11); [explained variance](#) (p. 129).

coverage

concept. Entry id coverage.

DEFINITION

The fraction of decisions a certificate clears, reported beside its false-clear rate, since a certificate that clears nothing has a false-clear rate of zero and no use. Equation 0.26.

EQUATION

Book equation 0.26.

$$FC = \Pr [W \text{ refutes } | \mathcal{C}_t \text{ clears}], \quad \text{coverage} = \Pr [\mathcal{C}_t \text{ clears}].$$

LEDGER

none

FIRST STATED

Data Mining as Observation, chapter 0 section 0.13 and chapter 13, introduced on 2026-09-03 when the false-clear rate was restated as a conditional rate, and Volume 14 chapter 19 at geometric-observation 7d91883.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 12 section 12.6	XPROTO-LLM, benchmark 0.909, 0.920, 0.909, thirty slices, target 0.8, naive 0.333, aware 0.033, spread 0.380, deployment mean 0.736, six bars on three seeds, sealed 2026-08-25 at b61f7f1	observation-theory-campaigns\experiments\LLM-EVAL-TRACK.md:1-50; observation-theory-campaigns\analysis\llm\XPROTO-LLM-graded.json; observation-theory-campaigns\experiments\SEALS.md:85
chapter 13 section 13.3	ZooKeeper hot 0.99 cold 0.01 witnessed 0.0, Postgres 0.50 to 0.06, MongoDB 0.47 to 0.03, production Postgres 0.47 to 0.02, real substrates, disjoint seeds	observation-theory-campaigns\experiments\DATABASE-FRESHNESS-TRACK.md:1-45

FAILURES AND CORRECTIONS

none

CONDITIONS

- A naive certificate clears every decision, so its coverage is one and its conditional and joint false-clear rates coincide.
- A witnessed certificate clears fewer decisions, and its false-clear rate is read beside its coverage, never alone.
- The word also names, in chapter 14, what a moderation instrument can read of the signal its inputs carry, which is a different quantity and is not this entry.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/Certificate.lean`, theorems `falseClear_mul_coverage`, `coverage_empty`, `falseClear_mem_unit`, `minOverStrata_passes_iff`, `minOverStrata_le_weighted_mean`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 10, 12, 13, 14.

RELATED

[false-clear rate](#) (p. 133); [certificate](#) (p. 55); [witness](#) (p. 370).

cross-corpus gate

instrument. Entry id `cross-corpus-gate`.

DEFINITION

The validation rule that an encoder votes only if its held-out AUROC on a corpus it was not trained on clears a preregistered margin over a bag-of-words null. Chapters 12 and 14.

EQUATION

Book equation 12.3.

$$\text{validated} \iff \text{AUROC}_{\text{cross}} - \max(\text{AUROC}_{\text{untrained}}, \text{AUROC}_{\text{BoW}}) \geq 0.10.$$

Book equation 0.16.

$$\text{AUROC} = \Pr[s^+ > s^-] + \frac{1}{2} \Pr[s^+ = s^-].$$

LEDGER

none

FIRST STATED

The moral-embedding program, `xbse/README.md:104-130,160-172`, and chapter 12 section 12.5 and chapter 14 section 14.2 of *Data Mining as Observation*.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 8 section 8.6	rights encoder 0.467 below untrained baseline, the AUROC lesson, cross-corpus same-sign fix	<code>xbse\README.md:160-172</code> ; <code>xbse\experiments\rights_r6_summary.json</code>
chapter 12 section 12.5	within 0.75 to 0.955 collapsing to 0.47 to 0.55, cross-corpus positives as the fix, adversary not load-bearing	<code>xbse\README.md:104-130,160-172</code>

FAILURES AND CORRECTIONS

none

CONDITIONS

- An encoder votes only if its held-out AUROC on a corpus it was not trained on clears both nulls, the untrained encoder and the bag-of-words baseline, by the preregistered margin of one tenth. A validated encoder therefore beats each null by the margin.
- Since AUROC is at most one, a null above nine tenths cannot be cleared by any encoder, and the gate's verdict is unchanged by any strictly monotone recalibration of the score.
- The rights encoder fell below the untrained baseline and was retired rather than tuned, and the within-corpus scores that had collapsed on a second corpus were the reason the gate is cross-corpus.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/CrossCorpusGate.lean`, `theorems clears_bow`, `clears_untrained`, `not_validated_of_saturated`, `margin_example`, `validated_comp`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 8, 12, 14.

RELATED

[reliability weight](#) (p. 298); [harness](#) (p. 159); [leakage](#) (p. 203); [preregistration](#) (p. 258).

cross-support ratio

concept. Entry id `cross-support-ratio`.

DEFINITION

The smallest item support in an itemset over the largest, with a threshold below which the itemset is discarded. Chapter 5.

EQUATION

Book equation 5.3.

$$X \subseteq Y \implies s(X) \geq s(Y).$$

Book equation 5.1.

$$s(X) = \frac{\sigma(X)}{N}, \quad c(X \rightarrow Y) = \frac{\sigma(X \cup Y)}{\sigma(X)} = \frac{s(X \cup Y)}{s(X)}.$$

LEDGER

none

FIRST STATED

TSK chapter 5 on cross-support patterns, as chapter 5 section 5.5 of *Data Mining as Observation* states it.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 5 section 5.1 to 5.3, 5.5	support, confidence, lift, the Apriori principle, FP-growth, closed and maximal itemsets, objective measures, Simpson's paradox, cross-support	TSK 2e chapter 5

FAILURES AND CORRECTIONS

none

CONDITIONS

- The smallest item support in an itemset over the largest. It lies in the unit interval, is one exactly when every item has the same support, and adding an item can only lower it, so a threshold on it prunes supersets safely in chapter 5's sense.
- It is a property of the item supports alone and says nothing about the itemset's own support, so it discards an itemset for pairing a common item with a rare one, not for being rare.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/CrossSupport.lean`, theorems `ratio_mem_unit`, `ratio_eq_one_iff`, `ratio_anti`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 5.

RELATED

[Apriori principle](#) (p. 22); [safe pruning](#) (p. 311); [lift](#) (p. 205); [multiple comparisons](#) (p. 221).

cross-validation

instrument. Entry id `cross-validation`.

DEFINITION

Splitting the data into folds and evaluating on each fold a model trained on the others. Repeated cross-validation multiplies confidence it did not earn, chapter 8 section 8.5.

EQUATION

Book equation 8.2.

$$\widehat{\text{Var}}_{\text{NB}} = \left(\frac{1}{J} + \frac{n_{\text{test}}}{n_{\text{train}}} \right) \hat{\sigma}^2,$$

$$J = 1000, \frac{n_{\text{test}}}{n_{\text{train}}} = \frac{1}{4} \Rightarrow 1 + 250 = 251, \sqrt{251} = 15.84.$$

Book equation 0.18.

$$\widehat{\text{Var}}_{\text{NB}} = \left(\frac{1}{J} + \frac{n_{\text{test}}}{n_{\text{train}}} \right) \hat{\sigma}^2, \quad \frac{\widehat{\text{Var}}_{\text{NB}}}{\hat{\sigma}^2/J} = 1 + J \frac{n_{\text{test}}}{n_{\text{train}}}.$$

LEDGER

none

FIRST STATED

Stone, cross-validators choice, 1974, as chapter 8 section 8.5 of *Data Mining as Observation* reads it, with the program's fold accounting in `constraint-gap/review/FINDINGS.md:1-35`.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 6 section 6.4	the uncorrected t stored as sigma, 251 and 15.84, nine wins to three, seven, eight, 3 wins 17 ties 11 losses on 31 datasets, the three-way inconsistency	<code>constraint-gap\review\FINDINGS.md:1-35</code> ; <code>constraint-gap\README.md:57-76</code>
chapter 8 section 8.5	variance inflation 251, SE inflation 15.84, J_{eff} 3.98, about 62 needed, 3 of 9 at full, 7 at half, 8 at a third	<code>constraint-gap\review\FINDINGS.md:1-35</code>

FAILURES AND CORRECTIONS

none

CONDITIONS

- Splitting the rows into folds and evaluating on each fold a model trained on the others, so every row is tested exactly once. The overall accuracy is the size-weighted mean of the fold accuracies, and with equal folds it is their plain mean.
- The folds share training data, so repeated cross-validation multiplies confidence it did not earn, which the Nadeau and Bengio correction prices at one plus the fold count times the test-to-train ratio.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/CrossValidation.lean`, `theorems sizes_sum`, `accuracy_weighted`, `accuracy_mean_of_equal`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 1, 2, 4, 5, 6, 7, 8, 14.

RELATED

[Nadeau and Bengio correction](#) (p. 225); [harness](#) (p. 159); [standard error](#) (p. 337); [leakage](#) (p. 203).

curvature

concept. Entry id curvature.

DEFINITION

The second derivative of a consumer, which the finite difference of equation 0.8 does not read and which sets the error of the linear model at a step. Chapter 0 section 0.5 and chapter 7 section 7.2.

EQUATION

Book equation 0.8.

$$g_j \approx \frac{C(x + h e_j) - C(x - h e_j)}{2h}, \quad j = 1, \dots, d.$$

Book equation 0.9.

$$P_C = \mathbb{E}[g g^\top], \quad g = \nabla C(x).$$

Book equation 6.1.

$$s(x) = w \cdot x + b, \quad P_C = \mathbb{E}[\sigma'(s)^2] w w^\top, \quad \text{rank } P_C = 1.$$

LEDGER

- GO-B-optim-D4 (034 · D4). Optimization — gradient compression, curvature (Hessian) read operator, on a REAL model (logistic regression); optional stretch [predicted]. `geometric-observation/claims/LEDGER.md:122` at 9f3829f.

FIRST STATED

Chapter 0 section 0.5 and chapter 7 section 7.2 of *Data Mining as Observation*, with the curvature reader's flip in `geometric-observation/claims/LEDGER.md` row GO-B-optim-D4.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 7 section 7.2	second-order leaf values, logistic curvature	ESL 2e 10.9 to 10.13; XGBoost introduction to boosted trees
chapter 7 section 7.2	real logistic model, exact Hessian, anti 300 of 300, flip 82 of 300, coupling diagnosis, bound not refutation	<code>geometric-observation\claims\LEDGER.md</code> row GO-B-optim-D4

FAILURES AND CORRECTIONS

none

CONDITIONS

- The second derivative of a consumer, which the finite difference of equation 0.8 does not read. For a quadratic the second difference recovers it exactly at every step, the error of the linear model at a step is the curvature times the step squared, and the gradient at the two ends of a step differs by the curvature times the step. An affine consumer has none.

- The reader of a gradient is the optimizer, and its read operator is the loss curvature, which chapter 4's flip was measured against on a real logistic model with its exact Hessian, anti three hundred of three hundred and the flip in 82 of 300.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/Curvature.lean`, `theorems second_quad`, `second_affine`, `linear_error`, `gradient_change`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 3, 4, 7.

RELATED

[Hessian](#) (p. 164); [finite difference](#) (p. 137); [sensitivity](#) (p. 319); [boosting](#) (p. 44); [read operator](#) (p. 282).

D

DBSCAN

instrument. Entry id dbscan.

DEFINITION

A clustering that calls a point a core point when at least a minimum number of points lie within a radius, grows clusters by reachability from core points, and leaves the rest as noise. Chapter 9.

EQUATION

Book equation 9.1.

$$\text{SSE} = \sum_{k=1}^K \sum_{i \in \mathcal{C}_k} \|x_i - c_k\|^2, \quad \text{the } P_C = I \text{ distortion summed within clusters.}$$

Book equation 0.19.

$$L = I - D^{-1/2} A D^{-1/2}, \quad D = \text{diag}(d_1, \dots, d_n), \quad L u_k = \lambda_k u_k, \quad 0 = \lambda_1 \leq \lambda_2 \leq \dots.$$

Book equation 10.3.

$$N_k(x \mid Q) = |\{q \in Q : x \in \text{top}_k(q)\}|, \quad \text{anti-hub : } N_k(x) = 0.$$

LEDGER

- NEG-11. (GO-5, prospective ×4) The $\alpha=1$ density/hubness quotient decisively and density-specifically restores invariant fidelity in a non-spectral domain. [refuted]. [geometric-observation/claims/LEDGER.md:104](#) at 9f3829f.

FIRST STATED

Ester, Kriegel, Sander, and Xu, a density-based algorithm for discovering clusters, 1996, as chapter 9 section 9.1 of *Data Mining as Observation* reads it.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 9 section 9.1	the geodesic-rank reader discards the radius, row normalization is the angular projection	geometric-observation\chapters\ch09_legibility.md:10-41 ; the-angular-observer\theorem.md:110-146
chapter 10 section 10.1	the four detector families	TSK 2e chapter 9

FAILURES AND CORRECTIONS

- NEG-11, [refuted]. (GO-5, prospective ×4) The $\alpha=1$ density/hubness quotient decisively and density-specifically restores invariant fidelity in a non-spectral domain.

CONDITIONS

- A clustering that calls a point a core point when at least a minimum number of points lie within a radius, grows a cluster by reachability from core points, and leaves the rest as noise. A wider radius or a smaller

count makes more core points, every reachability chain starts at a core point, and a point that is not a core point reaches nothing.

- Its read subspace is the density coordinate, the one the geodesic reader of a spectral embedding discards, so two clustering consumers of one embedding read complementary coordinates.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/Dbscan.lean`, `theorems ball_mono`, `core_mono`, `core_anti`, `reach_from_core`, `noise_unreachable`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 9, 10.

RELATED

[density](#) (p. 99); [k-means](#) (p. 189); [hierarchical clustering](#) (p. 165); [spectral clustering](#) (p. 331); [outlier](#) (p. 242).

decision boundary

concept. Entry id `decision-boundary`.

DEFINITION

The set of inputs whose score equals the threshold. Chapter 6.

EQUATION

Book equation 6.1.

$$s(x) = w \cdot x + b, \quad P_C = \mathbb{E}[\sigma'(s)^2] w w^\top, \quad \text{rank } P_C = 1.$$

Book equation 6.2.

$$\max_{\tau, \text{dir}} F_1\left(\mathbf{1}[\text{dir}(g(f(X)), \tau)], y\right) = \max_{\tau, \text{dir}} F_1\left(\mathbf{1}[\text{dir}(f(X), \tau)], y\right)$$

for every strictly monotone g .

LEDGER

- GO-1. The consumer's invariant/nuisance split is identifiable ex ante from the consumer functional. [predicted]. `geometric-observation/claims/LEDGER.md:62` at `9f3829f`.

FIRST STATED

Chapter 6 section 6.1 of *Data Mining as Observation*, where the classifier's read direction and its threshold meet.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

CONDITIONS

- The set of inputs whose score equals the threshold. For a linear classifier it is an affine hyperplane. Moving orthogonally to the weights never crosses it, moving along the weights crosses it exactly once, and every boundary point has the same projection on the weights.
- The boundary is where the read direction and the threshold meet and nothing else about the input enters it, which is the classifier's nuisance drawn as a picture.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

lean/DataMiningAsObservation/DecisionBoundary.lean, theorems orth_stays, cross_once, same_projection, at observation-data-mining af776fd.

USED IN

Data Mining as Observation chapters 0.

RELATED

[classifier](#) (p. 60); [threshold](#) (p. 350); [margin](#) (p. 213); [nuisance](#) (p. 233).

decision tree

instrument. Entry id decision-tree.

DEFINITION

A classifier that scores by a sequence of axis-aligned splits, flat within each leaf, so that its sensitivity is zero almost everywhere and its importances count splits. Chapter 6.

EQUATION

Book equation 6.1.

$$s(x) = w \cdot x + b, \quad P_C = \mathbb{E}[\sigma'(s)^2] w w^\top, \quad \text{rank } P_C = 1.$$

Book equation 0.8.

$$g_j \approx \frac{C(x + h e_j) - C(x - h e_j)}{2h}, \quad j = 1, \dots, d.$$

Book equation 14.5.

$$\text{FC}_g = \Pr[y = \text{violation} \mid \hat{y} = \text{clear}, g], \quad \text{verdict} = \max_{g: n_g \geq n_{\min}} \text{FC}_g, \quad \text{ABSTAIN otherwise.}$$

LEDGER

- GO-1. The consumer's invariant/nuisance split is identifiable ex ante from the consumer functional. [predicted]. [geometric-observation/claims/LEDGER.md:62 at 9f3829f](#).

FIRST STATED

Breiman, Friedman, Olshen, and Stone, classification and regression trees, 1984, as chapter 6 section 6.1 of *Data Mining as Observation* reads it, with the selection-consumer regime in [readscope/readscope/regimes.py:1-60](#).

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 2 section 2.3	refusal regimes, selection consumers read order, recurrences compound	readscope/readscope/regimes.py:1-60
chapter 6 section 6.1	selection consumers have zero sensitivity almost everywhere	readscope/readscope/regimes.py:1-60
chapter 6 section 6.3	the five-dataset table, breast cancer 0.955 to 0.963, the pattern, DOI 10.5281/zenodo.20660206	theory-radar/README.md:100-140
chapter 7 section 7.3	the five-dataset outcomes	theory-radar/README.md:100-140

FAILURES AND CORRECTIONS

none

CONDITIONS

- A classifier that scores by a sequence of axis-aligned splits and is flat within each leaf. Its finite difference is zero at every row that does not straddle a split, a split on one coordinate reads nothing of the others, and the Gini impurity it splits on is at most one half for two classes and zero exactly for a pure leaf.
- Its read subspace is spanned by the coordinates it splits on and a feature that appears in no split is in its nuisance exactly, which is why a finite-difference probe of a tree returns zero at most rows and importances that count splits are the right sensitivity for this reader.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/DecisionTree.lean`, `theorems stump_flat_left`, `stump_flat_right`, `split_reads_one_axis`, `gini_le_half`, `gini_eq_zero_iff`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 4, 5, 6, 7, 8, 9, 11, 14.

RELATED

[classifier](#) (p. 60); [importance](#) (p. 172); [boosting](#) (p. 44); [ensemble](#) (p. 121); [sensitivity](#) (p. 319).

declaration

instrument. Entry id `declaration`.

DEFINITION

A sealed statement of what a study will count, drop, or treat as a failure, written before the run. Chapter 8 section 8.8.

EQUATION

Book equation 8.1.

$$O_{\text{harness}} = (C_{\text{score}}, G_{\text{metric}}, B = \text{folds} \times \text{samples} \times \text{seeds}).$$

Book equation 8.3.

$$\Pr[\text{at least one of } m \text{ null tests passes}] = 1 - (1 - \alpha)^m, \quad \alpha_{\text{Bonferroni}} = \frac{\alpha}{m}.$$

LEDGER

- OT-11. Feedback-free staleness: streaming-retrieval damage tracks measured drift; derived-cadence re-allocation removes it. `[void].geometric-observation/claims/LEDGER.md:48` at `9f3829f`.

FIRST STATED

Chapter 8 section 8.8 of *Data Mining as Observation*, with the sealed declaration in `geometric-observation/crucible/DECLARATION-V1.md:1-20`.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 2 section 2.3	the missing-data rule as a required preregistration field	observation-theory-campaigns\experiments\PREREG-TEMPLATE.md:47-52
chapter 8 section 8.8	declaration drafted 2026-08-17, sealed 2026-08-18, one count corrected, G2	geometric-observation\crucible\DECLARATION-V1.md:1-20; geometric-observation\crucible\OT-CRUCIBLE-4.md:31-35

FAILURES AND CORRECTIONS

none

CONDITIONS

- A statement of what a study will count, drop, or treat as a failure, sealed before the run. A changed digest would prove a changed declaration, and a declaration drafted on one day and sealed the next is the record's normal shape.
- The declaration for the fourth crucible was drafted 2026-08-17, sealed 2026-08-18, and had one count corrected, and the missing-data rule is a required declared field.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/Seal.lean`, `theorems changed_of_hash_ne`, `hash_eq_of_eq`, `exists_collision`, `at observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 2, 7, 8, 10, 11, 12.

RELATED

[registered](#) (p. 296); [sealed](#) (p. 316); [preregistration](#) (p. 258); [commit hash](#) (p. 67).

degree

concept. Entry id degree.

DEFINITION

Of a node in a graph, the number of edges it has. Chapter 0 section 0.10.

EQUATION

Book equation 0.19.

$$L = I - D^{-1/2} A D^{-1/2}, \quad D = \text{diag}(d_1, \dots, d_n), \quad L u_k = \lambda_k u_k, \quad 0 = \lambda_1 \leq \lambda_2 \leq \dots.$$

Book equation 3.3.

$$X_i = r_i \theta_i, \quad r_i \rightarrow \frac{1}{\sqrt{d_i}}, \quad \theta_i = \frac{X_i}{\|X_i\|} \in S^{m-1}.$$

LEDGER

- NEG-1. Fixed-scale, uniform-in-m bi-Lipschitz for the commute filter. [refuted]. `geometric-observation/claims/LEDGER.md:94 at 9f3829f`.

FIRST STATED

Chapter 0 section 0.10 and chapter 3 section 3.3 of *Data Mining as Observation*, with the radius-against-degree measurement in the-angular-observer, the-angular-observer/README.md:135-139.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 3 section 3.2	condition (A2), tangential displacement	turboquant-pro\docs\KV_KEYS_FINDING.md:61-86; geometric-observation\chapters\ch09_legibility.md:42-54
chapter 3 section 3.3	resistance converges to 1 over d_i plus 1 over d_j , simplex argument, radius converges to 1 over root d_i	the-angular-observer\theorem.md:110-146
chapter 3 section 3.3	radius against degree correlation 0.92 to 0.99	the-angular-observer\README.md:135-139
chapter 3 section 3.3	angle 0.92 flat in m, magnitude 0.03, full 0.75 decaying	wolfram-observer-bridge\theorem.md:1-20; the-angular-observer\theorem.md:187-224
chapter 3 section 3.3	substrate table	the-angular-observer\README.md:99-109
chapter 3 section 3.3	row normalization is the projection onto the read subspace of the geodesic-rank consumer	geometric-observation\chapters\ch09_legibility.md:31-41; geometric-observation\chapters\ch03_historical_precursors.md:110
chapter 3 section 3.3	proven versus conjectured ledger, d at least 3, d equals 2 borderline, fixed-m conditional theorem, heat filter uniform, commute metric form false, Green-kernel rank limit for d at most 3, obstruction at d equals 4	the-angular-observer\theorem.md:84-107, 147-224; the-angular-observer\README.md:150-185
chapter 3 section 3.3	the self-refuted v0.8 claim, NEG-1	the-angular-observer\README.md:170-175; geometric-observation\chapters\ch16_honest_negatives.md NEG-1
chapter 3 section 3.3	hyperbolic rejected, curvature negative 0.98 to negative 0.14, trend 1.09 minus 0.157 d across 20 manifolds and 5 families	the-angular-observer\README.md:111-147, 183-185
chapter 9 section 9.1	the geodesic-rank reader discards the radius, row normalization is the angular projection	geometric-observation\chapters\ch09_legibility.md:10-41; the-angular-observer\theorem.md:110-146

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 9 section 9.2	the battery, three new templates, frozen ratios, code hash, bars at least 10 of 12 and the dimension ordering, eccentricity scope	the-angular-observer\PREREG_RECOGNIZER_BATTERY.md:1-80
chapter 9 section 9.2	12 of 12, dimensions 1.81, 2.80, 1.74, angular Spearman ranges, eccentricity spreads, verdict confirmed, GO-P-2026-041	the-angular-observer\experiments\manifold-recovery\battery_result.json
chapter 9 section 9.2	causal set 0.57 with dimension 2.9 to 10.7, small world 0.45, control 0.01, rung 4 decay 0.747 to 0.679 below 0.75	the-angular-observer\README.md:99-109; wolfram-observer-bridge\rung4_scaledm_result.json; wolfram-observer-bridge\review-2.txt
chapter 11 section 11.1	condition (A2), cosine satisfies it, post-rotary keys do not, the cone below cell size	turboquant-pro\docs\KV_KEYS_FINDING.md:61-86; the-angular-observer\README.md:26-31

FAILURES AND CORRECTIONS

- NEG-1, [refuted]. Fixed-scale, uniform-in-m bi-Lipschitz for the commute filter.

CONDITIONS

- The number of edges a node has. Summed over the graph the degrees count every edge twice, the sum is even, and no node has degree more than the number of other nodes.
- The radius of a spectral embedding tracks the degree at correlation 0.92 to 0.99 on the substrates tested, which is the measured half of chapter 3's claim that the magnitude is degree noise and the angle is what the geodesic-rank reader reads.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/Degree.lean`, `theorems sum_degrees`, `degree_lt_card`, `sum_degrees_even`, `average_degree`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 1, 3, 4, 8, 9.

RELATED

[Laplacian](#) (p. 199); [geodesic distance](#) (p. 154); [hubness](#) (p. 168); [recognizer](#) (p. 288).

density

concept. Entry id density.

DEFINITION

The local crowding of rows around a point, the coordinate the geodesic reader of a spectral embedding discards and DBSCAN and the density detector read. Chapter 3 section 3.3 and chapter 9.

EQUATION

Book equation 3.3.

$$X_i = r_i \theta_i, \quad r_i \rightarrow \frac{1}{\sqrt{d_i}}, \quad \theta_i = \frac{X_i}{\|X_i\|} \in S^{m-1}.$$

Book equation 0.19.

$$L = I - D^{-1/2} A D^{-1/2}, \quad D = \text{diag}(d_1, \dots, d_n), \quad L u_k = \lambda_k u_k, \quad 0 = \lambda_1 \leq \lambda_2 \leq \dots.$$

Book equation 9.1.

$$\text{SSE} = \sum_{k=1}^K \sum_{i \in \mathcal{C}_k} \|x_i - c_k\|^2, \quad \text{the } P_C = I \text{ distortion summed within clusters.}$$

LEDGER

- NEG-1. Fixed-scale, uniform-in-m bi-Lipschitz for the commute filter. [refuted]. `geometric-observation/claims/LEDGER.md:94` at 9f3829f.
- NEG-11. (GO-5, prospective ×4) The $\alpha=1$ density/hubness quotient decisively and density-specifically restores invariant fidelity in a non-spectral domain. [refuted]. `geometric-observation/claims/LEDGER.md:104` at 9f3829f.

FIRST STATED

Chapter 3 section 3.3 of *Data Mining as Observation*, with the radius-against-degree measurement in the-`angular-observer/README.md:135-139`.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 3 section 3.3	radius against degree correlation 0.92 to 0.99	<code>the-angular-observer\README.md:135-139</code>
chapter 9 section 9.1	the geodesic-rank reader discards the radius, row normalization is the angular projection	<code>geometric-observation\chapters\ch09_legibility.md:10-41</code> ; <code>the-angular-observer\theorem.md:110-146</code>
chapter 10 section 10.1	the four detector families	TSK 2e chapter 9

FAILURES AND CORRECTIONS

- NEG-1, [refuted]. Fixed-scale, uniform-in-m bi-Lipschitz for the commute filter.
- NEG-11, [refuted]. (GO-5, prospective ×4) The $\alpha=1$ density/hubness quotient decisively and density-specifically restores invariant fidelity in a non-spectral domain.

CONDITIONS

- The local crowding of rows around a point, which on a neighbourhood graph is the degree. The radius of the spectral embedding converges to one over the square root of the degree, so the geodesic reader discards the density and DBSCAN and the density detector read it. A wider radius or a smaller count makes more core points, and the sum of degrees is twice the edge count.
- The radius-against-degree correlation was 0.92 to 0.99 across substrates, and the quotient that removed density to restore invariant fidelity was refuted four times, NEG-11.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

lean/DataMiningAsObservation/Dbscan.lean, theorems ball_mono, core_mono, core_anti, reach_from_core, noise_unreachable, at observation-data-mining af776fd.

lean/DataMiningAsObservation/Degree.lean, theorems sum_degrees, degree_lt_card, sum_degrees_even, average_degree, at observation-data-mining af776fd.

USED IN

Data Mining as Observation chapters 3, 8, 9, 10, 11.

RELATED

[degree](#) (p. 97); [spectral embedding](#) (p. 332); [DBSCAN](#) (p. 93); [geodesic distance](#) (p. 154); [hubness](#) (p. 168).

deployment mismatch

concept. Entry id deployment-mismatch.

DEFINITION

The failure in which the consumer that was evaluated is not the consumer that was deployed, or time moved between the two. Chapters 1 and 13.

EQUATION

Book equation 13.1.

$$\text{FC} = \Pr[W \text{ refutes} \mid \mathcal{C}_t \text{ clears}], \quad d_O(\Delta) = \text{tr}(P_C M_{\text{drift}}(\Delta)), \quad M_{\text{drift}}(\Delta) = \mathbb{E}[\delta_\Delta \delta_\Delta^\top].$$

LEDGER

- OT-4. Operator drift predicts a real long-generation degradation and a derived refresh intervention moves it. [refuted]. [geometric-observation/claims/LEDGER.md:36](#) at 9f3829f.
- GO-2/GO-12/GO-13 operational (KV serving, 077). Consumer-relative access width measured on a production serving stack (Qwen2.5-7B KV-cache eviction, matched budget): task quality tracks measured predictive uncertainty u about the consumer's future reads, not nominal scorer width — the 32-query snapshot beats the 1024-query scorer $+0.4375 \pm 0.070$ at 5% keep (bar 0.30) and survives 97% eviction with zero drop, while wide-window scoring is statistically indistinguishable from random eviction at extreme budgets; the recency-hoarding starvation signature replicated across three disjoint prompt sets (oracle-miss gap 0.370 vs bar 0.25). The novel equal-uncertainty analytic-equality control (degradation-titrated, constructible by design) REFUTED its own equality prediction on the pre-registered branch: with calibration health $4 \times -130 \times$ inside gates, equal scalar u did NOT give equal quality (V4 $+0.078$ vs 0.0625 tolerance; the pre-registered $p=0.03$ contrast firmed to $+0.359 \pm 0.068$, 5.3 SE) — equal scalar uncertainty is insufficient, error structure matters, consistent with GO-13 Theorem 1's own $r \geq 2$ scoping of equal-q universality (a scalar-context privilege). Successor arc: 056 honest miss $\rightarrow$ 075 ID burned on a disclosed design failure $\rightarrow$ 077 sealed and split-verdict. [demonstrated]. [geometric-observation/claims/LEDGER.md:81](#) at 9f3829f.

FIRST STATED

Chapter 1 section 1.7 and chapter 13 of *Data Mining as Observation*, with the program's case in the radio slice sweep, [observation-theory-campaigns/analysis/llm/PREREG-XPROTO-LLM.md](#), and the serving-stack measurement of ledger row GO-2/GO-12/GO-13 operational.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 12 section 12.6	XPROTO-LLM, benchmark 0.909, 0.920, 0.909, thirty slices, target 0.8, naive 0.333, aware 0.033, spread 0.380, deployment mean 0.736, six bars on three seeds, sealed 2026-08-25 at b61f7f1	observation-theory-campaigns\experiments\LLM-EVAL-TRACK.md:1-50; observation-theory-campaigns\analysis\llm\XPROTO-LLM-graded.json; observation-theory-campaigns\experiments\SEALS.md:85
chapter 13 section 13.6	attempt two, broken proxy, control did not exist, identifier burned, 16 prompts, 1360 s	geometric-observation\prereg\GO-P-2026-077-kv-consumer-relative.md:1-15; geometric-observation\results\G013-kvaw-pilot-disclosed.json
chapter 13 section 13.6	attempt three, windows 1024, 256, 32, uncertainty 0.982 to 0.892, 5 of 6, 0.4375 with SE 0.070 at 5 percent keep vs 0.30, 97 percent eviction, oracle-miss 0.370 vs 0.25, V4 0.078 vs 0.0625, contrast 0.359 with SE 0.068, n 64, seed 20260812, 89 duty cycles	geometric-observation\claims\LEDGER.md row GO-2/GO-12/GO-13 operational; geometric-observation\prereg\GO-P-2026-077-kv-consumer-relative.md; geometric-observation\results\G013-kvaw2-governed.json

FAILURES AND CORRECTIONS

- OT-4, [refuted]. Operator drift predicts a real long-generation degradation and a derived refresh intervention moves it.

CONDITIONS

- The failure in which the consumer that was evaluated is not the consumer that was deployed, or time moved between the two. A deployment score is a weighted mean over the slices actually served, so it lies between the worst slice and the best, and a benchmark drawn from the best slice overstates it unless every weighted slice matches.
- The chapter's case scores 0.909 on the benchmark against a target of 0.8 and 0.736 on the deployment mean, and the six bars that separate the two are preregistered.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/DeploymentMismatch.lean`, `theorems deployment_le_max`, `min_le_deployment`, `deployment_lt_max_of_gap`, `book_numbers`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 1, 12.

RELATED

[observer](#) (p. 236); [certificate](#) (p. 55); [coherence time](#) (p. 63); [min-over-strata](#) (p. 217).

detector

instrument. Entry id detector.

DEFINITION

A scorer that calls a row an outlier, of four families reading distance from the mean, distance to neighbours, density relative to neighbours, or cluster membership. Chapter 10 section 10.1.

EQUATION

Book equation 0.33.

$$d_M(x) = \sqrt{(x - \mu)^\top \Sigma^{-1} (x - \mu)}.$$

Book equation 10.3.

$$N_k(x \mid Q) = |\{q \in Q : x \in \text{top}_k(q)\}|, \quad \text{anti-hub} : N_k(x) = 0.$$

Book equation 9.1.

$$\text{SSE} = \sum_{k=1}^K \sum_{i \in \mathcal{C}_k} \|x_i - c_k\|^2, \quad \text{the } P_C = I \text{ distortion summed within clusters.}$$

LEDGER

- NEG-14. (GO-P-2026-037, prospective) `a2_probe.median_unit_displacement` is a single-statistic predictor of the flip regime (`unit_disp` $\geq 1.0 \Rightarrow$ generic-polar-flip vs needs-blind-probe). [refuted]. `geometric-observation/claims/LEDGER.md:108` at 9f3829f.

FIRST STATED

Chapter 10 section 10.1 of *Data Mining as Observation*, after TSK chapter 9, with the hierarchical typing in `turboquant-pro/turboquant_pro/anatomy.py:98-170`.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 10 section 10.1	the four detector families	TSK 2e chapter 9
chapter 10 section 10.3	hierarchical typing, tails 0.95 and 0.85, prescriptions, two designs that died, correlation above 0.8 on an isotropic Gaussian	<code>turboquant-pro\turboquant_pro\anatomy.py:98-170</code>

FAILURES AND CORRECTIONS

- NEG-14, [refuted]. (GO-P-2026-037, prospective) `a2_probe.median_unit_displacement` is a single-statistic predictor of the flip regime (`unit_disp` $\geq 1.0 \Rightarrow$ generic-polar-flip vs needs-blind-probe).

CONDITIONS

- A scorer that calls a row an outlier, of four families. The statistical detector reads distance from the mean in units of spread, the proximity detector reads distance to neighbours, the density detector reads density relative to neighbours, and the clustering detector reads membership. Each reads a coordinate the others discard.
- No detector of the data alone sees the observer's outliers, the anti-hubs, because they are a property of the queries, and a detector's positives are few by Chebyshev's arithmetic.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

lean/DataMiningAsObservation/Outlier.lean, theorems zscore_mean, zscore_affine, outlier_iff, zscore_reflect, chebyshev, at observation-data-mining af776fd.

lean/DataMiningAsObservation/Mahalanobis.lean, theorems dm2_nonneg, dm2_mean, dm2_scale, dm2_eq_whitened, dm2_antitone_in_variance, at observation-data-mining af776fd.

lean/DataMiningAsObservation/Dbscan.lean, theorems ball_mono, core_mono, core_anti, reach_from_core, noise_unreachable, at observation-data-mining af776fd.

USED IN

Data Mining as Observation chapters 3, 8, 10, 13.

RELATED

[outlier](#) (p. 242); [Mahalanobis distance](#) (p. 210); [density](#) (p. 99); [DBSCAN](#) (p. 93); [anti-hub](#) (p. 17).

direction-only quantizer

instrument. Entry id direction-only-quantizer.

DEFINITION

A quantizer that stores each vector’s length exactly and rounds its direction. Chapter 11.

EQUATION

Book equation 11.1.

$$\cos(k, \hat{k}) = 0.995 \quad \text{while} \quad \text{PPL} : 12.24 \rightarrow 10643.$$

Book equation 0.1.

$$x \cdot y = \sum_{i=1}^d x_i y_i, \quad \|x\| = \sqrt{x \cdot x}, \quad \cos \theta = \frac{x \cdot y}{\|x\| \|y\|}.$$

LEDGER

- NEG-2. Reconstruction cosine as a proxy for key quality. [refuted]. geometric-observation/claims/LEDGER.md:95 at 9f3829f.
- GO-B-LOCATA. Real microphone-array recordings (LOCATA), DOA consumer — held-out confirmation with the PolarQuant compressor [predicted]. geometric-observation/claims/LEDGER.md:117 at 9f3829f.

FIRST STATED

Chapter 11 section 11.2 of Data Mining as Observation, with PolarQuant in turboquant-pro/docs/KV_KEYS_FINDING.md:1-49 and the LOCATA comparison of the ledger.

MEASUREMENTS

Where the book states it	Numbers, as the book’s sources table records them	Source
chapter 1 section 1.4	cosine 0.995 and perplexity of order ten thousand, the recalibration negative	geometric-observation\chapters\ch02_failure_of_observer_free60; geometric-observation\chapters\ch16_honest_negatives.md NEG-2 and NEG-4; turboquant-pro\docs\KV_KEYS_FINDING.md:1-49

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 2 section 2.5	cosine 0.995 and the softmax reader	turboquant-pro\docs\KV_KEYS_FINDING.md:1-49
chapter 3 section 3.2	condition (A2), tangential displacement	turboquant-pro\docs\KV_KEYS_FINDING.md:61-86; geometric-observation\chapters\ch09_legibility.md:42-54
chapter 8 section 8.2	cosine 0.995, perplexity near 1e4	turboquant-pro\docs\KV_KEYS_FINDING.md:1-49; geometric-observation\chapters\ch16_honest_negatives.md NEG-2
chapter 11 section 11.1	condition (A2), cosine satisfies it, post-rotary keys do not, the cone below cell size	turboquant-pro\docs\KV_KEYS_FINDING.md:61-86; the-angular-observer\README.md:26-31
chapter 11 section 11.2	fp16 12.24, values-only 13.12, PolarQuant K4 10643 and 0.095, per-channel uniform K4 14.91 and 0.062, per-channel NUQ K3 15.77 and 0.148, 2.4x and 670x, pre-rotary near 22000, 2 key heads serve 12 query heads	turboquant-pro\docs\KV_KEYS_FINDING.md:1-49

FAILURES AND CORRECTIONS

- NEG-2, [refuted]. Reconstruction cosine as a proxy for key quality.

CONDITIONS

- A quantizer that stores each vector's length exactly and rounds its direction to a unit codeword. The quantized vector keeps the original's length, its cosine with any query is the codeword's cosine, and its score against a query is the length times the query's dot product with the codeword, so the score error is the length times the query's dot product with the direction error.
- A key's cosine with its own quantized form can be near one while its score against a query moves by a multiple of its length, which is why keys at cosine 0.995 raised the perplexity by three orders of magnitude.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/DirectionQuantizer.lean`, theorems `length_preserved`, `score_eq`, `score_error`, `cosine_eq`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 3, 4, 11.

RELATED

[quantization](#) (p. 266); [dot product](#) (p. 110); [KV cache](#) (p. 190); [attention](#) (p. 23); [the flip](#) (p. 138).

discretization

instrument. Entry id discretization.

DEFINITION

Collapsing an ordered quantity to a few labels by binning it, so that values in one cell become indistinguishable. It forms a quotient. Chapter 2 section 2.5.

EQUATION

Book equation 2.1.

$$P_{C_2 \circ C_1}(x) = J_1(x)^\top P_{C_2}(C_1(x)) J_1(x), \quad \text{rank } P_{C_2 \circ C_1}(x) \leq \text{rank } P_{C_2}(C_1(x)) \quad \text{at each row } x.$$

Book equation 0.12a.

$$x \sim_C x' \iff d_G(C(x), C(x')) = 0.$$

LEDGER

none

FIRST STATED

Chapter 2 section 2.5 of *Data Mining as Observation*, after TSK section 2.3.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 2 section 2.1	the four attribute types and permitted transformations	TSK 2e section 2.1
chapter 2 section 2.3	MCAR, MAR, MNAR and the remedies, imputation before splitting	TSK 2e section 2.2; instructor working documents, not public [@bond2026course], ECE_514-01_FA26_session-outlines.md:64-72

FAILURES AND CORRECTIONS

none

CONDITIONS

- Collapsing an ordered quantity to a few labels by binning it. The bin index is nondecreasing in the value, every value in a cell gets the cell's index, two values in the same cell become indistinguishable, and a sample confined to m cells has at most m distinct labels.
- A discretization forms a quotient before anyone has said who reads the result, and a consumer that read the values it merged has lost them for every stage after.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/Discretization.lean`, `theorems bin_mono`, `bin_of_mem`, `bin_eq_of_same_cell`, `card_bins_le`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 2.

RELATED

[quotient](#) (p. 271); [standardization](#) (p. 338); [attribute type](#) (p. 25); [bit](#) (p. 38); [quantization](#) (p. 266).

disparate impact ratio

concept. Entry id `disparate-impact-ratio`.

DEFINITION

The rate of a favourable decision in one group divided by the rate in another. Equation 0.37.

EQUATION

Book equation 0.37.

$$DI = \frac{\Pr[\hat{y} = 1 \mid g = a]}{\Pr[\hat{y} = 1 \mid g = b]}, \quad \text{equalized odds : } TPR_a = TPR_b, FPR_a = FPR_b.$$

Book equation 14.5.

$$FC_g = \Pr[y = \text{violation} \mid \hat{y} = \text{clear}, g], \quad \text{verdict} = \max_{g: n_g \geq n_{\min}} FC_g, \quad \text{ABSTAIN otherwise.}$$

LEDGER

none

FIRST STATED

Chapter 0 section 0.18 and chapter 14 section 14.6 of *Data Mining as Observation*, with the per-group verdict of equation 14.5.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 14 section 14.1	the re-gate table, prediction before result, both reports kept, binding nulls, 7 of 9 inside the interval, rights as a hard rule	gtc-prototype\docs\REGATE.md:1-70

FAILURES AND CORRECTIONS

none

CONDITIONS

- The ratio of the favourable-decision rate in one group to the rate in another. It is one exactly when the rates agree, it inverts when the groups are swapped, and the four-fifths rule is a statement about the rate gap.
- The book's verdict is per group with a null, the worst group's false-clear rate over the groups large enough to score, which clears a bar exactly when every scored group does and is never below the pooled rate.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/DisparateImpact.lean`, theorems `ratio_eq_one_iff`, `ratio_swap`, `four_fifths_iff`, `verdict_le_iff`, `weighted_le_verdict`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 14.

RELATED

[equalized odds](#) (p. 122); [min-over-strata](#) (p. 217); [false-clear rate](#) (p. 133); [abstention](#) (p. 9).

distance concentration

concept. Entry id `distance-concentration`.

DEFINITION

The narrowing of the spread of pairwise distances as dimension grows, read in the book as the reader running out of resolution. Chapter 3.

EQUATION

Book equation 0.7.

$$r_{\text{eff}} = \frac{(\sum_i \lambda_i)^2}{\sum_i \lambda_i^2}.$$

LEDGER

- NEG-11. (GO-5, prospective ×4) The $\alpha=1$ density/hubness quotient decisively and density-specifically restores invariant fidelity in a non-spectral domain. [refuted]. `geometric-observation/claims/LEDGER.md:104` at 9f3829f.

FIRST STATED

The classical result on nearest neighbours in high dimension, as TSK chapter 2 presents it, and chapter 3 section 3.4 of *Data Mining as Observation*, where it is read as the reader running out of resolution.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 3 section 3.5	Poisson null, hub excess, budget parameter	<code>openvector-bench\openvector_bench\hubness.py:41-100</code>
chapter 4 section 4.2	allocation report, gain over uniform, concentration caution below effective rank 2	<code>turboquant-pro\turboquant_pro\read_allocation.py:244-307</code>

FAILURES AND CORRECTIONS

- NEG-11, [refuted]. (GO-5, prospective ×4) The $\alpha=1$ density/hubness quotient decisively and density-specifically restores invariant fidelity in a non-spectral domain.

CONDITIONS

- When a squared distance is a sum of independent coordinate contributions with a common mean and variance, its relative spread is the variance over the dimension times the squared mean, which falls with the dimension and tends to zero. The independence is the model's assumption.
- The book reads the narrowing as a property of the reader, the identity reader on all coordinates, and not of the data. A consumer that reads a low-dimensional subspace does not see it, which is why the effective rank and not the ambient dimension is the number that matters.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/DistanceConcentration.lean`, `theorems relSpread_eq`, `relSpread_antitone`, `relSpread_tendsto_zero`, `exists_dim_relSpread_lt`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 3.

RELATED

[hubness](#) (p. 168); [effective rank](#) (p. 115); [read subspace](#) (p. 285); [rank certificate](#) (p. 275).

distortion

concept. Entry id distortion.

DEFINITION

The error a code costs a reader, the trace of the read operator against the error covariance. For the identity reader it is the reconstruction error. Chapter 4 section 4.2.

EQUATION

Book equation 4.2.

$$D(b) = \sum_i s_i \sigma_i^2 2^{-2b_i}, \quad s_i = v_i^\top P_C v_i, \quad b_i^* = \max\left(0, \frac{1}{2} \log_2 \frac{s_i \sigma_i^2}{\theta}\right), \quad \sum_i b_i^* = B,$$

Book equation 2.1.

$$P_{C_2 \circ C_1}(x) = J_1(x)^\top P_{C_2}(C_1(x)) J_1(x), \quad \text{rank } P_{C_2 \circ C_1}(x) \leq \text{rank } P_{C_2}(C_1(x)) \quad \text{at each row } x.$$

Book equation 0.11.

$$P_C(x) = J(x)^\top G(C(x)) J(x), \quad J(x) = \frac{\partial C}{\partial x}(x), \quad \bar{P}_{C,\mu} = \mathbb{E}_\mu[P_C(x)].$$

LEDGER

- OT-7. The damage form, trace pairing, rank, and loading covariance are GL(d)-invariant under $P' = A^{-\top} P A^{-1}$, while spectrum, effective rank, principal angles, and water-filling are O(d)-only — consumer-weighted damage is a geometric scalar, and P3's cliff cannot be bought down by reparameterization. [demonstrated]. [geometric-observation/claims/LEDGER.md:30 at 9f3829f](#).
- GO-2 (neg. half: not reconstruction). At matched bits, downstream preservation is not controlled by reconstruction error. [demonstrated]. [geometric-observation/claims/LEDGER.md:63 at 9f3829f](#).
- GO-6. At matched rate, output coding $\leq$ surrogate $\leq$ reconstruction on the consumer metric; the output-reconstruction gap is governed by the ker P_C entropy share, and the surrogate-output gap vanishes as rate grows. [demonstrated]. [geometric-observation/claims/LEDGER.md:68 at 9f3829f](#).

FIRST STATED

Chapter 1 section 1.4 and chapter 4 section 4.2 of *Data Mining as Observation*, with the read distortion in [geometric-observation/chapters/ch05_the_read_metric_and_the_quotient.md:49](#).

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 2 section 2.2	read distortion controls but is not a complete rank statistic, twelve of twelve, one middle pair misordered, NEG-9	geometric-observation\chapters\ch05_the_read_metric_and_the_75
chapter 3 section 3.2	traces 2.0, diagonals 0.3 and 1.7, consumers at 15 and 75 degrees, distortion 0.39 vs 1.61, 4.1 to one, computed not drawn	observation-theory\assets\make_flip_figure.py:4-24,107-119
chapter 4 section 4.2	GO-6, d 8 and r 4, output at or below surrogate at or below reconstruction at every rate, about 500 times, gap 0.41 to 0.005, isotropic control collapses	geometric-observation\claims\LEDGER.md row GO-6; geometric-observation\chapters\ch07_cost.md

FAILURES AND CORRECTIONS

none

CONDITIONS

- The error a code costs a reader, the trace of the read operator against the error covariance. For the identity reader it is the reconstruction error, and for a reader with one direction it is the error along that direction alone, so two codes with equal reconstruction error can differ four to one for a reader at fifteen degrees.
- At matched bits the downstream preservation is not controlled by reconstruction and is controlled by the consumer-projected error, twelve of twelve, and read distortion controls but is not a complete rank statistic.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/ReadDistortion.lean`, theorems `read_distortion`, `identity_reader`, `quad_one`, at `observation-data-mining af776fd`.

`lean/DataMiningAsObservation/Quantization.lean`, theorems `error_le_half_step`, `quantize_level`, `half_step_bound`, `sq_error_le`, `finer_step`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 1, 2, 3, 4, 7, 8, 9, 11, 13.

RELATED

[read distortion](#) (p. 281); [reconstruction error](#) (p. 290); [identity reader](#) (p. 171); [quantization](#) (p. 266).

dot product

concept. Entry id dot-product.

DEFINITION

The sum of the coordinatewise products of two vectors. Equation 0.1.

EQUATION

Book equation 0.1.

$$x \cdot y = \sum_{i=1}^d x_i y_i, \quad \|x\| = \sqrt{x \cdot x}, \quad \cos \theta = \frac{x \cdot y}{\|x\| \|y\|}.$$

Book equation 11.1.

$$\cos(k, \hat{k}) = 0.995 \quad \text{while} \quad \text{PPL} : 12.24 \rightarrow 10643.$$

LEDGER

- NEG-2. Reconstruction cosine as a proxy for key quality. [refuted]. `geometric-observation/claims/LEDGER.md:95` at `9f3829f`.

FIRST STATED

Chapter 0 section 0.1 and chapter 3 section 3.2 of *Data Mining as Observation*, with the program's cosine-versus-consumer case in `turboquant-pro/docs/KV_KEYS_FINDING.md:1-49`.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 1 section 1.4	cosine 0.995 and perplexity of order ten thousand, the recalibration negative	geometric-observation\chapters\ch02_failure_of_observer_free_60; geometric-observation\chapters\ch16_honest_negatives.md NEG-2 and NEG-4; turboquant-pro\docs\KV_KEYS_FINDING.md:1-49
chapter 2 section 2.5	cosine 0.995 and the softmax reader	turboquant-pro\docs\KV_KEYS_FINDING.md:1-49
chapter 3 section 3.2	condition (A2), tangential displacement	turboquant-pro\docs\KV_KEYS_FINDING.md:61-86; geometric-observation\chapters\ch09_legibility.md:42-54
chapter 8 section 8.2	cosine 0.995, perplexity near $1e4$	turboquant-pro\docs\KV_KEYS_FINDING.md:1-49; geometric-observation\chapters\ch16_honest_negatives.md NEG-2
chapter 11 section 11.1	condition (A2), cosine satisfies it, post-rotary keys do not, the cone below cell size	turboquant-pro\docs\KV_KEYS_FINDING.md:61-86; the-angular-observer\README.md:26-31
chapter 11 section 11.2	fp16 12.24, values-only 13.12, PolarQuant K4 10643 and 0.095, per-channel uniform K4 14.91 and 0.062, per-channel NUQ K3 15.77 and 0.148, 2.4x and 670x, pre-rotary near 22000, 2 key heads serve 12 query heads	turboquant-pro\docs\KV_KEYS_FINDING.md:1-49

FAILURES AND CORRECTIONS

- NEG-2, [refuted]. Reconstruction cosine as a proxy for key quality.

CONDITIONS

- The sum of the coordinatewise products of two vectors, symmetric and bilinear. The cosine, the dot product over the two lengths, lies between minus one and one by Cauchy–Schwarz and is unchanged when either vector is scaled by a positive factor, while the dot product scales with the vector.
- A cosine reader has declared length a nuisance and a dot-product reader has not, which is the reader difference of chapter 3, and cosine 0.995 between keys and their reconstruction said nothing about the softmax that read them.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/DotProduct.lean`, `theorems self_nonneg, dot_comm, dot_sq_le, cosine_mem, dot_smul, cosine_smul`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 1, 2, 3, 8, 11, 12.

RELATED

[projection](#) (p. 263); [Euclidean distance](#) (p. 124); [quotient](#) (p. 271); [nuisance](#) (p. 233).

drift

concept. Entry id drift.

DEFINITION

A change over time in what a consumer reads or in the data it reads. Chapter 8 tests a drift claim against its null and finds half of it was noise.

EQUATION

Book equation 13.1.

$$\text{FC} = \Pr[W \text{ refutes} \mid \mathcal{C}_t \text{ clears}], \quad d_O(\Delta) = \text{tr}(P_C M_{\text{drift}}(\Delta)), \quad M_{\text{drift}}(\Delta) = \mathbb{E}[\delta_\Delta \delta_\Delta^\top].$$

Book equation 0.10.

$$d_O = \text{tr}(P_C M_\delta) = \mathbb{E}[\delta^\top P_C \delta], \quad M_\delta = \mathbb{E}[\delta \delta^\top], \quad P_C = I \Rightarrow d_O = \text{tr} M_\delta.$$

LEDGER

- OT-4. Operator drift predicts a real long-generation degradation and a derived refresh intervention moves it. [refuted]. [geometric-observation/claims/LEDGER.md:36 at 9f3829f](#).
- OT-11. Feedback-free staleness: streaming-retrieval damage tracks measured drift; derived-cadence re-allocation removes it. [void]. [geometric-observation/claims/LEDGER.md:48 at 9f3829f](#).

FIRST STATED

Volume 14, chapter 19, [geometric-observation/chapters/ch19_the_certificate_that_ages.md](#), DOI 10.5281/zenodo.21776291, and the operator-drift record in [readscope](#), [readscope/calibration/records/c11c-operator-drift.json](#).

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 8 section 8.4	C-11c paired null, rank 2 positional 0.385 vs null 0.572, 0.615 vs 0.224, 14 of 16 cells, scope 16 cells one 3B model 192 positions	readscope/SPEC.md:806-857 ; readscope/calibration/records/c11c-operator-drift.json
chapter 11 section 11.9	drift at rank one 0.667 vs null 0.933, sixteen cells	readscope/CALIBRATION.md:600-660 F-24 ; readscope/calibration/records/c11c-operator-drift.json
chapter 11 section 11.9	C-12 four bars, 40 documents, 512 tokens, 13.4 point difference, teacher forcing removes it, negative 0.015 vs 0.005, Spearman negative 0.13 at p 0.45, sign test p 0.42, verdict FAIL, feedback compounding	readscope/calibration/records/c12-longgen-drift-sym.json ; readscope/calibration/DECLARATION-C12.md at commit 90e2ce2 ; readscope/SPEC.md:806-825

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 13 section 13.2	the grammar, certificate, witness, refresh floor, false-clear rate, vacuity, the witness table	geometric-observation\chapters\ch19_the_certificate_that_ages 95; observation-theory-campaigns\experiments\FRESHNESS-PROGRAM.md:1-40

FAILURES AND CORRECTIONS

- OT-4, [refuted]. Operator drift predicts a real long-generation degradation and a derived refresh intervention moves it.

CONDITIONS

- A drift is a change over time in what a consumer reads or in the data it reads, and its damage is the read distortion of the drift, so it is consumer-relative. The same drift is read as its full squared size by one consumer and as nothing by another, and a drift confined to the nuisance is read as zero however large.
- The operator-drift claim was tested against a paired null and half of it was noise. The rank-one drift held in fourteen of sixteen cells on one model, and the refresh intervention derived from it was refuted, which the ledger carries.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/Drift.lean`, theorems `damage_rank_one`, `same_drift_two_consumers`, `damage_eq_zero_of_nuisance`, `damage_nonneg`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 2, 8, 11, 13, 14.

RELATED

[coherence time](#) (p. 63); [freshness](#) (p. 149); [certificate](#) (p. 55); [read distortion](#) (p. 281).

E

early stopping

instrument. Entry id early-stopping.

DEFINITION

Halting a boosting run when a held-out score stops improving. The rule reads the held-out fold, so it is a harness and the round count is a fitted parameter. Chapter 7 section 7.5.

EQUATION

Book equation 8.1.

$$O_{\text{harness}} = (C_{\text{score}}, G_{\text{metric}}, B = \text{folds} \times \text{samples} \times \text{seeds}).$$

Book equation 8.2.

$$\widehat{\text{Var}}_{\text{NB}} = \left(\frac{1}{J} + \frac{n_{\text{test}}}{n_{\text{train}}} \right) \hat{\sigma}^2,$$

$$J = 1000, \frac{n_{\text{test}}}{n_{\text{train}}} = \frac{1}{4} \Rightarrow 1 + 250 = 251, \sqrt{251} = 15.84.$$

LEDGER

- NEG-4. Lightweight online (Lloyd) key calibration beats the calibration-free default on softmax-KL. [refuted]. [geometric-observation/claims/LEDGER.md:97 at 9f3829f](#).

FIRST STATED

Chapter 7 section 7.5 of *Data Mining as Observation*, after ESL 10.12.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 6 section 6.4	the reviewer's revision plan, drop sigma, bounded claim, 19 at 200 by 5 vs 11 at 20 by 5, rerun in progress	theory-radar\paper\REVISION_PLAN.md:1-60
chapter 6 section 6.5	loading weights and stability across folds requested	theory-radar\paper\REVISION_PLAN.md issue 6
chapter 7 section 7.2	second-order leaf values, logistic curvature	ESL 2e 10.9 to 10.13; XGBoost introduction to boosted trees

FAILURES AND CORRECTIONS

- NEG-4, [refuted]. Lightweight online (Lloyd) key calibration beats the calibration-free default on softmax-KL.

CONDITIONS

- Halting a boosting or gradient run when a held-out score stops improving. The stopping rule reads the held-out fold, so the fold it reads is no longer held out for the reported score, and a scorer that has read the test fold scores it perfectly.
- Early stopping is a harness. The round count it picks is a fitted parameter, and the report states which fold picked it and which fold scored it.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/Leakage.lean`, `theorems errors_lookup_eq_zero`, `lookup_default`, at `observation-data-mining af776fd`.

`lean/DataMiningAsObservation/CrossValidation.lean`, `theorems sizes_sum`, `accuracy_weighted`, `accuracy_mean_of_equal`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 7.

RELATED

[harness](#) (p. 159); [leakage](#) (p. 203); [cross-validation](#) (p. 89); [boosting](#) (p. 44); [split](#) (p. 335).

effective rank

concept. Entry `id effective-rank`.

DEFINITION

The number of directions a matrix really uses, the square of the sum of its eigenvalues over the sum of their squares, with no cutoff to choose. Equation 0.7.

EQUATION

Book equation 0.7.

$$r_{\text{eff}} = \frac{(\sum_i \lambda_i)^2}{\sum_i \lambda_i^2}.$$

Book equation 4.2.

$$D(b) = \sum_i s_i \sigma_i^2 2^{-2b_i}, \quad s_i = v_i^\top P_C v_i, \quad b_i^* = \max\left(0, \frac{1}{2} \log_2 \frac{s_i \sigma_i^2}{\theta}\right), \quad \sum_i b_i^* = B,$$

LEDGER

- OT-7. The damage form, trace pairing, rank, and loading covariance are $\text{GL}(d)$ -invariant under $P' = A^{-\top} P A^{-1}$, while spectrum, effective rank, principal angles, and water-filling are $O(d)$ -only — consumer-weighted damage is a geometric scalar, and P3’s cliff cannot be bought down by reparameterization. [demonstrated]. `geometric-observation/claims/LEDGER.md:30 at 9f3829f`.

FIRST STATED

The participation ratio of a spectrum, applied to read operators in `readscope`, `readscope/readscope/spectrum.py:35-70`, and chapter 0 section 0.4 of *Data Mining as Observation*.

MEASUREMENTS

Where the book states it	Numbers, as the book’s sources table records them	Source
chapter 4 section 4.2	effective rank as participation ratio, energy rank	<code>readscope\readscope\spectrum.py:35-70</code>
chapter 4 section 4.2	allocation report, gain over uniform, concentration caution below effective rank 2	<code>turboquant-pro\turboquant_pro\read_allocation.py:244-307</code>

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 9 section 9.4	explained ratios, effective rank 5.19 of 8, convergence with a second method, property of the representation not the space	gtc- prototype\docs\SPECTRUM_FINDINGS.md:10-18
chapter 14 section 14.3	floor 0.08 to 0.12, balanced resample 3001 items, identity attack 0.237 with interval 0.20 to 0.28 on 289, sexual 0.204, threat 0.093 retracted, first pass 18 positives, 87.7 and 1.4 percent	gtc- prototype\docs\SPECTRUM_FINDINGS.md:20-64
chapter 14 section 14.3	contraction 0.779 to 0.863, 77 of 1600, 4.8 percent, 0.872 to 0.863, weight 2.69	gtc- prototype\docs\SPECTRUM_FINDINGS.md:66-99
chapter 14 section 14.5	the contraction formula fairness minus the general component	gtc- prototype\docs\SPECTRUM_FINDINGS.md:90-99
chapter 14 section 14.7	51 percent moderated at 80 and 95 percent precision on the balanced set	gtc- prototype\docs\SPECTRUM_FINDINGS.md:84-88

FAILURES AND CORRECTIONS

none

CONDITIONS

- The effective rank is the square of the sum of the eigenvalues over the sum of their squares, between one and the dimension for a nonnegative spectrum, with no cutoff to choose.
- It is not invariant under a change of basis of the input, unlike the trace pairing and the rank, and the ledger row that says so is the one that licenses comparing read operators across coordinate systems.
- An allocation over a spectrum with effective rank below two concentrates on one direction, and the allocation report says so before giving a gain.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/EffectiveRank.lean`, `theorems sq_sum_le, effRank_le, sum_sq_le_sq_sum, one_le_effRank, effRank_const, effRank_single`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 4, 9, 11, 14.

RELATED

[read operator](#) (p. 282); [read subspace](#) (p. 285); [water-filling](#) (p. 365); [recognizer](#) (p. 288).

eigenvalue, eigenvector

concept. Entry id `eigenvalue-eigenvector`.

DEFINITION

A direction a symmetric matrix only stretches, and the factor by which it stretches it. The eigenvectors of a covariance are its principal directions. Equation 0.5.

EQUATION

Book equation 0.5.

$$\Sigma v_i = \lambda_i v_i, \quad \Sigma = \sum_{i=1}^d \lambda_i v_i v_i^\top, \quad v_i \cdot v_j = 0 \ (i \neq j).$$

Book equation 0.7.

$$r_{\text{eff}} = \frac{(\sum_i \lambda_i)^2}{\sum_i \lambda_i^2}.$$

Book equation 9.2.

$$N(\lambda) = \#\{k : \lambda_k \leq \lambda\} \sim C_d \lambda^{d/2} \quad \Rightarrow \quad d = 2 \frac{d \log N}{d \log \lambda}.$$

LEDGER

- OT-7. The damage form, trace pairing, rank, and loading covariance are GL(d)-invariant under $P' = A^{-\top} P A^{-1}$, while spectrum, effective rank, principal angles, and water-filling are O(d)-only — consumer-weighted damage is a geometric scalar, and P3's cliff cannot be bought down by reparameterization. [demonstrated].
geometric-observation/claims/LEDGER.md:30 at 9f3829f.

FIRST STATED

Chapter 0 section 0.4 of *Data Mining as Observation*, with the program's spectrum records in readscope and the recognizer battery.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 4 section 4.2	effective rank as participation ratio, energy rank	readscope\readscope\spectrum.py:35-70
chapter 9 section 9.4	explained ratios, effective rank 5.19 of 8, convergence with a second method, property of the representation not the space	gtc-prototype\docs\SPECTRUM_FINDINGS.md:10-18
chapter 14 section 14.3	floor 0.08 to 0.12, balanced resample 3001 items, identity attack 0.237 with interval 0.20 to 0.28 on 289, sexual 0.204, threat 0.093 retracted, first pass 18 positives, 87.7 and 1.4 percent	gtc-prototype\docs\SPECTRUM_FINDINGS.md:20-64
chapter 14 section 14.3	contraction 0.779 to 0.863, 77 of 1600, 4.8 percent, 0.872 to 0.863, weight 2.69	gtc-prototype\docs\SPECTRUM_FINDINGS.md:66-99
chapter 14 section 14.5	the contraction formula fairness minus the general component	gtc-prototype\docs\SPECTRUM_FINDINGS.md:90-99
chapter 14 section 14.7	51 percent moderated at 80 and 95 percent precision on the balanced set	gtc-prototype\docs\SPECTRUM_FINDINGS.md:84-88

FAILURES AND CORRECTIONS

none

CONDITIONS

- A direction a symmetric matrix only stretches, and the factor by which it stretches it. Eigenvectors with distinct eigenvalues are orthogonal, every eigenvalue of a semidefinite matrix is nonnegative, and the quadratic form along an eigenvector is the eigenvalue times the squared length.
- The eigenvectors of a covariance are its principal directions, the basis in which chapter 4 pairs the covariance with the read operator, and the eigenvalues of a Laplacian are what the recognizer reads.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/Eigen.lean`, `theorems pairing_symm`, `orthogonal_of_ne`, `eigenvalue_nonneg`, `quad_eigen`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 3, 4, 9, 11.

RELATED

[covariance matrix](#) (p. 85); [effective rank](#) (p. 115); [whitening](#) (p. 368); [recognizer](#) (p. 288); [Laplacian](#) (p. 199).

embedding

concept. Entry id embedding.

DEFINITION

A vector assigned to an object, a word, a sentence, a node, a document, so that nearness in the vector space stands for a relation between the objects. Chapter 11.

EQUATION

Book equation 11.1.

$$\cos(k, \hat{k}) = 0.995 \quad \text{while} \quad \text{PPL} : 12.24 \rightarrow 10643.$$

Book equation 11.2.

$$r = \frac{d_{\text{compressed}}}{d_{\text{exact}}}, \quad \kappa_{\text{strict}} = \frac{\max r}{\min r}, \quad \tau \geq 1 - 2\hat{\mu}(\kappa_{\text{strict}}), \quad \rho_S \geq 1 - 3\hat{\mu}(\kappa_{\text{strict}}),$$

$$\kappa_{97.5/2.5} = \frac{q_{97.5}(r)}{q_{2.5}(r)} \text{ gives the same two expressions as estimates, not floors.}$$

LEDGER

- NEG-2. Reconstruction cosine as a proxy for key quality. [refuted]. `geometric-observation/claims/LEDGER.md:95` at 9f3829f.
- GO-B-legal (035→036). Legal-citation retrieval (CourtListener), cosine-ranking consumer, LaBSE embeddings — real large corpus, non-physical consumer [predicted]. `geometric-observation/claims/LEDGER.md:119` at 9f3829f.

FIRST STATED

Chapter 11 section 11.1 of *Data Mining as Observation*, with the program's embeddings in the legal-citation flip, the GloVe corpus of readscape, and the attention keys of the KV finding.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 1 section 1.4	cosine 0.995 and perplexity of order ten thousand, the recalibration negative	geometric-observation\chapters\ch02_failure_of_observer_free-60; geometric-observation\chapters\ch16_honest_negatives.md NEG-2 and NEG-4; turboquant-pro\docs\KV_KEYS_FINDING.md:1-49
chapter 2 section 2.5	cosine 0.995 and the softmax reader	turboquant-pro\docs\KV_KEYS_FINDING.md:1-49
chapter 3 section 3.2	condition (A2), tangential displacement	turboquant-pro\docs\KV_KEYS_FINDING.md:61-86; geometric-observation\chapters\ch09_legibility.md:42-54
chapter 4 section 4.5	GloVe table, 73 percent at 64 components, 0.685 vs 0.862 and 0.866, 0.906 at matched bytes, 0.989 at 37 bytes, 768 to 256 keeps about 99 percent, the 95 percent rule	turboquant-pro\benchmarks\RESULTS_glove.md:1-40
chapter 8 section 8.2	cosine 0.995, perplexity near 1e4	turboquant-pro\docs\KV_KEYS_FINDING.md:1-49; geometric-observation\chapters\ch16_honest_negatives.md NEG-2
chapter 11 section 11.1	condition (A2), cosine satisfies it, post-rotary keys do not, the cone below cell size	turboquant-pro\docs\KV_KEYS_FINDING.md:61-86; the-angular-observer\README.md:26-31
chapter 11 section 11.2	fp16 12.24, values-only 13.12, PolarQuant K4 10643 and 0.095, per-channel uniform K4 14.91 and 0.062, per-channel NUQ K3 15.77 and 0.148, 2.4x and 670x, pre-rotary near 22000, 2 key heads serve 12 query heads	turboquant-pro\docs\KV_KEYS_FINDING.md:1-49

FAILURES AND CORRECTIONS

- NEG-2, [refuted]. Reconstruction cosine as a proxy for key quality.

CONDITIONS

- A vector assigned to an object so that nearness in the vector space stands for a relation between the objects. Objects with the same embedding are indistinguishable to every consumer of the embedding, so the embedding is a quotient, and a cosine reader takes a further quotient in which the nearest neighbour is unchanged when the query is rescaled, where a dot-product reader does not.
- What an embedding is worth to a consumer is the consumer's number. The keys that reconstructed at cosine 0.995 and raised the perplexity by three orders of magnitude are the case, and the rank certificate is the instrument that says which neighbour rankings a compressed embedding preserved.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

lean/DataMiningAsObservation/Embedding.lean, theorems quotient, cosine_scale_free, nearest_scale_free, dot_not_scale_free, at observation-data-mining af776fd.

USED IN

Data Mining as Observation chapters 0, 1, 3, 4, 8, 9, 10, 11, 12, 13, 14.

RELATED

[encoder](#) (p. 120); [dot product](#) (p. 110); [rank certificate](#) (p. 275); [hubness](#) (p. 168); [the flip](#) (p. 138).

encoder

concept. Entry id encoder.

DEFINITION

A model that maps an input to an embedding. Chapter 12.

EQUATION

Book equation 12.3.

$$\text{validated} \iff \text{AUROC}_{\text{cross}} - \max(\text{AUROC}_{\text{untrained}}, \text{AUROC}_{\text{BoW}}) \geq 0.10.$$

Book equation 14.1.

$$S = \frac{\sum_i w_i s_i}{\sum_i w_i}, \quad w_i = \max(0, 2 \cdot \text{AUROC}_i - 1).$$

LEDGER

- GO-B-legal (035→036). Legal-citation retrieval (CourtListener), cosine-ranking consumer, LaBSE embeddings — real large corpus, non-physical consumer [predicted]. `geometric-observation/claims/LEDGER.md:119` at 9f3829f.

FIRST STATED

Chapter 12 section 12.5 of *Data Mining as Observation*, with the program's encoders in `xbse` and the calibrated authority of `gtc-prototype/docs/CALIBRATED_AUTHORITY.md:19-63`.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 8 section 8.6	rights encoder 0.467 below untrained baseline, the AUROC lesson, cross-corpus same-sign fix	<code>xbse\README.md:160-172</code> ; <code>xbse\experiments\rights_r6_summary.json</code>
chapter 12 section 12.5	within 0.75 to 0.955 collapsing to 0.47 to 0.55, cross-corpus positives as the fix, adversary not load-bearing	<code>xbse\README.md:104-130,160-172</code>
chapter 12 section 12.5	the gate, both nulls, margin 0.10, scorecard 0.622 to 0.853, bag of words 0.46 to 0.54, rights 0.509 vs 0.512 and 0.467	<code>xbse\README.md:130-160,195-219</code>
chapter 12 section 12.5	ECE 0.018 to 0.101 vs raw up to 0.223, reliability weight, audit binding	<code>xbse\README.md:175-195</code> ; <code>gtc-prototype/docs/CALIBRATED_AUTHORITY.md:19-63</code>

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 14 section 14.1	the three questions, the trust beats, the audit binding	gtc-prototype\README.md:10-58; xbse\README.md:195-219
chapter 14 section 14.2	reliability weights and calibration errors per axis, the collapsed family's mean weight 0.559 against the general valence channel's own 0.735, the three design rules, 0.048 to 0.049 and 0.089 to 0.101 at 696 pairs	gtc-prototype\docs\CALIBRATED_AUTHORITY.md:1-65

FAILURES AND CORRECTIONS

none

CONDITIONS

- A model that maps an input to an embedding. An encoder votes only when its held-out AUROC on a corpus it was not trained on clears both nulls by the preregistered margin, its authority is its reliability weight, and a strictly monotone recalibration of its score changes neither the verdict nor the weight.
- The rights encoder fell below the untrained baseline and was retired rather than tuned, and the legal-citation encoder's flip is the ledger's demonstrated row.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/CrossCorpusGate.lean`, `theorems clears_bow`, `clears_untrained`, `not_validated_of_saturated`, `margin_example`, `validated_comp`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 7, 8, 9, 10, 11, 12, 14.

RELATED

[embedding](#) (p. 118); [cross-corpus gate](#) (p. 87); [reliability weight](#) (p. 298); [retrieval-augmented pipeline](#) (p. 303).

ensemble

instrument. Entry id `ensemble`.

DEFINITION

A scorer that averages or votes several scorers. Its average lies between its members and its squared error is at most their mean squared error. Chapter 7.

EQUATION

Book equation 6.1.

$$s(x) = w \cdot x + b, \quad P_C = \mathbb{E}[\sigma'(s)^2] w w^\top, \quad \text{rank } P_C = 1.$$

Book equation 0.18.

$$\widehat{\text{Var}}_{\text{NB}} = \left(\frac{1}{J} + \frac{n_{\text{test}}}{n_{\text{train}}} \right) \hat{\sigma}^2, \quad \widehat{\frac{\text{Var}}{\hat{\sigma}^2}}_{\text{NB}} = 1 + J \frac{n_{\text{test}}}{n_{\text{train}}}.$$

Book equation 8.2.

$$\widehat{\text{Var}}_{\text{NB}} = \left(\frac{1}{J} + \frac{n_{\text{test}}}{n_{\text{train}}} \right) \hat{\sigma}^2,$$

$$J = 1000, \frac{n_{\text{test}}}{n_{\text{train}}} = \frac{1}{4} \Rightarrow 1 + 250 = 251, \sqrt{251} = 15.84.$$

LEDGER

none

FIRST STATED

Breiman, bagging predictors, 1996, as chapter 7 section 7.1 of *Data Mining as Observation* reads it, with the fair comparison in `constraint-gap/review/INDETERMINATES.md:1-40`.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 6 section 6.3	the five-dataset table, breast cancer 0.955 to 0.963, the pattern, DOI 10.5281/zenodo.20660206	<code>theory-radar\README.md:100-140</code>
chapter 7 section 7.1	bagging variance, out-of-bag estimation, random forests, importances, AdaBoost	Hastie, Tibshirani, Friedman, ESL 2e chapters 15 and 10.1; TSK 2e 4.10
chapter 7 section 7.3	the five-dataset outcomes	<code>theory-radar\README.md:100-140</code>
chapter 7 section 7.4	nine to three, seventeen indeterminate, five analyses zero resolved, three re-encode the correction	<code>constraint-gap\review\INDETERMINATES.md:1-40</code>

FAILURES AND CORRECTIONS

none

CONDITIONS

- A scorer that averages or votes several scorers. Its average lies between the smallest and the largest member, its squared error is at most the mean of the members' squared errors, and a constant ensemble is its constant. That arithmetic is why bagging reduces variance.
- Whether an ensemble beats a formula is a measured question and not a principle, and the comparison is fair only inside one fold protocol with the Nadeau and Bengio correction applied, where nine wins to three became three wins, seventeen ties, and eleven losses.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/Ensemble.lean`, `theorems average_between`, `sq_average_le`, `average_const`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 6, 7.

RELATED

[boosting](#) (p. 44); [decision tree](#) (p. 95); [standard error](#) (p. 337); [Nadeau and Bengio correction](#) (p. 225).

equalized odds

concept. Entry id `equalized-odds`.

DEFINITION

The requirement that the true-positive rate and the false-positive rate be the same across groups. Equation 0.37.

EQUATION

Book equation 0.37.

$$DI = \frac{\Pr[\hat{y} = 1 \mid g = a]}{\Pr[\hat{y} = 1 \mid g = b]}, \quad \text{equalized odds : } TPR_a = TPR_b, FPR_a = FPR_b.$$

LEDGER

none

FIRST STATED

Chapter 0 section 0.18 and chapter 14 section 14.6 of *Data Mining as Observation*, where fairness cannot be had three ways.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 14 section 14.1	the re-gate table, prediction before result, both reports kept, binding nulls, 7 of 9 inside the interval, rights as a hard rule	gtc-prototype\docs\REGATE.md:1-70

FAILURES AND CORRECTIONS

none

CONDITIONS

- Equal true-positive and false-positive rates across groups. A group's favourable rate is its true-positive rate times its base rate plus its false-positive rate times the rest, so under equalized odds the favourable rates agree exactly when the base rates do, given that the classifier separates at all.
- Equalized odds and parity of outcomes therefore conflict whenever the base rates differ. With rates 0.8 and 0.2 and base rates one half and one tenth the favourable rates are 0.5 and 0.26, a ratio below four fifths.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

lean/DataMiningAsObservation/EqualizedOdds.lean, theorems parity_of_equal_base, favourable_sub, parity_iff_equal_base, example_conflict, at observation-data-mining af776fd.

USED IN

Data Mining as Observation chapters 0, 3, 10, 12, 14.

RELATED

[disparate impact ratio](#) (p. 106); [calibration](#) (p. 52); [min-over-strata](#) (p. 217); [certificate](#) (p. 55).

escalation

concept. Entry id escalation.

DEFINITION

The decision to hand an item to a person rather than decide it, the single-decision form of abstention. Chapter 14.

EQUATION

Book equation 14.5.

$$\text{FC}_g = \Pr[y = \text{violation} \mid \hat{y} = \text{clear}, g], \quad \text{verdict} = \max_{g: n_g \geq n_{\min}} \text{FC}_g, \quad \text{ABSTAIN otherwise.}$$

Book equation 10.7.

$$\text{verdict} = \min_{i: n_i \geq n_{\min}, |Q_i| \geq q_{\min}} \text{score}_i, \quad \text{ABSTAIN otherwise.}$$

Book equation 0.26.

$$\text{FC} = \Pr[W \text{ refutes} \mid \mathcal{C}_t \text{ clears}], \quad \text{coverage} = \Pr[\mathcal{C}_t \text{ clears}].$$

LEDGER

none

FIRST STATED

Chapter 14 section 14.9 of *Data Mining as Observation*, the single-decision form of abstention, with the refusal counts of the re-description test.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

CONDITIONS

- The decision to hand an item to a person rather than decide it. A rule that escalates every item whose margin is below a bar decides a set that can only shrink as the bar rises, so coverage can only fall, every decided item carries at least the bar's margin, and a bar of zero escalates nothing.
- Coverage times the decided false-clear rate is the fraction of all items cleared and wrong, so an escalation policy trades coverage for a false-clear rate and a deployment report states both.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/Escalation.lean`, `theorems decided_anti`, `coverage_anti`, `decided_margin`, `decided_zero`, `joint_rate`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 2, 8, 9, 10, 11, 14.

RELATED

[abstention](#) (p. 9); [margin](#) (p. 213); [coverage](#) (p. 86); [false-clear rate](#) (p. 133); [certificate](#) (p. 55).

Euclidean distance

concept. Entry id `euclidean-distance`.

DEFINITION

The straight-line distance between two vectors, which reads every coordinate at the scale it arrives in. Chapter 3.

EQUATION

Book equation 0.1.

$$x \cdot y = \sum_{i=1}^d x_i y_i, \quad \|x\| = \sqrt{x \cdot x}, \quad \cos \theta = \frac{x \cdot y}{\|x\| \|y\|}.$$

Book equation 4.1.

$$d_O = \text{tr}(P_C M_\delta) \quad \text{against} \quad \text{tr } M_\delta = d_O|_{P_C=I}.$$

LEDGER

- NEG-2. Reconstruction cosine as a proxy for key quality. [refuted]. `geometric-observation/claims/LEDGER.md:95` at 9f3829f.

FIRST STATED

Chapter 3 section 3.2 of *Data Mining as Observation*, where a distance is a choice of what to ignore.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 1 section 1.4	cosine 0.995 and perplexity of order ten thousand, the recalibration negative	<code>geometric-observation\chapters\ch02_failure_of_observer_free-60;</code> <code>geometric-observation\chapters\ch16_honest_negatives.md</code> NEG-2 and NEG-4; <code>turboquant-pro\docs\KV_KEYS_FINDING.md:1-49</code>
chapter 2 section 2.5	cosine 0.995 and the softmax reader	<code>turboquant-pro\docs\KV_KEYS_FINDING.md:1-49</code>
chapter 3 section 3.2	condition (A2), tangential displacement	<code>turboquant-pro\docs\KV_KEYS_FINDING.md:61-86;</code> <code>geometric-observation\chapters\ch09_legibility.md:42-54</code>
chapter 8 section 8.2	cosine 0.995, perplexity near 1e4	<code>turboquant-pro\docs\KV_KEYS_FINDING.md:1-49;</code> <code>geometric-observation\chapters\ch16_honest_negatives.md</code> NEG-2
chapter 11 section 11.1	condition (A2), cosine satisfies it, post-rotary keys do not, the cone below cell size	<code>turboquant-pro\docs\KV_KEYS_FINDING.md:61-86;</code> <code>the-angular-observer\README.md:26-31</code>
chapter 11 section 11.2	fp16 12.24, values-only 13.12, PolarQuant K4 10643 and 0.095, per-channel uniform K4 14.91 and 0.062, per-channel NUQ K3 15.77 and 0.148, 2.4x and 670x, pre-rotary near 22000, 2 key heads serve 12 query heads	<code>turboquant-pro\docs\KV_KEYS_FINDING.md:1-49</code>

FAILURES AND CORRECTIONS

- NEG-2, [refuted]. Reconstruction cosine as a proxy for key quality.

CONDITIONS

- The straight-line distance between two vectors, whose square is the identity reader's quadratic form on the difference, so it reads every coordinate at the scale it arrives in. It is symmetric and zero exactly between a row and itself.
- Rescaling one coordinate changes which of two rows is nearer. A choice of scale is a choice of reader, and reconstruction error, the Euclidean distance between a row and its code, is the identity reader's number and not the consumer's.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/EuclideanDistance.lean`, `theorems distSq_eq_quad_one`, `distSq_comm`, `distSq_eq_zero_iff`, `ranking_flips`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 1, 2, 3, 6, 10, 11, 12.

RELATED

[identity reader](#) (p. 171); [dot product](#) (p. 110); [Mahalanobis distance](#) (p. 210); [distance concentration](#) (p. 107).

eviction

concept. Entry id eviction.

DEFINITION

Choosing what to drop when a cache is full. Chapter 13.

EQUATION

Book equation 13.1.

$$\text{FC} = \Pr [W \text{ refutes} \mid \mathcal{C}_t \text{ clears}], \quad d_O(\Delta) = \text{tr} (P_C M_{\text{drift}}(\Delta)), \quad M_{\text{drift}}(\Delta) = \mathbb{E}[\delta_\Delta \delta_\Delta^\top].$$

Book equation 0.23.

$$\text{softmax}(z)_i = \frac{e^{z_i}}{\sum_j e^{z_j}}, \quad \text{output} = \sum_i \text{softmax}\left(\frac{q \cdot k_i}{\sqrt{d}}\right)_i v_i.$$

LEDGER

- GO-12. The dynamic region's opening control — staleness is access width, not delay: with full context-path access, pure-delay aging is information-free (the recoding identity $(Y, V, S^{(\Delta)}) \stackrel{d}{=} (Y, V, P^\Delta S^{(0)})$ makes every $\sigma(S)$ -measurable conditional functional exactly Δ -invariant; circulant-exact, $O(1/n)$ edge leakage on finite windows); with time-local slice access the tax is GO-11's static quadratic with the substitution set by the *encoder's* access — single-letter (Y_t, V_t) records at $(\rho, s/a^{2\Delta})$, context-epoch-latent records at the strictly smaller $(\rho a^\Delta, \tau^2)$ (gap up to 0.053 bits) — both strictly increasing in Δ with common limit $\frac{1}{2} \log_2(1/D)$. GO-8's age-dependence is the slice regime's operational face. Theorem 1 (the conditional-variance reduction) settles the causal-prefix eraser in the single-record setting: every observation-subset σ -algebra enters through $q_{\mathcal{G}} = \text{Var}(V_t \mid \mathcal{G})$ alone; slice/prefix/path = single-sample/Kalman-fixed-lag/noncausal-Wiener variances; strict interpolation at every finite lag with exact gap $C^{2\Delta}(P_f - q_{\text{path}})$; $W(\Delta) = kT \ln 2 \cdot L_{\text{prefix}}$ is the dynamic conditional-Landauer curve. Theorem 2-spectral (070): the spectral conditional RDF — at the work endpoint, $L(D)$ is the equal-slope allocation of per-frequency static quadratics over the circulant spectrum, with classical reverse water-filling ($\tau^2 \rightarrow \infty$) and the static theorem (flat spectrum) exact; per-mode convexity proven; the third promotion of the water level completed for the single-consumer face; Toeplitz transfer scoped imported-with-lemmas, $O(1/n)$ numerics. The weighted spectral theorem (071) closes Conjecture 1 at circulant

scope: J_w decomposes per-frequency at every w , each mode carrying Theorem 3's two-water-level system $(\gamma_0(\omega), \gamma_1(\omega))$ under ONE common distortion price (the conjectured 'pair of prices' refuted-as-phrased, rescoped to $\$(w, \mu)\$$); convexity via the full-region moment argument + perturbation. The third promotion is complete on BOTH faces. [predicted].geometric-observation/claims/LEDGER.md:78' at 9f3829f.

- GO-2/GO-12/GO-13 operational (KV serving, 077). Consumer-relative access width measured on a production serving stack (Qwen2.5-7B KV-cache eviction, matched budget): task quality tracks measured predictive uncertainty u about the consumer's future reads, not nominal scorer width — the 32-query snapshot beats the 1024-query scorer $+0.4375 \pm 0.070$ at 5% keep (bar 0.30) and survives 97% eviction with zero drop, while wide-window scoring is statistically indistinguishable from random eviction at extreme budgets; the recency-hoarding starvation signature replicated across three disjoint prompt sets (oracle-miss gap 0.370 vs bar 0.25). The novel equal-uncertainty analytic-equality control (degradation-titrated, constructible by design) REFUTED its own equality prediction on the pre-registered branch: with calibration health $4 \times -130 \times$ inside gates, equal scalar u did NOT give equal quality (V4 $+0.078$ vs 0.0625 tolerance; the pre-registered $\rho=0.03$ contrast firmed to $+0.359 \pm 0.068$, 5.3 SE) — equal scalar uncertainty is insufficient, error structure matters, consistent with GO-13 Theorem 1's own $r \geq 2$ scoping of equal-q universality (a scalar-context privilege). Successor arc: 056 honest miss $\rightarrow$ 075 ID burned on a disclosed design failure $\rightarrow$ 077 sealed and split-verdict. [demonstrated].geometric-observation/claims/LEDGER.md:81 at 9f3829f.

FIRST STATED

Chapter 13 section 13.6 of *Data Mining as Observation*, with the program's serving-stack case in geometric-observation/prereg/GO-P-2026-077-kv-consumer-relative-eviction.md and the ledger row that carries it.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 13 section 13.6	attempt two, broken proxy, control did not exist, identifier burned, 16 prompts, 1360 s	geometric-observation\prereg\GO-P-2026-077-kv-consumer-relative.md:1-15; geometric-observation\results\GO13-kvaw-pilot-disclosed.json
chapter 13 section 13.6	attempt three, windows 1024, 256, 32, uncertainty 0.982 to 0.892, 5 of 6, 0.4375 with SE 0.070 at 5 percent keep vs 0.30, 97 percent eviction, oracle-miss 0.370 vs 0.25, V4 0.078 vs 0.0625, contrast 0.359 with SE 0.068, n 64, seed 20260812, 89 duty cycles	geometric-observation\claims\LEDGER.md row GO-2/GO-12/GO-13 operational; geometric-observation\prereg\GO-P-2026-077-kv-consumer-relative.md; geometric-observation\results\GO13-kvaw2-governed.json

FAILURES AND CORRECTIONS

none

CONDITIONS

- Choosing what to drop when a cache is full. Evicting tokens replaces attention over all tokens with attention over the kept subset, so the output lies between the kept values, equals them when they agree, and the whole cache is the case of no eviction.
- Which tokens can go is consumer-relative, since a head reads a key only through its score, and the serving-stack measurement of ninety-seven percent eviction at a five percent keep against an oracle is the ledger's number, after one attempt whose control did not exist.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/Eviction.lean`, `theorems kept_weights_sum`, `keptOutput_le_max`, `min_le_keptOutput`, `keptOutput_const`, `keptOutput_univ`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 13.

RELATED

[KV cache](#) (p. 190); [attention](#) (p. 23); [coherence time](#) (p. 63); [budget](#) (p. 46).

expected calibration error

instrument. Entry id `expected-calibration-error`.

DEFINITION

The average over score bins of the absolute difference between the mean score and the fraction of positives in the bin. Equation 0.29.

EQUATION

Book equation 0.29.

$$\text{ECE} = \sum_b \frac{n_b}{n} |\bar{s}_b - \bar{y}_b|.$$

Book equation 0.38.

$$w = \max(0, 2 \cdot \text{AUROC} - 1).$$

Book equation 14.2.

$$\text{coverage difference}_c = \text{AUROC}_c(\text{embedding}) - \text{AUROC}_c(\text{validated axes}), \quad \text{floor} \approx 0.08 \text{ to } 0.12.$$

LEDGER

- NEG-4. Lightweight online (Lloyd) key calibration beats the calibration-free default on softmax-KL. [re-futed]. `geometric-observation/claims/LEDGER.md:97` at `9f3829f`.

FIRST STATED

Chapter 0 section 0.14 of *Data Mining as Observation*, equation 0.29, with the program's per-encoder errors in `xbse/README.md:175-195` and `gtc-prototype/docs/CALIBRATED_AUTHORITY.md:19-63`.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 12 section 12.5	ECE 0.018 to 0.101 vs raw up to 0.223, reliability weight, audit binding	<code>xbse\README.md:175-195</code> ; <code>gtc-prototype\docs\CALIBRATED_AUTHORITY.md:19-63</code>

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 14 section 14.2	reliability weights and calibration errors per axis, the collapsed family's mean weight 0.559 against the general valence channel's own 0.735, the three design rules, 0.048 to 0.049 and 0.089 to 0.101 at 696 pairs	gtc- prototype\docs\CALIBRATED_AUTHORITY.md:1-65

FAILURES AND CORRECTIONS

- NEG-4, [refuted]. Lightweight online (Lloyd) key calibration beats the calibration-free default on softmax-KL.

CONDITIONS

- The average over score bins, weighted by bin size, of the absolute difference between the bin's mean score and its fraction of positives. It lies in the unit interval, it is zero exactly when every bin's mean score equals its positive fraction, and a calibrated scorer has error zero.
- The program's validated encoders carry errors of 0.018 to 0.101 on held-out splits against raw values up to 0.223, and the reliability weight that gates an encoder's authority is computed beside it.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/Calibration.lean`, `theorems ece_nonneg, ece_le_one, ece_eq_zero_iff, ece_of_calibrated`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14.

RELATED

[calibration](#) (p. 52); [reliability weight](#) (p. 298); [AUROC](#) (p. 29); [threshold](#) (p. 350).

explained variance

concept. Entry `id` explained-variance.

DEFINITION

The fraction of total variance retained by a set of principal components, the identity reader's criterion for a reduction. Chapter 4.

EQUATION

Book equation 4.1.

$$d_O = \text{tr}(P_C M_\delta) \quad \text{against} \quad \text{tr } M_\delta = d_O|_{P_C=I}.$$

Book equation 0.7.

$$r_{\text{eff}} = \frac{(\sum_i \lambda_i)^2}{\sum_i \lambda_i^2}.$$

LEDGER

- GO-1. The consumer's invariant/nuisance split is identifiable ex ante from the consumer functional. [pre-dicted]. `geometric-observation/claims/LEDGER.md:62` at 9f3829f.

FIRST STATED

TSK appendix B on principal component analysis, as chapter 4 section 4.1 of *Data Mining as Observation* reads it, the identity reader's criterion for a reduction, with the program's spectrum record in `gtc-prototype/docs/SPECTRUM_FINDINGS.md:10-18`.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 6 section 6.2	planted probe, overlap 0.936 vs 0.059, twelve of twelve, reconstruction 0.40, five of five	geometric-observation\claims\LEDGER.md row GO-1
chapter 9 section 9.4	explained ratios, effective rank 5.19 of 8, convergence with a second method, property of the representation not the space	gtc-prototype\docs\SPECTRUM_FINDINGS.md:10-18
chapter 11 section 11.7	GO-1 overlap 0.936 vs 0.059, flip 12 of 12, reconstruction 0.40	geometric-observation\claims\LEDGER.md row GO-1; geometric-observation\chapters\ch10_the_blind_probe.md:34-52
chapter 14 section 14.3	floor 0.08 to 0.12, balanced resample 3001 items, identity attack 0.237 with interval 0.20 to 0.28 on 289, sexual 0.204, threat 0.093 retracted, first pass 18 positives, 87.7 and 1.4 percent	gtc-prototype\docs\SPECTRUM_FINDINGS.md:20-64
chapter 14 section 14.3	contraction 0.779 to 0.863, 77 of 1600, 4.8 percent, 0.872 to 0.863, weight 2.69	gtc-prototype\docs\SPECTRUM_FINDINGS.md:66-99
chapter 14 section 14.5	the contraction formula fairness minus the general component	gtc-prototype\docs\SPECTRUM_FINDINGS.md:90-99
chapter 14 section 14.7	51 percent moderated at 80 and 95 percent precision on the balanced set	gtc-prototype\docs\SPECTRUM_FINDINGS.md:84-88

FAILURES AND CORRECTIONS

none

CONDITIONS

- The fraction of total variance the first k components carry. It lies in the unit interval, cannot fall as k grows, and reaches one at the full dimension. With unit sensitivities it is the fraction a consumer retains, which is why it is the identity reader's criterion.
- A consumer that reads other directions retains a different fraction. With variances 9 and 1 and a consumer reading only the second direction, keeping the first component explains nine tenths of the variance and none of what the consumer reads, which is the flip in its smallest form.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/ExplainedVariance.lean`, `theorems explained_mem_unit`, `explained_mono`, `explained_full`, `retained_identity`, `retained_example`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 4, 8.

RELATED

[identity reader](#) (p. 171); [the flip](#) (p. 138); [effective rank](#) (p. 115); [water-filling](#) (p. 365).

F

F1

concept. Entry id f1.

DEFINITION

Twice the product of precision and recall over their sum. The optimal thresholded F1 of a score is the best F1 over every threshold and depends only on the score's ranking. Equation 0.28.

EQUATION

Book equation 0.28.

$$P = \frac{TP}{TP + FP}, \quad R = \frac{TP}{TP + FN}, \quad F_1 = \frac{2PR}{P + R}.$$

Book equation 6.2.

$$\max_{\tau, \text{dir}} F_1(\mathbf{1}[\text{dir}(g(f(X)), \tau)], y) = \max_{\tau, \text{dir}} F_1(\mathbf{1}[\text{dir}(f(X), \tau)], y)$$

for every strictly monotone g .

Book equation 5.4.

$$F_1^{\max} \leq \sup_{t \in [J, 1]} \frac{2t\pi}{t\pi + \pi + (t - J)(1 - \pi)}, \quad J = \max_{\tau} (\text{TPR} - \text{FPR}), \quad \pi = \text{prevalence},$$

$J \leq 2A - 1$ when the ROC curve is concave, and not in general.

LEDGER

none

FIRST STATED

Chapter 0 section 0.14 and chapter 6 section 6.3 of *Data Mining as Observation*, with the program's optimal thresholded F1 in theory-radar, DOI 10.5281/zenodo.20660206.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 5 section 5.4	Monotone Invariance and AUROC invariance theorems, the AUROC to F1 bound via the Youden index, stated in the source for every ROC curve and corrected in the book to concave curves, with the counterexample at AUROC 0.75 and F1 0.857	theory-radar\paper\astar_paper.tex:94-170; theory-radar\paper\theory_radar_paper.tex:290-304
chapter 6 section 6.3	Monotone Invariance Theorem and its proof	theory-radar\paper\theory_radar_paper.tex:290-304; theory-radar\paper\astar_paper.tex:94-110

FAILURES AND CORRECTIONS

none

CONDITIONS

- Twice the product of precision and recall over their sum. The optimal thresholded F1 of a score is the best F1 over every threshold in both directions, and it depends only on the score's ranking, so a strictly monotone transform of the score leaves it unchanged.
- Its ceiling from the Youden index holds for every score, and the AUROC form of that ceiling only for concave ROC curves, which is the theory-radar erratum.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/MonotoneInvariance.lean`, `theorems aurocNum_comp`, `auroc_comp`, `predicted_comp`, `predictedBelow_comp`, `sweptF1_comp`, `sweptF1Below_comp`, `optF1_comp`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 5, 6, 7.

RELATED

[precision, recall](#) (p. 257); [Monotone Invariance Theorem](#) (p. 220); [Youden F1 bound](#) (p. 372); [formula search](#) (p. 145); [threshold](#) (p. 350).

false-clear rate

concept. Entry id `false-clear-rate`.

DEFINITION

The fraction of the decisions a certificate cleared in which the witness said otherwise, conditional on clearing and reported beside the certificate's coverage. Equation 0.26.

EQUATION

Book equation 0.26.

$$\text{FC} = \Pr [W \text{ refutes } | \mathcal{C}_t \text{ clears}], \quad \text{coverage} = \Pr [\mathcal{C}_t \text{ clears}].$$

Book equation 13.1.

$$\text{FC} = \Pr [W \text{ refutes } | \mathcal{C}_t \text{ clears}], \quad d_O(\Delta) = \text{tr} (P_C M_{\text{drift}}(\Delta)), \quad M_{\text{drift}}(\Delta) = \mathbb{E}[\delta_\Delta \delta_\Delta^\top].$$

Book equation 12.4.

$$\text{FC}_{\text{naive}} = \Pr [\text{acc}_s < \text{target} | \text{benchmark} \geq \text{target}] = \frac{1}{S} \sum_{s=1}^S \mathbf{1}[\text{acc}_s < \text{target}]$$

when the benchmark clears, so that the naive certificate's coverage is 1.

LEDGER

none

FIRST STATED

Volume 14 chapter 19, `geometric-observation/chapters/ch19_the_certificate_that_ages.md:61-76`, and the freshness program's umbrella document `observation-theory-campaigns/experiments/FRESHNESS-PROGRAM.md`, re-homed into the campaigns repository on 2026-08-24.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 12 section 12.6	XPROTO-LLM, benchmark 0.909, 0.920, 0.909, thirty slices, target 0.8, naive 0.333, aware 0.033, spread 0.380, deployment mean 0.736, six bars on three seeds, sealed 2026-08-25 at b61f7f1	observation-theory-campaigns\experiments\LLM-EVAL-TRACK.md:1-50; observation-theory-campaigns\analysis\llm\XPROTO-LLM-graded.json; observation-theory-campaigns\experiments\SEALS.md:85
chapter 13 section 13.3	ZooKeeper hot 0.99 cold 0.01 witnessed 0.0, Postgres 0.50 to 0.06, MongoDB 0.47 to 0.03, production Postgres 0.47 to 0.02, real substrates, disjoint seeds	observation-theory-campaigns\experiments\DATABASE-FRESHNESS-TRACK.md:1-45
chapter 13 section 13.3	BGP 0.351, IS-IS 0.184, OSPF 0.083, second collector	observation-theory-campaigns\analysis\mongo\PREREG-XPROTO-MG.md:42; observation-theory-campaigns\experiments\ROUTING-TELEMETRY-TRACK.md:130-150; geometric-observation\BOOK-OUTLINE.md:75
chapter 13 section 13.3	radio 0.27 to 0.42 naive to 0.055 to 0.13, neural reconstruction 0.28 to 0.13	observation-theory-campaigns\experiments\RADIO-FRESHNESS-TRACK.md:20-35
chapter 13 section 13.4	refuted fit 0.177 T_coh R ² 0.915, 0.15 threshold vs 0.10 claim, fresh baseline 0.113, do not cite	observation-theory-campaigns\analysis\csi\CSI-refreshfloor.json:2,111-114; observation-theory-campaigns\experiments\RADIO-FRESHNESS-TRACK.md:41-47
chapter 13 section 13.4	sealed correction, 0.5 dB calibration, floors 4/6/4, 2/2/3, 1, slopes 0.1008, 0.1398, 0.1086, R ² 0.7376, 0.9218, 0.6174, bars B1 to B4 and MC1 to MC4 all true, seeds 20260827 to 20260829, sealed 1d3de4a	observation-theory-campaigns\analysis\csi\PREREG-XPROTO-CSI-SWEEP2.md:1-60; observation-theory-campaigns\analysis\csi\XPROTO-CSI-SWEEP2-graded.json; observation-theory-campaigns\experiments\SEALS.md:90

From observation-theory-campaigns/experiments/DATABASE-FRESHNESS-TRACK.md at 9d86211.

- line 23. XPROTO-PG (analysis/pgrep) | Postgres, recovery_min_apply_delay | WAL LSN | ~0.50 → ~0.06
- line 24. XPROTO-MG (analysis/mongo) | MongoDB delayed secondary | oplog ts | ~0.47 → ~0.03
- line 25. XPROTO-PGX (analysis/pgx) | production PG, netem lag | WAL LSN, pg_stat_statements footprint | ~0.47 → ~0.02
- line 27. XPROTO-ZK (analysis/zk) | ZooKeeper 3.9 ensemble | zxid, sync() | hot 0.99 / cold 0.01, witnessed 0.0

From observation-theory-campaigns/experiments/RADIO-FRESHNESS-TRACK.md at 9d86211.

- line 25. XPROTO-CSI (analysis/csi) | CQI → MCS | HARQ | 0.34–0.37 → 0.10 (OLLA) | pass 08-23

- line 26. XPROTO-BEAM (analysis/beam) | mmWave beam index | HARQ | 0.31 → 0.02 (BFR) | pass 08-23
- line 27. XPROTO-AICSI (analysis/aicsi) | neural-CSI recon (turboquant bridge) | precoder/HARQ | recon wins yet 0.28 → 0.13 | pass 08-23, scope-corrected 08-25 warning
- line 28. XPROTO-HO (analysis/ho) | RSRP → serving cell | RLF | 0.31–0.44 → 0.09–0.12 | pass 08-23
- line 30. XPROTO-PHY (analysis/phy) | PMI / RI / TA | HARQ | 0.27–0.42 → 0.055–0.13 | pass 08-24

FAILURES AND CORRECTIONS

- observation-theory-campaigns/experiments/RADIO-FRESHNESS-TRACK.md:33-39 at 9d86211. warning **XPROTO-AICSI scope correction (2026-08-25)**. The v1 seal stands for what it tested, but the reconstruction-vs-consumer dissociation does **not** survive the community-standard substrate. On real 3GPP CDL-C with a CsiNet-class codec the NMSE-optimal codec reconstructs near-perfectly (NMSE $\approx$ 0.03) and false-clears 0.0 on all seeds; the pre-registered kill fired. See analysis/aicsi/PREREG-XPROTO-AICSI-V2.md (REFUTED AT SHAKEDOWN, NOT SEALED, kept negative). This is a scope correction, not a retraction. Do not headline the AICSI row; WCNC §III-C is dropped.
- observation-theory-campaigns/experiments/RADIO-FRESHNESS-TRACK.md:41-50 at 9d86211. **The age horizon** (analysis/csi/, XPROTO-CSI-SWEEP2 sealed 2026-08-27, graded PASS): at the calibrated 0.10 budget the largest compliant report period is a few TTI at 10 Hz and 1 TTI at 50 Hz and above, so there is no proportionality law to fit. The earlier CSI-refreshfloor.* claim of a floor $\approx 0.177 \cdot T_{\text{coh}}$ ($R^2=0.915$) was an **unsealed exploration** measured at a relaxed 0.15 threshold against the 0.10 target, with a fresh baseline that never met the budget. It is refuted; the record is kept, not cited. Separately, the optimal linear predictor cannot beat the one-coherence-time wall (Gaussian fading → Wiener optimal), and that wall sits well outside the budget horizon. The **RAN governor** (analysis/ran) is the reference O-RAN rApp core: observe→measure false-clear→refresh at the floor→escalate (diversity / re-route).
- geometric-observation/chapters/ch19_the_certificate_that_ages.md:69-76 at 9f3829f. — the rate at which a clearance is wrong, conditional on the certificate having cleared, reported beside the certificate's coverage. (An earlier draft wrote the joint probability $\Pr[\text{clears} \wedge \text{refutes}]$; the joint rate is the product of the two and is what a naive certificate that clears everything reports, since its coverage is one. Corrected 2026-09-03.) A certificate whose false-clear rate exceeds the target it purports to guarantee is **vacuous** for that consumer, however confident it looks. FC is measurable wherever a witness exists, which — per the table — is nearly everywhere it matters.

CONDITIONS

- A witness must exist. Where none exists the rate is not measurable and the certificate is unassessed, not safe.
- The rate is per consumer, given per read operator, slice, group, or footprint, never as one aggregate.
- Coverage is reported with it. A witnessed certificate clears fewer decisions than a naive one, and the two numbers are read together.

Conditions are curated in entries.toml rather than read from a record.

MACHINE CHECKED

lean/DataMiningAsObservation/Certificate.lean, theorems falseClear_mul_coverage, coverage_empty, falseClear_mem_unit, minOverStrata_passes_iff, minOverStrata_le_weighted_mean, at observation-data-mining af776fd.

USED IN

Data Mining as Observation chapters 0, 12, 13, 14.

RELATED

[certificate](#) (p. 55); [witness](#) (p. 370); [coverage](#) (p. 86); [coherence time](#) (p. 63); [refresh floor](#) (p. 292); [min-over-strata](#) (p. 217).

fine-tuning

instrument. Entry id fine-tuning.

DEFINITION

Continuing to train an encoder on a new objective or corpus, so that its quotient changes. Chapter 11 section 11.8 and chapter 12 section 12.4.

EQUATION

Book equation 12.3.

$$\text{validated} \iff \text{AUROC}_{\text{cross}} - \max(\text{AUROC}_{\text{untrained}}, \text{AUROC}_{\text{BoW}}) \geq 0.10.$$

Book equation 11.1.

$$\cos(k, \hat{k}) = 0.995 \quad \text{while} \quad \text{PPL} : 12.24 \rightarrow 10643.$$

LEDGER

- GO-B-Llama. Trained frontier LLM (Llama-3.2-3B), softmax-attention consumer — blind probe on real post-RoPE keys [predicted]. `geometric-observation/claims/LEDGER.md:115` at 9f3829f.
- GO-B-Llama-rematch. Trained frontier LLM (Llama-3.2-3B), softmax-attention consumer — reconnected dissociation on real post-RoPE keys [predicted]. `geometric-observation/claims/LEDGER.md:118` at 9f3829f.
- GO-B-legal (035→036). Legal-citation retrieval (CourtListener), cosine-ranking consumer, LaBSE embeddings — real large corpus, non-physical consumer [predicted]. `geometric-observation/claims/LEDGER.md:119` at 9f3829f.

FIRST STATED

Chapter 11 section 11.8 and chapter 12 section 12.4 of *Data Mining as Observation*, with the translation-trained encoder in `turboquant-pro/docs/RESULTS_multilingual_strata.md:55-90`.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 10 section 10.4	first run, 350000 of 2391361 rows, 7 of 14 eligible, abstentions with 2 and 1 rows, Robin Hood 0.3447 to 0.4469 ratio 1.30 against 1.5, skew 2.68 to 4.40 ratio 1.64 against 3, 63 times sample range	<code>turboquant-pro/docs/RESULTS_multilingual_strata.md:1-55</code>
chapter 10 section 10.4	second prediction inverted, transit 0.389 vs 0.291, centrality difference signs, share 0.473 vs 0.391, seven of thirteen backbone	<code>turboquant-pro/docs/RESULTS_multilingual_strata.md:55-90</code> ; <code>turboquant-pro/docs/STRATA_RFC.md:98-130</code>
chapter 11 section 11.8	0.765 to 0.971 with interval 0.190 to 0.223, 0.545 to 0.562 with interval 0.004 to 0.031, v1 0.340 with interval negative 0.214 to negative 0.081, anisotropy 0.570 to 0.259	<code>lebse/README.md:40-75</code> ; <code>lebse/PAPER.md:1-15,60-66</code> ; <code>lebse/MODEL_CARD.md:48-52</code>
chapter 12 section 12.4	0.765 to 0.971, 0.545 to 0.562, v1 0.340, cosine 0.570 to 0.259	<code>lebse/README.md:40-75</code> ; <code>lebse/PAPER.md:1-15,60-66</code>
chapter 12 section 12.4	seven of thirteen backbone areas under the translation-trained encoder, all local-hub under the general one	<code>turboquant-pro/docs/RESULTS_multilingual_strata.md:55-90</code>

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 12 section 12.5	within 0.75 to 0.955 collapsing to 0.47 to 0.55, cross-corpus positives as the fix, adversary not load-bearing	xbse\README.md:104-130,160-172

FAILURES AND CORRECTIONS

none

CONDITIONS

- Continuing to train an encoder on a new objective or corpus, so that its quotient changes. A contrastive objective pulls declared pairs together, and the cross-corpus gate asks whether the tuned encoder still separates on a corpus it never saw.
- An encoder fine-tuned on one relation is evaluated on that relation and on one it never saw, with a paired interval on each, and the within-corpus score of 0.75 to 0.955 collapsed to 0.47 to 0.55 across corpora until cross-corpus positives were added.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/Contrastive.lean`, `theorems loss_nonneg, loss_eq_zero_iff, not_both, loss_mono_margin`, at `observation-data-mining af776fd`.

`lean/DataMiningAsObservation/CrossCorpusGate.lean`, `theorems clears_bow, clears_untrained, not_validated_of_saturated, margin_example, validated_comp`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 11, 12.

RELATED

[contrastive objective](#) (p. 76); [encoder](#) (p. 120); [cross-corpus gate](#) (p. 87); [leakage](#) (p. 203); [paraphrase class](#) (p. 246).

finite difference

instrument. Entry id `finite-difference`.

DEFINITION

An estimate of a derivative from two evaluations of the function at points a small step apart. Equation 0.8.

EQUATION

Book equation 0.8.

$$g_j \approx \frac{C(x + h e_j) - C(x - h e_j)}{2h}, \quad j = 1, \dots, d.$$

Book equation 0.9.

$$P_C = \mathbb{E}[g g^\top], \quad g = \nabla C(x).$$

Book equation 11.4.

$$\text{directions resolved}(k) = \begin{cases} 1 \text{ or } 2, & k < d \\ \text{rank } P_C, & k \geq d \end{cases} \quad \text{cost} = 2d \text{ consumer calls per operating point.}$$

LEDGER

- GO-1. The consumer's invariant/nuisance split is identifiable ex ante from the consumer functional. [predicted]. [geometric-observation/claims/LEDGER.md:62 at 9f3829f](#).

FIRST STATED

Chapter 0 section 0.5 of *Data Mining as Observation*, equation 0.8, with the budget law measured in [readscope/README.md:118-141](#) and [readscope/CALIBRATION.md:419-434](#).

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 11 section 11.7	k over d table 1, 2, 2, 16, 16, 16	readscope\README.md:118-141 ; readscope\SPEC.md:243-285 , record readscope\calibration\records\c2e-budget-law.json
chapter 11 section 11.7	rank-independent, 0.646 vs 0.366 at half dimension, never 0.90 below k equals d	readscope\CALIBRATION.md:419-434 F-15
chapter 11 section 11.7	GO-1 overlap 0.936 vs 0.059, flip 12 of 12, reconstruction 0.40	geometric-observation\claims\LEDGER.md row GO-1; geometric-observation\chapters\ch10_the_blind_probe.md:34-52

FAILURES AND CORRECTIONS

none

CONDITIONS

- An estimate of a derivative from two evaluations of the function at points a small step apart. The central difference recovers the derivative of a quadratic exactly at every step size, while the one-sided difference is off by the curvature times the step, which is why doing it in both directions is more accurate.
- Doing it once per coordinate costs d evaluations and in both directions costs 2d, which chapter 11 shows is the price at one operating point for a probe confined to the directions it chooses. The measurement instrument for that is the blind probe.

Conditions are curated in [entries.toml](#) rather than read from a record.

MACHINE CHECKED

[lean/DataMiningAsObservation/FiniteDifference.lean](#), theorems [central_quad](#), [forward_quad](#), [forward_error](#), [central_cost](#), at [observation-data-mining af776fd](#).

USED IN

Data Mining as Observation chapters 0, 1, 2, 4, 6, 7, 11, 12, 14.

RELATED

[sensitivity](#) (p. 319); [read operator](#) (p. 282); [blind probe](#) (p. 39); [budget](#) (p. 46); [budget cliff](#) (p. 49).

the flip

result. Entry id `flip`.

DEFINITION

At matched bits, the outcome in which the code with the worse reconstruction error scores better on the consumer's task, while a code built to destroy the read subspace scores worst. The book's central empirical claim.

Chapter 4.

EQUATION

Book equation 4.5.

$$\text{flip} : \quad \text{task}(O) > \text{task}(R) \text{ and } \text{tr } M_\delta^O > \text{tr } M_\delta^R, \quad \text{task}(\text{anti}) < \text{task}(R), \quad \text{bits}(O) = \text{bits}(R).$$

Book equation 3.1.

$$d_O(u) = u^\top \Sigma u, \quad u = (\cos 15^\circ, \sin 15^\circ), \\ \Sigma_1 = \text{diag}(0.3, 1.7), \Sigma_2 = \text{diag}(1.7, 0.3), \quad d_O = 0.394 \text{ vs } 1.606.$$

Book equation 0.10.

$$d_O = \text{tr}(P_C M_\delta) = \mathbb{E}[\delta^\top P_C \delta], \quad M_\delta = \mathbb{E}[\delta \delta^\top], \quad P_C = I \Rightarrow d_O = \text{tr } M_\delta.$$

LEDGER

- GO-2 (neg. half: not reconstruction). At matched bits, downstream preservation is not controlled by reconstruction error. [demonstrated]. [geometric-observation/claims/LEDGER.md:63 at 9f3829f](#).
- GO-2 (pos. half: consumer-projected covariance *controls*). Downstream preservation is controlled by the error covariance projected on the consumer's read subspace, $\text{tr}(P_C \Sigma_\delta)$. [replicated]. [geometric-observation/claims/LEDGER.md:64 at 9f3829f](#).
- GO-4. Fixed-budget verdicts invert under budget-matched observation, per the wavelength mechanism. [replicated]. [geometric-observation/claims/LEDGER.md:66 at 9f3829f](#).
- NEG-16 (KV serving, end-task). *At matched bits and matched reconstruction error, steering KV quantization error into the attention read subspace degrades LongBench task score by the registered floors on a deployed-class model*. Refuted at its registered effect sizes on this model and task. [refuted]. [geometric-observation/claims/LEDGER.md:92 at 9f3829f](#).

FIRST STATED

Volume 14, chapter 8, [geometric-observation/chapters/ch08_value.md:1-30](#), DOI [10.5281/zenodo.21776291](#). The two-dimensional case is the cover and equation 3.1 of *Data Mining as Observation*.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 1 section 1.4	cosine 0.995 and perplexity of order ten thousand, the recalibration negative	geometric-observation\chapters\ch02_failure_of_observer_free-60 ; geometric-observation\chapters\ch16_honest_negatives.md NEG-2 and NEG-4; turboquant-pro\docs\KV_KEYS_FINDING.md:1-49
chapter 1 section 1.4	the flip sealed in twelve domains and three physics, held in at least five domains and all three	geometric-observation\chapters\ch08_value.md:40-70 ; geometric-observation\claims\LEDGER.md rows GO-2, GO-B-AV163, D3, 038

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 2 section 2.5	whitened code wins at every budget	geometric-observation\chapters\ch08_value.md:84-97
chapter 3 section 3.3	the self-refuted v0.8 claim, NEG-1	the-angular-observer\README.md:170-175; geometric-observation\chapters\ch16_honest_negatives.md NEG-1
chapter 4 section 4.3	the flip definition, flip versus (A2) verdict	geometric-observation\chapters\ch08_value.md:1-30,100-108
chapter 4 section 4.3	GO-2 0.0934 vs 0.0938, 2.53 times, 12 of 12, anti 21 times; retrieval 0.0964, negative 4.70 and positive 4.65, recon 0.40	geometric-observation\claims\LEDGER.md rows GO-2 negative and positive halves
chapter 4 section 4.3	acoustic 148 of 201, 201 of 201, 152 of 201; seismic 13 of 17, 17 of 17, 13 of 17; whale 0.934 vs 0.883, 2 times, 300 of 300; at least 5 domains and 3 physics; battery prediction met	geometric-observation\chapters\ch08_value.md:40-70; geometric-observation\claims\LEDGER.md rows GO-B-AV163, D3, 038
chapter 4 section 4.3	whitened code on whale 0.83, 0.85, 0.97 vs 0.41, 0.80, 0.89	geometric-observation\chapters\ch08_value.md:84-97
chapter 4 section 4.4	gradient compression anti 300 of 300, flip 27 percent, coupling boundary	geometric-observation\chapters\ch08_value.md:108-116
chapter 4 section 4.4	GO-4 budget inversion, fixed m 10 rises, matched m 121, 126, 159 collapses, 3 seeds	geometric-observation\claims\LEDGER.md row GO-4
chapter 4 section 4.6	NEG-4 through NEG-10, the 25 percent codebook confound, Spearman 0.80, the recon-matched precondition	geometric-observation\chapters\ch16_honest_negatives.md NEG-4 to NEG-10
chapter 6 section 6.2	real model, median 0.567 vs bar 0.60, about 4.5 times chance, heads at 0.815 and 0.958, sixteen of sixteen both, two of three triggers, NEG-12, later rematch four of four	geometric-observation\claims\LEDGER.md rows GO-B-Llama and NEG-12; geometric-observation\chapters\ch16_honest_negatives.md:73-81
chapter 6 section 6.2	whale clan classifier, 8718 codas, 0.934 vs 0.883, reconstructs twice as well, 300 of 300	geometric-observation\chapters\ch08_value.md:40-70; geometric-observation\claims\LEDGER.md row 038
chapter 8 section 8.2	cosine 0.995, perplexity near 1e4	turboquant-pro\docs\KV_KEYS_FINDING.md:1-49; geometric-observation\chapters\ch16_honest_negatives.md NEG-2

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 8 section 8.2	recalibration improves reconstruction and worsens the consumer	geometric-observation\chapters\ch16_honest_negatives.md NEG-4

FAILURES AND CORRECTIONS

- NEG-16 (KV serving, end-task), [refuted]. *At matched bits and matched reconstruction error, steering KV quantization error into the attention read subspace degrades LongBench task score by the registered floors on a deployed-class model.* Refuted at its registered effect sizes on this model and task.
- geometric-observation/chapters/ch16_honest_negatives.md:28-30 at 9f3829f. **NEG-4 — lightweight on-line (Lloyd) key calibration beats the default.** [refuted]. *It improves reconstruction and is worse on the consumer metric* — reconstruction-is-not-the- target, in its purest, most anti-correlated form.
- geometric-observation/chapters/ch16_honest_negatives.md:32-34 at 9f3829f. **NEG-5 ... NEG-9 — the KV-key ladder that found the right statistic.** Five refuted proxies, and together the most instructive sequence in the ledger, because each refutation *located* the next hypothesis:
- geometric-observation/chapters/ch16_honest_negatives.md:36-50 at 9f3829f. - **NEG-5:** the originally-registered test missed (2/4 bars); the apparent reversal was a matched-bits confound (an uncounted per-block codebook, $\sim +25\%$ bits). Refutes the *test*, not GO-2 — and forced a bit-matched redesign. - **NEG-6:** a demeaned-error-norm proxy fails to order the arms. The controlling quantity is the across-token *variance* of the query-projected error, not an error-magnitude scalar. *Magnitude $\neq$ downstream-relevant structure* (Chapter 6's lemma, learned here). - **NEG-7:** at *fixed reconstruction*, the downstream ranking **flips** with the consumer. Invariant-preservation is a property of the (compressor, consumer) *pair*, not the compressor — the sharpest form of GO-2's negative half. - **NEG-8:** the variance-ratio proxy reaches only Spearman 0.80; it is a noisy linear estimator of $\text{tr}(P_C \Sigma_\delta)$. Gate on the *direct* projection instead. - **NEG-9:** even the direct trace $\text{tr}(P_C \Sigma_\delta)$, while nailing the discriminating flip 12/12, is *not a complete rank statistic* — it misorders a middle pair whose per-token error has structure the trace misses. The mechanism is real; the scalar is not exhaustive.
- geometric-observation/chapters/ch16_honest_negatives.md:56-61 at 9f3829f. **NEG-10 — the flip appears on any independent representation (prospective).** [refuted]. On embedding retrieval, arm b reconstructed strictly better *and* Pareto-dominated downstream — no flip — so reconstruction was not falsified there. **Precondition discovered: the flip is observable only for reconstruction-matched arms.** The anti-probe half still transferred. This negative is why every later flip row is built on recon-matched arms; the miss defined the protocol.
- geometric-observation/chapters/ch16_honest_negatives.md:106-112 at 9f3829f. - **Identifiability** (fixable): NEG-5, NEG-10, legal-035, NEG-12 — the read operator was mis-estimated or the arms weren't recon-matched. Fixed by the blind probe and by protocol. - **Coupling** (a true boundary): D4 — read and signal energy intrinsically aligned; not rehabilitatable by any read-operator recovery. - **Precondition** (needs a working consumer): moral-on-frozen-embeddings — no read direction to protect until the consumer is competent. - **Mechanism-absent** (a genuine refutation): NEG-11 — the claimed effect does not exist.

CONDITIONS

- Matched bits, and reconstruction matched or reported, for every comparison of two codes.
- A read operator misaligned with where the signal has its energy. The alignment of book equation 4.6 is the dial, and near one, the coupling null, no flip is possible.
- The read operator is the average of local linearizations, and the ordering it predicts is the ordering on average over the workload.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

lean/DataMiningAsObservation/Flip.lean, theorems trace_eq, reading_difference, reader_swap_reverses, flip, cos_sq_pi_div_twelve, sin_sq_pi_div_twelve, reading_15_first, reading_15_second, sqrt_three_bounds, readings_15_to_three_decimals, reader_75_is_swap, four_to_one_flip, at observation-data-mining af776fd.

USED IN

Data Mining as Observation chapters 2, 3, 4, 6, 7, 11, 12, 13, 14.

RELATED

[read distortion](#) (p. 281); [alignment](#) (p. 11); [anti arm](#) (p. 15); [rank certificate](#) (p. 275); [coupling null](#) (p. 84).

floor, ceiling

concept. Entry id floor-ceiling.

DEFINITION

A floor is a lower bound a quantity cannot fall below and a ceiling an upper bound it cannot exceed, each proved or measured and named as such. Chapter 0 section 0.9 and chapter 13.

EQUATION

Book equation 0.17.

$$\Pr[X \geq c] = 1 - \sum_{i < c} e^{-\mu} \frac{\mu^i}{i!}, \quad c^* = \max\{c : n \Pr[X \geq c] \geq 1\}, \quad \mu = \frac{n_q k}{n}.$$

Book equation 5.4.

$$F_1^{\max} \leq \sup_{t \in [J, 1]} \frac{2t\pi}{t\pi + \pi + (t - J)(1 - \pi)}, \quad J = \max_{\tau} (\text{TPR} - \text{FPR}), \quad \pi = \text{prevalence},$$

$J \leq 2A - 1$ when the ROC curve is concave, and not in general.

Book equation 13.2.

$$T_{\text{coh}} = \frac{0.423}{f_D},$$

refuted fit: $\phi \approx 0.177 T_{\text{coh}} (R^2 = 0.915)$, sealed: $\phi \leq 6$ TTI at 10 Hz, 1 TTI at ≥ 50 Hz.

LEDGER

- OT-4. Operator drift predicts a real long-generation degradation and a derived refresh intervention moves it. [refuted]. [geometric-observation/claims/LEDGER.md:36 at 9f3829f](#).
- NEG-13 → resolved. (GO-P-2026-026, prospective, real LLM) Appendix-E's omission floor is a rate-irreducible downstream wall on trained Llama read operators. [missed]. [geometric-observation/claims/LEDGER.md:106 at 9f3829f](#).

FIRST STATED

Chapter 0 section 0.9 and chapter 13 section 13.4 of *Data Mining as Observation*, with the omission floor in [geometric-observation/chapters/ch07_cost.md](#) and the Youden ceiling in [theory-radar/paper/astar_paper.tex:94-110](#).

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 4 section 4.2	GO-6, d 8 and r 4, output at or below surrogate at or below reconstruction at every rate, about 500 times, gap 0.41 to 0.005, isotropic control collapses	geometric-observation\claims\LEDGER.md row GO-6 ; geometric-observation\chapters\ch07_cost.md

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 4 section 4.2	commission tax of order half $r \log M$ over m , omission floor, unbounded corrected to floor, naive projector 30 percent over, VI-6	geometric-observation\chapters\ch07_cost.md section on mismatch; geometric-observation\claims\LEDGER.md VI-6
chapter 4 section 4.2	floor measured on 16 of 16 heads, naive assumption off 20 to 60 percent, about 100000 times, 0.10 nats against about $6e-7$, first gate missed 0 of 16, NEG-13	geometric-observation\chapters\ch07_cost.md; geometric-observation\claims\LEDGER.md GO-P-2026-027 and NEG-13
chapter 6 section 6.3	Monotone Invariance Theorem and its proof	theory-radar\paper\theory_radar_paper.tex:290-304; theory-radar\paper\astar_paper.tex:94-110
chapter 13 section 13.3	radio 0.27 to 0.42 naive to 0.055 to 0.13, neural reconstruction 0.28 to 0.13	observation-theory-campaigns\experiments\RADIO-FRESHNESS-TRACK.md:20-35
chapter 13 section 13.4	refuted fit 0.177 T_{coh} R^2 0.915, 0.15 threshold vs 0.10 claim, fresh baseline 0.113, do not cite	observation-theory-campaigns\analysis\csi\CSI-refreshfloor.json:2,111-114; observation-theory-campaigns\experiments\RADIO-FRESHNESS-TRACK.md:41-47

FAILURES AND CORRECTIONS

- OT-4, [refuted]. Operator drift predicts a real long-generation degradation and a derived refresh intervention moves it.

CONDITIONS

- A floor is a lower bound a quantity cannot fall below, and a ceiling an upper bound it cannot exceed, each proved or measured and named as such. The refresh floor is the shortest renewal that keeps a certificate within its error, the Poisson ceiling is the count chance allows, and the Youden ceiling bounds the best F1.
- A floor that was fitted as a fraction of the coherence time was refuted, and the omission floor that was first read as unbounded was corrected to a floor, so each is a measured object with a record.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/CoherenceTime.lean`, `theorems clarke_to_three_decimals`, `examples`, `refuted_law_exceeds_sealed`, at `observation-data-mining af776fd`.

`lean/DataMiningAsObservation/PoissonCeiling.lean`, `theorems mass_nonneg`, `tail_antitone`, `tail_zero`, `tail_le_one`, `expectedAtLeast_antitone`, `example_mean`, at `observation-data-mining af776fd`.

`lean/DataMiningAsObservation/YoudenF1.lean`, `theorems f1_eq`, `f1_le_of_youden`, `auroc_eq`, `youden_eq`, `youden_exceeds_auroc_form`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 3, 4, 5, 6, 7, 8, 9, 10, 11, 13, 14.

RELATED

[refresh floor](#) (p. 292); [Poisson ceiling](#) (p. 253); [Youden index](#) (p. 374); [budget cliff](#) (p. 49); [bar](#) (p. 34).

footprint

concept. Entry id footprint.

DEFINITION

The set of entries a reader reads. What the reader reads decides how warm and how stale the reader is. Chapter 13 section 13.3.

EQUATION

Book equation 0.26.

$$FC = \Pr [W \text{ refutes } | \mathcal{C}_t \text{ clears}], \quad \text{coverage} = \Pr [\mathcal{C}_t \text{ clears}].$$

Book equation 13.2.

$$T_{\text{coh}} = \frac{0.423}{f_D},$$

refuted fit: $\phi \approx 0.177 T_{\text{coh}} (R^2 = 0.915)$, sealed: $\phi \leq 6 \text{ TTI at } 10 \text{ Hz, } 1 \text{ TTI at } \geq 50 \text{ Hz}.$

Book equation 13.1.

$$FC = \Pr [W \text{ refutes } | \mathcal{C}_t \text{ clears}], \quad d_O(\Delta) = \text{tr} (P_C M_{\text{drift}}(\Delta)), \quad M_{\text{drift}}(\Delta) = \mathbb{E}[\delta_\Delta \delta_\Delta^\top].$$

LEDGER

- OT-11. Feedback-free staleness: streaming-retrieval damage tracks measured drift; derived-cadence re-allocation removes it. [void]. `geometric-observation/claims/LEDGER.md:48 at 9f3829f`.
- GO-12. The dynamic region's opening control — staleness is access width, not delay: with full context-path access, pure-delay aging is information-free (the recoding identity $(Y, V, S^{(\Delta)}) \stackrel{d}{=} (Y, V, P^\Delta S^{(0)})$ makes every $\sigma(S)$ -measurable conditional functional exactly Δ -invariant; circulant-exact, $O(1/n)$ edge leakage on finite windows); with time-local slice access the tax is GO-11's static quadratic with the substitution set by the *encoder's* access — single-letter (Y_t, V_t) records at $(\rho, s/a^{2\Delta})$, context-epoch-latent records at the strictly smaller $(\rho a^\Delta, \tau^2)$ (gap up to 0.053 bits) — both strictly increasing in Δ with common limit $\frac{1}{2} \log_2(1/D)$. GO-8's age-dependence is the slice regime's operational face. Theorem 1 (the conditional-variance reduction) settles the causal-prefix eraser in the single-record setting: every observation-subset σ -algebra enters through $q_{\mathcal{G}} = \text{Var}(V_t | \mathcal{G})$ alone; slice/prefix/path = single-sample/Kalman-fixed-lag/noncausal-Wiener variances; strict interpolation at every finite lag with exact gap $C^{2\Delta}(P_f - q_{\text{path}})$; $W(\Delta) = kT \ln 2 \cdot L_{\text{prefix}}$ is the dynamic conditional-Landauer curve. Theorem 2-spectral (070): the spectral conditional RDF — at the work endpoint, $L(D)$ is the equal-slope allocation of per-frequency static quadratics over the circulant spectrum, with classical reverse water-filling ($\tau^2 \rightarrow \infty$) and the static theorem (flat spectrum) exact; per-mode convexity proven; the third promotion of the water level completed for the single-consumer face; Toeplitz transfer scoped imported-with-lemmas, $O(1/n)$ numerics. The weighted spectral theorem (071) closes Conjecture 1 at circulant scope: J_w decomposes per-frequency at every w , each mode carrying Theorem 3's two-water-level system $(\gamma_0(\omega), \gamma_1(\omega))$ under ONE common distortion price (the conjectured 'pair of prices' refuted-as-phrased, rescoped to $\$(w, \mu)\$$); convexity via the full-region moment argument + perturbation. The third promotion is complete on BOTH faces. [predicted]. `geometric-observation/claims/LEDGER.md:78' at 9f3829f`.

FIRST STATED

Chapter 13 section 13.3 of *Data Mining as Observation*, with the reader's footprint in `geometric-observation/chapters/ch19_the_center` 95.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 13 section 13.2	the grammar, certificate, witness, refresh floor, false-clear rate, vacuity, the witness table	geometric-observation\chapters\ch19_the_certificate_that_ages95; observation-theory-campaigns\experiments\FRESHNESS-PROGRAM.md:1-40
chapter 13 section 13.3	ZooKeeper hot 0.99 cold 0.01 witnessed 0.0, Postgres 0.50 to 0.06, MongoDB 0.47 to 0.03, production Postgres 0.47 to 0.02, real substrates, disjoint seeds	observation-theory-campaigns\experiments\DATABASE-FRESHNESS-TRACK.md:1-45

FAILURES AND CORRECTIONS

none

CONDITIONS

- The set of entries a reader reads. The hit rate is one when the cache holds the footprint and does not move when entries outside it are added, so what the reader reads decides how warm, and how stale, the reader is.
- A replica is stale because of what the reader asks for. The hot reader read a false-clear rate of 0.99 and the cold reader 0.01 on the same store, and the aggregate described neither.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/Cache.lean`, `theorems hitRate_mem_unit`, `hitRate_cold`, `hitRate_warm`, `hitRate_mono`, `hitRate_footprint`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 2, 11, 12, 13, 14.

RELATED

[cache](#) (p. 51); [cold](#), [warm](#) (p. 65); [replica](#) (p. 299); [false-clear rate](#) (p. 133); [read subspace](#) (p. 285).

formula search

instrument. Entry id `formula-search`.

DEFINITION

A score written as a short expression in the features, such as the smaller of two features plus a third, with a threshold. Chapter 6.

EQUATION

Book equation 6.2.

$$\max_{\tau, \text{dir}} F_1\left(\mathbf{1}[\text{dir}(g(f(X)), \tau)], y\right) = \max_{\tau, \text{dir}} F_1\left(\mathbf{1}[\text{dir}(f(X), \tau)], y\right)$$

for every strictly monotone g .

Book equation 5.4.

$$F_1^{\max} \leq \sup_{t \in [J, 1]} \frac{2t\pi}{t\pi + \pi + (t - J)(1 - \pi)}, \quad J = \max_{\tau} (\text{TPR} - \text{FPR}), \quad \pi = \text{prevalence},$$

$J \leq 2A - 1$ when the ROC curve is concave, and not in general.

Book equation 7.4.

$$\Pr[\text{formula beats boosting on a new dataset}] = 0.345 \text{ overall,}$$

$$0.589 \text{ for } N < 800, \quad \text{predictive interval } [-0.165, +0.125].$$

Book equation 7.6.

$$\Delta F_1^{(i)}(n) = \alpha_i + \beta \log n + \varepsilon,$$

$n_{\text{cross}}^{(i)}$ = the training size at which the ensemble overtakes the formula.

LEDGER

none

FIRST STATED

theory-radar, DOI 10.5281/zenodo.20660206, theory-radar/README.md, with the corrected comparison in the constraint-gap review, constraint-gap/review/REDESIGN.md:1-110.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 5 section 5.4	Monotone Invariance and AUROC invariance theorems, the AUROC to F1 bound via the Youden index, stated in the source for every ROC curve and corrected in the book to concave curves, with the counterexample at AUROC 0.75 and F1 0.857	theory-radar\paper\astar_paper.tex:94-170; theory-radar\paper\theory_radar_paper.tex:290-304
chapter 5 section 5.4	seven thresholds up to 0.75, eight datasets, 56 conditions, zero admissibility violations, reductions 70 to 99.1 percent	theory-radar\paper\astar_paper.tex:170-200
chapter 5 section 5.4	the rerun of 2026-09-03 and 2026-09-04, pre-filter counts 2576 of 8700 and 413 of 1560, ceiling violations 812 and 412, Youden pruning 199 and 38, A* optima 0.958 and 0.694 in strict and fast modes, Youden mode short by 0.006 and 0.002	theory-radar\results\rerun_youden_2026_09_03.json; theory-radar\ERRATA.md
chapter 5 section 5.4	learned pruning 88 to 99.6 percent with zero false negatives	theory-radar\README.md:88-100
chapter 6 section 6.3	depth three, ten binary and eight unary operations, exact optimal F1 by sort and sweep, beam, projections	theory-radar\README.md:50-100, 140-200
chapter 6 section 6.3	Monotone Invariance Theorem and its proof	theory-radar\paper\theory_radar_paper.tex:290-304; theory-radar\paper\astar_paper.tex:94-110

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 6 section 6.3	88 to 99.6 percent pruned with zero false negatives, fair protocol, 200 by 5 folds	theory-radar\README.md:88-100
chapter 6 section 6.3	the five-dataset table, breast cancer 0.955 to 0.963, the pattern, DOI 10.5281/zenodo.20660206	theory-radar\README.md:100-140
chapter 6 section 6.4	the uncorrected t stored as sigma, 251 and 15.84, nine wins to three, seven, eight, 3 wins 17 ties 11 losses on 31 datasets, the three-way inconsistency	constraint-gap\review\FINDINGS.md:1-35; constraint-gap\README.md:57-76
chapter 6 section 6.4	the reviewer's revision plan, drop sigma, bounded claim, 19 at 200 by 5 vs 11 at 20 by 5, rerun in progress	theory-radar\paper\REVISION_PLAN.md:1-60
chapter 6 section 6.4	five analyses negative, 0.2 percent of floor, twelve times the band, learning-curve redesign	constraint-gap\README.md:57-76; constraint-gap\review\REDESIGN.md:1-45
chapter 6 section 6.5	loading weights and stability across folds requested	theory-radar\paper\REVISION_PLAN.md issue 6
chapter 7 section 7.3	the five-dataset outcomes	theory-radar\README.md:100-140
chapter 7 section 7.3	correlation negative 0.59 at p 4.8e-4 with size, negative 0.562 at p 0.001 with the ensemble's score, predictive 0.345 and 0.589, interval negative 0.165 to 0.125, the one-line summary	constraint-gap\review\INDETERMINATES.md:60-80; constraint-gap\review\REDESIGN.md:45-70
chapter 7 section 7.4	nine to three, seventeen indeterminate, five analyses zero resolved, three re-encode the correction	constraint-gap\review\INDETERMINATES.md:1-40
chapter 7 section 7.4	the floor within 0.2 percent, fold ratios 1.334 and 1.937, width 0.945 and 0.921, 8 to 11 percent, closest at 1.68	constraint-gap\review\INDETERMINATES.md:40-60
chapter 7 section 7.4	heterogeneity 93 to 95 percent, Q 446 on 30, scale 0.063, band 0.005, ROPE 0.073, shrinkage 0.1746 to 0.0997 at 42.9 percent	constraint-gap\review\INDETERMINATES.md:55-95
chapter 7 section 7.4	the redesign, grid 50 to N, ten folds, twenty repetitions, crossing size, the 800-row boundary declared in advance, seventeen at median 0.006 and p 0.964, store the folds	constraint-gap\review\REDESIGN.md:1-110
chapter 8 section 8.4	real 1.038 [0.958, 1.118], rotated 1.019 [0.903, 1.353], smooth 2.804 withdrawn, second attempt 0.184	constraint-gap-measurements\notes\negative_control.md:1-50
chapter 8 section 8.5	variance inflation 251, SE inflation 15.84, J_eff 3.98, about 62 needed, 3 of 9 at full, 7 at half, 8 at a third	constraint-gap\review\FINDINGS.md:1-35
chapter 8 section 8.5	9/3/19 to 3/17/11, five negative analyses, 0.2 percent of floor, twelve times the band, learning-curve redesign	constraint-gap\README.md:57-76; constraint-gap\review\REDESIGN.md

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 8 section 8.5	three-way inconsistency, lines 544, 548 to 556, 755	constraint-gap\review\FINDINGS.md:7-31 against theory-radar\paper\theory_radar_v6_submission.tex
chapter 8 section 8.9	twenty-seven errors, four in checking tools	constraint-gap-measurements\README.md:273-282

FAILURES AND CORRECTIONS

- theory-radar/ERRATA.md:3-75 at 37c4e6c. ## 2026-09-03. The AUROC to F1 bound holds only for concave ROC curves **What was claimed.** Theorem “AUROC-F1 Bound” in paper/astar_paper.tex, the derivation in src/symbolic_search/_auroc_proof.py, and the pruning bounds in run_astar_v2.py and src/symbolic_search/_heuristic_dag.py all rested on the statement that any ROC curve with area A has maximum Youden index $J = \max(\text{TPR} - \text{FPR})$ at most $2A - 1$. **Why it is wrong.** The statement holds for concave ROC curves (equivalently, scores whose likelihood ratio is monotone), because a concave curve lies above the two chords through its Youden point and so has area at least $(1 + J)/2$. It fails for non-concave curves. Counterexample, found by an external reviewer of the companion textbook: rank three quarters of the positives above every negative and the remaining quarter below every negative. AUROC is 0.75, the maximum Youden index is 0.75, and at prevalence one half the best thresholded F1 is 0.857, above the 0.80 the AUROC form of the ceiling allows. The sound ceiling for that score, from $J = 0.75$, is 0.889, and the score sits under it. The `auroc_f1_bound` formula in `run_astar_v2.py`, $2A\pi/(A\pi + (1 - A)(1 - \pi))$, is not a valid upper bound either: at prevalence 0.1 the same score reaches F1 0.857 against a “bound” of 0.5. **Consequence for reported results.** The searches that used these bounds to prune could have discarded a candidate formula whose ROC curve was not concave and whose true best F1 exceeded the computed ceiling. The formulas and wins reported by those runs are therefore what a pruned search found, and are lower bounds on what an unpruned search would find. The Monotone Invariance Theorem and the results that do not depend on the AUROC bound are unaffected. **Correction.** The sound form of the ceiling substitutes the measured maximum Youden index of the candidate's own scores for $2A - 1$. It holds for every score, costs the same sort as AUROC, and is what `max_f1_for_youden` and `youden_f1_bound` now compute. The AUROC form is retained under its original names with a docstring stating the concavity condition, and the pruning sites now use the measured index. A rerun of the published searches under the corrected bound was done on 2026-09-03 and 2026-09-04, `rerun_youden_2026_09_03.py`, results in `results/rerun_youden_2026_09_03.json`, run on the Atlas workstation, CPU only. **Rerun, part A, the pre-filter admissibility table (tab:auroc).** Depth-2 pairwise enumeration on the eight datasets, exhaustive against the AUROC pre-filter at the seven published thresholds, with the sound Youden ceiling beside it. The evaluation counts under the pre-filter reproduce the published table (Breast Cancer 8700 pairs, 2576 evaluated at 0.75; Wine 1560, 413; Circles 20, 4; Moons 20, 2). The pre-filter lost no optimum on any dataset at any threshold, which is what the paper reported. That was luck of the data, not the theorem: the retracted ceiling sat below the F1 a pair formula actually reach
- theory-radar/paper/REVISION_PLAN.md:39-45 at 37c4e6c. **Fix:** - Drop “ σ significance” everywhere - Report: mean $\Delta F1$, std, 95% CI, and corrected resampled t-statistic - Table columns: “Test F1”, “GB F1”, “ $\Delta F1$ ”, “95% CI”, “Direction” - Text: “the formula outperforms GB by $\Delta F1=0.031$, 95% CI [0.028, 0.034]” - Add Nadeau-Bengio corrected t-test (accounts for CV fold correlation) - Keep effect size as supplementary but don't call it “ σ ”

CONDITIONS

- The search enumerates short expressions over the features and scores each by its optimal thresholded F1, pruning monotone unary nodes by the Monotone Invariance Theorem and, in the A* companion, leaves by an F1 ceiling.
- Its first comparison reported nine wins that became three under the Nadeau and Bengio correction, with seventeen datasets indeterminate, and the record carries the retraction at the size of the result.
- The main comparison used a fold-local meta-learned pruning rule and never rested on the AUROC form of

the ceiling. The A* companion did, and the rerun shows what that cost, nothing on those datasets. Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/MonotoneInvariance.lean`, `theorems aurocNum_comp`, `auroc_comp`, `predicted_comp`, `predictedBelow_comp`, `sweptF1_comp`, `sweptF1Below_comp`, `optF1_comp`, at `observation-data-mining af776fd`.

`lean/DataMiningAsObservation/YoudenF1.lean`, `theorems f1_eq`, `f1_le_of_youden`, `auroc_eq`, `youden_eq`, `youden_exceeds_auroc_form`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 5, 6, 7, 14.

RELATED

[Monotone Invariance Theorem](#) (p. 220); [safe pruning](#) (p. 311); [Youden F1 bound](#) (p. 372); [read operator](#) (p. 282).

freshness

concept. Entry id freshness.

DEFINITION

Whether a stored value is still within its coherence time when it is read. Consumer-relative, since two readers of the same replica can have different freshness. Chapter 13.

EQUATION

Book equation 0.26.

$$FC = \Pr [W \text{ refutes } | \mathcal{C}_t \text{ clears}], \quad \text{coverage} = \Pr [\mathcal{C}_t \text{ clears}].$$

Book equation 0.25.

$$T_{\text{coh}} = \frac{0.423}{f_D}.$$

Book equation 13.2.

$$T_{\text{coh}} = \frac{0.423}{f_D},$$

refuted fit: $\phi \approx 0.177 T_{\text{coh}} (R^2 = 0.915)$, sealed: $\phi \leq 6 \text{ TTI at } 10 \text{ Hz}, 1 \text{ TTI at } \geq 50 \text{ Hz}$.

LEDGER

- OT-4. Operator drift predicts a real long-generation degradation and a derived refresh intervention moves it. [refuted]. `geometric-observation/claims/LEDGER.md:36` at `9f3829f`.
- OT-11. Feedback-free staleness: streaming-retrieval damage tracks measured drift; derived-cadence re-allocation removes it. [void]. `geometric-observation/claims/LEDGER.md:48` at `9f3829f`.

FIRST STATED

Volume 14, chapter 19, `geometric-observation/chapters/ch19_the_certificate_that_ages.md`, DOI [10.5281/zenodo.21776291](https://doi.org/10.5281/zenodo.21776291), and the freshness track, `observation-theory-campaigns/experiments/RADIO-FRESHNESS-TRACK.md:20-35`.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 13 section 13.2	the grammar, certificate, witness, refresh floor, false-clear rate, vacuity, the witness table	geometric-observation\chapters\ch19_the_certificate_that_ages95; observation-theory-campaigns\experiments\FRESHNESS-PROGRAM.md:1-40
chapter 13 section 13.3	radio 0.27 to 0.42 naive to 0.055 to 0.13, neural reconstruction 0.28 to 0.13	observation-theory-campaigns\experiments\RADIO-FRESHNESS-TRACK.md:20-35
chapter 13 section 13.4	refuted fit 0.177 T_{coh} R^2 0.915, 0.15 threshold vs 0.10 claim, fresh baseline 0.113, do not cite	observation-theory-campaigns\analysis\csi\CSI-refreshfloor.json:2,111-114; observation-theory-campaigns\experiments\RADIO-FRESHNESS-TRACK.md:41-47

FAILURES AND CORRECTIONS

- OT-4, [refuted]. Operator drift predicts a real long-generation degradation and a derived refresh intervention moves it.
- observation-theory-campaigns/experiments/RADIO-FRESHNESS-TRACK.md:41-50 at 9d86211. **The age horizon** (analysis/csi/, XPROTO-CSI-SWEEP2 sealed 2026-08-27, graded PASS): at the calibrated 0.10 budget the largest compliant report period is a few TTI at 10 Hz and 1 TTI at 50 Hz and above, so there is no proportionality law to fit. The earlier CSI-refreshfloor.* claim of a floor $\approx 0.177 \cdot T_{\text{coh}}$ ($R^2=0.915$) was an **unsealed exploration** measured at a relaxed 0.15 threshold against the 0.10 target, with a fresh baseline that never met the budget. It is refuted; the record is kept, not cited. Separately, the optimal linear predictor cannot beat the one-coherence-time wall (Gaussian fading $\rightarrow$ Wiener optimal), and that wall sits well outside the budget horizon. The **RAN governor** (analysis/ran) is the reference O-RAN rApp core: observe $\rightarrow$ measure false-clear $\rightarrow$ refresh at the floor $\rightarrow$ escalate (diversity / re-route).

CONDITIONS

- A stored value is fresh for a reader when its age is within that reader's coherence time. Freshness is consumer-relative. Two readers of the same replica with different coherence times disagree about every age between the two, and the staleness rate of a mixed population is a weighted mean of the two readers' rates, strictly between them when they differ, so it describes neither.
- The measured false-clear rates on the named substrates, ZooKeeper, Postgres, MongoDB, and the radio link, are the freshness program's numbers, and the refuted refresh law is its standing negative.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

lean/DataMiningAsObservation/Freshness.lean, theorems disagree, mixedRate_between, mixedRate_eq_left_iff, stale_for_all, at observation-data-mining af776fd.

USED IN

Data Mining as Observation chapters 0, 1, 2, 4, 8, 11, 12, 13, 14.

RELATED

[coherence time](#) (p. 63); [certificate](#) (p. 55); [false-clear rate](#) (p. 133); [refresh floor](#) (p. 292); [drift](#) (p. 112).

G

gate

instrument. Entry id gate.

DEFINITION

A bar that a build or a claim must pass before it proceeds, with the bar and the null written down first. Chapter 8 section 8.3 and chapter 11 section 11.6.

EQUATION

Book equation 12.3.

$$\text{validated} \iff \text{AUROC}_{\text{cross}} - \max(\text{AUROC}_{\text{untrained}}, \text{AUROC}_{\text{BoW}}) \geq 0.10.$$

Book equation 10.7.

$$\text{verdict} = \min_{i: n_i \geq n_{\min}, |Q_i| \geq q_{\min}} \text{score}_i, \quad \text{ABSTAIN otherwise.}$$

Book equation 8.3.

$$\Pr[\text{at least one of } m \text{ null tests passes}] = 1 - (1 - \alpha)^m, \quad \alpha_{\text{Bonferroni}} = \frac{\alpha}{m}.$$

LEDGER

- NEG-14. (GO-P-2026-037, prospective) a2_probe.median_unit_displacement is a single-statistic predictor of the flip regime ($\text{unit_disp} \geq 1.0 \Rightarrow \text{generic-polar-flip vs needs-blind-probe}$). [refuted]. geometric-observation/claims/LEDGER.md:108 at 9f3829f.
- GO-B-legal (035→036). Legal-citation retrieval (CourtListener), cosine-ranking consumer, LaBSE embeddings — real large corpus, non-physical consumer [predicted]. geometric-observation/claims/LEDGER.md:119 at 9f3829f.

FIRST STATED

Chapter 8 section 8.3 of *Data Mining as Observation*, with the build gate in turboquant-pro/docs/HUBNESS_PRIMER.md:86-131 and the cross-corpus gate in xbse/README.md:130-160.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 10 section 10.2	anti-hubs as where compressed indexes fail first, aggregate recall barely moves	turboquant-pro/docs/HUBNESS_PRIMER.md:86-131
chapter 10 section 10.5	Gate A design error, A prime skew 3.970 to 3.177, max 287 to 213, Robin Hood 0.372 to 0.261, fraction 0.117 to 0.079, compressed path 0.663 vs 0.90, seven strata, 0.62 to 0.69 vs 0.76 to 0.84	turboquant-pro/docs/RESULTS_strata_phase23_gates.md:1-45
chapter 10 section 10.5	rerank identical 0.6627, candidate coverage, depth curve 51 to 501, coverage 0.696 to 0.926, fidelity 0.663 to 0.702, originals rerank 0.923, 4096 bytes on 148	turboquant-pro/docs/RESULTS_strata_phase23_gates.md:45-110

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 10 section 10.5	fragile-first allocation 0.7118 vs 0.7251, targets up 0.004 to 0.034, redesigned allocator passed	turboquant-pro\docs\RESULTS_strata_phase23_gates.md:110-125
chapter 11 section 11.6	anti-hub recall, p05, hub-rank correlation, hub-set overlap, the build gate	turboquant-pro\docs\HUBNESS_PRIMER.md:86-131
chapter 12 section 12.5	the gate, both nulls, margin 0.10, scorecard 0.622 to 0.853, bag of words 0.46 to 0.54, rights 0.509 vs 0.512 and 0.467	xbse\README.md:130-160, 195-219

FAILURES AND CORRECTIONS

- NEG-14, [refuted]. (GO-P-2026-037, prospective) `a2_probe.median_unit_displacement` is a single-statistic predictor of the flip regime ($\text{unit_disp} \geq 1.0 \Rightarrow \text{generic-polar-flip vs needs-blind-probe}$).

CONDITIONS

- A bar that a build or a claim must pass before it proceeds, with the bar and the null written down first. Passing a higher bar passes every lower one, a gate discriminates only when the null fails it and the real system passes it, and a gate the null passes is vacuous.
- The hubness gate was predicted to be blind to the category mix and was not, so the instrument claim was withdrawn, and the gate meta-rule says relative contrast discriminates nothing.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/Bar.lean`, `theorems passes_anti`, `passes_mono`, `discriminates_iff`, `no_bar_of_null_ge`, `exists_bar_of_lt`, `vacuous_of_null_passes`, at `observation-data-mining af776fd`.

`lean/DataMiningAsObservation/CrossCorpusGate.lean`, `theorems clears_bow`, `clears_untrained`, `not_validated_of_saturated`, `margin_example`, `validated_comp`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 4, 8, 9, 10, 11, 12, 13, 14.

RELATED

[bar](#) (p. 34); [cross-corpus gate](#) (p. 87); [anti-hub recall](#) (p. 19); [preregistration](#) (p. 258); [null model](#) (p. 234).

Gaussian

concept. Entry `id gaussian`.

DEFINITION

The normal distribution, in many dimensions the one whose density falls with the Mahalanobis distance. An isotropic Gaussian has every direction the same. Chapter 9 and chapter 10.

EQUATION

Book equation 0.33.

$$d_M(x) = \sqrt{(x - \mu)^\top \Sigma^{-1} (x - \mu)}.$$

Book equation 0.6.

$$x_w = \Sigma^{-1/2} (x - \mu), \quad \Sigma^{-1/2} = \sum_i \lambda_i^{-1/2} v_i v_i^\top.$$

Book equation 9.3.

$$\hat{\mu} = \frac{\bar{s}_{\text{true}} - \bar{s}_{\text{distr}}}{\sigma_{\text{distr}}}, \quad \mu_{\text{crit}} = \mathbb{E} \left[\max_{N-1} \mathcal{N}(0, 1) \right], \quad \rho = \frac{\hat{\mu}}{\mu_{\text{crit}}}, \quad \rho = 1 \text{ vacuous.}$$

LEDGER

- GO-3. The certificate's vacuity threshold predicts where single-stage retrieval dies. [demonstrated]. `geometric-observation/claims/LEDGER.md:65` at 9f3829f.
- GO-8. On two independent source families (binary Markov; Gaussian AR(1)), a fixed stored record's operational reset threshold rises with the age of the retained side information exactly as the staleness-work complement prices it: same record, same bins, same decoder — the decodable bin rate climbs $0.10 \rightarrow 0.55$ bits/symbol across ages 0–64 of a $p = 0.05$ Markov chain, tracking $R_c - 1 + h_2(\hat{d} * q_t)$ within one grid step at every age, and a fixed bin rate flips from 1% error (age 0) to 100% (age 32). [replicated]. `geometric-observation/claims/LEDGER.md:71` at 9f3829f.

FIRST STATED

Chapter 9 section 9.3 and chapter 10 section 10.3 of *Data Mining as Observation*, with the isotropic Gaussian control in `turboquant-pro/turboquant_pro/anatomy.py:98-170`.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 10 section 10.3	hierarchical typing, tails 0.95 and 0.85, prescriptions, two designs that died, correlation above 0.8 on an isotropic Gaussian	<code>turboquant-pro\turboquant_pro\anatomy.py:98-170</code>
chapter 13 section 13.5	GO-8, 0.10 to 0.55 across ages 0 to 64, flip probability 0.05, 1 percent to 100 percent at age 32, Gaussian pass 5 of 5, the control-statistic caveat	<code>geometric-observation\claims\LEDGER.md</code> row GO-8; <code>geometric-observation\experiments\GO-landauer-gaussian-secondsettings-NOTES.md</code>

FAILURES AND CORRECTIONS

none

CONDITIONS

- The normal distribution, in many dimensions the one with density falling with the Mahalanobis distance. An isotropic Gaussian has every direction the same, so every reader reads the same variance and no code can flip, and whitening turns any Gaussian into an isotropic one.
- The isotropic Gaussian is the control on which hub typing correlated above 0.8 with nothing, and the Gaussian pass of GO-8 was five of five with the control-statistic caveat.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/Isotropy.lean`, theorems `isotropic_reads_same`, `isotropic_no_flip`, `anisotropic_readers_differ`, `flip_iff_anisotropic`, at `observation-data-mining` af776fd.

`lean/DataMiningAsObservation/Mahalanobis.lean`, theorems `dm2_nonneg`, `dm2_mean`, `dm2_scale`, `dm2_eq_whitened`, `dm2_antitone_in_variance`, at `observation-data-mining` af776fd.

USED IN

Data Mining as Observation chapters 9, 10, 11, 13.

RELATED

[isotropic](#) (p. 179); [Mahalanobis distance](#) (p. 210); [null model](#) (p. 234); [whitening](#) (p. 368); [standard error](#) (p. 337).

geodesic distance

concept. Entry id geodesic-distance.

DEFINITION

The length of the shortest path along edges between two nodes of a graph. Chapter 0 section 0.10.

EQUATION

Book equation 3.2.

$$R(i, j) = \sum_{k \geq 2} \frac{1}{\lambda_k} \left(\frac{u_k(i)}{\sqrt{d_i}} - \frac{u_k(j)}{\sqrt{d_j}} \right)^2 \rightarrow \frac{1}{d_i} + \frac{1}{d_j} \quad (n \rightarrow \infty, \dim \geq 3).$$

Book equation 3.3.

$$X_i = r_i \theta_i, \quad r_i \rightarrow \frac{1}{\sqrt{d_i}}, \quad \theta_i = \frac{X_i}{\|X_i\|} \in S^{m-1}.$$

Book equation 0.20.

$$\Psi_i = \left(\frac{u_k(i)}{\sqrt{\lambda_k d_i}} \right)_{k \geq 2}, \quad \|\Psi_i - \Psi_j\|^2 = R(i, j) = \frac{C(i, j)}{\text{vol}(G)}.$$

LEDGER

- GO-3. The certificate's vacuity threshold predicts where single-stage retrieval dies. [demonstrated]. [geometric-observation/claims/LEDGER.md:65 at 9f3829f](#).
- NEG-1. Fixed-scale, uniform-in-m bi-Lipschitz for the commute filter. [refuted]. [geometric-observation/claims/LEDGER.md:94 at 9f3829f](#).

FIRST STATED

Chapter 0 section 0.10 and chapter 3 section 3.3 of *Data Mining as Observation*, with the geodesic-rank reader of Volume 14 chapter 9, [geometric-observation/chapters/ch09_legibility.md:10-41](#).

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 3 section 3.2	condition (A2), tangential displacement	turboquant-pro\docs\KV_KEYS_FINDING.md:61-86; geometric-observation\chapters\ch09_legibility.md:42-54
chapter 3 section 3.3	resistance converges to 1 over d_i plus 1 over d_j, simplex argument, radius converges to 1 over root d_i	the-angular-observer\theorem.md:110-146
chapter 3 section 3.3	radius against degree correlation 0.92 to 0.99	the-angular-observer\README.md:135-139

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 3 section 3.3	angle 0.92 flat in m, magnitude 0.03, full 0.75 decaying	wolfram-observer-bridge\theorem.md:1-20; the-angular-observer\theorem.md:187-224
chapter 3 section 3.3	substrate table	the-angular-observer\README.md:99-109
chapter 3 section 3.3	row normalization is the projection onto the read subspace of the geodesic-rank consumer	geometric-observation\chapters\ch09_legibility.md:31-41; geometric-observation\chapters\ch03_historical_precursors.md:110
chapter 3 section 3.3	proven versus conjectured ledger, d at least 3, d equals 2 borderline, fixed-m conditional theorem, heat filter uniform, commute metric form false, Green-kernel rank limit for d at most 3, obstruction at d equals 4	the-angular-observer\theorem.md:84-107, 147-224; the-angular-observer\README.md:150-185
chapter 3 section 3.3	the self-refuted v0.8 claim, NEG-1	the-angular-observer\README.md:170-175; geometric-observation\chapters\ch16_honest_negatives.md NEG-1
chapter 3 section 3.3	hyperbolic rejected, curvature negative 0.98 to negative 0.14, trend 1.09 minus 0.157 d across 20 manifolds and 5 families	the-angular-observer\README.md:111-147, 183-185
chapter 9 section 9.1	the geodesic-rank reader discards the radius, row normalization is the angular projection	geometric-observation\chapters\ch09_legibility.md:10-41; the-angular-observer\theorem.md:110-146
chapter 9 section 9.2	the battery, three new templates, frozen ratios, code hash, bars at least 10 of 12 and the dimension ordering, eccentricity scope	the-angular-observer\PREREG_RECOGNIZER_BATTERY.md:1-80
chapter 9 section 9.2	12 of 12, dimensions 1.81, 2.80, 1.74, angular Spearman ranges, eccentricity spreads, verdict confirmed, GO-P-2026-041	the-angular-observer\experiments\manifold-recovery\battery_result.json
chapter 9 section 9.2	causal set 0.57 with dimension 2.9 to 10.7, small world 0.45, control 0.01, rung 4 decay 0.747 to 0.679 below 0.75	the-angular-observer\README.md:99-109; wolfram-observer-bridge\rung4_scaledm_result.json; wolfram-observer-bridge\review-2.txt

chapter 11 section 11.1	condition (A2), cosine satisfies it, post-rotary keys do not, the cone below cell size	turboquant- pro\docs\KV_KEYS_FINDING.md:61-86; the-angular-observer\README.md:26- 31
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FAILURES AND CORRECTIONS

- NEG-1, [refuted]. Fixed-scale, uniform-in-m bi-Lipschitz for the commute filter.

CONDITIONS

- The length of the shortest path along edges between two nodes. It is zero from a node to itself, symmetric, one between adjacent nodes, and on a connected graph obeys the triangle inequality.
- The geodesic-rank reader reads only the ordering of these distances, so a strictly increasing transform of them leaves its nearest neighbour unchanged. That the angle of the spectral embedding carries the ordering and the radius carries degree is the measured claim of chapter 3, with the refuted commute-filter row beside it.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/GeodesicDistance.lean`, theorems `dist_self`, `dist_comm`, `dist_triangle`, `dist_adj`, `geodesic_rank_invariant`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 3, 9, 10.

RELATED

[rank-faithful](#) (p. 277); [Laplacian](#) (p. 199); [recognizer](#) (p. 288); [bi-Lipschitz](#) (p. 37).

graph

concept. Entry id graph.

DEFINITION

A set of nodes and a set of edges joining pairs of them. A neighbourhood graph joins each row of a dataset to its nearest rows. Chapter 0 section 0.10.

EQUATION

Book equation 0.19.

$$L = I - D^{-1/2} A D^{-1/2}, \quad D = \text{diag}(d_1, \dots, d_n), \quad L u_k = \lambda_k u_k, \quad 0 = \lambda_1 \leq \lambda_2 \leq \dots.$$

Book equation 0.20.

$$\Psi_i = \left(\frac{u_k(i)}{\sqrt{\lambda_k d_i}} \right)_{k \geq 2}, \quad \|\Psi_i - \Psi_j\|^2 = R(i, j) = \frac{C(i, j)}{\text{vol}(G)}.$$

LEDGER

- GO-3. The certificate's vacuity threshold predicts where single-stage retrieval dies. [demonstrated]. `geometric-observation/claims/LEDGER.md:65` at 9f3829f.
- NEG-1. Fixed-scale, uniform-in-m bi-Lipschitz for the commute filter. [refuted]. `geometric-observation/claims/LEDGER.md:94` at 9f3829f.

FIRST STATED

Chapter 0 section 0.10 of *Data Mining as Observation*, with the neighbourhood graphs of the-angular-observer and the recognizer battery.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 3 section 3.2	condition (A2), tangential displacement	turboquant- pro\docs\KV_KEYS_FINDING.md:61-86; geometric- observation\chapters\ch09_legibility.md:42-54
chapter 3 section 3.3	resistance converges to 1 over d_i plus 1 over d_j , simplex argument, radius converges to 1 over root d_i	the-angular- observer\theorem.md:110-146
chapter 3 section 3.3	radius against degree correlation 0.92 to 0.99	the-angular- observer\README.md:135-139
chapter 3 section 3.3	angle 0.92 flat in m, magnitude 0.03, full 0.75 decaying	wolfram-observer- bridge\theorem.md:1-20; the-angular- observer\theorem.md:187-224
chapter 3 section 3.3	substrate table	the-angular-observer\README.md:99-109
chapter 3 section 3.3	row normalization is the projection onto the read subspace of the geodesic-rank consumer	geometric- observation\chapters\ch09_legibility.md:31-41; geometric- observation\chapters\ch03_historical_precursors.md:110
chapter 3 section 3.3	proven versus conjectured ledger, d at least 3, d equals 2 borderline, fixed-m conditional theorem, heat filter uniform, commute metric form false, Green-kernel rank limit for d at most 3, obstruction at d equals 4	the-angular- observer\theorem.md:84-107, 147-224; the-angular- observer\README.md:150-185
chapter 3 section 3.3	the self-refuted v0.8 claim, NEG-1	the-angular- observer\README.md:170-175; geometric- observation\chapters\ch16_honest_negatives.md NEG-1
chapter 3 section 3.3	hyperbolic rejected, curvature negative 0.98 to negative 0.14, trend 1.09 minus 0.157 d across 20 manifolds and 5 families	the-angular- observer\README.md:111-147, 183-185
chapter 9 section 9.1	the geodesic-rank reader discards the radius, row normalization is the angular projection	geometric- observation\chapters\ch09_legibility.md:10-41; the-angular- observer\theorem.md:110-146

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 9 section 9.2	the battery, three new templates, frozen ratios, code hash, bars at least 10 of 12 and the dimension ordering, eccentricity scope	the-angular-observer\PREREG_RECOGNIZER_BATTERY.md:1-80
chapter 9 section 9.2	12 of 12, dimensions 1.81, 2.80, 1.74, angular Spearman ranges, eccentricity spreads, verdict confirmed, GO-P-2026-041	the-angular-observer\experiments\manifold-recovery\battery_result.json
chapter 9 section 9.2	causal set 0.57 with dimension 2.9 to 10.7, small world 0.45, control 0.01, rung 4 decay 0.747 to 0.679 below 0.75	the-angular-observer\README.md:99-109; wolfram-observer-bridge\rung4_scaledm_result.json; wolfram-observer-bridge\review-2.txt
chapter 11 section 11.1	condition (A2), cosine satisfies it, post-rotary keys do not, the cone below cell size	turboquant-pro\docs\KV_KEYS_FINDING.md:61-86; the-angular-observer\README.md:26-31

FAILURES AND CORRECTIONS

- NEG-1, [refuted]. Fixed-scale, uniform-in-m bi-Lipschitz for the commute filter.

CONDITIONS

- A set of nodes and a set of edges joining pairs of them. A neighbourhood graph joins each row to its k nearest rows, so every node has exactly k out-neighbours. The mutual form keeps an edge only when each node is among the other's neighbours, is symmetric, is a subgraph of the directed form, and gives no node more than k mutual neighbours.
- The Laplacian of the graph carries the dimension in its eigenvalue count and the geodesic ranking in its low eigenvectors' angles, which is what chapters 3 and 9 read.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

lean/DataMiningAsObservation/Graph.lean, theorems `outDegree`, `mutual_symm`, `mutual_sub`, `mutualNbrs_card_le`, `mem_mutualNbrs`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 3, 9, 10, 11.

RELATED

[degree](#) (p. 97); [Laplacian](#) (p. 199); [geodesic distance](#) (p. 154); [spectral embedding](#) (p. 332); [manifold](#) (p. 210).

H

harness

concept. Entry id harness.

DEFINITION

The code that evaluates a model. It is itself a consumer with a read subspace, and chapter 8 shows how it reads the test fold when allowed to.

EQUATION

Book equation 8.1.

$$O_{\text{harness}} = (C_{\text{score}}, G_{\text{metric}}, B = \text{folds} \times \text{samples} \times \text{seeds}).$$

Book equation 8.2.

$$\widehat{\text{Var}}_{\text{NB}} = \left(\frac{1}{J} + \frac{n_{\text{test}}}{n_{\text{train}}} \right) \hat{\sigma}^2,$$

$$J = 1000, \frac{n_{\text{test}}}{n_{\text{train}}} = \frac{1}{4} \Rightarrow 1 + 250 = 251, \sqrt{251} = 15.84.$$

Book equation 8.3.

$$\Pr[\text{at least one of } m \text{ null tests passes}] = 1 - (1 - \alpha)^m, \quad \alpha_{\text{Bonferroni}} = \frac{\alpha}{m}.$$

LEDGER

- OT-11. Feedback-free staleness: streaming-retrieval damage tracks measured drift; derived-cadence re-allocation removes it. [void]. [geometric-observation/claims/LEDGER.md:48 at 9f3829f](#).
- NEG-2. Reconstruction cosine as a proxy for key quality. [refuted]. [geometric-observation/claims/LEDGER.md:95 at 9f3829f](#).
- NEG-4. Lightweight online (Lloyd) key calibration beats the calibration-free default on softmax-KL. [refuted]. [geometric-observation/claims/LEDGER.md:97 at 9f3829f](#).

FIRST STATED

Chapter 8 section 8.1 of *Data Mining as Observation*, with the program's own harness failures in [openvector-bench/results/QUERY_COUPLING_ARTIFACT.md:1-20](#) and [constraint-gap/review/FINDINGS.md:1-35](#).

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 1 section 1.4	cosine 0.995 and perplexity of order ten thousand, the recalibration negative	geometric-observation\chapters\ch02_failure_of_observer_free-60; geometric-observation\chapters\ch16_honest_negatives.md NEG-2 and NEG-4; turboquant-pro\docs\KV_KEYS_FINDING.md:1-49
chapter 2 section 2.5	cosine 0.995 and the softmax reader	turboquant-pro\docs\KV_KEYS_FINDING.md:1-49

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 3 section 3.2	condition (A2), tangential displacement	turboquant-pro\docs\KV_KEYS_FINDING.md:61-86; geometric-observation\chapters\ch09_legibility.md:42-54
chapter 6 section 6.4	the uncorrected t stored as sigma, 251 and 15.84, nine wins to three, seven, eight, 3 wins 17 ties 11 losses on 31 datasets, the three-way inconsistency	constraint-gap\review\FINDINGS.md:1-35; constraint-gap\README.md:57-76
chapter 8 section 8.1	G1 339.9 vs 71.9 same corpus, real about 61, rounds 1 to 4 at 300 to 420, three families falsified	openvector-bench\results\QUERY_COUPLING_ARTIFACT.md:1-20
chapter 8 section 8.2	cosine 0.995, perplexity near 1e4	turboquant-pro\docs\KV_KEYS_FINDING.md:1-49; geometric-observation\chapters\ch16_honest_negatives.md NEG-2
chapter 8 section 8.4	real 1.038 [0.958, 1.118], rotated 1.019 [0.903, 1.353], smooth 2.804 withdrawn, second attempt 0.184	constraint-gap-measurements\notes\negative_control.md:1-50
chapter 8 section 8.4	gate meta-rule, relative contrast discriminates nothing	openvector-bench\openvector_bench\score_rc1.py:1-14; turboquant-pro\docs\RESEARCH_ROADMAP.md:133-160
chapter 8 section 8.5	variance inflation 251, SE inflation 15.84, J_eff 3.98, about 62 needed, 3 of 9 at full, 7 at half, 8 at a third	constraint-gap\review\FINDINGS.md:1-35
chapter 11 section 11.1	condition (A2), cosine satisfies it, post-rotary keys do not, the cone below cell size	turboquant-pro\docs\KV_KEYS_FINDING.md:61-86; the-angular-observer\README.md:26-31
chapter 11 section 11.2	fp16 12.24, values-only 13.12, PolarQuant K4 10643 and 0.095, per-channel uniform K4 14.91 and 0.062, per-channel NUQ K3 15.77 and 0.148, 2.4x and 670x, pre-rotary near 22000, 2 key heads serve 12 query heads	turboquant-pro\docs\KV_KEYS_FINDING.md:1-49

FAILURES AND CORRECTIONS

- NEG-2, [refuted]. Reconstruction cosine as a proxy for key quality.
- NEG-4, [refuted]. Lightweight online (Lloyd) key calibration beats the calibration-free default on softmax-KL.
- theory-radar/paper/REVISION_PLAN.md:39-45 at 37c4e6c. **Fix:** - Drop " σ significance" everywhere - Report: mean $\Delta F1$, std, 95% CI, and corrected resampled t-statistic - Table columns: "Test F1", "GB F1", " $\Delta F1$ ", "95% CI", "Direction" - Text: "the formula outperforms GB by $\Delta F1=0.031$, 95% CI [0.028, 0.034]" - Add Nadeau-Bengio corrected t-test (accounts for CV fold correlation) - Keep effect size as supplementary but don't call it " σ "

CONDITIONS

- The harness is the code that evaluates a model, and it is a consumer with its own read subspace, its output metric, and a budget of folds, samples, and seeds. What it reads decides what the reported score measures.
- A harness that is allowed to see the test fold, through feature selection before the split, through repeated selection on one split, or through a metric that discriminates nothing, reports a score of itself.
- Repeated cross-validation folds are not independent, so the variance of a fold mean is understated unless corrected, and the program's own retracted comparison is the case.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/Bonferroni.lean`, `theorems family_error_le`, `bonferroni`, `one_look`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 1, 2, 4, 6, 7, 8, 10, 12, 14.

RELATED

[leakage](#) (p. 203); [observer](#) (p. 236); [preregistration](#) (p. 258); [ledger class](#) (p. 204).

hash

instrument. Entry id hash.

DEFINITION

A short fixed-length fingerprint of a file, computed so that any change to the file changes the fingerprint. Chapter 0 section 0.12.

EQUATION

Book equation 8.3.

$$\Pr[\text{at least one of } m \text{ null tests passes}] = 1 - (1 - \alpha)^m, \quad \alpha_{\text{Bonferroni}} = \frac{\alpha}{m}.$$

LEDGER

- OT-7. The damage form, trace pairing, rank, and loading covariance are $GL(d)$ -invariant under $P' = A^{-T}PA^{-1}$, while spectrum, effective rank, principal angles, and water-filling are $O(d)$ -only — consumer-weighted damage is a geometric scalar, and P3's cliff cannot be bought down by reparameterization. [demonstrated]. `geometric-observation/claims/LEDGER.md:30` at `9f3829f`.

FIRST STATED

Chapter 0 section 0.12 of *Data Mining as Observation*, with the program's seal ledger, `observation-theory-campaigns/experiments/SEALS.md:1-10`, where every seal records a SHA-256.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

CONDITIONS

- A short fixed-length fingerprint of a file, computed so that any change to the file changes the fingerprint. A changed hash proves a changed file, an unchanged file has an unchanged hash, and a digest with fewer values than there are files sends two distinct files to the same digest, so an equal hash is evidence and not proof.
- The seal ledger records the hash beside the commit, and the recognizer's templates and the probe's records

carry theirs, so that a reader with the repository can check what was fixed before the measurement. Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/Seal.lean`, theorems `changed_of_hash_ne`, `hash_eq_of_eq`, `exists_collision`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14.

RELATED

[sealed](#) (p. 316); [preregistration](#) (p. 258); [ledger class](#) (p. 204); [certificate](#) (p. 55).

head

concept. Entry id head.

DEFINITION

One attention operation. A model has many, each with its own queries, keys, and values. Chapter 0 section 0.11.

EQUATION

Book equation 0.23.

$$\text{softmax}(z)_i = \frac{e^{z_i}}{\sum_j e^{z_j}}, \quad \text{output} = \sum_i \text{softmax}\left(\frac{q \cdot k_i}{\sqrt{d}}\right)_i v_i.$$

Book equation 11.1.

$$\cos(k, \hat{k}) = 0.995 \quad \text{while} \quad \text{PPL} : 12.24 \rightarrow 10643.$$

LEDGER

- G0-2/G0-12/G0-13 operational (KV serving, 077). Consumer-relative access width measured on a production serving stack (Qwen2.5-7B KV-cache eviction, matched budget): task quality tracks measured predictive uncertainty u about the consumer's future reads, not nominal scorer width — the 32-query snapshot beats the 1024-query scorer $+0.4375 \pm 0.070$ at 5% keep (bar 0.30) and survives 97% eviction with zero drop, while wide-window scoring is statistically indistinguishable from random eviction at extreme budgets; the recency-hoarding starvation signature replicated across three disjoint prompt sets (oracle-miss gap 0.370 vs bar 0.25). The novel equal-uncertainty analytic-equality control (degradation-titrated, constructible by design) REFUTED its own equality prediction on the pre-registered branch: with calibration health $4 \times -130 \times$ inside gates, equal scalar u did NOT give equal quality (V4 $+0.078$ vs 0.0625 tolerance; the pre-registered $\rho=0.03$ contrast firmed to $+0.359 \pm 0.068$, 5.3 SE) — equal scalar uncertainty is insufficient, error structure matters, consistent with G0-13 Theorem 1's own $r \geq 2$ scoping of equal-q universality (a scalar-context privilege). Successor arc: 056 honest miss $\rightarrow$ 075 ID burned on a disclosed design failure $\rightarrow$ 077 sealed and split-verdict. [demonstrated]. `geometric-observation/claims/LEDGER.md:81` at 9f3829f.
- NEG-2. Reconstruction cosine as a proxy for key quality. [refuted]. `geometric-observation/claims/LEDGER.md:95` at 9f3829f.

FIRST STATED

Chapter 0 section 0.11 of *Data Mining as Observation*, with the program's head-level measurements in `turboquant-pro/docs/KV_KEYS_FINDING.md:1-49` and the planted probe of chapter 6.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 1 section 1.4	cosine 0.995 and perplexity of order ten thousand, the recalibration negative	geometric-observation\chapters\ch02_failure_of_observer_free_60; geometric-observation\chapters\ch16_honest_negatives.md NEG-2 and NEG-4; turboquant-pro\docs\KV_KEYS_FINDING.md:1-49
chapter 2 section 2.5	cosine 0.995 and the softmax reader	turboquant-pro\docs\KV_KEYS_FINDING.md:1-49
chapter 3 section 3.2	condition (A2), tangential displacement	turboquant-pro\docs\KV_KEYS_FINDING.md:61-86; geometric-observation\chapters\ch09_legibility.md:42-54
chapter 8 section 8.2	cosine 0.995, perplexity near $1e4$	turboquant-pro\docs\KV_KEYS_FINDING.md:1-49; geometric-observation\chapters\ch16_honest_negatives.md NEG-2
chapter 11 section 11.1	condition (A2), cosine satisfies it, post-rotary keys do not, the cone below cell size	turboquant-pro\docs\KV_KEYS_FINDING.md:61-86; the-angular-observer\README.md:26-31
chapter 11 section 11.2	fp16 12.24, values-only 13.12, PolarQuant K4 10643 and 0.095, per-channel uniform K4 14.91 and 0.062, per-channel NUQ K3 15.77 and 0.148, 2.4x and 670x, pre-rotary near 22000, 2 key heads serve 12 query heads	turboquant-pro\docs\KV_KEYS_FINDING.md:1-49
chapter 11 section 11.2	the calibration-time probe and the streaming monitor	turboquant-pro\turboquant_pro\A2_probe.py:315-438; turboquant-pro\tests\test_A2_probe.py

FAILURES AND CORRECTIONS

- NEG-2, [refuted]. Reconstruction cosine as a proxy for key quality.

CONDITIONS

- One attention operation, with its own queries, keys, and values. A head's output lies between the smallest and largest value, it reads the keys only through their scores against its query, and a key change the query does not read leaves its output unchanged however large the change.
- Each head has its own read subspace, a few query-weighted directions of each key, so a compression that serves one head can fail another, and the serving-stack measurement of chapter 13 is per head.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

lean/DataMiningAsObservation/Attention.lean, theorems `output_le_max`, `min_le_output`, `output_congr`, `output_nuisance`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 1, 2, 4, 6, 7, 8, 11, 12, 13.

RELATED

[attention](#) (p. 23); [KV cache](#) (p. 190); [read subspace](#) (p. 285); [softmax](#) (p. 327).

Hessian

concept. Entry id hessian.

DEFINITION

The matrix of second derivatives of a score, whose off-diagonal entries are the interactions. Naive Bayes has a diagonal one. Chapter 6.

EQUATION

Book equation 6.1.

$$s(x) = w \cdot x + b, \quad P_C = \mathbb{E}[\sigma'(s)^2] w w^\top, \quad \text{rank } P_C = 1.$$

Book equation 0.9.

$$P_C = \mathbb{E}[g g^\top], \quad g = \nabla C(x).$$

LEDGER

- GO-B-optim-D4 (034 · D4). Optimization — gradient compression, curvature (Hessian) read operator, on a REAL model (logistic regression); optional stretch [predicted]. `geometric-observation/claims/LEDGER.md:122` at 9f3829f.

FIRST STATED

Chapter 6 section 6.1 and chapter 7 section 7.2 of *Data Mining as Observation*, with the exact Hessian of a real logistic model in `geometric-observation/claims/LEDGER.md` row GO-B-optim-D4.

MEASUREMENTS

Where the book states it	Numbers, as the book’s sources table records them	Source
chapter 7 section 7.2	second-order leaf values, logistic curvature	ESL 2e 10.9 to 10.13; XGBoost introduction to boosted trees
chapter 7 section 7.2	real logistic model, exact Hessian, anti 300 of 300, flip 82 of 300, coupling diagnosis, bound not refutation	<code>geometric-observation/claims/LEDGER.md</code> row GO-B-optim-D4

FAILURES AND CORRECTIONS

none

CONDITIONS

- The matrix of second derivatives of a score, whose off-diagonal entries are the interactions. The second difference recovers a quadratic’s curvature exactly, and a score that is additive across features, naive Bayes, has a diagonal Hessian because one feature’s contribution is the same whatever the others are.
- The read operator is not the Hessian. Two features’ sensitivities can rise and fall together across rows even when neither changes the other’s effect, so a diagonal Hessian does not give a diagonal read operator.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

lean/DataMiningAsObservation/Curvature.lean, theorems second_quad, second_affine, linear_error, gradient_change, at observation-data-mining af776fd.

lean/DataMiningAsObservation/NaiveBayes.lean, theorems log_likelihood_sum, log_odds_sum, contribution_independent, at observation-data-mining af776fd.

USED IN

Data Mining as Observation chapters 6, 7, 14.

RELATED

[curvature](#) (p. 91); [naive Bayes](#) (p. 226); [boosting](#) (p. 44); [read operator](#) (p. 282); [Jacobian](#) (p. 183).

hierarchical clustering

instrument. Entry id hierarchical-clustering.

DEFINITION

A clustering that merges the closest pair of clusters at each step, under a linkage that reads the nearest pair, the farthest, or the average, so that merge heights never decrease. Chapter 9.

EQUATION

Book equation 9.1.

$$\text{SSE} = \sum_{k=1}^K \sum_{i \in \mathcal{C}_k} \|x_i - c_k\|^2, \quad \text{the } P_C = I \text{ distortion summed within clusters.}$$

Book equation 3.1.

$$d_O(u) = u^\top \Sigma u, \quad u = (\cos 15^\circ, \sin 15^\circ), \\ \Sigma_1 = \text{diag}(0.3, 1.7), \Sigma_2 = \text{diag}(1.7, 0.3), \quad d_O = 0.394 \text{ vs } 1.606.$$

LEDGER

none

FIRST STATED

Chapter 9 section 9.1 of *Data Mining as Observation*, after TSK chapter 7.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 9 section 9.1	the geodesic-rank reader discards the radius, row normalization is the angular projection	geometric-observation\chapters\ch09_legibility.md:10-41; the-angular-observer\theorem.md:110-146
chapter 10 section 10.1	the four detector families	TSK 2e chapter 9

FAILURES AND CORRECTIONS

none

CONDITIONS

- A clustering that merges the closest pair of clusters at each step under a linkage that reads the nearest pair, the farthest pair, or the average. When the merged pair was the closest at some height, no updated distance falls below it, so the merge heights never decrease, single linkage is never above complete, and average linkage lies between them.
- The dendrogram inherits the distance's quotient at every level, so the tree is a certificate about the chosen quotient and not about the data.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/Hierarchical.lean`, `theorems single_ge, complete_ge, average_ge, single_le_complete, average_between`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 9.

RELATED

[k-means](#) (p. 189); [DBSCAN](#) (p. 93); [Euclidean distance](#) (p. 124); [validity index](#) (p. 360); [quotient](#) (p. 271).

hub

concept. Entry id hub.

DEFINITION

A row retrieved as a nearest neighbour far more often than chance allows. The book shows hubness is almost entirely a property of the queries and the reader, not of the corpus. Chapters 3 and 10.

EQUATION

Book equation 10.3.

$$N_k(x \mid Q) = |\{q \in Q : x \in \text{top}_k(q)\}|, \quad \text{anti-hub} : N_k(x) = 0.$$

Book equation 0.17.

$$\Pr[X \geq c] = 1 - \sum_{i < c} e^{-\mu} \frac{\mu^i}{i!}, \quad c^* = \max\{c : n \Pr[X \geq c] \geq 1\}, \quad \mu = \frac{n_q k}{n}.$$

Book equation 10.5.

$$\begin{aligned} \text{central} &: \text{pct}_{\text{centrality}} \geq 0.95; \\ \text{dense} &: \neg \text{central} \wedge \text{pct}_{\text{density}} \geq 0.85; \quad \text{prescribe by } f_{\text{central}}, f_{\text{dense}} \text{ at } 0.75. \end{aligned}$$

LEDGER

- NEG-11. (GO-5, prospective ×4) The $\alpha=1$ density/hubness quotient decisively and density-specifically restores invariant fidelity in a non-spectral domain. [refuted]. `geometric-observation/claims/LEDGER.md:104` at 9f3829f.
- NEG-14. (GO-P-2026-037, prospective) `a2_probe.median_unit_displacement` is a single-statistic predictor of the flip regime ($\text{unit_disp} \geq 1.0 \Rightarrow \text{generic-polar-flip vs needs-blind-probe}$). [refuted]. `geometric-observation/claims/LEDGER.md:108` at 9f3829f.

FIRST STATED

Radovanović, Nanopoulos, and Ivanović, hubs in space, 2010, as chapter 3 section 3.5 of *Data Mining as Observation* reads it, with the program's finding in `openvector-bench` that hubness is a query property, `openvector-bench/results/QUERY_COUPLING_ARTIFACT.md:1-20`.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 3 section 3.5	78 vs 369, density correlation about 0.67, 8 percent vs 34 percent, centering vs mutual-proximity rescaling	turboquant-pro\docs\HUBNESS_PRIMER.md:38-59,140-160
chapter 3 section 3.5	Poisson null, hub excess, budget parameter	openvector-bench\openvector_bench\hubness.py:41-100
chapter 8 section 8.1	G1 339.9 vs 71.9 same corpus, real about 61, rounds 1 to 4 at 300 to 420, three families falsified	openvector-bench\results\QUERY_COUPLING_ARTIFACT.md:1-20
chapter 10 section 10.2	anti-hubs as where compressed indexes fail first, aggregate recall barely moves	turboquant-pro\docs\HUBNESS_PRIMER.md:86-131
chapter 10 section 10.3	abstain below 2.5 k	turboquant-pro\docs\HUBNESS_PRIMER.md:140-160
chapter 11 section 11.6	count of ten, hubs and anti-hubs, max 78 vs 369, density correlation about 0.67, 8 percent vs 34 percent, abstain below 2.5k, centering vs mutual-proximity rescaling	turboquant-pro\docs\HUBNESS_PRIMER.md:1-60,60-170
chapter 11 section 11.6	anti-hub recall, p05, hub-rank correlation, hub-set overlap, the build gate	turboquant-pro\docs\HUBNESS_PRIMER.md:86-131
chapter 11 section 11.6	the centroid-injection attack	turboquant-pro\docs\HUBNESS_PRIMER.md:168-197, citing arXiv 2604.05480

FAILURES AND CORRECTIONS

- NEG-11, [refuted]. (GO-5, prospective $\times 4$) The $\alpha=1$ density/hubness quotient decisively and density-specifically restores invariant fidelity in a non-spectral domain.
- NEG-14, [refuted]. (GO-P-2026-037, prospective) `a2_probe.median_unit_displacement` is a single-statistic predictor of the flip regime ($\text{unit_disp} \geq 1.0 \Rightarrow \text{generic-polar-flip vs needs-blind-probe}$).

CONDITIONS

- A row retrieved as a nearest neighbour far more often than chance allows, a count above the Poisson ceiling. Since the counts sum to the slots handed out, the number of rows with count above a ceiling is at most the slots over the ceiling plus one, so hubs are few by arithmetic, and a higher ceiling names fewer of them.
- Whether a row is a hub depends on the queries and the retrieval rule alone, which is the sense in which hubness is a query property, and the program's first hub counts fell by a factor of several when query coupling was removed.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/Hub.lean`, theorems `card_hubs_le`, `hubs_congr`, `hubs_anti`, `hubs_empty`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 1, 3, 5, 10, 11, 12, 13.

RELATED

[hubness](#) (p. 168); [Poisson ceiling](#) (p. 253); [anti-hub](#) (p. 17); [Robin Hood index](#) (p. 305); [null model](#) (p. 234).

hubness

result. Entry id hubness.

DEFINITION

The excess of a few rows in the nearest-neighbour lists of many queries over what the Poisson ceiling allows, measured by the busiest count and the Robin Hood index, a property of the queries. Chapter 3 section 3.5 and chapter 10.

EQUATION

Book equation 0.17.

$$\Pr[X \geq c] = 1 - \sum_{i < c} e^{-\mu} \frac{\mu^i}{i!}, \quad c^* = \max\{c : n \Pr[X \geq c] \geq 1\}, \quad \mu = \frac{n_q k}{n}.$$

Book equation 0.35.

$$\text{RH} = \sum_i \max\left(0, \frac{N_i}{\sum_j N_j} - \frac{1}{n}\right).$$

Book equation 10.3.

$$N_k(x \mid Q) = |\{q \in Q : x \in \text{top}_k(q)\}|, \quad \text{anti-hub} : N_k(x) = 0.$$

LEDGER

- GO-5. An $\alpha=1$ density/hubness quotient restores invariant fidelity in ≥ 1 non-spectral domain. [refuted]. [geometric-observation/claims/LEDGER.md:67 at 9f3829f](#).
- NEG-15 (Bell boundary). *Query-conditioned hubness supplies a mechanism for Bell-inequality violation without action at a distance*. Refuted as a mechanism; the settings-as-queries reframing survives only as vocabulary. [demonstrated]. [geometric-observation/claims/LEDGER.md:93 at 9f3829f](#).
- NEG-11. (GO-5, prospective $\times 4$) The $\alpha=1$ density/hubness quotient decisively and density-specifically restores invariant fidelity in a non-spectral domain. [refuted]. [geometric-observation/claims/LEDGER.md:104 at 9f3829f](#).

FIRST STATED

Radovanović, Nanopoulos, and Ivanović for the phenomenon. The observation program's reading, that hubness is almost entirely a property of the queries and the reader rather than the corpus, in `openvector-bench` and `turboquant-pro`, and in chapter 3 section 3.5 and chapter 10 of *Data Mining as Observation*.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 2 section 2.6	ball-growth heat a missing-mean artifact, eleven campaigns, 3 of 12 to 11 of 12 cells	<code>openvector-bench\README.md:160-170;</code> <code>openvector-bench\results\RC13_VERDICT.md:1-20</code>
chapter 3 section 3.5	Poisson null, hub excess, budget parameter	<code>openvector-bench\openvector_bench\hubness.py:41-100</code>

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 3 section 3.5	query mass best single feature at every K for all four responses, seven features add at most 20 percent, threshold 1.25 on three of four, zero of four, 1000 real queries, 1024 dimensions	openvector-bench\results\R13_STAGE0_RESULT.md:40-50; openvector-bench\results\R13_STAGE1_RESULT.md:1-40
chapter 3 section 3.6	Bell audit, max S 2.00000 across 72 configurations, d 3 to 128, correlation negative 0.036, post-selected 2.7308, controls 2.748, 2.386, 3.174, seed 20260817, sealed at 6e825d8	geometric-observation\articles\2026-08-03-hubness-does-not-weaken-bell.md:1-60; geometric-observation\claims\LEDGER.md row NEG-15; geometric-observation\results\G0-bell-geometry-audit.json
chapter 8 section 8.1	G1 339.9 vs 71.9 same corpus, real about 61, rounds 1 to 4 at 300 to 420, three families falsified	openvector-bench\results\QUERY_COUPLING_ARTIFACT.md:1-20
chapter 8 section 8.4	gate meta-rule, relative contrast discriminates nothing	openvector-bench\openvector_bench\score_rc1.py:1-14; turboquant-pro\docs\RESEARCH_ROADMAP.md:133-160
chapter 8 section 8.7	RC-7 five of ten, band 0.076 vs 0.19, gate missed by 0.007, 2.4 times, eight-block rule	openvector-bench\paper\profile\PROFILE_PAPER.md:576-598; openvector-bench\results\RC7_VERDICT.md
chapter 10 section 10.2	five kinds, balanced accuracy 0.611, 0.718, 0.684, the category table, per-category 0.57 to 0.82, the sweep, pigeonhole floor	openvector-bench\results\R13_STAGE1_RESULT.md:40-75
chapter 10 section 10.2	half 2 failed, at most 1.28 times against 2, withdrawal and its scope	openvector-bench\results\R13_STAGE1_RESULT.md:75-110
chapter 10 section 10.4	area map, boundary rule, hash, refuse not warn, intra and transit counts, area classes, abstention rule	turboquant-pro\docs\STRATA_RFC.md:24-98
chapter 10 section 10.4	second prediction inverted, transit 0.389 vs 0.291, centrality difference signs, share 0.473 vs 0.391, seven of thirteen backbone	turboquant-pro\docs\RESULTS_multilingual_strata.md:55-90; turboquant-pro\docs\STRATA_RFC.md:98-130
chapter 11 section 11.4	0.999 against own ranking vs 0.592 against fp32 truth, three truth layers, difficulty strata	openvector-bench\README.md:60-96

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 11 section 11.4	R80 table, 25x probe depth, 92 percent own cell, one third cross-article, 0.53 matches same-article share, relative contrast discriminates nothing	openvector-bench\results\R80_ANN.md:1-40; turboquant-pro\docs\RESEARCH_ROADMAP.md:133-160
chapter 11 section 11.6	query mass best single feature at every K for all four responses, seven features add at most 20 percent at 12 leaves, threshold 1.25 on three of four, zero of four, 1000 real queries, 1024 dimensions	openvector-bench\results\R13_STAGE0_RESULT.md:40-50; openvector-bench\results\R13_STAGE1_RESULT.md:1-40
chapter 12 section 12.2	recall 0.999 vs 0.592, the derived death point within 6 percent across fourteen corpora	openvector-bench\README.md:60-96; geometric-observation\claims\LEDGER.md row GO-3
chapter 13 section 13.1	one hundred billion vectors on a preemptible fleet, systems result not a tier, corpus rejected by the admission battery	openvector-bench\README.md:250-263
chapter 13 section 13.1	twelve-figure corpus about 128 TB, kilobyte manifest	openvector-bench\README.md:60-80

FAILURES AND CORRECTIONS

- GO-5, [refuted]. An $\alpha=1$ density/hubness quotient restores invariant fidelity in ≥ 1 non-spectral domain.
- NEG-11, [refuted]. (GO-5, prospective $\times 4$) The $\alpha=1$ density/hubness quotient decisively and density-specifically restores invariant fidelity in a non-spectral domain.

CONDITIONS

- A hub is a row retrieved as a nearest neighbour far more often than the Poisson null allows, and the ceiling of that null is derived from the query count, the neighbourhood size, and the corpus size.
- Hubness is measured against a query workload and belongs to it. The two corpus-side mechanisms that were registered to explain it were refuted under seal.
- Anti-hubs come in kinds, and one kind is manufactured by the query budget through the pigeonhole floor, so a count of never-retrieved rows changes with the number of slots per row.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

lean/DataMiningAsObservation/Hubness.lean, theorems `sum_count`, `sum_count_eq`, `count_congr`, `antiHub_iff`, at `observation-data-mining af776fd`.

lean/DataMiningAsObservation/Hub.lean, theorems `card_hubs_le`, `hubs_congr`, `hubs_anti`, `hubs_empty`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 1, 3, 5, 10, 11, 12, 13.

RELATED

[hub](#) (p. 166); [anti-hub](#) (p. 17); [Poisson ceiling](#) (p. 253); [quotient](#) (p. 271); [certificate](#) (p. 55); [rank certificate](#) (p. 275); [min-over-strata](#) (p. 217).

I

identity reader

concept. Entry id identity-reader.

DEFINITION

The consumer whose read operator is the identity, for which read distortion is mean squared error. Every consumer-blind metric assumes it. Chapter 0 section 0.5.

EQUATION

Book equation 0.10.

$$d_O = \text{tr}(P_C M_\delta) = \mathbb{E}[\delta^\top P_C \delta], \quad M_\delta = \mathbb{E}[\delta \delta^\top], \quad P_C = I \Rightarrow d_O = \text{tr} M_\delta.$$

Book equation 1.2.

$$d_O = \text{tr}(P_C M_\delta), \quad (d_O = \text{tr} M_\delta \text{ for every admissible } M_\delta) \Leftrightarrow P_C = I.$$

Book equation 4.1.

$$d_O = \text{tr}(P_C M_\delta) \quad \text{against} \quad \text{tr} M_\delta = d_O|_{P_C=I}.$$

LEDGER

- GO-2 (neg. half: not reconstruction). At matched bits, downstream preservation is not controlled by reconstruction error. [demonstrated]. [geometric-observation/claims/LEDGER.md:63 at 9f3829f](#).

FIRST STATED

Volume 14, chapter 2, [geometric-observation/chapters/ch02_failure_of_observer_free_measurement.md](#), DOI 10.5281/zenodo.21776291, where reconstruction error is the distortion of the reader that reads every direction equally.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 1 section 1.4	three virtues one vice, the identity slice	geometric-observation\chapters\ch02_failure_of_observer_free_40
chapter 1 section 1.4	cosine 0.995 and perplexity of order ten thousand, the recalibration negative	geometric-observation\chapters\ch02_failure_of_observer_free_60 ; geometric-observation\chapters\ch16_honest_negatives.md NEG-2 and NEG-4; turboquant-pro\docs\KV_KEYS_FINDING.md:1-49
chapter 1 section 1.4	the flip sealed in twelve domains and three physics, held in at least five domains and all three	geometric-observation\chapters\ch08_value.md:40-70 ; geometric-observation\claims\LEDGER.md ROWS GO-2, GO-B-AV163, D3, 038

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 4 section 4.3	GO-2 0.0934 vs 0.0938, 2.53 times, 12 of 12, anti 21 times; retrieval 0.0964, negative 4.70 and positive 4.65, recon 0.40	geometric-observation\claims\LEDGER.md rows GO-2 negative and positive halves
chapter 4 section 4.5	GloVe table, 73 percent at 64 components, 0.685 vs 0.862 and 0.866, 0.906 at matched bytes, 0.989 at 37 bytes, 768 to 256 keeps about 99 percent, the 95 percent rule	turboquant-pro\benchmarks\RESULTS_glove.md:1-40

FAILURES AND CORRECTIONS

none

CONDITIONS

- Read distortion equals the trace of the error's second moment for every admissible error exactly when the read operator is the identity. A particular error can make the two agree by accident.
- Explained variance and the rule of picking rank at 95 percent are the identity reader's criteria, and the book records them as the benchmark's rules rather than its own.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

lean/DataMiningAsObservation/ReadDistortion.lean, theorems `read_distortion`, `identity_reader`, `quad_one`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 1, 2, 3, 4, 8, 9, 10, 12, 13.

RELATED

[read operator](#) (p. 282); [read distortion](#) (p. 281); [the flip](#) (p. 138); [quotient](#) (p. 271).

importance

instrument. Entry `id importance`.

DEFINITION

A per-feature score of how much a classifier reads it. For a tree it counts splits, which is the right sensitivity for a reader that is flat within leaves. Permutation importance and partial dependence answer related questions. Chapter 7 and chapter 14.

EQUATION

Book equation 14.5.

$$FC_g = \Pr[y = \text{violation} \mid \hat{y} = \text{clear}, g], \quad \text{verdict} = \max_{g: n_g \geq n_{\min}} FC_g, \quad \text{ABSTAIN otherwise.}$$

Book equation 0.9.

$$P_C = \mathbb{E}[g g^\top], \quad g = \nabla C(x).$$

Book equation 6.1.

$$s(x) = w \cdot x + b, \quad P_C = \mathbb{E}[\sigma'(s)^2] w w^\top, \quad \text{rank } P_C = 1.$$

LEDGER

- GO-EC-3. A read operator recovered from a black-box consumer by query-only finite-difference probing, composed with the Kalman covariance as $\text{tr}(\hat{P}_C \Sigma)$, prospectively selects sensors that improve the held-out consumer at matched budgets with probe cost charged — capturing 94.6% of the known analytic optimum’s gain on the positive-control arm (gate $\geq 75\%$) and improving 16.3% over the best consumer-agnostic policy on non-analytic consumers (gate $\geq 8\%$), with trace-matched ordering carried by the composition at 86.9% over 61 pairs (gate $\geq 65\%$). [predicted]. `geometric-observation/claims/LEDGER.md:162 at 9f3829f`.

FIRST STATED

Breiman, random forests, 2001, as chapter 7 section 7.1 and chapter 14 section 14.5 of *Data Mining as Observation* read it, with the selection-consumer regime in `readscope/readscope/regimes.py:1-60`.

MEASUREMENTS

Where the book states it	Numbers, as the book’s sources table records them	Source
chapter 2 section 2.3	refusal regimes, selection consumers read order, recurrences compound	<code>readscope/readscope/regimes.py:1-60</code>
chapter 6 section 6.1	selection consumers have zero sensitivity almost everywhere	<code>readscope/readscope/regimes.py:1-60</code>
chapter 7 section 7.1	bagging variance, out-of-bag estimation, random forests, importances, AdaBoost	Hastie, Tibshirani, Friedman, ESL 2e chapters 15 and 10.1; TSK 2e 4.10
chapter 14 section 14.5	the contraction formula fairness minus the general component	<code>gtc-prototype/docs/SPECTRUM_FINDINGS.md:90-99</code>

FAILURES AND CORRECTIONS

none

CONDITIONS

- A per-feature score of how much a classifier reads it. For a gradient reader the averaged squared sensitivity is the read operator’s diagonal, and for a tree, whose finite difference is zero at every row that does not straddle a split, counting splits is the right sensitivity.
- Permutation importance averages the consumer over rows that may never occur together; and partial dependence reads it at inputs it was never trained on, so the three answer different questions and the report names which.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/Attribution.lean`, theorems `attr_sum_affine`, `attr_unread`, `sq_sensitivity_eq_readOp_diag`, at `observation-data-mining af776fd`.

`lean/DataMiningAsObservation/DecisionTree.lean`, theorems `stump_flat_left`, `stump_flat_right`, `split_reads_one_axis`, `gini_le_half`, `gini_eq_zero_iff`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 4, 6, 7, 14.

RELATED

[attribution](#) (p. 26); [decision tree](#) (p. 95); [sensitivity](#) (p. 319); [read operator](#) (p. 282); [ensemble](#) (p. 121).

imputation

instrument. Entry id imputation.

DEFINITION

Writing a value into a missing cell that the consumer will read. Mean imputation leaves the column mean unchanged and shrinks its variance. Chapter 2.

EQUATION

Book equation 2.1.

$$P_{C_2 \circ C_1}(x) = J_1(x)^\top P_{C_2}(C_1(x)) J_1(x), \quad \text{rank } P_{C_2 \circ C_1}(x) \leq \text{rank } P_{C_2}(C_1(x)) \quad \text{at each row } x.$$

Book equation 0.9.

$$P_C = \mathbb{E}[g g^\top], \quad g = \nabla C(x).$$

LEDGER

none

FIRST STATED

Chapter 2 section 2.3 of *Data Mining as Observation*, with the missing-data rule as a required preregistration field in observation-theory-campaigns/experiments/PREREG-TEMPLATE.md:47-52.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 2 section 2.3	MCAR, MAR, MNAR and the remedies, imputation before splitting	TSK 2e section 2.2; instructor working documents, not public [@bond2026course], ECE_514-01_FA26_session-outlines.md:64-72
chapter 2 section 2.3	the missing-data rule as a required preregistration field	observation-theory-campaigns\experiments\PREREG-TEMPLATE.md:47-52

FAILURES AND CORRECTIONS

none

CONDITIONS

- Writing a value into a missing cell that the consumer will read. Mean imputation leaves the column mean unchanged, the filled cells add nothing to the sum of squared deviations, and the variance shrinks because the same sum of squares is spread over more rows.
- An imputation is a claim about the mechanism that produced the gap, and it is fit inside the fold and stated in the preregistration, because an imputation fit on all rows leaks the test fold into the training fold.

Conditions are curated in entries.toml rather than read from a record.

MACHINE CHECKED

lean/DataMiningAsObservation/Imputation.lean, theorems mean_imputed, sum_sq_imputed, variance_imputed_le, at observation-data-mining af776fd.

USED IN

Data Mining as Observation chapters 1, 2, 8.

RELATED

[standardization](#) (p. 338); [read operator](#) (p. 282); [leakage](#) (p. 203); [preregistration](#) (p. 258).

intrinsic dimension

concept. Entry id intrinsic-dimension.

DEFINITION

The number of directions a dataset actually varies along, whatever the number of its coordinates. Chapter 0 section 0.10.

EQUATION

Book equation 0.31.

$$N(\lambda) = \#\{k : \lambda_k \leq \lambda\} \sim C_d \lambda^{d/2}.$$

Book equation 9.2.

$$N(\lambda) = \#\{k : \lambda_k \leq \lambda\} \sim C_d \lambda^{d/2} \quad \Rightarrow \quad d = 2 \frac{d \log N}{d \log \lambda}.$$

Book equation 0.7.

$$r_{\text{eff}} = \frac{(\sum_i \lambda_i)^2}{\sum_i \lambda_i^2}.$$

LEDGER

- G0-3. The certificate’s vacuity threshold predicts where single-stage retrieval dies. [demonstrated]. [geometric-observation/claims/LEDGER.md:65 at 9f3829f](#).

FIRST STATED

Weyl’s law, 1911, as chapter 0 section 0.15 states it, applied in the recognizer of Volume 14 chapter 11, [geometric-observation/chapters/ch11_the_recognizer.md:1-95](#), and chapter 9 section 9.2 of *Data Mining as Observation*.

MEASUREMENTS

Where the book states it	Numbers, as the book’s sources table records them	Source
chapter 3 section 3.3	resistance converges to 1 over d_i plus 1 over d_j, simplex argument, radius converges to 1 over root d_i	the-angular-observer\theorem.md:110-146
chapter 3 section 3.3	radius against degree correlation 0.92 to 0.99	the-angular-observer\README.md:135-139
chapter 3 section 3.3	angle 0.92 flat in m, magnitude 0.03, full 0.75 decaying	wolfram-observer-bridge\theorem.md:1-20; the-angular-observer\theorem.md:187-224
chapter 3 section 3.3	substrate table	the-angular-observer\README.md:99-109

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 3 section 3.3	proven versus conjectured ledger, d at least 3, d equals 2 borderline, fixed-m conditional theorem, heat filter uniform, commute metric form false, Green-kernel rank limit for d at most 3, obstruction at d equals 4	the-angular-observer\theorem.md:84-107,147-224; the-angular-observer\README.md:150-185
chapter 3 section 3.3	the self-refuted v0.8 claim, NEG-1	the-angular-observer\README.md:170-175; geometric-observation\chapters\ch16_honest_negatives.md NEG-1
chapter 3 section 3.3	hyperbolic rejected, curvature negative 0.98 to negative 0.14, trend 1.09 minus 0.157 d across 20 manifolds and 5 families	the-angular-observer\README.md:111-147,183-185
chapter 9 section 9.1	the geodesic-rank reader discards the radius, row normalization is the angular projection	geometric-observation\chapters\ch09_legibility.md:10-41; the-angular-observer\theorem.md:110-146
chapter 9 section 9.2	the recognizer's mechanism, low multiplets and angular distances, dimension before shape by Weyl's law, refusal, the growth gotcha	geometric-observation\chapters\ch11_the_recognizer.md:1-95
chapter 9 section 9.2	the battery, three new templates, frozen ratios, code hash, bars at least 10 of 12 and the dimension ordering, eccentricity scope	the-angular-observer\PREREG_RECOGNIZER_BATTERY.md:1-80
chapter 9 section 9.2	12 of 12, dimensions 1.81, 2.80, 1.74, angular Spearman ranges, eccentricity spreads, verdict confirmed, GO-P-2026-041	the-angular-observer\experiments\manifold-recovery\battery_result.json
chapter 9 section 9.2	causal set 0.57 with dimension 2.9 to 10.7, small world 0.45, control 0.01, rung 4 decay 0.747 to 0.679 below 0.75	the-angular-observer\README.md:99-109; wolfram-observer-bridge\rung4_scaledm_result.json; wolfram-observer-bridge\review-2.txt
chapter 9 section 9.4	explained ratios, effective rank 5.19 of 8, convergence with a second method, property of the representation not the space	gtc-prototype\docs\SPECTRUM_FINDINGS.md:10-18
chapter 11 section 11.1	condition (A2), cosine satisfies it, post-rotary keys do not, the cone below cell size	turboquant-pro\docs\KV_KEYS_FINDING.md:61-86; the-angular-observer\README.md:26-31

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 14 section 14.3	floor 0.08 to 0.12, balanced resample 3001 items, identity attack 0.237 with interval 0.20 to 0.28 on 289, sexual 0.204, threat 0.093 retracted, first pass 18 positives, 87.7 and 1.4 percent	gtc- prototype\docs\SPECTRUM_FINDINGS.md:20-64
chapter 14 section 14.3	contraction 0.779 to 0.863, 77 of 1600, 4.8 percent, 0.872 to 0.863, weight 2.69	gtc- prototype\docs\SPECTRUM_FINDINGS.md:66-99
chapter 14 section 14.5	the contraction formula fairness minus the general component	gtc- prototype\docs\SPECTRUM_FINDINGS.md:90-99
chapter 14 section 14.7	51 percent moderated at 80 and 95 percent precision on the balanced set	gtc- prototype\docs\SPECTRUM_FINDINGS.md:84-88

FAILURES AND CORRECTIONS

none

CONDITIONS

- The intrinsic dimension is the number of directions a dataset varies along, whatever its coordinate count. Under Weyl's law the count of eigenvalues below a level grows like the level to half the dimension, so the dimension is twice the slope of log count against log level between any two levels, whatever the constant.
- Dimension emerges before shape. The recognizer reports it with confidence before it names a manifold, and the law is an asymptotic statement about manifolds that a finite sample approximates.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/IntrinsicDimension.lean`, `theorems log_weyl`, `dimension_from_slope`, `weyl_double`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 3, 8, 9, 11.

RELATED

[recognizer](#) (p. 288); [effective rank](#) (p. 115); [distance concentration](#) (p. 107); [vacuity threshold](#) (p. 359).

inverted file

instrument. Entry `id` `inverted-file`.

DEFINITION

An index that partitions vectors into cells by k-means and answers a query by searching only the cells nearest to it. The number of cells searched is the probe count. Chapter 11.

EQUATION

Book equation 11.3.

$$\text{recall}@k = \frac{|\text{returned top-}k \cap \text{true top-}k|}{k},$$

true top- k computed from the uncompressed vectors.

Book equation 11.4.

$$\text{directions resolved}(k) = \begin{cases} 1 \text{ or } 2, & k < d \\ \text{rank } P_C, & k \geq d \end{cases} \quad \text{cost} = 2d \text{ consumer calls per operating point.}$$

LEDGER

- GO-3. The certificate's vacuity threshold predicts where single-stage retrieval dies. [demonstrated]. [geometric-observation/claims/LEDGER.md:65 at 9f3829f](#).
- NEG-14. (GO-P-2026-037, prospective) `a2_probe.median_unit_displacement` is a single-statistic predictor of the flip regime (`unit_disp ≥ 1.0 ⇒ generic-polar-flip vs needs-blind-probe`). [refuted]. [geometric-observation/claims/LEDGER.md:108 at 9f3829f](#).

FIRST STATED

Sivic and Zisserman's video Google and the IVF index of Jégou, Douze, and Schmid, as chapter 11 section 11.3 of *Data Mining as Observation* presents them, with the program's probe sweeps in `openvector-bench`.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 11 section 11.2	the calibration-time probe and the streaming monitor	<code>turboquant-pro\turboquant_pro\</code> <code>a2_probe.py:315-438</code> ; <code>turboquant-pro\tests\test_a2_probe.py</code>
chapter 11 section 11.7	GO-1 overlap 0.936 vs 0.059, flip 12 of 12, reconstruction 0.40	<code>geometric-observation\claims\LEDGER.md</code> row GO-1; <code>geometric-observation\chapters\ch10_the_blind_probe.md:34-52</code>
chapter 12 section 12.3	035 miss 0.773 vs 0.757, identifiability diagnosis, 036 blind probe r 32, 0.779 vs 0.771, 200 of 200, recon 0.220 vs 0.584, 039 virgin split 0.796 vs 0.780, margin 0.008 to 0.016, edges baseline	<code>geometric-observation\chapters\ch10_the_blind_probe.md:53-90</code> ; <code>geometric-observation\claims\LEDGER.md</code> rows GO-B-legal 035 and 036, GO-P-2026-039
chapter 12 section 12.3	041 non-oracle, frozen LSA TF-IDF to SVD 100 train-only, AUROC 0.975 vs 0.910, flip tied, magnitude overshoot, partial	<code>geometric-observation\chapters\ch10_the_blind_probe.md:100-118</code> ; <code>geometric-observation\claims\LEDGER.md</code> row GO-B-blind 041
chapter 14 section 14.8	quantum falsified, 320 probes, contextuality zero, DOI 10.5281/zenodo.20660110	<code>non-abelian-sqnd\README.md:5-22</code> ; <code>sqnd-probe\README.md:12-20</code> ; <code>sqnd-probe\CITATION.cff:16</code>

FAILURES AND CORRECTIONS

- NEG-14, [refuted]. (GO-P-2026-037, prospective) `a2_probe.median_unit_displacement` is a single-statistic predictor of the flip regime (`unit_disp ≥ 1.0 ⇒ generic-polar-flip vs needs-blind-probe`).

CONDITIONS

- An index that partitions vectors into cells by k-means and searches only the cells nearest the query. Probing more cells never loses a candidate, a neighbour is found exactly when its cell is probed, the scanned fraction is the probed cells' size over the corpus, and with equal cells it is the probe count over the cell count.
- The recall at one probe against eight in the chapter's table is a measurement, and the vacuity threshold of the certificate predicts where single-stage retrieval dies, which is the ledger's demonstrated row.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/InvertedFile.lean`, theorems `scanned_mono`, `found_iff`, `scannedFraction_mem_unit`, `equal_cells`, `scanned_univ`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 11, 13, 14.

RELATED

[recall at k](#) (p. 286); [product quantization](#) (p. 261); [vacuity threshold](#) (p. 359); [anti-hub](#) (p. 17).

isotropic

concept. Entry id `isotropic`.

DEFINITION

Of a covariance, having the same variance in every direction. Chapter 0.

EQUATION

Book equation 3.1.

$$d_O(u) = u^\top \Sigma u, \quad u = (\cos 15^\circ, \sin 15^\circ), \\ \Sigma_1 = \text{diag}(0.3, 1.7), \Sigma_2 = \text{diag}(1.7, 0.3), \quad d_O = 0.394 \text{ vs } 1.606.$$

Book equation 4.6.

$$\kappa = \text{tr}(\bar{P}_C \bar{\Sigma}_x) \in [0, 1], \quad \kappa \rightarrow 1 \text{ is the coupling null (no flip).}$$

LEDGER

- GO-2 (neg. half: not reconstruction). At matched bits, downstream preservation is not controlled by reconstruction error. [demonstrated]. `geometric-observation/claims/LEDGER.md:63` at `9f3829f`.

FIRST STATED

Chapter 0 section 0.3 and chapter 3 section 3.2 of *Data Mining as Observation*, with the isotropic-Gaussian control of chapter 10.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

CONDITIONS

- An isotropic covariance has the same variance in every direction, so every unit reader reads it the same and no flip is possible. An anisotropic one has different variances in different directions, and two unit readers can read it differently.
- The flip needs anisotropy. Two codes with the variances exchanged are read the same by every reader exactly when the two variances agree, which is the boundary the coupling null names from the other side.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/Isotropy.lean`, `theorems isotropic_reads_same`, `isotropic_no_flip`, `anisotropic_readers_differ`, `flip_iff_anisotropic`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 2, 3, 4, 7, 9, 10, 11, 12.

RELATED

[the flip](#) (p. 138); [coupling null](#) (p. 84); [covariance matrix](#) (p. 85); [alignment](#) (p. 11).

itemset

concept. Entry `id itemset`.

DEFINITION

A set of items that appear together in a transaction. A closed itemset has no superset with the same support and a maximal one has no frequent superset. Chapter 5.

EQUATION

Book equation 5.1.

$$s(X) = \frac{\sigma(X)}{N}, \quad c(X \rightarrow Y) = \frac{\sigma(X \cup Y)}{\sigma(X)} = \frac{s(X \cup Y)}{s(X)}.$$

Book equation 5.3.

$$X \subseteq Y \implies s(X) \geq s(Y).$$

LEDGER

none

FIRST STATED

Agrawal, Imieliński, and Swami, mining association rules, 1993, as TSK chapter 5 presents it and chapter 5 section 5.1 of *Data Mining as Observation* states it.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 5 section 5.1 to 5.3, 5.5	support, confidence, lift, the Apriori principle, FP-growth, closed and maximal itemsets, objective measures, Simpson's paradox, cross-support	TSK 2e chapter 5

FAILURES AND CORRECTIONS

none

CONDITIONS

- A set of items that appear together in a transaction. Its support lies in the unit interval, the empty itemset has support one, a superset has support at most that of a subset, and the support of a union is at most the smaller of the two supports.

- A closed itemset has no strict superset with the same support and a maximal frequent itemset has no frequent strict superset, and every maximal frequent itemset is closed, since a superset with the same support would be frequent too.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/ItemSet.lean`, theorems `supportFrac_mem_unit`, `supportFrac_empty`, `supportFrac_anti`, `supportFrac_union_le`, `closed_of_maximal`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 5.

RELATED

[support](#) (p. 343); [Apriori principle](#) (p. 22); [safe pruning](#) (p. 311); [confidence](#) (p. 70).

J

Jaccard

concept. Entry id jaccard.

DEFINITION

The size of the intersection of two sets over the size of their union, between zero and one, zero exactly when the sets are disjoint, which discards joint absence. Chapter 0 section 0.6 and chapter 3.

EQUATION

Book equation 0.1.

$$x \cdot y = \sum_{i=1}^d x_i y_i, \quad \|x\| = \sqrt{x \cdot x}, \quad \cos \theta = \frac{x \cdot y}{\|x\| \|y\|}.$$

Book equation 3.1.

$$d_O(u) = u^\top \Sigma u, \quad u = (\cos 15^\circ, \sin 15^\circ), \\ \Sigma_1 = \text{diag}(0.3, 1.7), \quad \Sigma_2 = \text{diag}(1.7, 0.3), \quad d_O = 0.394 \text{ vs } 1.606.$$

LEDGER

none

FIRST STATED

Jaccard, the distribution of the flora in the alpine zone, 1912, as chapter 0 section 0.6 and chapter 3 section 3.1 of *Data Mining as Observation* read it.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 3 section 3.2	traces 2.0, diagonals 0.3 and 1.7, consumers at 15 and 75 degrees, distortion 0.39 vs 1.61, 4.1 to one, computed not drawn	observation-theory/assets/make_flip_figure.py:4-24,107-119

FAILURES AND CORRECTIONS

none

CONDITIONS

- The size of the intersection of two sets over the size of their union, between zero and one, one for a nonempty set against itself, symmetric, and zero exactly when the sets are disjoint. A subset's coefficient is its share of the larger set.
- It reads nothing outside the union, so its distance lives on the quotient that discards joint absence, which is the right quotient for a consumer that reads presence and the wrong one for a consumer that reads counts.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

lean/DataMiningAsObservation/Jaccard.lean, theorems jaccard_nonneg, jaccard_le_one, jaccard_self, jaccard_comm, jaccard_eq_zero_iff, jaccard_subset, at observation-data-mining af776fd.

USED IN

Data Mining as Observation chapters 0, 3.

RELATED

[cosine](#) (p. 82); [quotient](#) (p. 271); [bag of words](#) (p. 32); [Euclidean distance](#) (p. 124); [itemset](#) (p. 180).

Jacobian

concept. Entry id jacobian.

DEFINITION

The matrix of partial derivatives of a vector-valued consumer, whose transpose times itself averages to the read operator. A discrete stage has none. Chapter 0 section 0.5 and chapter 12.

EQUATION

Book equation 0.9.

$$P_C = \mathbb{E}[g g^\top], \quad g = \nabla C(x).$$

Book equation 2.1.

$$P_{C_2 \circ C_1}(x) = J_1(x)^\top P_{C_2}(C_1(x)) J_1(x), \quad \text{rank } P_{C_2 \circ C_1}(x) \leq \text{rank } P_{C_2}(C_1(x)) \quad \text{at each row } x.$$

Book equation 0.11.

$$P_C(x) = J(x)^\top G(C(x)) J(x), \quad J(x) = \frac{\partial C}{\partial x}(x), \quad \bar{P}_{C,\mu} = \mathbb{E}_\mu[P_C(x)].$$

LEDGER

- OT-7. The damage form, trace pairing, rank, and loading covariance are GL(d)-invariant under $P' = A^{-\top} P A^{-1}$, while spectrum, effective rank, principal angles, and water-filling are O(d)-only — consumer-weighted damage is a geometric scalar, and P3's cliff cannot be bought down by reparameterization. [demonstrated].
geometric-observation/claims/LEDGER.md:30 at 9f3829f.

FIRST STATED

Chapter 0 section 0.5 and chapter 2 section 2.2 of *Data Mining as Observation*, with the pullback composition and the rank bound in geometric-observation/chapters/ch06_mathematical_preliminaries.md:10-27.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 2 section 2.2	read subspace small, operator local, pullback composition, rank cannot increase	geometric-observation\chapters\ch05_the_read_metric_and_the_c48; geometric-observation\chapters\ch06_mathematical_preliminaries27
chapter 12 section 12.2	pullback composition and the rank bound	geometric-observation\chapters\ch06_mathematical_preliminaries27; chapter 2 of this book

FAILURES AND CORRECTIONS

none

CONDITIONS

- The matrix of partial derivatives of a vector-valued consumer, whose transpose times itself averages to the read operator. Along a chain of stages the rank of the composed operator is at most the rank of either stage, and a stage that identifies two inputs identifies them for every stage after it.
- Chunking and indexing have no Jacobian, since a document goes to a set of pieces and a query to a candidate list, and what survives without a derivative is the statement about distinctions.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/Pipeline.lean`, theorems `quotient_inherited`, `quotient_inherited_chain`, `rank_comp_le_first`, `rank_comp_le_second`, at `observation-data-mining af776fd`.

`lean/DataMiningAsObservation/ReadOperator.lean`, theorems `rank_one_reads_one_direction`, `readOp_mulVec`, `quad_readOp`, `quad_readOp_nonneg`, `readOp_mulVec_eq_zero_iff`, `readOp_diag`, `readOp_offdiag`, `readOp_symm`, `readOp_neg`, `affine_const_along_nuisance`, `readOp_affine`, `readOp_sqLength_basis`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 2, 12.

RELATED

[read operator](#) (p. 282); [sensitivity](#) (p. 319); [Hessian](#) (p. 164); [retrieval-augmented pipeline](#) (p. 303).

K

Kendall correlation

concept. Entry id `kendall-correlation`.

DEFINITION

The fraction of item pairs two rankings order the same way, minus the fraction they order differently. Equation 0.15.

EQUATION

Book equation 0.15.

$$\tau = \frac{\#\{\text{concordant pairs}\} - \#\{\text{discordant pairs}\}}{n(n-1)/2}.$$

Book equation 11.2.

$$r = \frac{d_{\text{compressed}}}{d_{\text{exact}}}, \quad \kappa_{\text{strict}} = \frac{\max r}{\min r}, \quad \tau \geq 1 - 2\hat{\mu}(\kappa_{\text{strict}}), \quad \rho_S \geq 1 - 3\hat{\mu}(\kappa_{\text{strict}}),$$

$$\kappa_{97.5/2.5} = \frac{q_{97.5}(r)}{q_{2.5}(r)} \text{ gives the same two expressions as estimates, not floors.}$$

LEDGER

- GO-B-Llama. Trained frontier LLM (Llama-3.2-3B), softmax-attention consumer — blind probe on real post-RoPE keys [predicted]. `geometric-observation/claims/LEDGER.md:115` at 9f3829f.
- GO-B-Llama-rematch. Trained frontier LLM (Llama-3.2-3B), softmax-attention consumer — reconnected dissociation on real post-RoPE keys [predicted]. `geometric-observation/claims/LEDGER.md:118` at 9f3829f.

FIRST STATED

Kendall, a new measure of rank correlation, 1938, as chapter 0 section 0.9 states it, and the rank certificate's statistic in `readscope` and `openvector-bench`.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

CONDITIONS

- The fraction of item pairs two rankings order the same way minus the fraction they order differently. It lies between minus one and one, it is one when no pair disagrees, and since it depends only on the orderings a strictly increasing transform of either score leaves it unchanged.
- It is the rank-agreement statistic the rank certificate bounds from below, and a floor on it is a floor on the consumer's rankings, not on any distance.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/Kendall.lean`, theorems `not_both`, `card_add_le`, `tau_mem`, `concordant_comp`, `discordant_comp`, `tau_comp`, `discordant_self`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 1, 8, 11.

RELATED

[rank certificate](#) (p. 275); [rank-faithful](#) (p. 277); [Monotone Invariance Theorem](#) (p. 220); [recall at k](#) (p. 286).

kernel

concept. Entry id kernel.

DEFINITION

The directions a matrix sends to zero. The kernel of the read operator is the nuisance. Chapter 0 section 0.5.

EQUATION

Book equation 0.9.

$$P_C = \mathbb{E}[g g^\top], \quad g = \nabla C(x).$$

Book equation 0.12a.

$$x \sim_C x' \iff d_G(C(x), C(x')) = 0.$$

Book equation 0.12b.

$$x \sim_{\bar{P}_{C,\mu}} x' \iff x - x' \in \ker \bar{P}_{C,\mu}.$$

LEDGER

- OT-7. The damage form, trace pairing, rank, and loading covariance are $GL(d)$ -invariant under $P' = A^{-\top} P A^{-1}$, while spectrum, effective rank, principal angles, and water-filling are $O(d)$ -only — consumer-weighted damage is a geometric scalar, and P3's cliff cannot be bought down by reparameterization. [demonstrated]. [geometric-observation/claims/LEDGER.md:30 at 9f3829f](#).
- GO-6. At matched rate, output coding $\leq$ surrogate $\leq$ reconstruction on the consumer metric; the output-reconstruction gap is governed by the $\ker P_C$ entropy share, and the surrogate-output gap vanishes as rate grows. [demonstrated]. [geometric-observation/claims/LEDGER.md:68 at 9f3829f](#).

FIRST STATED

Chapter 0 section 0.5 and section 0.6 of *Data Mining as Observation*, with the kernel entropy share in [geometric-observation/claims/LEDGER.md](#) row GO-6.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 2 section 2.2	read subspace small, operator local, pullback composition, rank cannot increase	geometric-observation\chapters\ch05_the_read_metric_and_the_o 48; geometric-observation\chapters\ch06_mathematical_preliminaries 27
chapter 4 section 4.2	GO-6, d 8 and r 4, output at or below surrogate at or below reconstruction at every rate, about 500 times, gap 0.41 to 0.005, isotropic control collapses	geometric-observation\claims\LEDGER.md row GO-6; geometric-observation\chapters\ch07_cost.md

FAILURES AND CORRECTIONS

none

CONDITIONS

- The directions a matrix sends to zero. The kernel of the read operator is the nuisance, the directions the consumer never reads, and a direction is in it exactly when every row's sensitivity is orthogonal to it. The kernel's dimension is the dimension less the rank.
- The reconstruction coder spends bits on the kernel and the output coder does not, and the gap between them is governed by the share of the error's entropy that lies in the kernel.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/ReadOperator.lean`, theorems `rank_one_reads_one_direction`, `readOp_mulVec`, `quad_readOp`, `quad_readOp_nonneg`, `readOp_mulVec_eq_zero_iff`, `readOp_diag`, `readOp_offdiag`, `readOp_symm`, `readOp_neg`, `affine_const_along_nuisance`, `readOp_affine`, `readOp_sqLength_basis`, at `observation-data-mining af776fd`.

`lean/DataMiningAsObservation/Rank.lean`, theorems `rank_le_width`, `rank_le_height`, `rank_outer_le_one`, `rank_mul_le`, `rank_zero`, `rank_transpose`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 1, 2, 4, 6, 11.

RELATED

[nuisance](#) (p. 233); [read operator](#) (p. 282); [read subspace](#) (p. 285); [quotient](#) (p. 271); [rank](#) (p. 274).

KL divergence

concept. Entry id `kl-divergence`.

DEFINITION

A measure of how far one probability distribution is from another, zero when they are identical. Equation 0.24.

EQUATION

Book equation 0.24.

$$\text{KL}(p \parallel q) = \sum_i p_i \ln \frac{p_i}{q_i} \geq 0.$$

Book equation 0.22.

$$\text{PPL} = 2^H, \quad H = -\frac{1}{T} \sum_{t=1}^T \log_2 p(w_t \mid w_{<t}).$$

LEDGER

- NEG-2. Reconstruction cosine as a proxy for key quality. [refuted]. `geometric-observation/claims/LEDGER.md:95` at `9f3829f`.

FIRST STATED

Kullback and Leibler, on information and sufficiency, 1951, as chapter 0 section 0.11 states it beside perplexity, with the program's case in `turboquant-pro/docs/KV_KEYS_FINDING.md:1-49`.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 1 section 1.4	cosine 0.995 and perplexity of order ten thousand, the recalibration negative	geometric-observation\chapters\ch02_failure_of_observer_free_60; geometric-observation\chapters\ch16_honest_negatives.md NEG-2 and NEG-4; turboquant-pro\docs\KV_KEYS_FINDING.md:1-49
chapter 2 section 2.5	cosine 0.995 and the softmax reader	turboquant-pro\docs\KV_KEYS_FINDING.md:1-49
chapter 3 section 3.2	condition (A2), tangential displacement	turboquant-pro\docs\KV_KEYS_FINDING.md:61-86; geometric-observation\chapters\ch09_legibility.md:42-54
chapter 8 section 8.2	cosine 0.995, perplexity near $1e4$	turboquant-pro\docs\KV_KEYS_FINDING.md:1-49; geometric-observation\chapters\ch16_honest_negatives.md NEG-2
chapter 11 section 11.1	condition (A2), cosine satisfies it, post-rotary keys do not, the cone below cell size	turboquant-pro\docs\KV_KEYS_FINDING.md:61-86; the-angular-observer\README.md:26-31
chapter 11 section 11.2	fp16 12.24, values-only 13.12, PolarQuant K4 10643 and 0.095, per-channel uniform K4 14.91 and 0.062, per-channel NUQ K3 15.77 and 0.148, 2.4x and 670x, pre-rotary near 22000, 2 key heads serve 12 query heads	turboquant-pro\docs\KV_KEYS_FINDING.md:1-49

FAILURES AND CORRECTIONS

- NEG-2, [refuted]. Reconstruction cosine as a proxy for key quality.

CONDITIONS

- The expected log ratio of two distributions' masses under the first. It is nonnegative by Gibbs' inequality, zero when the distributions agree, and not symmetric, so the direction of the comparison is part of the claim.
- The book's use is through perplexity. A reconstruction at cosine 0.995 raised the perplexity by three orders of magnitude, which is the case that reconstruction error is not the consumer's error.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

lean/DataMiningAsObservation/KL.lean, theorems `kl_nonneg`, `kl_self`, `kl_not_symm`, at observation-data-mining af776fd.

USED IN

Data Mining as Observation chapters 0, 1, 8, 11.

RELATED

[identity reader](#) (p. 171); [read distortion](#) (p. 281); [calibration](#) (p. 52); [the flip](#) (p. 138).

k-means

instrument. Entry id k-means.

DEFINITION

A clustering that minimizes the sum of squared errors to the cluster centres. The mean minimizes each cluster's error and assigning each row to its nearest centre never raises the total, so it is the identity reader's clustering. Chapter 9.

EQUATION

Book equation 9.1.

$$\text{SSE} = \sum_{k=1}^K \sum_{i \in \mathcal{C}_k} \|x_i - c_k\|^2, \quad \text{the } P_C = I \text{ distortion summed within clusters.}$$

Book equation 0.30.

$$\text{SSE} = \sum_k \sum_{i \in \mathcal{C}_k} \|x_i - c_k\|^2, \quad s_i = \frac{b_i - a_i}{\max(a_i, b_i)}.$$

Book equation 4.2.

$$D(b) = \sum_i s_i \sigma_i^2 2^{-2b_i}, \quad s_i = v_i^\top P_C v_i, \quad b_i^* = \max\left(0, \frac{1}{2} \log_2 \frac{s_i \sigma_i^2}{\theta}\right), \quad \sum_i b_i^* = B,$$

LEDGER

- GO-3. The certificate's vacuity threshold predicts where single-stage retrieval dies. [demonstrated].
geometric-observation/claims/LEDGER.md:65 at 9f3829f.

FIRST STATED

Lloyd, least squares quantization in PCM, 1957, as chapter 9 section 9.1 of *Data Mining as Observation* reads it.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 9 section 9.4	explained ratios, effective rank 5.19 of 8, convergence with a second method, property of the representation not the space	gtc- prototype\docs\SPECTRUM_FINDINGS.md:10-18
chapter 10 section 10.1	the four detector families	TSK 2e chapter 9

FAILURES AND CORRECTIONS

none

CONDITIONS

- A clustering that minimizes the sum of squared errors to the cluster centres. The error about any centre is the error about the mean plus the row count times the squared distance between the two, so the mean minimizes each cluster's error, and assigning each row to its nearest centre never raises the total.
- It is the identity reader's clustering. It weighs every direction equally, and a direction with large variance and no group structure pulls its centres exactly as variance-based reduction pulls its components.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

lean/DataMiningAsObservation/KMeans.lean, theorems sse_decomposition, mean_minimizes, assign_nearest, at observation-data-mining af776fd.

USED IN

Data Mining as Observation chapters 0, 9, 10, 11.

RELATED

[identity reader](#) (p. 171); [validity index](#) (p. 360); [silhouette](#) (p. 324); [inverted file](#) (p. 177); [DBSCAN](#) (p. 93).

KV cache

concept. Entry id kv-cache.

DEFINITION

The stored keys and values of earlier tokens, kept so they need not be recomputed. Compressing it is where several of the book's examples go wrong. Chapters 0 and 11.

EQUATION

Book equation 0.23.

$$\text{softmax}(z)_i = \frac{e^{z_i}}{\sum_j e^{z_j}}, \quad \text{output} = \sum_i \text{softmax}\left(\frac{q \cdot k_i}{\sqrt{d}}\right)_i v_i.$$

Book equation 11.1.

$$\cos(k, \hat{k}) = 0.995 \quad \text{while} \quad \text{PPL} : 12.24 \rightarrow 10643.$$

LEDGER

- GO-2/GO-12/GO-13 operational (KV serving, 077). Consumer-relative access width measured on a production serving stack (Qwen2.5-7B KV-cache eviction, matched budget): task quality tracks measured predictive uncertainty u about the consumer's future reads, not nominal scorer width — the 32-query snapshot beats the 1024-query scorer +0.4375±0.070 at 5% keep (bar 0.30) and survives 97% eviction with zero drop, while wide-window scoring is statistically indistinguishable from random eviction at extreme budgets; the recency-hoarding starvation signature replicated across three disjoint prompt sets (oracle-miss gap 0.370 vs bar 0.25). The novel equal-uncertainty analytic-equality control (degradation-titrated, constructible by design) REFUTED its own equality prediction on the pre-registered branch: with calibration health 4×-130× inside gates, equal scalar u did NOT give equal quality (V4 +0.078 vs 0.0625 tolerance; the pre-registered p=0.03 contrast firmed to +0.359±0.068, 5.3 SE) — equal scalar uncertainty is insufficient, error structure matters, consistent with GO-13 Theorem 1's own r≥2 scoping of equal-q universality (a scalar-context privilege). Successor arc: 056 honest miss → 075 ID burned on a disclosed design failure → 077 sealed and split-verdict. [demonstrated]. [geometric-observation/claims/LEDGER.md:81 at 9f3829f](#).
- NEG-2. Reconstruction cosine as a proxy for key quality. [refuted]. [geometric-observation/claims/LEDGER.md:95 at 9f3829f](#).

FIRST STATED

Chapter 0 section 0.11 and chapter 11 section 11.1 of *Data Mining as Observation*, with the program's measurements in [turboquant-pro/docs/KV_KEYS_FINDING.md:1-49](#) and the serving-stack ledger row.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 1 section 1.4	cosine 0.995 and perplexity of order ten thousand, the recalibration negative	geometric-observation\chapters\ch02_failure_of_observer_free_60; geometric-observation\chapters\ch16_honest_negatives.md NEG-2 and NEG-4; turboquant-pro\docs\KV_KEYS_FINDING.md:1-49
chapter 2 section 2.5	cosine 0.995 and the softmax reader	turboquant-pro\docs\KV_KEYS_FINDING.md:1-49
chapter 3 section 3.2	condition (A2), tangential displacement	turboquant-pro\docs\KV_KEYS_FINDING.md:61-86; geometric-observation\chapters\ch09_legibility.md:42-54
chapter 8 section 8.2	cosine 0.995, perplexity near 1e4	turboquant-pro\docs\KV_KEYS_FINDING.md:1-49; geometric-observation\chapters\ch16_honest_negatives.md NEG-2
chapter 8 section 8.9	13.7 ROUGE-L, 0.25/4.19/9.60/13.7 at 64/128/256/512, re-validation negative 0.31 n=40, 26.64 under symmetric nf4, _quant_nf4a_group unchanged since 289bdfc before 4f7baab	turboquant-pro\CLAIMS.md:63-81; turboquant-pro\benchmarks\kvquant_matrix\REVAL-2026-08-08.md
chapter 11 section 11.1	condition (A2), cosine satisfies it, post-rotary keys do not, the cone below cell size	turboquant-pro\docs\KV_KEYS_FINDING.md:61-86; the-angular-observer\README.md:26-31
chapter 11 section 11.2	fp16 12.24, values-only 13.12, PolarQuant K4 10643 and 0.095, per-channel uniform K4 14.91 and 0.062, per-channel NUQ K3 15.77 and 0.148, 2.4x and 670x, pre-rotary near 22000, 2 key heads serve 12 query heads	turboquant-pro\docs\KV_KEYS_FINDING.md:1-49

FAILURES AND CORRECTIONS

- NEG-2, [refuted]. Reconstruction cosine as a proxy for key quality.

CONDITIONS

- The stored keys and values of every earlier token for every layer, so its size is the product of layers, tokens, two, the head width, and the bits per number. Compressing it changes the keys the heads read.
- A compression that preserves every query-key score preserves every head output, and one that preserves the keys' cosine need not, since two keys of the same direction and different lengths have cosine one and different scores. That is the attention-key finding in one sentence.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/KVCache.lean`, `theorems cacheBits_tokens`, `cacheBits_half`, `ratios`, `output_of_scores_preserved`, `cosine_not_score`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 11.

RELATED

[attention](#) (p. 23); [eviction](#) (p. 126); [the flip](#) (p. 138); [budget](#) (p. 46).

L

lag

concept. Entry id lag.

DEFINITION

How far a replica trails the primary. The same lag gives two readers two staleness rates. Chapter 0 section 0.13 and chapter 13.

EQUATION

Book equation 0.26.

$$FC = \Pr [W \text{ refutes } | \mathcal{C}_t \text{ clears}], \qquad \text{coverage} = \Pr [\mathcal{C}_t \text{ clears}].$$

Book equation 13.2.

$$T_{\text{coh}} = \frac{0.423}{f_D},$$

refuted fit: $\phi \approx 0.177 T_{\text{coh}} (R^2 = 0.915),$

sealed: $\phi \leq 6$ TTI at 10 Hz, 1 TTI at ≥ 50 Hz.

Book equation 0.25.

$$T_{\text{coh}} = \frac{0.423}{f_D}.$$

LEDGER

- OT-11. Feedback-free staleness: streaming-retrieval damage tracks measured drift; derived-cadence re-allocation removes it. [void]. geometric-observation/claims/LEDGER.md:48 at 9f3829f.

FIRST STATED

Chapter 0 section 0.13 of *Data Mining as Observation*, with the replica measurements in observation-theory-campaigns/experiments/DATABASE-FRESHNESS-TRACK.md.

MEASUREMENTS

Where the book states it	Numbers, as the book’s sources table records them	Source
chapter 13 section 13.2	the grammar, certificate, witness, refresh floor, false-clear rate, vacuity, the witness table	geometric-observation\chapters\ch19_the_certificate_that_ages95; observation-theory-campaigns\experiments\FRESHNESS-PROGRAM.md:1-40
chapter 13 section 13.3	ZooKeeper hot 0.99 cold 0.01 witnessed 0.0, Postgres 0.50 to 0.06, MongoDB 0.47 to 0.03, production Postgres 0.47 to 0.02, real substrates, disjoint seeds	observation-theory-campaigns\experiments\DATABASE-FRESHNESS-TRACK.md:1-45

FAILURES AND CORRECTIONS

none

CONDITIONS

- How far a replica trails the primary, in writes or in time. A read of the replica is stale for a consumer whose footprint the lag has touched and fresh for one it has not, so the same lag gives two readers two staleness rates, and a mixed workload's rate lies between them.
- Postgres read 0.50 to 0.06, MongoDB 0.47 to 0.03, and production Postgres 0.47 to 0.02 across readers on the same lag, with disjoint seeds.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/Replica.lean`, theorems `stale_mono`, `stale_zero`, `naive_certificate`, `witnessed_certificate`, `witnessed_coverage`, at `observation-data-mining af776fd`.

`lean/DataMiningAsObservation/Freshness.lean`, theorems `disagree`, `mixedRate_between`, `mixedRate_eq_left_iff`, `stale_for_all`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 13.

RELATED

[replica](#) (p. 299); [freshness](#) (p. 149); [coherence time](#) (p. 63); [false-clear rate](#) (p. 133); [footprint](#) (p. 144).

Landauer's principle

result. Entry id `landauers-principle`.

DEFINITION

Erasing one bit costs at least Boltzmann's constant times the absolute temperature times the natural log of two. Equation 0.27.

EQUATION

Book equation 0.27.

$$E_{\min} = k_B T \ln 2 \approx 2.87 \times 10^{-21} \text{ J at 300 K.}$$

Book equation 13.3.

$$W_{\text{reset}} \geq k_B T \ln 2 \cdot H(M | S), \quad H(M | S) \leq H(M).$$

LEDGER

- GO-7. A stored description's description rate and its conditional Landauer reset content are operationally separate resources: the same finite- n code index needing $\hat{R} \approx 0.67$ bits/symbol to describe is fully recoverable from retained side information at bin rate $0.26 = 0.39\hat{R}$, fails increasingly below its conditional content, and fails absolutely (err 1.00 at every bin rate) without S . [replicated]. `geometric-observation/claims/LEDGER.md:69 at 9f3829f`.
- GO-8. On two independent source families (binary Markov; Gaussian AR(1)), a fixed stored record's operational reset threshold rises with the age of the retained side information exactly as the staleness-work complement prices it: same record, same bins, same decoder — the decodable bin rate climbs $0.10 \rightarrow 0.55$ bits/symbol across ages 0–64 of a $p = 0.05$ Markov chain, tracking $R_c - 1 + h_2(\hat{d} * q_t)$ within one grid step at every age, and a fixed bin rate flips from 1% error (age 0) to 100% (age 32). [replicated]. `geometric-observation/claims/LEDGER.md:71 at 9f3829f`.
- GO-9. Coordinated reset is operationally cheaper than independent reset by the records' shared-structure information: with two consumer records sharing a component, recovering either record's bin residual with the *other record intact* lowers the decodable threshold by $\text{gap}_{TC} = 1 - h_2(\hat{d} * \hat{d})$ (measured 0.60 and 0.45 vs 0.476 predicted) — including on the record whose own reset side information is useless; mis-

matched pairing saves nothing. Held on two independent source families (binary; Gaussian). [replicated].
geometric-observation/claims/LEDGER.md:73 at 9f3829f.

FIRST STATED

Landauer, irreversibility and heat generation in the computing process, 1961, as chapter 0 section 0.13 states it, and its consumer-relative form in Volume 14 and the rate-work paper of ledger rows GO-7 to GO-9.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 4 section 4.2	GO-6, d 8 and r 4, output at or below surrogate at or below reconstruction at every rate, about 500 times, gap 0.41 to 0.005, isotropic control collapses	geometric-observation\claims\LEDGER.md row GO-6; geometric-observation\chapters\ch07_cost.md
chapter 4 section 4.4	GO-4 budget inversion, fixed m 10 rises, matched m 121, 126, 159 collapses, 3 seeds	geometric-observation\claims\LEDGER.md row GO-4
chapter 6 section 6.2	planted probe, overlap 0.936 vs 0.059, twelve of twelve, reconstruction 0.40, five of five	geometric-observation\claims\LEDGER.md row GO-1
chapter 7 section 7.2	curvature reader, identical reconstruction 0.298, flip twelve of twelve both, projected variance twelve of twelve, reconstruction Spearman negative 0.20, identity curvature tie	geometric-observation\experiments\G02-gradient-curvature-NOTES.md:1-40; geometric-observation\claims\LEDGER.md row GO-B
chapter 7 section 7.2	real logistic model, exact Hessian, anti 300 of 300, flip 82 of 300, coupling diagnosis, bound not refutation	geometric-observation\claims\LEDGER.md row GO-B-optim-D4
chapter 9 section 9.3	the margin certificate, mu crit as the expected maximum of N minus 1 standard normals, rho, death at 0.948 within 6 percent, Spearman 0.991 vs 0.873, fourteen corpora, six gates, the v1 to v3 path, the standing correction	geometric-observation\experiments\G03-certificate-vacuity-v3-NOTES.md:1-60; geometric-observation\claims\LEDGER.md row GO-3
chapter 11 section 11.7	GO-1 overlap 0.936 vs 0.059, flip 12 of 12, reconstruction 0.40	geometric-observation\claims\LEDGER.md row GO-1; geometric-observation\chapters\ch10_the_blind_probe.md:34-52
chapter 12 section 12.2	recall 0.999 vs 0.592, the derived death point within 6 percent across fourteen corpora	openvector-bench\README.md:60-96; geometric-observation\claims\LEDGER.md row GO-3
chapter 12 section 12.3	uncompressed 0.79, centred 0.84	geometric-observation\claims\LEDGER.md row GO-B-legal

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 12 section 12.3	041 non-oracle, frozen LSA TF-IDF to SVD 100 train-only, AUROC 0.975 vs 0.910, flip tied, magnitude overshoot, partial	geometric-observation\chapters\ch10_the_blind_probe.md:100-118; geometric-observation\claims\LEDGER.md row GO-B-blind 041
chapter 13 section 13.5	GO-7, 0.67 bits per symbol, bin rate 0.26 equals 0.39, error 1.00 without side information, two families, six codebooks	geometric-observation\claims\LEDGER.md row GO-7
chapter 13 section 13.5	GO-8, 0.10 to 0.55 across ages 0 to 64, flip probability 0.05, 1 percent to 100 percent at age 32, Gaussian pass 5 of 5, the control-statistic caveat	geometric-observation\claims\LEDGER.md row GO-8; geometric-observation\experiments\GO-landauer-gaussian-secondsettings-NOTES.md
chapter 13 section 13.6	attempt three, windows 1024, 256, 32, uncertainty 0.982 to 0.892, 5 of 6, 0.4375 with SE 0.070 at 5 percent keep vs 0.30, 97 percent eviction, oracle-miss 0.370 vs 0.25, V4 0.078 vs 0.0625, contrast 0.359 with SE 0.068, n 64, seed 20260812, 89 duty cycles	geometric-observation\claims\LEDGER.md row GO-2/GO-12/GO-13 operational; geometric-observation\prereg\GO-P-2026-077-kv-consumer-relative.md; geometric-observation\results\GO13-kvaw2-governed.json

FAILURES AND CORRECTIONS

none

CONDITIONS

- Erasing one bit costs at least Boltzmann's constant times the absolute temperature times the natural logarithm of two, 2.87 times ten to the minus twenty-one joules at 300 kelvin. The reset bound of chapter 13 charges that price per bit of the record's conditional entropy given what the consumer keeps.
- Conditioning cannot raise entropy, so the consumer-relative bound is never above the consumer-free one, and a record that keeps nothing about the erased part costs nothing to reset. The entropies themselves are the ledger's measurements, two source families and six codebooks.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/Landauer.lean`, `theorems eMin_300`, `resetBound_le`, `resetBound_zero`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 13.

RELATED

[budget](#) (p. 46); [certificate](#) (p. 55); [read operator](#) (p. 282); [coherence time](#) (p. 63).

language model

concept. Entry id `language-model`.

DEFINITION

A model that reads a sequence of tokens and outputs a probability for the next one, scored by perplexity. Chapter 0 section 0.11.

EQUATION

Book equation 0.22.

$$\text{PPL} = 2^H, \quad H = -\frac{1}{T} \sum_{t=1}^T \log_2 p(w_t | w_{<t}).$$

Book equation 0.23.

$$\text{softmax}(z)_i = \frac{e^{z_i}}{\sum_j e^{z_j}}, \quad \text{output} = \sum_i \text{softmax}\left(\frac{q \cdot k_i}{\sqrt{d}}\right)_i v_i.$$

Book equation 11.1.

$$\cos(k, \hat{k}) = 0.995 \quad \text{while} \quad \text{PPL} : 12.24 \rightarrow 10643.$$

LEDGER

- NEG-16 (KV serving, end-task). *At matched bits and matched reconstruction error, steering KV quantization error into the attention read subspace degrades LongBench task score by the registered floors on a deployed-class model.* Refuted at its registered effect sizes on this model and task. [refuted]. geometric-observation/claims/LEDGER.md:92 at 9f3829f.
- NEG-2. Reconstruction cosine as a proxy for key quality. [refuted]. geometric-observation/claims/LEDGER.md:95 at 9f3829f.

FIRST STATED

Chapter 0 section 0.11 of *Data Mining as Observation*, with the serving measurements in geometric-observation/experiments/GO-kv-serving-flip-NOTES.md.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 1 section 1.4	cosine 0.995 and perplexity of order ten thousand, the recalibration negative	geometric-observation\chapters\ch02_failure_of_observer_free60; geometric-observation\chapters\ch16_honest_negatives.md NEG-2 and NEG-4; turboquant-pro\docs\KV_KEYS_FINDING.md:1-49
chapter 2 section 2.5	cosine 0.995 and the softmax reader	turboquant-pro\docs\KV_KEYS_FINDING.md:1-49
chapter 8 section 8.2	cosine 0.995, perplexity near 1e4	turboquant-pro\docs\KV_KEYS_FINDING.md:1-49; geometric-observation\chapters\ch16_honest_negatives.md NEG-2

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 11 section 11.2	fp16 12.24, values-only 13.12, PolarQuant K4 10643 and 0.095, per-channel uniform K4 14.91 and 0.062, per-channel NUQ K3 15.77 and 0.148, 2.4x and 670x, pre-rotary near 22000, 2 key heads serve 12 query heads	turboquant-pro\docs\KV_KEYS_FINDING.md:1-49
chapter 13 section 13.6	attempt one, six gates, 4 pass, K3 0.075 vs 0.15, K4 0.050 vs 0.10, arms 0.925, 0.925, 0.900, 0.850, recon ratio 1.000000, n 40, mechanism 15 to 36 times	geometric-observation\experiments\GO-kv-serving-flip-NOTES.md:1-40; geometric-observation\prereg\GO-P-2026-056-kv-serving-flip.md:1-40; geometric-observation\results\GO-KV-serving-flip-7B.json
chapter 13 section 13.6	attempt two, broken proxy, control did not exist, identifier burned, 16 prompts, 1360 s	geometric-observation\prereg\GO-P-2026-077-kv-consumer-relative.md:1-15; geometric-observation\results\GO13-kvaw-pilot-disclosed.json
chapter 13 section 13.6	attempt three, windows 1024, 256, 32, uncertainty 0.982 to 0.892, 5 of 6, 0.4375 with SE 0.070 at 5 percent keep vs 0.30, 97 percent eviction, oracle-miss 0.370 vs 0.25, V4 0.078 vs 0.0625, contrast 0.359 with SE 0.068, n 64, seed 20260812, 89 duty cycles	geometric-observation\claims\LEDGER.md row GO-2/GO-12/GO-13 operational; geometric-observation\prereg\GO-P-2026-077-kv-consumer-relative.md; geometric-observation\results\GO13-kvaw2-governed.json

FAILURES AND CORRECTIONS

- NEG-16 (KV serving, end-task), [refuted]. *At matched bits and matched reconstruction error, steering KV quantization error into the attention read subspace degrades LongBench task score by the registered floors on a deployed-class model.* Refuted at its registered effect sizes on this model and task.
- NEG-2, [refuted]. Reconstruction cosine as a proxy for key quality.

CONDITIONS

- A model that reads a sequence of tokens and outputs a probability for the next one. Its perplexity is two to the power of the average bits per token, at least one and equal to the vocabulary size for a uniform guess, and its attention weights are a softmax, positive and summing to one.
- A seven-billion-parameter model served passage retrieval over contexts of fourteen thousand tokens across three registered attempts, and a key quantizer with reconstruction cosine 0.995 broke it to a perplexity near ten thousand.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/Perplexity.lean`, theorems `bits_nonneg`, `one_le_perplexity`, `perplexity_uniform`, `perplexity_mono`, `bits_of_finding`, at `observation-data-mining af776fd`.

`lean/DataMiningAsObservation/Softmax.lean`, theorems `denom_pos`, `softmax_pos`, `softmax_sum`, `softmax_shift`, `softmax_lt_iff`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 1, 4, 6, 8, 11, 12, 13.

RELATED

[token](#) (p. 351); [perplexity](#) (p. 250); [attention](#) (p. 23); [KV cache](#) (p. 190); [softmax](#) (p. 327).

Laplacian

concept. Entry id laplacian.

DEFINITION

A matrix built from a graph’s edges whose eigenvectors, from the smallest eigenvalue up, are the smoothest functions on the graph. Equation 0.19.

EQUATION

Book equation 0.19.

$$L = I - D^{-1/2}AD^{-1/2}, \quad D = \text{diag}(d_1, \dots, d_n), \quad Lu_k = \lambda_k u_k, \quad 0 = \lambda_1 \leq \lambda_2 \leq \dots.$$

Book equation 0.20.

$$\Psi_i = \left(\frac{u_k(i)}{\sqrt{\lambda_k d_i}} \right)_{k \geq 2}, \quad \|\Psi_i - \Psi_j\|^2 = R(i, j) = \frac{C(i, j)}{\text{vol}(G)}.$$

Book equation 9.2.

$$N(\lambda) = \#\{k : \lambda_k \leq \lambda\} \sim C_d \lambda^{d/2} \quad \Rightarrow \quad d = 2 \frac{d \log N}{d \log \lambda}.$$

LEDGER

- NEG-1. Fixed-scale, uniform-in-m bi-Lipschitz for the commute filter. [refuted]. [geometric-observation/claims/LEDGER.md:94 at 9f3829f](#).

FIRST STATED

Chapter 0 section 0.10 of *Data Mining as Observation*, with the program’s spectral work in Volume 14 chapter 9 and the-angular-observer.

MEASUREMENTS

Where the book states it	Numbers, as the book’s sources table records them	Source
chapter 3 section 3.2	condition (A2), tangential displacement	turboquant-pro\docs\KV_KEYS_FINDING.md:61-86; geometric-observation\chapters\ch09_legibility.md:42-54
chapter 3 section 3.3	resistance converges to 1 over d_i plus 1 over d_j, simplex argument, radius converges to 1 over root d_i	the-angular-observer\theorem.md:110-146
chapter 3 section 3.3	radius against degree correlation 0.92 to 0.99	the-angular-observer\README.md:135-139

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 3 section 3.3	angle 0.92 flat in m, magnitude 0.03, full 0.75 decaying	wolfram-observer-bridge\theorem.md:1-20; the-angular-observer\theorem.md:187-224
chapter 3 section 3.3	substrate table	the-angular-observer\README.md:99-109
chapter 3 section 3.3	row normalization is the projection onto the read subspace of the geodesic-rank consumer	geometric-observation\chapters\ch09_legibility.md:31-41; geometric-observation\chapters\ch03_historical_precursors.md:110
chapter 3 section 3.3	proven versus conjectured ledger, d at least 3, d equals 2 borderline, fixed-m conditional theorem, heat filter uniform, commute metric form false, Green-kernel rank limit for d at most 3, obstruction at d equals 4	the-angular-observer\theorem.md:84-107, 147-224; the-angular-observer\README.md:150-185
chapter 3 section 3.3	the self-refuted v0.8 claim, NEG-1	the-angular-observer\README.md:170-175; geometric-observation\chapters\ch16_honest_negatives.md NEG-1
chapter 3 section 3.3	hyperbolic rejected, curvature negative 0.98 to negative 0.14, trend 1.09 minus 0.157 d across 20 manifolds and 5 families	the-angular-observer\README.md:111-147, 183-185
chapter 9 section 9.1	the geodesic-rank reader discards the radius, row normalization is the angular projection	geometric-observation\chapters\ch09_legibility.md:10-41; the-angular-observer\theorem.md:110-146
chapter 9 section 9.2	the battery, three new templates, frozen ratios, code hash, bars at least 10 of 12 and the dimension ordering, eccentricity scope	the-angular-observer\PREREG_RECOGNIZER_BATTERY.md:1-80
chapter 9 section 9.2	12 of 12, dimensions 1.81, 2.80, 1.74, angular Spearman ranges, eccentricity spreads, verdict confirmed, GO-P-2026-041	the-angular-observer\experiments\manifold-recovery\battery_result.json
chapter 9 section 9.2	causal set 0.57 with dimension 2.9 to 10.7, small world 0.45, control 0.01, rung 4 decay 0.747 to 0.679 below 0.75	the-angular-observer\README.md:99-109; wolfram-observer-bridge\rung4_scaledm_result.json; wolfram-observer-bridge\review-2.txt
chapter 9 section 9.4	explained ratios, effective rank 5.19 of 8, convergence with a second method, property of the representation not the space	gtc-prototype\docs\SPECTRUM_FINDINGS.md:10-18

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 11 section 11.1	condition (A2), cosine satisfies it, post-rotary keys do not, the cone below cell size	turboquant-pro\docs\KV_KEYS_FINDING.md:61-86; the-angular-observer\README.md:26-31
chapter 14 section 14.3	floor 0.08 to 0.12, balanced resample 3001 items, identity attack 0.237 with interval 0.20 to 0.28 on 289, sexual 0.204, threat 0.093 retracted, first pass 18 positives, 87.7 and 1.4 percent	gtc-prototype\docs\SPECTRUM_FINDINGS.md:20-64
chapter 14 section 14.3	contraction 0.779 to 0.863, 77 of 1600, 4.8 percent, 0.872 to 0.863, weight 2.69	gtc-prototype\docs\SPECTRUM_FINDINGS.md:66-99
chapter 14 section 14.5	the contraction formula fairness minus the general component	gtc-prototype\docs\SPECTRUM_FINDINGS.md:90-99
chapter 14 section 14.7	51 percent moderated at 80 and 95 percent precision on the balanced set	gtc-prototype\docs\SPECTRUM_FINDINGS.md:84-88

FAILURES AND CORRECTIONS

- NEG-1, [refuted]. Fixed-scale, uniform-in-m bi-Lipschitz for the commute filter.

CONDITIONS

- A matrix built from a graph's edges whose quadratic form is half the weighted sum of squared differences across edges, so it is nonnegative, zero on every constant, and positive on any function that differs across an edge of positive weight. The eigenvectors from the smallest eigenvalue up are therefore the smoothest functions on the graph.
- The file states the unnormalized form, and equation 0.19's normalized form is the same quadratic form in rescaled coordinates. The commute-time scaling of the embedding and the uniform bi-Lipschitz claim for the commute filter are in the ledger, the second refuted.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

lean/DataMiningAsObservation/Laplacian.lean, theorems quad_eq, quad_nonneg, quad_const, quad_pos_of_edge, at observation-data-mining af776fd.

USED IN

Data Mining as Observation chapters 0, 3, 4, 9, 11.

RELATED

[intrinsic dimension](#) (p. 175); [recognizer](#) (p. 288); [rank-faithful](#) (p. 277); [bi-Lipschitz](#) (p. 37).

latent semantic analysis

instrument. Entry id latent-semantic-analysis.

DEFINITION

The singular value decomposition of the TF-IDF matrix with the top components kept, which is principal components on documents. Chapter 12.

EQUATION

Book equation 0.36.

$$w_{t,d} = \text{tf}_{t,d} \cdot \ln \frac{N}{\text{df}_t}, \quad \text{tf}_{t,d} = \frac{\text{count of } t \text{ in } d}{\text{length of } d}, \quad \text{df}_t = \text{documents containing } t.$$

Book equation 0.5.

$$\Sigma v_i = \lambda_i v_i, \quad \Sigma = \sum_{i=1}^d \lambda_i v_i v_i^\top, \quad v_i \cdot v_j = 0 \ (i \neq j).$$

Book equation 4.2.

$$D(b) = \sum_i s_i \sigma_i^2 2^{-2b_i}, \quad s_i = v_i^\top P_C v_i, \quad b_i^* = \max\left(0, \frac{1}{2} \log_2 \frac{s_i \sigma_i^2}{\theta}\right), \quad \sum_i b_i^* = B,$$

LEDGER

- GO-B-legal (035→036). Legal-citation retrieval (CourtListener), cosine-ranking consumer, LaBSE embeddings — real large corpus, non-physical consumer [predicted]. `geometric-observation/claims/LEDGER.md:119` at 9f3829f.

FIRST STATED

Deerwester, Dumais, Furnas, Landauer, and Harshman, indexing by latent semantic analysis, 1990, as chapter 12 section 12.1 of *Data Mining as Observation* reads it, with the program's frozen embedding in `geometric-observation/chapters/ch10_the_blind_probe.md:100-118`.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 12 section 12.3	041 non-oracle, frozen LSA TF-IDF to SVD 100 train-only, AUROC 0.975 vs 0.910, flip tied, magnitude overshoot, partial	<code>geometric-observation/chapters/ch10_the_blind_probe.md:100-118</code> ; <code>geometric-observation/claims/LEDGER.md</code> row GO-B-blind 041

FAILURES AND CORRECTIONS

none

CONDITIONS

- The singular value decomposition of the TF-IDF matrix with the top components kept, which is principal components on documents. The fraction of variance kept grows with the components kept and reaches one at full rank, and the components dropped cost the sum of their eigenvalues.
- It is the identity reader on term-document variance, and the components it keeps are the directions of largest variance whether or not the consumer reads them. The non-oracle test fit a frozen embedding of 100 components on the training split only and recovered the reader blind.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/PCA.lean`, `theorems varAlong_basis`, `varAlong_le`, `varAlong_ge`, `dropped_eq`, `dropped_nonneg`, at `observation-data-mining af776fd`.

lean/DataMiningAsObservation/ExplainedVariance.lean, theorems explained_mem_unit, explained_mono, explained_full, retained_identity, retained_example, at observation-data-mining af776fd.

lean/DataMiningAsObservation/TFIDF.lean, theorems weight_everywhere, weight_nonneg, weight_antitone, tf_mem_unit, at observation-data-mining af776fd.

USED IN

Data Mining as Observation chapters 0, 12.

RELATED

[TF-IDF](#) (p. 350); [principal component analysis](#) (p. 260); [explained variance](#) (p. 129); [bag of words](#) (p. 32); [blind probe](#) (p. 39).

leakage

concept. Entry id leakage.

DEFINITION

Any information in the training data that could only be known after the decision, or any identifier that lets the model recognize a row it will be tested on. Chapter 8.

EQUATION

Book equation 8.1.

$$O_{\text{harness}} = (C_{\text{score}}, G_{\text{metric}}, B = \text{folds} \times \text{samples} \times \text{seeds}).$$

LEDGER

- NEG-2. Reconstruction cosine as a proxy for key quality. [refuted]. [geometric-observation/claims/LEDGER.md:95](#) at 9f3829f.

FIRST STATED

Chapter 8 section 8.1 of *Data Mining as Observation*, with the ESL example of selection before the split, Hastie, Tibshirani, and Friedman, section 7.10.2.

MEASUREMENTS

Where the book states it	Numbers, as the book’s sources table records them	Source
chapter 4 section 4.7	ESL 7.10.2 numbers	Hastie, Tibshirani, Friedman, ESL 2e, section 7.10.2; chapter 8 of this book
chapter 8 section 8.1	ESL 7.10.2 numbers (N=50, p=5000, 100 selected, 3 percent vs 50 percent)	Hastie, Tibshirani, Friedman, ESL 2e, section 7.10.2
chapter 8 section 8.1	G1 339.9 vs 71.9 same corpus, real about 61, rounds 1 to 4 at 300 to 420, three families falsified	openvector-bench/results/QUERY_COUPLING_ARTIFACT.md:1-20

FAILURES AND CORRECTIONS

- NEG-2, [refuted]. Reconstruction cosine as a proxy for key quality.

CONDITIONS

- Leakage is any information in the training data that could only be known after the decision, or any identifier that lets the model recognize a row it will be tested on. It is a property of the harness’s read subspace, not of the model.

- Selecting features on the whole data and then cross-validating leaks the test fold into the selection, and the reported error can fall from the true rate to a fraction of it, as the ESL example shows.
- Coupling between the queries and the corpus of a retrieval benchmark is leakage of the same kind, and the program's hubness numbers moved by a factor of several when the coupling was removed.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/Leakage.lean`, `theorems errors_lookup_eq_zero`, `lookup_default`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 1, 2, 4, 8.

RELATED

[harness](#) (p. 159); [preregistration](#) (p. 258); [sealed](#) (p. 316); [hubness](#) (p. 168).

ledger class

concept. Entry id `ledger-class`.

DEFINITION

One of six labels every headline claim carries, proved, demonstrated, replicated, predicted, exploratory, or refuted, defined in the evidence ledger of Volume 14. Chapter 1.

EQUATION

Book equation 1.1.

$$O = (C, G, B).$$

LEDGER

- OT-11. Feedback-free staleness: streaming-retrieval damage tracks measured drift; derived-cadence re-allocation removes it. [void]. `geometric-observation/claims/LEDGER.md:48` at 9f3829f.
- GO-2 (neg. half: not reconstruction). At matched bits, downstream preservation is not controlled by reconstruction error. [demonstrated]. `geometric-observation/claims/LEDGER.md:63` at 9f3829f.
- NEG-15 (Bell boundary). *Query-conditioned hubness supplies a mechanism for Bell-inequality violation without action at a distance.* Refuted as a mechanism; the settings-as-queries reframing survives only as vocabulary. [demonstrated]. `geometric-observation/claims/LEDGER.md:93` at 9f3829f.

FIRST STATED

Volume 14, `geometric-observation/PROTOCOL.md:58-75` and `geometric-observation/OBSERVATION.md:43-45`, DOI 10.5281/zenodo.21776291, with the ledger itself at `geometric-observation/claims/LEDGER.md`.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 1 section 1.5	the six classes and the ledger rule	<code>geometric-observation\PROTOCOL.md:58-75</code> ; <code>geometric-observation\OBSERVATION.md:43-45</code>
chapter 8 section 8.3	OT-11 void, strata 0.81 to 0.87	<code>geometric-observation\claims\LEDGER.md:48</code>

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 8 section 8.8	sixteen standing negatives	geometric-observation\claims\LEDGER.md NEG-1 to NEG-16
chapter 8 section 8.10	the six classes	geometric-observation\PROTOCOL.md:58-75
chapter 11 section 11.5	CI fails when CLAIMS.md and claims.yaml disagree	turboquant-pro\CLAIMS.md:1-27; turboquant-pro\tests\test_claims_ledger.py

FAILURES AND CORRECTIONS

none

CONDITIONS

- Every headline claim carries one of six labels, proved, demonstrated, replicated, predicted, exploratory, or refuted. A claim's class can only rise by a sealed pass on held-out data and only fall by a sealed miss, and a refuted claim is reported at the prominence of a pass.
- A number produced without a sealed bar is exploratory whatever its size. It may motivate a claim and may not carry one.
- The authority for a claim's class is the ledger, and a chapter or an encyclopedia entry that disagrees with it is rebuilt.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/Ledger.lean`, `theorems step_none`, `step_miss_le`, `step_pass_ge`, `step_miss_refuted`, `exploratory_stays`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 8, 11.

RELATED

[preregistration](#) (p. 258); [sealed](#) (p. 316); [certificate](#) (p. 55); [false-clear rate](#) (p. 133).

lift

concept. Entry id `lift`.

DEFINITION

Confidence divided by the consequent's support, so that lift one is the independence baseline. It fails at low support. Chapter 5.

EQUATION

Book equation 5.2.

$$\text{lift}(X \rightarrow Y) = \frac{c(X \rightarrow Y)}{s(Y)} = \frac{s(X \cup Y)}{s(X) s(Y)}.$$

Book equation 5.1.

$$s(X) = \frac{\sigma(X)}{N}, \quad c(X \rightarrow Y) = \frac{\sigma(X \cup Y)}{\sigma(X)} = \frac{s(X \cup Y)}{s(X)}.$$

LEDGER

none

FIRST STATED

TSK chapter 5 on objective measures, as chapter 5 section 5.1 of *Data Mining as Observation* states it.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 5 section 5.1 to 5.3, 5.5	support, confidence, lift, the Apriori principle, FP-growth, closed and maximal itemsets, objective measures, Simpson's paradox, cross-support	TSK 2e chapter 5

FAILURES AND CORRECTIONS

none

CONDITIONS

- Confidence over the consequent's support, the joint support over the product of the two supports. It is one under independence, symmetric in the two itemsets, and bounded by the reciprocal of the consequent's support.
- It fails at low support. One transaction in N containing both items, and neither elsewhere, gives lift N , which is why the rules with the highest lift are the ones chapter 8's multiple-comparison rule applies to.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/Lift.lean`, theorems `lift_indep`, `lift_symm`, `lift_le_inv`, `lift_single`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 5.

RELATED

[Apriori principle](#) (p. 22); [multiple comparisons](#) (p. 221); [safe pruning](#) (p. 311); [harness](#) (p. 159).

linear classifier

instrument. Entry id `linear-classifier`.

DEFINITION

A classifier that scores by a weighted sum, logistic regression or a linear support vector machine. Its read subspace is the one direction of its weight vector. Chapter 6.

EQUATION

Book equation 6.1.

$$s(x) = w \cdot x + b, \quad P_C = \mathbb{E}[\sigma'(s)^2] w w^\top, \quad \text{rank } P_C = 1.$$

Book equation 0.9.

$$P_C = \mathbb{E}[g g^\top], \quad g = \nabla C(x).$$

Book equation 0.28.

$$P = \frac{TP}{TP + FP}, \quad R = \frac{TP}{TP + FN}, \quad F_1 = \frac{2PR}{P + R}.$$

LEDGER

- GO-1. The consumer's invariant/nuisance split is identifiable ex ante from the consumer functional. [predicted]. `geometric-observation/claims/LEDGER.md:62 at 9f3829f`.

FIRST STATED

Chapter 6 section 6.1 of *Data Mining as Observation*, with the planted affine consumer in `geometric-observation/claims/LEDGER.md` row GO-1.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 6 section 6.1	the classifier row of the consumer table, the output metric makes a different observer	<code>geometric-observation\chapters\ch04_the_observer_triple.md:60-135</code>
chapter 6 section 6.2	planted probe, overlap 0.936 vs 0.059, twelve of twelve, reconstruction 0.40, five of five	<code>geometric-observation\claims\LEDGER.md</code> row GO-1
chapter 11 section 11.7	GO-1 overlap 0.936 vs 0.059, flip 12 of 12, reconstruction 0.40	<code>geometric-observation\claims\LEDGER.md</code> row GO-1; <code>geometric-observation\chapters\ch10_the_blind_probe.md:34-52</code>

FAILURES AND CORRECTIONS

none

CONDITIONS

- A classifier that scores by a weighted sum, logistic regression or a linear support vector machine. Its decision boundary is the hyperplane where the weighted sum is zero, its read subspace is the one direction of its weight vector, and its read operator is the outer product of that vector with itself, scaled by the score's slope.
- Its nuisance is everything orthogonal to the weight vector, and a planted linear consumer is the case a probe is tested against before it is trusted on a real model.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/Logistic.lean`, theorems `sigmoid_pos`, `sigmoid_lt_one`, `sigmoid_zero`, `sigmoid_neg`, `sigmoid_strictMono`, `decision_iff`, `decision_linear`, at `observation-data-mining af776fd`.

`lean/DataMiningAsObservation/ReadOperator.lean`, theorems `rank_one_reads_one_direction`, `readOp_mulVec`, `quad_readOp`, `quad_readOp_nonneg`, `readOp_mulVec_eq_zero_iff`, `readOp_diag`, `readOp_offdiag`, `readOp_symm`, `readOp_neg`, `affine_const_along_nuisance`, `readOp_affine`, `readOp_sqLength_basis`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 2, 6, 7, 11.

RELATED

[logistic regression](#) (p. 208); [decision boundary](#) (p. 94); [read subspace](#) (p. 285); [margin](#) (p. 213); [planted](#) (p. 252).

logistic regression

instrument. Entry id logistic-regression.

DEFINITION

A linear classifier whose score is the sigmoid of a weighted sum, strictly between zero and one, above one half exactly on one side of the hyperplane. Chapter 6.

EQUATION

Book equation 6.1.

$$s(x) = w \cdot x + b, \quad P_C = \mathbb{E}[\sigma'(s)^2] w w^\top, \quad \text{rank } P_C = 1.$$

Book equation 0.9.

$$P_C = \mathbb{E}[g g^\top], \quad g = \nabla C(x).$$

Book equation 0.28.

$$P = \frac{TP}{TP + FP}, \quad R = \frac{TP}{TP + FN}, \quad F_1 = \frac{2PR}{P + R}.$$

LEDGER

- GO-1. The consumer's invariant/nuisance split is identifiable ex ante from the consumer functional. [predicted]. [geometric-observation/claims/LEDGER.md:62 at 9f3829f](#).
- GO-B-optim-D4 (034 · D4). Optimization — gradient compression, curvature (Hessian) read operator, on a REAL model (logistic regression); optional stretch [predicted]. [geometric-observation/claims/LEDGER.md:122 at 9f3829f](#).

FIRST STATED

Cox, the regression analysis of binary sequences, 1958, as chapter 6 section 6.1 of *Data Mining as Observation* reads it, with the real logistic model's Hessian in [geometric-observation/claims/LEDGER.md](#) row GO-B-optim-D4.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 6 section 6.1	the classifier row of the consumer table, the output metric makes a different observer	geometric-observation\chapters\ch04_the_observer_triple.md:60 at 135
chapter 7 section 7.2	real logistic model, exact Hessian, anti 300 of 300, flip 82 of 300, coupling diagnosis, bound not refutation	geometric-observation\claims\LEDGER.md row GO-B-optim-D4

FAILURES AND CORRECTIONS

none

CONDITIONS

- A linear classifier whose score is the sigmoid of a weighted sum. The sigmoid lies strictly between zero and one, is one half at zero, is symmetric about one half, and is strictly increasing, so the decision at threshold one half is the sign of the weighted sum and the decision boundary is the hyperplane.
- Its score moves only along its weight vector, so its read subspace is one direction and its nuisance is everything orthogonal, and its read operator is the outer product of the weight vector with itself scaled by how steep the sigmoid is at each row.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/Logistic.lean`, theorems `sigmoid_pos`, `sigmoid_lt_one`, `sigmoid_zero`, `sigmoid_neg`, `sigmoid_strictMono`, `decision_iff`, `decision_linear`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 6, 7, 11, 12.

RELATED

[classifier](#) (p. 60); [decision boundary](#) (p. 94); [threshold](#) (p. 350); [read subspace](#) (p. 285); [Hessian](#) (p. 164).

M

Mahalanobis distance

concept. Entry id mahalanobis-distance.

DEFINITION

The Euclidean distance of a row from the mean after whitening. Equation 0.33.

EQUATION

Book equation 0.33.

$$d_M(x) = \sqrt{(x - \mu)^\top \Sigma^{-1} (x - \mu)}.$$

Book equation 10.1.

$$z(x) = \frac{x - \mu}{\sigma}, \quad d_M(x)^2 = (x - \mu)^\top \Sigma^{-1} (x - \mu) = \sum_i \frac{(v_i \cdot (x - \mu))^2}{\lambda_i}.$$

LEDGER

none

FIRST STATED

Mahalanobis, on the generalised distance in statistics, 1936, as chapter 0 section 0.16 states it, and chapter 10 section 10.1 of *Data Mining as Observation*, where the statistical detector is the identity reader on whitened data.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

CONDITIONS

- The Euclidean distance of a row from the mean after whitening. In the eigenbasis its square is the sum of the squared centred coordinates over the eigenvalues, nonnegative, zero at the mean, unchanged when data and covariance are rescaled together, and a deviation along a direction of larger variance counts for less.
- The detector that scores by it treats every whitened direction equally, so it is the identity reader on whitened data, and a consumer that reads a subspace calls different rows outliers.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/Mahalanobis.lean`, theorems `dM2_nonneg`, `dM2_mean`, `dM2_scale`, `dM2_eq_whitened`, `dM2_antitone_in_variance`, at observation-data-mining af776fd.

USED IN

Data Mining as Observation chapters 0, 10.

RELATED

[whitening](#) (p. 368); [identity reader](#) (p. 171); [hubness](#) (p. 168); [abstention](#) (p. 9).

manifold

concept. Entry id manifold.

DEFINITION

A curved surface of some dimension that looks flat when viewed closely. A dataset lies on one when its rows vary along fewer directions than they have coordinates. Chapter 0 section 0.10.

EQUATION

Book equation 0.31.

$$N(\lambda) = \#\{k : \lambda_k \leq \lambda\} \sim C_d \lambda^{d/2}.$$

Book equation 9.2.

$$N(\lambda) = \#\{k : \lambda_k \leq \lambda\} \sim C_d \lambda^{d/2} \quad \Rightarrow \quad d = 2 \frac{d \log N}{d \log \lambda}.$$

Book equation 0.32.

$$\text{dist}(m, t) = \sqrt{\frac{1}{8} \sum_{k=1}^8 (\ln m_k - \ln t_k)^2}.$$

LEDGER

- GO-3. The certificate's vacuity threshold predicts where single-stage retrieval dies. [demonstrated].
geometric-observation/claims/LEDGER.md:65 at 9f3829f.

FIRST STATED

Chapter 0 section 0.10 and chapter 9 section 9.2 of *Data Mining as Observation*, with the recognizer of Volume 14 chapter 11 and its battery in the-angular-observer.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 3 section 3.3	resistance converges to 1 over d_i plus 1 over d_j, simplex argument, radius converges to 1 over root d_i	the-angular-observer\theorem.md:110-146
chapter 3 section 3.3	radius against degree correlation 0.92 to 0.99	the-angular-observer\README.md:135-139
chapter 3 section 3.3	angle 0.92 flat in m, magnitude 0.03, full 0.75 decaying	wolfram-observer-bridge\theorem.md:1-20; the-angular-observer\theorem.md:187-224
chapter 3 section 3.3	substrate table	the-angular-observer\README.md:99-109
chapter 3 section 3.3	proven versus conjectured ledger, d at least 3, d equals 2 borderline, fixed-m conditional theorem, heat filter uniform, commute metric form false, Green-kernel rank limit for d at most 3, obstruction at d equals 4	the-angular-observer\theorem.md:84-107, 147-224; the-angular-observer\README.md:150-185

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 3 section 3.3	the self-refuted v0.8 claim, NEG-1	the-angular-observer\README.md:170-175; geometric-observation\chapters\ch16_honest_negatives.md NEG-1
chapter 3 section 3.3	hyperbolic rejected, curvature negative 0.98 to negative 0.14, trend 1.09 minus 0.157 d across 20 manifolds and 5 families	the-angular-observer\README.md:111-147,183-185
chapter 9 section 9.1	the geodesic-rank reader discards the radius, row normalization is the angular projection	geometric-observation\chapters\ch09_legibility.md:10-41; the-angular-observer\theorem.md:110-146
chapter 9 section 9.2	the recognizer's mechanism, low multiplets and angular distances, dimension before shape by Weyl's law, refusal, the growth gotcha	geometric-observation\chapters\ch11_the_recognizer.md:1-95
chapter 9 section 9.2	the battery, three new templates, frozen ratios, code hash, bars at least 10 of 12 and the dimension ordering, eccentricity scope	the-angular-observer\PREREG_RECOGNIZER_BATTERY.md:1-80
chapter 9 section 9.2	12 of 12, dimensions 1.81, 2.80, 1.74, angular Spearman ranges, eccentricity spreads, verdict confirmed, GO-P-2026-041	the-angular-observer\experiments\manifold-recovery\battery_result.json
chapter 9 section 9.2	causal set 0.57 with dimension 2.9 to 10.7, small world 0.45, control 0.01, rung 4 decay 0.747 to 0.679 below 0.75	the-angular-observer\README.md:99-109; wolfram-observer-bridge\rung4_scaledm_result.json; wolfram-observer-bridge\review-2.txt
chapter 11 section 11.1	condition (A2), cosine satisfies it, post-rotary keys do not, the cone below cell size	turboquant-pro\docs\KV_KEYS_FINDING.md:61-86; the-angular-observer\README.md:26-31

FAILURES AND CORRECTIONS

none

CONDITIONS

- A curved surface of some dimension that looks flat when viewed closely. A dataset lies on one when its rows vary along fewer directions than they have coordinates, and under Weyl's law the count of Laplacian eigenvalues below a level grows like the level to half that dimension, so the dimension is twice the slope of log count against log level.
- The recognizer names the manifold from the eigenvalue multiplets or certifies that none is present, and the twelve of twelve battery is the ledger's demonstrated row.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

lean/DataMiningAsObservation/IntrinsicDimension.lean, theorems log_weyl, dimension_from_slope, weyl_double, at observation-data-mining af776fd.

USED IN

Data Mining as Observation chapters 0, 3, 9.

RELATED

[intrinsic dimension](#) (p. 175); [recognizer](#) (p. 288); [graph](#) (p. 156); [Laplacian](#) (p. 199); [template match](#) (p. 348).

margin

concept. Entry id margin.

DEFINITION

How far a row's score sits from the threshold. Chapter 6.

EQUATION

Book equation 9.3.

$$\hat{\mu} = \frac{\bar{s}_{\text{true}} - \bar{s}_{\text{distr}}}{\sigma_{\text{distr}}}, \quad \mu_{\text{crit}} = \mathbb{E} \left[\max_{N-1} \mathcal{N}(0, 1) \right], \quad \rho = \frac{\hat{\mu}}{\mu_{\text{crit}}}, \quad \rho = 1 \text{ vacuous.}$$

Book equation 11.2.

$$r = \frac{d_{\text{compressed}}}{d_{\text{exact}}}, \quad \kappa_{\text{strict}} = \frac{\max r}{\min r}, \quad \tau \geq 1 - 2\hat{\mu}(\kappa_{\text{strict}}), \quad \rho_S \geq 1 - 3\hat{\mu}(\kappa_{\text{strict}}),$$

$$\kappa_{97.5/2.5} = \frac{q_{97.5}(r)}{q_{2.5}(r)} \text{ gives the same two expressions as estimates, not floors.}$$

LEDGER

- G0-3. The certificate's vacuity threshold predicts where single-stage retrieval dies. [demonstrated].
geometric-observation/claims/LEDGER.md:65 at 9f3829f.

FIRST STATED

Chapter 6 section 6.1 of *Data Mining as Observation*, with the margin certificate of chapter 9 section 9.3 and the neighbour margins of chapter 11.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

CONDITIONS

- How far a row's score sits from the threshold. A perturbation whose length times the weight length is smaller than the margin cannot move a linear classifier's score across the threshold, by Cauchy–Schwarz, and a step along the weights of exactly that size reaches the boundary.
- The margin is therefore a certificate on a decision, and chapter 9's margin certificate is the same idea with a noise model in place of a perturbation bound and a vacuity threshold derived from it.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

lean/DataMiningAsObservation/Margin.lean, theorems score_add, decision_stable, tight_along_weights, at observation-data-mining af776fd.

USED IN

Data Mining as Observation chapters 0, 6, 7, 9, 11, 12, 14.

RELATED

[classifier](#) (p. 60); [decision boundary](#) (p. 94); [vacuity threshold](#) (p. 359); [rank certificate](#) (p. 275); [escalation](#) (p. 123).

matched bits

instrument. Entry id matched-bits.

DEFINITION

The rule that two codes are compared only at the same bit count. At matched bits two codes with equal reconstruction error can still differ for a reader, which is the flip. Chapter 4 section 4.3.

EQUATION

Book equation 4.2.

$$D(b) = \sum_i s_i \sigma_i^2 2^{-2b_i}, \quad s_i = v_i^\top P_C v_i, \quad b_i^* = \max\left(0, \frac{1}{2} \log_2 \frac{s_i \sigma_i^2}{\theta}\right), \quad \sum_i b_i^* = B,$$

Book equation 4.6.

$$\kappa = \text{tr}(\bar{P}_C \bar{\Sigma}_x) \in [0, 1], \quad \kappa \rightarrow 1 \text{ is the coupling null (no flip).}$$

Book equation 0.15.

$$\tau = \frac{\#\{\text{concordant pairs}\} - \#\{\text{discordant pairs}\}}{n(n-1)/2}.$$

LEDGER

- G0-2 (neg. half: not reconstruction). At matched bits, downstream preservation is not controlled by reconstruction error: [demonstrated]. [geometric-observation/claims/LEDGER.md:63](#) at 9f3829f.
- G0-2 (pos. half: consumer-projected covariance *controls*). Downstream preservation is controlled by the error covariance projected on the consumer's read subspace, $\text{tr}(P_C \Sigma_\delta)$. [replicated]. [geometric-observation/claims/LEDGER.md:64](#) at 9f3829f.
- NEG-4. Lightweight online (Lloyd) key calibration beats the calibration-free default on softmax-KL. [refuted]. [geometric-observation/claims/LEDGER.md:97](#) at 9f3829f.

FIRST STATED

Chapter 4 section 4.3 of *Data Mining as Observation*, with the bit-matched redesign in [geometric-observation/chapters/ch08_value](#). 30.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 1 section 1.4	cosine 0.995 and perplexity of order ten thousand, the recalibration negative	geometric-observation\chapters\ch02_failure_of_observer_free_60; geometric-observation\chapters\ch16_honest_negatives.md NEG-2 and NEG-4; turboquant-pro\docs\KV_KEYS_FINDING.md:1-49
chapter 3 section 3.3	the self-refuted v0.8 claim, NEG-1	the-angular-observer\README.md:170-175; geometric-observation\chapters\ch16_honest_negatives.md NEG-1
chapter 4 section 4.3	the flip definition, flip versus (A2) verdict	geometric-observation\chapters\ch08_value.md:1-30,100-108
chapter 4 section 4.3	GO-2 0.0934 vs 0.0938, 2.53 times, 12 of 12, anti 21 times; retrieval 0.0964, negative 4.70 and positive 4.65, recon 0.40	geometric-observation\claims\LEDGER.md rows GO-2 negative and positive halves
chapter 4 section 4.6	NEG-4 through NEG-10, the 25 percent codebook confound, Spearman 0.80, the recon-matched precondition	geometric-observation\chapters\ch16_honest_negatives.md NEG-4 to NEG-10
chapter 6 section 6.2	real model, median 0.567 vs bar 0.60, about 4.5 times chance, heads at 0.815 and 0.958, sixteen of sixteen both, two of three triggers, NEG-12, later rematch four of four	geometric-observation\claims\LEDGER.md rows GO-B-Llama and NEG-12; geometric-observation\chapters\ch16_honest_negatives.md:73-81
chapter 8 section 8.2	cosine 0.995, perplexity near 1e4	turboquant-pro\docs\KV_KEYS_FINDING.md:1-49; geometric-observation\chapters\ch16_honest_negatives.md NEG-2
chapter 8 section 8.2	recalibration improves reconstruction and worsens the consumer	geometric-observation\chapters\ch16_honest_negatives.md NEG-4

FAILURES AND CORRECTIONS

- NEG-4, [refuted]. Lightweight online (Lloyd) key calibration beats the calibration-free default on softmax-KL.

CONDITIONS

- The rule that two codes are compared only at the same bit count, since each added bit halves the step and quarters the squared error. At matched bits two codes with equal reconstruction error can still differ for a reader, and the flip is that difference.
- A comparison that was not bit-matched was refuted by the twenty-five percent codebook confound, and the redesign matched bits before any verdict was read.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

lean/DataMiningAsObservation/Bit.lean, theorems levels_succ, levels_add, step_succ, sqError_succ, sqError_antitone, levels_zero, at observation-data-mining af776fd.

lean/DataMiningAsObservation/Isotropy.lean, theorems isotropic_reads_same, isotropic_no_flip, anisotropic_readers_differ, flip_iff_anisotropic, at observation-data-mining af776fd.

USED IN

Data Mining as Observation chapters 4, 6, 12.

RELATED

[bit](#) (p. 38); [budget](#) (p. 46); [quantization](#) (p. 266); [control](#) (p. 77).

metric

concept. Entry id metric.

DEFINITION

A rule for the distance between two rows or two outputs. A distance is a choice of what to ignore, and a local metric on an output is a positive semidefinite matrix. Chapter 3 section 3.1 and chapter 0 section 0.5.

EQUATION

Book equation 3.1.

$$d_O(u) = u^\top \Sigma u, \quad u = (\cos 15^\circ, \sin 15^\circ), \\ \Sigma_1 = \text{diag}(0.3, 1.7), \Sigma_2 = \text{diag}(1.7, 0.3), \quad d_O = 0.394 \text{ vs } 1.606.$$

Book equation 0.11.

$$P_C(x) = J(x)^\top G(C(x)) J(x), \quad J(x) = \frac{\partial C}{\partial x}(x), \quad \bar{P}_{C,\mu} = \mathbb{E}_\mu[P_C(x)].$$

Book equation 1.1.

$$O = (C, G, B).$$

LEDGER

- GO-EC-3. A read operator recovered from a black-box consumer by query-only finite-difference probing, composed with the Kalman covariance as $\text{tr}(\hat{P}_C \Sigma)$, prospectively selects sensors that improve the held-out consumer at matched budgets with probe cost charged — capturing 94.6% of the known analytic optimum's gain on the positive-control arm (gate $\geq 75\%$) and improving 16.3% over the best consumer-agnostic policy on non-analytic consumers (gate $\geq 8\%$), with trace-matched ordering carried by the composition at 86.9% over 61 pairs (gate $\geq 65\%$). [predicted]. [geometric-observation/claims/LEDGER.md:162](#) at 9f3829f.

FIRST STATED

Chapter 3 section 3.1 and chapter 0 section 0.5 of *Data Mining as Observation*, with the read metric in [geometric-observation/chapters/ch05_the_read_metric_and_the_quotient.md:7-48](#).

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 2 section 2.2	read subspace small, operator local, pullback composition, rank cannot increase	geometric-observation\chapters\ch05_the_read_metric_and_the_c48; geometric-observation\chapters\ch06_mathematical_preliminaries27
chapter 3 section 3.2	traces 2.0, diagonals 0.3 and 1.7, consumers at 15 and 75 degrees, distortion 0.39 vs 1.61, 4.1 to one, computed not drawn	observation-theory\assets\make_flip_figure.py:4-24,107-119

FAILURES AND CORRECTIONS

none

CONDITIONS

- A rule for the distance between two rows, or between two outputs. A distance function is a choice of what to ignore, and the local metric on a consumer's output is the positive semidefinite matrix that says how a small change is scored. The squared Euclidean distance is symmetric and zero exactly between equal rows.
- A dataset-level loss, accuracy, F1, or a rank correlation is not a local metric and cannot be inserted into the read-operator formula, and the same two codes get opposite verdicts from two metrics.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/EuclideanDistance.lean`, theorems `distSq_eq_quad_one`, `distSq_comm`, `distSq_eq_zero_iff`, `ranking_flips`, at `observation-data-mining af776fd`.

`lean/DataMiningAsObservation/OutputMetric.lean`, theorems `neg_reverses`, `readOp_of_neg`, `cost_flips`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14.

RELATED

[output metric](#) (p. 243); [Euclidean distance](#) (p. 124); [read operator](#) (p. 282); [quotient](#) (p. 271); [geodesic distance](#) (p. 154).

min-over-strata

concept. Entry id `min-over-strata`.

DEFINITION

The rule that a verdict over several groups is the worst group's verdict, with abstention counted as a verdict, never the average. Chapter 10.

EQUATION

Book equation 10.7.

$$\text{verdict} = \min_{i: n_i \geq n_{\min}, |Q_i| \geq q_{\min}} \text{score}_i, \quad \text{ABSTAIN otherwise.}$$

Book equation 14.5.

$$FC_g = \Pr[y = \text{violation} \mid \hat{y} = \text{clear}, g], \quad \text{verdict} = \max_{g: n_g \geq n_{\min}} FC_g, \quad \text{ABSTAIN otherwise.}$$

LEDGER

- NEG-14. (GO-P-2026-037, prospective) a2_probe.median_unit_displacement is a single-statistic predictor of the flip regime ($\text{unit_disp} \geq 1.0 \Rightarrow \text{generic-polar-flip vs needs-blind-probe}$). [refuted]. geometric-observation/claims/LEDGER.md:108 at 9f3829f.

FIRST STATED

The compression program's stratified evaluation, STRATA, in turboquant-pro, DOI 10.5281/zenodo.20660087, and chapter 10 section 10.4 of *Data Mining as Observation*.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 2 section 2.6	ball-growth heat a missing-mean artifact, eleven campaigns, 3 of 12 to 11 of 12 cells	openvector-bench\README.md:160-170; openvector-bench\results\RC13_VERDICT.md:1-20
chapter 3 section 3.5	Poisson null, hub excess, budget parameter	openvector-bench\openvector_bench\hubness.py:41-100
chapter 3 section 3.5	query mass best single feature at every K for all four responses, seven features add at most 20 percent, threshold 1.25 on three of four, zero of four, 1000 real queries, 1024 dimensions	openvector-bench\results\R13_STAGE0_RESULT.md:40-50; openvector-bench\results\R13_STAGE1_RESULT.md:1-40
chapter 8 section 8.1	G1 339.9 vs 71.9 same corpus, real about 61, rounds 1 to 4 at 300 to 420, three families falsified	openvector-bench\results\QUERY_COUPLING_ARTIFACT.md:1-20
chapter 8 section 8.4	gate meta-rule, relative contrast discriminates nothing	openvector-bench\openvector_bench\score_rc1.py:1-14; turboquant-pro\docs\RESEARCH_ROADMAP.md:133-160
chapter 8 section 8.7	RC-7 five of ten, band 0.076 vs 0.19, gate missed by 0.007, 2.4 times, eight-block rule	openvector-bench\paper\profile\PROFILE_PAPER.md:576-598; openvector-bench\results\RC7_VERDICT.md
chapter 8 section 8.10	the AI-assistant boundary, fresh-context passes	geometric-observation\chapters\ch15_registration_first.md; geometric-observation\chapters\ch17_open_problems_and_the_key 96
chapter 10 section 10.2	five kinds, balanced accuracy 0.611, 0.718, 0.684, the category table, per-category 0.57 to 0.82, the sweep, pigeonhole floor	openvector-bench\results\R13_STAGE1_RESULT.md:40-75

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 10 section 10.2	half 2 failed, at most 1.28 times against 2, withdrawal and its scope	openvector-bench\results\R13_STAGE1_RESULT.md:75-110
chapter 10 section 10.4	area map, boundary rule, hash, refuse not warn, intra and transit counts, area classes, abstention rule	turboquant-pro\docs\STRATA_RFC.md:24-98
chapter 10 section 10.4	first run, 350000 of 2391361 rows, 7 of 14 eligible, abstentions with 2 and 1 rows, Robin Hood 0.3447 to 0.4469 ratio 1.30 against 1.5, skew 2.68 to 4.40 ratio 1.64 against 3, 63 times sample range	turboquant-pro\docs\RESULTS_multilingual_strata.md:1-55
chapter 10 section 10.4	second prediction inverted, transit 0.389 vs 0.291, centrality difference signs, share 0.473 vs 0.391, seven of thirteen backbone	turboquant-pro\docs\RESULTS_multilingual_strata.md:55-90; turboquant-pro\docs\STRATA_RFC.md:98-130
chapter 10 section 10.5	Gate A design error; A prime skew 3.970 to 3.177, max 287 to 213, Robin Hood 0.372 to 0.261, fraction 0.117 to 0.079, compressed path 0.663 vs 0.90, seven strata, 0.62 to 0.69 vs 0.76 to 0.84	turboquant-pro\docs\RESULTS_strata_phase23_gates.md:1-45
chapter 10 section 10.5	rerank identical 0.6627, candidate coverage, depth curve 51 to 501, coverage 0.696 to 0.926, fidelity 0.663 to 0.702, originals rerank 0.923, 4096 bytes on 148	turboquant-pro\docs\RESULTS_strata_phase23_gates.md:45-110
chapter 10 section 10.5	fragile-first allocation 0.7118 vs 0.7251, targets up 0.004 to 0.034, redesigned allocator passed	turboquant-pro\docs\RESULTS_strata_phase23_gates.md:110-125
chapter 11 section 11.4	0.999 against own ranking vs 0.592 against fp32 truth, three truth layers, difficulty strata	openvector-bench\README.md:60-96
chapter 11 section 11.4	R80 table, 25x probe depth, 92 percent own cell, one third cross-article, 0.53 matches same-article share, relative contrast discriminates nothing	openvector-bench\results\R80_ANN.md:1-40; turboquant-pro\docs\RESEARCH_ROADMAP.md:133-160
chapter 11 section 11.6	query mass best single feature at every K for all four responses, seven features add at most 20 percent at 12 leaves, threshold 1.25 on three of four, zero of four, 1000 real queries, 1024 dimensions	openvector-bench\results\R13_STAGE0_RESULT.md:40-50; openvector-bench\results\R13_STAGE1_RESULT.md:1-40
chapter 12 section 12.2	recall 0.999 vs 0.592, the derived death point within 6 percent across fourteen corpora	openvector-bench\README.md:60-96; geometric-observation\claims\LEDGER.md row GO-3
chapter 12 section 12.4	seven of thirteen backbone areas under the translation-trained encoder, all local-hub under the general one	turboquant-pro\docs\RESULTS_multilingual_strata.md:55-90

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 13 section 13.1	one hundred billion vectors on a preemptible fleet, systems result not a tier, corpus rejected by the admission battery	openvector-bench\README.md:250-263
chapter 13 section 13.1	twelve-figure corpus about 128 TB, kilobyte manifest	openvector-bench\README.md:60-80

FAILURES AND CORRECTIONS

- NEG-14, [refuted]. (GO-P-2026-037, prospective) `a2_probe.median_unit_displacement` is a single-statistic predictor of the flip regime ($\text{unit_disp} \geq 1.0 \Rightarrow \text{generic-polar-flip vs needs-blind-probe}$).

CONDITIONS

- A verdict over several groups is the worst group's verdict, with a group too small to score counted as an abstention and reported, never averaged away.
- The first stratified run made two wrong predictions, and the record carries them at the size of the result.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/Certificate.lean`, `theorems falseClear_mul_coverage`, `coverage_empty`, `falseClear_mem_unit`, `minOverStrata_passes_iff`, `minOverStrata_le_weighted_mean`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 10, 11, 14.

RELATED

[false-clear rate](#) (p. 133); [coverage](#) (p. 86); [hubness](#) (p. 168); [certificate](#) (p. 55).

Monotone Invariance Theorem

result. Entry id `monotone-invariance`.

DEFINITION

A strictly monotone transform of a score cannot change its optimal thresholded F1 or its AUROC, because both depend only on the ranking. Equation 6.2.

EQUATION

Book equation 6.2.

$$\max_{\tau, \text{dir}} F_1\left(\mathbf{1}[\text{dir}(g(f(X)), \tau)], y\right) = \max_{\tau, \text{dir}} F_1\left(\mathbf{1}[\text{dir}(f(X), \tau)], y\right)$$

for every strictly monotone g .

LEDGER

none

FIRST STATED

`theory-radar`, `theory-radar/paper/theory_radar_paper.tex:290-304` and `theory-radar/paper/astar_paper.tex:94-110`, DOI 10.5281/zenodo.20660206, and chapter 6 section 6.3 of *Data Mining as Observation*.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 5 section 5.4	Monotone Invariance and AUROC invariance theorems, the AUROC to F1 bound via the Youden index, stated in the source for every ROC curve and corrected in the book to concave curves, with the counterexample at AUROC 0.75 and F1 0.857	theory- radar\paper\astar_paper.tex:94-170; theory- radar\paper\theory_radar_paper.tex:290-304
chapter 6 section 6.3	Monotone Invariance Theorem and its proof	theory- radar\paper\theory_radar_paper.tex:290-304; theory- radar\paper\astar_paper.tex:94-110

FAILURES AND CORRECTIONS

none

CONDITIONS

- A strictly monotone transform of a score leaves AUROC, defined pairwise with ties counted one half, and optimal thresholded F1, defined by sweeping the threshold over every score value in both directions, unchanged, because both depend only on the ranking.
- The theorem licenses pruning every monotone unary node of a formula search with zero loss. It says nothing about non-monotone transforms, which can change both quantities.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/MonotoneInvariance.lean`, `theorems aurocNum_comp`, `auroc_comp`, `predicted_comp`, `predictedBelow_comp`, `sweptF1_comp`, `sweptF1Below_comp`, `optF1_comp`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 5, 6, 12.

RELATED

[formula search](#) (p. 145); [safe pruning](#) (p. 311); [Youden F1 bound](#) (p. 372).

multiple comparisons

concept. Entry id `multiple-comparisons`.

DEFINITION

The inflation of false positives when many hypotheses are tested and the best is reported. The Bonferroni correction divides the threshold by the number of tests. Chapter 0 section 0.9.

EQUATION

Book equation 8.3.

$$\Pr[\text{at least one of } m \text{ null tests passes}] = 1 - (1 - \alpha)^m, \quad \alpha_{\text{Bonferroni}} = \frac{\alpha}{m}.$$

LEDGER

none

FIRST STATED

Chapter 0 section 0.9 and chapter 8 section 8.5 of *Data Mining as Observation*, with the program's own case in `constraint-gap/review/FINDINGS.md:1-35`.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 6 section 6.4	the uncorrected t stored as sigma, 251 and 15.84, nine wins to three, seven, eight, 3 wins 17 ties 11 losses on 31 datasets, the three-way inconsistency	<code>constraint-gap\review\FINDINGS.md:1-35</code> ; <code>constraint-gap\README.md:57-76</code>
chapter 6 section 6.4	five analyses negative, 0.2 percent of floor, twelve times the band, learning-curve redesign	<code>constraint-gap\README.md:57-76</code> ; <code>constraint-gap\review\REDESIGN.md:1-45</code>
chapter 7 section 7.3	correlation negative 0.59 at p 4.8e-4 with size, negative 0.562 at p 0.001 with the ensemble's score, predictive 0.345 and 0.589, interval negative 0.165 to 0.125, the one-line summary	<code>constraint-gap\review\INDETERMINATES.md:60-80</code> ; <code>constraint-gap\review\REDESIGN.md:45-70</code>
chapter 7 section 7.4	the redesign, grid 50 to N, ten folds, twenty repetitions, crossing size, the 800-row boundary declared in advance, seventeen at median 0.006 and p 0.964, store the folds	<code>constraint-gap\review\REDESIGN.md:1-110</code>
chapter 8 section 8.5	variance inflation 251, SE inflation 15.84, J_{eff} 3.98, about 62 needed, 3 of 9 at full, 7 at half, 8 at a third	<code>constraint-gap\review\FINDINGS.md:1-35</code>
chapter 8 section 8.5	9/3/19 to 3/17/11, five negative analyses, 0.2 percent of floor, twelve times the band, learning-curve redesign	<code>constraint-gap\README.md:57-76</code> ; <code>constraint-gap\review\REDESIGN.md</code>

FAILURES AND CORRECTIONS

none

CONDITIONS

- Test many hypotheses and report the best, and the reported score is the maximum of many noisy draws, at least any single draw. Ten thousand null tests at five percent pass five hundred by chance, forty pass two, and the expected passes grow without bound in the number of tests.
- The Bonferroni correction divides the threshold by the number of tests and keeps the family-wise rate under the target, and a preregistration that names one look needs no correction.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/MultipleComparisons.lean`, `theoremsmax_ge_each`, `book_numbers`, `corrected_family_rate`, `expectedPasses_unbounded`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 5, 8.

RELATED

[harness](#) (p. 159); [Nadeau and Bengio correction](#) (p. 225); [preregistration](#) (p. 258); [formula search](#) (p. 145).

multiplet

concept. Entry id multiplet.

DEFINITION

A group of eigenvalues that are equal or nearly so, whose pattern names the shape a dataset lies on. Chapter 9.

EQUATION

Book equation 0.19.

$$L = I - D^{-1/2} A D^{-1/2}, \quad D = \text{diag}(d_1, \dots, d_n), \quad L u_k = \lambda_k u_k, \quad 0 = \lambda_1 \leq \lambda_2 \leq \dots.$$

Book equation 0.31.

$$N(\lambda) = \#\{k : \lambda_k \leq \lambda\} \sim C_d \lambda^{d/2}.$$

Book equation 9.2.

$$N(\lambda) = \#\{k : \lambda_k \leq \lambda\} \sim C_d \lambda^{d/2} \quad \Rightarrow \quad d = 2 \frac{d \log N}{d \log \lambda}.$$

LEDGER

none

FIRST STATED

Chapter 0 section 0.15 and chapter 9 section 9.2 of *Data Mining as Observation*, with the program's recognizer in `geometric-observation/chapters/ch11_the_recognizer.md:1-95`.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 9 section 9.2	the recognizer's mechanism, low multiplets and angular distances, dimension before shape by Weyl's law, refusal, the growth gotcha	<code>geometric-observation\chapters\ch11_the_recognizer.md:1-95</code>
chapter 9 section 9.2	the battery, three new templates, frozen ratios, code hash, bars at least 10 of 12 and the dimension ordering, eccentricity scope	<code>the-angular-observer\PREREG_RECOGNIZER_BATTERY.md:1-80</code>
chapter 9 section 9.2	12 of 12, dimensions 1.81, 2.80, 1.74, angular Spearman ranges, eccentricity spreads, verdict confirmed, GO-P-2026-041	<code>the-angular-observer\experiments\manifold-recovery\battery_result.json</code>

FAILURES AND CORRECTIONS

none

CONDITIONS

- A group of eigenvalues that are equal or nearly so, whose pattern names the shape a dataset lies on. The cycle's Laplacian eigenvalue at k equals the one at n minus k , so every eigenvalue of the discrete circle other than the constant mode and, for even n , the alternating mode appears twice.

- Symmetric shapes produce multiplets, the sphere in groups of three, five, seven, and a product of shapes has the product of their patterns. The recognizer reads the pattern of the lowest eigenvalues against stored templates and refuses when none matches.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/Multiplet.lean`, theorems `cycleEig_zero`, `cycleEig_nonneg`, `cycleEig_le_four`, `cycleEig_pair`, `pair_distinct`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 9.

RELATED

[Laplacian](#) (p. 199); [recognizer](#) (p. 288); [manifold](#) (p. 210); [Weyl's law](#) (p. 367); [eigenvalue, eigenvector](#) (p. 116).

N

Nadeau and Bengio correction

correction. Entry id nadeau-and-bengio-correction.

DEFINITION

The inflation of the variance of a repeated cross-validation estimate by one plus the ratio of test to training size times the number of repeats, because the folds are not independent. Equation 0.18.

EQUATION

Book equation 0.18.

$$\widehat{\text{Var}}_{\text{NB}} = \left(\frac{1}{J} + \frac{n_{\text{test}}}{n_{\text{train}}}\right)\hat{\sigma}^2, \quad \frac{\widehat{\text{Var}}_{\text{NB}}}{\hat{\sigma}^2/J} = 1 + J \frac{n_{\text{test}}}{n_{\text{train}}}.$$

Book equation 8.2.

$$\widehat{\text{Var}}_{\text{NB}} = \left(\frac{1}{J} + \frac{n_{\text{test}}}{n_{\text{train}}}\right)\hat{\sigma}^2, \\ J = 1000, \frac{n_{\text{test}}}{n_{\text{train}}} = \frac{1}{4} \Rightarrow 1 + 250 = 251, \sqrt{251} = 15.84.$$

LEDGER

none

FIRST STATED

Nadeau and Bengio, inference for the generalization error; 2003, as chapter 8 cites it, applied to the program's own comparison in `constraint-gap/review/FINDINGS.md:1-35` and chapter 8 section 8.5 of *Data Mining as Observation*.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 6 section 6.4	the uncorrected t stored as sigma, 251 and 15.84, nine wins to three, seven, eight, 3 wins 17 ties 11 losses on 31 datasets, the three-way inconsistency	<code>constraint-gap\review\FINDINGS.md:1-35</code> ; <code>constraint-gap\README.md:57-76</code>
chapter 6 section 6.4	the reviewer's revision plan, drop sigma, bounded claim, 19 at 200 by 5 vs 11 at 20 by 5, rerun in progress	<code>theory-radar\paper\REVISION_PLAN.md:1-60</code>
chapter 6 section 6.4	five analyses negative, 0.2 percent of floor, twelve times the band, learning-curve redesign	<code>constraint-gap\README.md:57-76</code> ; <code>constraint-gap\review\REDESIGN.md:1-45</code>
chapter 6 section 6.5	loading weights and stability across folds requested	<code>theory-radar\paper\REVISION_PLAN.md</code> issue 6

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 7 section 7.3	correlation negative 0.59 at p 4.8e-4 with size, negative 0.562 at p 0.001 with the ensemble's score, predictive 0.345 and 0.589, interval negative 0.165 to 0.125, the one-line summary	constraint-gap\review\INDETERMINATES.md:60-80; constraint-gap\review\REDESIGN.md:45-70
chapter 7 section 7.4	the redesign, grid 50 to N, ten folds, twenty repetitions, crossing size, the 800-row boundary declared in advance, seventeen at median 0.006 and p 0.964, store the folds	constraint-gap\review\REDESIGN.md:1-110
chapter 8 section 8.5	variance inflation 251, SE inflation 15.84, J_eff 3.98, about 62 needed, 3 of 9 at full, 7 at half, 8 at a third	constraint-gap\review\FINDINGS.md:1-35
chapter 8 section 8.5	9/3/19 to 3/17/11, five negative analyses, 0.2 percent of floor, twelve times the band, learning-curve redesign	constraint-gap\README.md:57-76; constraint-gap\review\REDESIGN.md
chapter 8 section 8.5	three-way inconsistency, lines 544, 548 to 556, 755	constraint-gap\review\FINDINGS.md:7-31 against theory-radar\paper\theory_radar_v6_submission.tex

FAILURES AND CORRECTIONS

- theory-radar/paper/REVISION_PLAN.md:39-45 at 37c4e6c. **Fix:** - Drop “ σ significance” everywhere - Report: mean $\Delta F1$, std, 95% CI, and corrected resampled t-statistic - Table columns: “Test F1”, “GB F1”, “ $\Delta F1$ ”, “95% CI”, “Direction” - Text: “the formula outperforms GB by $\Delta F1=0.031$, 95% CI [0.028, 0.034]” - Add Nadeau-Bengio corrected t-test (accounts for CV fold correlation) - Keep effect size as supplementary but don't call it “ σ ”

CONDITIONS

- Repeated cross-validation folds share training data, so the naive variance of a mean difference is multiplied by one plus the fold count times the test-to-train ratio. The corrected variance is never below the naive one and the inflation grows without bound in the fold count.
- With a thousand folds at a ratio of one quarter the inflation is 251, the standard error grows by 15.84, and about four folds' worth of independent evidence remain. The program's nine wins became three under it, with seventeen indeterminate.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

lean/DataMiningAsObservation/NadeauBengio.lean, theorems correctedVar_eq, naiveVar_le_correctedVar, inflation_unbounded, book_numbers, at observation-data-mining af776fd.

USED IN

Data Mining as Observation chapters 0, 6, 7, 8.

RELATED

[harness](#) (p. 159); [formula search](#) (p. 145); [preregistration](#) (p. 258); [multiple comparisons](#) (p. 221).

naive Bayes

instrument. Entry id naive-bayes.

DEFINITION

A classifier that multiplies one-dimensional likelihoods and adds their logs, so that each feature's contribution is the same whatever the others are and it cannot read an interaction. Chapter 6.

EQUATION

Book equation 6.1.

$$s(x) = w \cdot x + b, \quad P_C = \mathbb{E}[\sigma'(s)^2] w w^\top, \quad \text{rank } P_C = 1.$$

Book equation 0.9.

$$P_C = \mathbb{E}[g g^\top], \quad g = \nabla C(x).$$

LEDGER

none

FIRST STATED

Chapter 6 section 6.1 of *Data Mining as Observation*, after TSK chapter 4.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 2 section 2.3	refusal regimes, selection consumers read order, recurrences compound	readscope\readscope\regimes.py:1-60
chapter 6 section 6.1	the classifier row of the consumer table, the output metric makes a different observer	geometric-observation\chapters\ch04_the_observer_triple.md:60-135
chapter 6 section 6.1	selection consumers have zero sensitivity almost everywhere	readscope\readscope\regimes.py:1-60

FAILURES AND CORRECTIONS

none

CONDITIONS

- A classifier that multiplies one-dimensional likelihoods and adds their logs. The log of the product is the sum of the logs, the log odds add across features, and one feature's contribution is the same whatever the others are, so the score has no interaction terms and its Hessian is diagonal.
- What naive Bayes cannot read is the interaction, which is a limit of the model class rather than a direction in its kernel, and its read operator is still not diagonal in general because two sensitivities can rise and fall together across rows.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/NaiveBayes.lean`, `theorems log_likelihood_sum, log_odds_sum, contribution_independent`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 6.

RELATED

[classifier](#) (p. 60); [Hessian](#) (p. 164); [confidence](#) (p. 70); [logistic regression](#) (p. 208).

nat

concept. Entry id nat.

DEFINITION

The unit of information measured with the natural logarithm, between 1.4426 and 1.4427 bits. Chapter 0 section 0.7.

EQUATION

Book equation 0.15.

$$\tau = \frac{\#\{\text{concordant pairs}\} - \#\{\text{discordant pairs}\}}{n(n-1)/2}.$$

Book equation 4.2.

$$D(b) = \sum_i s_i \sigma_i^2 2^{-2b_i}, \quad s_i = v_i^\top P_C v_i, \quad b_i^* = \max\left(0, \frac{1}{2} \log_2 \frac{s_i \sigma_i^2}{\theta}\right), \quad \sum_i b_i^* = B,$$

Book equation 0.16.

$$\text{AUROC} = \Pr[s^+ > s^-] + \frac{1}{2} \Pr[s^+ = s^-].$$

LEDGER

- GO-7. A stored description's description rate and its conditional Landauer reset content are operationally separate resources: the same finite- n code index needing $\hat{R} \approx 0.67$ bits/symbol to describe is fully recoverable from retained side information at bin rate $0.26 = 0.39\hat{R}$, fails increasingly below its conditional content, and fails absolutely (err 1.00 at every bin rate) without S . [replicated]. `geometric-observation/claims/LEDGER.md:69 at 9f3829f`.
- NEG-13 → resolved. (GO-P-2026-026, prospective, real LLM) Appendix-E's omission floor is a rate-irreducible downstream wall on trained Llama read operators. [missed]. `geometric-observation/claims/LEDGER.md:106 at 9f3829f`.

FIRST STATED

Chapter 0 section 0.7 of *Data Mining as Observation*, with the omission floor in nats in `geometric-observation/chapters/ch07_cost.md`.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 2 section 2.4	C-7, sixteen points in 128 dimensions, rank fifteen, near ten to the eleventh nats, refuses when samples do not exceed dimension, warns below five per dimension, loading is a property of two distributions	<code>readscope\SPEC.md:653-680</code>
chapter 2 section 2.4	C-7b, 92.1 percent, fit range 0.89 to 91.64, 15 percent inside, maximum 1.9e12, endpoint attenuation 0.437, 202 of 240 at 1.0, refuses to extrapolate, loading is a warning	<code>readscope\SPEC.md:680-720</code> ; <code>readscope\readscope\loading.py:136-219</code>

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 2 section 2.6	0.647 published, 1.000 recovered, weighted vs unweighted median 0.796 range 0.678 to 0.985, probe 1.000 on every cell, 36 cells, one cell at 0.985	readscope\SPEC.md:410-450
chapter 2 section 2.6	C-10, 24 to 576 queries, reference rank 24 to 128, 0.821 to 0.703 vs 0.647, 68 percent of distance closed, 0.174 to 0.056, probe 1.000000 on 16 cells, residual uncontrolled	readscope\SPEC.md:450-500
chapter 4 section 4.2	GO-6, d 8 and r 4, output at or below surrogate at or below reconstruction at every rate, about 500 times, gap 0.41 to 0.005, isotropic control collapses	geometric-observation\claims\LEDGER.md row GO-6; geometric-observation\chapters\ch07_cost.md
chapter 4 section 4.2	commission tax of order half r log M over m, omission floor, unbounded corrected to floor, naive projector 30 percent over, VI-6	geometric-observation\chapters\ch07_cost.md section on mismatch; geometric-observation\claims\LEDGER.md VI-6
chapter 4 section 4.2	floor measured on 16 of 16 heads, naive assumption off 20 to 60 percent, about 100000 times, 0.10 nats against about 6e-7, first gate missed 0 of 16, NEG-13	geometric-observation\chapters\ch07_cost.md; geometric-observation\claims\LEDGER.md GO-P-2026-027 and NEG-13
chapter 8 section 8.4	C-11c paired null, rank 2 positional 0.385 vs null 0.572, 0.615 vs 0.224, 14 of 16 cells, scope 16 cells one 3B model 192 positions	readscope\SPEC.md:806-857; readscope\calibration\records\c11c-operator-drift.json
chapter 11 section 11.3	tau floor 1 minus 2 mu, Spearman floor 1 minus 3 mu, example kappa 1.0148, mu 0.0664, floor 0.8671, 19900 pairs, max certifiable 1.83, vacuous means exact rerank, sha256 binding, reference field and the 0.3 overlap difference	turboquant-pro\docs\CERTIFICATE_SPEC.md:1-80
chapter 11 section 11.7	k over d table 1, 2, 2, 16, 16, 16	readscope\README.md:118-141; readscope\SPEC.md:243-285, record readscope\calibration\records\c2e-budget-law.json
chapter 11 section 11.7	C-15 equal budget, 3072 observations, medians 0.02, 0.11, 0.05, 0.04, 1.0, 1.0, margin 0.883 vs 0.3, 0.062 to 0.057 at 8x	readscope\SPEC.md:285-323, record readscope\calibration\records\c15-budget-surface.json, readscope\calibration\DECLARATION-C15.md
chapter 11 section 11.9	C-12 four bars, 40 documents, 512 tokens, 13.4 point difference, teacher forcing removes it, negative 0.015 vs 0.005, Spearman negative 0.13 at p 0.45, sign test p 0.42, verdict FAIL, feedback compounding	readscope\calibration\records\c12-longgen-drift-sym.json; readscope\calibration\DECLARATION-C12.md at commit 90e2ce2; readscope\SPEC.md:806-825

FAILURES AND CORRECTIONS

none

CONDITIONS

- The unit of information measured with the natural logarithm, as a bit is measured with the logarithm to base two. The two conversions undo each other, the conversion is monotone, and one nat is between 1.4426 and 1.4427 bits.
- The omission floor was measured at 0.10 nats against about six in ten million on sixteen of sixteen heads, and Landauer's bound is stated per nat as kT and per bit as kT times the log of two.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/NatUnit.lean`, theorems `log_two_pos`, `nats_bits`, `bits_nats`, `bitsOfNats_mono`, `one_nat_in_bits`, at `observation-data-mining af776fd`.

`lean/DataMiningAsObservation/Bit.lean`, theorems `levels_succ`, `levels_add`, `step_succ`, `sqError_succ`, `sqError_antitone`, `levels_zero`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 2, 4.

RELATED

[bit](#) (p. 38); [budget](#) (p. 46); [Landauer's principle](#) (p. 194); [perplexity](#) (p. 250); [KL divergence](#) (p. 187).

nearest neighbour

concept. Entry id nearest-neighbour.

DEFINITION

A row at minimal distance from a query. A strictly increasing transform of the distance names the same nearest neighbours, so the classifier reads the distance's order and nothing else. Chapter 6.

EQUATION

Book equation 3.1.

$$d_O(u) = u^\top \Sigma u, \quad u = (\cos 15^\circ, \sin 15^\circ), \\ \Sigma_1 = \text{diag}(0.3, 1.7), \Sigma_2 = \text{diag}(1.7, 0.3), \quad d_O = 0.394 \text{ vs } 1.606.$$

Book equation 0.19.

$$L = I - D^{-1/2} A D^{-1/2}, \quad D = \text{diag}(d_1, \dots, d_n), \quad L u_k = \lambda_k u_k, \quad 0 = \lambda_1 \leq \lambda_2 \leq \dots.$$

Book equation 10.3.

$$N_k(x \mid Q) = |\{q \in Q : x \in \text{top}_k(q)\}|, \quad \text{anti-hub} : N_k(x) = 0.$$

LEDGER

- NEG-11. (GO-5, prospective ×4) The $\alpha=1$ density/hubness quotient decisively and density-specifically restores invariant fidelity in a non-spectral domain. [refuted]. `geometric-observation/claims/LEDGER.md:104` at 9f3829f.

FIRST STATED

Cover and Hart, nearest neighbor pattern classification, 1967, as chapter 6 section 6.1 of *Data Mining as Observation* reads it, with the neighbour lists of `openvector-bench/openvector_bench/hubness.py:41-100`.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 3 section 3.5	78 vs 369, density correlation about 0.67, 8 percent vs 34 percent, centering vs mutual-proximity rescaling	turboquant-pro\docs\HUBNESS_PRIMER.md:38-59,140-160
chapter 3 section 3.5	Poisson null, hub excess, budget parameter	openvector-bench\openvector_bench\hubness.py:41-100

FAILURES AND CORRECTIONS

- NEG-11, [refuted]. (GO-5, prospective ×4) The $\alpha=1$ density/hubness quotient decisively and density-specifically restores invariant fidelity in a non-spectral domain.

CONDITIONS

- A row at minimal distance from a query under a chosen distance. A strictly increasing transform of the distance names the same nearest neighbours, so a nearest-neighbour classifier reads the distance's order and nothing else, a row is its own nearest neighbour, and two nearest neighbours sit at the same distance.
- Under Euclidean distance the classifier reads every standardized coordinate equally and under cosine it reads the angle, so it inherits the distance's quotient, and the rows that appear in many neighbour lists are the hubs.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/NearestNeighbour.lean`, theorems `nearest_comp`, `nearest_smul`, `self_nearest`, `nearest_same_distance`, at `observation-data-mining af776fd`.

`lean/DataMiningAsObservation/Graph.lean`, theorems `outDegree`, `mutual_symm`, `mutual_sub`, `mutualNbrs_card_le`, `mem_mutualNbrs`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 3, 6, 8, 10, 11, 12.

RELATED

[neighbourhood graph](#) (p. 231); [hub](#) (p. 166); [recall at k](#) (p. 286); [Euclidean distance](#) (p. 124); [cosine](#) (p. 82).

neighbourhood graph

concept. Entry id `neighbourhood-graph`.

DEFINITION

A graph on a dataset that joins each row to its k nearest rows, or to every row within a radius. Chapter 0 section 0.10.

EQUATION

Book equation 0.19.

$$L = I - D^{-1/2} A D^{-1/2}, \quad D = \text{diag}(d_1, \dots, d_n), \quad L u_k = \lambda_k u_k, \quad 0 = \lambda_1 \leq \lambda_2 \leq \dots$$

Book equation 0.20.

$$\Psi_i = \left(\frac{u_k(i)}{\sqrt{\lambda_k d_i}} \right)_{k \geq 2}, \quad \|\Psi_i - \Psi_j\|^2 = R(i, j) = \frac{C(i, j)}{\text{vol}(G)}.$$

Book equation 3.3.

$$X_i = r_i \theta_i, \quad r_i \rightarrow \frac{1}{\sqrt{d_i}}, \quad \theta_i = \frac{X_i}{\|X_i\|} \in S^{m-1}.$$

LEDGER

- NEG-1. Fixed-scale, uniform-in-m bi-Lipschitz for the commute filter. [refuted]. `geometric-observation/claims/LEDGER.md:94` at 9f3829f.

FIRST STATED

Chapter 0 section 0.10 of *Data Mining as Observation*, with the substrate table in `the-angular-observer/README.md:99-109`.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 3 section 3.3	resistance converges to 1 over d_i plus 1 over d_j, simplex argument, radius converges to 1 over root d_i	<code>the-angular-observer\theorem.md:110-146</code>
chapter 3 section 3.3	substrate table	<code>the-angular-observer\README.md:99-109</code>
chapter 9 section 9.1	the geodesic-rank reader discards the radius, row normalization is the angular projection	<code>geometric-observation\chapters\ch09_legibility.md:10-41</code> ; <code>the-angular-observer\theorem.md:110-146</code>
chapter 9 section 9.2	causal set 0.57 with dimension 2.9 to 10.7, small world 0.45, control 0.01, rung 4 decay 0.747 to 0.679 below 0.75	<code>the-angular-observer\README.md:99-109</code> ; <code>wolfram-observer-bridge\rung4_scaledm_result.json</code> ; <code>wolfram-observer-bridge\review-2.txt</code>

FAILURES AND CORRECTIONS

- NEG-1, [refuted]. Fixed-scale, uniform-in-m bi-Lipschitz for the commute filter.

CONDITIONS

- A graph on a dataset that joins each row to its k nearest rows, or to every row within a radius. Every node of a k-nearest graph has out-degree k, the mutual neighbours of a node are at most k, and being mutual neighbours is symmetric.
- As the graph grows on a manifold its low Laplacian eigenvectors converge to the smoothest functions on the manifold, so the embedding becomes a picture of the manifold, and the hubs of the graph are a property of the queries that built it.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/Graph.lean`, `theorems outDegree`, `mutual_symm`, `mutual_sub`, `mutualNbrs_card_le`, `mem_mutualNbrs`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 3, 9, 11.

RELATED

[graph](#) (p. 156); [degree](#) (p. 97); [Laplacian](#) (p. 199); [spectral embedding](#) (p. 332); [geodesic distance](#) (p. 154); [hub](#) (p. 166).

nuisance

concept. Entry id nuisance.

DEFINITION

The directions of the data a consumer cannot distinguish, the kernel of its read operator. Chapter 1.

EQUATION

Book equation 0.12a.

$$x \sim_C x' \iff d_G(C(x), C(x')) = 0.$$

Book equation 0.12b.

$$x \sim_{\bar{P}_{C,\mu}} x' \iff x - x' \in \ker \bar{P}_{C,\mu}.$$

LEDGER

- GO-1. The consumer's invariant/nuisance split is identifiable ex ante from the consumer functional. [pre-dicted]. [geometric-observation/claims/LEDGER.md:62](#) at 9f3829f.
- NEG-6. Relative per-channel-demeaned error norm is the quotient-tangential quantity that controls softmax-KL. [refuted]. [geometric-observation/claims/LEDGER.md:99](#) at 9f3829f.
- NEG-7. Per-channel quantization is universally the invariant-preserving arm for softmax-key attention (a compressor property). [refuted]. [geometric-observation/claims/LEDGER.md:100](#) at 9f3829f.

FIRST STATED

Volume 14, chapter 5, [geometric-observation/chapters/ch05_the_read_metric_and_the_quotient.md:7-48](#), DOI 10.5281/zenodo.21776291, where the nuisance is the kernel of the read operator.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 2 section 2.2	read subspace small, operator local, pullback composition, rank cannot increase	geometric-observation\chapters\ch05_the_read_metric_and_the_quotient.md:48 ; geometric-observation\chapters\ch06_mathematical_preliminaries.md:27
chapter 2 section 2.2	read distortion controls but is not a complete rank statistic, twelve of twelve, one middle pair misordered, NEG-9	geometric-observation\chapters\ch05_the_read_metric_and_the_quotient.md:75
chapter 6 section 6.2	planted probe, overlap 0.936 vs 0.059, twelve of twelve, reconstruction 0.40, five of five	geometric-observation\claims\LEDGER.md row GO-1

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 11 section 11.7	GO-1 overlap 0.936 vs 0.059, flip 12 of 12, reconstruction 0.40	geometric-observation\claims\LEDGER.md row GO-1; geometric-observation\chapters\ch10_the_blind_probe.md:34-52

FAILURES AND CORRECTIONS

- NEG-6, [refuted]. Relative per-channel-demeaned error norm is the quotient-tangential quantity that controls softmax-KL.
- NEG-7, [refuted]. Per-channel quantization is universally the invariant-preserving arm for softmax-key attention (a compressor property).

CONDITIONS

- The nuisance is the kernel of the workload-averaged read operator, the set of directions unread at almost every row of that workload, since the kernel of the average is the intersection of the local kernels.
- For a nonlinear consumer a null direction can turn with the row, so the nuisance of the average is not a global equivalence of inputs, and cosine similarity's radial direction is the case.
- A transform is safe when the reader it produces still resolves the read subspace, and discretization is safe only when the values it merges differ in nuisance directions.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/ReadOperator.lean`, `theorems rank_one_reads_one_direction`, `readOp_mulVec`, `quad_readOp`, `quad_readOp_nonneg`, `readOp_mulVec_eq_zero_iff`, `readOp_diag`, `readOp_offdiag`, `readOp_symm`, `readOp_neg`, `affine_const_along_nuisance`, `readOp_affine`, `readOp_sqLength_basis`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 1, 2, 3, 6, 9, 11, 12, 13.

RELATED

[read subspace](#) (p. 285); [read operator](#) (p. 282); [quotient](#) (p. 271); [identity reader](#) (p. 171).

null model

instrument. Entry id `null-model`.

DEFINITION

A version of the data with the structure under test removed and everything else kept, the right comparison for any measured statistic. Chapter 0 section 0.9.

EQUATION

Book equation 0.17.

$$\Pr[X \geq c] = 1 - \sum_{i < c} e^{-\mu} \frac{\mu^i}{i!}, \quad c^* = \max\{c : n \Pr[X \geq c] \geq 1\}, \quad \mu = \frac{n_q k}{n}.$$

Book equation 8.3.

$$\Pr[\text{at least one of } m \text{ null tests passes}] = 1 - (1 - \alpha)^m, \quad \alpha_{\text{Bonferroni}} = \frac{\alpha}{m}.$$

LEDGER

- OT-4. Operator drift predicts a real long-generation degradation and a derived refresh intervention moves it. [refuted]. [geometric-observation/claims/LEDGER.md:36 at 9f3829f](#).
- NEG-15 (Bell boundary). *Query-conditioned hubness supplies a mechanism for Bell-inequality violation without action at a distance*. Refuted as a mechanism; the settings-as-queries reframing survives only as vocabulary. [demonstrated]. [geometric-observation/claims/LEDGER.md:93 at 9f3829f](#).

FIRST STATED

Chapter 0 section 0.9 and chapter 8 section 8.4 of *Data Mining as Observation*, with the program's Poisson, paired, and permutation nulls in `openvector-bench`, `readscope`, and the encoder gate.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

- OT-4, [refuted]. Operator drift predicts a real long-generation degradation and a derived refresh intervention moves it.

CONDITIONS

- A version of the data with the structure under test removed and everything else kept. A label permutation keeps every class size and the bag of scores, so any statistic that reads only class sizes or only scores is unchanged by it, and only a statistic that reads the pairing between score and label can move.
- That is what makes the permuted statistic the right comparison, and the excess over the null mean is the number a chapter reports. A number without its null is not yet a finding, and the drift claim that lost half its size to a paired null is the ledger's case.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/NullModel.lean`, `theorems card_pos_perm`, `scores_perm`, `stat_of_counts_const`, `excess_eq_zero_iff`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 2, 3, 4, 5, 8, 9, 10, 11, 12, 14.

RELATED

[Poisson ceiling](#) (p. 253); [chance level](#) (p. 58); [cross-corpus gate](#) (p. 87); [p-value](#) (p. 264); [harness](#) (p. 159).

O

observer

concept. Entry id observer.

DEFINITION

A consumer, its output metric, and its budget, written as the triple in equation 1.1. Naming all three is what every later chapter checks.

EQUATION

Book equation 1.1.

$$O = (C, G, B).$$

Book equation 0.9.

$$P_C = \mathbb{E}[g g^\top], \quad g = \nabla C(x).$$

Book equation 0.11.

$$P_C(x) = J(x)^\top G(C(x)) J(x), \quad J(x) = \frac{\partial C}{\partial x}(x), \quad \bar{P}_{C,\mu} = \mathbb{E}_\mu[P_C(x)].$$

LEDGER

- OT-7. The damage form, trace pairing, rank, and loading covariance are GL(d)-invariant under $P' = A^{-\top} P A^{-1}$, while spectrum, effective rank, principal angles, and water-filling are O(d)-only — consumer-weighted damage is a geometric scalar, and P3's cliff cannot be bought down by reparameterization. [demonstrated]. [geometric-observation/claims/LEDGER.md:30 at 9f3829f](#).
- GO-1. The consumer's invariant/nuisance split is identifiable ex ante from the consumer functional. [predicted]. [geometric-observation/claims/LEDGER.md:62 at 9f3829f](#).

FIRST STATED

Volume 14, chapter 4, [geometric-observation/chapters/ch04_the_observer_triple.md:9-60](#), and [geometric-observation/OBSERVATION.md:1-10](#), DOI 10.5281/zenodo.21776291. Version 1.0 of the theory was declared on 2026-08-18 in [geometric-observation/crucible/DECLARATION-V1.md](#).

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 1 section 1.2	the observer triple, read subspace, nuisance, same read operator means same read geometry	geometric-observation\chapters\ch04_the_observer_triple.md:9-60 ; geometric-observation\OBSERVATION.md:1-10
chapter 1 section 1.2	the consumer table	geometric-observation\OBSERVATION.md:19-29
chapter 1 section 1.3	no-consumer is a precondition, coupling is a true null, every verdict budget-relative	geometric-observation\OBSERVATION.md:31-40

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 1 section 1.5	the Bell boundary	geometric-observation\OBSERVATION.md:35-40; geometric-observation\claims\LEDGER.md row NEG-15
chapter 1 section 1.5	the six classes and the ledger rule	geometric-observation\PROTOCOL.md:58-75; geometric-observation\OBSERVATION.md:43-45
chapter 6 section 6.1	the classifier row of the consumer table, the output metric makes a different observer	geometric-observation\chapters\ch04_the_observer_triple.md:60-135
chapter 8 section 8.8	declaration drafted 2026-08-17, sealed 2026-08-18, one count corrected, G2	geometric-observation\crucible\DECLARATION-V1.md:1-20; geometric-observation\crucible\OT-CRUCIBLE-4.md:31-35
chapter 8 section 8.10	the six classes	geometric-observation\PROTOCOL.md:58-75
chapter 11 section 11.7	C-15 equal budget, 3072 observations, medians 0.02, 0.11, 0.05, 0.04, 1.0, 1.0, margin 0.883 vs 0.3, 0.062 to 0.057 at 8x	readscope\SPEC.md:285-323, record readscope\calibration\records\c15-budget-surface.json, readscope\calibration\DECLARATION-C15.md
chapter 11 section 11.9	C-12 four bars, 40 documents, 512 tokens, 13.4 point difference, teacher forcing removes it, negative 0.015 vs 0.005, Spearman negative 0.13 at p 0.45, sign test p 0.42, verdict FAIL, feedback compounding	readscope\calibration\records\c12-longgen-drift-sym.json; readscope\calibration\DECLARATION-C12.md at commit 90e2ce2; readscope\SPEC.md:806-825

FAILURES AND CORRECTIONS

none

CONDITIONS

- An observer is a consumer, its output metric, and its budget. The consumer and the local geometry of its output metric determine the read operator. The budget bounds what of it can be measured and used and changes it only by changing the consumer.
- Two consumers with the same read operator on the same workload share a read geometry and are not thereby the same observer, since the consumer stays part of the triple and a sign change reverses every ranking.
- The output metric is a loss on the consumer's output. Where it has a local quadratic representation, that local geometry enters the read operator. A dataset-level loss has none, and the read operator is then taken on the score with the identity geometry.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

lean/DataMiningAsObservation/ReadOperator.lean, theorems rank_one_reads_one_direction, readOp_mulVec, quad_readOp, quad_readOp_nonneg, readOp_mulVec_eq_zero_iff, readOp_diag, readOp_offdiag, readOp_symm, readOp_neg, affine_const_along_nuisance, readOp_affine, readOp_sqLength_basis, at observation-data-mining

af776fd.

USED IN

Data Mining as Observation chapters 0, 1, 2, 3, 4, 5, 6, 8, 9, 10, 11, 12, 13.

RELATED

[read operator](#) (p. 282); [quotient](#) (p. 271); [read distortion](#) (p. 281); [certificate](#) (p. 55).

operating point

concept. Entry id operating-point.

DEFINITION

The row at which a consumer is read, or the threshold at which a classifier is scored. A probe pays $2d$ evaluations for one. Chapter 0 section 0.5 and chapter 11 section 11.7.

EQUATION

Book equation 0.8.

$$g_j \approx \frac{C(x + h e_j) - C(x - h e_j)}{2h}, \quad j = 1, \dots, d.$$

Book equation 11.4.

$$\text{directions resolved}(k) = \begin{cases} 1 \text{ or } 2, & k < d \\ \text{rank } P_C, & k \geq d \end{cases} \quad \text{cost} = 2d \text{ consumer calls per operating point.}$$

Book equation 0.28.

$$P = \frac{TP}{TP + FP}, \quad R = \frac{TP}{TP + FN}, \quad F_1 = \frac{2PR}{P + R}.$$

LEDGER

- OT-3. Under subspace-confined second-order transcripts, fewer than d directions cannot identify a hidden leading eigenspace (theorem, adaptive to $d-2$ / oblivious to $d-1$); a known k_0 -dim exclusion moves the cliff to exactly $d-k_0$ and never softens it. [demonstrated]. [geometric-observation/claims/LEDGER.md:31 at 9f3829f](#).
- GO-4. Fixed-budget verdicts invert under budget-matched observation, per the wavelength mechanism. [replicated]. [geometric-observation/claims/LEDGER.md:66 at 9f3829f](#).

FIRST STATED

Chapter 0 section 0.5 and chapter 11 section 11.7 of *Data Mining as Observation*, with the budget law in [readscope/README.md:118-141](#).

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 11 section 11.7	k over d table 1, 2, 2, 16, 16, 16	readscope\README.md:118-141 ; readscope\SPEC.md:243-285 , record readscope\calibration\records\c2e-budget-law.json
chapter 11 section 11.7	rank-independent, 0.646 vs 0.366 at half dimension, never 0.90 below k equals d	readscope\CALIBRATION.md:419-434 F-15

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 11 section 11.7	C-15 equal budget, 3072 observations, medians 0.02, 0.11, 0.05, 0.04, 1.0, 1.0, margin 0.883 vs 0.3, 0.062 to 0.057 at 8x	readscope\SPEC.md:285-323, record readscope\calibration\records\c15-budget-surface.json, readscope\calibration\DECLARATION-C15.md

FAILURES AND CORRECTIONS

none

CONDITIONS

- The row at which a consumer is read, or the threshold at which a classifier is scored. A central-difference probe pays two evaluations per coordinate for one operating point, and raising a classifier's threshold can only shrink the set it calls positive.
- Whether many cheaper operating points can average their way back to the population operator was measured within a budget range and not proved, and the cliff at k equal to d does not soften for a probe confined to the directions it chooses.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/FiniteDifference.lean`, theorems `central_quad`, `forward_quad`, `forward_error`, `central_cost`, at `observation-data-mining af776fd`.

`lean/DataMiningAsObservation/Threshold.lean`, theorems `predicted_anti`, `tp_anti`, `fp_anti`, `decision_comp`, `predicted_extremes`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 2, 5, 6, 11, 14.

RELATED

[finite difference](#) (p. 137); [budget](#) (p. 46); [budget cliff](#) (p. 49); [threshold](#) (p. 350); [blind probe](#) (p. 39).

orthogonal

concept. Entry `id orthogonal`.

DEFINITION

Of two vectors, having dot product zero. The squared length of a sum of orthogonal vectors is the sum of the squared lengths. Chapter 0 section 0.1.

EQUATION

Book equation 0.1.

$$x \cdot y = \sum_{i=1}^d x_i y_i, \quad \|x\| = \sqrt{x \cdot x}, \quad \cos \theta = \frac{x \cdot y}{\|x\| \|y\|}.$$

Book equation 0.3.

$$\Sigma_{ij} = \mathbb{E}[(x_i - \mu_i)(x_j - \mu_j)], \quad \text{tr } \Sigma = \sum_i \Sigma_{ii}.$$

Book equation 0.9.

$$P_C = \mathbb{E}[g g^\top], \quad g = \nabla C(x).$$

LEDGER

- OT-7. The damage form, trace pairing, rank, and loading covariance are $GL(d)$ -invariant under $P' = A^{-T}PA^{-1}$, while spectrum, effective rank, principal angles, and water-filling are $O(d)$ -only — consumer-weighted damage is a geometric scalar, and P3’s cliff cannot be bought down by reparameterization. [demonstrated].
geometric-observation/claims/LEDGER.md:30 at 9f3829f.

FIRST STATED

Chapter 0 section 0.1 and section 0.2 of *Data Mining as Observation*.

MEASUREMENTS

Where the book states it	Numbers, as the book’s sources table records them	Source
chapter 2 section 2.2	read subspace small, operator local, pullback composition, rank cannot increase	geometric-observation\chapters\ch05_the_read_metric_and_the_o48; geometric-observation\chapters\ch06_mathematical_preliminaries27

FAILURES AND CORRECTIONS

none

CONDITIONS

- Of two vectors, having dot product zero. The relation is symmetric, survives rescaling, holds between any vector and zero, the squared length of a sum of orthogonal vectors is the sum of the squared lengths, and the residual of a projection is orthogonal to the direction projected on.
- A linear classifier’s nuisance is everything orthogonal to its weight vector, and the eigenvectors of a symmetric matrix with distinct eigenvalues are orthogonal, so the read subspace and the nuisance split every row.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/Orthogonal.lean`, theorems `orth_symm`, `pythagoras`, `orth_smul`, `orth_zero`, `residual_orth`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 3, 4, 6, 9, 11, 12.

RELATED

[dot product](#) (p. 110); [projection](#) (p. 263); [eigenvalue, eigenvector](#) (p. 116); [nuisance](#) (p. 233); [read subspace](#) (p. 285).

outer product

concept. Entry id `outer-product`.

DEFINITION

The matrix whose entry in row i and column j is the product of a vector’s i -th and j -th coordinates, of rank at most one and positive semidefinite for a vector with itself. Chapter 0 section 0.5.

EQUATION

Book equation 0.9.

$$P_C = \mathbb{E}[g g^\top], \quad g = \nabla C(x).$$

Book equation 6.1.

$$s(x) = w \cdot x + b, \quad P_C = \mathbb{E}[\sigma'(s)^2] w w^\top, \quad \text{rank } P_C = 1.$$

Book equation 0.3.

$$\Sigma_{ij} = \mathbb{E}[(x_i - \mu_i)(x_j - \mu_j)], \quad \text{tr } \Sigma = \sum_i \Sigma_{ii}.$$

LEDGER

- OT-7. The damage form, trace pairing, rank, and loading covariance are $\text{GL}(d)$ -invariant under $P' = A^{-\top} P A^{-1}$, while spectrum, effective rank, principal angles, and water-filling are $O(d)$ -only — consumer-weighted damage is a geometric scalar, and P3’s cliff cannot be bought down by reparameterization. [demonstrated].
geometric-observation/claims/LEDGER.md:30 at 9f3829f.

FIRST STATED

Chapter 0 section 0.5 of *Data Mining as Observation*.

MEASUREMENTS

Where the book states it	Numbers, as the book’s sources table records them	Source
chapter 2 section 2.2	read subspace small, operator local, pullback composition, rank cannot increase	geometric-observation\chapters\ch05_the_read_metric_and_the_o48; geometric-observation\chapters\ch06_mathematical_preliminaries27
chapter 6 section 6.1	the classifier row of the consumer table, the output metric makes a different observer	geometric-observation\chapters\ch04_the_observer_triple.md:60135

FAILURES AND CORRECTIONS

none

CONDITIONS

- The matrix whose entry in row i and column j is the product of the i -th and j -th coordinates of a vector. Its action on any vector is a multiple of the vector it was built from, it has rank at most one, and the outer product of a vector with itself is positive semidefinite.
- The read operator is the weighted mean of the outer product of the sensitivity with itself, and a linear classifier’s read operator is one outer product scaled by how steep the score is.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/PositiveSemidefinite.lean`, `theorems psd_add, psd_smul, psd_outer, psd_readOp, psd_diag_nonneg`, at `observation-data-mining af776fd`.
`lean/DataMiningAsObservation/ReadOperator.lean`, `theorems rank_one_reads_one_direction, readOp_mulVec, quad_readOp, quad_readOp_nonneg, readOp_mulVec_eq_zero_iff, readOp_diag, readOp_offdiag, readOp_symm`,

readOp_neg, affine_const_along_nuisance, readOp_affine, readOp_sqlength_basis, at observation-data-mining af776fd.

lean/DataMiningAsObservation/Rank.lean, theorems rank_le_width, rank_le_height, rank_outer_le_one, rank_mul_le, rank_zero, rank_transpose, at observation-data-mining af776fd.

USED IN

Data Mining as Observation chapters 0, 2, 6.

RELATED

[read operator](#) (p. 282); [rank](#) (p. 274); [positive semidefinite](#) (p. 256); [covariance matrix](#) (p. 85); [sensitivity](#) (p. 319).

outlier

concept. Entry id outlier.

DEFINITION

A row the reader cannot place. The statistical detector calls a row an outlier when its distance from the mean in units of spread passes a threshold, and no more than one over the threshold squared of the weight can. Chapter 10.

EQUATION

Book equation 0.33.

$$d_M(x) = \sqrt{(x - \mu)^\top \Sigma^{-1} (x - \mu)}.$$

Book equation 10.3.

$$N_k(x \mid Q) = |\{q \in Q : x \in \text{top}_k(q)\}|, \quad \text{anti-hub} : N_k(x) = 0.$$

Book equation 0.6.

$$x_w = \Sigma^{-1/2}(x - \mu), \quad \Sigma^{-1/2} = \sum_i \lambda_i^{-1/2} v_i v_i^\top.$$

LEDGER

- NEG-14. (GO-P-2026-037, prospective) a2_probe.median_unit_displacement is a single-statistic predictor of the flip regime (unit_disp $\geq 1.0 \Rightarrow$ generic-polar-flip vs needs-blind-probe). [refuted]. [geometric-observation/claims/LEDGER.md:108](#) at 9f3829f.

FIRST STATED

Chapter 0 section 0.16 and chapter 10 section 10.1 of *Data Mining as Observation*, after TSK chapter 9, with the hierarchical typing in [turboquant-pro/turboquant_pro/anatomy.py:98-170](#).

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 10 section 10.1	the four detector families	TSK 2e chapter 9
chapter 10 section 10.3	hierarchical typing, tails 0.95 and 0.85, prescriptions, two designs that died, correlation above 0.8 on an isotropic Gaussian	turboquant-pro\turboquant_pro\anatomy.py:98-170

FAILURES AND CORRECTIONS

- NEG-14, [refuted]. (GO-P-2026-037, prospective) `a2_probe.median_unit_displacement` is a single-statistic predictor of the flip regime ($\text{unit_disp} \geq 1.0 \Rightarrow \text{generic-polar-flip vs needs-blind-probe}$).

CONDITIONS

- A row the reader cannot place. The statistical detector scores a row by its distance from the mean in units of spread, which is zero at the mean, unchanged when data, mean, and spread are rescaled and shifted together, and past a threshold t exactly when the row lies more than t spreads from the mean. By Chebyshev the weight of rows beyond t is at most the mean squared score over t squared.
- The four detector families read different coordinates, distance from the mean, distance to neighbours, density relative to neighbours, and cluster membership, and the observer's outliers, the anti-hubs, are the rows no detector of the data alone can see.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/Outlier.lean`, theorems `zscore_mean`, `zscore_affine`, `outlier_iff`, `zscore_reflect`, `chebyshev`, at `observation-data-mining af776fd`.

`lean/DataMiningAsObservation/Mahalanobis.lean`, theorems `dm2_nonneg`, `dm2_mean`, `dm2_scale`, `dm2_eq_whitened`, `dm2_antitone_in_variance`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 2, 3, 10.

RELATED

[Mahalanobis distance](#) (p. 210); [anti-hub](#) (p. 17); [abstention](#) (p. 9); [whitening](#) (p. 368); [density](#) (p. 99).

output metric

concept. Entry id `output-metric`.

DEFINITION

The rule by which a consumer's mistakes are scored, the second element of an observer. Chapter 1.

EQUATION

Book equation 1.1.

$$O = (C, G, B).$$

Book equation 0.9.

$$P_C = \mathbb{E}[g g^\top], \quad g = \nabla C(x).$$

Book equation 6.1.

$$s(x) = w \cdot x + b, \quad P_C = \mathbb{E}[\sigma'(s)^2] w w^\top, \quad \text{rank } P_C = 1.$$

LEDGER

- GO-2 (neg. half: not reconstruction). At matched bits, downstream preservation is not controlled by reconstruction error. [demonstrated]. `geometric-observation/claims/LEDGER.md:63` at `9f3829f`.
- GO-2 (pos. half: consumer-projected covariance *controls*). Downstream preservation is controlled by the error covariance projected on the consumer's read subspace, `tr(P_C \cdot \Sigma_\delta)`. [replicated]. `geometric-observation/claims/LEDGER.md:64` at `9f3829f`.

FIRST STATED

Chapter 1 section 1.2 of *Data Mining as Observation*, with the flip's verdict inversion in `geometric-observation/chapters/ch08_value.md:1-30`.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 3 section 3.2	traces 2.0, diagonals 0.3 and 1.7, consumers at 15 and 75 degrees, distortion 0.39 vs 1.61, 4.1 to one, computed not drawn	<code>observation-theory/assets/make_flip_figure.py:4-24,107-119</code>
chapter 4 section 4.3	the flip definition, flip versus (A2) verdict	<code>geometric-observation/chapters/ch08_value.md:1-30,100-108</code>
chapter 4 section 4.3	GO-2 0.0934 vs 0.0938, 2.53 times, 12 of 12, anti 21 times; retrieval 0.0964, negative 4.70 and positive 4.65, recon 0.40	<code>geometric-observation/claims/LEDGER.md</code> rows GO-2 negative and positive halves

FAILURES AND CORRECTIONS

none

CONDITIONS

- The rule by which a consumer's mistakes are scored, the second element of an observer. A consumer and its negation have the same read operator and reverse every comparison, so the read operator alone does not fix the observer, and two cost matrices score the same pair of classifiers in opposite orders, so naming the output metric is naming what a downstream mistake is.
- A dataset-level loss, accuracy, F1, a rank correlation, or dollars lost is not a local geometry on the score and cannot be inserted into the read-operator formula. Each interestingness measure of chapter 5 and each fairness metric of chapter 14 is an output metric, right for some consumer and wrong for others.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/OutputMetric.lean`, theorems `neg_reverses`, `readOp_of_neg`, `cost_flips`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 1, 3, 5, 6, 8, 9, 11, 14.

RELATED

[observer](#) (p. 236); [consumer](#) (p. 73); [read operator](#) (p. 282); [budget](#) (p. 46).

P

paired

instrument. Entry id paired.

DEFINITION

Of a comparison, made on the same rows or seeds for both arms. Pairing helps exactly when the two arms covary. Chapter 0 section 0.9.

EQUATION

Book equation 0.18.

$$\widehat{\text{Var}}_{\text{NB}} = \left(\frac{1}{J} + \frac{n_{\text{test}}}{n_{\text{train}}} \right) \hat{\sigma}^2, \quad \frac{\widehat{\text{Var}}_{\text{NB}}}{\hat{\sigma}^2/J} = 1 + J \frac{n_{\text{test}}}{n_{\text{train}}}.$$

Book equation 8.2.

$$\widehat{\text{Var}}_{\text{NB}} = \left(\frac{1}{J} + \frac{n_{\text{test}}}{n_{\text{train}}} \right) \hat{\sigma}^2, \\ J = 1000, \frac{n_{\text{test}}}{n_{\text{train}}} = \frac{1}{4} \Rightarrow 1 + 250 = 251, \sqrt{251} = 15.84.$$

Book equation 0.29.

$$\text{ECE} = \sum_b \frac{n_b}{n} |\bar{s}_b - \bar{y}_b|.$$

LEDGER

- NEG-14. (GO-P-2026-037, prospective) a2_probe.median_unit_displacement is a single-statistic predictor of the flip regime (unit_disp $\geq 1.0 \Rightarrow$ generic-polar-flip vs needs-blind-probe). [refuted]. geometric-observation/claims/LEDGER.md:108 at 9f3829f.
- GO-B-Llama. Trained frontier LLM (Llama-3.2-3B), softmax-attention consumer — blind probe on real post-RoPE keys [predicted]. geometric-observation/claims/LEDGER.md:115 at 9f3829f.
- GO-B-Llama-rematch. Trained frontier LLM (Llama-3.2-3B), softmax-attention consumer — reconnected dissociation on real post-RoPE keys [predicted]. geometric-observation/claims/LEDGER.md:118 at 9f3829f.

FIRST STATED

Chapter 0 section 0.9 and chapter 8 section 8.4 of *Data Mining as Observation*, with the paired null in readscope/SPEC.md:806-857.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 8 section 8.4	C-11c paired null, rank 2 positional 0.385 vs null 0.572, 0.615 vs 0.224, 14 of 16 cells, scope 16 cells one 3B model 192 positions	readscope\SPEC.md:806-857; readscope\calibration\records\c11c-operator-drift.json

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 11 section 11.8	0.765 to 0.971 with interval 0.190 to 0.223, 0.545 to 0.562 with interval 0.004 to 0.031, v1 0.340 with interval negative 0.214 to negative 0.081, anisotropy 0.570 to 0.259	lebse\README.md:40-75; lebse\PAPER.md:1-15,60-66; lebse\MODEL_CARD.md:48-52
chapter 12 section 12.4	0.765 to 0.971, 0.545 to 0.562, v1 0.340, cosine 0.570 to 0.259	lebse\README.md:40-75; lebse\PAPER.md:1-15,60-66

FAILURES AND CORRECTIONS

- NEG-14, [refuted]. (GO-P-2026-037, prospective) `a2_probe.median_unit_displacement` is a single-statistic predictor of the flip regime ($\text{unit_disp} \geq 1.0 \Rightarrow \text{generic-polar-flip vs needs-blind-probe}$).

CONDITIONS

- Of a comparison, made on the same rows or the same seeds for both arms, so that the difference is scored row by row. The variance of the difference is the sum of the variances less twice the covariance, so pairing helps exactly when the two arms covary.
- The paired null at rank two read 0.385 against 0.572 on fourteen of sixteen cells, and the encoder comparisons carry a paired bootstrap interval on each relation.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/Bootstrap.lean`, `theorems mean_sub`, `var_sub`, `paired_lt_iff`, `cov_comm`, `at observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 4, 6, 8, 11, 12.

RELATED

[bootstrap](#) (p. 45); [confidence interval](#) (p. 71); [seed](#) (p. 318); [standard error](#) (p. 337); [control](#) (p. 77).

paraphrase class

concept. Entry id `paraphrase-class`.

DEFINITION

A set of rewordings of one input that preserve its meaning. A decision is invariant to re-description when it does not move across the class. Chapter 14.

EQUATION

Book equation 14.3.

$$\theta_d = \frac{\text{mean}_i |S_A(i) - S_B(i)|}{\text{mean}_{i \neq j} |S_A(i) - S_A(j)|}, \quad \text{bar } \theta_d \leq 0.5.$$

LEDGER

none

FIRST STATED

Chapter 14 section 14.4 of *Data Mining as Observation*, with the program's re-description test in `gtc-prototype/docs/REGATE.md:1-70` and the back-translation run.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

CONDITIONS

- A set of rewordings of one input that preserve its meaning. A verdict is invariant to re-description when it does not move across the class, and the ratio of equation 14.3 measures the movement under rewording over the spread between inputs. It is nonnegative, zero exactly when no item's score moves, and at most one half exactly when the movement is at most half the spread.
- The measured ratios, 0.407 falling to 0.219 on sixty items and 0.482 to 0.301 under back-translation, with three of six gold items still flipping, are the program's numbers, and the items that flip are routed to escalation.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/ParaphraseClass.lean`, `theorems movement_nonneg`, `ratio_nonneg`, `movement_eq_zero_iff`, `ratio_invariant`, `bar_iff`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 12, 14.

RELATED

[quotient](#) (p. 271); [escalation](#) (p. 123); [certificate](#) (p. 55); [deployment mismatch](#) (p. 101).

percentile

concept. Entry id percentile.

DEFINITION

The value below which a given fraction of the rows fall. The fraction at or below a value is nondecreasing in the value. Chapter 11 section 11.3.

EQUATION

Book equation 11.2.

$$r = \frac{d_{\text{compressed}}}{d_{\text{exact}}}, \quad \kappa_{\text{strict}} = \frac{\max r}{\min r}, \quad \tau \geq 1 - 2\hat{\mu}(\kappa_{\text{strict}}), \quad \rho_S \geq 1 - 3\hat{\mu}(\kappa_{\text{strict}}),$$

$$\kappa_{97.5/2.5} = \frac{q_{97.5}(r)}{q_{2.5}(r)} \text{ gives the same two expressions as estimates, not floors.}$$

Book equation 0.35.

$$\text{RH} = \sum_i \max\left(0, \frac{N_i}{\sum_j N_j} - \frac{1}{n}\right).$$

Book equation 10.3.

$$N_k(x \mid Q) = |\{q \in Q : x \in \text{top}_k(q)\}|, \quad \text{anti-hub} : N_k(x) = 0.$$

LEDGER

- NEG-14. (GO-P-2026-037, prospective) `a2_probe.median_unit_displacement` is a single-statistic predictor of the flip regime (`unit_disp` $\gtrless$ 1.0 $\Rightarrow$ generic-polar-flip vs needs-blind-probe). [refuted]. `geometric-`

observation/claims/LEDGER.md:108 at 9f3829f.

FIRST STATED

Chapter 11 section 11.3 of *Data Mining as Observation*, with the percentile setting of the rank certificate in turboquant-pro/turboquant_pro.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 10 section 10.2	anti-hubs as where compressed indexes fail first, aggregate recall barely moves	turboquant-pro\docs\HUBNESS_PRIMER.md:86-131
chapter 10 section 10.3	hierarchical typing, tails 0.95 and 0.85, prescriptions, two designs that died, correlation above 0.8 on an isotropic Gaussian	turboquant-pro\turboquant_pro\anatomy.py:98-170
chapter 11 section 11.2	the calibration-time probe and the streaming monitor	turboquant-pro\turboquant_pro\ a2_probe.py:315-438; turboquant-pro\tests\test_a2_probe.py
chapter 11 section 11.6	anti-hub recall, p05, hub-rank correlation, hub-set overlap, the build gate	turboquant-pro\docs\HUBNESS_PRIMER.md:86-131

FAILURES AND CORRECTIONS

- NEG-14, [refuted]. (GO-P-2026-037, prospective) a2_probe.median_unit_displacement is a single-statistic predictor of the flip regime ($\text{unit_disp} \geq 1.0 \Rightarrow \text{generic-polar-flip vs needs-blind-probe}$).

CONDITIONS

- The value below which a given fraction of the rows fall. The fraction at or below a value is nonnegative, at most one, nondecreasing in the value, one at or above the largest row and zero below the smallest, and an upper percentile over a lower one is at least one.
- The rank certificate's percentile setting reads the 97.5 over 2.5 percentile ratio as a robust estimate where the strict setting reads the max over min, and the anti-hub gate reads the fifth percentile of recall.

Conditions are curated in entries.toml rather than read from a record.

MACHINE CHECKED

lean/DataMiningAsObservation/Percentile.lean, theorems cdf_nonneg, cdf_le_one, cdf_mono, cdf_of_all, cdf_of_none, ratio_ge_one, at observation-data-mining af776fd.

USED IN

Data Mining as Observation chapters 3, 10, 11.

RELATED

[rank certificate](#) (p. 275); [anti-hub recall](#) (p. 19); [skewness](#) (p. 326); [threshold](#) (p. 350); [Robin Hood index](#) (p. 305).

per-channel quantizer

instrument. Entry id per-channel-quantizer.

DEFINITION

A quantizer that chooses a scale per coordinate. Chapter 11.

EQUATION

Book equation 11.1.

$$\cos(k, \hat{k}) = 0.995 \quad \text{while} \quad \text{PPL} : 12.24 \rightarrow 10643.$$

Book equation 0.13.

$$D(b) = \sum_i s_i v_i 4^{-b_i}, \quad b_i = \max\left(0, \frac{1}{2} \log_2 \frac{s_i v_i}{\theta}\right), \quad \sum_i b_i = B.$$

LEDGER

- NEG-2. Reconstruction cosine as a proxy for key quality. [refuted]. `geometric-observation/claims/LEDGER.md:95` at 9f3829f.

FIRST STATED

Chapter 11 section 11.2 of *Data Mining as Observation*, with the per-channel rows of the quantizer table in `turboquant-pro/docs/KV_KEYS_FINDING.md:1-49`.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 1 section 1.4	cosine 0.995 and perplexity of order ten thousand, the recalibration negative	<code>geometric-observation\chapters\ch02_failure_of_observer_free-60</code> ; <code>geometric-observation\chapters\ch16_honest_negatives.md</code> NEG-2 and NEG-4; <code>turboquant-pro\docs\KV_KEYS_FINDING.md:1-49</code>
chapter 2 section 2.5	cosine 0.995 and the softmax reader	<code>turboquant-pro\docs\KV_KEYS_FINDING.md:1-49</code>
chapter 3 section 3.2	condition (A2), tangential displacement	<code>turboquant-pro\docs\KV_KEYS_FINDING.md:61-86</code> ; <code>geometric-observation\chapters\ch09_legibility.md:42-54</code>
chapter 8 section 8.2	cosine 0.995, perplexity near 1e4	<code>turboquant-pro\docs\KV_KEYS_FINDING.md:1-49</code> ; <code>geometric-observation\chapters\ch16_honest_negatives.md</code> NEG-2
chapter 11 section 11.1	condition (A2), cosine satisfies it, post-rotary keys do not, the cone below cell size	<code>turboquant-pro\docs\KV_KEYS_FINDING.md:61-86</code> ; <code>the-angular-observer\README.md:26-31</code>
chapter 11 section 11.2	fp16 12.24, values-only 13.12, PolarQuant K4 10643 and 0.095, per-channel uniform K4 14.91 and 0.062, per-channel NUQ K3 15.77 and 0.148, 2.4x and 670x, pre-rotary near 22000, 2 key heads serve 12 query heads	<code>turboquant-pro\docs\KV_KEYS_FINDING.md:1-49</code>

FAILURES AND CORRECTIONS

- NEG-2, [refuted]. Reconstruction cosine as a proxy for key quality.

CONDITIONS

- A quantizer that chooses a step per coordinate. The squared error of a vector is at most the sum of the squared half steps, and a finer step on one coordinate lowers that coordinate's bound alone, which is the allocation water-filling makes by sensitivity.
- At four bits the per-channel uniform quantizer held the perplexity at 14.91 against 12.24 uncompressed, where the direction-only quantizer at the same bits did not.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/Quantization.lean`, `theorems error_le_half_step`, `quantize_level`, `half_step_bound`, `sq_error_le`, `finer_step`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 3, 4, 11.

RELATED

[quantization](#) (p. 266); [direction-only quantizer](#) (p. 104); [water-filling](#) (p. 365); [KV cache](#) (p. 190).

perplexity

concept. Entry id perplexity.

DEFINITION

Two to the power of the average number of bits a language model needs per token of a test text. Lower is better. Equation 0.22.

EQUATION

Book equation 0.22.

$$\text{PPL} = 2^H, \quad H = -\frac{1}{T} \sum_{t=1}^T \log_2 p(w_t \mid w_{<t}).$$

Book equation 11.1.

$$\cos(k, \hat{k}) = 0.995 \quad \text{while} \quad \text{PPL} : 12.24 \rightarrow 10643.$$

LEDGER

- NEG-2. Reconstruction cosine as a proxy for key quality. [refuted]. `geometric-observation/claims/LEDGER.md:95` at 9f3829f.
- NEG-4. Lightweight online (Lloyd) key calibration beats the calibration-free default on softmax-KL. [refuted]. `geometric-observation/claims/LEDGER.md:97` at 9f3829f.

FIRST STATED

Chapter 0 section 0.11 of *Data Mining as Observation*, with the program's case in `turboquant-pro/docs/KV_KEYS_FINDING.md:1-49`.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 1 section 1.4	cosine 0.995 and perplexity of order ten thousand, the recalibration negative	geometric-observation\chapters\ch02_failure_of_observer_free_60; geometric-observation\chapters\ch16_honest_negatives.md NEG-2 and NEG-4; turboquant-pro\docs\KV_KEYS_FINDING.md:1-49
chapter 2 section 2.5	cosine 0.995 and the softmax reader	turboquant-pro\docs\KV_KEYS_FINDING.md:1-49
chapter 3 section 3.2	condition (A2), tangential displacement	turboquant-pro\docs\KV_KEYS_FINDING.md:61-86; geometric-observation\chapters\ch09_legibility.md:42-54
chapter 8 section 8.2	cosine 0.995, perplexity near 1e4	turboquant-pro\docs\KV_KEYS_FINDING.md:1-49; geometric-observation\chapters\ch16_honest_negatives.md NEG-2
chapter 11 section 11.1	condition (A2), cosine satisfies it, post-rotary keys do not, the cone below cell size	turboquant-pro\docs\KV_KEYS_FINDING.md:61-86; the-angular-observer\README.md:26-31
chapter 11 section 11.2	fp16 12.24, values-only 13.12, PolarQuant K4 10643 and 0.095, per-channel uniform K4 14.91 and 0.062, per-channel NUQ K3 15.77 and 0.148, 2.4x and 670x, pre-rotary near 22000, 2 key heads serve 12 query heads	turboquant-pro\docs\KV_KEYS_FINDING.md:1-49

FAILURES AND CORRECTIONS

- NEG-2, [refuted]. Reconstruction cosine as a proxy for key quality.
- NEG-4, [refuted]. Lightweight online (Lloyd) key calibration beats the calibration-free default on softmax-KL.

CONDITIONS

- Two to the power of the average number of bits a language model needs per token of a test text. It is at least one, equals the vocabulary size for a model that spreads its mass evenly, and is monotone in the bits.
- The attention-key finding raised it from 12.24 to 10643 at cosine 0.995, a rise of more than nine bits per token, and the recalibration that improved the reconstruction worsened it, the two standing negatives that keep reconstruction error off the list of acceptance metrics.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/Perplexity.lean`, `theorems bits_nonneg`, `one_le_perplexity`, `perplexity_uniform`, `perplexity_mono`, `bits_of_finding`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 1, 8, 11.

RELATED

[KL divergence](#) (p. 187); [the flip](#) (p. 138); [identity reader](#) (p. 171); [read distortion](#) (p. 281).

planted

instrument. Entry id planted.

DEFINITION

Of a case, built so that the answer is known before the instrument reads it, such as an affine consumer whose read operator is its weight vector's outer product. Chapter 6 section 6.2.

EQUATION

Book equation 0.9.

$$P_C = \mathbb{E}[g g^\top], \quad g = \nabla C(x).$$

Book equation 6.1.

$$s(x) = w \cdot x + b, \quad P_C = \mathbb{E}[\sigma'(s)^2] w w^\top, \quad \text{rank } P_C = 1.$$

Book equation 11.4.

$$\text{directions resolved}(k) = \begin{cases} 1 \text{ or } 2, & k < d \\ \text{rank } P_C, & k \geq d \end{cases} \quad \text{cost} = 2d \text{ consumer calls per operating point.}$$

LEDGER

- GO-1. The consumer's invariant/nuisance split is identifiable ex ante from the consumer functional. [predicted]. [geometric-observation/claims/LEDGER.md:62](#) at 9f3829f.
- NEG-12. (Gate B, prospective, real LLM) On a trained frontier attention layer the blind probe recovers the read operator above the sealed bar *and* projection beats reconstruction. [refuted]. [geometric-observation/claims/LEDGER.md:105](#) at 9f3829f.

FIRST STATED

Chapter 6 section 6.2 of *Data Mining as Observation*, with the planted probe in [geometric-observation/claims/LEDGER.md](#) row GO-1.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 6 section 6.2	planted probe, overlap 0.936 vs 0.059, twelve of twelve, reconstruction 0.40, five of five	geometric-observation\claims\LEDGER.md row GO-1
chapter 11 section 11.7	GO-1 overlap 0.936 vs 0.059, flip 12 of 12, reconstruction 0.40	geometric-observation\claims\LEDGER.md row GO-1; geometric-observation\chapters\ch10_the_blind_probe.md:34-52

FAILURES AND CORRECTIONS

- NEG-12, [refuted]. (Gate B, prospective, real LLM) On a trained frontier attention layer the blind probe recovers the read operator above the sealed bar *and* projection beats reconstruction.

CONDITIONS

- Of a case, built so that the answer is known before the instrument reads it. An affine consumer has a known read operator, the outer product of its weight vector, so a probe that recovers it has been tested against a truth.
- The planted probe recovered the read subspace at overlap 0.936 against chance 0.059, twelve of twelve, and the real model then missed its bar at 0.567 against 0.60, which is the order the book keeps.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/ReadOperator.lean`, `theorems rank_one_reads_one_direction, readOp_mulVec, quad_readOp, quad_readOp_nonneg, readOp_mulVec_eq_zero_iff, readOp_diag, readOp_offdiag, readOp_symm, readOp_neg, affine_const_along_nuisance, readOp_affine, readOp_sqLength_basis`, at `observation-data-mining af776fd`.

`lean/DataMiningAsObservation/ProbeCliff.lean`, `theorems centralDiff_affine, centralDiff_basis, exists_blind_direction, indistinguishable, budget_cliff`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 3, 6, 8, 11.

RELATED

[blind probe](#) (p. 39); [read operator](#) (p. 282); [control](#) (p. 77); [budget cliff](#) (p. 49).

Poisson ceiling

instrument. Entry id `poisson-ceiling`.

DEFINITION

The largest neighbour count that at least one point in a dataset would reach by chance under a Poisson null, above which a count is evidence of a hub. Equation 0.17.

EQUATION

Book equation 0.17.

$$\Pr[X \geq c] = 1 - \sum_{i < c} e^{-\mu} \frac{\mu^i}{i!}, \quad c^* = \max\{c : n \Pr[X \geq c] \geq 1\}, \quad \mu = \frac{n_q k}{n}.$$

Book equation 10.3.

$$N_k(x \mid Q) = |\{q \in Q : x \in \text{top}_k(q)\}|, \quad \text{anti-hub} : N_k(x) = 0.$$

LEDGER

- NEG-11. (GO-5, prospective ×4) The $\alpha=1$ density/hubness quotient decisively and density-specifically restores invariant fidelity in a non-spectral domain. [refuted]. `geometric-observation/claims/LEDGER.md:104` at 9f3829f.

FIRST STATED

`openvector-bench, openvector-bench/openvector_bench/hubness.py:41-100`, and chapter 0 section 0.8 and chapter 3 section 3.5 of *Data Mining as Observation*.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 3 section 3.5	78 vs 369, density correlation about 0.67, 8 percent vs 34 percent, centering vs mutual-proximity rescaling	turboquant-pro\docs\HUBNESS_PRIMER.md:38-59,140-160
chapter 3 section 3.5	Poisson null, hub excess, budget parameter	openvector-bench\openvector_bench\hubness.py:41-100
chapter 8 section 8.1	G1 339.9 vs 71.9 same corpus, real about 61, rounds 1 to 4 at 300 to 420, three families falsified	openvector-bench\results\QUERY_COUPLING_ARTIFACT.md:1-20
chapter 10 section 10.2	anti-hubs as where compressed indexes fail first, aggregate recall barely moves	turboquant-pro\docs\HUBNESS_PRIMER.md:86-131
chapter 10 section 10.3	abstain below 2.5 k	turboquant-pro\docs\HUBNESS_PRIMER.md:140-160
chapter 11 section 11.6	count of ten, hubs and anti-hubs, max 78 vs 369, density correlation about 0.67, 8 percent vs 34 percent, abstain below 2.5k, centering vs mutual-proximity rescaling	turboquant-pro\docs\HUBNESS_PRIMER.md:1-60,60-170
chapter 11 section 11.6	anti-hub recall, p05, hub-rank correlation, hub-set overlap, the build gate	turboquant-pro\docs\HUBNESS_PRIMER.md:86-131
chapter 11 section 11.6	the centroid-injection attack	turboquant-pro\docs\HUBNESS_PRIMER.md:168-197, citing arXiv 2604.05480

FAILURES AND CORRECTIONS

- NEG-11, [refuted]. (GO-5, prospective ×4) The $\alpha=1$ density/hubness quotient decisively and density-specifically restores invariant fidelity in a non-spectral domain.

CONDITIONS

- Under the null that the slots are handed out at random, a row's count is Poisson with mean the slots per row, and the ceiling is the largest count at least one row would reach by chance. A count above it is evidence of a hub, and a count below it is not evidence of anything.
- The ceiling depends on the query set, through the mean and through the rows the queries reach, so it is recomputed when the queries change, and the program's first hubness numbers moved by a factor of several when query coupling was removed.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/PoissonCeiling.lean`, `theoremsmass_nonneg`, `tail_antitone`, `tail_zero`, `tail_le_one`, `expectedAtLeast_antitone`, `example_mean`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 3.

RELATED

[hubness](#) (p. 168); [anti-hub](#) (p. 17); [harness](#) (p. 159); [min-over-strata](#) (p. 217).

posited versus measured

concept. Entry id `posited-versus-measured`.

DEFINITION

The division of every claim in a report into those asserted by design and those backed by an artifact. Chapter 14.

EQUATION

Book equation 1.1.

$$O = (C, G, B).$$

LEDGER

- OT-11. Feedback-free staleness: streaming-retrieval damage tracks measured drift; derived-cadence re-allocation removes it. [void]. `geometric-observation/claims/LEDGER.md:48` at 9f3829f.
- NEG-15 (Bell boundary). *Query-conditioned hubness supplies a mechanism for Bell-inequality violation without action at a distance.* Refuted as a mechanism; the settings-as-queries reframing survives only as vocabulary. [demonstrated]. `geometric-observation/claims/LEDGER.md:93` at 9f3829f.

FIRST STATED

The epistemic banners of the four posited volumes, `geometric-ai/README.md`, `geometric-cognition/README.md`, `geometric-reasoning/README.md`, `geometric-education/README.md`, and chapter 14 section 14.8 of *Data Mining as Observation*.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 14 section 14.8	registry rules, V4 proved in Lean and verified 2026-08-17 at commit ea7ee82, D4 posited with the drift list, reliability weight as one formula one source, 8 of 9	<code>erisml-lib/docs/CONCEPT_REGISTRY.md:1-160</code>
chapter 14 section 14.8	the four posited volumes	<code>geometric-ai/README.md</code> ; <code>geometric-cognition/README.md</code> ; <code>geometric-reasoning/README.md</code> ; <code>geometric-education/README.md</code> epistemic banners

FAILURES AND CORRECTIONS

none

CONDITIONS

- Every claim in a report is either posited, asserted by design with no artifact behind it, or measured, backed by a committed artifact the reader can open. The division is stated at the top of the report and not inferred from the prose.
- A posited claim carries no ledger class above exploratory, and it stays there until a sealed measurement moves it. The ledger rule that says so is the one machine-checked in the book.
- Four volumes of the program are posited in their entirety and say so on their first page. Their claims are cited only as proposals.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/Ledger.lean`, `theorems step_none`, `step_miss_le`, `step_pass_ge`, `step_miss_refuted`, `exploratory_stays`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 14.

RELATED

[ledger class](#) (p. 204); [preregistration](#) (p. 258); [sealed](#) (p. 316); [reliability weight](#) (p. 298).

positive semidefinite

concept. Entry id positive-semidefinite.

DEFINITION

Of a symmetric matrix, having every quadratic form nonnegative, as the read operator and the covariance are. Chapter 0 section 0.3.

EQUATION

Book equation 0.9.

$$P_C = \mathbb{E}[g g^\top], \quad g = \nabla C(x).$$

Book equation 0.11.

$$P_C(x) = J(x)^\top G(C(x)) J(x), \quad J(x) = \frac{\partial C}{\partial x}(x), \quad \bar{P}_{C,\mu} = \mathbb{E}_\mu[P_C(x)].$$

Book equation 0.3.

$$\Sigma_{ij} = \mathbb{E}[(x_i - \mu_i)(x_j - \mu_j)], \quad \text{tr } \Sigma = \sum_i \Sigma_{ii}.$$

LEDGER

- OT-7. The damage form, trace pairing, rank, and loading covariance are $\text{GL}(d)$ -invariant under $P' = A^{-\top} P A^{-1}$, while spectrum, effective rank, principal angles, and water-filling are $O(d)$ -only — consumer-weighted damage is a geometric scalar, and P3’s cliff cannot be bought down by reparameterization. [demonstrated]. [geometric-observation/claims/LEDGER.md:30 at 9f3829f](#).

FIRST STATED

Chapter 0 section 0.3 and section 0.5 of *Data Mining as Observation*.

MEASUREMENTS

Where the book states it	Numbers, as the book’s sources table records them	Source
chapter 2 section 2.2	read subspace small, operator local, pullback composition, rank cannot increase	geometric-observation\chapters\ch05_the_read_metric_and_the_c48; geometric-observation\chapters\ch06_mathematical_preliminaries27

FAILURES AND CORRECTIONS

none

CONDITIONS

- Of a symmetric matrix, having every quadratic form nonnegative. A sum of such matrices is one, a nonnegative multiple is one, an outer product of a vector with itself is one, the read operator is one whenever the weights are nonnegative, and the diagonal entries are nonnegative.

- The local metric on a consumer's output is positive semidefinite and exists only where the output metric has a local quadratic representation, and every eigenvalue of a positive semidefinite matrix is nonnegative.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/PositiveSemidefinite.lean`, theorems `psd_add`, `psd_smul`, `psd_outer`, `psd_readOp`, `psd_diag_nonneg`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 2.

RELATED

[read operator](#) (p. 282); [covariance matrix](#) (p. 85); [outer product](#) (p. 240); [eigenvalue, eigenvector](#) (p. 116); [metric](#) (p. 216).

precision, recall

concept. Entry id `precision-recall`.

DEFINITION

The fraction of predicted positives that are truly positive, and the fraction of true positives that were predicted. Equation 0.28.

EQUATION

Book equation 0.28.

$$P = \frac{TP}{TP + FP}, \quad R = \frac{TP}{TP + FN}, \quad F_1 = \frac{2PR}{P + R}.$$

Book equation 6.2.

$$\max_{\tau, \text{dir}} F_1\left(\mathbf{1}[\text{dir}(g(f(X)), \tau)], y\right) = \max_{\tau, \text{dir}} F_1\left(\mathbf{1}[\text{dir}(f(X), \tau)], y\right)$$

for every strictly monotone g .

LEDGER

none

FIRST STATED

Chapter 0 section 0.14 of *Data Mining as Observation*, with the program's precision targets in `gtc-prototype/docs/SPECTRUM_FINDINGS.md`: 84–88.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 14 section 14.7	51 percent moderated at 80 and 95 percent precision on the balanced set	<code>gtc-prototype/docs/SPECTRUM_FINDINGS.md</code> : 84–88

FAILURES AND CORRECTIONS

none

CONDITIONS

- The fraction of predicted positives that are truly positive, and the fraction of true positives that were predicted, both in the unit interval. F1 is their harmonic mean, between the smaller of the two and their arithmetic mean, equal to both when they agree, and zero when either is zero.
- In counts F1 is twice the true positives over twice the true positives plus the false positives and false negatives, the form the Youden ceiling bounds.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/PrecisionRecall.lean`, `theorems precision_mem_unit, recall_mem_unit, f1_le_mean, min_le_f1, f1_self, f1_zero, f1_counts`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 1, 4, 6, 8, 9, 10, 11, 12, 14.

RELATED

[F1](#) (p. 132); [threshold](#) (p. 350); [recall at k](#) (p. 286); [Youden F1 bound](#) (p. 372).

preregistration

instrument. Entry `id` preregistration.

DEFINITION

Committing the hypothesis, the bar, and the analysis before the measurement is run, so that the record shows what was predicted. Chapter 8.

EQUATION

Book equation 8.3.

$$\Pr[\text{at least one of } m \text{ null tests passes}] = 1 - (1 - \alpha)^m, \quad \alpha_{\text{Bonferroni}} = \frac{\alpha}{m}.$$

LEDGER

- OT-7. The damage form, trace pairing, rank, and loading covariance are $GL(d)$ -invariant under $P' = A^{-T}PA^{-1}$, while spectrum, effective rank, principal angles, and water-filling are $O(d)$ -only — consumer-weighted damage is a geometric scalar, and P3's cliff cannot be bought down by reparameterization. [demonstrated]. `geometric-observation/claims/LEDGER.md:30` at 9f3829f.
- GO-2 (neg. half: not reconstruction). At matched bits, downstream preservation is not controlled by reconstruction error. [demonstrated]. `geometric-observation/claims/LEDGER.md:63` at 9f3829f.

FIRST STATED

Chapter 8 section 8.8 of *Data Mining as Observation*, with the program's template `observation-theory-campaigns/experiments/PREREG-TEMPLATE.md:1-71` and Volume 14's protocol `geometric-observation/PROTOCOL.md:58-75`.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 1 section 1.1	the four tasks, the churn framing, the framing workshop	TSK chapter 1; instructor working documents, not public [<code>@bond2026course</code>], <code>ECE_514-01_FA26_session-outlines.md:29-60</code>

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 1 section 1.3	the four failure modes	ECE_514-01_FA26_session-outlines.md:38
chapter 1 section 1.6	the course map	ECE_514-01_FA26_session-outlines.md:29-332
chapter 1 section 1.7	the field guide, the decision test, the red flags	ECE_514-01_FA26_session-outlines.md:39-46
chapter 2 section 2.3	MCAR, MAR, MNAR and the remedies, imputation before splitting	TSK 2e section 2.2; instructor working documents, not public [@bond2026course], ECE_514-01_FA26_session-outlines.md:64-72
chapter 2 section 2.3	the missing-data rule as a required preregistration field	observation-theory-campaigns\experiments\PREREG-TEMPLATE.md:47-52
chapter 5 section 5.3	the transferable idea as a course learning outcome	instructor working documents, not public [@bond2026course], ECE_514-01_FA26_session-outlines.md:117-136
chapter 7 section 7.6	the preregistration fields	instructor working documents, not public [@bond2026course], ECE_514-01_FA26_session-outlines.md:157-176; chapter 8 section 8.8 of this book
chapter 8 section 8.3	anti-vacuity bar, quoted	observation-theory-campaigns\experiments\PREREG-PF5-002.md:79-86
chapter 8 section 8.8	template fields	observation-theory-campaigns\experiments\PREREG-TEMPLATE.md:1-71
chapter 8 section 8.8	declaration drafted 2026-08-17, sealed 2026-08-18, one count corrected, G2	geometric-observation\crucible\DECLARATION-V1.md:1-20; geometric-observation\crucible\OT-CRUCIBLE-4.md:31-35

FAILURES AND CORRECTIONS

- observation-theory-campaigns/ERRATA.md:90-110 at 9d86211. ## E3. Two wrong bar values in the campaign notes **Found** 2026-08-07, same pass. Not a published error. experiments/CAMPAIGN.md twice described the PF4-006 entry-point and timestep clause as having a bar of 1e-10. The sealed declaration and the committed record both set that bar at 1e-8. The 1e-10 figure is the separate translation-covariance bar. The narrative was wrong and the verdict was not, since the clause failed at 1.3e-2 against either number. The same document described PREREG-PF4-009 as passing all six bars. The sealed document groups its requirements into six bars and the governed runner reports eight items, and the mapping between them was never declared. Both statements are now made explicitly. Corrected in place, since these are working notes rather than sealed documents or published records. —

CONDITIONS

- A preregistration commits the hypothesis, the bar, the analysis, and the missing-data rule before the measurement is run, so that the record shows what was predicted rather than what was found.
- Its bar must be anti-vacuous. A pass that any instrument would have produced, a threshold met by every

reading, proves nothing, and the program's own first PF5 pass is the case.

- Multiple looks at the data need a corrected threshold, and a preregistration that names one look needs none.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/Bonferroni.lean`, theorems `family_error_le`, `bonferroni`, `one_look`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14.

RELATED

[sealed](#) (p. 316); [ledger class](#) (p. 204); [harness](#) (p. 159); [certificate](#) (p. 55).

principal component analysis

instrument. Entry `id principal-component-analysis`.

DEFINITION

Projection of the data onto the eigenvectors of its covariance with the largest eigenvalues, the reduction that minimizes reconstruction error for the identity reader. Chapter 4.

EQUATION

Book equation 0.5.

$$\Sigma v_i = \lambda_i v_i, \quad \Sigma = \sum_{i=1}^d \lambda_i v_i v_i^\top, \quad v_i \cdot v_j = 0 \ (i \neq j).$$

Book equation 0.7.

$$r_{\text{eff}} = \frac{(\sum_i \lambda_i)^2}{\sum_i \lambda_i^2}.$$

Book equation 4.2.

$$D(b) = \sum_i s_i \sigma_i^2 2^{-2b_i}, \quad s_i = v_i^\top P_C v_i, \quad b_i^* = \max\left(0, \frac{1}{2} \log_2 \frac{s_i \sigma_i^2}{\theta}\right), \quad \sum_i b_i^* = B,$$

LEDGER

- GO-1. The consumer's invariant/nuisance split is identifiable ex ante from the consumer functional. [predicted]. `geometric-observation/claims/LEDGER.md:62` at `9f3829f`.
- GO-6. At matched rate, output coding $\leq$ surrogate $\leq$ reconstruction on the consumer metric; the output-reconstruction gap is governed by the $\ker P_C$ entropy share, and the surrogate-output gap vanishes as rate grows. [demonstrated]. `geometric-observation/claims/LEDGER.md:68` at `9f3829f`.

FIRST STATED

Pearson, on lines and planes of closest fit, 1901, and Hotelling, 1933, as chapter 4 section 4.1 of *Data Mining as Observation* reads them, with the program's truncation record in `turboquant-pro/benchmarks/RESULTS_glove.md:1-40`.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 4 section 4.2	effective rank as participation ratio, energy rank	readscope\readscope\spectrum.py:35-70
chapter 4 section 4.5	GloVe table, 73 percent at 64 components, 0.685 vs 0.862 and 0.866, 0.906 at matched bytes, 0.989 at 37 bytes, 768 to 256 keeps about 99 percent, the 95 percent rule	turboquant-pro\benchmarks\RESULTS_glove.md:1-40
chapter 4 section 4.5	truncation claim reproducible, loses on compact sets	turboquant-pro\CLAIMS.md:28-49
chapter 11 section 11.5	9.6x at recall 0.999 CI-gated on GloVe 1.18M; 32x at 0.9993 on private 199k LaBSE, ties OPQ, beats RaBitQ, 20x build; 27.7x and 114x reported; PCA truncation loses on compact sets; 20x at 199k and 4x at 1M over OPQ; RaBitQ builds in under a second	turboquant-pro\CLAIMS.md:28-49

FAILURES AND CORRECTIONS

none

CONDITIONS

- Projection of the data onto the eigenvectors of its covariance with the largest eigenvalues. The variance along a unit direction lies between the smallest and the largest eigenvalue, a basis direction attains its own eigenvalue, and the error of keeping the first k components is the sum of the dropped eigenvalues.
- It is the reduction that minimizes reconstruction error for the identity reader, which weighs all directions equally, which is to say for no particular reader at all. At matched reconstruction the consumer-aware code beat it in twelve of twelve domains, and truncation loses on compact sets.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

lean/DataMiningAsObservation/PCA.lean, theorems `varAlong_basis`, `varAlong_le`, `varAlong_ge`, `dropped_eq`, `dropped_nonneg`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 1, 4, 6, 12.

RELATED

[covariance matrix](#) (p. 85); [explained variance](#) (p. 129); [effective rank](#) (p. 115); [projection](#) (p. 263); [identity reader](#) (p. 171).

product quantization

instrument. Entry `id product-quantization`.

DEFINITION

Splitting a vector into pieces and quantizing each piece with its own codebook. Chapter 11.

EQUATION

Book equation 11.3.

$$\text{recall}@k = \frac{|\text{returned top-}k \cap \text{true top-}k|}{k},$$

true top- k computed from the uncompressed vectors.

Book equation 4.2.

$$D(b) = \sum_i s_i \sigma_i^2 2^{-2b_i}, \quad s_i = v_i^\top P_C v_i, \quad b_i^* = \max\left(0, \frac{1}{2} \log_2 \frac{s_i \sigma_i^2}{\theta}\right), \quad \sum_i b_i^* = B,$$

LEDGER

- NEG-14. (GO-P-2026-037, prospective) a2_probe.median_unit_displacement is a single-statistic predictor of the flip regime (unit_disp $\geq 1.0 \Rightarrow$ generic-polar-flip vs needs-blind-probe). [refuted]. geometric-observation/claims/LEDGER.md:108 at 9f3829f.

FIRST STATED

Jégou, Douze, and Schmid, product quantization for nearest neighbor search, 2011, as chapter 11 section 11.3 of *Data Mining as Observation* presents it, with the program's comparisons in openvector-bench and turboquant-pro.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 10 section 10.4	area map, boundary rule, hash, refuse not warn, intra and transit counts, area classes, abstention rule	turboquant-pro\docs\STRATA_RFC.md:24-98
chapter 10 section 10.4	second prediction inverted, transit 0.389 vs 0.291, centrality difference signs, share 0.473 vs 0.391, seven of thirteen backbone	turboquant-pro\docs\RESULTS_multilingual_strata.md:55-90; turboquant-pro\docs\STRATA_RFC.md:98-130

FAILURES AND CORRECTIONS

- NEG-14, [refuted]. (GO-P-2026-037, prospective) a2_probe.median_unit_displacement is a single-statistic predictor of the flip regime (unit_disp $\geq 1.0 \Rightarrow$ generic-polar-flip vs needs-blind-probe).

CONDITIONS

- Splitting a vector into pieces and quantizing each with its own codebook. The squared error of the whole is the sum of the pieces' errors, so the best code for the whole is the best code for each piece separately, and K entries per piece over M pieces address K to the M cells in $M \log_2 K$ bits.
- It is evaluated by recall at k per stratum and by the rank certificate, never by reconstruction error, and the anti-hub stratum is where it fails first.

Conditions are curated in entries.toml rather than read from a record.

MACHINE CHECKED

lean/DataMiningAsObservation/ProductQuantization.lean, theorems sq_error_add, inf_add, bits_of_codebooks, table_row, at observation-data-mining af776fd.

USED IN

Data Mining as Observation chapters 0, 4, 8, 11, 13.

RELATED

[recall at k](#) (p. 286); [anti-hub](#) (p. 17); [water-filling](#) (p. 365); [rank certificate](#) (p. 275).

projection

concept. Entry id projection.

DEFINITION

The component of a vector along a unit direction. Equation 0.2.

EQUATION

Book equation 0.2.

$$\text{proj}_u(x) = (u \cdot x) u, \qquad \|u\| = 1.$$

Book equation 0.12a.

$$x \sim_C x' \iff d_G(C(x), C(x')) = 0.$$

Book equation 0.12b.

$$x \sim_{\bar{P}_{C,\mu}} x' \iff x - x' \in \ker \bar{P}_{C,\mu}.$$

LEDGER

- GO-1. The consumer’s invariant/nuisance split is identifiable ex ante from the consumer functional. [predicted]. `geometric-observation/claims/LEDGER.md:62` at 9f3829f.

FIRST STATED

Chapter 0 section 0.2 of *Data Mining as Observation*, with the program’s angular projection in Volume 14 chapter 9, `geometric-observation/chapters/ch09_legibility.md:31-41`.

MEASUREMENTS

Where the book states it	Numbers, as the book’s sources table records them	Source
chapter 6 section 6.2	planted probe, overlap 0.936 vs 0.059, twelve of twelve, reconstruction 0.40, five of five	<code>geometric-observation/claims/LEDGER.md</code> row GO-1
chapter 11 section 11.7	GO-1 overlap 0.936 vs 0.059, flip 12 of 12, reconstruction 0.40	<code>geometric-observation/claims/LEDGER.md</code> row GO-1; <code>geometric-observation/chapters/ch10_the_blind_probe.md:34-52</code>

FAILURES AND CORRECTIONS

none

CONDITIONS

- The component of a vector along a unit direction, the direction scaled by their dot product. Projecting twice is projecting once, the remainder is orthogonal to the direction, the projection is no longer than the vector, and adding anything orthogonal to the direction leaves it unchanged.
- The last is the quotient a one-direction reader takes. Row normalization is the projection onto the read subspace of the geodesic-rank consumer, and the invariant-nuisance split of the ledger is the same statement for a general consumer.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

lean/DataMiningAsObservation/Projection.lean, theorems proj_proj, dot_sub_proj, proj_sq_le, proj_add_orth, proj_add, at observation-data-mining af776fd.

USED IN

Data Mining as Observation chapters 0, 1, 3, 4, 6, 7, 9, 11.

RELATED

[dot product](#) (p. 110); [read subspace](#) (p. 285); [quotient](#) (p. 271); [nuisance](#) (p. 233).

p-value

concept. Entry id p-value.

DEFINITION

The probability of seeing a difference at least as large as the one observed if the true difference were zero. Chapter 0 section 0.9.

EQUATION

Book equation 8.3.

$$\Pr[\text{at least one of } m \text{ null tests passes}] = 1 - (1 - \alpha)^m, \quad \alpha_{\text{Bonferroni}} = \frac{\alpha}{m}.$$

LEDGER

none

FIRST STATED

Chapter 0 section 0.9 of *Data Mining as Observation*, with the program's own p-values in the ensemble comparison of chapter 7 and the C-12 record in readscope.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 6 section 6.4	the uncorrected t stored as sigma, 251 and 15.84, nine wins to three, seven, eight, 3 wins 17 ties 11 losses on 31 datasets, the three-way inconsistency	constraint-gap\review\FINDINGS.md:1-35; constraint-gap\README.md:57-76
chapter 8 section 8.5	variance inflation 251, SE inflation 15.84, J_eff 3.98, about 62 needed, 3 of 9 at full, 7 at half, 8 at a third	constraint-gap\review\FINDINGS.md:1-35

FAILURES AND CORRECTIONS

none

CONDITIONS

- Against a finite null sample, the fraction of null draws at least as large as the observation. It lies in the unit interval, never rises as the observation grows, and under the null the fraction of draws with p-value at or below a level is at most that level, which is what makes a threshold on it a false-positive rate.
- A p-value counts one look. Forty looks at five percent pass two by chance, and the book's rule is to count every comparison made and correct for it.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

lean/DataMiningAsObservation/PValue.lean, theorems pvalue_mem_unit, pvalue_antitone, uniform_bound, at observation-data-mining af776fd.

USED IN

Data Mining as Observation chapters 0, 11.

RELATED

[multiple comparisons](#) (p. 221); [confidence interval](#) (p. 71); [harness](#) (p. 159); [preregistration](#) (p. 258).

Q

quantization

concept. Entry id quantization.

DEFINITION

Replacing each number with one of a small set of allowed values so that it can be stored in a few bits. Chapter 0 section 0.11.

EQUATION

Book equation 0.13.

$$D(b) = \sum_i s_i v_i 4^{-b_i}, \quad b_i = \max\left(0, \frac{1}{2} \log_2 \frac{s_i v_i}{\theta}\right), \quad \sum_i b_i = B.$$

Book equation 11.1.

$$\cos(k, \hat{k}) = 0.995 \quad \text{while} \quad \text{PPL} : 12.24 \rightarrow 10643.$$

LEDGER

- GO-2 (neg. half: not reconstruction). At matched bits, downstream preservation is not controlled by reconstruction error. [demonstrated]. `geometric-observation/claims/LEDGER.md:63` at 9f3829f.
- NEG-2. Reconstruction cosine as a proxy for key quality. [refuted]. `geometric-observation/claims/LEDGER.md:95` at 9f3829f.

FIRST STATED

Chapter 0 section 0.11 and chapter 11 section 11.2 of *Data Mining as Observation*, with the program’s quantizer table in `turboquant-pro/docs/KV_KEYS_FINDING.md:1-49`.

MEASUREMENTS

Where the book states it	Numbers, as the book’s sources table records them	Source
chapter 1 section 1.4	cosine 0.995 and perplexity of order ten thousand, the recalibration negative	<code>geometric-observation\chapters\ch02_failure_of_observer_free-60;</code> <code>geometric-observation\chapters\ch16_honest_negatives.md</code> NEG-2 and NEG-4; <code>turboquant-pro\docs\KV_KEYS_FINDING.md:1-49</code>
chapter 2 section 2.5	cosine 0.995 and the softmax reader	<code>turboquant-pro\docs\KV_KEYS_FINDING.md:1-49</code>
chapter 3 section 3.2	condition (A2), tangential displacement	<code>turboquant-pro\docs\KV_KEYS_FINDING.md:61-86;</code> <code>geometric-observation\chapters\ch09_legibility.md:42-54</code>

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 8 section 8.2	cosine 0.995, perplexity near 1e4	turboquant-pro\docs\KV_KEYS_FINDING.md:1-49; geometric-observation\chapters\ch16_honest_negatives.md NEG-2
chapter 8 section 8.9	13.7 ROUGE-L, 0.25/4.19/9.60/13.7 at 64/128/256/512, re-validation negative 0.31 n=40, 26.64 under symmetric nf4, _quant_nf4a_group unchanged since 289bdfc before 4f7baab	turboquant-pro\CLAIMS.md:63-81; turboquant-pro\benchmarks\kvquant_matrix\REVAL-2026-08-08.md
chapter 11 section 11.1	condition (A2), cosine satisfies it, post-rotary keys do not, the cone below cell size	turboquant-pro\docs\KV_KEYS_FINDING.md:61-86; the-angular-observer\README.md:26-31
chapter 11 section 11.2	fp16 12.24, values-only 13.12, PolarQuant K4 10643 and 0.095, per-channel uniform K4 14.91 and 0.062, per-channel NUQ K3 15.77 and 0.148, 2.4x and 670x, pre-rotary near 22000, 2 key heads serve 12 query heads	turboquant-pro\docs\KV_KEYS_FINDING.md:1-49

FAILURES AND CORRECTIONS

- NEG-2, [refuted]. Reconstruction cosine as a proxy for key quality.

CONDITIONS

- Replacing each number with one of a small set of allowed values so that it can be stored in a few bits. A uniform quantizer with a given step has error at most half the step and zero error on a level, and halving the step halves the bound.
- Those are the identity reader's bounds. What a quantizer does to a consumer is the flip's question, and the key quantizer table of chapter 11 is the case in which the smaller reconstruction error was the worse code.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/Quantization.lean`, `theorems error_1e_half_step`, `quantize_level`, `half_step_bound`, `sq_error_1e`, `finer_step`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 2, 3, 4, 7, 8, 10, 11.

RELATED

[bit](#) (p. 38); [codebook](#) (p. 61); [per-channel quantizer](#) (p. 248); [direction-only quantizer](#) (p. 104); [the flip](#) (p. 138).

query, key, value

concept. Entry id query-key-value.

DEFINITION

The three vectors each token produces inside attention. The query is compared to earlier keys and the result weights the values. Chapter 0 section 0.11.

EQUATION

Book equation 0.23.

$$\text{softmax}(z)_i = \frac{e^{z_i}}{\sum_j e^{z_j}}, \quad \text{output} = \sum_i \text{softmax}\left(\frac{q \cdot k_i}{\sqrt{d}}\right)_i v_i.$$

Book equation 11.1.

$$\cos(k, \hat{k}) = 0.995 \quad \text{while} \quad \text{PPL} : 12.24 \rightarrow 10643.$$

LEDGER

- NEG-2. Reconstruction cosine as a proxy for key quality. [refuted]. `geometric-observation/claims/LEDGER.md:95` at 9f3829f.

FIRST STATED

Chapter 0 section 0.11 of *Data Mining as Observation*, with the program's key-side measurements in `turboquant-pro/docs/KV_KEYS_FINDING.md:1-49`.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 1 section 1.4	cosine 0.995 and perplexity of order ten thousand, the recalibration negative	<code>geometric-observation\chapters\ch02_failure_of_observer_free_60;</code> <code>geometric-observation\chapters\ch16_honest_negatives.md</code> NEG-2 and NEG-4; <code>turboquant-pro\docs\KV_KEYS_FINDING.md:1-49</code>
chapter 2 section 2.5	cosine 0.995 and the softmax reader	<code>turboquant-pro\docs\KV_KEYS_FINDING.md:1-49</code>
chapter 3 section 3.2	condition (A2), tangential displacement	<code>turboquant-pro\docs\KV_KEYS_FINDING.md:61-86;</code> <code>geometric-observation\chapters\ch09_legibility.md:42-54</code>
chapter 8 section 8.2	cosine 0.995, perplexity near 1e4	<code>turboquant-pro\docs\KV_KEYS_FINDING.md:1-49;</code> <code>geometric-observation\chapters\ch16_honest_negatives.md</code> NEG-2
chapter 11 section 11.1	condition (A2), cosine satisfies it, post-rotary keys do not, the cone below cell size	<code>turboquant-pro\docs\KV_KEYS_FINDING.md:61-86;</code> <code>the-angular-observer\README.md:26-31</code>
chapter 11 section 11.2	fp16 12.24, values-only 13.12, PolarQuant K4 10643 and 0.095, per-channel uniform K4 14.91 and 0.062, per-channel NUQ K3 15.77 and 0.148, 2.4x and 670x, pre-rotary near 22000, 2 key heads serve 12 query heads	<code>turboquant-pro\docs\KV_KEYS_FINDING.md:1-49</code>

FAILURES AND CORRECTIONS

- NEG-2, [refuted]. Reconstruction cosine as a proxy for key quality.

CONDITIONS

- The three vectors each token produces inside attention. The query is compared to earlier keys and the result weights the values, so a head reads the keys only through their scores against the query and its output lies between the smallest and largest value.
- Keys are read along the query and values are averaged, so a compression that preserves query-key scores preserves every output, and one that preserves the keys' cosine need not.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/Attention.lean`, `theorems output_le_max`, `min_le_output`, `output_congr`, `output_nuisance`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14.

RELATED

[attention](#) (p. 23); [head](#) (p. 162); [KV cache](#) (p. 190); [softmax](#) (p. 327).

query set

concept. Entry id `query-set`.

DEFINITION

The rows a retrieval benchmark asks about. Hubness and the never-retrieved floor are properties of it. Chapter 3 section 3.5 and chapter 10 section 10.2.

EQUATION

Book equation 10.3.

$$N_k(x \mid Q) = |\{q \in Q : x \in \text{top}_k(q)\}|, \quad \text{anti-hub} : N_k(x) = 0.$$

Book equation 10.4.

$$\text{never retrieved} \geq 1 - \frac{|Q|k}{n} \quad \text{whenever} \quad |Q|k < n.$$

Book equation 0.17.

$$\Pr[X \geq c] = 1 - \sum_{i < c} e^{-\mu} \frac{\mu^i}{i!}, \quad c^* = \max\{c : n \Pr[X \geq c] \geq 1\}, \quad \mu = \frac{n_q k}{n}.$$

LEDGER

- NEG-15 (Bell boundary). *Query-conditioned hubness supplies a mechanism for Bell-inequality violation without action at a distance.* Refuted as a mechanism; the settings-as-queries reframing survives only as vocabulary. [demonstrated]. `geometric-observation/claims/LEDGER.md:93` at 9f3829f.
- NEG-11. (GO-5, prospective ×4) The $\alpha=1$ density/hubness quotient decisively and density-specifically restores invariant fidelity in a non-spectral domain. [refuted]. `geometric-observation/claims/LEDGER.md:104` at 9f3829f.

FIRST STATED

Chapter 3 section 3.5 and chapter 10 section 10.2 of *Data Mining as Observation*, with the query-coupling artifact in `openvector-bench/results/QUERY_COUPLING_ARTIFACT.md:1-20`.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 3 section 3.5	Poisson null, hub excess, budget parameter	openvector-bench\openvector_bench\hubness.py:41-100
chapter 3 section 3.5	query mass best single feature at every K for all four responses, seven features add at most 20 percent, threshold 1.25 on three of four, zero of four, 1000 real queries, 1024 dimensions	openvector-bench\results\R13_STAGE0_RESULT.md:40-50; openvector-bench\results\R13_STAGE1_RESULT.md:1-40
chapter 8 section 8.1	G1 339.9 vs 71.9 same corpus, real about 61, rounds 1 to 4 at 300 to 420, three families falsified	openvector-bench\results\QUERY_COUPLING_ARTIFACT.md:1-20
chapter 10 section 10.2	five kinds, balanced accuracy 0.611, 0.718, 0.684, the category table, per-category 0.57 to 0.82, the sweep, pigeonhole floor	openvector-bench\results\R13_STAGE1_RESULT.md:40-75
chapter 10 section 10.2	half 2 failed, at most 1.28 times against 2, withdrawal and its scope	openvector-bench\results\R13_STAGE1_RESULT.md:75-110
chapter 11 section 11.6	query mass best single feature at every K for all four responses, seven features add at most 20 percent at 12 leaves, threshold 1.25 on three of four, zero of four, 1000 real queries, 1024 dimensions	openvector-bench\results\R13_STAGE0_RESULT.md:40-50; openvector-bench\results\R13_STAGE1_RESULT.md:1-40

FAILURES AND CORRECTIONS

- NEG-11, [refuted]. (GO-5, prospective $\times 4$) The $\alpha=1$ density/hubness quotient decisively and density-specifically restores invariant fidelity in a non-spectral domain.

CONDITIONS

- The rows a retrieval benchmark asks about. Two query sets that retrieve the same lists have the same hubs, no queries give no hubs, and the rows never retrieved number at least the row count less the queries times k, so hubness and the never-retrieved floor are properties of the query set.
- Queries drawn from the corpus inflated the first hub counts by a factor of several, and a generator matched on the query budget would be matched on an artifact.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

lean/DataMiningAsObservation/Hub.lean, theorems `card_hubs_le`, `hubs_congr`, `hubs_anti`, `hubs_empty`, at `observation-data-mining af776fd`.

lean/DataMiningAsObservation/Hubness.lean, theorems `sum_count`, `sum_count_eq`, `count_congr`, `antiHub_iff`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 1, 2, 3, 4, 8, 10, 11, 12, 13.

RELATED

[hub](#) (p. 166); [hubness](#) (p. 168); [Poisson ceiling](#) (p. 253); [recall at k](#) (p. 286); [harness](#) (p. 159).

quotient

concept. Entry id quotient.

DEFINITION

The space of data with a set of transformations declared not to matter, so that two rows differing only by such a transformation are the same point. Every distance is a distance on some quotient. Equation 0.12.

EQUATION

Book equation 0.12a.

$$x \sim_C x' \iff d_G(C(x), C(x')) = 0.$$

Book equation 0.12b.

$$x \sim_{\bar{P}_{C,\mu}} x' \iff x - x' \in \ker \bar{P}_{C,\mu}.$$

LEDGER

- GO-1. The consumer's invariant/nuisance split is identifiable ex ante from the consumer functional. [pre-dicted]. [geometric-observation/claims/LEDGER.md:62 at 9f3829f](#).
- GO-5. An $\alpha=1$ density/hubness quotient restores invariant fidelity in ≥ 1 non-spectral domain. [refuted]. [geometric-observation/claims/LEDGER.md:67 at 9f3829f](#).
- NEG-6. Relative per-channel-demeaned error norm is the quotient-tangential quantity that controls softmax-KL. [refuted]. [geometric-observation/claims/LEDGER.md:99 at 9f3829f](#).

FIRST STATED

Volume 14, chapter 5, [geometric-observation/chapters/ch05_the_read_metric_and_the_quotient.md:78-90](#), DOI 10.5281/zenodo.21776291, and chapter 3 section 3.1 of *Data Mining as Observation*, where every similarity measure induces a distance on some quotient.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 2 section 2.2	read subspace small, operator local, pullback composition, rank cannot increase	geometric-observation\chapters\ch05_the_read_metric_and_the_quotient.md:48 ; geometric-observation\chapters\ch06_mathematical_preliminaries.md:27
chapter 2 section 2.2	read distortion controls but is not a complete rank statistic, twelve of twelve, one middle pair misordered, NEG-9	geometric-observation\chapters\ch05_the_read_metric_and_the_quotient.md:75
chapter 3 section 3.6	GO-5 refuted in four attempts, under 2 percent, control 0.015 vs 0.019, alpha hurts in direct and quantized retrieval, NEG-11	geometric-observation\claims\LEDGER.md rows GO-5 and NEG-11; geometric-observation\experiments\GO5-diffusion-distance-NOTES.md
chapter 6 section 6.2	planted probe, overlap 0.936 vs 0.059, twelve of twelve, reconstruction 0.40, five of five	geometric-observation\claims\LEDGER.md row GO-1

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 11 section 11.7	GO-1 overlap 0.936 vs 0.059, flip 12 of 12, reconstruction 0.40	geometric-observation\claims\LEDGER.md row GO-1; geometric-observation\chapters\ch10_the_blind_probe.md:34-52

FAILURES AND CORRECTIONS

- GO-5, [refuted]. An $\alpha=1$ density/hubness quotient restores invariant fidelity in ≥ 1 non-spectral domain.
- NEG-6, [refuted]. Relative per-channel-demeaned error norm is the quotient-tangential quantity that controls softmax-KL.

CONDITIONS

- The equivalence x related to x' when their difference lies in the kernel of the read operator is exact for an affine consumer with a fixed output geometry.
- Three things are kept apart. Exact equivalence, that the consumer's outputs agree under its output metric. A local null direction, in the kernel of the operator at one row. A workload-common null direction, in the kernel of the workload average, unread at almost every row. For a nonlinear consumer the third does not deliver the first, and cosine similarity is the case, whose radial direction turns with the row.
- An invertible transform identifies no two rows and forms no quotient. It changes the geometry a reader sees. Discretization changes the quotient.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

lean/DataMiningAsObservation/ReadOperator.lean, theorems rank_one_reads_one_direction, readOp_mulVec, quad_readOp, quad_readOp_nonneg, readOp_mulVec_eq_zero_iff, readOp_diag, readOp_offdiag, readOp_symm, readOp_neg, affine_const_along_nuisance, readOp_affine, readOp_sqLength_basis, at observation-data-mining af776fd.

USED IN

Data Mining as Observation chapters 0, 1, 2, 3, 5, 6, 8, 9, 11, 12.

RELATED

[read operator](#) (p. 282); [identity reader](#) (p. 171); [the flip](#) (p. 138); [hubness](#) (p. 168).

R

random forest

instrument. Entry id random-forest.

DEFINITION

An ensemble of decision trees, each fit on a bootstrap sample with a random subset of features at each split. Chapter 7.

EQUATION

Book equation 6.1.

$$s(x) = w \cdot x + b, \quad P_C = \mathbb{E}[\sigma'(s)^2] w w^\top, \quad \text{rank } P_C = 1.$$

Book equation 14.5.

$$\text{FC}_g = \Pr[y = \text{violation} \mid \hat{y} = \text{clear}, g], \quad \text{verdict} = \max_{g: n_g \geq n_{\min}} \text{FC}_g, \quad \text{ABSTAIN otherwise.}$$

LEDGER

none

FIRST STATED

Breiman, random forests, 2001, as chapter 7 section 7.1 of *Data Mining as Observation* reads it.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 6 section 6.3	the five-dataset table, breast cancer 0.955 to 0.963, the pattern, DOI 10.5281/zenodo.20660206	theory-radar\README.md:100-140
chapter 7 section 7.1	bagging variance, out-of-bag estimation, random forests, importances, AdaBoost	Hastie, Tibshirani, Friedman, ESL 2e chapters 15 and 10.1; TSK 2e 4.10
chapter 7 section 7.3	the five-dataset outcomes	theory-radar\README.md:100-140

FAILURES AND CORRECTIONS

none

CONDITIONS

- An ensemble of decision trees, each fit on a bootstrap sample with a random subset of features at each split, that averages or votes. The average lies between its members and its squared error is at most their mean squared error, and each tree is flat within a leaf so the forest's importances count splits.
- On the five-dataset benchmark the forest and the boosted trees win when the boundary needs many features and lose to a formula when it needs few.

Conditions are curated in entries.toml rather than read from a record.

MACHINE CHECKED

lean/DataMiningAsObservation/Ensemble.lean, theorems average_between, sq_average_le, average_const, at observation-data-mining af776fd.

lean/DataMiningAsObservation/DecisionTree.lean, theorems stump_flat_left, stump_flat_right, split_reads_one_axis, gini_le_half, gini_eq_zero_iff, at observation-data-mining af776fd.

USED IN

Data Mining as Observation chapters 6, 7.

RELATED

[ensemble](#) (p. 121); [decision tree](#) (p. 95); [bagging](#) (p. 31); [importance](#) (p. 172); [boosting](#) (p. 44).

rank

concept. Entry id rank.

DEFINITION

The dimension of a matrix's column space, at most the number of rows and of columns. An outer product has rank at most one, and a product has rank at most either factor's. Chapter 0 section 0.2.

EQUATION

Book equation 0.7.

$$r_{\text{eff}} = \frac{(\sum_i \lambda_i)^2}{\sum_i \lambda_i^2}.$$

Book equation 2.1.

$$P_{C_2 \circ C_1}(x) = J_1(x)^\top P_{C_2}(C_1(x)) J_1(x), \quad \text{rank } P_{C_2 \circ C_1}(x) \leq \text{rank } P_{C_2}(C_1(x)) \quad \text{at each row } x.$$

Book equation 6.1.

$$s(x) = w \cdot x + b, \quad P_C = \mathbb{E}[\sigma'(s)^2] w w^\top, \quad \text{rank } P_C = 1.$$

LEDGER

- OT-7. The damage form, trace pairing, rank, and loading covariance are GL(d)-invariant under $P' = A^{-\top} P A^{-1}$, while spectrum, effective rank, principal angles, and water-filling are O(d)-only — consumer-weighted damage is a geometric scalar, and P3's cliff cannot be bought down by reparameterization. [demonstrated]. [geometric-observation/claims/LEDGER.md:30 at 9f3829f](#).
- GO-1. The consumer's invariant/nuisance split is identifiable ex ante from the consumer functional. [predicted]. [geometric-observation/claims/LEDGER.md:62 at 9f3829f](#).

FIRST STATED

Chapter 0 section 0.2 and chapter 2 section 2.2 of *Data Mining as Observation*, with the rank bound of the chain rule in [geometric-observation/chapters/ch06_mathematical_preliminaries.md:10-27](#).

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 2 section 2.2	read subspace small, operator local, pullback composition, rank cannot increase	geometric-observation\chapters\ch05_the_read_metric_and_the_o 48; geometric-observation\chapters\ch06_mathematical_preliminaries 27
chapter 12 section 12.2	pullback composition and the rank bound	geometric-observation\chapters\ch06_mathematical_preliminaries 27; chapter 2 of this book

FAILURES AND CORRECTIONS

none

CONDITIONS

- The dimension of a matrix's column space. It is at most the number of rows and of columns, an outer product has rank at most one, a product has rank at most the rank of either factor, and transposing does not change it.
- The read operator of a linear classifier has rank one, the composed read operator of a chain has rank at most either stage's, and the effective rank of a spectrum is the continuous version chapter 4 allocates bits by.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/Rank.lean`, `theorems rank_le_width`, `rank_le_height`, `rank_outer_le_one`, `rank_mul_le`, `rank_zero`, `rank_transpose`, at `observation-data-mining af776fd`.

`lean/DataMiningAsObservation/Pipeline.lean`, `theorems quotient_inherited`, `quotient_inherited_chain`, `rank_comp_le_first`, `rank_comp_le_second`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14.

RELATED

[effective rank](#) (p. 115); [rank certificate](#) (p. 275); [read subspace](#) (p. 285); [Jacobian](#) (p. 183); [outer product](#) (p. 240).

rank certificate

instrument. Entry id `rank-certificate`.

DEFINITION

A bound, computed from a compressed representation, that certifies which neighbour rankings the compression preserved for a consumer and which it did not. In its strict setting the floor is a guarantee, in its percentile setting an estimate. A vacuous result proves nothing about the corpus and is an instruction to fall back to exact reranking. Chapter 11.

EQUATION

Book equation 11.2.

$$r = \frac{d_{\text{compressed}}}{d_{\text{exact}}}, \quad \kappa_{\text{strict}} = \frac{\max r}{\min r}, \quad \tau \geq 1 - 2\hat{\mu}(\kappa_{\text{strict}}), \quad \rho_S \geq 1 - 3\hat{\mu}(\kappa_{\text{strict}}),$$

$$\kappa_{97.5/2.5} = \frac{q_{97.5}(r)}{q_{2.5}(r)} \text{ gives the same two expressions as estimates, not floors.}$$

Book equation 11.3.

$$\text{recall}@k = \frac{|\text{returned top-}k \cap \text{true top-}k|}{k},$$

true top- k computed from the uncompressed vectors.

LEDGER

- G0-3. The certificate's vacuity threshold predicts where single-stage retrieval dies. [demonstrated]. `geometric-observation/claims/LEDGER.md:65` at `9f3829f`.

FIRST STATED

The compression program's certificate specification, `turboquant-pro/docs/CERTIFICATE_SPEC.md`, DOI 10.5281/zenodo.20660087, and chapter 11 section 11.3 of *Data Mining as Observation*.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 4 section 4.5	truncation claim reproducible, loses on compact sets	<code>turboquant-pro\CLAIMS.md:28-49</code>
chapter 8 section 8.9	13.7 ROUGE-L, 0.25/4.19/9.60/13.7 at 64/128/256/512, re-validation negative 0.31 n=40, 26.64 under symmetric nf4, <code>_quant_nf4a_group</code> unchanged since 289bdfc before 4f7baab	<code>turboquant-pro\CLAIMS.md:63-81</code> ; <code>turboquant-pro\benchmarks\kvquant_matrix\REVAL-2026-08-08.md</code>
chapter 9 section 9.3	the margin certificate, μ crit as the expected maximum of N minus 1 standard normals, ρ , death at 0.948 within 6 percent, Spearman 0.991 vs 0.873, fourteen corpora, six gates, the v1 to v3 path, the standing correction	<code>geometric-observation\experiments\G03-certificate-vacuity-v3-NOTES.md:1-60</code> ; <code>geometric-observation\claims\LEDGER.md</code> row G0-3
chapter 11 section 11.2	cosine 0.995	<code>turboquant-pro\README.md:86-88</code> ; <code>turboquant-pro\CLAIMS.md</code> row <code>kv_keys_per_channel</code>
chapter 11 section 11.3	τ floor 1 minus 2 μ , Spearman floor 1 minus 3 μ , example κ 1.0148, μ 0.0664, floor 0.8671, 19900 pairs, max certifiable 1.83, vacuous means exact rerank, sha256 binding, reference field and the 0.3 overlap difference	<code>turboquant-pro\docs\CERTIFICATE_SPEC.md:1-80</code>
chapter 11 section 11.5	9.6x at recall 0.999 CI-gated on GloVe 1.18M; 32x at 0.9993 on private 199k LaBSE, ties OPQ, beats RaBitQ, 20x build; 27.7x and 114x reported; PCA truncation loses on compact sets; 20x at 199k and 4x at 1M over OPQ; RaBitQ builds in under a second	<code>turboquant-pro\CLAIMS.md:28-49</code>
chapter 11 section 11.5	CI fails when <code>CLAIMS.md</code> and <code>claims.yaml</code> disagree	<code>turboquant-pro\CLAIMS.md:1-27</code> ; <code>turboquant-pro\tests\test_claims_ledger.py</code>

FAILURES AND CORRECTIONS

- `turboquant-pro/turboquant_pro/rank_certificate.py:25-36` at 856c4cb. * **Strict** ($l_0=0$, $h_i=100$): κ is the true worst-case distortion (max/min per-pair ratio). The bound is then unconditional – a genuine distribution-free floor over *all* pairs – but a single collapsed or near-duplicate pair (ratio $\rightarrow 0$ or huge) sends $\kappa \rightarrow \infty$ and makes the certificate vacuous, so it is brittle to data-artifact pairs. * **Robust** (the $l_0=2.5$, $h_i=97.5$ default): κ is the percentile-robust distortion, trimming the most-distorted ~5% of pairs. This is the sensible default and matches the source paper's torus protocol, but the resulting floor is **conditional**: it holds for the central 95% of pairs, *not* unconditionally over all of them. The trimmed tail can invert arbitrarily, so the reported floor is a robust estimate, not a hard worst-case guarantee. Use the strict regime when you need the unconditional bound.

CONDITIONS

- In the strict setting, the zeroth and hundredth percentiles of the distance ratio, the floor is a guarantee for the anchor sample, broken by a single collapsed pair.
- In the percentile setting the implementation computes the distortion on the retained central pairs and the concentration over all pairs, and the floor is a robust estimate, not a guarantee on either set.
- The floor is about the global ordering of the sampled pairwise distances. It does not directly certify a query's top-k recall, one query's neighbour order, or the trimmed tail.
- A floor on a sample of anchors is a statement about that sample. Carrying it to the corpus needs a sampling argument the certificate records the inputs for and does not supply.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/RankCertificate.lean`, `theorems nn_preserved`, `nn_preserved_of_kappa_one`, `kappa_ge_one`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 11.

RELATED

[certificate](#) (p. 55); [vacuity threshold](#) (p. 359); [the flip](#) (p. 138); [hubness](#) (p. 168).

rank-faithful

concept. Entry id `rank-faithful`.

DEFINITION

Of a map, preserving the ordering of distances without any bound on the stretch. Chapter 0 section 0.10.

EQUATION

Book equation 0.21.

$$\frac{1}{K} d(x, y) \leq d'(f(x), f(y)) \leq K d(x, y) \quad \text{for all } x, y.$$

Book equation 11.2.

$$r = \frac{d_{\text{compressed}}}{d_{\text{exact}}}, \quad \kappa_{\text{strict}} = \frac{\max r}{\min r}, \quad \tau \geq 1 - 2\hat{\mu}(\kappa_{\text{strict}}), \quad \rho_S \geq 1 - 3\hat{\mu}(\kappa_{\text{strict}}),$$

$$\kappa_{97.5/2.5} = \frac{q_{97.5}(r)}{q_{2.5}(r)} \text{ gives the same two expressions as estimates, not floors.}$$

LEDGER

- NEG-1. Fixed-scale, uniform-in-m bi-Lipschitz for the commute filter. [refuted]. `geometric-observation/claims/LEDGER.md:94` at `9f3829f`.

FIRST STATED

Volume 14, chapter 9, `geometric-observation/chapters/ch09_legibility.md:10-41`, DOI [10.5281/zenodo.21776291](https://doi.org/10.5281/zenodo.21776291), and chapter 0 section 0.10 and chapter 3 section 3.3 of *Data Mining as Observation*.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 3 section 3.2	condition (A2), tangential displacement	turboquant-pro\docs\KV_KEYS_FINDING.md:61-86; geometric-observation\chapters\ch09_legibility.md:42-54
chapter 3 section 3.3	resistance converges to 1 over d_i plus 1 over d_j , simplex argument, radius converges to 1 over root d_i	the-angular-observer\theorem.md:110-146
chapter 3 section 3.3	radius against degree correlation 0.92 to 0.99	the-angular-observer\README.md:135-139
chapter 3 section 3.3	angle 0.92 flat in m , magnitude 0.03, full 0.75 decaying	wolfram-observer-bridge\theorem.md:1-20; the-angular-observer\theorem.md:187-224
chapter 3 section 3.3	substrate table	the-angular-observer\README.md:99-109
chapter 3 section 3.3	row normalization is the projection onto the read subspace of the geodesic-rank consumer	geometric-observation\chapters\ch09_legibility.md:31-41; geometric-observation\chapters\ch03_historical_precursors.md:110
chapter 3 section 3.3	proven versus conjectured ledger, d at least 3, d equals 2 borderline, fixed- m conditional theorem, heat filter uniform, commute metric form false, Green-kernel rank limit for d at most 3, obstruction at d equals 4	the-angular-observer\theorem.md:84-107, 147-224; the-angular-observer\README.md:150-185
chapter 3 section 3.3	the self-refuted v0.8 claim, NEG-1	the-angular-observer\README.md:170-175; geometric-observation\chapters\ch16_honest_negatives.md NEG-1
chapter 3 section 3.3	hyperbolic rejected, curvature negative 0.98 to negative 0.14, trend 1.09 minus 0.157 d across 20 manifolds and 5 families	the-angular-observer\README.md:111-147, 183-185
chapter 9 section 9.1	the geodesic-rank reader discards the radius, row normalization is the angular projection	geometric-observation\chapters\ch09_legibility.md:10-41; the-angular-observer\theorem.md:110-146
chapter 9 section 9.2	the battery, three new templates, frozen ratios, code hash, bars at least 10 of 12 and the dimension ordering, eccentricity scope	the-angular-observer\PREREG_RECOGNIZER_BATTERY.md:1-80
chapter 9 section 9.2	12 of 12, dimensions 1.81, 2.80, 1.74, angular Spearman ranges, eccentricity spreads, verdict confirmed, GO-P-2026-041	the-angular-observer\experiments\manifold-recovery\battery_result.json

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 9 section 9.2	causal set 0.57 with dimension 2.9 to 10.7, small world 0.45, control 0.01, rung 4 decay 0.747 to 0.679 below 0.75	the-angular-observer\README.md:99-109; wolfram-observer-bridge\rung4_scaledm_result.json; wolfram-observer-bridge\review-2.txt
chapter 11 section 11.1	condition (A2), cosine satisfies it, post-rotary keys do not, the cone below cell size	turboquant-pro\docs\KV_KEYS_FINDING.md:61-86; the-angular-observer\README.md:26-31

FAILURES AND CORRECTIONS

- NEG-1, [refuted]. Fixed-scale, uniform-in-m bi-Lipschitz for the commute filter.

CONDITIONS

- A map is rank-faithful when it preserves the ordering of distances with no bound on the stretch. Every strictly increasing transform of the distance is rank-faithful, so a reader that ranks, the nearest-neighbour reader among them, returns the same answer under it.
- Rank-faithful and bi-Lipschitz are separate words. The square preserves every ordering and stretches large distances beyond any fixed factor, and the substrate table measures rank agreement, not stretch.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/RankFaithful.lean`, theorems `isRankFaithful_of_strictMono`, `closer_set_eq`, `nearest_eq`, `sq_rankFaithful_on_nonneg`, `sq_not_bi_lipschitz`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 3.

RELATED

[bi-Lipschitz](#) (p. 37); [rank certificate](#) (p. 275); [recognizer](#) (p. 288); [quotient](#) (p. 271).

read direction

concept. Entry id `read-direction`.

DEFINITION

An eigenvector of the read operator with a nonzero eigenvalue, a direction the consumer is sensitive to. Chapter 2 section 2.2.

EQUATION

Book equation 0.9.

$$P_C = \mathbb{E}[g g^\top], \quad g = \nabla C(x).$$

Book equation 6.1.

$$s(x) = w \cdot x + b, \quad P_C = \mathbb{E}[\sigma'(s)^2] w w^\top, \quad \text{rank } P_C = 1.$$

Book equation 4.2.

$$D(b) = \sum_i s_i \sigma_i^2 2^{-2b_i}, \quad s_i = v_i^\top P_C v_i, \quad b_i^* = \max\left(0, \frac{1}{2} \log_2 \frac{s_i \sigma_i^2}{\theta}\right), \quad \sum_i b_i^* = B,$$

LEDGER

- GO-1. The consumer's invariant/nuisance split is identifiable ex ante from the consumer functional. [predicted]. `geometric-observation/claims/LEDGER.md:62` at 9f3829f.
- GO-2 (pos. half: consumer-projected covariance *controls*). Downstream preservation is controlled by the error covariance projected on the consumer's read subspace, $\text{tr}(P_C \Sigma_\delta)$. [replicated]. `geometric-observation/claims/LEDGER.md:64` at 9f3829f.

FIRST STATED

Chapter 2 section 2.2 and chapter 4 section 4.3 of *Data Mining as Observation*, with the control that destroys the read direction in `geometric-observation/chapters/ch08_value.md:40-70`.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 1 section 1.4	the flip sealed in twelve domains and three physics, held in at least five domains and all three	<code>geometric-observation/chapters/ch08_value.md:40-70</code> ; <code>geometric-observation/claims/LEDGER.md</code> ROWS GO-2, GO-B-AV163, D3, 038
chapter 4 section 4.3	acoustic 148 of 201, 201 of 201, 152 of 201; seismic 13 of 17, 17 of 17, 13 of 17; whale 0.934 vs 0.883, 2 times, 300 of 300; at least 5 domains and 3 physics; battery prediction met	<code>geometric-observation/chapters/ch08_value.md:40-70</code> ; <code>geometric-observation/claims/LEDGER.md</code> ROWS GO-B-AV163, D3, 038
chapter 6 section 6.2	planted probe, overlap 0.936 vs 0.059, twelve of twelve, reconstruction 0.40, five of five	<code>geometric-observation/claims/LEDGER.md</code> ROW GO-1
chapter 6 section 6.2	whale clan classifier, 8718 codas, 0.934 vs 0.883, reconstructs twice as well, 300 of 300	<code>geometric-observation/chapters/ch08_value.md:40-70</code> ; <code>geometric-observation/claims/LEDGER.md</code> ROW 038
chapter 11 section 11.7	GO-1 overlap 0.936 vs 0.059, flip 12 of 12, reconstruction 0.40	<code>geometric-observation/claims/LEDGER.md</code> ROW GO-1; <code>geometric-observation/chapters/ch10_the_blind_probe.md:34-52</code>

FAILURES AND CORRECTIONS

none

CONDITIONS

- An eigenvector of the read operator with a nonzero eigenvalue, a direction the consumer is sensitive to. An affine consumer has exactly one, its weight vector; the projection onto a read direction is idempotent, and its residual is orthogonal to it.

- The code that puts the whole error on the read direction is the worst, the control that destroys the read direction is confirmed worst, and no code of a given total error can be read as more than the total.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/ReadOperator.lean`, theorems `rank_one_reads_one_direction`, `readOp_mulVec`, `quad_readOp`, `quad_readOp_nonneg`, `readOp_mulVec_eq_zero_iff`, `readOp_diag`, `readOp_offdiag`, `readOp_symm`, `readOp_neg`, `affine_const_along_nuisance`, `readOp_affine`, `readOp_sqLength_basis`, at `observation-data-mining af776fd`.

`lean/DataMiningAsObservation/Projection.lean`, theorems `proj_proj`, `dot_sub_proj`, `proj_sq_le`, `proj_add_orth`, `proj_add`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 2, 3, 4, 6, 8, 11, 12.

RELATED

[read subspace](#) (p. 285); [read operator](#) (p. 282); [sensitivity](#) (p. 319); [nuisance](#) (p. 233); [projection](#) (p. 263).

read distortion

concept. Entry id `read-distortion`.

DEFINITION

The error as a consumer experiences it, the trace of the read operator times the error's second-moment matrix, which is its covariance when the error is centered. For the identity reader it is mean squared error. Equation 0.10.

EQUATION

Book equation 0.10.

$$d_O = \text{tr}(P_C M_\delta) = \mathbb{E}[\delta^\top P_C \delta], \quad M_\delta = \mathbb{E}[\delta \delta^\top], \quad P_C = I \Rightarrow d_O = \text{tr} M_\delta.$$

Book equation 1.2.

$$d_O = \text{tr}(P_C M_\delta), \quad (d_O = \text{tr} M_\delta \text{ for every admissible } M_\delta) \iff P_C = I.$$

Book equation 4.1.

$$d_O = \text{tr}(P_C M_\delta) \quad \text{against} \quad \text{tr} M_\delta = d_O|_{P_C=I}.$$

LEDGER

- GO-2 (pos. half: consumer-projected covariance *controls*). Downstream preservation is controlled by the error covariance projected on the consumer's read subspace, $\text{tr}(P_C \Sigma_\delta)$. [replicated]. `geometric-observation/claims/LEDGER.md:64` at `9f3829f`.

FIRST STATED

Volume 14, chapter 5, `geometric-observation/chapters/ch05_the_read_metric_and_the_quotient.md:49-75`, DOI `10.5281/zenodo.21776291`, and chapter 2 of the same volume for the failure of observer-free measurement.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 1 section 1.4	the flip sealed in twelve domains and three physics, held in at least five domains and all three	geometric-observation\chapters\ch08_value.md:40-70; geometric-observation\claims\LEDGER.md rows GO-2, GO-B-AV163, D3, 038
chapter 2 section 2.2	read subspace small, operator local, pullback composition, rank cannot increase	geometric-observation\chapters\ch05_the_read_metric_and_the_c 48; geometric-observation\chapters\ch06_mathematical_preliminaries 27
chapter 2 section 2.2	read distortion controls but is not a complete rank statistic, twelve of twelve, one middle pair misordered, NEG-9	geometric-observation\chapters\ch05_the_read_metric_and_the_c 75
chapter 4 section 4.3	GO-2 0.0934 vs 0.0938, 2.53 times, 12 of 12, anti 21 times; retrieval 0.0964, negative 4.70 and positive 4.65, recon 0.40	geometric-observation\claims\LEDGER.md rows GO-2 negative and positive halves

FAILURES AND CORRECTIONS

none

CONDITIONS

- Exact, with no centering, for a fixed read operator, which is the affine case. For a nonlinear consumer whose error depends on the row it is a first-order factorized surrogate, exact when the local operator and the error's outer product are uncorrelated across rows.
- The second-moment matrix of the error, not its covariance, unless the error is centered by construction.
- The trace of the second moment is the mean squared vector norm, which is the dimension times the per-coordinate mean squared error.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/ReadDistortion.lean`, `theorems read_distortion`, `identity_reader`, `quad_one`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 1, 2, 4.

RELATED

[the flip](#) (p. 138); [alignment](#) (p. 11); [identity reader](#) (p. 171); [quotient](#) (p. 271).

read operator

concept. Entry `id read-operator`.

DEFINITION

The average outer product of a consumer's sensitivity over a dataset, an average of local linearizations whose kernel is the set of directions unread at almost every row of the workload. Its range is the read subspace and its kernel is the nuisance. Equations 0.9 and 0.11.

EQUATION

Book equation 0.9.

$$P_C = \mathbb{E}[g g^\top], \quad g = \nabla C(x).$$

Book equation 0.11.

$$P_C(x) = J(x)^\top G(C(x)) J(x), \quad J(x) = \frac{\partial C}{\partial x}(x), \quad \bar{P}_{C,\mu} = \mathbb{E}_\mu[P_C(x)].$$

Book equation 2.1.

$$P_{C_2 \circ C_1}(x) = J_1(x)^\top P_{C_2}(C_1(x)) J_1(x), \quad \text{rank } P_{C_2 \circ C_1}(x) \leq \text{rank } P_{C_2}(C_1(x)) \quad \text{at each row } x.$$

Book equation 6.1.

$$s(x) = w \cdot x + b, \quad P_C = \mathbb{E}[\sigma'(s)^2] w w^\top, \quad \text{rank } P_C = 1.$$

LEDGER

- OT-7. The damage form, trace pairing, rank, and loading covariance are $\text{GL}(d)$ -invariant under $P' = A^{-\top} P A^{-1}$, while spectrum, effective rank, principal angles, and water-filling are $O(d)$ -only — consumer-weighted damage is a geometric scalar, and P3's cliff cannot be bought down by reparameterization. [demonstrated]. [geometric-observation/claims/LEDGER.md:30 at 9f3829f](#).
- GO-1. The consumer's invariant/nuisance split is identifiable ex ante from the consumer functional. [predicted]. [geometric-observation/claims/LEDGER.md:62 at 9f3829f](#).
- GO-EC-3. A read operator recovered from a black-box consumer by query-only finite-difference probing, composed with the Kalman covariance as $\text{tr}(\hat{P}_C \Sigma)$, prospectively selects sensors that improve the held-out consumer at matched budgets with probe cost charged — capturing 94.6% of the known analytic optimum's gain on the positive-control arm (gate $\geq 75\%$) and improving 16.3% over the best consumer-agnostic policy on non-analytic consumers (gate $\geq 8\%$), with trace-matched ordering carried by the composition at 86.9% over 61 pairs (gate $\geq 65\%$). [predicted]. [geometric-observation/claims/LEDGER.md:162 at 9f3829f](#).

FIRST STATED

Volume 14, chapters 4 and 5, [geometric-observation/chapters/ch04_the_observer_triple.md:9-60](#) and [geometric-observation/chapters/ch05_the_read_metric_and_the_quotient.md:7-48](#), DOI 10.5281/zenodo.21776291.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 1 section 1.2	the observer triple, read subspace, nuisance, same read operator means same read geometry	geometric-observation\chapters\ch04_the_observer_triple.md:9-60 ; geometric-observation\OBSERVATION.md:1-10
chapter 2 section 2.2	read subspace small, operator local, pullback composition, rank cannot increase	geometric-observation\chapters\ch05_the_read_metric_and_the_quotient.md:7-48 ; geometric-observation\chapters\ch06_mathematical_preliminaries.md:27

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 2 section 2.2	read distortion controls but is not a complete rank statistic, twelve of twelve, one middle pair misordered, NEG-9	geometric-observation\chapters\ch05_the_read_metric_and_the_75
chapter 2 section 2.3	refusal regimes, selection consumers read order, recurrences compound	readscope\readscope\regimes.py:1-60
chapter 6 section 6.1	the classifier row of the consumer table, the output metric makes a different observer	geometric-observation\chapters\ch04_the_observer_triple.md:60-135
chapter 6 section 6.1	selection consumers have zero sensitivity almost everywhere	readscope\readscope\regimes.py:1-60
chapter 6 section 6.2	planted probe, overlap 0.936 vs 0.059, twelve of twelve, reconstruction 0.40, five of five	geometric-observation\claims\LEDGER.md row GO-1
chapter 11 section 11.7	GO-1 overlap 0.936 vs 0.059, flip 12 of 12, reconstruction 0.40	geometric-observation\claims\LEDGER.md row GO-1; geometric-observation\chapters\ch10_the_blind_probe.md:34-52

FAILURES AND CORRECTIONS

- `geometric-observation/chapters/ch05_the_read_metric_and_the_quotient.md:42-47` at 9f3829f. **Operating-point dependence.** P_C is a *local* object — it depends on x_0 through J . For a linear consumer it is global; for a nonlinear one it varies over X , and the honest version of every result carries P_C as a field, not a constant. The blind probe of Chapter 10 recovers P_C at an operating point precisely because it is local; averaging it over a data distribution gives the $\bar{P}_C$ that enters the alignment law κ (Chapter 12).

CONDITIONS

- Each local operator is positive semidefinite, so the kernel of the workload average is the intersection of the local kernels up to sets of rows of measure zero, and an average-null direction is locally unread at almost every row of that workload.
- The average depends on the workload it was taken over and is written with that workload where the difference matters.
- Its off-diagonal entries are co-sensitivities, how two features' sensitivities vary together across rows, and not interactions, which are cross-partials and live in the Hessian.
- The metric in the vector-output form is the local geometry of the output metric where that metric has a local quadratic representation. A dataset-level loss has none, and the operator is then taken on the score with the identity geometry.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/ReadOperator.lean`, `theorems rank_one_reads_one_direction`, `readOp_mulVec`, `quad_readOp`, `quad_readOp_nonneg`, `readOp_mulVec_eq_zero_iff`, `readOp_diag`, `readOp_offdiag`, `readOp_symm`, `readOp_neg`, `affine_const_along_nuisance`, `readOp_affine`, `readOp_sqLength_basis`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 1, 2, 4, 6, 7, 8, 11, 12, 13, 14.

RELATED

[read distortion](#) (p. 281); [quotient](#) (p. 271); [identity reader](#) (p. 171); [blind probe](#) (p. 39); [the flip](#) (p. 138).

read subspace

concept. Entry id read-subspace.

DEFINITION

The directions of the data a consumer can distinguish, spanned by the eigenvectors of its read operator with nonzero eigenvalue. Chapter 0 section 0.5.

EQUATION

Book equation 0.9.

$$P_C = \mathbb{E}[g g^\top], \quad g = \nabla C(x).$$

Book equation 2.1.

$$P_{C_2 \circ C_1}(x) = J_1(x)^\top P_{C_2}(C_1(x)) J_1(x), \quad \text{rank } P_{C_2 \circ C_1}(x) \leq \text{rank } P_{C_2}(C_1(x)) \quad \text{at each row } x.$$

Book equation 6.1.

$$s(x) = w \cdot x + b, \quad P_C = \mathbb{E}[\sigma'(s)^2] w w^\top, \quad \text{rank } P_C = 1.$$

LEDGER

- OT-3. Under subspace-confined second-order transcripts, fewer than d directions cannot identify a hidden leading eigenspace (theorem, adaptive to $d-2$ / oblivious to $d-1$); a known k_0 -dim exclusion moves the cliff to exactly $d-k_0$ and never softens it. [demonstrated]. [geometric-observation/claims/LEDGER.md:31 at 9f3829f](#).
- NEG-12. (Gate B, prospective, real LLM) On a trained frontier attention layer the blind probe recovers the read operator above the sealed bar *and* projection beats reconstruction. [refuted]. [geometric-observation/claims/LEDGER.md:105 at 9f3829f](#).
- GO-B-Llama. Trained frontier LLM (Llama-3.2-3B), softmax-attention consumer — blind probe on real post-RoPE keys [predicted]. [geometric-observation/claims/LEDGER.md:115 at 9f3829f](#).
- GO-B-Llama-rematch. Trained frontier LLM (Llama-3.2-3B), softmax-attention consumer — reconnected dissociation on real post-RoPE keys [predicted]. [geometric-observation/claims/LEDGER.md:118 at 9f3829f](#).
- GO-B-Llama-rematch. Trained frontier LLM (Llama-3.2-3B), softmax-attention consumer — reconnected dissociation on real post-RoPE keys [predicted]. [geometric-observation/claims/LEDGER.md:118 at 9f3829f](#).

FIRST STATED

Volume 14, chapter 5, [geometric-observation/chapters/ch05_the_read_metric_and_the_quotient.md:7-48](#), DOI 10.5281/zenodo.21776291, where the read subspace is the range of the read operator.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 2 section 2.2	read subspace small, operator local, pullback composition, rank cannot increase	geometric-observation\chapters\ch05_the_read_metric_and_the_quotient.md:48 ; geometric-observation\chapters\ch06_mathematical_preliminaries.md:27

Where the book states it	Numbers, as the book’s sources table records them	Source
chapter 2 section 2.2	read distortion controls but is not a complete rank statistic, twelve of twelve, one middle pair misordered, NEG-9	geometric-observation\chapters\ch05_the_read_metric_and_the_c75
chapter 6 section 6.2	real model, median 0.567 vs bar 0.60, about 4.5 times chance, heads at 0.815 and 0.958, sixteen of sixteen both, two of three triggers, NEG-12, later rematch four of four	geometric-observation\claims\LEDGER.md rows GO-B-Llama and NEG-12; geometric-observation\chapters\ch16_honest_negatives.md:73-81
chapter 11 section 11.7	confinement theorem, side information moves the cliff to d minus k0, noisy cliff proved then measured	readscope\PRINCIPLES.md:112-142; geometric-observation\crucible\OT3-THEOREM.md; geometric-observation\crucible\OT3-NOISY-THEOREM.md

FAILURES AND CORRECTIONS

- NEG-12, [refuted]. (Gate B, prospective, real LLM) On a trained frontier attention layer the blind probe recovers the read operator above the sealed bar *and* projection beats reconstruction.

CONDITIONS

- The read subspace is spanned by the eigenvectors of the read operator with nonzero eigenvalue, and it is small. A linear classifier reads one direction, an attention head a handful, and the input has hundreds or thousands.
- Pointwise, a scalar-output consumer reads one direction at each row, and the averaged operator can still have full rank, as the consumer that reads the squared length of a row shows.
- Recovering it from a black box costs two consumer calls per input dimension per row and is a cliff at the full dimension.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/ReadOperator.lean`, `theorems rank_one_reads_one_direction`, `readOp_mulVec`, `quad_readOp`, `quad_readOp_nonneg`, `readOp_mulVec_eq_zero_iff`, `readOp_diag`, `readOp_offdiag`, `readOp_symm`, `readOp_neg`, `affine_const_along_nuisance`, `readOp_affine`, `readOp_sqLength_basis`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 1, 2, 3, 4, 6, 7, 9, 10, 11, 12, 13.

RELATED

[nuisance](#) (p. 233); [read operator](#) (p. 282); [blind probe](#) (p. 39); [budget cliff](#) (p. 49).

recall at k

concept. Entry id `recall-at-k`.

DEFINITION

The fraction of a query’s true k nearest neighbours that an index returned. Equation 11.3.

EQUATION

Book equation 11.3.

$$\text{recall}@k = \frac{|\text{returned top-}k \cap \text{true top-}k|}{k},$$

true top- k computed from the uncompressed vectors.

Book equation 10.7.

$$\text{verdict} = \min_{i: n_i \geq n_{\min}, |Q_i| \geq q_{\min}} \text{score}_i, \quad \text{ABSTAIN otherwise.}$$

LEDGER

- NEG-14. (GO-P-2026-037, prospective) a2_probe.median_unit_displacement is a single-statistic predictor of the flip regime ($\text{unit_disp} \geq 1.0 \Rightarrow \text{generic-polar-flip vs needs-blind-probe}$). [refuted]. geometric-observation/claims/LEDGER.md:108 at 9f3829f.
- GO-B-Llama. Trained frontier LLM (Llama-3.2-3B), softmax-attention consumer — blind probe on real post-RoPE keys [predicted]. geometric-observation/claims/LEDGER.md:115 at 9f3829f.
- GO-B-Llama-rematch. Trained frontier LLM (Llama-3.2-3B), softmax-attention consumer — reconnected dissociation on real post-RoPE keys [predicted]. geometric-observation/claims/LEDGER.md:118 at 9f3829f.

FIRST STATED

Chapter 11 section 11.3 of *Data Mining as Observation*, with the program's stratified form in turboquant-pro, turboquant-pro/docs/HUBNESS_PRIMER.md:86-131.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 3 section 3.5	78 vs 369, density correlation about 0.67, 8 percent vs 34 percent, centering vs mutual-proximity rescaling	turboquant-pro/docs/HUBNESS_PRIMER.md:38-59,140-160
chapter 10 section 10.2	anti-hubs as where compressed indexes fail first, aggregate recall barely moves	turboquant-pro/docs/HUBNESS_PRIMER.md:86-131
chapter 10 section 10.3	abstain below 2.5 k	turboquant-pro/docs/HUBNESS_PRIMER.md:140-160
chapter 11 section 11.6	count of ten, hubs and anti-hubs, max 78 vs 369, density correlation about 0.67, 8 percent vs 34 percent, abstain below 2.5k, centering vs mutual-proximity rescaling	turboquant-pro/docs/HUBNESS_PRIMER.md:1-60,60-170
chapter 11 section 11.6	anti-hub recall, p05, hub-rank correlation, hub-set overlap, the build gate	turboquant-pro/docs/HUBNESS_PRIMER.md:86-131
chapter 11 section 11.6	the centroid-injection attack	turboquant-pro/docs/HUBNESS_PRIMER.md:168-197, citing arXiv 2604.05480

FAILURES AND CORRECTIONS

- NEG-14, [refuted]. (GO-P-2026-037, prospective) a2_probe.median_unit_displacement is a single-statistic predictor of the flip regime ($\text{unit_disp} \geq 1.0 \Rightarrow \text{generic-polar-flip vs needs-blind-probe}$).

CONDITIONS

- The fraction of a query's true k nearest neighbours, computed from the uncompressed vectors, that the index returned. It lies in the unit interval and is one exactly when the returned list is the true list.

- The aggregate over queries is a weighted mean over strata, so a stratum that fails entirely moves it by no more than its weight. That is why the anti-hub stratum is reported on its own, and why aggregate recall is never an acceptance metric on its own.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/RecallAtK.lean`, `theorems recallAtK_mem_unit`, `recallAtK_eq_one_iff`, `aggregate_le_of_failing`, `aggregate_example`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 10, 11, 12.

RELATED

[anti-hub](#) (p. 17); [min-over-strata](#) (p. 217); [rank certificate](#) (p. 275); [hubness](#) (p. 168).

recognizer

instrument. Entry id recognizer.

DEFINITION

The instrument that decides whether a clustering certificate is right, wrong, or vacuous, by naming the manifold from the eigenvalue multiplets or certifying that none is present. Chapter 9.

EQUATION

Book equation 9.3.

$$\hat{\mu} = \frac{\bar{s}_{\text{true}} - \bar{s}_{\text{distr}}}{\sigma_{\text{distr}}}, \quad \mu_{\text{crit}} = \mathbb{E} \left[\max_{N-1} \mathcal{N}(0, 1) \right], \quad \rho = \frac{\hat{\mu}}{\mu_{\text{crit}}}, \quad \rho = 1 \text{ vacuous.}$$

Book equation 0.32.

$$\text{dist}(m, t) = \sqrt{\frac{1}{8} \sum_{k=1}^8 (\ln m_k - \ln t_k)^2}.$$

Book equation 9.2.

$$N(\lambda) = \#\{k : \lambda_k \leq \lambda\} \sim C_d \lambda^{d/2} \quad \Rightarrow \quad d = 2 \frac{d \log N}{d \log \lambda}.$$

LEDGER

- GO-3. The certificate's vacuity threshold predicts where single-stage retrieval dies. [demonstrated]. `geometric-observation/claims/LEDGER.md:65` at `9f3829f`.

FIRST STATED

Volume 14, chapter 11, `geometric-observation/chapters/ch11_the_recognizer.md:1-95`, DOI [10.5281/zenodo.21776291](https://doi.org/10.5281/zenodo.21776291), and the-angular-observer's recognizer battery, `the-angular-observer/PREREG_RECOGNIZER_BATTERY.md:1-80`.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 3 section 3.3	resistance converges to 1 over d_i plus 1 over d_j , simplex argument, radius converges to 1 over root d_i	the-angular-observer\theorem.md:110-146
chapter 3 section 3.3	radius against degree correlation 0.92 to 0.99	the-angular-observer\README.md:135-139
chapter 3 section 3.3	angle 0.92 flat in m, magnitude 0.03, full 0.75 decaying	wolfram-observer-bridge\theorem.md:1-20; the-angular-observer\theorem.md:187-224
chapter 3 section 3.3	substrate table	the-angular-observer\README.md:99-109
chapter 3 section 3.3	proven versus conjectured ledger, d at least 3, d equals 2 borderline, fixed-m conditional theorem, heat filter uniform, commute metric form false, Green-kernel rank limit for d at most 3, obstruction at d equals 4	the-angular-observer\theorem.md:84-107, 147-224; the-angular-observer\README.md:150-185
chapter 3 section 3.3	the self-refuted v0.8 claim, NEG-1	the-angular-observer\README.md:170-175; geometric-observation\chapters\ch16_honest_negatives.md NEG-1
chapter 3 section 3.3	hyperbolic rejected, curvature negative 0.98 to negative 0.14, trend 1.09 minus 0.157 d across 20 manifolds and 5 families	the-angular-observer\README.md:111-147, 183-185
chapter 9 section 9.1	the geodesic-rank reader discards the radius, row normalization is the angular projection	geometric-observation\chapters\ch09_legibility.md:10-41; the-angular-observer\theorem.md:110-146
chapter 9 section 9.2	the recognizer's mechanism, low multiplets and angular distances, dimension before shape by Weyl's law, refusal, the growth gotcha	geometric-observation\chapters\ch11_the_recognizer.md:1-95
chapter 9 section 9.2	the battery, three new templates, frozen ratios, code hash, bars at least 10 of 12 and the dimension ordering, eccentricity scope	the-angular-observer\PREREG_RECOGNIZER_BATTERY.md:1-80
chapter 9 section 9.2	12 of 12, dimensions 1.81, 2.80, 1.74, angular Spearman ranges, eccentricity spreads, verdict confirmed, GO-P-2026-041	the-angular-observer\experiments\manifold-recovery\battery_result.json
chapter 9 section 9.2	causal set 0.57 with dimension 2.9 to 10.7, small world 0.45, control 0.01, rung 4 decay 0.747 to 0.679 below 0.75	the-angular-observer\README.md:99-109; wolfram-observer-bridge\rung4_scaledm_result.json; wolfram-observer-bridge\review-2.txt

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 9 section 9.3	the margin certificate, mu crit as the expected maximum of N minus 1 standard normals, rho, death at 0.948 within 6 percent, Spearman 0.991 vs 0.873, fourteen corpora, six gates, the v1 to v3 path, the standing correction	geometric-observation\experiments\G03-certificate-vacuity-v3-NOTES.md:1-60; geometric-observation\claims\LEDGER.md row GO-3
chapter 11 section 11.1	condition (A2), cosine satisfies it, post-rotary keys do not, the cone below cell size	turboquant-pro\docs\KV_KEYS_FINDING.md:61-86; the-angular-observer\README.md:26-31

FAILURES AND CORRECTIONS

none

CONDITIONS

- The recognizer names a manifold from the low eigenvalue multiplets of a neighbourhood graph's Laplacian by template match, or certifies that none is present, and it can only name shapes in its template set.
- It was held to a preregistered held-out battery, and the vacuity threshold below which it certifies nothing is derived from the problem rather than tuned.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/Recognizer.lean`, `theorems templateDist_self`, `templateDist_comm`, `templateDist_scale`, `templateDist_nonneg`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 9.

RELATED

[certificate](#) (p. 55); [vacuity threshold](#) (p. 359); [hubness](#) (p. 168).

reconstruction error

concept. Entry id `reconstruction-error`.

DEFINITION

The squared length of the difference between a row and its approximation, the identity reader's distortion. Chapter 1 section 1.4.

EQUATION

Book equation 4.2.

$$D(b) = \sum_i s_i \sigma_i^2 2^{-2b_i}, \quad s_i = v_i^\top P_C v_i, \quad b_i^* = \max\left(0, \frac{1}{2} \log_2 \frac{s_i \sigma_i^2}{\theta}\right), \quad \sum_i b_i^* = B,$$

Book equation 1.1.

$$O = (C, G, B).$$

Book equation 0.5.

$$\Sigma v_i = \lambda_i v_i, \quad \Sigma = \sum_{i=1}^d \lambda_i v_i v_i^\top, \quad v_i \cdot v_j = 0 \ (i \neq j).$$

LEDGER

- GO-2 (neg. half: not reconstruction). At matched bits, downstream preservation is not controlled by reconstruction error. [demonstrated]. `geometric-observation/claims/LEDGER.md:63 at 9f3829f`.
- GO-6. At matched rate, output coding $\leq$ surrogate $\leq$ reconstruction on the consumer metric; the output-reconstruction gap is governed by the ker P_C entropy share, and the surrogate-output gap vanishes as rate grows. [demonstrated]. `geometric-observation/claims/LEDGER.md:68 at 9f3829f`.
- NEG-4. Lightweight online (Lloyd) key calibration beats the calibration-free default on softmax-KL. [refuted]. `geometric-observation/claims/LEDGER.md:97 at 9f3829f`.

FIRST STATED

Chapter 1 section 1.4 and chapter 4 section 4.1 of *Data Mining as Observation*, with the identity slice in `geometric-observation/chapters/ch02_failure_of_observer_free_measurement.md:1-40`.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 1 section 1.4	three virtues one vice, the identity slice	<code>geometric-observation\chapters\ch02_failure_of_observer_free_40</code>
chapter 3 section 3.2	traces 2.0, diagonals 0.3 and 1.7, consumers at 15 and 75 degrees, distortion 0.39 vs 1.61, 4.1 to one, computed not drawn	<code>observation-theory\assets\make_flip_figure.py:4-24,107-119</code>
chapter 4 section 4.2	GO-6, d 8 and r 4, output at or below surrogate at or below reconstruction at every rate, about 500 times, gap 0.41 to 0.005, isotropic control collapses	<code>geometric-observation\claims\LEDGER.md</code> row GO-6; <code>geometric-observation\chapters\ch07_cost.md</code>

FAILURES AND CORRECTIONS

- NEG-4, [refuted]. Lightweight online (Lloyd) key calibration beats the calibration-free default on softmax-KL.

CONDITIONS

- The squared length of the difference between a row and its approximation, the identity reader's distortion. It is nonnegative, zero exactly when the approximation is the row, a sum of per-coordinate squared errors, and for a projection onto a unit direction it is the squared length less the squared component along it.
- A reconstruction cosine of 0.995 sat beside a perplexity of ten thousand, recalibration improved reconstruction and worsened the consumer, and at matched reconstruction the consumer-aware code won in twelve of twelve domains.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/ReconstructionError.lean`, theorems `recon_nonneg`, `recon_eq_zero_iff`, `recon_sum`, `recon_proj`, at `observation-data-mining af776fd`.

`lean/DataMiningAsObservation/ReadDistortion.lean`, theorems `read_distortion`, `identity_reader`, `quad_one`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 1, 4, 6, 7, 8, 11, 12, 13.

RELATED

[identity reader](#) (p. 171); [read distortion](#) (p. 281); [distortion](#) (p. 109); [principal component analysis](#) (p. 260).

refresh floor

correction. Entry id refresh-floor.

DEFINITION

A claimed lower bound on how often a certificate must be renewed. The proportional-to-coherence-time version was refuted and replaced by a sealed horizon. Chapter 13.

EQUATION

Book equation 13.2.

$$T_{\text{coh}} = \frac{0.423}{f_D},$$

refuted fit: $\phi \approx 0.177 T_{\text{coh}} (R^2 = 0.915)$, sealed: $\phi \leq 6 \text{ TTI at } 10 \text{ Hz, } 1 \text{ TTI at } \geq 50 \text{ Hz.}$

LEDGER

- OT-4. Operator drift predicts a real long-generation degradation and a derived refresh intervention moves it. [refuted]. `geometric-observation/claims/LEDGER.md:36 at 9f3829f`.
- OT-11. Feedback-free staleness: streaming-retrieval damage tracks measured drift; derived-cadence re-allocation removes it. [void]. `geometric-observation/claims/LEDGER.md:48 at 9f3829f`.

FIRST STATED

An unsealed exploration in the radio freshness track, `observation-theory-campaigns/analysis/csi/CSI-refreshfloor.json`, refuted and replaced by the sealed sweep `observation-theory-campaigns/analysis/csi/PREREG-XPROTO-CSI-SWEEP2.md`.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 13 section 13.3	BGP 0.351, IS-IS 0.184, OSPF 0.083, second collector	<code>observation-theory-campaigns\analysis\mongo\PREREG-XPROTO-MG.md:42</code> ; <code>observation-theory-campaigns\experiments\ROUTING-TELEMETRY-TRACK.md:130-150</code> ; <code>geometric-observation\BOOK-OUTLINE.md:75</code>
chapter 13 section 13.3	radio 0.27 to 0.42 naive to 0.055 to 0.13, neural reconstruction 0.28 to 0.13	<code>observation-theory-campaigns\experiments\RADIO-FRESHNESS-TRACK.md:20-35</code>
chapter 13 section 13.4	refuted fit 0.177 T_{coh} R^2 0.915, 0.15 threshold vs 0.10 claim, fresh baseline 0.113, do not cite	<code>observation-theory-campaigns\analysis\csi\CSI-refreshfloor.json:2,111-114</code> ; <code>observation-theory-campaigns\experiments\RADIO-FRESHNESS-TRACK.md:41-47</code>

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 13 section 13.4	sealed correction, 0.5 dB calibration, floors 4/6/4, 2/2/3, 1, slopes 0.1008, 0.1398, 0.1086, R^2 0.7376, 0.9218, 0.6174, bars B1 to B4 and MC1 to MC4 all true, seeds 20260827 to 20260829, sealed 1d3de4a	observation-theory-campaigns\analysis\csi\PREREG-XPROTO-CSI-SWEEP2.md:1-60; observation-theory-campaigns\analysis\csi\XPROTO-CSI-SWEEP2-graded.json; observation-theory-campaigns\experiments\SEALS.md:90
chapter 13 section 13.4	freshness sweep classes, sensing refuted at about 1.6 times	geometric-observation\BOOK-OUTLINE.md:70-90

FAILURES AND CORRECTIONS

- OT-4, [refuted]. Operator drift predicts a real long-generation degradation and a derived refresh intervention moves it.
- observation-theory-campaigns/experiments/RADIO-FRESHNESS-TRACK.md:41-50 at 9d86211. **The age horizon** (analysis/csi/, XPROTO-CSI-SWEEP2 sealed 2026-08-27, graded PASS): at the calibrated 0.10 budget the largest compliant report period is a few TTI at 10 Hz and 1 TTI at 50 Hz and above, so there is no proportionality law to fit. The earlier CSI-refreshfloor.* claim of a floor $\approx 0.177 \cdot T_{\text{coh}}$ ($R^2=0.915$) was an **unsealed exploration** measured at a relaxed 0.15 threshold against the 0.10 target, with a fresh baseline that never met the budget. It is refuted; the record is kept, not cited. Separately, the optimal linear predictor cannot beat the one-coherence-time wall (Gaussian fading $\rightarrow$ Wiener optimal), and that wall sits well outside the budget horizon. The **RAN governor** (analysis/ran) is the reference O-RAN rApp core: observe $\rightarrow$ measure false-clear $\rightarrow$ refresh at the floor $\rightarrow$ escalate (diversity / re-route).

CONDITIONS

- The proportional law, a floor of about 0.177 times the coherence time, was fit at a relaxed threshold against a fresh baseline that never met the budget. It is refuted, the record is kept, and it is not cited.
- The sealed replacement calibrated the baseline to the budget first and found floors of a few transmission intervals at 10 hertz and one at 50 hertz and above, with the optimal linear predictor unable to beat the one-coherence-time wall.
- Volume 14's book outline still cites the refuted law, an open item.

Conditions are curated in entries.toml rather than read from a record.

MACHINE CHECKED

lean/DataMiningAsObservation/CoherenceTime.lean, theorems clarke_to_three_decimals, examples, refuted_law_exceeds_sealed, at observation-data-mining af776fd.

USED IN

Data Mining as Observation chapters 13.

RELATED

[coherence time](#) (p. 63); [certificate](#) (p. 55); [false-clear rate](#) (p. 133).

refresh interval

instrument. Entry id refresh-interval.

DEFINITION

The longest renewal period that keeps a certificate within its promised error, a measured property of the system and not a convention. Chapter 13.

EQUATION

Book equation 13.2.

$$T_{\text{coh}} = \frac{0.423}{f_D},$$

refuted fit: $\phi \approx 0.177 T_{\text{coh}} (R^2 = 0.915)$, sealed: $\phi \leq 6$ TTI at 10 Hz, 1 TTI at ≥ 50 Hz.

Book equation 0.25.

$$T_{\text{coh}} = \frac{0.423}{f_D}.$$

Book equation 0.26.

$$\text{FC} = \Pr [W \text{ refutes } | \mathcal{C}_t \text{ clears}], \quad \text{coverage} = \Pr [\mathcal{C}_t \text{ clears}].$$

LEDGER

- OT-4. Operator drift predicts a real long-generation degradation and a derived refresh intervention moves it. [refuted]. `geometric-observation/claims/LEDGER.md:36` at 9f3829f.

FIRST STATED

Chapter 13 section 13.2 of *Data Mining as Observation*, with the sealed floors in `observation-theory-campaigns/experiments/RADIO-FRESHNESS-TRACK.md:20-35` and the refuted fit in `observation-theory-campaigns/analysis/csi/CSI-refreshfloor.json`.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 13 section 13.3	radio 0.27 to 0.42 naive to 0.055 to 0.13, neural reconstruction 0.28 to 0.13	<code>observation-theory-campaigns/experiments/RADIO-FRESHNESS-TRACK.md:20-35</code>
chapter 13 section 13.4	refuted fit 0.177 T_{coh} R^2 0.915, 0.15 threshold vs 0.10 claim, fresh baseline 0.113, do not cite	<code>observation-theory-campaigns/analysis/csi/CSI-refreshfloor.json:2,111-114</code> ; <code>observation-theory-campaigns/experiments/RADIO-FRESHNESS-TRACK.md:41-47</code>
chapter 13 section 13.4	freshness sweep classes, sensing refuted at about 1.6 times	<code>geometric-observation/BOOK-OUTLINE.md:70-90</code>

FAILURES AND CORRECTIONS

- OT-4, [refuted]. Operator drift predicts a real long-generation degradation and a derived refresh intervention moves it.

CONDITIONS

- The longest renewal period that keeps a certificate within its promised error, a measured property of the system and not a convention. A certificate older than the interval is stale for every consumer whose floor is shorter, and the floors are counted in slots that add over consecutive durations.
- There is no proportionality law between the interval and the coherence time. The fit of the floor as 0.177 of the coherence time was refuted and is not to be cited, and the sealed floors were 4, 1, and 1 slots at 10, 50, and 400 hertz.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

lean/DataMiningAsObservation/CoherenceTime.lean, theorems clark_e_to_three_decimals, examples, refuted_law_exceeds_sealed, at observation-data-mining af776fd.

lean/DataMiningAsObservation/TTI.lean, theorems slots_of_ms, slots_add, slots_mono, slots_coherence, at observation-data-mining af776fd.

USED IN

Data Mining as Observation chapters 13, 14.

RELATED

[refresh floor](#) (p. 292); [coherence time](#) (p. 63); [certificate](#) (p. 55); [false-clear rate](#) (p. 133); [transmission time interval](#) (p. 355).

refusal

concept. Entry id refusal.

DEFINITION

An instrument's declining to report a number it cannot support. A refusal is a verdict, and it is counted. Chapter 2 section 2.4 and chapter 9 section 9.2.

EQUATION

Book equation 10.7.

$$\text{verdict} = \min_{i: n_i \geq n_{\min}, |Q_i| \geq q_{\min}} \text{score}_i, \quad \text{ABSTAIN otherwise.}$$

Book equation 14.4.

$$\text{attribution}_j(x) \approx g_j(x) \delta_j, \quad \overline{\text{importance}}_j \approx (P_C)_{jj} = \mathbb{E}[g_j^2].$$

Book equation 9.3.

$$\hat{\mu} = \frac{\bar{s}_{\text{true}} - \bar{s}_{\text{distr}}}{\sigma_{\text{distr}}}, \quad \mu_{\text{crit}} = \mathbb{E} \left[\max_{N-1} \mathcal{N}(0, 1) \right], \quad \rho = \frac{\hat{\mu}}{\mu_{\text{crit}}}, \quad \rho = 1 \text{ vacuous.}$$

LEDGER

- GO-3. The certificate's vacuity threshold predicts where single-stage retrieval dies. [demonstrated]. [geometric-observation/claims/LEDGER.md:65 at 9f3829f](#).
- NEG-14. (GO-P-2026-037, prospective) `a2_probe.median_unit_displacement` is a single-statistic predictor of the flip regime ($\text{unit_disp} \geq 1.0 \Rightarrow \text{generic-polar-flip vs needs-blind-probe}$). [refuted]. [geometric-observation/claims/LEDGER.md:108 at 9f3829f](#).

FIRST STATED

Chapter 2 section 2.4 and chapter 9 section 9.2 of *Data Mining as Observation*, with the refusal regimes in [readscope/readscope/regimes.py:1-60](#).

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 2 section 2.3	refusal regimes, selection consumers read order, recurrences compound	readscope\readscope\regimes.py:1-60

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 2 section 2.6	0.647 published, 1.000 recovered, weighted vs unweighted median 0.796 range 0.678 to 0.985, probe 1.000 on every cell, 36 cells, one cell at 0.985	readscope\SPEC.md:410-450
chapter 6 section 6.1	selection consumers have zero sensitivity almost everywhere	readscope\readscope\regimes.py:1-60
chapter 10 section 10.4	area map, boundary rule, hash, refuse not warn, intra and transit counts, area classes, abstention rule	turboquant-pro\docs\STRATA_RFC.md:24-98
chapter 10 section 10.4	second prediction inverted, transit 0.389 vs 0.291, centrality difference signs, share 0.473 vs 0.391, seven of thirteen backbone	turboquant-pro\docs\RESULTS_multilingual_strata.md:55-90; turboquant-pro\docs\STRATA_RFC.md:98-130

FAILURES AND CORRECTIONS

- NEG-14, [refuted]. (GO-P-2026-037, prospective) `a2_probe.median_unit_displacement` is a single-statistic predictor of the flip regime ($\text{unit_disp} \geq 1.0 \Rightarrow \text{generic-polar-flip vs needs-blind-probe}$).

CONDITIONS

- An instrument's declining to report a number it cannot support, when samples do not exceed the dimension, when no template matches, or when a group is too thin. A refusal is a verdict, it is counted, and a mean never hides it.
- The recognizer refuses when the ratios match no stored template, the probe refuses to extrapolate past its fit range, and the strata design refuses rather than warns.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/Abstention.lean`, `theorems abstain_not_passes`, `verdict_eq_none_iff`, `passes_verdict_iff`, `inf_le_of_subset`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 2, 8, 9, 10, 11, 14.

RELATED

[abstention](#) (p. 9); [vacuity threshold](#) (p. 359); [recognizer](#) (p. 288); [verdict](#) (p. 362); [coverage](#) (p. 86).

registered

concept. Entry id registered.

DEFINITION

Of a claim, bar, null, or budget, written into a sealed file before the measurement. Chapter 8 section 8.8.

EQUATION

Book equation 8.1.

$$O_{\text{harness}} = (C_{\text{score}}, G_{\text{metric}}, B = \text{folds} \times \text{samples} \times \text{seeds}).$$

Book equation 8.3.

$$\Pr[\text{at least one of } m \text{ null tests passes}] = 1 - (1 - \alpha)^m, \quad \alpha_{\text{Bonferroni}} = \frac{\alpha}{m}.$$

LEDGER

- GO-1. The consumer's invariant/nuisance split is identifiable ex ante from the consumer functional. [predicted]. `geometric-observation/claims/LEDGER.md:62` at 9f3829f.
- GO-3. The certificate's vacuity threshold predicts where single-stage retrieval dies. [demonstrated]. `geometric-observation/claims/LEDGER.md:65` at 9f3829f.
- NEG-11. (GO-5, prospective ×4) The $\alpha=1$ density/hubness quotient decisively and density-specifically restores invariant fidelity in a non-spectral domain. [refuted]. `geometric-observation/claims/LEDGER.md:104` at 9f3829f.

FIRST STATED

Chapter 8 section 8.8 of *Data Mining as Observation*, with the template in `observation-theory-campaigns/experiments/PREREG-TEMPLATE.md:1-71` and the seal rule in `observation-theory-campaigns/experiments/SEALS.md:1-10`.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 8 section 8.8	template fields	<code>observation-theory-campaigns/experiments/PREREG-TEMPLATE.md:1-71</code>
chapter 8 section 8.8	seal = commit plus SHA-256	<code>observation-theory-campaigns/experiments/SEALS.md:1-10</code>
chapter 9 section 9.2	the battery, three new templates, frozen ratios, code hash, bars at least 10 of 12 and the dimension ordering, eccentricity scope	<code>the-angular-observer\PREREG_RECOGNIZER_BATTERY.md:1-80</code>

FAILURES AND CORRECTIONS

- NEG-11, [refuted]. (GO-5, prospective ×4) The $\alpha=1$ density/hubness quotient decisively and density-specifically restores invariant fidelity in a non-spectral domain.

CONDITIONS

- Of a claim, bar, null, or budget, written into a sealed file before the measurement, so that a changed digest would prove a changed prediction. A registered bar discriminates only when the null fails it and the real system passes it.
- A registered miss is kept as a miss, an identifier burned at pilot is not reused, and every number in the book was registered before it was measured or is labelled as not having been.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/Seal.lean`, `theorems changed_of_hash_ne`, `hash_eq_of_eq`, `exists_collision`, at `observation-data-mining af776fd`.

`lean/DataMiningAsObservation/Bar.lean`, `theorems passes_anti`, `passes_mono`, `discriminates_iff`, `no_bar_of_null_ge`, `exists_bar_of_lt`, `vacuous_of_null_passes`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14.

RELATED

[preregistration](#) (p. 258); [sealed](#) (p. 316); [bar](#) (p. 34); [ledger class](#) (p. 204); [declaration](#) (p. 96).

reliability weight

concept. Entry id reliability-weight.

DEFINITION

Zero or twice a validated encoder's held-out AUROC minus one, whichever is larger, used as that encoder's authority. It is a discrimination weight, and the calibration evidence is the expected calibration error reported beside it. Equation 0.38.

EQUATION

Book equation 0.38.

$$w = \max(0, 2 \cdot \text{AUROC} - 1).$$

Book equation 0.16.

$$\text{AUROC} = \Pr[s^+ > s^-] + \frac{1}{2} \Pr[s^+ = s^-].$$

LEDGER

none

FIRST STATED

The moral-embedding program's calibrated authority, `gtc-prototype/docs/CALIBRATED_AUTHORITY.md:19-63` and `xbse/README.md:175-195`, and chapter 12 section 12.5 and chapter 14 section 14.2 of *Data Mining as Observation*.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 12 section 12.5	ECE 0.018 to 0.101 vs raw up to 0.223, reliability weight, audit binding	<code>xbse\README.md:175-195</code> ; <code>gtc-prototype/docs/CALIBRATED_AUTHORITY.md:19-63</code>
chapter 14 section 14.2	reliability weights and calibration errors per axis, the collapsed family's mean weight 0.559 against the general valence channel's own 0.735, the three design rules, 0.048 to 0.049 and 0.089 to 0.101 at 696 pairs	<code>gtc-prototype/docs/CALIBRATED_AUTHORITY.md:1-65</code>

FAILURES AND CORRECTIONS

none

CONDITIONS

- The weight is zero or twice the held-out AUROC minus one, whichever is larger, so it lies in the unit interval, is zero at chance and below, and is unchanged by any strictly monotone recalibration of the score.
- It is a discrimination weight and not a calibration weight. The expected calibration error is reported beside it, and an encoder below the untrained baseline gets weight zero whatever its calibration.
- One formula and one source. The registry that consumes the weights computes them from the recorded held-out AUROC and nothing else.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

lean/DataMiningAsObservation/ReliabilityWeight.lean, theorems pair_le_one, aurocNum_le, auroc_le_one, weight_mem_unit, weight_eq_zero_of_le_half, weight_comp, at observation-data-mining af776fd.

USED IN

Data Mining as Observation chapters 0, 12, 14.

RELATED

[Monotone Invariance Theorem](#) (p. 220); [harness](#) (p. 159); [posited versus measured](#) (p. 254); [certificate](#) (p. 55).

replica

concept. Entry id replica.

DEFINITION

A copy of a database kept on another machine. Its lag is how far it trails the primary. Chapter 13.

EQUATION

Book equation 0.26.

$$\text{FC} = \Pr[W \text{ refutes} \mid \mathcal{C}_t \text{ clears}], \quad \text{coverage} = \Pr[\mathcal{C}_t \text{ clears}].$$

Book equation 13.1.

$$\text{FC} = \Pr[W \text{ refutes} \mid \mathcal{C}_t \text{ clears}], \quad d_O(\Delta) = \text{tr}(P_C M_{\text{drift}}(\Delta)), \quad M_{\text{drift}}(\Delta) = \mathbb{E}[\delta_\Delta \delta_\Delta^\top].$$

LEDGER

- OT-11. Feedback-free staleness: streaming-retrieval damage tracks measured drift; derived-cadence re-allocation removes it. [void]. `geometric-observation/claims/LEDGER.md:48` at 9f3829f.

FIRST STATED

Chapter 13 section 13.3 of *Data Mining as Observation*, with the program's substrate measurements in `observation-theory-campaigns/analysis/mongo/PREREG-XPROTO-MG.md` and the database rows of the freshness track.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 13 section 13.3	BGP 0.351, IS-IS 0.184, OSPF 0.083, second collector	<code>observation-theory-campaigns\analysis\mongo\PREREG-XPROTO-MG.md:42</code> ; <code>observation-theory-campaigns\experiments\ROUTING-TELEMETRY-TRACK.md:130-150</code> ; <code>geometric-observation\BOOK-OUTLINE.md:75</code>

FAILURES AND CORRECTIONS

none

CONDITIONS

- A copy of a database kept on another machine, trailing the primary by a lag. The keys stale under a lag grow with the lag and vanish at no lag. A naive certificate that clears every read has coverage one and a

false-clear rate equal to the stale fraction, and a witness-backed certificate that clears only keys outside the lag has a false-clear rate of zero at the price of coverage.

- Which reads are stale is consumer-relative, since two readers of the same replica have different coherence times, and the measured false-clear rates on ZooKeeper, Postgres, and MongoDB are the ledger's numbers.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/Replica.lean`, `theorems stale_mono`, `stale_zero`, `naive_certificate`, `witnessed_certificate`, `witnessed_coverage`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 1, 4, 13.

RELATED

[freshness](#) (p. 149); [certificate](#) (p. 55); [witness](#) (p. 370); [false-clear rate](#) (p. 133); [coverage](#) (p. 86).

rerank

instrument. Entry id `rerank`.

DEFINITION

Reordering a candidate list with a second scorer. Its recall is at most the candidate coverage. Chapter 10 section 10.5.

EQUATION

Book equation 11.3.

$$\text{recall}@k = \frac{|\text{returned top-}k \cap \text{true top-}k|}{k},$$

true top- k computed from the uncompressed vectors.

Book equation 10.7.

$$\text{verdict} = \min_{i: n_i \geq n_{\min}, |Q_i| \geq q_{\min}} \text{score}_i, \quad \text{ABSTAIN otherwise.}$$

Book equation 12.3.

$$\text{validated} \iff \text{AUROC}_{\text{cross}} - \max(\text{AUROC}_{\text{untrained}}, \text{AUROC}_{\text{BoW}}) \geq 0.10.$$

LEDGER

- NEG-14. (GO-P-2026-037, prospective) `a2_probe.median_unit_displacement` is a single-statistic predictor of the flip regime (`unit_disp` $\geq 1.0 \Rightarrow$ generic-polar-flip vs needs-blind-probe). [refuted]. `geometric-observation/claims/LEDGER.md:108` at 9f3829f.
- GO-B-Llama. Trained frontier LLM (Llama-3.2-3B), softmax-attention consumer — blind probe on real post-RoPE keys [predicted]. `geometric-observation/claims/LEDGER.md:115` at 9f3829f.
- GO-B-Llama-rematch. Trained frontier LLM (Llama-3.2-3B), softmax-attention consumer — reconnected dissociation on real post-RoPE keys [predicted]. `geometric-observation/claims/LEDGER.md:118` at 9f3829f.

FIRST STATED

Chapter 10 section 10.5 of *Data Mining as Observation*, with the candidate-coverage finding in `turboquant-pro/docs/RESULTS_strata_phase23_gates.md`.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 4 section 4.5	GloVe table, 73 percent at 64 components, 0.685 vs 0.862 and 0.866, 0.906 at matched bytes, 0.989 at 37 bytes, 768 to 256 keeps about 99 percent, the 95 percent rule	turboquant-pro\benchmarks\RESULTS_glove.md:1-40
chapter 10 section 10.4	first run, 350000 of 2391361 rows, 7 of 14 eligible, abstentions with 2 and 1 rows, Robin Hood 0.3447 to 0.4469 ratio 1.30 against 1.5, skew 2.68 to 4.40 ratio 1.64 against 3, 63 times sample range	turboquant-pro\docs\RESULTS_multilingual_strata.md:1-55
chapter 10 section 10.4	second prediction inverted, transit 0.389 vs 0.291, centrality difference signs, share 0.473 vs 0.391, seven of thirteen backbone	turboquant-pro\docs\RESULTS_multilingual_strata.md:55-90; turboquant-pro\docs\STRATA_RFC.md:98-130
chapter 10 section 10.5	Gate A design error, A prime skew 3.970 to 3.177, max 287 to 213, Robin Hood 0.372 to 0.261, fraction 0.117 to 0.079, compressed path 0.663 vs 0.90, seven strata, 0.62 to 0.69 vs 0.76 to 0.84	turboquant-pro\docs\RESULTS_strata_phase23_gates.md:1-45
chapter 10 section 10.5	rerank identical 0.6627, candidate coverage, depth curve 51 to 501, coverage 0.696 to 0.926, fidelity 0.663 to 0.702, originals rerank 0.923, 4096 bytes on 148	turboquant-pro\docs\RESULTS_strata_phase23_gates.md:45-110
chapter 10 section 10.5	fragile-first allocation 0.7118 vs 0.7251, targets up 0.004 to 0.034, redesigned allocator passed	turboquant-pro\docs\RESULTS_strata_phase23_gates.md:110-125
chapter 11 section 11.4	R80 table, 25x probe depth, 92 percent own cell, one third cross-article, 0.53 matches same-article share, relative contrast discriminates nothing	openvector-bench\results\R80_ANN.md:1-40; turboquant-pro\docs\RESEARCH_ROADMAP.md:133-160
chapter 12 section 12.4	seven of thirteen backbone areas under the translation-trained encoder, all local-hub under the general one	turboquant-pro\docs\RESULTS_multilingual_strata.md:55-90

FAILURES AND CORRECTIONS

- NEG-14, [refuted]. (GO-P-2026-037, prospective) a2_probe.median_unit_displacement is a single-statistic predictor of the flip regime ($\text{unit_disp} \geq 1.0 \Rightarrow \text{generic-polar-flip vs needs-blind-probe}$).

CONDITIONS

- Reordering a candidate list with a second scorer. A rerank cannot return a row the list does not hold, so its recall is at most the candidate coverage, an oracle rerank reaches the coverage exactly, and a deeper list can only raise the coverage.
- Reranking the compressed candidates read 0.6627, identical to the unranked recall, because candidate coverage was the wall, and the depth curve from 51 to 501 moved coverage from 0.696 to 0.926.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

lean/DataMiningAsObservation/Rerank.lean, theorems hits_le_coverage, recall_le_coverage, oracle_rerank, coverage_mono, at observation-data-mining af776fd.

USED IN

Data Mining as Observation chapters 4, 10, 11, 12.

RELATED

[recall at k](#) (p. 286); [anti-hub recall](#) (p. 19); [inverted file](#) (p. 177); [rank certificate](#) (p. 275); [retrieval-augmented pipeline](#) (p. 303).

residualization

instrument. Entry id residualization.

DEFINITION

Removing from a column the least-squares multiple of another. The residual is orthogonal to the column removed. Chapter 8 section 8.6 and chapter 12 section 12.5.

EQUATION

Book equation 0.1.

$$x \cdot y = \sum_{i=1}^d x_i y_i, \quad \|x\| = \sqrt{x \cdot x}, \quad \cos \theta = \frac{x \cdot y}{\|x\| \|y\|}.$$

Book equation 12.3.

$$\text{validated} \iff \text{AUROC}_{\text{cross}} - \max(\text{AUROC}_{\text{untrained}}, \text{AUROC}_{\text{BoW}}) \geq 0.10.$$

Book equation 14.2.

$$\text{coverage difference}_c = \text{AUROC}_c(\text{embedding}) - \text{AUROC}_c(\text{validated axes}), \quad \text{floor} \approx 0.08 \text{ to } 0.12.$$

LEDGER

- GO-B-legal (035→036). Legal-citation retrieval (CourtListener), cosine-ranking consumer, LaBSE embeddings — real large corpus, non-physical consumer [predicted]. [geometric-observation/claims/LEDGER.md:119](#) at 9f3829f.

FIRST STATED

Chapter 8 section 8.6 and chapter 12 section 12.5 of *Data Mining as Observation*, with the bifactor readout in [xbse/docs/BIFACTOR_READOUT.md:14-100](#).

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 8 section 8.6	books 0.241 at 17 sigma, residual 0.093, R2 0.009, z 6.5, p 5.7e-11, 85 percent, fiction 0.131 on 2250	geometric-aesthetics\book\src\chapter-17-empirical-evidence-for-geometric-aesthetics.md:95-106

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 12 section 12.5	valence channel 0.856 plus or minus 0.008 on 51319 rows, P1 failed 4 of 11, the G-share table, 12 by 12 gate five own-axis six demoted, bimodal G row, 0.875 vs 0.811, display rule	xbse\docs\BIFACTOR_READOUT.md:14-100
chapter 14 section 14.8	0.241 to 0.093, 4998 books, 6.5 standard errors, one to two percent	geometric-aesthetics\book\src\chapter-17-empirical-evidence-for-geometric-aesthetics.md:95-106

FAILURES AND CORRECTIONS

none

CONDITIONS

- Removing from a column the least-squares multiple of another. The residual is orthogonal to the column removed, its squared length is the original less the squared dot product over the removed column's squared length, a column orthogonal to the removed one is unchanged, and a column residualized on itself vanishes.
- Whether named categories survive residualization on the general valence channel is chapter 12's test, and the book-length signal fell from 0.241 to a residual 0.093 at six and a half standard errors.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/Residual.lean`, `theorems residual_orth`, `residual_sq`, `residual_of_orth`, `residual_self`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 8, 12, 14.

RELATED

[correlation](#) (p. 80); [orthogonal](#) (p. 239); [confound](#) (p. 72); [contraction](#) (p. 75); [projection](#) (p. 263).

retrieval-augmented pipeline

concept. Entry id `retrieval-augmented-pipeline`.

DEFINITION

A system that answers a question by chunking documents, embedding the chunks, indexing them, retrieving the nearest to the embedded question, and handing them to a generator. A chain of observers. Chapter 12.

EQUATION

Book equation 12.1.

$$x \sim_{\text{stage}} x' \implies x \sim_{\text{pipeline}} x' \quad \text{for every stage, differentiable or not,}$$

$$\text{rank } P_{\text{run}}(x) \leq \min_{\text{stages in the run}} \text{rank } J_{\text{stage}}(x) \quad \text{on a differentiable run of stages, at each row.}$$

Book equation 13.1.

$$\text{FC} = \Pr[W \text{ refutes } | \mathcal{C}_t \text{ clears}], \quad d_O(\Delta) = \text{tr}(P_C M_{\text{drift}}(\Delta)), \quad M_{\text{drift}}(\Delta) = \mathbb{E}[\delta_\Delta \delta_\Delta^\top].$$

LEDGER

- OT-11. Feedback-free staleness: streaming-retrieval damage tracks measured drift; derived-cadence re-allocation removes it. [void]. `geometric-observation/claims/LEDGER.md:48` at 9f3829f.
- GO-B-legal (035→036). Legal-citation retrieval (CourtListener), cosine-ranking consumer, LaBSE embeddings — real large corpus, non-physical consumer [predicted]. `geometric-observation/claims/LEDGER.md:119` at 9f3829f.

FIRST STATED

Chapter 12 of *Data Mining as Observation*, with the program's streaming-retrieval staleness row OT-11 and the legal-citation retrieval flip.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 12 section 12.3	035 miss 0.773 vs 0.757, identifiability diagnosis, 036 blind probe r 32, 0.779 vs 0.771, 200 of 200, recon 0.220 vs 0.584, 039 virgin split 0.796 vs 0.780, margin 0.008 to 0.016, edges baseline	<code>geometric-observation\chapters\ch10_the_blind_probe.md:53-90</code> ; <code>geometric-observation\claims\LEDGER.md</code> rows GO-B-legal 035 and 036, GO-P-2026-039
chapter 12 section 12.3	uncompressed 0.79, centred 0.84	<code>geometric-observation\claims\LEDGER.md</code> row GO-B-legal
chapter 12 section 12.6	XPROTO-LLM, benchmark 0.909, 0.920, 0.909, thirty slices, target 0.8, naive 0.333, aware 0.033, spread 0.380, deployment mean 0.736, six bars on three seeds, sealed 2026-08-25 at b61f7f1	<code>observation-theory-campaigns\experiments\LLM-EVAL-TRACK.md:1-50</code> ; <code>observation-theory-campaigns\analysis\llm\XPROTO-LLM-graded.json</code> ; <code>observation-theory-campaigns\experiments\SEALS.md:85</code>

FAILURES AND CORRECTIONS

none

CONDITIONS

- A system that chunks documents, embeds the chunks, indexes them, retrieves the nearest to the embedded question, and hands them to a generator. Whatever one stage declares the same, every later stage and the pipeline declare the same, and for linear stages the rank of the composition is at most the rank of any stage, so the pipeline's read subspace is no larger than its narrowest stage's.
- Both hold for every stage, differentiable or not. The benchmark score of the whole is a certificate whose false-clear rate on deployment slices is the chapter's measurement.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/Pipeline.lean`, `theorems quotient_inherited`, `quotient_inherited_chain`, `rank_comp_le_first`, `rank_comp_le_second`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 1, 2, 3, 8, 10, 11, 12, 13, 14.

RELATED

[quotient](#) (p. 271); [read subspace](#) (p. 285); [deployment mismatch](#) (p. 101); [freshness](#) (p. 149).

Robin Hood index

instrument. Entry id robin-hood-index.

DEFINITION

The fraction of a total count that would have to move from points above the mean to points below it to make every count equal. Equation 0.35.

EQUATION

Book equation 0.35.

$$RH = \sum_i \max\left(0, \frac{N_i}{\sum_j N_j} - \frac{1}{n}\right).$$

Book equation 10.4.

$$\text{never retrieved} \geq 1 - \frac{|Q|k}{n} \quad \text{whenever} \quad |Q|k < n.$$

LEDGER

- NEG-14. (GO-P-2026-037, prospective) a2_probe.median_unit_displacement is a single-statistic predictor of the flip regime (unit_disp ≥ 1.0 ⇒ generic-polar-flip vs needs-blind-probe). [refuted]. geometric-observation/claims/LEDGER.md:108 at 9f3829f.

FIRST STATED

Hoover’s index of concentration, applied to neighbour counts in turboquant-pro’s strata design, turboquant-pro/docs/STRATA_RFC.md:24-98, and chapter 0 section 0.16 of *Data Mining as Observation*.

MEASUREMENTS

Where the book states it	Numbers, as the book’s sources table records them	Source
chapter 10 section 10.4	area map, boundary rule, hash, refuse not warn, intra and transit counts, area classes, abstention rule	turboquant-pro/docs/STRATA_RFC.md:24-98
chapter 10 section 10.4	second prediction inverted, transit 0.389 vs 0.291, centrality difference signs, share 0.473 vs 0.391, seven of thirteen backbone	turboquant-pro/docs/RESULTS_multilingual_strata.md:55-90; turboquant-pro/docs/STRATA_RFC.md:98-130

FAILURES AND CORRECTIONS

- NEG-14, [refuted]. (GO-P-2026-037, prospective) a2_probe.median_unit_displacement is a single-statistic predictor of the flip regime (unit_disp ≥ 1.0 ⇒ generic-polar-flip vs needs-blind-probe).

CONDITIONS

- The fraction of the total count that would have to move from points above the mean to points below it to make every count equal, the excess above the mean over the total, which is half the total absolute deviation over the total. It is nonnegative, zero exactly at equality, at most one, and unchanged by a common scaling of the counts.

- The strata design reports it per area beside the skew, and the first run's ratio of 1.30 against a bar of 1.5 is the number the abstention rule was written around.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/RobinHood.lean`, `theorems sum_dev`, `excess_eq_half_abs`, `robinHood_nonneg`, `robinHood_eq_zero_iff`, `robinHood_le_one`, `robinHood_scale`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 10.

RELATED

[hubness](#) (p. 168); [anti-hub](#) (p. 17); [Poisson ceiling](#) (p. 253); [abstention](#) (p. 9).

ROC curve

instrument. Entry id `roc-curve`.

DEFINITION

The path of true positive rate against false positive rate as the threshold sweeps, whose area is the AUROC. Chapter 5 section 5.4.

EQUATION

Book equation 5.4.

$$F_1^{\max} \leq \sup_{t \in [J, 1]} \frac{2t\pi}{t\pi + \pi + (t - J)(1 - \pi)}, \quad J = \max_{\tau} (\text{TPR} - \text{FPR}), \quad \pi = \text{prevalence},$$

$$J \leq 2A - 1 \text{ when the ROC curve is concave, and not in general.}$$

Book equation 0.28.

$$P = \frac{TP}{TP + FP}, \quad R = \frac{TP}{TP + FN}, \quad F_1 = \frac{2PR}{P + R}.$$

Book equation 0.38.

$$w = \max(0, 2 \cdot \text{AUROC} - 1).$$

LEDGER

none

FIRST STATED

Chapter 5 section 5.4 of *Data Mining as Observation*, with the Youden ceiling and its correction to concave curves in `theory-radar/paper/astar_paper.tex:94-110`.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 5 section 5.4	seven thresholds up to 0.75, eight datasets, 56 conditions, zero admissibility violations, reductions 70 to 99.1 percent	<code>theory-radar\paper\astar_paper.tex:170-200</code>

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 6 section 6.3	Monotone Invariance Theorem and its proof	theory- radar\paper\theory_radar_paper.tex:290-304; theory- radar\paper\astar_paper.tex:94-110

FAILURES AND CORRECTIONS

none

CONDITIONS

- The path of true positive rate against false positive rate as the threshold sweeps. Its area is the AUROC, which lies in the unit interval, is one half for a constant score, is one for a perfect ranking, and is unchanged by any strictly increasing transform of the score.
- The bound from AUROC to the best F1 through the Youden index holds for concave curves and not for every curve, with the counterexample at AUROC 0.75, and the book corrects the source on that point.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/Auroc.lean`, `theorems pair_nonneg`, `auroc_nonneg`, `auroc_le_one`, `auroc_perfect`, `auroc_reversed`, `auroc_chance`, `auroc_monotone_invariant`, at `observation-data-mining af776fd`.

`lean/DataMiningAsObservation/YoudenF1.lean`, `theorems f1_eq`, `f1_le_of_youden`, `auroc_eq`, `youden_eq`, `youden_exceeds_auroc_form`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 1, 2, 3, 5, 6, 7, 8, 11, 12, 13, 14.

RELATED

[AUROC](#) (p. 29); [Youden index](#) (p. 374); [threshold](#) (p. 350); [precision, recall](#) (p. 257); [F1](#) (p. 132).

rotary position embedding

concept. Entry id rotary-position-embedding.

DEFINITION

A rotation applied to queries and keys that encodes where each token sits in the sequence. Chapter 0 section 0.11.

EQUATION

Book equation 0.23.

$$\text{softmax}(z)_i = \frac{e^{z_i}}{\sum_j e^{z_j}}, \quad \text{output} = \sum_i \text{softmax}\left(\frac{q \cdot k_i}{\sqrt{d}}\right)_i v_i.$$

Book equation 11.1.

$$\cos(k, \hat{k}) = 0.995 \quad \text{while} \quad \text{PPL} : 12.24 \rightarrow 10643.$$

LEDGER

- NEG-2. Reconstruction cosine as a proxy for key quality. [refuted]. `geometric-observation/claims/LEDGER.md:95` at 9f3829f.

FIRST STATED

Su and others, RoFormer, 2021, as chapter 0 section 0.11 of *Data Mining as Observation* states it, with the program's key-side measurements in `turboquant-pro/docs/KV_KEYS_FINDING.md:1-49`.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 1 section 1.4	cosine 0.995 and perplexity of order ten thousand, the recalibration negative	geometric-observation\chapters\ch02_failure_of_observer_free60; geometric-observation\chapters\ch16_honest_negatives.md NEG-2 and NEG-4; turboquant-pro\docs\KV_KEYS_FINDING.md:1-49
chapter 2 section 2.5	cosine 0.995 and the softmax reader	turboquant-pro\docs\KV_KEYS_FINDING.md:1-49
chapter 3 section 3.2	condition (A2), tangential displacement	turboquant-pro\docs\KV_KEYS_FINDING.md:61-86; geometric-observation\chapters\ch09_legibility.md:42-54
chapter 8 section 8.2	cosine 0.995, perplexity near 1e4	turboquant-pro\docs\KV_KEYS_FINDING.md:1-49; geometric-observation\chapters\ch16_honest_negatives.md NEG-2
chapter 11 section 11.1	condition (A2), cosine satisfies it, post-rotary keys do not, the cone below cell size	turboquant-pro\docs\KV_KEYS_FINDING.md:61-86; the-angular-observer\README.md:26-31
chapter 11 section 11.2	fp16 12.24, values-only 13.12, PolarQuant K4 10643 and 0.095, per-channel uniform K4 14.91 and 0.062, per-channel NUQ K3 15.77 and 0.148, 2.4x and 670x, pre-rotary near 22000, 2 key heads serve 12 query heads	turboquant-pro\docs\KV_KEYS_FINDING.md:1-49

FAILURES AND CORRECTIONS

- NEG-2, [refuted]. Reconstruction cosine as a proxy for key quality.

CONDITIONS

- A rotation applied to queries and keys by an angle proportional to position. A rotation preserves length, and the score of a query at one position against a key at another depends only on the difference of positions, so a head reads relative position and shifting both positions by the same amount changes no score.
- A key quantizer that rounds the rotated key rounds its position as well as its content, which is part of why the direction-only quantizer moved the scores.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/Rope.lean`, theorems `rot_length`, `rot_dot`, `relative_position`, `shift_invariant`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14.

RELATED

[attention](#) (p. 23); [query, key, value](#) (p. 267); [KV cache](#) (p. 190); [dot product](#) (p. 110).

ROUGE

concept. Entry id rouge.

DEFINITION

A family of scores for generated text against a reference. Chapter 0 section 0.11.

EQUATION

Book equation 0.22.

$$\text{PPL} = 2^H, \quad H = -\frac{1}{T} \sum_{t=1}^T \log_2 p(w_t \mid w_{<t}).$$

LEDGER

- NEG-4. Lightweight online (Lloyd) key calibration beats the calibration-free default on softmax-KL. [refuted]. `geometric-observation/claims/LEDGER.md:97` at 9f3829f.

FIRST STATED

Lin, ROUGE, 2004, as chapter 0 section 0.11 states it, with the program's 13.7 ROUGE-L number and its re-validation in `turboquant-pro/CLAIMS.md:63-81`.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 8 section 8.9	13.7 ROUGE-L, 0.25/4.19/9.60/13.7 at 64/128/256/512, re-validation negative 0.31 n=40, 26.64 under symmetric nf4, <code>_quant_nf4a_group</code> unchanged since 289bdfc before 4f7baab	<code>turboquant-pro/CLAIMS.md:63-81</code> ; <code>turboquant-pro/benchmarks/kvquant_matrix/REVAL-2026-08-08.md</code>

FAILURES AND CORRECTIONS

- NEG-4, [refuted]. Lightweight online (Lloyd) key calibration beats the calibration-free default on softmax-KL.

CONDITIONS

- A family of scores for generated text against a reference. ROUGE-1 recall counts the reference's words the candidate contains, with multiplicity, over the reference's length. It lies in the unit interval, is one for the reference itself, and is one for any rearrangement of the reference, since it reads bags and not order.
- A ROUGE score therefore certifies less than it seems to. The 13.7 ROUGE-L difference at 512 tokens re-validated at negative 0.31 on forty documents, and the claims ledger carries both numbers.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/Rouge.lean`, `theorems overlap_le`, `rouge1_mem_unit`, `rouge1_self`, `rouge1_perm`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 8.

RELATED

[bag of words](#) (p. 32); [perplexity](#) (p. 250); [harness](#) (p. 159); [certificate](#) (p. 55).

S

safe pruning

concept. Entry id safe-pruning.

DEFINITION

A rule for discarding candidates during a search that is guaranteed to discard no solution. The Apriori principle and the Monotone Invariance Theorem are the book's two instances. Chapter 5.

EQUATION

Book equation 5.3.

$$X \subseteq Y \implies s(X) \geq s(Y).$$

Book equation 5.4.

$$F_1^{\max} \leq \sup_{t \in [J, 1]} \frac{2t\pi}{t\pi + \pi + (t - J)(1 - \pi)}, \quad J = \max_{\tau} (\text{TPR} - \text{FPR}), \quad \pi = \text{prevalence},$$

$J \leq 2A - 1$ when the ROC curve is concave, and not in general.

LEDGER

none

FIRST STATED

Chapter 5 section 5.4 of *Data Mining as Observation*, with the program's two owned instances in theory-radar and arc-equivariant-search.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 5 section 5.4	Monotone Invariance and AUROC invariance theorems, the AUROC to F1 bound via the Youden index, stated in the source for every ROC curve and corrected in the book to concave curves, with the counterexample at AUROC 0.75 and F1 0.857	theory-radar\paper\astar_paper.tex:94-170; theory-radar\paper\theory_radar_paper.tex:290-304
chapter 5 section 5.4	seven thresholds up to 0.75, eight datasets, 56 conditions, zero admissibility violations, reductions 70 to 99.1 percent	theory-radar\paper\astar_paper.tex:170-200
chapter 5 section 5.4	the rerun of 2026-09-03 and 2026-09-04, pre-filter counts 2576 of 8700 and 413 of 1560, ceiling violations 812 and 412, Youden pruning 199 and 38, A* optima 0.958 and 0.694 in strict and fast modes, Youden mode short by 0.006 and 0.002	theory-radar\results\rerun_youden_2026_09_03.json; theory-radar\ERRATA.md
chapter 5 section 5.4	learned pruning 88 to 99.6 percent with zero false negatives	theory-radar\README.md:88-100

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 5 section 5.4	naive 53 equals pruned 53, budget exhaustion 94 percent to 0, 29 operations, depth three, 0 of 172 held out, learned ranking AUC 0.899 against 0.560, reachability as the wall	arc-equivariant-search\EXPERIMENTS.md:20-35,60-80; arc-equivariant-search\README.md:13; arc-equivariant-search\paper\arc_equivariant_search.md:20-30
chapter 6 section 6.3	depth three, ten binary and eight unary operations, exact optimal F1 by sort and sweep, beam, projections	theory-radar\README.md:50-100,140-200
chapter 6 section 6.3	Monotone Invariance Theorem and its proof	theory-radar\paper\theory_radar_paper.tex:290-304; theory-radar\paper\astar_paper.tex:94-110
chapter 6 section 6.3	88 to 99.6 percent pruned with zero false negatives, fair protocol, 200 by 5 folds	theory-radar\README.md:88-100
chapter 6 section 6.3	the five-dataset table, breast cancer 0.955 to 0.963, the pattern, DOI 10.5281/zenodo.20660206	theory-radar\README.md:100-140
chapter 6 section 6.4	the reviewer's revision plan, drop sigma, bounded claim, 19 at 200 by 5 vs 11 at 20 by 5, rerun in progress	theory-radar\paper\REVISION_PLAN.md:1-60
chapter 6 section 6.5	loading weights and stability across folds requested	theory-radar\paper\REVISION_PLAN.md issue 6
chapter 7 section 7.3	the five-dataset outcomes	theory-radar\README.md:100-140
chapter 8 section 8.5	three-way inconsistency, lines 544, 548 to 556, 755	constraint-gap\review\FINDINGS.md:7-31 against theory-radar\paper\theory_radar_v6_submission.tex

FAILURES AND CORRECTIONS

- theory-radar/ERRATA.md:3-75 at 37c4e6c. ## 2026-09-03. The AUROC to F1 bound holds only for concave ROC curves **What was claimed.** Theorem “AUROC-F1 Bound” in paper/astar_paper.tex, the derivation in src/symbolic_search/_auroc_proof.py, and the pruning bounds in run_astar_v2.py and src/symbolic_search/_heuristic_dag.py all rested on the statement that any ROC curve with area A has maximum Youden index $J = \max(\text{TPR} - \text{FPR})$ at most $2A - 1$. **Why it is wrong.** The statement holds for concave ROC curves (equivalently, scores whose likelihood ratio is monotone), because a concave curve lies above the two chords through its Youden point and so has area at least $(1 + J)/2$. It fails for non-concave curves. Counterexample, found by an external reviewer of the companion textbook: rank three quarters of the positives above every negative and the remaining quarter below every negative. AUROC is 0.75, the maximum Youden index is 0.75, and at prevalence one half the best thresholded F1 is 0.857, above the 0.80 the AUROC form of the ceiling allows. The sound ceiling for that score, from $J = 0.75$, is 0.889, and the score sits under it. The `auroc_f1_bound` formula in `run_astar_v2.py`, $2A\pi/(A\pi + (1 - A)(1 - \pi))$, is not a valid upper bound either: at prevalence 0.1 the same score reaches F1 0.857 against a “bound” of 0.5. **Consequence for reported results.** The searches that used these bounds to prune could have discarded a candidate formula whose ROC curve was not concave and whose true best F1 exceeded the computed ceiling. The formulas and wins reported by those runs are therefore what a pruned search

found, and are lower bounds on what an unpruned search would find. The Monotone Invariance Theorem and the results that do not depend on the AUROC bound are unaffected. **Correction.** The sound form of the ceiling substitutes the measured maximum Youden index of the candidate's own scores for $2A - 1$. It holds for every score, costs the same sort as AUROC, and is what `max_f1_for_youden` and `youden_f1_bound` now compute. The AUROC form is retained under its original names with a docstring stating the concavity condition, and the pruning sites now use the measured index. A rerun of the published searches under the corrected bound was done on 2026-09-03 and 2026-09-04, `rerun_youden_2026_09_03.py`, results in `results/rerun_youden_2026_09_03.json`, run on the Atlas workstation, CPU only. **Rerun, part A, the pre-filter admissibility table (tab:auroc).** Depth-2 pairwise enumeration on the eight datasets, exhaustive against the AUROC pre-filter at the seven published thresholds, with the sound Youden ceiling beside it. The evaluation counts under the pre-filter reproduce the published table (Breast Cancer 8700 pairs, 2576 evaluated at 0.75; Wine 1560, 413; Circles 20, 4; Moons 20, 2). The pre-filter lost no optimum on any dataset at any threshold, which is what the paper reported. That was luck of the data, not the theorem: the retracted ceiling sat below the F1 a pair formula actually reach

CONDITIONS

- A pruning rule is safe when the pruned search returns the same answer as the unpruned one on every input. The Apriori principle and the Monotone Invariance Theorem are safe by proof.
- A learned or heuristic rule cannot earn that word from a finite test. It can be empirically lossless on the held-out cases, and the record says which cases.
- The AUROC form of the F1 ceiling is a safe pruning bound only for concave ROC curves. The Youden form is sound for every score at the leaf, and pruning a subtree by its root's ceiling is a heuristic in both forms.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/SafePruning.lean`, theorems `support_anti`, `apriori`, `subset_of_frequent`, at `observation-data-mining af776fd`.

`lean/DataMiningAsObservation/YoudenF1.lean`, theorems `f1_eq`, `f1_le_of_youden`, `auroc_eq`, `youden_eq`, `youden_exceeds_auroc_form`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 5.

RELATED

[Apriori principle](#) (p. 22); [Monotone Invariance Theorem](#) (p. 220); [Youden F1 bound](#) (p. 372); [formula search](#) (p. 145).

sampling

instrument. Entry `id sampling`.

DEFINITION

Choosing which rows to read. The standard error of a sample mean falls as one over the square root of the sample size. Chapter 2 section 2.7.

EQUATION

Book equation 0.18.

$$\widehat{\text{Var}}_{\text{NB}} = \left(\frac{1}{J} + \frac{n_{\text{test}}}{n_{\text{train}}} \right) \hat{\sigma}^2, \quad \frac{\widehat{\text{Var}}_{\text{NB}}}{\hat{\sigma}^2/J} = 1 + J \frac{n_{\text{test}}}{n_{\text{train}}}.$$

Book equation 8.2.

$$\widehat{\text{Var}}_{\text{NB}} = \left(\frac{1}{J} + \frac{n_{\text{test}}}{n_{\text{train}}} \right) \hat{\sigma}^2,$$

$$J = 1000, \frac{n_{\text{test}}}{n_{\text{train}}} = \frac{1}{4} \Rightarrow 1 + 250 = 251, \sqrt{251} = 15.84.$$

Book equation 2.1.

$$P_{C_2 \circ C_1}(x) = J_1(x)^\top P_{C_2}(C_1(x)) J_1(x), \quad \text{rank } P_{C_2 \circ C_1}(x) \leq \text{rank } P_{C_2}(C_1(x)) \quad \text{at each row } x.$$

LEDGER

- NEG-14. (GO-P-2026-037, prospective) a2_probe.median_unit_displacement is a single-statistic predictor of the flip regime (unit_disp $\geq 1.0 \Rightarrow$ generic-polar-flip vs needs-blind-probe). [refuted]. geometric-observation/claims/LEDGER.md:108 at 9f3829f.

FIRST STATED

Chapter 2 section 2.7 of *Data Mining as Observation*, with the first strata run's sample in turboquant-pro/docs/STRATA_RFC.md:24-98.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 4 section 4.7	ESL 7.10.2 numbers	Hastie, Tibshirani, Friedman, ESL 2e, section 7.10.2; chapter 8 of this book
chapter 8 section 8.1	ESL 7.10.2 numbers (N=50, p=5000, 100 selected, 3 percent vs 50 percent)	Hastie, Tibshirani, Friedman, ESL 2e, section 7.10.2
chapter 10 section 10.4	area map, boundary rule, hash, refuse not warn, intra and transit counts, area classes, abstention rule	turboquant-pro/docs/STRATA_RFC.md:24-98
chapter 10 section 10.4	second prediction inverted, transit 0.389 vs 0.291, centrality difference signs, share 0.473 vs 0.391, seven of thirteen backbone	turboquant-pro/docs/RESULTS_multilingual_strata.md:55-90; turboquant-pro/docs/STRATA_RFC.md:98-130

FAILURES AND CORRECTIONS

- NEG-14, [refuted]. (GO-P-2026-037, prospective) a2_probe.median_unit_displacement is a single-statistic predictor of the flip regime (unit_disp $\geq 1.0 \Rightarrow$ generic-polar-flip vs needs-blind-probe).

CONDITIONS

- Choosing which rows to read. A sample's mean lies between its smallest and largest row, the standard error of the mean falls as one over the square root of the sample size, and a sample too thin to score is abstained on rather than averaged in.
- Sampling and aggregation are design choices that decide what the reader can see, and the first strata run read 350000 of 2391361 rows and abstained on the strata with two rows and one.

Conditions are curated in entries.toml rather than read from a record.

MACHINE CHECKED

lean/DataMiningAsObservation/StandardError.lean, theorems se_pos, se_quarter, se_antitone, se_tendsto_zero, thousand_folds, at observation-data-mining af776fd.

lean/DataMiningAsObservation/Ensemble.lean, theorems average_between, sq_average_le, average_const, at observation-data-mining af776fd.

USED IN

Data Mining as Observation chapters 0, 2, 3, 4, 5, 6, 8, 9, 10, 11, 12, 13, 14.

RELATED

[standard error](#) (p. 337); [aggregation](#) (p. 10); [bootstrap](#) (p. 45); [seed](#) (p. 318); [stratification](#) (p. 339).

score

concept. Entry id score.

DEFINITION

The number a classifier produces per row before a threshold turns it into a decision. Chapter 6.

EQUATION

Book equation 6.1.

$$s(x) = w \cdot x + b, \quad P_C = \mathbb{E}[\sigma'(s)^2] w w^\top, \quad \text{rank } P_C = 1.$$

Book equation 6.2.

$$\max_{\tau, \text{dir}} F_1\left(\mathbf{1}[\text{dir}(g(f(X)), \tau)], y\right) = \max_{\tau, \text{dir}} F_1\left(\mathbf{1}[\text{dir}(f(X), \tau)], y\right)$$

for every strictly monotone g .

LEDGER

none

FIRST STATED

Chapter 6 section 6.1 of *Data Mining as Observation*, with the program's threshold sweeps in theory-radar.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 5 section 5.4	Monotone Invariance and AUROC invariance theorems, the AUROC to F1 bound via the Youden index, stated in the source for every ROC curve and corrected in the book to concave curves, with the counterexample at AUROC 0.75 and F1 0.857	theory-radar\paper\astar_paper.tex:94-170; theory-radar\paper\theory_radar_paper.tex:290-304
chapter 6 section 6.3	Monotone Invariance Theorem and its proof	theory-radar\paper\theory_radar_paper.tex:290-304; theory-radar\paper\astar_paper.tex:94-110

FAILURES AND CORRECTIONS

none

CONDITIONS

- The number a classifier produces per row before a threshold turns it into a decision. Every decision at every threshold is a function of the scores' ordering, so a strictly increasing recalibration of the scores with the matching recalibration of the threshold changes no decision.

- What a recalibration can change is calibration, which is why the reliability weight reads the ordering and the expected calibration error reads the values, and neither stands in for the other.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/Threshold.lean`, theorems `predicted_anti`, `tp_anti`, `fp_anti`, `decision_comp`, `predicted_extremes`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14.

RELATED

[classifier](#) (p. 60); [threshold](#) (p. 350); [Monotone Invariance Theorem](#) (p. 220); [calibration](#) (p. 52); [reliability weight](#) (p. 298).

sealed

instrument. Entry id sealed.

DEFINITION

Of a prediction, committed with its hash recorded before the measurement was run, so that anyone with the repository can verify it has not changed. Chapter 0 section 0.12.

EQUATION

Book equation 8.1.

$$O_{\text{harness}} = (C_{\text{score}}, G_{\text{metric}}, B = \text{folds} \times \text{samples} \times \text{seeds}).$$

LEDGER

- OT-7. The damage form, trace pairing, rank, and loading covariance are $GL(d)$ -invariant under $P' = A^{-T}PA^{-1}$, while spectrum, effective rank, principal angles, and water-filling are $O(d)$ -only — consumer-weighted damage is a geometric scalar, and P3's cliff cannot be bought down by reparameterization. [demonstrated]. `geometric-observation/claims/LEDGER.md:30 at 9f3829f`.
- OT-11. Feedback-free staleness: streaming-retrieval damage tracks measured drift; derived-cadence re-allocation removes it. [void]. `geometric-observation/claims/LEDGER.md:48 at 9f3829f`.

FIRST STATED

The program's seal ledger, `observation-theory-campaigns/experiments/SEALS.md:1-10`, a seal being the registration id, the sealing commit, and the SHA-256 of the sealed file at that commit, and chapter 0 section 0.12 of *Data Mining as Observation*.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 8 section 8.8	seal = commit plus SHA-256	<code>observation-theory-campaigns/experiments/SEALS.md:1-10</code>
chapter 8 section 8.8	Second Crucible 1 of 5, four instrument deaths, rate-limit rule; Third Crucible 2 of 3	<code>geometric-observation/claims/LEDGER.md:50</code> ; <code>geometric-observation/crucible/OT-CRUCIBLE-3-VERDICT.md:1-20</code>

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 8 section 8.8	declaration drafted 2026-08-17, sealed 2026-08-18, one count corrected, G2	geometric-observation\crucible\DECLARATION-V1.md:1-20; geometric-observation\crucible\OT-CRUCIBLE-4.md:31-35
chapter 12 section 12.6	XPROTO-LLM, benchmark 0.909, 0.920, 0.909, thirty slices, target 0.8, naive 0.333, aware 0.033, spread 0.380, deployment mean 0.736, six bars on three seeds, sealed 2026-08-25 at b61f7f1	observation-theory-campaigns\experiments\LLM-EVAL-TRACK.md:1-50; observation-theory-campaigns\analysis\llm\XPROTO-LLM-graded.json; observation-theory-campaigns\experiments\SEALS.md:85
chapter 13 section 13.4	sealed correction, 0.5 dB calibration, floors 4/6/4, 2/2/3, 1, slopes 0.1008, 0.1398, 0.1086, R^2 0.7376, 0.9218, 0.6174, bars B1 to B4 and MC1 to MC4 all true, seeds 20260827 to 20260829, sealed 1d3de4a	observation-theory-campaigns\analysis\csi\PREREG-XPROTO-CSI-SWEEP2.md:1-60; observation-theory-campaigns\analysis\csi\XPROTO-CSI-SWEEP2-graded.json; observation-theory-campaigns\experiments\SEALS.md:90

FAILURES AND CORRECTIONS

- observation-theory-campaigns/ERRATA.md:111-120 at 9d86211. ## Standing note on records and runners
Two failures this session shared one shape. A record was committed from a runner later found defective, and the correction was left uncommitted where nothing reading the repository could see it. The rule adopted in response is that a rerun writes to a new path and never over a committed record, and that the superseded record stays in the repository naming what replaced it. See the PF4-007 subsection of experiments/CAMPAIGN.md.

CONDITIONS

- A prediction is sealed when its file is committed and the commit and the content hash are recorded in the seal ledger before the measurement is run, so that anyone with the repository can verify it has not changed.
- A seal fixes the text, not its correctness. The version 1.0 declaration was sealed with one count wrong, and the correction is recorded beside the seal rather than by editing the sealed file.
- An unsealed exploration can be reported and cannot carry a claim, and the refuted refresh-floor law is the program's standing example.

Conditions are curated in entries.toml rather than read from a record.

MACHINE CHECKED

lean/DataMiningAsObservation/Seal.lean, theorems changed_of_hash_ne, hash_eq_of_eq, exists_collision, at observation-data-mining af776fd.

USED IN

Data Mining as Observation chapters 0, 1, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14.

RELATED

[preregistration](#) (p. 258); [ledger class](#) (p. 204); [certificate](#) (p. 55); [refresh floor](#) (p. 292).

seed

concept. Entry id seed.

DEFINITION

The number that fixes a run's randomness, so that two runs with the same seed are one run and only runs with different seeds measure variance. Chapter 8.

EQUATION

Book equation 0.18.

$$\widehat{\text{Var}}_{\text{NB}} = \left(\frac{1}{J} + \frac{n_{\text{test}}}{n_{\text{train}}} \right) \hat{\sigma}^2, \quad \frac{\widehat{\text{Var}}_{\text{NB}}}{\hat{\sigma}^2/J} = 1 + J \frac{n_{\text{test}}}{n_{\text{train}}}.$$

Book equation 8.2.

$$\widehat{\text{Var}}_{\text{NB}} = \left(\frac{1}{J} + \frac{n_{\text{test}}}{n_{\text{train}}} \right) \hat{\sigma}^2, \\ J = 1000, \frac{n_{\text{test}}}{n_{\text{train}}} = \frac{1}{4} \Rightarrow 1 + 250 = 251, \sqrt{251} = 15.84.$$

Book equation 8.1.

$$O_{\text{harness}} = (C_{\text{score}}, G_{\text{metric}}, B = \text{folds} \times \text{samples} \times \text{seeds}).$$

LEDGER

- NEG-14. (GO-P-2026-037, prospective) a2_probe.median_unit_displacement is a single-statistic predictor of the flip regime (unit_disp $\geq 1.0 \Rightarrow$ generic-polar-flip vs needs-blind-probe). [refuted]. geometric-observation/claims/LEDGER.md:108 at 9f3829f.

FIRST STATED

Chapter 8 section 8.1 of *Data Mining as Observation*, with the disjoint-seed rule of the database freshness track in observation-theory-campaigns/experiments/DATABASE-FRESHNESS-TRACK.md.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 4 section 4.4	GO-4 budget inversion, fixed m 10 rises, matched m 121, 126, 159 collapses, 3 seeds	geometric-observation\claims\LEDGER.md row GO-4
chapter 13 section 13.3	ZooKeeper hot 0.99 cold 0.01 witnessed 0.0, Postgres 0.50 to 0.06, MongoDB 0.47 to 0.03, production Postgres 0.47 to 0.02, real substrates, disjoint seeds	observation-theory-campaigns\experiments\DATABASE-FRESHNESS-TRACK.md:1-45

FAILURES AND CORRECTIONS

- NEG-14, [refuted]. (GO-P-2026-037, prospective) a2_probe.median_unit_displacement is a single-statistic predictor of the flip regime (unit_disp $\geq 1.0 \Rightarrow$ generic-polar-flip vs needs-blind-probe).

CONDITIONS

- The number that fixes a run's randomness, so that two runs with the same seed are one run and only runs with different seeds measure variance. The standard error over n seeds falls as one over the square root

of n , and a paired comparison across seeds has smaller variance than an unpaired one exactly when the two arms covary.

- A preregistration names its seeds before the run, the freshness tracks used disjoint seeds for the two arms, and a claim replicated on one seed is replicated once.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/StandardError.lean`, theorems `se_pos`, `se_quarter`, `se_antitone`, `se_tendsto_zero`, `thousand_folds`, at `observation-data-mining af776fd`.

`lean/DataMiningAsObservation/Bootstrap.lean`, theorems `mean_sub`, `var_sub`, `paired_lt_iff`, `cov_comm`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 4, 6, 7, 8, 9, 11, 12, 13.

RELATED

[harness](#) (p. 159); [standard error](#) (p. 337); [bootstrap](#) (p. 45); [preregistration](#) (p. 258); [sealed](#) (p. 316).

sensitivity

concept. Entry id sensitivity.

DEFINITION

The vector of partial derivatives of a consumer with respect to each coordinate of its input, at one point. Equation 0.8.

EQUATION

Book equation 0.8.

$$g_j \approx \frac{C(x + h e_j) - C(x - h e_j)}{2h}, \quad j = 1, \dots, d.$$

Book equation 0.11.

$$P_C(x) = J(x)^\top G(C(x)) J(x), \quad J(x) = \frac{\partial C}{\partial x}(x), \quad \bar{P}_{C,\mu} = \mathbb{E}_\mu[P_C(x)].$$

Book equation 14.4.

$$\text{attribution}_j(x) \approx g_j(x) \delta_j, \quad \overline{\text{importance}}_j \approx (P_C)_{jj} = \mathbb{E}[g_j^2].$$

LEDGER

- NEG-14. (GO-P-2026-037, prospective) `a2_probe.median_unit_displacement` is a single-statistic predictor of the flip regime (`unit_disp` $\geq 1.0 \Rightarrow$ generic-polar-flip vs needs-blind-probe). [refuted]. `geometric-observation/claims/LEDGER.md:108` at 9f3829f.
- GO-EC-3. A read operator recovered from a black-box consumer by query-only finite-difference probing, composed with the Kalman covariance as $\text{tr}(\hat{P}_C \Sigma)$, prospectively selects sensors that improve the held-out consumer at matched budgets with probe cost charged — capturing 94.6% of the known analytic optimum's gain on the positive-control arm (gate $\geq 75\%$) and improving 16.3% over the best consumer-agnostic policy on non-analytic consumers (gate $\geq 8\%$), with trace-matched ordering carried by the composition at 86.9% over 61 pairs (gate $\geq 65\%$). [predicted]. `geometric-observation/claims/LEDGER.md:162` at 9f3829f.

FIRST STATED

Volume 14, chapter 5, `geometric-observation/chapters/ch05_the_read_metric_and_the_quotient.md:7-48`, DOI `10.5281/zenodo.21776291`, as the gradient of the consumer at a row.

MEASUREMENTS

Where the book states it	Numbers, as the book’s sources table records them	Source
chapter 2 section 2.2	read subspace small, operator local, pullback composition, rank cannot increase	<code>geometric-observation\chapters\ch05_the_read_metric_and_the_</code> <code>48;</code> <code>geometric-observation\chapters\ch06_mathematical_preliminaries</code> <code>27</code>
chapter 2 section 2.2	read distortion controls but is not a complete rank statistic, twelve of twelve, one middle pair misordered, NEG-9	<code>geometric-observation\chapters\ch05_the_read_metric_and_the_</code> <code>75</code>
chapter 2 section 2.3	refusal regimes, selection consumers read order, recurrences compound	<code>readscope\readscope\regimes.py:1-</code> <code>60</code>
chapter 6 section 6.1	selection consumers have zero sensitivity almost everywhere	<code>readscope\readscope\regimes.py:1-</code> <code>60</code>

FAILURES AND CORRECTIONS

- NEG-14, [refuted]. (GO-P-2026-037, prospective) `a2_probe.median_unit_displacement` is a single-statistic predictor of the flip regime ($\text{unit_disp} \geq 1.0 \Rightarrow \text{generic-polar-flip vs needs-blind-probe}$).

CONDITIONS

- The sensitivity of a consumer at a row is the vector of partial derivatives of its output with respect to each coordinate, measurable without the formula by a central finite difference at two calls per coordinate.
- The read operator is the workload average of the sensitivity’s outer product, so its diagonal is the expected squared sensitivity to each feature and its off-diagonal entries are co-sensitivities, not interactions.
- A gradient-based attribution estimates the sensitivity. Permutation importance, partial dependence, and Shapley values measure other things.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/ReadOperator.lean`, `theorems rank_one_reads_one_direction`, `readOp_mulVec`, `quad_readOp`, `quad_readOp_nonneg`, `readOp_mulVec_eq_zero_iff`, `readOp_diag`, `readOp_offdiag`, `readOp_symm`, `readOp_neg`, `affine_const_along_nuisance`, `readOp_affine`, `readOp_sqLength_basis`, at `observation-data-mining/af776fd`.

USED IN

Data Mining as Observation chapters 0, 1, 2, 4, 6, 7, 8, 11, 14.

RELATED

[read operator](#) (p. 282); [consumer](#) (p. 73); [blind probe](#) (p. 39); [read subspace](#) (p. 285).

session deck

reference. Entry id `session-deck`.

DEFINITION

The lecture deck of ECE 514, cited by line range from the instructor's session outlines, which are not public. Chapter 1 section 1.6.

EQUATION

Book equation 1.1.

$$O = (C, G, B).$$

LEDGER

none

FIRST STATED

Chapter 1 section 1.6 of *Data Mining as Observation*, citing the instructor's working documents, ECE_514-01_FA26_session-outlines.md, which are not public.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 1 section 1.1	the four tasks, the churn framing, the framing workshop	TSK chapter 1; instructor working documents, not public [@bond2026course], ECE_514-01_FA26_session-outlines.md:29-60
chapter 1 section 1.3	the four failure modes	ECE_514-01_FA26_session-outlines.md:38
chapter 1 section 1.6	the course map	ECE_514-01_FA26_session-outlines.md:29-332
chapter 1 section 1.7	the field guide, the decision test, the red flags	ECE_514-01_FA26_session-outlines.md:39-46
chapter 2 section 2.3	MCAR, MAR, MNAR and the remedies, imputation before splitting	TSK 2e section 2.2; instructor working documents, not public [@bond2026course], ECE_514-01_FA26_session-outlines.md:64-72
chapter 5 section 5.3	the transferable idea as a course learning outcome	instructor working documents, not public [@bond2026course], ECE_514-01_FA26_session-outlines.md:117-136
chapter 7 section 7.6	the preregistration fields	instructor working documents, not public [@bond2026course], ECE_514-01_FA26_session-outlines.md:157-176; chapter 8 section 8.8 of this book

FAILURES AND CORRECTIONS

none

CONDITIONS

- The lecture deck of the course the book accompanies, ECE 514, cited by line range from the instructor's session outlines. A row that names it names a file and lines like every other source, and a changed digest would prove a changed deck.
- The deck is not public. The book quotes what it says and marks each such row as instructor working documents, and the reader who wants the deck asks the instructor.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/Seal.lean`, `theorems changed_of_hash_ne`, `hash_eq_of_eq`, `exists_collision`, `at observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 1, 2, 3, 4, 5, 6, 8, 10, 11, 12, 14.

RELATED

[sources table](#) (p. 329); [commit hash](#) (p. 67); [harness](#) (p. 159); [preregistration](#) (p. 258).

shard

instrument. Entry id `shard`.

DEFINITION

One partition of an index that is searched separately, with the results merged. The global top k lies inside the union of the shards' top k . Chapter 13.

EQUATION

Book equation 11.3.

$$\text{recall}@k = \frac{|\text{returned top-}k \cap \text{true top-}k|}{k},$$

true top- k computed from the uncompressed vectors.

Book equation 11.4.

$$\text{directions resolved}(k) = \begin{cases} 1 \text{ or } 2, & k < d \\ \text{rank } P_C, & k \geq d \end{cases} \quad \text{cost} = 2d \text{ consumer calls per operating point.}$$

Book equation 13.2.

$$T_{\text{coh}} = \frac{0.423}{f_D},$$

refuted fit: $\phi \approx 0.177 T_{\text{coh}} (R^2 = 0.915)$, sealed: $\phi \leq 6 \text{ TTI at } 10 \text{ Hz, } 1 \text{ TTI at } \geq 50 \text{ Hz}.$

LEDGER

- GO-12. The dynamic region's opening control — staleness is access width, not delay: with full context-path access, pure-delay aging is information-free (the recoding identity $(Y, V, S^{(\Delta)}) \stackrel{d}{=} (Y, V, P^\Delta S^{(0)})$ makes every $\sigma(S)$ -measurable conditional functional exactly Δ -invariant; circulant-exact, $O(1/n)$ edge leakage on finite windows); with time-local slice access the tax is GO-11's static quadratic with the substitution set by the *encoder's* access — single-letter (Y_t, V_t) records at $(\rho, s/a^{2\Delta})$, context-epoch-latent records at the strictly smaller $(\rho a^\Delta, \tau^2)$ (gap up to 0.053 bits) — both strictly increasing in Δ with common limit $\frac{1}{2} \log_2(1/D)$. GO-8's age-dependence is the slice regime's operational face. Theorem 1 (the conditional-variance reduction) settles the causal-prefix eraser in the single-record setting: every observation-subset σ -algebra enters through $q_{\mathcal{G}} = \text{Var}(V_t \mid \mathcal{G})$ alone; slice/prefix/path = single-sample/Kalman-fixed-lag/noncausal-Wiener variances; strict interpolation at every finite lag with exact gap $C^{2\Delta}(P_f - q_{\text{path}})$; $W(\Delta) = kT \ln 2 \cdot L_{\text{prefix}}$ is the dynamic conditional-Landauer curve. Theorem 2-spectral (070): the spectral conditional RDF — at the work endpoint, $L(D)$ is the equal-slope allocation of per-frequency static quadratics over the circulant spectrum, with classical reverse water-filling ($\tau^2 \rightarrow \infty$) and the static theorem (flat spectrum) exact; per-mode convexity proven; the third

promotion of the water level completed for the single-consumer face; Toeplitz transfer scoped imported-with-lemmas, $O(1/n)$ numerics. The weighted spectral theorem (071) closes Conjecture 1 at circulant scope: J_w decomposes per-frequency at every w , each mode carrying Theorem 3's two-water-level system $(\gamma_0(\omega), \gamma_1(\omega))$ under ONE common distortion price (the conjectured 'pair of prices' refuted-as-phrased, rescoped to $\$(w, \mu)\$$); convexity via the full-region moment argument + perturbation. The third promotion is complete on BOTH faces.[predicted].geometric-observation/claims/LEDGER.md:78' at 9f3829f.

FIRST STATED

Chapter 13 section 13.1 of *Data Mining as Observation*, with the placement study in turboquant-pro/docs/RESEARCH_ROADMAP.md:133-160 and the fleet run in openvector-bench/README.md:250-263.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 8 section 8.4	gate meta-rule, relative contrast discriminates nothing	openvector-bench\openvector_bench\score_rc1.py:1-14; turboquant-pro\docs\RESEARCH_ROADMAP.md:133-160
chapter 11 section 11.4	R80 table, 25x probe depth, 92 percent own cell, one third cross-article, 0.53 matches same-article share, relative contrast discriminates nothing	openvector-bench\results\R80_ANN.md:1-40; turboquant-pro\docs\RESEARCH_ROADMAP.md:133-160
chapter 13 section 13.1	22.30 s cold vs 0.835 s warm at eight probes, every prior number warm, fragmentation 63.7 bracketing 42 to 47, 64 shards, routing sparsity computed and discarded	turboquant-pro\docs\RESEARCH_ROADMAP.md:133-160
chapter 13 section 13.1	one hundred billion vectors on a preemptible fleet, systems result not a tier, corpus rejected by the admission battery	openvector-bench\README.md:250-263

FAILURES AND CORRECTIONS

none

CONDITIONS

- One partition of an index that is searched separately, with the results merged. If each shard returns its own top k and the global top k is strict, the global top k lies inside the union of the shards' answers, so the merge loses nothing and reads at most the number of shards times k candidates.
- A placement study over 64 shards reported search times taken with the file cache already holding the shards, and a speedup of 63.7 times was fragmentation, so a sharded number names whether it is warm and what the shards were.

Conditions are curated in entries.toml rather than read from a record.

MACHINE CHECKED

lean/DataMiningAsObservation/Shard.lean, theorems global_in_local, global_in_union, merge_cost, at observation-data-mining af776fd.

USED IN

Data Mining as Observation chapters 11, 13.

RELATED

[inverted file](#) (p. 177); [recall at k](#) (p. 286); [cold, warm](#) (p. 65); [cache](#) (p. 51); [budget](#) (p. 46).

silhouette

concept. Entry id silhouette.

DEFINITION

For a row, the difference between its mean distance to the nearest other cluster and its mean distance to its own cluster, divided by the larger of the two. Equation 0.30.

EQUATION

Book equation 0.30.

$$\text{SSE} = \sum_k \sum_{i \in \mathcal{C}_k} \|x_i - c_k\|^2, \quad s_i = \frac{b_i - a_i}{\max(a_i, b_i)}.$$

Book equation 9.1.

$$\text{SSE} = \sum_{k=1}^K \sum_{i \in \mathcal{C}_k} \|x_i - c_k\|^2, \quad \text{the } P_C = I \text{ distortion summed within clusters.}$$

LEDGER

none

FIRST STATED

Rousseeuw, silhouettes, 1987, as TSK chapter 7 presents it and chapter 0 section 0.15 states it, and chapter 9 of *Data Mining as Observation*, where it is one validity index among those the recognizer replaces.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

CONDITIONS

- For a row, the difference between its mean distance to the nearest other cluster and its mean distance to its own cluster, over the larger of the two. It lies between minus one and one, is positive exactly when the row is closer to its own cluster, and is unchanged when every distance is scaled by the same factor.
- It is computed under the identity reader on the distances it is given, so it validates a clustering for that reader and not for a consumer that reads other directions, which is why chapter 9 asks the recognizer to name the manifold instead.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/Silhouette.lean`, `theorems silhouette_mem`, `silhouette_pos_iff`, `silhouette_scale`, `silhouette_self`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 9, 10.

RELATED

[recognizer](#) (p. 288); [vacuity threshold](#) (p. 359); [identity reader](#) (p. 171); [quotient](#) (p. 271).

Simpson's paradox

concept. Entry id simpsons-paradox.

DEFINITION

A reversal of a comparison when groups are combined, which the book reads as an aggregation quotient. Chapter 5.

EQUATION

Book equation 10.7.

$$\text{verdict} = \min_{i: n_i \geq n_{\min}, |Q_i| \geq q_{\min}} \text{score}_i, \quad \text{ABSTAIN otherwise.}$$

LEDGER

none

FIRST STATED

TSK chapter 5 on objective measures, as chapter 5 section 5.5 of *Data Mining as Observation* reads it, an aggregation quotient.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 5 section 5.1 to 5.3, 5.5	support, confidence, lift, the Apriori principle, FP-growth, closed and maximal itemsets, objective measures, Simpson's paradox, cross-support	TSK 2e chapter 5

FAILURES AND CORRECTIONS

none

CONDITIONS

- A pooled rate is the size-weighted mean of the group rates, so pooling is a quotient that declares the group labels irrelevant, and it can reverse a comparison that holds in every group when the groups have different sizes under the two arms.
- With equal group sizes under both arms the pooled comparison agrees with a comparison that holds in every group. The book's answer is the min-over-strata verdict, which never pools.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/Simpson.lean`, `theorems pooled_eq_weighted`, `reversal`, `no_reversal_of_equal_sizes`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 5.

RELATED

[quotient](#) (p. 271); [min-over-strata](#) (p. 217); [abstention](#) (p. 9); [deployment mismatch](#) (p. 101).

skewness

concept. Entry id skewness.

DEFINITION

The mean cubed deviation of a column from its mean over the cube of its spread, zero for a symmetric column. The hubness gate reads it. Chapter 3 section 3.5.

EQUATION

Book equation 0.35.

$$RH = \sum_i \max\left(0, \frac{N_i}{\sum_j N_j} - \frac{1}{n}\right).$$

Book equation 10.3.

$$N_k(x \mid Q) = |\{q \in Q : x \in \text{top}_k(q)\}|, \quad \text{anti-hub} : N_k(x) = 0.$$

Book equation 0.17.

$$\Pr[X \geq c] = 1 - \sum_{i < c} e^{-\mu} \frac{\mu^i}{i!}, \quad c^* = \max\{c : n \Pr[X \geq c] \geq 1\}, \quad \mu = \frac{n_q k}{n}.$$

LEDGER

- NEG-14. (GO-P-2026-037, prospective) a2_probe.median_unit_displacement is a single-statistic predictor of the flip regime (unit_disp $\geq 1.0 \Rightarrow$ generic-polar-flip vs needs-blind-probe). [refuted]. geometric-observation/claims/LEDGER.md:108 at 9f3829f.

FIRST STATED

Chapter 3 section 3.5 and chapter 10 section 10.2 of *Data Mining as Observation*, with the hubness gate's skew in openvector-bench/results/R13_STAGE1_RESULT.md:75-110.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 10 section 10.2	half 2 failed, at most 1.28 times against 2, withdrawal and its scope	openvector-bench/results/R13_STAGE1_RESULT.md:75-110
chapter 10 section 10.4	area map, boundary rule, hash, refuse not warn, intra and transit counts, area classes, abstention rule	turboquant-pro/docs/STRATA_RFC.md:24-98
chapter 10 section 10.4	second prediction inverted, transit 0.389 vs 0.291, centrality difference signs, share 0.473 vs 0.391, seven of thirteen backbone	turboquant-pro/docs/RESULTS_multilingual_strata.md:55-90; turboquant-pro/docs/STRATA_RFC.md:98-130
chapter 10 section 10.5	Gate A design error, A prime skew 3.970 to 3.177, max 287 to 213, Robin Hood 0.372 to 0.261, fraction 0.117 to 0.079, compressed path 0.663 vs 0.90, seven strata, 0.62 to 0.69 vs 0.76 to 0.84	turboquant-pro/docs/RESULTS_strata_phase23_gates.md:1-45

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 10 section 10.5	rerank identical 0.6627, candidate coverage, depth curve 51 to 501, coverage 0.696 to 0.926, fidelity 0.663 to 0.702, originals rerank 0.923, 4096 bytes on 148	turboquant-pro\docs\RESULTS_strata_phase23_gates.md:45-110
chapter 10 section 10.5	fragile-first allocation 0.7118 vs 0.7251, targets up 0.004 to 0.034, redesigned allocator passed	turboquant-pro\docs\RESULTS_strata_phase23_gates.md:110-125

FAILURES AND CORRECTIONS

- NEG-14, [refuted]. (GO-P-2026-037, prospective) `a2_probe.median_unit_displacement` is a single-statistic predictor of the flip regime ($\text{unit_disp} \geq 1.0 \Rightarrow \text{generic-polar-flip vs needs-blind-probe}$).

CONDITIONS

- The weighted mean cubed deviation of a column from its mean over the cube of its spread. It is unchanged by a shift and by a positive rescaling, changes sign when the column is negated, and is zero for a column symmetric about its mean.
- The hubness gate reads the skew of the neighbour-count distribution, and the remedy that was to fix it moved the skew from 3.970 to 3.177 while the compressed path still failed every stratum.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/Skewness.lean`, `theorems skew_shift`, `skew_scale`, `skew_neg`, `skew_symmetric`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 3, 5, 10, 11.

RELATED

[hubness](#) (p. 168); [Robin Hood index](#) (p. 305); [variance](#) (p. 361); [gate](#) (p. 151); [percentile](#) (p. 247).

softmax

concept. Entry id softmax.

DEFINITION

The function that exponentiates a list of numbers and divides by their sum, so that they are positive and add to one. Equation 0.23.

EQUATION

Book equation 0.23.

$$\text{softmax}(z)_i = \frac{e^{z_i}}{\sum_j e^{z_j}}, \quad \text{output} = \sum_i \text{softmax}\left(\frac{q \cdot k_i}{\sqrt{d}}\right)_i v_i.$$

Book equation 0.24.

$$\text{KL}(p \parallel q) = \sum_i p_i \ln \frac{p_i}{q_i} \geq 0.$$

LEDGER

- NEG-8. The Var-ratio tang_qproj is a ≥ 0.9 -Spearman rank proxy for softmax-KL under every consumer. [refuted]. `geometric-observation/claims/LEDGER.md:101` at 9f3829f.
- NEG-9. The projected-variance trace $\text{tr}(\mathbf{P}_C \cdot \Sigma_\delta)$ is a complete rank statistic for softmax-KL across all arms. [refuted]. `geometric-observation/claims/LEDGER.md:102` at 9f3829f.

FIRST STATED

Chapter 0 section 0.11 of *Data Mining as Observation*, with the program's softmax reader in `turboquant-pro/docs/KV_KEYS_FINDING.md:1-49`.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 1 section 1.4	cosine 0.995 and perplexity of order ten thousand, the recalibration negative	<code>geometric-observation/chapters/ch02_failure_of_observer_free_60</code> ; <code>geometric-observation/chapters/ch16_honest_negatives.md</code> NEG-2 and NEG-4; <code>turboquant-pro/docs/KV_KEYS_FINDING.md:1-49</code>
chapter 2 section 2.5	cosine 0.995 and the softmax reader	<code>turboquant-pro/docs/KV_KEYS_FINDING.md:1-49</code>
chapter 3 section 3.2	condition (A2), tangential displacement	<code>turboquant-pro/docs/KV_KEYS_FINDING.md:61-86</code> ; <code>geometric-observation/chapters/ch09_legibility.md:42-54</code>
chapter 8 section 8.2	cosine 0.995, perplexity near $1e4$	<code>turboquant-pro/docs/KV_KEYS_FINDING.md:1-49</code> ; <code>geometric-observation/chapters/ch16_honest_negatives.md</code> NEG-2
chapter 11 section 11.1	condition (A2), cosine satisfies it, post-rotary keys do not, the cone below cell size	<code>turboquant-pro/docs/KV_KEYS_FINDING.md:61-86</code> ; <code>the-angular-observer/README.md:26-31</code>
chapter 11 section 11.2	fp16 12.24, values-only 13.12, PolarQuant K4 10643 and 0.095, per-channel uniform K4 14.91 and 0.062, per-channel NUQ K3 15.77 and 0.148, 2.4x and 670x, pre-rotary near 22000, 2 key heads serve 12 query heads	<code>turboquant-pro/docs/KV_KEYS_FINDING.md:1-49</code>

FAILURES AND CORRECTIONS

- NEG-8, [refuted]. The Var-ratio tang_qproj is a ≥ 0.9 -Spearman rank proxy for softmax-KL under every consumer.
- NEG-9, [refuted]. The projected-variance trace $\text{tr}(\mathbf{P}_C \cdot \Sigma_\delta)$ is a complete rank statistic for softmax-KL across all arms.

CONDITIONS

- The softmax exponentiates a list of scores and divides by the sum, so the weights are positive and add to one. Adding the same constant to every score leaves it unchanged, and a larger score gets a larger weight.

- As a consumer it reads the scores through their differences, which is why the two proxies for softmax-KL in the ledger, the variance ratio and the projected-variance trace, were refuted.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/Softmax.lean`, `theorems` `denom_pos`, `softmax_pos`, `softmax_sum`, `softmax_shift`, `softmax_lt_iff`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 2, 3, 4, 8, 11.

RELATED

[attention](#) (p. 23); [KL divergence](#) (p. 187); [perplexity](#) (p. 250); [consumer](#) (p. 73).

sources table

instrument. Entry id `sources-table`.

DEFINITION

The table at the end of every chapter naming, for each number, the repository, file, and line range it came from. Chapter 0 section 0.21.

EQUATION

Book equation 8.1.

$$O_{\text{harness}} = (C_{\text{score}}, G_{\text{metric}}, B = \text{folds} \times \text{samples} \times \text{seeds}).$$

LEDGER

none

FIRST STATED

Chapter 0 section 0.21 of *Data Mining as Observation*, with the program's claims ledger that the same discipline enforces in `turboquant-pro/CLAIMS.md:1-27`.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 11 section 11.5	CI fails when <code>CLAIMS.md</code> and <code>claims.yaml</code> disagree	<code>turboquant-pro/CLAIMS.md:1-27</code> ; <code>turboquant-pro/tests/test_claims_ledger.py</code>

FAILURES AND CORRECTIONS

none

CONDITIONS

- The table at the end of every chapter naming, for each number, the repository, file, and line range it came from, so that no number in the book has to be taken on trust. A row that names a commit hash binds the number to one state of the file, since a changed digest proves a changed file.
- The table is a citation and not a proof. The book's checker verifies that every cited path exists, and the encyclopedia's checker verifies the numbers of its hand-filled entries against the generated ones, but the numbers themselves are the records' and the reader who wants them opens the file.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

lean/DataMiningAsObservation/Seal.lean, theorems changed_of_hash_ne, hash_eq_of_eq, exists_collision, at observation-data-mining af776fd.

USED IN

Data Mining as Observation chapters 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14.

RELATED

[commit hash](#) (p. 67); [sealed](#) (p. 316); [ledger class](#) (p. 204); [preregistration](#) (p. 258); [certificate](#) (p. 55).

Spearman correlation

concept. Entry id spearman-correlation.

DEFINITION

The ordinary correlation between two lists of ranks. Equation 0.14.

EQUATION

Book equation 0.14.

$$\rho_S = 1 - \frac{6 \sum_i d_i^2}{n(n^2 - 1)}, \quad d_i = \text{difference of the two ranks of item } i.$$

Book equation 11.2.

$$r = \frac{d_{\text{compressed}}}{d_{\text{exact}}}, \quad \kappa_{\text{strict}} = \frac{\max r}{\min r}, \quad \tau \geq 1 - 2\hat{\mu}(\kappa_{\text{strict}}), \quad \rho_S \geq 1 - 3\hat{\mu}(\kappa_{\text{strict}}),$$

$$\kappa_{97.5/2.5} = \frac{q_{97.5}(r)}{q_{2.5}(r)} \text{ gives the same two expressions as estimates, not floors.}$$

LEDGER

- GO-3. The certificate's vacuity threshold predicts where single-stage retrieval dies. [demonstrated]. [geometric-observation/claims/LEDGER.md:65 at 9f3829f](#).
- NEG-8. The Var-ratio tang_qproj is a ≥ 0.9 -Spearman rank proxy for softmax-KL under every consumer. [refuted]. [geometric-observation/claims/LEDGER.md:101 at 9f3829f](#).

FIRST STATED

Spearman, the proof and measurement of association between two things, 1904, as chapter 0 section 0.9 states it, and the rank statistic of the recognizer battery in the-angular-observer and of the rank certificate in readscope.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

- NEG-8, [refuted]. The Var-ratio tang_qproj is a ≥ 0.9 -Spearman rank proxy for softmax-KL under every consumer.

CONDITIONS

- The ordinary correlation between two lists of ranks, one minus six times the sum of squared rank differences over $n(n^2 - 1)$. It is at most one, one when the rankings agree, and minus one for three items in opposite orders, and since ranks depend on the ordering alone a strictly increasing transform of either score leaves it unchanged.
- A high Spearman between a proxy and a consumer's loss is not a certificate. The variance-ratio proxy with Spearman above 0.9 is a refuted ledger row, and the rank certificate's floor on Spearman is the strict setting's guarantee, not a measured correlation.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/Spearman.lean`, `theorems rho_le_one`, `rho_identical`, `rho_reversed_three`, `rank_comp`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 3, 4, 7, 9, 11.

RELATED

[Kendall correlation](#) (p. 185); [rank certificate](#) (p. 275); [recognizer](#) (p. 288); [Monotone Invariance Theorem](#) (p. 220).

spectral clustering

instrument. Entry id `spectral-clustering`.

DEFINITION

Clustering the rows of the row-normalized spectral embedding, which reads the angle and discards the density. Chapter 3 section 3.3 and chapter 9.

EQUATION

Book equation 0.19.

$$L = I - D^{-1/2} A D^{-1/2}, \quad D = \text{diag}(d_1, \dots, d_n), \quad L u_k = \lambda_k u_k, \quad 0 = \lambda_1 \leq \lambda_2 \leq \dots.$$

Book equation 0.20.

$$\Psi_i = \left(\frac{u_k(i)}{\sqrt{\lambda_k d_i}} \right)_{k \geq 2}, \quad \|\Psi_i - \Psi_j\|^2 = R(i, j) = \frac{C(i, j)}{\text{vol}(G)}.$$

Book equation 3.3.

$$X_i = r_i \theta_i, \quad r_i \rightarrow \frac{1}{\sqrt{d_i}}, \quad \theta_i = \frac{X_i}{\|X_i\|} \in S^{m-1}.$$

LEDGER

- GO-3. The certificate's vacuity threshold predicts where single-stage retrieval dies. [demonstrated]. `geometric-observation/claims/LEDGER.md:65 at 9f3829f`.
- NEG-1. Fixed-scale, uniform-in-m bi-Lipschitz for the commute filter. [refuted]. `geometric-observation/claims/LEDGER.md:94 at 9f3829f`.

FIRST STATED

Ng, Jordan, and Weiss, on spectral clustering, 2002, as chapter 3 section 3.3 and chapter 9 section 9.1 of *Data Mining as Observation* read it, with the angular reading in `the-angular-observer/theorem.md:110-146`.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 3 section 3.3	resistance converges to 1 over d_i plus 1 over d_j , simplex argument, radius converges to 1 over root d_i	<code>the-angular-observer\theorem.md:110-146</code>

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 9 section 9.1	the geodesic-rank reader discards the radius, row normalization is the angular projection	geometric-observation\chapters\ch09_legibility.md:10-41; the-angular-observer\theorem.md:110-146
chapter 9 section 9.2	the recognizer's mechanism, low multiplets and angular distances, dimension before shape by Weyl's law, refusal, the growth gotcha	geometric-observation\chapters\ch11_the_recognizer.md:1-95

FAILURES AND CORRECTIONS

- NEG-1, [refuted]. Fixed-scale, uniform-in-m bi-Lipschitz for the commute filter.

CONDITIONS

- Clustering the rows of the row-normalized spectral embedding, which reads the angle and discards the density. Row normalization puts every row on the unit sphere, keeps the cosine between rows, and ignores a positive rescaling, and the Laplacian's quadratic form is nonnegative, zero on constants, and positive across any edge.
- Ng, Jordan, and Weiss normalized because it improved results, and chapter 3 says why. The Euclidean distance in the embedding converges to degree noise and the angle keeps the geodesics.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/SpectralEmbedding.lean`, `theorems dot_self_nonneg`, `rowNormalize_unit`, `dot_rowNormalize`, `rowNormalize_smul`, at `observation-data-mining af776fd`.

`lean/DataMiningAsObservation/Laplacian.lean`, `theorems quad_eq`, `quad_nonneg`, `quad_const`, `quad_pos_of_edge`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 3, 9.

RELATED

[spectral embedding](#) (p. 332); [Laplacian](#) (p. 199); [k-means](#) (p. 189); [recognizer](#) (p. 288); [geodesic distance](#) (p. 154).

spectral embedding

instrument. Entry id `spectral-embedding`.

DEFINITION

The coordinates a node receives from its values in the first few nontrivial eigenvectors of the graph Laplacian. Clustering those coordinates is spectral clustering. Chapter 3.

EQUATION

Book equation 0.19.

$$L = I - D^{-1/2} A D^{-1/2}, \quad D = \text{diag}(d_1, \dots, d_n), \quad L u_k = \lambda_k u_k, \quad 0 = \lambda_1 \leq \lambda_2 \leq \dots.$$

Book equation 0.20.

$$\Psi_i = \left(\frac{u_k(i)}{\sqrt{\lambda_k d_i}} \right)_{k \geq 2}, \quad \|\Psi_i - \Psi_j\|^2 = R(i, j) = \frac{C(i, j)}{\text{vol}(G)}.$$

Book equation 3.3.

$$X_i = r_i \theta_i, \quad r_i \rightarrow \frac{1}{\sqrt{d_i}}, \quad \theta_i = \frac{X_i}{\|X_i\|} \in S^{m-1}.$$

LEDGER

- NEG-1. Fixed-scale, uniform-in-m bi-Lipschitz for the commute filter. [refuted]. `geometric-observation/claims/LEDGER.md:94 at 9f3829f`.

FIRST STATED

Ng, Jordan, and Weiss, on spectral clustering, 2002, as chapter 3 section 3.3 of *Data Mining as Observation* reads it, with the program's angular reading in `the-angular-observer/theorem.md:110-146`.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 3 section 3.3	resistance converges to 1 over d_i plus 1 over d_j, simplex argument, radius converges to 1 over root d_i	<code>the-angular-observer/theorem.md:110-146</code>
chapter 3 section 3.3	row normalization is the projection onto the read subspace of the geodesic-rank consumer	<code>geometric-observation/chapters/ch09_legibility.md:31-41;</code> <code>geometric-observation/chapters/ch03_historical_precursors.md:110</code>
chapter 9 section 9.1	the geodesic-rank reader discards the radius, row normalization is the angular projection	<code>geometric-observation/chapters/ch09_legibility.md:10-41;</code> <code>the-angular-observer/theorem.md:110-146</code>

FAILURES AND CORRECTIONS

- NEG-1, [refuted]. Fixed-scale, uniform-in-m bi-Lipschitz for the commute filter.

CONDITIONS

- The coordinates a node receives from its values in the first few nontrivial eigenvectors of the graph Laplacian. Row normalization puts every embedded row on the unit sphere, leaves the cosine between rows unchanged, and ignores a positive rescaling of a row, so it is the projection onto the read subspace of the geodesic-rank consumer.
- Its magnitude converges to degree noise and its angle keeps the geodesics, so the cosine between embedded rows is the right similarity and the Euclidean distance the wrong one. Clustering the normalized coordinates is spectral clustering, and a density reader of the same embedding reads the complementary coordinate.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/SpectralEmbedding.lean`, `theorems dot_self_nonneg`, `rowNormalize_unit`, `dot_rowNormalize`, `rowNormalize_smul`, at `observation-data-mining af776fd`.

lean/DataMiningAsObservation/Laplacian.lean, theorems quad_eq, quad_nonneg, quad_const, quad_pos_of_edge, at observation-data-mining af776fd.

USED IN

Data Mining as Observation chapters 0, 3, 9.

RELATED

[Laplacian](#) (p. 199); [commute time](#) (p. 68); [geodesic distance](#) (p. 154); [degree](#) (p. 97); [recognizer](#) (p. 288).

spectrum

concept. Entry id spectrum.

DEFINITION

The list of a matrix's eigenvalues. Chapter 0 section 0.4.

EQUATION

Book equation 0.5.

$$\Sigma v_i = \lambda_i v_i, \quad \Sigma = \sum_{i=1}^d \lambda_i v_i v_i^\top, \quad v_i \cdot v_j = 0 \ (i \neq j).$$

Book equation 0.7.

$$r_{\text{eff}} = \frac{(\sum_i \lambda_i)^2}{\sum_i \lambda_i^2}.$$

Book equation 9.2.

$$N(\lambda) = \#\{k : \lambda_k \leq \lambda\} \sim C_d \lambda^{d/2} \quad \Rightarrow \quad d = 2 \frac{d \log N}{d \log \lambda}.$$

LEDGER

- OT-7. The damage form, trace pairing, rank, and loading covariance are GL(d)-invariant under $P' = A^{-\top} P A^{-1}$, while spectrum, effective rank, principal angles, and water-filling are O(d)-only — consumer-weighted damage is a geometric scalar, and P3's cliff cannot be bought down by reparameterization. [demonstrated]. [geometric-observation/claims/LEDGER.md:30 at 9f3829f](#).
- GO-3. The certificate's vacuity threshold predicts where single-stage retrieval dies. [demonstrated]. [geometric-observation/claims/LEDGER.md:65 at 9f3829f](#).

FIRST STATED

Chapter 0 section 0.4 of *Data Mining as Observation*, with the program's spectrum records in `readscope`, `readscope/readscope/spectrum.py:35-70`, and the recognizer battery.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 4 section 4.2	effective rank as participation ratio, energy rank	<code>readscope\readscope\spectrum.py:35-70</code>

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 9 section 9.4	explained ratios, effective rank 5.19 of 8, convergence with a second method, property of the representation not the space	gtc- prototype\docs\SPECTRUM_FINDINGS.md:10-18
chapter 14 section 14.3	floor 0.08 to 0.12, balanced resample 3001 items, identity attack 0.237 with interval 0.20 to 0.28 on 289, sexual 0.204, threat 0.093 retracted, first pass 18 positives, 87.7 and 1.4 percent	gtc- prototype\docs\SPECTRUM_FINDINGS.md:20-64
chapter 14 section 14.3	contraction 0.779 to 0.863, 77 of 1600, 4.8 percent, 0.872 to 0.863, weight 2.69	gtc- prototype\docs\SPECTRUM_FINDINGS.md:66-99
chapter 14 section 14.5	the contraction formula fairness minus the general component	gtc- prototype\docs\SPECTRUM_FINDINGS.md:90-99
chapter 14 section 14.7	51 percent moderated at 80 and 95 percent precision on the balanced set	gtc- prototype\docs\SPECTRUM_FINDINGS.md:84-88

FAILURES AND CORRECTIONS

none

CONDITIONS

- The list of a matrix's eigenvalues. Its effective rank, the square of its sum over the sum of its squares, lies between one and the dimension, is the dimension for a flat spectrum, and is one for a single eigenvalue.
- The spectrum is not invariant under a change of basis of the input where the trace pairing and the rank are, and the Laplacian's spectrum is what the recognizer reads for dimension and shape.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/EffectiveRank.lean`, `theorems sq_sum_le`, `effRank_le`, `sum_sq_le_sq_sum`, `one_le_effRank`, `effRank_const`, `effRank_single`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 3, 4, 9, 11, 14.

RELATED

[eigenvalue](#), [eigenvector](#) (p. 116); [effective rank](#) (p. 115); [recognizer](#) (p. 288); [Laplacian](#) (p. 199); [isotropic](#) (p. 179).

split

instrument. Entry id `split`.

DEFINITION

A partition of the rows into a part the model is fit on and a part it is scored on. A scorer that has read the test part scores it perfectly. Chapter 8 section 8.1.

EQUATION

Book equation 8.1.

$$O_{\text{harness}} = (C_{\text{score}}, G_{\text{metric}}, B = \text{folds} \times \text{samples} \times \text{seeds}).$$

Book equation 8.2.

$$\widehat{\text{Var}}_{\text{NB}} = \left(\frac{1}{J} + \frac{n_{\text{test}}}{n_{\text{train}}} \right) \hat{\sigma}^2,$$

$$J = 1000, \frac{n_{\text{test}}}{n_{\text{train}}} = \frac{1}{4} \Rightarrow 1 + 250 = 251, \sqrt{251} = 15.84.$$

Book equation 12.3.

$$\text{validated} \iff \text{AUROC}_{\text{cross}} - \max(\text{AUROC}_{\text{untrained}}, \text{AUROC}_{\text{BoW}}) \geq 0.10.$$

LEDGER

- NEG-4. Lightweight online (Lloyd) key calibration beats the calibration-free default on softmax-KL. [refuted]. [geometric-observation/claims/LEDGER.md:97 at 9f3829f](#).
- GO-B-legal (035→036). Legal-citation retrieval (CourtListener), cosine-ranking consumer, LaBSE embeddings — real large corpus, non-physical consumer [predicted]. [geometric-observation/claims/LEDGER.md:119 at 9f3829f](#).

FIRST STATED

Chapter 8 section 8.1 of *Data Mining as Observation*, with the virgin split in [geometric-observation/chapters/ch10_the_blind_probe.md:118](#).

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 4 section 4.7	ESL 7.10.2 numbers	Hastie, Tibshirani, Friedman, ESL 2e, section 7.10.2; chapter 8 of this book
chapter 6 section 6.4	the uncorrected t stored as sigma, 251 and 15.84, nine wins to three, seven, eight, 3 wins 17 ties 11 losses on 31 datasets, the three-way inconsistency	constraint-gap\review\FINDINGS.md:1-35 ; constraint-gap\README.md:57-76
chapter 8 section 8.1	ESL 7.10.2 numbers (N=50, p=5000, 100 selected, 3 percent vs 50 percent)	Hastie, Tibshirani, Friedman, ESL 2e, section 7.10.2
chapter 8 section 8.5	variance inflation 251, SE inflation 15.84, J_eff 3.98, about 62 needed, 3 of 9 at full, 7 at half, 8 at a third	constraint-gap\review\FINDINGS.md:1-35
chapter 12 section 12.3	041 non-oracle, frozen LSA TF-IDF to SVD 100 train-only, AUROC 0.975 vs 0.910, flip tied, magnitude overshoot, partial	geometric-observation\chapters\ch10_the_blind_probe.md:100-118 ; geometric-observation\claims\LEDGER.md row GO-B-blind 041

FAILURES AND CORRECTIONS

- NEG-4, [refuted]. Lightweight online (Lloyd) key calibration beats the calibration-free default on softmax-KL.

CONDITIONS

- A partition of the rows into a part the model is fit on and a part it is scored on. The parts' sizes add to the row count, a row in the test part is not in the training part, and a scorer that has read the test part scores it perfectly, which is leakage.
- The virgin split of the non-oracle test raised the flip's margin to 0.796 against 0.780, and the split is fixed before any transform is fit, since an imputation fit on all rows reads the test part.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/CrossValidation.lean`, `theorems sizes_sum`, `accuracy_weighted`, `accuracy_mean_of_equal`, at `observation-data-mining af776fd`.

`lean/DataMiningAsObservation/Leakage.lean`, `theorems errors_lookup_eq_zero`, `lookup_default`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14.

RELATED

[cross-validation](#) (p. 89); [leakage](#) (p. 203); [harness](#) (p. 159); [seed](#) (p. 318); [stratification](#) (p. 339).

standard error

concept. Entry id `standard-error`.

DEFINITION

The spread of an estimate across repeated samples. Chapter 0 section 0.9.

EQUATION

Book equation 0.18.

$$\widehat{\text{Var}}_{\text{NB}} = \left(\frac{1}{J} + \frac{n_{\text{test}}}{n_{\text{train}}} \right) \hat{\sigma}^2, \quad \frac{\widehat{\text{Var}}_{\text{NB}}}{\hat{\sigma}^2/J} = 1 + J \frac{n_{\text{test}}}{n_{\text{train}}}.$$

Book equation 8.2.

$$\widehat{\text{Var}}_{\text{NB}} = \left(\frac{1}{J} + \frac{n_{\text{test}}}{n_{\text{train}}} \right) \hat{\sigma}^2,$$

$$J = 1000, \frac{n_{\text{test}}}{n_{\text{train}}} = \frac{1}{4} \Rightarrow 1 + 250 = 251, \sqrt{251} = 15.84.$$

LEDGER

none

FIRST STATED

Chapter 0 section 0.9 of *Data Mining as Observation*, with the program's own inflation in `constraint-gap/review/FINDINGS.md:1-35`.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 6 section 6.4	the uncorrected t stored as sigma, 251 and 15.84, nine wins to three, seven, eight, 3 wins 17 ties 11 losses on 31 datasets, the three-way inconsistency	constraint-gap\review\FINDINGS.md:1-35; constraint-gap\README.md:57-76
chapter 8 section 8.5	variance inflation 251, SE inflation 15.84, J_eff 3.98, about 62 needed, 3 of 9 at full, 7 at half, 8 at a third	constraint-gap\review\FINDINGS.md:1-35

FAILURES AND CORRECTIONS

none

CONDITIONS

- The spread of an estimate across repeated samples, the spread of one draw over the square root of the number of independent draws. It is positive, falls as the draws grow, halves only when the draws quadruple, and tends to zero.
- Dependent draws do not shrink it this way. A thousand repeated folds shrink the naive standard error by a factor near 31.6 and the corrected one by 15.84 less, which is the Nadeau and Bengio correction.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/StandardError.lean`, theorems `se_pos`, `se_quarter`, `se_antitone`, `se_tendsto_zero`, `thousand_folds`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 6, 7, 8, 13, 14.

RELATED

[Nadeau and Bengio correction](#) (p. 225); [harness](#) (p. 159); [multiple comparisons](#) (p. 221); [preregistration](#) (p. 258).

standardization

instrument. Entry id standardization.

DEFINITION

Subtracting a column's mean and dividing by its spread, so that it has mean zero and variance one. It is invertible and preserves every ordering, so it changes the reader and not the data. Chapter 2.

EQUATION

Book equation 2.1.

$$P_{C_2 \circ C_1}(x) = J_1(x)^\top P_{C_2}(C_1(x)) J_1(x), \quad \text{rank } P_{C_2 \circ C_1}(x) \leq \text{rank } P_{C_2}(C_1(x)) \quad \text{at each row } x.$$

Book equation 0.6.

$$x_w = \Sigma^{-1/2}(x - \mu), \quad \Sigma^{-1/2} = \sum_i \lambda_i^{-1/2} v_i v_i^\top.$$

Book equation 3.1.

$$d_O(u) = u^\top \Sigma u, \quad u = (\cos 15^\circ, \sin 15^\circ),$$

$$\Sigma_1 = \text{diag}(0.3, 1.7), \Sigma_2 = \text{diag}(1.7, 0.3), \quad d_O = 0.394 \text{ vs } 1.606.$$

LEDGER

none

FIRST STATED

Chapter 2 section 2.5 of *Data Mining as Observation*, with the whitened code's record in `geometric-observation/chapters/ch08_value.md:84-97`.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 2 section 2.1	the four attribute types and permitted transformations	TSK 2e section 2.1
chapter 2 section 2.5	whitened code wins at every budget	<code>geometric-observation\chapters\ch08_value.md:84-97</code>
chapter 4 section 4.3	whitened code on whale 0.83, 0.85, 0.97 vs 0.41, 0.80, 0.89	<code>geometric-observation\chapters\ch08_value.md:84-97</code>

FAILURES AND CORRECTIONS

none

CONDITIONS

- Subtracting a column's mean and dividing by its spread. The standardized column has weighted mean zero and weighted variance one, the transform is invertible, and it preserves every ordering of the rows, so it changes the reader's geometry and not what the table holds.
- An invertible transform identifies no two rows and forms no quotient. For a consumer that reads distances across columns the declaration that scale is irrelevant is almost always right, and for a consumer that reads one column it does nothing.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/Standardization.lean`, theorems `mean_zero`, `variance_one`, `standardize_inv`, `standardize_lt_iff`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 1, 2, 3, 4, 6, 9, 10, 11.

RELATED

[whitening](#) (p. 368); [imputation](#) (p. 174); [Euclidean distance](#) (p. 124); [quotient](#) (p. 271); [outlier](#) (p. 242).

stratification

instrument. Entry `id` stratification.

DEFINITION

Splitting rows into strata, by difficulty or by group, and scoring each separately, so that the report can take the minimum over strata. Chapter 10 and chapter 12.

EQUATION

Book equation 10.7.

$$\text{verdict} = \min_{i: n_i \geq n_{\min}, |Q_i| \geq q_{\min}} \text{score}_i, \quad \text{ABSTAIN otherwise.}$$

Book equation 14.5.

$$\text{FC}_g = \Pr[y = \text{violation} \mid \hat{y} = \text{clear}, g], \quad \text{verdict} = \max_{g: n_g \geq n_{\min}} \text{FC}_g, \quad \text{ABSTAIN otherwise.}$$

Book equation 8.2.

$$\widehat{\text{Var}}_{\text{NB}} = \left(\frac{1}{J} + \frac{n_{\text{test}}}{n_{\text{train}}} \right) \hat{\sigma}^2,$$

$$J = 1000, \frac{n_{\text{test}}}{n_{\text{train}}} = \frac{1}{4} \Rightarrow 1 + 250 = 251, \sqrt{251} = 15.84.$$

LEDGER

- NEG-14. (GO-P-2026-037, prospective) a2_probe.median_unit_displacement is a single-statistic predictor of the flip regime (unit_disp $\geq 1.0 \Rightarrow$ generic-polar-flip vs needs-blind-probe). [refuted]. geometric-observation/claims/LEDGER.md:108 at 9f3829f.
- GO-B-Llama. Trained frontier LLM (Llama-3.2-3B), softmax-attention consumer — blind probe on real post-RoPE keys [predicted]. geometric-observation/claims/LEDGER.md:115 at 9f3829f.
- GO-B-Llama-rematch. Trained frontier LLM (Llama-3.2-3B), softmax-attention consumer — reon-matched dissociation on real post-RoPE keys [predicted]. geometric-observation/claims/LEDGER.md:118 at 9f3829f.

FIRST STATED

Chapter 10 section 10.4 of *Data Mining as Observation*, with the strata design in turboquant-pro/docs/STRATA_RFC.md:24-98.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 10 section 10.4	area map, boundary rule, hash, refuse not warn, intra and transit counts, area classes, abstention rule	turboquant-pro\docs\STRATA_RFC.md:24-98
chapter 10 section 10.4	first run, 350000 of 2391361 rows, 7 of 14 eligible, abstentions with 2 and 1 rows, Robin Hood 0.3447 to 0.4469 ratio 1.30 against 1.5, skew 2.68 to 4.40 ratio 1.64 against 3, 63 times sample range	turboquant-pro\docs\RESULTS_multilingual_strata.md:1-55
chapter 10 section 10.4	second prediction inverted, transit 0.389 vs 0.291, centrality difference signs, share 0.473 vs 0.391, seven of thirteen backbone	turboquant-pro\docs\RESULTS_multilingual_strata.md:55-90; turboquant-pro\docs\STRATA_RFC.md:98-130

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 10 section 10.5	Gate A design error, A prime skew 3.970 to 3.177, max 287 to 213, Robin Hood 0.372 to 0.261, fraction 0.117 to 0.079, compressed path 0.663 vs 0.90, seven strata, 0.62 to 0.69 vs 0.76 to 0.84	turboquant-pro\docs\RESULTS_strata_phase23_gates.md:1-45
chapter 10 section 10.5	rerank identical 0.6627, candidate coverage, depth curve 51 to 501, coverage 0.696 to 0.926, fidelity 0.663 to 0.702, originals rerank 0.923, 4096 bytes on 148	turboquant-pro\docs\RESULTS_strata_phase23_gates.md:45-110
chapter 10 section 10.5	fragile-first allocation 0.7118 vs 0.7251, targets up 0.004 to 0.034, redesigned allocator passed	turboquant-pro\docs\RESULTS_strata_phase23_gates.md:110-125
chapter 12 section 12.4	seven of thirteen backbone areas under the translation-trained encoder, all local-hub under the general one	turboquant-pro\docs\RESULTS_multilingual_strata.md:55-90

FAILURES AND CORRECTIONS

- NEG-14, [refuted]. (GO-P-2026-037, prospective) `a2_probe.median_unit_displacement` is a single-statistic predictor of the flip regime ($\text{unit_disp} \geq 1.0 \Rightarrow \text{generic-polar-flip vs needs-blind-probe}$).

CONDITIONS

- Splitting rows into strata, by difficulty or by group, and scoring each separately, so that the report can take the minimum over strata and count the strata too thin to score. A weighted aggregate over strata is at most the pass bar when one stratum fails by enough, and the folds of a stratified split test every row once.
- The first strata run scored seven of fourteen eligible strata and abstained on those with two rows and one, and the second prediction inverted, which the record keeps.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/Certificate.lean`, `theorems falseClear_mul_coverage, coverage_empty, falseClear_mem_unit, minOverStrata_passes_iff, minOverStrata_le_weighted_mean, at observation-data-mining af776fd`.

`lean/DataMiningAsObservation/CrossValidation.lean`, `theorems sizes_sum, accuracy_weighted, accuracy_mean_of_equal, at observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 5, 8, 10, 11, 12.

RELATED

[min-over-strata](#) (p. 217); [abstention](#) (p. 9); [anti-hub recall](#) (p. 19); [cross-validation](#) (p. 89); [harness](#) (p. 159).

sum of squared errors

concept. Entry id `sum-of-squared-errors`.

DEFINITION

The sum over rows of the squared distance to the row's cluster centre, the validity index k-means minimizes. Chapter 0 section 0.15.

EQUATION

Book equation 9.1.

$$\text{SSE} = \sum_{k=1}^K \sum_{i \in \mathcal{C}_k} \|x_i - c_k\|^2, \quad \text{the } P_C = I \text{ distortion summed within clusters.}$$

Book equation 0.30.

$$\text{SSE} = \sum_k \sum_{i \in \mathcal{C}_k} \|x_i - c_k\|^2, \quad s_i = \frac{b_i - a_i}{\max(a_i, b_i)}.$$

Book equation 4.2.

$$D(b) = \sum_i s_i \sigma_i^2 2^{-2b_i}, \quad s_i = v_i^\top P_C v_i, \quad b_i^* = \max\left(0, \frac{1}{2} \log_2 \frac{s_i \sigma_i^2}{\theta}\right), \quad \sum_i b_i^* = B,$$

LEDGER

- GO-3. The certificate's vacuity threshold predicts where single-stage retrieval dies. [demonstrated].
geometric-observation/claims/LEDGER.md:65 at 9f3829f.

FIRST STATED

Chapter 0 section 0.15 and chapter 9 section 9.1 of *Data Mining as Observation*.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 9 section 9.4	explained ratios, effective rank 5.19 of 8, convergence with a second method, property of the representation not the space	gtc- prototype/docs/SPECTRUM_FINDINGS.md:10-18
chapter 10 section 10.1	the four detector families	TSK 2e chapter 9

FAILURES AND CORRECTIONS

none

CONDITIONS

- The sum over rows of the squared distance to the row's cluster centre, the validity index k-means minimizes. The error about any centre is the error about the mean plus the count times the squared distance between the two, so the mean minimizes it, and it is the identity reader's distortion summed within clusters.
- Two clusterings of the same rows with the same sum of squared errors can be read oppositely by a consumer that weighs directions unequally.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

lean/DataMiningAsObservation/KMeans.lean, theorems sse_decomposition, mean_minimizes, assign_nearest, at observation-data-mining af776fd.

lean/DataMiningAsObservation/ReconstructionError.lean, theorems recon_nonneg, recon_eq_zero_iff, recon_sum, recon_proj, at observation-data-mining af776fd.

USED IN

Data Mining as Observation chapters 0, 9.

RELATED

[k-means](#) (p. 189); [validity index](#) (p. 360); [reconstruction error](#) (p. 290); [identity reader](#) (p. 171); [silhouette](#) (p. 324).

support

concept. Entry id support.

DEFINITION

Of an itemset, the fraction of transactions that contain it. Chapter 5.

EQUATION

Book equation 5.1.

$$s(X) = \frac{\sigma(X)}{N}, \quad c(X \rightarrow Y) = \frac{\sigma(X \cup Y)}{\sigma(X)} = \frac{s(X \cup Y)}{s(X)}.$$

Book equation 5.3.

$$X \subseteq Y \implies s(X) \geq s(Y).$$

LEDGER

none

FIRST STATED

TSK chapter 5, as chapter 5 section 5.1 of *Data Mining as Observation* states it, with anti-monotonicity as the Apriori license.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 5 section 5.1 to 5.3, 5.5	support, confidence, lift, the Apriori principle, FP-growth, closed and maximal itemsets, objective measures, Simpson's paradox, cross-support	TSK 2e chapter 5

FAILURES AND CORRECTIONS

none

CONDITIONS

- The fraction of transactions that contain an itemset. Support cannot increase when an itemset grows, so an infrequent itemset has only infrequent supersets and every subset of a frequent itemset is frequent, which is the license to prune the lattice.
- The count of candidates rather than the count of transactions decides whether a run finishes, and a rare consequent lets lift reach the transaction count on a single co-occurrence.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/SafePruning.lean`, theorems `support_anti`, `apriori`, `subset_of_frequent`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 2, 5, 6, 7, 8.

RELATED

[itemset](#) (p. 180); [Apriori principle](#) (p. 22); [lift](#) (p. 205); [cross-support ratio](#) (p. 88).

surrogate

instrument. Entry id surrogate.

DEFINITION

A quadratic stand-in for the consumer's output metric, the read distortion, used to allocate bits when the output itself cannot be. Chapter 4 section 4.2.

EQUATION

Book equation 4.2.

$$D(b) = \sum_i s_i \sigma_i^2 2^{-2b_i}, \quad s_i = v_i^\top P_C v_i, \quad b_i^* = \max\left(0, \frac{1}{2} \log_2 \frac{s_i \sigma_i^2}{\theta}\right), \quad \sum_i b_i^* = B,$$

Book equation 4.6.

$$\kappa = \text{tr}(\bar{P}_C \bar{\Sigma}_x) \in [0, 1], \quad \kappa \rightarrow 1 \text{ is the coupling null (no flip).}$$

Book equation 0.11.

$$P_C(x) = J(x)^\top G(C(x)) J(x), \quad J(x) = \frac{\partial C}{\partial x}(x), \quad \bar{P}_{C,\mu} = \mathbb{E}_\mu[P_C(x)].$$

LEDGER

- GO-6. At matched rate, output coding $\leq$ surrogate $\leq$ reconstruction on the consumer metric; the output-reconstruction gap is governed by the $\ker P_C$ entropy share, and the surrogate-output gap vanishes as rate grows. [demonstrated]. [geometric-observation/claims/LEDGER.md:68 at 9f3829f](#).
- NEG-9. The projected-variance trace $\text{tr}(P_C \Sigma_\delta)$ is a complete rank statistic for softmax-KL across all arms. [refuted]. [geometric-observation/claims/LEDGER.md:102 at 9f3829f](#).

FIRST STATED

Chapter 4 section 4.2 of *Data Mining as Observation*, with the surrogate ordering in [geometric-observation/claims/LEDGER.md](#) row GO-6.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 4 section 4.2	water-filling formula, directions below the water get no bits, the surrogate caveat	readscope/readscope/allocate.py:1-100
chapter 4 section 4.2	GO-6, d 8 and r 4, output at or below surrogate at or below reconstruction at every rate, about 500 times, gap 0.41 to 0.005, isotropic control collapses	geometric-observation/claims/LEDGER.md row GO-6; geometric-observation/chapters/ch07_cost.md

FAILURES AND CORRECTIONS

- NEG-9, [refuted]. The projected-variance trace $\text{tr}(P_C \Sigma_\delta)$ is a complete rank statistic for softmax-KL across all arms.

CONDITIONS

- A quadratic stand-in for the consumer's output metric, the read distortion, used to allocate bits when the output itself cannot be. The read distortion is a quadratic form in the read operator, and two readers at different angles rank the same two errors differently.
- At every rate the output coder is at or below the surrogate, which is at or below reconstruction, and the surrogate-to-output gap shrinks from 0.41 to 0.005 as the rate grows, while the read distortion is a control and not a complete rank statistic.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/ReadDistortion.lean`, `theorems read_distortion`, `identity_reader`, `quad_one`, at `observation-data-mining af776fd`.

`lean/DataMiningAsObservation/Isotropy.lean`, `theorems isotropic_reads_same`, `isotropic_no_flip`, `anisotropic_readers_differ`, `flip_iff_anisotropic`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 4, 8.

RELATED

[read distortion](#) (p. 281); [distortion](#) (p. 109); [output metric](#) (p. 243); [identity reader](#) (p. 171).

sweep

instrument. Entry id sweep.

DEFINITION

Running a measurement across a range of one parameter and reading the curve. Chapter 5 section 5.4 and chapter 9 section 9.3.

EQUATION

Book equation 6.2.

$$\max_{\tau, \text{dir}} F_1(\mathbf{1}[\text{dir}(g(f(X)), \tau)], y) = \max_{\tau, \text{dir}} F_1(\mathbf{1}[\text{dir}(f(X), \tau)], y)$$

for every strictly monotone g .

Book equation 0.28.

$$P = \frac{TP}{TP + FP}, \quad R = \frac{TP}{TP + FN}, \quad F_1 = \frac{2PR}{P + R}.$$

Book equation 9.3.

$$\hat{\mu} = \frac{\bar{s}_{\text{true}} - \bar{s}_{\text{distr}}}{\sigma_{\text{distr}}}, \quad \mu_{\text{crit}} = \mathbb{E}\left[\max_{N-1} \mathcal{N}(0, 1)\right], \quad \rho = \frac{\hat{\mu}}{\mu_{\text{crit}}}, \quad \rho = 1 \text{ vacuous.}$$

LEDGER

- G0-3. The certificate's vacuity threshold predicts where single-stage retrieval dies. [demonstrated]. `geometric-observation/claims/LEDGER.md:65 at 9f3829f`.
- G0-4. Fixed-budget verdicts invert under budget-matched observation, per the wavelength mechanism. [replicated]. `geometric-observation/claims/LEDGER.md:66 at 9f3829f`.

FIRST STATED

Chapter 5 section 5.4 and chapter 9 section 9.3 of *Data Mining as Observation*, with the degradation sweep in `openvector-bench/README.md:60-96`.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 5 section 5.4	seven thresholds up to 0.75, eight datasets, 56 conditions, zero admissibility violations, reductions 70 to 99.1 percent	theory-radar\paper\astar_paper.tex:170-200
chapter 11 section 11.4	0.999 against own ranking vs 0.592 against fp32 truth, three truth layers, difficulty strata	openvector-bench\README.md:60-96
chapter 11 section 11.7	C-15 equal budget, 3072 observations, medians 0.02, 0.11, 0.05, 0.04, 1.0, 1.0, margin 0.883 vs 0.3, 0.062 to 0.057 at 8x	readscope\SPEC.md:285-323, record readscope\calibration\records\c15-budget-surface.json, readscope\calibration\DECLARATION-C15.md
chapter 12 section 12.2	recall 0.999 vs 0.592, the derived death point within 6 percent across fourteen corpora	openvector-bench\README.md:60-96; geometric-observation\claims\LEDGER.md ROW GO-3

FAILURES AND CORRECTIONS

none

CONDITIONS

- Running a measurement across a range of one parameter, a threshold, a budget, or a degradation, and reading the curve. As a threshold rises the set called positive can only shrink, so true and false positives fall together and the ROC curve is traced.
- Fourteen corpora were swept through degradation until retrieval died, the budget sweep found the cliff at `k` equal to `d`, and the threshold sweep found the optimal F1 by sort.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/Threshold.lean`, `theorems predicted_anti`, `tp_anti`, `fp_anti`, `decision_comp`, `predicted_extremes`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 3, 4, 5, 6, 7, 8, 9, 10, 13.

RELATED

[threshold](#) (p. 350); [ROC curve](#) (p. 306); [budget cliff](#) (p. 49); [sensitivity](#) (p. 319); [harness](#) (p. 159).

T

teacher forcing

concept. Entry id teacher-forcing.

DEFINITION

Feeding a language model the correct text as context rather than its own earlier outputs. Chapter 0 section 0.11.

EQUATION

Book equation 0.22.

$$\text{PPL} = 2^H, \quad H = -\frac{1}{T} \sum_{t=1}^T \log_2 p(w_t \mid w_{<t}).$$

Book equation 11.1.

$$\cos(k, \hat{k}) = 0.995 \quad \text{while} \quad \text{PPL} : 12.24 \rightarrow 10643.$$

LEDGER

- OT-4. Operator drift predicts a real long-generation degradation and a derived refresh intervention moves it. [refuted]. [geometric-observation/claims/LEDGER.md:36 at 9f3829f](#).

FIRST STATED

Chapter 0 section 0.11 of *Data Mining as Observation*, with the program's C-12 record in [readscope](#), [readscope/CALIBRATION.md:600-660](#).

MEASUREMENTS

none

FAILURES AND CORRECTIONS

- OT-4, [refuted]. Operator drift predicts a real long-generation degradation and a derived refresh intervention moves it.

CONDITIONS

- Feeding a model the correct text as context rather than its own earlier outputs. The forced and free runs agree exactly as long as the forced predictions match the reference, and they part at the first token the model gets wrong, after which the free run's context is no longer the reference's.
- Perplexity is measured under teacher forcing, so a good perplexity does not certify a free run. The C-12 record's thirteen-point difference vanished under teacher forcing, and the operator-drift claim it was meant to support is refuted in the ledger.

Conditions are curated in [entries.toml](#) rather than read from a record.

MACHINE CHECKED

[lean/DataMiningAsObservation/TeacherForcing.lean](#), [theorems free_eq_take](#), [free_length](#), [free_ne_of_error](#), at [observation-data-mining af776fd](#).

USED IN

Data Mining as Observation chapters 0, 11.

RELATED

[perplexity](#) (p. 250); [drift](#) (p. 112); [harness](#) (p. 159); [deployment mismatch](#) (p. 101).

template match

instrument. Entry id template-match.

DEFINITION

Comparing a measured list of numbers, such as eigenvalue ratios, to a stored list for each candidate shape and picking the smallest distance between the logarithms. Equation 0.32.

EQUATION

Book equation 0.32.

$$\text{dist}(m, t) = \sqrt{\frac{1}{8} \sum_{k=1}^8 (\ln m_k - \ln t_k)^2}.$$

Book equation 9.3.

$$\hat{\mu} = \frac{\bar{s}_{\text{true}} - \bar{s}_{\text{distr}}}{\sigma_{\text{distr}}}, \quad \mu_{\text{crit}} = \mathbb{E} \left[\max_{N-1} \mathcal{N}(0, 1) \right], \quad \rho = \frac{\hat{\mu}}{\mu_{\text{crit}}}, \quad \rho = 1 \text{ vacuous}.$$

LEDGER

- GO-3. The certificate’s vacuity threshold predicts where single-stage retrieval dies. [demonstrated].
geometric-observation/claims/LEDGER.md:65 at 9f3829f.

FIRST STATED

Volume 14, chapter 11, geometric-observation/chapters/ch11_the_recognizer.md:1-95, DOI 10.5281/zenodo.21776291, and the recognizer battery in the-angular-observer.

MEASUREMENTS

Where the book states it	Numbers, as the book’s sources table records them	Source
chapter 3 section 3.3	resistance converges to 1 over d_i plus 1 over d_j, simplex argument, radius converges to 1 over root d_i	the-angular-observer\theorem.md:110-146
chapter 3 section 3.3	radius against degree correlation 0.92 to 0.99	the-angular-observer\README.md:135-139
chapter 3 section 3.3	angle 0.92 flat in m, magnitude 0.03, full 0.75 decaying	wolfram-observer-bridge\theorem.md:1-20; the-angular-observer\theorem.md:187-224
chapter 3 section 3.3	substrate table	the-angular-observer\README.md:99-109
chapter 3 section 3.3	proven versus conjectured ledger, d at least 3, d equals 2 borderline, fixed-m conditional theorem, heat filter uniform, commute metric form false, Green-kernel rank limit for d at most 3, obstruction at d equals 4	the-angular-observer\theorem.md:84-107, 147-224; the-angular-observer\README.md:150-185
chapter 3 section 3.3	the self-refuted v0.8 claim, NEG-1	the-angular-observer\README.md:170-175; geometric-observation\chapters\ch16_honest_negatives.md NEG-1

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 3 section 3.3	hyperbolic rejected, curvature negative 0.98 to negative 0.14, trend 1.09 minus 0.157 d across 20 manifolds and 5 families	the-angular-observer\README.md:111-147,183-185
chapter 9 section 9.1	the geodesic-rank reader discards the radius, row normalization is the angular projection	geometric-observation\chapters\ch09_legibility.md:10-41; the-angular-observer\theorem.md:110-146
chapter 9 section 9.2	the battery, three new templates, frozen ratios, code hash, bars at least 10 of 12 and the dimension ordering, eccentricity scope	the-angular-observer\PREREG_RECOGNIZER_BATTERY.md:1-80
chapter 9 section 9.2	12 of 12, dimensions 1.81, 2.80, 1.74, angular Spearman ranges, eccentricity spreads, verdict confirmed, GO-P-2026-041	the-angular-observer\experiments\manifold-recovery\battery_result.json
chapter 9 section 9.2	causal set 0.57 with dimension 2.9 to 10.7, small world 0.45, control 0.01, rung 4 decay 0.747 to 0.679 below 0.75	the-angular-observer\README.md:99-109; wolfram-observer-bridge\rung4_scaledm_result.json; wolfram-observer-bridge\review-2.txt
chapter 11 section 11.1	condition (A2), cosine satisfies it, post-rotary keys do not, the cone below cell size	turboquant-pro\docs\KV_KEYS_FINDING.md:61-86; the-angular-observer\README.md:26-31

FAILURES AND CORRECTIONS

none

CONDITIONS

- Comparing a measured list of numbers, the eigenvalue ratios, to a stored list for each candidate shape and picking the smallest root-mean-square distance between the logarithms. The distance is zero between a list and itself, symmetric, and unchanged when both lists are scaled by the same factor, so the match reads shape and not size.
- The templates are frozen before the battery is run, with their code hash recorded, and the match can only name shapes in its template set, which is why the recognizer also certifies that none is present.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/Recognizer.lean`, `theorems templateDist_self`, `templateDist_comm`, `templateDist_scale`, `templateDist_nonneg`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 2, 8, 9, 14.

RELATED

[recognizer](#) (p. 288); [vacuity threshold](#) (p. 359); [intrinsic dimension](#) (p. 175); [Laplacian](#) (p. 199).

TF-IDF

concept. Entry id `tf-idf`.

DEFINITION

A weighting of term counts by how rare the term is across the collection, so that a term in every document carries no weight. Equation 0.36.

EQUATION

Book equation 0.36.

$$w_{t,d} = \text{tf}_{t,d} \cdot \ln \frac{N}{\text{df}_t}, \quad \text{tf}_{t,d} = \frac{\text{count of } t \text{ in } d}{\text{length of } d}, \quad \text{df}_t = \text{documents containing } t.$$

Book equation 12.3.

$$\text{validated} \iff \text{AUROC}_{\text{cross}} - \max(\text{AUROC}_{\text{untrained}}, \text{AUROC}_{\text{BoW}}) \geq 0.10.$$

LEDGER

- GO-B-legal (035→036). Legal-citation retrieval (CourtListener), cosine-ranking consumer, LaBSE embeddings — real large corpus, non-physical consumer [predicted]. `geometric-observation/claims/LEDGER.md:119` at 9f3829f.

FIRST STATED

Spärck Jones, a statistical interpretation of term specificity, 1972, as chapter 0 section 0.17 states it, with the program's frozen LSA baseline in ledger row GO-B-legal.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

CONDITIONS

- A term's weight is its frequency in the document times the logarithm of the document count over the number containing it. A term in every document carries no weight, the weight is nonnegative and falls as the term spreads, and the frequency lies in the unit interval.
- It is a reader that reads rarity. The legal-citation flip used a frozen TF-IDF to SVD baseline trained on the training split only, and the flip tied while its magnitude overshot, which the ledger carries as partial.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/TFIDF.lean`, `theorems weight_everywhere`, `weight_nonneg`, `weight_antitone`, `tf_mem_unit`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 12.

RELATED

[bag of words](#) (p. 32); [retrieval-augmented pipeline](#) (p. 303); [cross-corpus gate](#) (p. 87); [quotient](#) (p. 271).

threshold

concept. Entry id `threshold`.

DEFINITION

The value at which a score becomes a positive decision. Chapter 6.

EQUATION

Book equation 6.2.

$$\max_{\tau, \text{dir}} F_1(\mathbf{1}[\text{dir}(g(f(X)), \tau)], y) = \max_{\tau, \text{dir}} F_1(\mathbf{1}[\text{dir}(f(X), \tau)], y)$$

for every strictly monotone g .

Book equation 0.28.

$$P = \frac{TP}{TP + FP}, \quad R = \frac{TP}{TP + FN}, \quad F_1 = \frac{2PR}{P + R}.$$

LEDGER

none

FIRST STATED

Chapter 6 section 6.1 of *Data Mining as Observation*, with the program's threshold sweeps in theory-radar.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

CONDITIONS

- The value at which a score becomes a positive decision. Raising it can only shrink the predicted set, so true positives and false positives can only fall, which is why the ROC curve is traced by a single sweep, and below every score everything is positive while above every score nothing is.
- A strictly increasing recalibration of the score with the matching recalibration of the threshold leaves every decision unchanged, which is the Monotone Invariance Theorem at the level of one decision.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/Threshold.lean`, theorems `predicted_anti`, `tp_anti`, `fp_anti`, `decision_comp`, `predicted_extremes`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14.

RELATED

[Monotone Invariance Theorem](#) (p. 220); [Youden F1 bound](#) (p. 372); [reliability weight](#) (p. 298); [calibration](#) (p. 52).

token

concept. Entry id token.

DEFINITION

A piece of text, roughly a word, the unit a language model reads and writes. Chapter 0 section 0.11.

EQUATION

Book equation 0.22.

$$\text{PPL} = 2^H, \quad H = -\frac{1}{T} \sum_{t=1}^T \log_2 p(w_t | w_{<t}).$$

Book equation 0.23.

$$\text{softmax}(z)_i = \frac{e^{z_i}}{\sum_j e^{z_j}}, \quad \text{output} = \sum_i \text{softmax}\left(\frac{q \cdot k_i}{\sqrt{d}}\right)_i v_i.$$

Book equation 11.1.

$$\cos(k, \hat{k}) = 0.995 \quad \text{while} \quad \text{PPL} : 12.24 \rightarrow 10643.$$

LEDGER

none

FIRST STATED

Chapter 0 section 0.11 of *Data Mining as Observation*, with the program's serving measurements in `geometric-observation/experiments/GO-kv-serving-flip-NOTES.md`.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 13 section 13.6	attempt one, six gates, 4 pass, K3 0.075 vs 0.15, K4 0.050 vs 0.10, arms 0.925, 0.925, 0.900, 0.850, recon ratio 1.000000, n 40, mechanism 15 to 36 times	<code>geometric-observation/experiments/GO-kv-serving-flip-NOTES.md:1-40</code> ; <code>geometric-observation/prereg/GO-P-2026-056-kv-serving-flip.md:1-40</code> ; <code>geometric-observation/results/GO-KV-serving-flip-7B.json</code>
chapter 13 section 13.6	attempt two, broken proxy, control did not exist, identifier burned, 16 prompts, 1360 s	<code>geometric-observation/prereg/GO-P-2026-077-kv-consumer-relative.md:1-15</code> ; <code>geometric-observation/results/G013-kvaw-pilot-disclosed.json</code>
chapter 13 section 13.6	attempt three, windows 1024, 256, 32, uncertainty 0.982 to 0.892, 5 of 6, 0.4375 with SE 0.070 at 5 percent keep vs 0.30, 97 percent eviction, oracle-miss 0.370 vs 0.25, V4 0.078 vs 0.0625, contrast 0.359 with SE 0.068, n 64, seed 20260812, 89 duty cycles	<code>geometric-observation/claims/LEDGER.md</code> row GO-2/GO-12/GO-13 operational; <code>geometric-observation/prereg/GO-P-2026-077-kv-consumer-relative.md</code> ; <code>geometric-observation/results/G013-kvaw2-governed.json</code>

FAILURES AND CORRECTIONS

none

CONDITIONS

- A piece of text, roughly a word, the unit a language model reads and writes. A model reads a sequence of tokens and outputs a probability for the next one, and perplexity is two to the power of the average bits per token, so it is at least one and equals the vocabulary size for a uniform guess.

- The key-value cache stores one key and one value per token per head, and a generation of 512 tokens under a wrong codebook was where the retracted degradation curve was measured. Token counts are the budget in every serving row of chapter 13.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/Perplexity.lean`, `theorems bits_nonneg`, `one_le_perplexity`, `perplexity_uniform`, `perplexity_mono`, `bits_of_finding`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 4, 8, 11, 12, 13.

RELATED

[perplexity](#) (p. 250); [attention](#) (p. 23); [KV cache](#) (p. 190); [rotary position embedding](#) (p. 307); [teacher forcing](#) (p. 347).

trace

concept. Entry id `trace`.

DEFINITION

The sum of a matrix's diagonal, which for a covariance is the total variance. Equation 0.3.

EQUATION

Book equation 0.3.

$$\Sigma_{ij} = \mathbb{E}[(x_i - \mu_i)(x_j - \mu_j)], \quad \text{tr } \Sigma = \sum_i \Sigma_{ii}.$$

Book equation 0.10.

$$d_O = \text{tr}(P_C M_\delta) = \mathbb{E}[\delta^\top P_C \delta], \quad M_\delta = \mathbb{E}[\delta \delta^\top], \quad P_C = I \Rightarrow d_O = \text{tr } M_\delta.$$

LEDGER

- OT-7. The damage form, trace pairing, rank, and loading covariance are $\text{GL}(d)$ -invariant under $P' = A^{-\top} P A^{-1}$, while spectrum, effective rank, principal angles, and water-filling are $O(d)$ -only — consumer-weighted damage is a geometric scalar, and P3's cliff cannot be bought down by reparameterization. [demonstrated]. `geometric-observation/claims/LEDGER.md:30` at 9f3829f.
- OT-2. Loading is a covariance, not a distance: reading error under a change of measure is priced by $\varepsilon \cdot \|E[h \cdot A]\|$ — predicted from the base measure alone — and a full-magnitude shift orthogonal to the operator's variation does nothing. [predicted]. `geometric-observation/claims/LEDGER.md:33` at 9f3829f.

FIRST STATED

Chapter 0 section 0.2 of *Data Mining as Observation*, with the trace pairing of Volume 14's invariance row OT-7.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 2 section 2.2	read subspace small, operator local, pullback composition, rank cannot increase	geometric-observation\chapters\ch05_the_read_metric_and_the_o48; geometric-observation\chapters\ch06_mathematical_preliminaries27
chapter 2 section 2.2	read distortion controls but is not a complete rank statistic, twelve of twelve, one middle pair misordered, NEG-9	geometric-observation\chapters\ch05_the_read_metric_and_the_o75

FAILURES AND CORRECTIONS

none

CONDITIONS

- The sum of a matrix's diagonal. It is linear, the trace of the identity is the dimension, the trace of an outer product is the vector's squared length, the trace of a product does not depend on the order, and the trace of a read operator times a rank-one error is the read distortion of that error.
- For a covariance the trace is the total variance, and the trace pairing of a read operator with a second moment is the invariant the ledger's row OT-7 names, unchanged by a change of basis where the spectrum and effective rank are not.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/Trace.lean`, `theorems trace_add`, `trace_smul`, `trace_one`, `trace_vecMulVec`, `trace_mul_comm`, `trace_read`, `trace_identity_read`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 1, 3, 4, 13.

RELATED

[covariance matrix](#) (p. 85); [read distortion](#) (p. 281); [identity reader](#) (p. 171); [effective rank](#) (p. 115).

transaction

concept. Entry id transaction.

DEFINITION

One row of a market-basket table, the set of items bought together. Chapter 5 section 5.1.

EQUATION

Book equation 5.1.

$$s(X) = \frac{\sigma(X)}{N}, \quad c(X \rightarrow Y) = \frac{\sigma(X \cup Y)}{\sigma(X)} = \frac{s(X \cup Y)}{s(X)}.$$

Book equation 5.3.

$$X \subseteq Y \implies s(X) \geq s(Y).$$

LEDGER

none

FIRST STATED

Agrawal, Imieliński, and Swami, mining association rules, 1993, as chapter 5 section 5.1 of *Data Mining as Observation* reads it.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 5 section 5.1 to 5.3, 5.5	support, confidence, lift, the Apriori principle, FP-growth, closed and maximal itemsets, objective measures, Simpson's paradox, cross-support	TSK 2e chapter 5

FAILURES AND CORRECTIONS

none

CONDITIONS

- One row of a market-basket table, the set of items bought together. The support of an itemset counts the transactions that contain it, it cannot rise as the itemset grows, and the support fraction lies in the unit interval.
- A transaction is the unit the consumer reads in chapter 5, and which items count as the same is the quotient the pattern report has to name.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/ItemSet.lean`, theorems `supportFrac_mem_unit`, `supportFrac_empty`, `supportFrac_anti`, `supportFrac_union_le`, `closed_of_maximal`, at `observation-data-mining af776fd`.

`lean/DataMiningAsObservation/SafePruning.lean`, theorems `support_anti`, `apriori`, `subset_of_frequent`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 1, 5, 13.

RELATED

[itemset](#) (p. 180); [support](#) (p. 343); [confidence](#) (p. 70); [anti-monotonicity](#) (p. 21).

transmission time interval

concept. Entry id `transmission-time-interval`.

DEFINITION

The one-millisecond slot in which a 5G tower schedules a transmission, abbreviated TTI. Chapter 13.

EQUATION

Book equation 0.25.

$$T_{\text{coh}} = \frac{0.423}{f_D}.$$

Book equation 13.2.

$$T_{\text{coh}} = \frac{0.423}{f_D},$$

refuted fit: $\phi \approx 0.177 T_{\text{coh}} (R^2 = 0.915)$, sealed: $\phi \leq 6$ TTI at 10 Hz, 1 TTI at ≥ 50 Hz.

LEDGER

none

FIRST STATED

Chapter 0 section 0.13 of *Data Mining as Observation*, with the program's sealed floors in observation-theory-campaigns/experiments/RADIO-FRESHNESS-TRACK.md:20-35.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 13 section 13.3	radio 0.27 to 0.42 naive to 0.055 to 0.13, neural reconstruction 0.28 to 0.13	observation-theory-campaigns\experiments\RADIO-FRESHNESS-TRACK.md:20-35
chapter 13 section 13.4	refuted fit 0.177 T_{coh} R^2 0.915, 0.15 threshold vs 0.10 claim, fresh baseline 0.113, do not cite	observation-theory-campaigns\analysis\csi\CSI-refreshfloor.json:2,111-114; observation-theory-campaigns\experiments\RADIO-FRESHNESS-TRACK.md:41-47

FAILURES AND CORRECTIONS

none

CONDITIONS

- The one-millisecond slot in which a 5G tower schedules a transmission, abbreviated TTI. A duration of m milliseconds spans m slots, slots add over consecutive durations, and a longer duration spans more of them, which is the arithmetic the refresh floors are counted in.
- The sealed floors were 4, 1, and 1 slots at Doppler frequencies of 10, 50, and 400 hertz, and the earlier fit of the floor as a fixed fraction of the coherence time was refuted and is not to be cited.

Conditions are curated in entries.toml rather than read from a record.

MACHINE CHECKED

lean/DataMiningAsObservation/TTI.lean, theorems slots_of_ms, slots_add, slots_mono, slots_coherence, at observation-data-mining af776fd.

USED IN

Data Mining as Observation chapters 0, 13.

RELATED

[block error rate](#) (p. 42); [coherence time](#) (p. 63); [refresh floor](#) (p. 292); [freshness](#) (p. 149).

truncation

instrument. Entry id truncation.

DEFINITION

Keeping the first k components of a spectrum and dropping the rest, a bet that the consumer reads the top of the spectrum. Chapter 4 section 4.5.

EQUATION

Book equation 4.2.

$$D(b) = \sum_i s_i \sigma_i^2 2^{-2b_i}, \quad s_i = v_i^\top P_C v_i, \quad b_i^* = \max\left(0, \frac{1}{2} \log_2 \frac{s_i \sigma_i^2}{\theta}\right), \quad \sum_i b_i^* = B,$$

Book equation 0.7.

$$r_{\text{eff}} = \frac{(\sum_i \lambda_i)^2}{\sum_i \lambda_i^2}.$$

Book equation 11.4.

$$\text{directions resolved}(k) = \begin{cases} 1 \text{ or } 2, & k < d \\ \text{rank } P_C, & k \geq d \end{cases} \quad \text{cost} = 2d \text{ consumer calls per operating point.}$$

LEDGER

- GO-4. Fixed-budget verdicts invert under budget-matched observation, per the wavelength mechanism. [replicated]. geometric-observation/claims/LEDGER.md:66 at 9f3829f.

FIRST STATED

Chapter 4 section 4.5 of *Data Mining as Observation*, with the truncation claim in turboquant-pro/CLAIMS.md:28-49 and the GloVe table in turboquant-pro/benchmarks/RESULTS_glove.md:1-40.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 4 section 4.5	GloVe table, 73 percent at 64 components, 0.685 vs 0.862 and 0.866, 0.906 at matched bytes, 0.989 at 37 bytes, 768 to 256 keeps about 99 percent, the 95 percent rule	turboquant-pro/benchmarks/RESULTS_glove.md:1-40
chapter 4 section 4.5	truncation claim reproducible, loses on compact sets	turboquant-pro/CLAIMS.md:28-49
chapter 11 section 11.5	9.6x at recall 0.999 CI-gated on GloVe 1.18M; 32x at 0.9993 on private 199k LaBSE, ties OPQ, beats RaBitQ, 20x build; 27.7x and 114x reported; PCA truncation loses on compact sets; 20x at 199k and 4x at 1M over OPQ; RaBitQ builds in under a second	turboquant-pro/CLAIMS.md:28-49

FAILURES AND CORRECTIONS

none

CONDITIONS

- Keeping the first k components of a spectrum and dropping the rest. The variance kept grows with k and reaches one at full rank, and the error is the sum of the dropped eigenvalues, so truncation is a bet that the consumer reads the top of the spectrum.
- Truncation to 64 components kept 73 percent of GloVe's variance and lost 0.685 against 0.862 downstream, is reproducible, and loses on compact sets.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/PCA.lean`, theorems `varAlong_basis`, `varAlong_le`, `varAlong_ge`, `dropped_eq`, `dropped_nonneg`, at `observation-data-mining af776fd`.

`lean/DataMiningAsObservation/ExplainedVariance.lean`, theorems `explained_mem_unit`, `explained_mono`, `explained_full`, `retained_identity`, `retained_example`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 4, 11.

RELATED

[principal component analysis](#) (p. 260); [explained variance](#) (p. 129); [budget](#) (p. 46); [spectrum](#) (p. 334); [concentrated](#) (p. 69).

V

vacuity threshold

result. Entry id vacuity-threshold.

DEFINITION

The value of a certificate’s statistic below which the certificate is uninformative, derived from the problem rather than tuned to the data. Chapter 9.

EQUATION

Book equation 9.3.

$$\hat{\mu} = \frac{\bar{s}_{\text{true}} - \bar{s}_{\text{distr}}}{\sigma_{\text{distr}}}, \quad \mu_{\text{crit}} = \mathbb{E}\left[\max_{N-1} \mathcal{N}(0, 1)\right], \quad \rho = \frac{\hat{\mu}}{\mu_{\text{crit}}}, \quad \rho = 1 \text{ vacuous.}$$

LEDGER

- GO-3. The certificate’s vacuity threshold predicts where single-stage retrieval dies. [demonstrated].
geometric-observation/claims/LEDGER.md:65 at 9f3829f.

FIRST STATED

Volume 14, the GO-3 registration and its notes, geometric-observation/experiments/GO3-certificate-vacuity-v3-NOTES.md:1-60, DOI 10.5281/zenodo.21776291, and chapter 9 section 9.3 of *Data Mining as Observation*.

MEASUREMENTS

Where the book states it	Numbers, as the book’s sources table records them	Source
chapter 9 section 9.3	the margin certificate, mu crit as the expected maximum of N minus 1 standard normals, rho, death at 0.948 within 6 percent, Spearman 0.991 vs 0.873, fourteen corpora, six gates, the v1 to v3 path, the standing correction	geometric-observation\experiments\GO3-certificate-vacuity-v3-NOTES.md:1-60; geometric-observation\claims\LEDGER.md row GO-3

FAILURES AND CORRECTIONS

none

CONDITIONS

- The threshold is derived from the problem, the expected maximum of the distractors’ margins, and is not tuned to the data.
- A certificate at or below the threshold proves nothing about the object. It has not shown that no structure exists, only that this test could not certify one at this resolution.
- Tested under seal on retrieval, which is recognition with one candidate per query, and the transition has a finite width.

Conditions are curated in entries.toml rather than read from a record.

MACHINE CHECKED

lean/DataMiningAsObservation/Vacuity.lean, theorems maxOver_mono, rho_eq_one_iff, rho_lt_one_of_lt, at observation-data-mining af776fd.

USED IN

Data Mining as Observation chapters 8, 9.

RELATED

[certificate](#) (p. 55); [rank certificate](#) (p. 275); [recognizer](#) (p. 288).

validity index

concept. Entry id validity-index.

DEFINITION

A score for a clustering computed without labels. Chapter 9.

EQUATION

Book equation 9.1.

$$\text{SSE} = \sum_{k=1}^K \sum_{i \in \mathcal{C}_k} \|x_i - c_k\|^2, \quad \text{the } P_C = I \text{ distortion summed within clusters.}$$

Book equation 0.30.

$$\text{SSE} = \sum_k \sum_{i \in \mathcal{C}_k} \|x_i - c_k\|^2, \quad s_i = \frac{b_i - a_i}{\max(a_i, b_i)}.$$

LEDGER

- GO-3. The certificate's vacuity threshold predicts where single-stage retrieval dies. [demonstrated]. [geometric-observation/claims/LEDGER.md:65 at 9f3829f](#).

FIRST STATED

TSK chapter 7 on cluster validity, as chapter 9 section 9.1 of *Data Mining as Observation* reads it, an output metric that needs a null.

MEASUREMENTS

none

FAILURES AND CORRECTIONS

none

CONDITIONS

- A score for a clustering computed without labels. The silhouette is the case, in the interval from minus one to one, positive exactly when a row is closer to its own cluster, and unchanged by a common rescaling of the distances.
- A validity index is an output metric and needs a null, since it is computed under the identity reader on the distances it is given, which is why chapter 9 asks the recognizer to name the manifold or certify that none is present.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/Silhouette.lean`, `theorems silhouette_mem`, `silhouette_pos_iff`, `silhouette_scale`, `silhouette_self`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 1, 2, 3, 4, 6, 7, 8, 9, 10, 11, 12, 13, 14.

RELATED

[silhouette](#) (p. 324); [recognizer](#) (p. 288); [vacuity threshold](#) (p. 359); [null model](#) (p. 234).

variance

concept. Entry id variance.

DEFINITION

The mean squared deviation of a column from its mean, the identity reader’s measure of spread. Chapter 0 section 0.3.

EQUATION

Book equation 0.5.

$$\Sigma v_i = \lambda_i v_i, \quad \Sigma = \sum_{i=1}^d \lambda_i v_i v_i^\top, \quad v_i \cdot v_j = 0 \ (i \neq j).$$

Book equation 0.7.

$$r_{\text{eff}} = \frac{(\sum_i \lambda_i)^2}{\sum_i \lambda_i^2}.$$

Book equation 9.1.

$$\text{SSE} = \sum_{k=1}^K \sum_{i \in \mathcal{C}_k} \|x_i - c_k\|^2, \quad \text{the } P_C = I \text{ distortion summed within clusters.}$$

LEDGER

- OT-7. The damage form, trace pairing, rank, and loading covariance are GL(d)-invariant under $P' = A^{-\top} P A^{-1}$, while spectrum, effective rank, principal angles, and water-filling are O(d)-only — consumer-weighted damage is a geometric scalar, and P3’s cliff cannot be bought down by reparameterization. [demonstrated]. [geometric-observation/claims/LEDGER.md:30 at 9f3829f](#).

FIRST STATED

Chapter 0 section 0.3 of *Data Mining as Observation*, with the program’s spectrum tools in `readscope/readscope/spectrum.py:35-70`.

MEASUREMENTS

Where the book states it	Numbers, as the book’s sources table records them	Source
chapter 4 section 4.2	effective rank as participation ratio, energy rank	<code>readscope\readscope\spectrum.py:35-70</code>
chapter 9 section 9.4	explained ratios, effective rank 5.19 of 8, convergence with a second method, property of the representation not the space	<code>gtc-prototype\docs\SPECTRUM_FINDINGS.md:10-18</code>

FAILURES AND CORRECTIONS

none

CONDITIONS

- The weighted mean of the squared deviation of a column from its weighted mean. It is nonnegative, equals the mean square less the squared mean, is zero for a constant column, scales with the square of a rescaling, and with positive weights is zero only for a constant column.
- Variance is the identity reader's measure of spread. A direction with large variance and no group structure pulls principal components and k-means centres alike, and whether a consumer reads that direction is a separate question.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/Variance.lean`, theorems `variance_nonneg`, `variance_eq`, `variance_const`, `variance_smul`, `eq_mean_of_variance_eq_zero`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 1, 2, 3, 4, 6, 7, 8, 9, 10, 12, 14.

RELATED

[covariance matrix](#) (p. 85); [explained variance](#) (p. 129); [standardization](#) (p. 338); [spectrum](#) (p. 334); [effective rank](#) (p. 115).

verdict

concept. Entry id `verdict`.

DEFINITION

The outcome of a registered test, pass, fail, or abstain, taken as the worst group with the groups too thin to score counted. Chapter 1 section 1.5 and chapter 10 section 10.4.

EQUATION

Book equation 10.7.

$$\text{verdict} = \min_{i: n_i \geq n_{\min}, |Q_i| \geq q_{\min}} \text{score}_i, \quad \text{ABSTAIN otherwise.}$$

Book equation 14.5.

$$\text{FC}_g = \Pr[y = \text{violation} \mid \hat{y} = \text{clear}, g], \quad \text{verdict} = \max_{g: n_g \geq n_{\min}} \text{FC}_g, \quad \text{ABSTAIN otherwise.}$$

Book equation 8.3.

$$\Pr[\text{at least one of } m \text{ null tests passes}] = 1 - (1 - \alpha)^m, \quad \alpha_{\text{Bonferroni}} = \frac{\alpha}{m}.$$

LEDGER

- GO-3. The certificate's vacuity threshold predicts where single-stage retrieval dies. [demonstrated]. `geometric-observation/claims/LEDGER.md:65` at `9f3829f`.
- NEG-14. (GO-P-2026-037, prospective) `a2_probe.median_unit_displacement` is a single-statistic predictor of the flip regime (`unit_disp ≥ 1.0 ⇒ generic-polar-flip` vs `needs-blind-probe`). [refuted]. `geometric-observation/claims/LEDGER.md:108` at `9f3829f`.

FIRST STATED

Chapter 1 section 1.5 and chapter 10 section 10.4 of *Data Mining as Observation*, with the six classes in `geometric-observation/PROTOCOL.md:58-75`.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 1 section 1.5	the six classes and the ledger rule	geometric-observation\PROTOCOL.md:58-75; geometric-observation\OBSERVATION.md:43-45
chapter 3 section 3.5	query mass best single feature at every K for all four responses, seven features add at most 20 percent, threshold 1.25 on three of four, zero of four, 1000 real queries, 1024 dimensions	openvector-bench\results\R13_STAGE0_RESULT.md:40-50; openvector-bench\results\R13_STAGE1_RESULT.md:1-40
chapter 8 section 8.10	the six classes	geometric-observation\PROTOCOL.md:58-75
chapter 9 section 9.2	12 of 12, dimensions 1.81, 2.80, 1.74, angular Spearman ranges, eccentricity spreads, verdict confirmed, GO-P-2026-041	the-angular-observer\experiments\manifold-recovery\battery_result.json
chapter 10 section 10.2	five kinds, balanced accuracy 0.611, 0.718, 0.684, the category table, per-category 0.57 to 0.82, the sweep, pigeonhole floor	openvector-bench\results\R13_STAGE1_RESULT.md:40-75
chapter 10 section 10.2	half 2 failed, at most 1.28 times against 2, withdrawal and its scope	openvector-bench\results\R13_STAGE1_RESULT.md:75-110
chapter 11 section 11.6	query mass best single feature at every K for all four responses, seven features add at most 20 percent at 12 leaves, threshold 1.25 on three of four, zero of four, 1000 real queries, 1024 dimensions	openvector-bench\results\R13_STAGE0_RESULT.md:40-50; openvector-bench\results\R13_STAGE1_RESULT.md:1-40

FAILURES AND CORRECTIONS

- NEG-14, [refuted]. (GO-P-2026-037, prospective) `a2_probe.median_unit_displacement` is a single-statistic predictor of the flip regime ($\text{unit_disp} \geq 1.0 \Rightarrow \text{generic-polar-flip vs needs-blind-probe}$).

CONDITIONS

- The outcome of a registered test, pass, fail, or abstain, taken as the worst group with the groups too thin to score counted. A verdict at one bar implies the verdict at every lower bar, and an abstention is reported as a verdict and not dropped.
- A verdict is relative to a budget and does not transfer to another, and the same two codes get opposite verdicts from two output metrics, so a verdict names its bar, its null, its budget, and its consumer.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/Bar.lean`, `theorems passes_anti, passes_mono, discriminates_iff, no_bar_of_null_ge, exists_bar_of_lt, vacuous_of_null_passes`, at `observation-data-mining af776fd`.

`lean/DataMiningAsObservation/Abstention.lean`, `theorems abstain_not_passes, verdict_eq_none_iff, passes_verdict_iff, inf_le_of_subset`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 1, 2, 3, 4, 6, 7, 8, 9, 10, 11, 12, 13, 14.

RELATED

[bar](#) (p. 34); [abstention](#) (p. 9); [ledger class](#) (p. 204); [min-over-strata](#) (p. 217); [certificate](#) (p. 55).

W

water-filling

concept. Entry id water-filling.

DEFINITION

The allocation of a bit budget across directions that gives each direction half the log of its sensitivity-weighted variance over a common water level, and nothing to directions under the water. Equation 0.13.

EQUATION

Book equation 0.13.

$$D(b) = \sum_i s_i v_i 4^{-b_i}, \quad b_i = \max\left(0, \frac{1}{2} \log_2 \frac{s_i v_i}{\theta}\right), \quad \sum_i b_i = B.$$

Book equation 4.2.

$$D(b) = \sum_i s_i \sigma_i^2 2^{-2b_i}, \quad s_i = v_i^\top P_C v_i, \quad b_i^* = \max\left(0, \frac{1}{2} \log_2 \frac{s_i \sigma_i^2}{\theta}\right), \quad \sum_i b_i^* = B,$$

LEDGER

none

FIRST STATED

Shannon’s power allocation across channels, applied with a consumer’s sensitivity in place of a signal’s power in readscope and turboquant-pro, and chapter 0 section 0.7 and chapter 4 section 4.2 of *Data Mining as Observation*.

MEASUREMENTS

Where the book states it	Numbers, as the book’s sources table records them	Source
chapter 2 section 2.3	refusal regimes, selection consumers read order, recurrences compound	readscope\readscope\regimes.py:1-60
chapter 2 section 2.4	C-7, sixteen points in 128 dimensions, rank fifteen, near ten to the eleventh nats, refuses when samples do not exceed dimension, warns below five per dimension, loading is a property of two distributions	readscope\SPEC.md:653-680
chapter 2 section 2.4	C-7b, 92.1 percent, fit range 0.89 to 91.64, 15 percent inside, maximum 1.9e12, endpoint attenuation 0.437, 202 of 240 at 1.0, refuses to extrapolate, loading is a warning	readscope\SPEC.md:680-720; readscope\readscope\loading.py:136-219
chapter 2 section 2.6	0.647 published, 1.000 recovered, weighted vs unweighted median 0.796 range 0.678 to 0.985, probe 1.000 on every cell, 36 cells, one cell at 0.985	readscope\SPEC.md:410-450

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 2 section 2.6	C-10, 24 to 576 queries, reference rank 24 to 128, 0.821 to 0.703 vs 0.647, 68 percent of distance closed, 0.174 to 0.056, probe 1.000000 on 16 cells, residual uncontrolled	readscope\SPEC.md:450-500
chapter 3 section 3.4	chance overlap rank over dim reported with every reading	readscope\README.md:200-222; readscope\readscope\metrics.py:31-99
chapter 4 section 4.2	water-filling formula, directions below the water get no bits, the surrogate caveat	readscope\readscope\allocate.py:1-100
chapter 4 section 4.2	effective rank as participation ratio, energy rank	readscope\readscope\spectrum.py:35-70
chapter 4 section 4.4	GO-4 budget inversion, fixed m 10 rises, matched m 121, 126, 159 collapses, 3 seeds	geometric-observation\claims\LEDGER.md row GO-4
chapter 4 section 4.5	GloVe table, 73 percent at 64 components, 0.685 vs 0.862 and 0.866, 0.906 at matched bytes, 0.989 at 37 bytes, 768 to 256 keeps about 99 percent, the 95 percent rule	turboquant-pro\benchmarks\RESULTS_glove.md:1-40
chapter 6 section 6.1	selection consumers have zero sensitivity almost everywhere	readscope\readscope\regimes.py:1-60
chapter 8 section 8.3	C-8 D5 mean absolute error 0.0000, every reading 1.000, C-7b saturation, the F-21 rule	readscope\CALIBRATION.md F-21 (about line 519)
chapter 8 section 8.4	C-11c paired null, rank 2 positional 0.385 vs null 0.572, 0.615 vs 0.224, 14 of 16 cells, scope 16 cells one 3B model 192 positions	readscope\SPEC.md:806-857; readscope\calibration\records\c11c-operator-drift.json
chapter 11 section 11.7	k over d table 1, 2, 2, 16, 16, 16	readscope\README.md:118-141; readscope\SPEC.md:243-285, record readscope\calibration\records\c2e-budget-law.json
chapter 11 section 11.7	rank-independent, 0.646 vs 0.366 at half dimension, never 0.90 below k equals d	readscope\CALIBRATION.md:419-434 F-15
chapter 11 section 11.7	C-15 equal budget, 3072 observations, medians 0.02, 0.11, 0.05, 0.04, 1.0, 1.0, margin 0.883 vs 0.3, 0.062 to 0.057 at 8x	readscope\SPEC.md:285-323, record readscope\calibration\records\c15-budget-surface.json, readscope\calibration\DECLARATION-C15.md
chapter 11 section 11.7	confinement theorem, side information moves the cliff to d minus k0, noisy cliff proved then measured	readscope\PRINCIPLES.md:112-142; geometric-observation\crucible\OT3-THEOREM.md; geometric-observation\crucible\OT3-NOISY-THEOREM.md
chapter 11 section 11.7	124 cells at resolution 1.000 at k over d 1.25, five families, linearity partial	readscope\README.md:200-222

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 11 section 11.9	drift at rank one 0.667 vs null 0.933, sixteen cells	readscope\CALIBRATION.md:600-660 F-24; readscope\calibration\records\c11c-operator-drift.json
chapter 11 section 11.9	C-12 four bars, 40 documents, 512 tokens, 13.4 point difference, teacher forcing removes it, negative 0.015 vs 0.005, Spearman negative 0.13 at p 0.45, sign test p 0.42, verdict FAIL, feedback compounding	readscope\calibration\records\c12-longgen-drift-sym.json; readscope\calibration\DECLARATION-C12.md at commit 90e2ce2; readscope\SPEC.md:806-825

FAILURES AND CORRECTIONS

none

CONDITIONS

- The closed form assumes a uniform quantizer at high rate, so that each bit quarters the squared error, and directions taken as the eigenvectors of the covariance so that their errors add.
- The sensitivity of each direction is read in that same eigenbasis, as the quadratic form of the read operator along the eigenvector, and pairing the eigenvalues of the two matrices is valid only when they share an eigenbasis.
- Coarse quantizers, unequal codebook steps, and errors not spread evenly depart from the formula, and chapter 4 reports where a measured allocation did.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/WaterFilling.lean`, `theorems contribution_eq_water`, `terms_mul`, `two_sqrt_le`, `distortion2_ge`, `distortion2_eq_of_equal`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 0, 4.

RELATED

[read operator](#) (p. 282); [read distortion](#) (p. 281); [alignment](#) (p. 11); [the flip](#) (p. 138).

Weyl's law

concept. Entry id `weyls-law`.

DEFINITION

The rule for how the number of eigenvalues below a value grows with that value, whose exponent reveals the dimension of the space. Equation 0.31.

EQUATION

Book equation 0.31.

$$N(\lambda) = \#\{k : \lambda_k \leq \lambda\} \sim C_d \lambda^{d/2}.$$

Book equation 9.2.

$$N(\lambda) = \#\{k : \lambda_k \leq \lambda\} \sim C_d \lambda^{d/2} \quad \Rightarrow \quad d = 2 \frac{d \log N}{d \log \lambda}.$$

Book equation 0.19.

$$L = I - D^{-1/2}AD^{-1/2}, \quad D = \text{diag}(d_1, \dots, d_n), \quad Lu_k = \lambda_k u_k, \quad 0 = \lambda_1 \leq \lambda_2 \leq \dots.$$

LEDGER

- GO-3. The certificate’s vacuity threshold predicts where single-stage retrieval dies. [demonstrated].
geometric-observation/claims/LEDGER.md:65 at 9f3829f.

FIRST STATED

Weyl, 1911, as chapter 0 section 0.15 of *Data Mining as Observation* states it, with the program’s dimension reading in geometric-observation/chapters/ch11_the_recognizer.md:1-95.

MEASUREMENTS

Where the book states it	Numbers, as the book’s sources table records them	Source
chapter 9 section 9.2	the recognizer’s mechanism, low multiplets and angular distances, dimension before shape by Weyl’s law, refusal, the growth gotcha	geometric-observation\chapters\ch11_the_recognizer.md:1-95
chapter 9 section 9.2	12 of 12, dimensions 1.81, 2.80, 1.74, angular Spearman ranges, eccentricity spreads, verdict confirmed, GO-P-2026-041	the-angular-observer\experiments\manifold-recovery\battery_result.json

FAILURES AND CORRECTIONS

none

CONDITIONS

- The rule that the number of eigenvalues below a value grows like that value to the power of half the dimension, so the dimension is in the exponent and is read from the slope of the count against the value on log axes. Doubling the value multiplies the count by two to the half dimension.
- Chapter 9 reads a dataset’s dimension from its spectrum before reading its shape, and the recognizer’s battery recovered dimensions of 1.81, 2.80, and 1.74 on three sealed templates. The finer geometry is in the constant and the multiplets, not the exponent.

Conditions are curated in entries.toml rather than read from a record.

MACHINE CHECKED

lean/DataMiningAsObservation/IntrinsicDimension.lean, theorems log_weyl, dimension_from_slope, weyl_double, at observation-data-mining af776fd.

USED IN

Data Mining as Observation chapters 0, 9.

RELATED

[intrinsic dimension](#) (p. 175); [manifold](#) (p. 210); [Laplacian](#) (p. 199); [multiplet](#) (p. 223); [recognizer](#) (p. 288).

whitening

concept. Entry id whitening.

DEFINITION

Rescaling data so that its covariance becomes the identity. Equation 0.6.

EQUATION

Book equation 0.6.

$$x_w = \Sigma^{-1/2}(x - \mu), \quad \Sigma^{-1/2} = \sum_i \lambda_i^{-1/2} v_i v_i^\top.$$

Book equation 0.5.

$$\Sigma v_i = \lambda_i v_i, \quad \Sigma = \sum_{i=1}^d \lambda_i v_i v_i^\top, \quad v_i \cdot v_j = 0 \ (i \neq j).$$

LEDGER

- GO-2 (neg. half: not reconstruction). At matched bits, downstream preservation is not controlled by reconstruction error. [demonstrated]. [geometric-observation/claims/LEDGER.md:63](#) at 9f3829f.

FIRST STATED

Chapter 0 section 0.2 of *Data Mining as Observation*, with the program's whitened code in Volume 14 chapter 8, [geometric-observation/chapters/ch08_value.md:84-97](#).

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 2 section 2.5	whitened code wins at every budget	geometric-observation\chapters\ch08_value.md:84-97
chapter 4 section 4.3	whitened code on whale 0.83, 0.85, 0.97 vs 0.41, 0.80, 0.89	geometric-observation\chapters\ch08_value.md:84-97

FAILURES AND CORRECTIONS

none

CONDITIONS

- Rescaling data so that its covariance becomes the identity. In the covariance's eigenbasis each coordinate is divided by the square root of its eigenvalue, each whitened coordinate then has unit variance, and the squared length of a whitened row is the Mahalanobis form.
- Whitening is a change of reader, not a change of data. The whitened code won at every budget on the whale corpus because the consumer read the whitened directions, and the same code under a different consumer has no such guarantee.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

[lean/DataMiningAsObservation/Whitening.lean](#), theorems `whiten_unit_variance`, `whiten_sq_sum`, at [observation-data-mining af776fd](#).

USED IN

Data Mining as Observation chapters 0, 2, 4, 10.

RELATED

[Mahalanobis distance](#) (p. 210); [identity reader](#) (p. 171); [read operator](#) (p. 282); [water-filling](#) (p. 365).

witness

concept. Entry id witness.

DEFINITION

An independent measurement of whether a certificate’s claim was true. Chapter 13.

EQUATION

Book equation 0.26.

$$FC = \Pr [W \text{ refutes } | \mathcal{C}_t \text{ clears}], \quad \text{coverage} = \Pr [\mathcal{C}_t \text{ clears}].$$

Book equation 13.1.

$$FC = \Pr [W \text{ refutes } | \mathcal{C}_t \text{ clears}], \quad d_O(\Delta) = \text{tr} (P_C M_{\text{drift}}(\Delta)), \quad M_{\text{drift}}(\Delta) = \mathbb{E}[\delta_\Delta \delta_\Delta^\top].$$

LEDGER

none

FIRST STATED

Volume 14, chapter 19, [geometric-observation/chapters/ch19_the_certificate_that_ages.md:61-76](#), and the freshness program’s umbrella document [observation-theory-campaigns/experiments/FRESHNESS-PROGRAM.md](#).

MEASUREMENTS

Where the book states it	Numbers, as the book’s sources table records them	Source
chapter 12 section 12.6	XPROTO-LLM, benchmark 0.909, 0.920, 0.909, thirty slices, target 0.8, naive 0.333, aware 0.033, spread 0.380, deployment mean 0.736, six bars on three seeds, sealed 2026-08-25 at b61f7f1	observation-theory-campaigns\experiments\LLM-EVAL-TRACK.md:1-50; observation-theory-campaigns\analysis\llm\XPROTO-LLM-graded.json; observation-theory-campaigns\experiments\SEALS.md:85
chapter 13 section 13.3	ZooKeeper hot 0.99 cold 0.01 witnessed 0.0, Postgres 0.50 to 0.06, MongoDB 0.47 to 0.03, production Postgres 0.47 to 0.02, real substrates, disjoint seeds	observation-theory-campaigns\experiments\DATABASE-FRESHNESS-TRACK.md:1-45
chapter 13 section 13.3	BGP 0.351, IS-IS 0.184, OSPF 0.083, second collector	observation-theory-campaigns\analysis\mongo\PREREG-XPROTO-MG.md:42; observation-theory-campaigns\experiments\ROUTING-TELEMETRY-TRACK.md:130-150; geometric-observation\BOOK-OUTLINE.md:75
chapter 13 section 13.3	radio 0.27 to 0.42 naive to 0.055 to 0.13, neural reconstruction 0.28 to 0.13	observation-theory-campaigns\experiments\RADIO-FRESHNESS-TRACK.md:20-35

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 13 section 13.4	refuted fit 0.177 T_coh R ² 0.915, 0.15 threshold vs 0.10 claim, fresh baseline 0.113, do not cite	observation-theory-campaigns\analysis\csi\CSI-refreshfloor.json:2,111-114; observation-theory-campaigns\experiments\RADIO-FRESHNESS-TRACK.md:41-47

From observation-theory-campaigns/experiments/DATABASE-FRESHNESS-TRACK.md at 9d86211.

- line 23. XPROTO-PG (analysis/pgrep) | Postgres, recovery_min_apply_delay | WAL LSN | ~0.50 → ~0.06
- line 24. XPROTO-MG (analysis/mongo) | MongoDB delayed secondary | oplog ts | ~0.47 → ~0.03
- line 25. XPROTO-PGX (analysis/pgx) | production PG, netem lag | WAL LSN, pg_stat_statements footprint | ~0.47 → ~0.02
- line 27. XPROTO-ZK (analysis/zk) | ZooKeeper 3.9 ensemble | zxid, sync() | hot 0.99 / cold 0.01, witnessed 0.0

From observation-theory-campaigns/experiments/RADIO-FRESHNESS-TRACK.md at 9d86211.

- line 25. XPROTO-CSI (analysis/csi) | CQI → MCS | HARQ | 0.34–0.37 → 0.10 (OLLA) | pass 08-23
- line 26. XPROTO-BEAM (analysis/beam) | mmWave beam index | HARQ | 0.31 → 0.02 (BFR) | pass 08-23
- line 27. XPROTO-AICSI (analysis/aicsi) | neural-CSI recon (turboquant bridge) | precoder/HARQ | recon wins yet 0.28 → 0.13 | pass 08-23, scope-corrected 08-25 warning
- line 28. XPROTO-HO (analysis/ho) | RSRP → serving cell | RLF | 0.31–0.44 → 0.09–0.12 | pass 08-23
- line 30. XPROTO-PHY (analysis/phy) | PMI / RI / TA | HARQ | 0.27–0.42 → 0.055–0.13 | pass 08-24

FAILURES AND CORRECTIONS

none

CONDITIONS

- A witness is independent of the certificate it grades. A certificate that grades itself has no witness.
- The witness is the consumer's own outcome or a measurement of it, so the same certificate has a different witness for each consumer, as the hot and cold readers of one replica show.

Conditions are curated in entries.toml rather than read from a record.

MACHINE CHECKED

lean/DataMiningAsObservation/Certificate.lean, theorems falseClear_mul_coverage, coverage_empty, falseClear_mem_unit, minOverStrata_passes_iff, minOverStrata_le_weighted_mean, at observation-data-mining af776fd.

USED IN

Data Mining as Observation chapters 0, 12, 13, 14.

RELATED

[certificate](#) (p. 55); [false-clear rate](#) (p. 133); [coverage](#) (p. 86).

Y

Youden F1 bound

correction. Entry id youden-f1-bound.

DEFINITION

none

EQUATION

Book equation 5.4.

$$F_1^{\max} \leq \sup_{t \in [J, 1]} \frac{2t\pi}{t\pi + \pi + (t - J)(1 - \pi)}, \quad J = \max_{\tau} (\text{TPR} - \text{FPR}), \quad \pi = \text{prevalence},$$
$$J \leq 2A - 1 \text{ when the ROC curve is concave, and not in general.}$$

LEDGER

none

FIRST STATED

The AUROC form, theorem “AUROC-F1 Bound” in theory-radar/paper/astar_paper.tex and the derivation in theory-radar/src/symbolic_search/_auroc_proof.py, repository DOI 10.5281/zenodo.20660206. The corrected form at theory-radar 1b3d105.

MEASUREMENTS

Where the book states it	Numbers, as the book’s sources table records them	Source
chapter 5 section 5.4	Monotone Invariance and AUROC invariance theorems, the AUROC to F1 bound via the Youden index, stated in the source for every ROC curve and corrected in the book to concave curves, with the counterexample at AUROC 0.75 and F1 0.857	theory-radar\paper\astar_paper.tex:94-170; theory-radar\paper\theory_radar_paper.tex:290-304
chapter 5 section 5.4	seven thresholds up to 0.75, eight datasets, 56 conditions, zero admissibility violations, reductions 70 to 99.1 percent	theory-radar\paper\astar_paper.tex:170-200
chapter 5 section 5.4	the rerun of 2026-09-03 and 2026-09-04, pre-filter counts 2576 of 8700 and 413 of 1560, ceiling violations 812 and 412, Youden pruning 199 and 38, A* optima 0.958 and 0.694 in strict and fast modes, Youden mode short by 0.006 and 0.002	theory-radar\results\rerun_youden_2026_09_03.json; theory-radar\ERRATA.md
chapter 6 section 6.3	Monotone Invariance Theorem and its proof	theory-radar\paper\theory_radar_paper.tex:290-304; theory-radar\paper\astar_paper.tex:94-110

Rerun of the A* paper’s pruning tables by theory-radar/rerun_youden_2026_09_03.py. Part A is depth-2 pairwise enumeration on eight datasets, exhaustive against the AUROC pre-filter at seven thresholds and against Youden

pruning. Part B is the depth-3 A* search with 50,000 expansions in strict mode (no pre-filter), the published fast mode (AUROC pre-filter at 0.52), and the youden mode.

Part A, recorded at commit aedc053 of theory-radar. | Dataset, depth 2 pairs | Pairs | Exhaustive F1 | AUROC pre-filter at 0.75, evaluated, admissible | Youden pruning, evaluated, admissible | Pairs whose F1 exceeds the AUROC ceiling | |---|---|---| | Circles (2) | 20 | 0.9960 | 4, yes | 7, yes | 14 | | Moons (2) | 20 | 0.8330 | 2, yes | 2, yes | 3 | | Breast Cancer (30) | 8700 | 0.9542 | 2576, yes | 199, yes | 812 | | Wine (13), class 0 vs rest | 1560 | 0.9204 | 413, yes | 38, yes | 412 | | Synthetic (10) | 900 | 0.9556 | 10, yes | 7, yes | 18 | | Synthetic (20) | 3800 | 0.9460 | 8, yes | 8, yes | 57 | | Synthetic (30) | 8700 | 0.9637 | 8, yes | 8, yes | 61 | | Synthetic (40) | 15600 | 0.9425 | 32, yes | 8, yes | 125 |

Part B. | Dataset | strict F1 and formula | fast F1 | youden F1 and formula | expansions strict, fast, youden | |---|---|---| | Circles (2) | 0.9965, (x1 hypot x2) | 0.9965 | 0.9965, (x1 hypot x2) | 1626, 540, 87 | | Moons (2) | 0.8927, (neg(x1) max x2) | 0.8927 | 0.8927, (neg(x1) max x2) | 1626, 1386, 86 | | Synthetic (20) | 1.0000, (x0 hypot x1) | 1.0000 | 1.0000, (x0 hypot x1) | 50000, 50000, 465 | | Breast Cancer (10) | 0.9578, ((f1 min f2) + f7) | 0.9578 | 0.9516, ((f3 + f7) min f6) | 50000, 50000, 841 | | Diabetes (8) | 0.6937, ((d5 min d7) + d1) | 0.6937 | 0.6921, ((d1 + d7) min d5) | 50000, 46307, 192 |

FAILURES AND CORRECTIONS

- theory-radar/ERRATA.md:3-75 at 37c4e6c. ## 2026-09-03. The AUROC to F1 bound holds only for concave ROC curves **What was claimed.** Theorem “AUROC-F1 Bound” in paper/astar_paper.tex, the derivation in src/symbolic_search/_auroc_proof.py, and the pruning bounds in run_astar_v2.py and src/symbolic_search/_heuristic_dag.py all rested on the statement that any ROC curve with area A has maximum Youden index $J = \max(\text{TPR} - \text{FPR})$ at most $2A - 1$. **Why it is wrong.** The statement holds for concave ROC curves (equivalently, scores whose likelihood ratio is monotone), because a concave curve lies above the two chords through its Youden point and so has area at least $(1 + J)/2$. It fails for non-concave curves. Counterexample, found by an external reviewer of the companion textbook: rank three quarters of the positives above every negative and the remaining quarter below every negative. AUROC is 0.75, the maximum Youden index is 0.75, and at prevalence one half the best thresholded F1 is 0.857, above the 0.80 the AUROC form of the ceiling allows. The sound ceiling for that score, from $J = 0.75$, is 0.889, and the score sits under it. The `auroc_f1_bound` formula in `run_astar_v2.py`, $2A\pi/(A\pi + (1 - A)(1 - \pi))$, is not a valid upper bound either: at prevalence 0.1 the same score reaches F1 0.857 against a “bound” of 0.5. **Consequence for reported results.** The searches that used these bounds to prune could have discarded a candidate formula whose ROC curve was not concave and whose true best F1 exceeded the computed ceiling. The formulas and wins reported by those runs are therefore what a pruned search found, and are lower bounds on what an unpruned search would find. The Monotone Invariance Theorem and the results that do not depend on the AUROC bound are unaffected. **Correction.** The sound form of the ceiling substitutes the measured maximum Youden index of the candidate’s own scores for $2A - 1$. It holds for every score, costs the same sort as AUROC, and is what `max_f1_for_youden` and `youden_f1_bound` now compute. The AUROC form is retained under its original names with a docstring stating the concavity condition, and the pruning sites now use the measured index. A rerun of the published searches under the corrected bound was done on 2026-09-03 and 2026-09-04, `rerun_youden_2026_09_03.py`, results in `results/rerun_youden_2026_09_03.json`, run on the Atlas workstation, CPU only. **Rerun, part A, the pre-filter admissibility table (tab:auroc).** Depth-2 pairwise enumeration on the eight datasets, exhaustive against the AUROC pre-filter at the seven published thresholds, with the sound Youden ceiling beside it. The evaluation counts under the pre-filter reproduce the published table (Breast Cancer 8700 pairs, 2576 evaluated at 0.75; Wine 1560, 413; Circles 20, 4; Moons 20, 2). The pre-filter lost no optimum on any dataset at any threshold, which is what the paper reported. That was luck of the data, not the theorem: the retracted ceiling sat below the F1 a pair formula actually reach

CONDITIONS

- The Youden form holds for every score with a finite sample.
- The AUROC form holds for concave ROC curves, and as a pruning bound it is safe only there.
- Inside a tree search either form bounds the formula itself and not its descendants, so pruning a subtree by its root’s ceiling is a heuristic in both cases.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/YoudenF1.lean`, `theorems f1_eq, f1_le_of_youden, auroc_eq, youden_eq, youden_exceeds_auroc_form`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 5.

RELATED

[Monotone Invariance Theorem](#) (p. 220); [formula search](#) (p. 145); [safe pruning](#) (p. 311); [Apriori principle](#) (p. 22).

Youden index

instrument. Entry `id youden-index`.

DEFINITION

True positive rate minus false positive rate at a threshold, whose maximum over thresholds is the pointwise ceiling that prunes the formula search. Chapter 5 section 5.4.

EQUATION

Book equation 5.4.

$$F_1^{\max} \leq \sup_{t \in [J, 1]} \frac{2t\pi}{t\pi + \pi + (t - J)(1 - \pi)}, \quad J = \max_{\tau} (\text{TPR} - \text{FPR}), \quad \pi = \text{prevalence},$$
$$J \leq 2A - 1 \text{ when the ROC curve is concave, and not in general.}$$

Book equation 0.28.

$$P = \frac{TP}{TP + FP}, \quad R = \frac{TP}{TP + FN}, \quad F_1 = \frac{2PR}{P + R}.$$

Book equation 0.38.

$$w = \max(0, 2 \cdot \text{AUROC} - 1).$$

LEDGER

none

FIRST STATED

Youden, index for rating diagnostic tests, 1950, as chapter 5 section 5.4 of *Data Mining as Observation* reads it, with the pointwise ceiling in `theory-radar/paper/astar_paper.tex:94-110`.

MEASUREMENTS

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 5 section 5.4	seven thresholds up to 0.75, eight datasets, 56 conditions, zero admissibility violations, reductions 70 to 99.1 percent	<code>theory-radar\paper\astar_paper.tex:170-200</code>
chapter 5 section 5.4	learned pruning 88 to 99.6 percent with zero false negatives	<code>theory-radar\README.md:88-100</code>

Where the book states it	Numbers, as the book's sources table records them	Source
chapter 6 section 6.3	Monotone Invariance Theorem and its proof	theory-radar\paper\theory_radar_paper.tex:290-304; theory-radar\paper\astar_paper.tex:94-110
chapter 6 section 6.3	88 to 99.6 percent pruned with zero false negatives, fair protocol, 200 by 5 folds	theory-radar\README.md:88-100

FAILURES AND CORRECTIONS

none

CONDITIONS

- True positive rate minus false positive rate at a threshold, whose maximum over thresholds bounds the best F1 the score can reach at any threshold. The bound is exact for concave ROC curves and fails for a curve at AUROC 0.75.
- The pointwise ceiling on a node of the formula search is proved, and using it to prune the node's descendants is a heuristic that missed the optimum on two of five datasets in the rerun.

Conditions are curated in `entries.toml` rather than read from a record.

MACHINE CHECKED

`lean/DataMiningAsObservation/YoudenF1.lean`, `theorems f1_eq, f1_le_of_youden, auroc_eq, youden_eq, youden_exceeds_auroc_form`, at `observation-data-mining af776fd`.

USED IN

Data Mining as Observation chapters 5.

RELATED

[ROC curve](#) (p. 306); [AUROC](#) (p. 29); [F1](#) (p. 132); [safe pruning](#) (p. 311).

Entries by kind

Concepts, 157. [abstention](#) (p. 9); [aggregation](#) (p. 10); [alignment](#) (p. 11); [anisotropic](#) (p. 14); [anti-hub](#) (p. 17); [anti-monotonicity](#) (p. 21); [attention](#) (p. 23); [attribute type](#) (p. 25); [AUROC](#) (p. 29); [bag of words](#) (p. 32); [balanced accuracy](#) (p. 33); [bi-Lipschitz](#) (p. 37); [bit](#) (p. 38); [block error rate](#) (p. 42); [budget](#) (p. 46); [cache](#) (p. 51); [calibration](#) (p. 52); [capacity](#) (p. 53); [certificate](#) (p. 55); [chance level](#) (p. 58); [chunk](#) (p. 59); [classifier](#) (p. 60); [codebook](#) (p. 61); [coherence time](#) (p. 63); [cold, warm](#) (p. 65); [commute time](#) (p. 68); [concentrated](#) (p. 69); [confidence](#) (p. 70); [confidence interval](#) (p. 71); [confound](#) (p. 72); [consumer](#) (p. 73); [contrastive objective](#) (p. 76); [correction](#) (p. 79); [correlation](#) (p. 80); [cosine](#) (p. 82); [coupling null](#) (p. 84); [covariance matrix](#) (p. 85); [coverage](#) (p. 86); [cross-support ratio](#) (p. 88); [curvature](#) (p. 91); [decision boundary](#) (p. 94); [degree](#) (p. 97); [density](#) (p. 99); [deployment mismatch](#) (p. 101); [disparate impact ratio](#) (p. 106); [distance concentration](#) (p. 107); [distortion](#) (p. 109); [dot product](#) (p. 110); [drift](#) (p. 112); [effective rank](#) (p. 115); [eigenvalue, eigenvector](#) (p. 116); [embedding](#) (p. 118); [encoder](#) (p. 120); [equalized odds](#) (p. 122); [escalation](#) (p. 123); [Euclidean distance](#) (p. 124); [eviction](#) (p. 126); [explained variance](#) (p. 129); [F1](#) (p. 132); [false-clear rate](#) (p. 133); [floor, ceiling](#) (p. 142); [footprint](#) (p. 144); [freshness](#) (p. 149); [Gaussian](#) (p. 152); [geodesic distance](#) (p. 154); [graph](#) (p. 156); [harness](#) (p. 159); [head](#) (p. 162); [Hessian](#) (p. 164); [hub](#) (p. 166); [identity reader](#) (p. 171); [intrinsic dimension](#) (p. 175); [isotropic](#) (p. 179); [item-set](#) (p. 180); [Jaccard](#) (p. 182); [Jacobian](#) (p. 183); [Kendall correlation](#) (p. 185); [kernel](#) (p. 186); [KL divergence](#) (p. 187); [KV cache](#) (p. 190); [lag](#) (p. 193); [language model](#) (p. 196); [Laplacian](#) (p. 199); [leakage](#) (p. 203); [ledger class](#) (p. 204); [lift](#) (p. 205); [Mahalanobis distance](#) (p. 210); [manifold](#) (p. 210); [margin](#) (p. 213); [metric](#) (p. 216); [min-over-strata](#) (p. 217); [multiple comparisons](#) (p. 221); [multiplet](#) (p. 223); [nat](#) (p. 228); [nearest neighbour](#) (p. 230); [neighbourhood graph](#) (p. 231); [nuisance](#) (p. 233); [observer](#) (p. 236); [operating point](#) (p. 238); [orthogonal](#) (p. 239); [outer product](#) (p. 240); [outlier](#) (p. 242); [output metric](#) (p. 243); [paraphrase class](#) (p. 246); [percentile](#) (p. 247); [perplexity](#) (p. 250); [posited versus measured](#) (p. 254); [positive semidefinite](#) (p. 256); [precision, recall](#) (p. 257); [projection](#) (p. 263); [p-value](#) (p. 264); [quantization](#) (p. 266); [query, key, value](#) (p. 267); [query set](#) (p. 269); [quotient](#) (p. 271); [rank](#) (p. 274); [rank-faithful](#) (p. 277); [read direction](#) (p. 279); [read distortion](#) (p. 281); [read operator](#) (p. 282); [read subspace](#) (p. 285); [recall at k](#) (p. 286); [reconstruction error](#) (p. 290); [refusal](#) (p. 295); [registered](#) (p. 296); [reliability weight](#) (p. 298); [replica](#) (p. 299); [retrieval-augmented pipeline](#) (p. 303); [rotary position embedding](#) (p. 307); [ROUGE](#) (p. 309); [safe pruning](#) (p. 311); [score](#) (p. 315); [seed](#) (p. 318); [sensitivity](#) (p. 319); [silhouette](#) (p. 324); [Simpson's paradox](#) (p. 325); [skewness](#) (p. 326); [softmax](#) (p. 327); [Spearman correlation](#) (p. 330); [spectrum](#) (p. 334); [standard error](#) (p. 337); [sum of squared errors](#) (p. 341); [support](#) (p. 343); [teacher forcing](#) (p. 347); [TF-IDF](#) (p. 350); [threshold](#) (p. 350); [token](#) (p. 351); [trace](#) (p. 353); [transaction](#) (p. 354); [transmission time interval](#) (p. 355); [validity index](#) (p. 360); [variance](#) (p. 361); [verdict](#) (p. 362); [water-filling](#) (p. 365); [Weyl's law](#) (p. 367); [whitening](#) (p. 368); [witness](#) (p. 370).

Instruments, 71. [allocation](#) (p. 12); [anti arm](#) (p. 15); [anti-hub recall](#) (p. 19); [attribution](#) (p. 26); [audit](#) (p. 27); [bagging](#) (p. 31); [bar](#) (p. 34); [baseline](#) (p. 35); [blind probe](#) (p. 39); [boosting](#) (p. 44); [bootstrap](#) (p. 45); [challenge set](#) (p. 57); [commit hash](#) (p. 67); [contraction](#) (p. 75); [control](#) (p. 77); [cross-corpus gate](#) (p. 87); [cross-validation](#) (p. 89); [DBSCAN](#) (p. 93); [decision tree](#) (p. 95); [declaration](#) (p. 96); [detector](#) (p. 102); [direction-only quantizer](#) (p. 104); [discretization](#) (p. 105); [early stopping](#) (p. 114); [ensemble](#) (p. 121); [expected calibration error](#) (p. 128); [fine-tuning](#) (p. 135); [finite difference](#) (p. 137); [formula search](#) (p. 145); [gate](#) (p. 151); [hash](#) (p. 161); [hierarchical clustering](#) (p. 165); [importance](#) (p. 172); [imputation](#) (p. 174); [inverted file](#) (p. 177); [k-means](#) (p. 189); [latent semantic analysis](#) (p. 201); [linear classifier](#) (p. 206); [logistic regression](#) (p. 208); [matched bits](#) (p. 214); [naive Bayes](#) (p. 226); [null model](#) (p. 234); [paired](#) (p. 245); [per-channel quantizer](#) (p. 248); [planted](#) (p. 252); [Poisson ceiling](#) (p. 253); [preregistration](#) (p. 258); [principal component analysis](#) (p. 260); [product quantization](#) (p. 261); [random forest](#) (p. 273); [rank certificate](#) (p. 275); [recognizer](#) (p. 288); [refresh interval](#) (p. 293); [rerank](#) (p. 300); [residualization](#) (p. 302); [Robin Hood index](#) (p. 305); [ROC curve](#) (p. 306); [sampling](#) (p. 313); [sealed](#) (p. 316); [shard](#) (p. 322); [sources table](#) (p. 329); [spectral clustering](#) (p. 331); [spectral embedding](#) (p. 332); [split](#) (p. 335); [standardization](#) (p. 338); [stratification](#) (p. 339); [surrogate](#) (p. 344); [sweep](#) (p. 345); [template match](#) (p. 348); [truncation](#) (p. 356); [Youden index](#) (p. 374).

Results, 7. [Apriori principle](#) (p. 22); [budget cliff](#) (p. 49); [the flip](#) (p. 138); [hubness](#) (p. 168); [Landauer's principle](#) (p. 194); [Monotone Invariance Theorem](#) (p. 220); [vacuity threshold](#) (p. 359).

Corrections, 3. [Nadeau and Bengio correction](#) (p. 225); [refresh floor](#) (p. 292); [Youden F1 bound](#) (p. 372).

References, 1. [session deck](#) (p. 320).

Entries by chapter of the book

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Provenance

This document was built on 2026-09-06 from commit 43db613 of ahb-sjsu/observation-theory-campaigns by `encyclopedia/build_pdf.py`. The entries were generated 2026-09-06 by `encyclopedia/generate.py` from geometric-observation 9f3829f, observation-theory-campaigns 9d86211, theory-radar 37c4e6c, observation-data-mining af776fd, turboquant-pro 856c4cb, gtc-prototype 328741f, readscope c8d0289.

The hand-filled entries, the flip, the false-clear rate, and the Youden F1 bound, were written on 2026-09-03 to fix the schema and are the generator's acceptance test. `encyclopedia/check.py` verifies that every number in each of them appears in its generated counterpart and that every cited path exists.